

# TFPT — Changelog

All changes and new results, by date

Stefan Hamann

Alessandro Rizzo

July 29, 2026

This file is the canonical, dated record of every change to the TFPT theory development: new verification modules (vN), status changes, editorial passes and infrastructure. It is maintained *in the same change* as the work it records, and it is also imported at the end of the `introduction` document. Module numbers vN refer to `verification/vN_*.py`; claim IDs refer to rows of `verification/status_ledger.csv`.

**2026-07-29 (II) (v547 — the level-lemma identities of phase 2 promoted; prime-front diary T146 (level lemma, LEVEL-CARRIES) + website sync; NO marker moves)**

- **New module v547 (PRIME.LEVEL.LEMMA.01).** Promotion of the a-priori core of the discovery part T146 (`level_lemma_probe.py`) — 20 checks,  $\sim 0.4$ s, everything recomputed on small frame-A windows, nothing cited from the sandbox; the companion to v546, whose item (5) certified  $S1'$  with a  $c_0$  *measured* at the minimiser's own layer cake — this module supplies the a-priori side, every factor a functional of the form alone. Scope fence: identities, per-instance theorems with checked hypotheses and per-window *certified* inequalities — no fit, no  $D$ -exponent, no zone-uniform statement, and explicitly *no statement for all  $D$*  (the asymptotic delocalisation of the Green columns, T146's remainder, stays open and typed open). Battery: 12 frame-A windows ( $n = 4 \dots 139$ ,  $m = 26 \dots 285 \leq 300$ ,  $D = 7.578 \times 10^{-3} \dots 2.789 \times 10^{-2}$ ), 9 random positive definite forms for the form-independent items, and a stress ladder  $m = 64 \dots 288$ ; each statement carries a mutation control that must fail by  $\geq 10^{-3}$ .
- **(1) The union identity for nested cakes (T146) [E].** For a nested decreasing chain with positive coefficients,  $\sum_{j,l} c_j c_l |S_j \cup S_l| = 2T \|\psi_t\|_1 - \|\psi_t\|^2$  with  $T = \sum_j c_j$  and  $\psi_t = \sum_j c_j \mathbf{1}_{S_j}$  — form-independent and  $O(m)$ , the reason the cake base is movable at all; exact (0.0) against T145's  $K \times K$  double sum on all 12 window minimisers and 9 random vectors, *and* against the union sizes computed from the sets themselves with no nestedness used (controls: one shuffled proper level set  $3.2 \times 10^{-2}$ , a 1% perturbation of the heaviest coefficient  $4.8 \times 10^{-3}$ ).
- **(2) The layer-cake counting lemma at a general base (T146) [E].**  $G(b) \leq 2b^2 \|v\|_\infty \|v\|_1 / \|v\|^2 + \varepsilon(b)$  for *any* vector at all 6 preregistered bases (margin  $\geq 1.08$  on the battery and on every window minimiser); at  $b = 2$  the leading constant is the classical dyadic 8 *exactly*, and the base is *free* because the chain consumes only the lower domination  $|v| \leq \psi_t$  — the a-priori constant gains  $4/b_*^2 = 3.7612$  over base 2 on every window (controls: the bound  $\times 10^{-3}$  fails on every draw and base, and the one-level-down cake breaks the domination on every draw — the +1 in the cake exponent is load-bearing).
- **(3) The resolvent delocalisation bound (T146) [E].**  $\psi = \lambda R \psi$  to  $9.0 \times 10^{-14}$  — an *identity*, so no Davis–Kahan transfer step appears anywhere — with  $\lambda_{\text{up}} = \min_j \text{Rayleigh}(Re_j) \geq \lambda_{\text{min}}$  (Rayleigh–Ritz; window ratio 1.0592–1.2883), the coordinate bound  $|\psi_j| \leq \lambda \|Re_j\|$  (Cauchy–Schwarz) and the column norms certified from above with the linear-solve residual paid for, giving  $\|\psi\|_\infty \leq \lambda_{\text{up}} \max_j \|Re_j\|$  on every window (within 1.061–1.337 of the truth) and  $\|\psi\|_1 \leq$

$\sqrt{m} \min(1, \Gamma_1)$ . The lever: the trivial ceiling  $\max_j \|Re_j\| \leq 1/\lambda_{\min}$  would give  $\Gamma \leq \sqrt{m} = 5.1$ – $16.9$  (useless); the certified columns sit a factor  $2.85$ – $10.17$  *below* it (preregistered bar  $2.0$ ), giving  $\Gamma = 1.8356$ – $2.2797$  — read off the form, never off the minimiser (controls: a  $1\%$   $\lambda$  perturbation breaks the identity at  $1.0 \times 10^{-2}$ , the sup bound  $\times 10^{-3}$  fails on every window).

- **(4) The composite level lemma (T146) [E].**  $c_0^{\text{ap}} = 2b^2 \Gamma \min(1, \Gamma_1) + \varepsilon = 3.9042$ – $4.8488$  dominates the measured level constant on all 12 windows (slack  $1.986$ – $2.418$ ) and stays under the preregistered ceiling  $16 = 4 \times \text{Maz'ya's Dirichlet value } 4$ ; the certified window inequality  $\text{cert\_lam\_max}(R) \leq c_0^{\text{ap}} \Psi_{\text{abs}}$  holds on 12/12 (both sides by completed Cholesky, the density the  $|R|$  one with Charikar's cited constant), closing the chain  $\hat{\lambda} \geq 1/(c_0^{\text{ap}} \Psi_{\text{abs}})$ ,  $\text{gap} \geq \hat{\lambda}/\kappa_{\text{up}}$  at loss factor  $0.0423$ – $0.1272$  of the exact gap. The built-in discriminator: on T145's no-go form  $R = aa^T + \varepsilon I$ ,  $a_i = i^{-1/2}$ ,  $\Gamma$  grows  $3.676 \rightarrow 6.798$  over  $m = 64 \dots 288$  (ratio  $1.85 \geq$  the preregistered  $1.5$ ) and  $c_0^{\text{ap}} = 11.10$  *exceeds* every window value — the bound must be worse there, since a candidate that did not fail on the no-go would prove a false statement — while the theorem half survives on every no-go size and the Dirichlet path-Laplacian control stays flat (max/min  $1.0063$ ; control:  $c_0^{\text{ap}} \times 10^{-3}$  breaks the certificate on every window).
- **Fences and status.** Per instance on small windows only — a finite list of certified window inequalities with an explicit maximum; nothing uniform in the zone index or in  $D$ ; only the sign-free / Green constant is bounded a priori, the energy-route  $\sigma_{\text{tot}}$  stays *measured*, and every T146 exponent is a fit that stays in the sandbox. Maz'ya 1985 (the strong-type half *not* used as a theorem anywhere), Rayleigh 1877 / Ritz 1909, Cauchy–Schwarz, Miclo 1999, Davis–Kahan 1970 (cited, deliberately *not* used), Perron–Frobenius (not applicable to  $E$ , not used), Fukushima–Ōshima–Takeda 1994, Charikar 2000, Goldberg 1984, Gershgorin 1931, Wilkinson 1968 / Higham 2002 named classical; Weil 1952 cited, never used as a criterion; zero-firewall AST-checked. Ledger row `PRIME.LEVEL.LEMMA.01` [E]; **no marker of any pre-existing contract moves**, and this is not RH evidence.
- **Diary, paper and website (T146, level-carries).** The T146 probe (`level_lemma_probe.py`, contract `LEVEL.LEMMA`, 30/30, 7.6 s, 64 windows  $m = 26 \dots 1173$ , 9/9 preregistered verdict conditions) closes L1 on the measurement surface as a chain of theorems and certified window inequalities with no step reading the minimiser: four hand-written profiles lose by 5–9 orders of magnitude, the Green column survives ( $\lambda_{\text{up}}/\hat{\lambda} = 1.0497$ – $1.3764$ ), Davis–Kahan is instrumented and discarded (informative on 0/64; the spectrum bottom is a near-degenerate block,  $\lambda_2/\hat{\lambda} = 1.2542$ – $3.4471$ ), the cake base proves free (the dyadic 8 falls to  $2b^2 \approx 2$ , gain 3.76 at zero cost) and  $c_0^{\text{ap}} = 3.9042$ – $4.8488$  on 64/64 windows lands at the size of Maz'ya's classical Dirichlet value 4, closing the chain at loss factor  $0.0422$ – $0.1586$ ; the one genuine remainder is  $D$ -uniformity for *all*  $D$  — the asymptotic delocalisation of the Green columns. `experiments/next.txt` part T146 (series totals 146 probes / 3689 checks); `tfpt_prime_front.tex` gains the T146 subsection with the “After T146” keybox, the v547 entry in the verified layer (now thirteen modules), the phase-2 section retitled T126–T146 and the updated probe index; `tfpt_1`, `tfpt_3` and `tfpt_research_contracts` gain the v547 companion blocks; `website/lib/primeFront.ts` gains the T146 and v547 feed entries; suite counter  $540 \rightarrow 541$  scripts. T147 (`green_asymptotic_probe.py`) is running at exactly the all- $D$  asymptotics and is *not* part of this change.

**2026-07-29 (I) (v546 — the capacity-chain identities of phase 2 promoted; prime-front diary T145 (Maz'ya proof, ONE-STEP-MISSING) + website sync; NO marker moves)**

- **New module v546 (PRIME.CAPCHAIN.IDENT.01).** Promotion of the identity-shaped core of the capacity cycle T142 (`conductance_profile_probe.py`), T143 (`sharp_capacity_probe.py`), T144 (`capacity_inequality_probe.py`) and T145 (`mazya_proof_probe.py`) — 26 checks,  $\sim 0.5$  s, everything recomputed on small frame-A windows, nothing cited from the sandbox. Scope fence: identities, per-instance theorems with checked hypotheses and per-window *certified*

inequalities — no fit, no  $D$ -exponent, no zone-uniform statement, and *no level lemma* (T145's L1 stays open, unclaimed). Battery: 12 windows ( $n = 4 \dots 139$ ,  $h = 26 \dots 285 \leq 300$ ,  $D = 7.578 \times 10^{-3} \dots 2.789 \times 10^{-2}$ , 32–5102 positive off-diagonal edges); the covering kernel is positive definite on 11, where items (1)–(2) run; each identity carries a mutation control that must fail by  $\geq 10^{-3}$ .

- **(1) The capacity decomposition (T142) [E].**  $K^{-1} = D^T J^{-1} D + x x^T / \text{cap}$  with  $J = D K D^T$  (assembled entrywise,  $3.8 \times 10^{-15}$ ),  $x = K^{-1} \mathbf{1}$  and  $\text{cap} = \mathbf{1}^T K^{-1} \mathbf{1}$ , as a matrix identity ( $1.4 \times 10^{-13}$ ); under the congruence  $u = K^{1/2} v$  the Dirichlet half is an *orthogonal projection* ( $P^2 = P$  at  $1.9 \times 10^{-12}$ ,  $\Omega = \lambda_{\max}(P) = 1$  to  $5.1 \times 10^{-12}$  — the T142 normalisation, where T141 had guessed 20.7–2724), its complement *is* the capacity rank-one and it annihilates  $K^{-1/2} \mathbf{1}$  (controls: dropping the rank-one  $3.4 \times 10^{-1}$ , a 1% cap perturbation  $3.4 \times 10^{-3}$ ).
- **(2) The exact capacity–Rayleigh identity (T143) [E].** With  $H = \text{diag}(s) + L_N$  ( $4.7 \times 10^{-16}$ ),  $L_N = D^T B D$  for the signed crossing kernel ( $4.9 \times 10^{-16}$ ) and  $\rho(W) = 1 - \lambda_{\min}(A, A_B)$  ( $5.4 \times 10^{-14}$ ), the assembled quotient  $[d^T (J^{-1} - B) d + (x^T v)^2 / \text{cap} - \sum_k s_k v_k^2] / [d^T J^{-1} d + (x^T v)^2 / \text{cap}]$  equals  $v^T (K^{-1} - H) v / v^T K^{-1} v$  for *every*  $v$  ( $9.5 \times 10^{-14}$  on 24 random vectors per window), reproduces  $1 - \rho(W)$  at the exact top eigenvector ( $1.7 \times 10^{-9}$  under the *declared* 1/gap conditioning bar  $5.4 \times 10^{-8}$ ), and every random quotient stays  $\geq 1 - \rho(W)$ ; on the constant vector the whole denominator *is* the capacity term (controls: dropping it,  $\text{rel} = 1$ ; the  $|N|$  crossing kernel,  $2.6 \times 10^{-1}$ ).
- **(3) Cholesky nestedness (T144) [E].**  $\text{cap}_E([a, a+j])$  equals the prefix sum of squared forward-substitution entries of *one* triangular solve: 3404 interval capacities from 36 solves at  $2.0 \times 10^{-14}$  against independent Schur solves, exhaustive on the median window (every prefix,  $2.8 \times 10^{-14}$ ), monotone in  $b$  by construction (control: the reversed ordering computes *suffix* capacities and deviates by  $\geq 4.6 \times 10^{-1}$ ).
- **(4) The whitening certificate (T144) [E].**  $\lambda_{\max}(W) \leq \kappa_{\text{up}}$  (completed Cholesky)  $\leq$  the Gershgorin ceiling on every window ( $\kappa_{\text{up}} = 1.2245\text{--}1.4236$  against  $\leq 2.1213$ ); the licensed chain direction  $\lambda_{\min}(E) / \kappa_{\text{up}} \leq \lambda_{\min}(A, A_B)$  holds on every window — a denominator comparison, untouched by T142's numerator obstruction — and the Cholesky refuses at the shrunk shift  $0.995 \lambda_{\max}(W)$ .
- **(5) The M4 split and the certified chain (T145) [E].** The mass half of Maz'ya's step M4:  $\sum_k f_k^2 \leq \psi^2$  *pointwise* (a theorem for nonnegative truncations, exact slack  $\geq 0$ ), the row-wise implication on rows with  $(E\mathbf{1})_i \geq 0$ , measured mass ratio  $\leq 0.2995$  and the full step  $\sigma_{\text{tot}} = 0.2197\text{--}0.4425 < 1$  on every window although the Markov hypothesis is absent. Licence 4 is stated with  $|R|$  and *not*  $R^+$ , with the witness  $R = [[1, -\frac{1}{2}], [-\frac{1}{2}, 1]]$ ,  $x = (1, -1)$  ( $3.0 > 2.0$ ) built in as a counterexample check. The sign-free chain is ordered on every window, and  $S1'$  is *certified per window on both routes*:  $\text{cert\_lam\_max}(R) \leq 12 \sigma_{\text{tot}} \Psi_{\text{pos}}$  (energy,  $c_0 = 2.6369\text{--}5.3099$ ) and  $\leq G_{\text{dy}} \Psi_{\text{abs}}$  (sign-free,  $c_0 = 2.5154\text{--}6.1449$ ); the better route gives  $c_0 = 2.5154\text{--}4.2265$  against the classical Dirichlet value 4. The all-sets density bound carries Charikar's cited constant, bracketed by exhaustive enumeration of all 4095 subsets of a small block (control:  $c_0 \times 10^{-3}$  breaks the certificate on every window).
- **Fences and status.** Per instance on small windows only; nothing uniform in the zone index or in  $D$ ; the  $c_0$  of item (5) is *measured* at the minimiser's own layer cake and the a-priori level lemma (T145's L1) stays open. Maz'ya 1985 (the strong-type half *not* used as a theorem anywhere — it is the thing the cycle transcribes), Muckenhoupt 1972, Fukushima–Ōshima–Takeda 1994, Miclo 1999, Charikar 2000, Goldberg 1984, Gershgorin 1931, Wilkinson 1968 / Higham 2002 named classical; Weil 1952 cited, never used as a criterion; zero-firewall AST-checked. Ledger row `PRIME.CAPCHAIN.IDENT.01` [E]; **no marker of any pre-existing contract moves**, and this is not RH evidence.
- **Diary, paper and website (T145, one-step-missing).** The T145 probe (`mazya_proof_probe.py`, contract `MAZYA.PROOF`, 33/33, 8.0s, 64 windows  $m = 26 \dots 1173$ ) transcribes Maz'ya's classical

proof step by step onto the non-Markovian form: M1, M2, M3 are theorems, the Markov property sits in exactly one line (M4), that line splits, its mass half is a theorem and dominates ( $\sigma_{\text{tot}} = 0.215\text{--}0.443 < 1$ ),  $c_0$  becomes explicit (2.248–4.227, fit  $D^{0.028 \pm 0.017}$ , zone-uniform) and S1' is certified on 64/64 windows; an explicit no-go proves a level-profile hypothesis is *necessary*; S2' retired, R5 downgraded, R4 closed — the mathematical rest is *one* named lemma, L1. `experiments/next.txt` part T145 (series totals 145 probes / 3659 checks); `tfpt_prime_front.tex` gains the T145 subsection, the v546 entry in the verified layer (now twelve modules) and the updated probe index; `website/lib/primeFront.ts` gains the T145 feed entry; suite counter 539  $\rightarrow$  540 scripts. T146 (`level_lemma_probe.py`) is running at exactly L1 and is *not* part of this change.

## 2026-07-28 (III) (v545 — the Hardy-core identities of phase 2 promoted; prime-front diary T140/T141 + website sync; NO marker moves)

- **New module v545 (PRIME.HARDY.IDENT.01).** Promotion of the identity blocks of the discovery parts T140 (`signed_band_probe.py`) and T141 (`discrete_hardy_probe.py`) — 36 checks,  $\sim 1$  s, everything recomputed on small frame-A windows, nothing cited from the sandbox. Scope fence: *identities*, closed-form quantities and the *validity* — never the *size* — of two certified upper-bound shapes, plus one negative typing. No fit is promoted, no  $D$ -exponent, no zone-uniform statement, and no bound that beats the target. Battery: 11 windows ( $n = 8 \dots 139$ ,  $h = 58 \dots 285 \leq 300$ , core  $m = 57 \dots 284$ , 193–5102 positive off-diagonal edges, 10–87 loaded stripes),  $D = 7.578 \times 10^{-3} \dots 2.388 \times 10^{-2}$  (a factor 3.15, so item (8) is a comparison and not a single point), exact  $\rho(W) = 0.997759\text{--}0.999981$ ; the explicit  $n_e \times n_e$  signed Gram is formed only on the 5 windows with  $n_e \leq 900$ .
- **(1)–(3) The form lifting, the exact finite core and the energy reordering [E].** With  $M$  the edge–interval incidence matrix and  $C = \text{diag}(\sqrt{\Delta})M$ , the T139 telescope lifts from the *entries* of the signed Gram to its *form*:  $\text{Gram} = CHC^T$  at  $3.0 \times 10^{-15}$ , hence  $\text{rank}(\text{Gram}) \leq h-1$  (measured 57–183 against 193–882 edges; control: unweighted incidence 1.6). The covering kernel in *closed geometric form*,  $K_{rs} = W([r \wedge s, r \vee s])$ , equals  $M^T \Delta M$  ( $1.3 \times 10^{-15}$ ) and the spectral radius is an *exact* eigenvalue of the finite core,  $\rho(W) = \lambda_{\max}(K^{1/2} H K^{1/2}) = 1 - \lambda_{\min}(A, A_B)$  ( $5.4 \times 10^{-14}$ ) =  $\lambda_{\max}(\text{Gram})$  ( $7.8 \times 10^{-16}$ ) — three independent routes, no truncation (control: the factorwise product  $\lambda_{\max}(K)\lambda_{\max}(H)$  misses by  $7.8 \times 10^2$ ). The energy reordering  $H = \text{diag}(s) + L_N$  holds at  $4.7 \times 10^{-16}$ : a mass term plus a long-range Dirichlet form, its weights positive on 0.676–0.872 of the off-diagonal area (a measurement, no law).
- **(4) The checkerboard split [E].** At group length exactly  $b$ ,  $\text{Band}_b = D + A_{\text{even}} + A_{\text{odd}}$  *entrywise* (0.0) on all 9 (window,  $b$ ) pairs over  $b = 4, 8, 16$ , and the resulting three-step Weyl sequence dominates  $\lambda_{\max}(\text{Band}_b)$  with 0 exceptions (worst ratio 1.6940) at a step count independent of the up to 9 groups — three steps instead of  $O(nb)$  (control:  $L = b/2$  loses entries,  $3.5 \times 10^{-1}$ ).
- **(5)–(6) The four Hardy identities and the closed Muckenhoupt quantity [E].** (i) the  $K^+$  Rayleigh form  $\rho(W) = u^T H u / u^T K^+ u$  at  $u = K^{1/2} v$  ( $6.6 \times 10^{-13}$ ); (ii) the signed crossing kernel  $L_N = D^T B D$  with  $B_{kl} = \sum_{r \leq k \wedge l, k \vee l \leq s} N_{rs}$  — the same closed covering formula as  $K$ , on the increment grid ( $4.9 \times 10^{-16}$ ); (iii)  $Q = B_+ \mathbf{1}$  ( $1.4 \times 10^{-15}$ ): the Cauchy–Schwarz weight T140 introduced by hand *is* the row sum of  $B_+$ , so that step is the Gershgorin diagonal step on  $B_+$ ; (iv)  $DKD^T = L_\Delta$  on the interior nodes ( $2.7 \times 10^{-14}$ ) with diagonal  $(\Delta \mathbf{1})_{k+1} = 0.4886\text{--}15.15$ , the endpoint edge mass — a *local* quantity and not  $\lambda_{\max}(K)$  (control: the crossing kernel of  $|N|$  does not reproduce  $L_N$ ,  $2.6 \times 10^{-1}$ ; the signs are load-bearing). The two-weight quantity  $A_{M0} = \max_k Q_k (\Delta \mathbf{1})_{k+1}$  is computed by two closed routes agreeing to  $8.3 \times 10^{-15}$ ; its value (65.5–4244) is *reported*, its size and zone behaviour are not promoted (control:  $\lambda_{\max}(K)$  in place of the local mass inflates it by  $\geq 358$ ).
- **(7) Two certified shapes, promoted as *valid* bounds only [E].** Every Loewner step is certified by a completed Cholesky ( $-1.5 \times 10^{-14}$ ,  $-2.3 \times 10^{-10}$ ). The additive Hardy

chain dominates  $\rho(W)$  on 11/11 windows (factor 3.074–10.758); the exact two-term Weyl split dominates too (1.691–2.503) and lies *below* the chain everywhere, so it is the floor of the whole additive family; the genuinely joint shape  $\Lambda(H, Y) \cdot \Omega(Y)$  with a positive definite Jacobi Hardy form  $Y$  dominates on 11/11 with the closed-form geometric profile (8.815–74.872 at  $\Omega = 8.33$ –36.2) and on 11/11 with the per-zone tridiagonal read-off of  $K^+$ . Promoted is the *dominance* on these instances and nothing about the factors. Controls: the same  $Y$  without its mass term is singular — the constant vector lies in its kernel ( $5.7 \times 10^{-17}$ ) while  $H$  does not vanish there — so the certificate *refuses* on 11/11; and dropping the *positive* part instead of the negative one falls below  $\rho(W)$  on 11/11.

- **(8) The sign-versus-absolute separation — a negative typing [X].** Over the measured  $D$  range the exactly bounded object (the signed Dirichlet share) spreads by  $1.36\times$ , while every absolute variant spreads by at least  $54.94\times$  (Loewner drop 62.03, Cauchy–Schwarz 54.94, product 56.92), and dropping the positive part of  $H$  alone costs  $9.96$ – $464.63\times$ . The corresponding power laws are *fits* and labelled as fits:  $D^{-0.138 \pm 0.094}$  (signed) against  $D^{-0.776 \pm 0.352}$  (certified Muckenhoupt) and  $D^{-2.748}$ ,  $D^{-2.820}$ ,  $D^{-2.918}$  (absolute). Promoted is the instance comparison of the measured spreads, not the exponents.
- **Fences and status.** Per instance on small windows only; nothing uniform in the zone index or in  $D$ ; item (2) is an identity and not a bound, so the zone-uniform discrete Hardy inequality stays *open*; item (7) promotes validity and not size — neither shape beats the target. Abel summation, Cauchy–Schwarz, Weyl 1912, Gershgorin 1931, Haynsworth 1968, Ostrowski 1937 / Fan 1958 / Berman–Plemmons 1979, Muckenhoupt 1972, Bradley 1978, Opic–Kufner 1990, Bennett 1987/1991, Gantmacher–Krein 1950/1960, Karlin 1968, Wilkinson 1968 / Higham 2002 named classical; Weil 1952 cited, never used as a criterion; zero-firewall AST-checked. Ledger row PRIME.HARDY.IDENT.01 [E] + [X]; **no marker of any pre-existing contract moves**, and this is not RH evidence.
- **Diary, paper and website.** experiments/next.txt part T141 (HARDY-RESISTS, 22/22; series totals 141 probes / 3547 checks); tfpt\_prime\_front.tex gains the T141 subsection, the v545 entry in the verified layer and the updated probe index; website/lib/primeFront.ts gains the T141 feed entry and the v545 promotion entry. T142 (conductance\_profile\_probe.py) is running at the one open ingredient and is *not* part of this change.

**2026-07-28 (II) (v543 + v544 — the lumped  $M$ -matrix pair and the long-lag support structure promoted; prime-front diary T136/T137 + website sync; NO marker moves)**

- **New module v543 (PRIME.MMATRIX.IDENT.01).** Promotion of the identity/theorem block of the discovery part T136 (35 checks,  $\sim 1$ s), recomputed on small frame-A windows — nothing cited from the sandbox. Scope fence: matrix identities, per-instance theorems with *checked* hypotheses, and one *negative* per-instance result; no fit, no  $D$ -exponent, no uniform-in-zone claim, no floor value. Battery: 20 seam windows ( $n = 4 \dots 59$ ,  $\rho = 1.25 \dots 4.00$ ,  $h = 25 \dots 299$ ; every factorised or diagonalised matrix  $\leq 300$ ) giving 40 symmetric matrices (pole-free odd sections + border Schur blocks), 39 with positive off-diagonals, plus 312 shifted ladder rungs and 8 whitened two-grid rungs.
- **(1)–(2) The lumped pair and the congruence [E].** With  $\Delta$  the positive off-diagonal part and  $L_\Delta = \text{diag}(\Delta \mathbf{1}) - \Delta$ , the pair  $A_B = A + L_\Delta$  is Stieltjes (off-diagonals exactly  $\leq 0$ ),  $L_\Delta$  is a graph Laplacian hence PSD with zero row sums, so  $A_B \mathbf{1} = A \mathbf{1}$  ( $1.6 \times 10^{-15}$ ) and  $A_B \succeq A$  — an *upper* bound on  $\lambda_{\min}(A)$  and a floor *never* (control: lumping the negative off-diagonals destroys Stieltjes,  $2.0 \times 10^{-1}$ ). The congruence  $A = A_B^{1/2}(I - W)A_B^{1/2}$ ,  $W = A_B^{-1/2}L_\Delta A_B^{-1/2} \succeq 0$ , holds as a *matrix* identity ( $4.4 \times 10^{-15}$ ) with  $\lambda_{\max}(W) = \rho(L_\Delta A_B^{-1})$  ( $4.4 \times 10^{-15}$ ), and the *declared circularity*  $\lambda_{\max}(W) < 1 \iff A \succ 0$  is checked in *both* directions (0 exceptions forward; 39/39

shifted controls in the indefinite range give  $\lambda_{\max}(W) \geq 1$  — the Loewner sandwich is vacuous by itself, and that is the content.

- **(3)–(4) The anchor and the empty shift ladder [E].** With  $x = A_B^{-1}\mathbf{1} \geq 0$  and  $A_B^{-1} \geq 0$  entrywise (Ostrowski 1937, *checked*),  $\lambda_{\min}(A_B) \geq 1/\max_r x_r$  on all 39 rows (min slack  $3.0 \times 10^{-3}$ , sharpest ratio 0.9070) — one solve, one sign check, no eigenvalue (control: a positive off-diagonal makes it false,  $1.9000 > 0.1000$ ). Lumping commutes with diagonal shifts,  $(A - sI)_B = A_B - sI$  ( $2.5 \times 10^{-16}$  on 312 rungs), and  $(A_B - sI)^{-1} \geq A_B^{-1} \geq 0$  entrywise (Berman–Plemmons 1979, violation 0), so  $\max_r x_r(s)$  and the a-priori margin  $\|L_\Delta\|_\infty \max_r x_r(s)$  are non-decreasing along the whole ladder: the shifted ladder is *structurally empty* — a theorem, not a measurement.
- **(5)–(6) Varga’s identity and the a-priori Jacobi radius [E].** For the regular splitting  $A_B = D_0 - N_0$  (all three hypotheses checked on 39/39),  $\rho(J) = \tau/(1 + \tau)$  with  $\tau = \rho(A_B^{-1}N_0)$  at  $4.8 \times 10^{-15}$  ( $\rho(J) = 0.7347\text{--}0.9860$ ); the Collatz–Wielandt bracket at the anchor vector gives  $\min_r (Tx)_r/x_r \leq \tau \leq \max_r (Tx)_r/x_r$  (0 exceptions) and hence an *a-priori*  $\rho(J) \leq 0.9861 < 1$  with no eigenvalue anywhere, sharp on the measured gap by 1.0062–1.0531. Controls: a non-regular splitting breaks the identity ( $1.1 \times 10^{-1}$ ); at a sign-changing test vector the quotient bound is false on 39/39.
- **(7) The  $k$ -scanned M18 majorant — a negative result [X].** The whitened mass identity  $\|L^{-1}\hat{s}\|^2 = \hat{s}^\top U^{-1}\hat{s}$  holds at  $7.2 \times 10^{-16}$  and the triangle chain is a *valid* majorant at every  $k$  separately (39 (rung,  $k$ ) pairs over  $k = 2 \dots 64$ , 0 violations), so the minimum over the ladder is legitimate — and it does *not* reach the preregistered T134 bar 1/2 on any rung (best 3.080; best over the three values T134 tried, also 3.080) while the *measured* bad mass is  $\leq 0.0120$ . The localisation term alone already exceeds the bar (0.542): the loss is the bound, and what is missing is a localisation statement. The bar is *not* moved and nothing is upgraded.
- **New module v544 (PRIME.LONGLAG.SUPP.01).** Promotion of the structure block of the discovery part T137 (24 checks,  $\sim 0.2$ s) on the same 20 windows (3–25 prime-power atoms, 20555 positive off-diagonal edges, 455 loaded stripes, 39 bracketed lumped comparisons).
- **(1)–(4) The support is a comb stripe set, and its amplitude is certified [E].** The kernel splits exactly as  $c = c_{\text{arch}} - c_{\text{comb}}$  with  $c_{\text{comb}} \geq 0$ , and the archimedean section alone has *all* off-diagonals  $\leq 0$  on 20/20 windows: every positive off-diagonal of  $A$  is comb-generated (control: the full section carries  $+2.0 \times 10^{-1}$ ). Every one of the 20555 positive edges has its Hankel index  $M - 1 - r - t$  inside the atom-hat index set — per-window fraction exactly 1.000000 — so  $\text{supp}(\Delta)$  is a union of anti-diagonal stripes at prime-power atom positions (the converse is reported, *not* claimed). Each stripe is a perfect matching (0 endpoint collisions), so its edge vectors  $b_e = e_r - e_t$  are *exactly* orthogonal and  $L_\Delta$  restricted to a stripe is a direct sum of  $2 \times 2$  blocks (control: edges across two stripes overlap). And  $A_{rs} = A_{rs}^{\text{arch}} - c_{\text{comb}}[|r - s|] + c_{\text{comb}}[\text{anti}]$  gives the amplitude certificate  $\Delta_{rt} \leq c_{\text{comb}}[M - 1 - r - t]$  on all 20555 edges (excess  $-1.9 \times 10^{-6}$ , sharpest ratio 0.99998532) — no eigenvalue, no norm; control: the Toeplitz lag in place of the Hankel index breaks it on 20415/20555. Honestly undecided and *not* promoted: at  $h \leq 300$  the hats never overlap, so the tighter one-atom refinement coincides with the certificate here.
- **(5) The Green function, bracketed two-sidedly [E].** For the lumped comparison  $B = D_0 - N_0$ ,  $J = D_0^{-1}N_0 \geq 0$ , the anchor identity  $q_J = \max_r (Jx)_r/x_r = 1 - \min_r 1/(d_r x_r) < 1$  ( $8.2 \times 10^{-16}$ ) needs no measurement; then the Neumann partial sum to  $K = 8$  is an *entrywise* lower bound for  $B^{-1}$  ( $-2.1 \times 10^{-4}$ ) *because the discarded terms are nonnegative*, and partial sum  $+(x_r/x_s)q_J^{K+1}/((1 - q_J)d_s)$  is an entrywise upper bound ( $-1.4 \times 10^{-1}$ ), on 39 rows (20 pole-free sections + 19 border Schur blocks). Control: dropping the tail makes the upper bound false. This delivers item (2) of the T136 rest list.
- **(6) The absolute-value envelope family is certified dead [X].** With  $E = B^{-1}L_\Delta$ , Gershgorin, every row-sum norm and every Collatz–Wielandt quotient at every *positive* weight are all  $\geq \rho(|E|)$ , so one rigorous *lower* bound decides the whole family at once: a Collatz–Wielandt quotient from below at a power-iterated weight gives  $\rho(|E|) \geq 1.3141 \geq 1$  on 20/20 rows

(the cheapest four members all  $\geq 1.5937$ ). No member of the family can ever certify  $\lambda_{\max}(W) < 1$  — a certificate of *death*, not a measurement. The *signed* radius  $\rho(E) = \lambda_{\max}(W) \leq 0.99995781$  on the same rows: the absolute value alone destroys the cancellation, and what survives is a sign-preserving bound that does not exist yet.

- **Named limits as load-bearing content.** Every statement is *per instance on small windows* — nothing uniform in the zone index, in  $D$  or in the gap. The T136  $D$ -exponents (anchor  $D^{-0.565}$ , margin gap  $D^{+2.72}$ , slack  $D^{0.116}$ ) are fits and stay in the sandbox; the structure bound assembled from the support and amplitude facts does *not* beat the margin (T137: 0 of 35 windows) and is *not* promoted; entrywise positivity of the *target* inverse at depth stays open. Stieltjes / Ostrowski 1937 / Fan 1958 / Berman–Plemmons 1979 / Varga 1962 / Frobenius 1912–Perron 1907 / Collatz 1942–Wielandt 1950 / Gershgorin 1931 / Neumann series / Haynsworth 1968 / Wilkinson 1968–Higham 2002 named classical; Demko–Moss–Smith 1984 cited as a model and *not* applied (the comparison is dense); Weil 1952 cited, never used as a criterion; zero-firewall AST-checked; NOT “almost RH”.
- **Diary Teile 136/137 (sandbox).** `m_matrix_pair_probe.py` (contract M.MATRIX.PAIR, 30/30, verdict ONE-CARRIES): the  $\rho(K)$  item closes with Varga’s regular splitting and the Collatz–Wielandt bracket (sharp 1.00–1.03 on 900/900), the exact three-way bookkeeping  $\lambda_{\min}(A) \sim D^{2.27} = \text{anchor } D^{-0.56} \times \text{gap } D^{2.72} \times \text{slack } D^{0.12}$  attributes the whole degradation to the margin (which is only  $2.5 \times 10^{-6}$ – $2.5 \times 10^{-2}$  wide while every cheap bound on  $\rho(E)$  sits at 1.82–3.11), and M17 closes *negatively* (the bad pencil subspace is delocalised, participation ratio 0.54–0.98, its dimension tracking the atom count at  $n_{\text{bad}}/n_{\text{at}} = 1.64$  rather than the grid, so  $\sigma_b(k) \geq 0.646$  everywhere); promotion list 15  $\rightarrow$  22 ready. `long_lag_probe.py` (contract LONG.LAG, 22/22, verdict BOTH-RESIST):  $L_{\Delta}$  is fully comb-generated (the archimedean half has a positive off-diagonal on 0 of 40 windows), the structure bound does *not* beat the margin (0/35, overshoot  $24$ – $7.1 \times 10^5$  gaps, second differences decaying only algebraically as  $\text{dist}^{-0.63}$  where Demko–Moss–Smith would need a geometric rate), the envelope family is certified dead from below ( $\rho(|E|) \geq 1.3218$  on 35/35), and the Green bracket closes two-sidedly (lower recovers 0.011–0.406 of the minimum entry, upper overshoots by 2.49–12.24). Series counter: 137 probes, 3438/3438 sandbox checks. Sandbox only for the probes: no ledger row, no marker move, not RH evidence.
- **Ledger/papers/website.** Two new ledger rows: PRIME.MMATRIX.IDENT.01 and PRIME.LONGLAG.SUPP.01; \veri citations in `tfpt_1`, `tfpt_3`, `tfpt_research_contracts` and the promotion place of `tfpt_prime_front` (verified layer now *ten* modules; phase 2 extended to T126–T137); website mirrors (/prime-front feed entries T136/T137, status surfaces Teile 11–137 / 3438 checks, verification DAG, papers data) updated in the same change.

**2026-07-28 (I) (v542 — the margin-chain identities of phase 2 promoted; v379 hardened to an exact positive-mixture Gram certificate; prime-front diary T132/T133 + website sync; NO marker moves)**

- **New module v542 (PRIME.MARGIN.IDENT.01).** Promotion of the *identity/theorem block* of the discovery parts T128–T131 into one narrow load-bearing module (44 checks,  $\sim 0.2$  s), recomputed on small frame-A windows — nothing cited from the sandbox. Scope fence: identities and *per-instance* theorems only; no fit, no graded floor, no Lanczos value, no uniform-in-zone statement. Battery: 24 seam instances ( $n = 3 \dots 97$ ,  $\rho = 1.25 \dots 4.00$ ,  $h_c = 24 \dots 214$ ,  $h_f = 36 \dots 299$ ; every inverted or diagonalised matrix  $\leq 300$ ), 43 graded/fine grid pairs ( $m \leq 172$ ) and 400 randomised profiles; each identity is a numerical residual against a *preregistered* tolerance plus at least one mutation control that must fail by  $\geq 10^{-3}$  relative.
- **(1)–(2) Border geometry and the (T) identity [E].**  $\sum_i (1 - g_i) = t_l + t_r$  from the exact interval geometry (max rel  $5.5 \times 10^{-13}$ ), so the  $\kappa$  weights are a probability vector and the flat border vector gives  $\kappa_{\text{flat}} = 1$  *exactly* (control: a 1% perturbation of  $g_i$  breaks it by  $7.9 \times 10^{-2}$ );

$\tau_{\text{dn}} = \|y\|^2 - m_{\text{prot}} + m_{\text{fill}} - V_{\text{bord}}$  at  $2.2 \times 10^{-16}$ , with  $\tau$  independently recomputed from the *full* odd overlap matrix ( $1.1 \times 10^{-13}$ ); dropping any of the three correction terms breaks the identity.

- **(3)–(4) The profile identity and the curvature chain [E].**  $p_{N-1} = 2 - p_0 + 2E$  at  $2.3 \times 10^{-16}$  (instances) /  $1.4 \times 10^{-15}$  (400 random profiles), hence the equivalence  $2E > p_0 \iff p_{N-1} > 2$  with 0 exceptions and both branches realised (60/400); the Abel/Hölder chain  $\kappa \leq 2 - p_0 + \frac{N-2}{N} \sum_j |\Delta p_j - \overline{\Delta p}| + (p_{\text{max}} - p_{N-1})$  holds per instance with 0 violations (min slack  $1.0 \times 10^{-1} / 3.7 \times 10^{-2}$ ), while the *same* chain without the curvature term is violated on 88/400 profiles — the curvature term is load-bearing, not decorative.
- **(5)–(6) Assembly and the C ea/Strang defect [E].** The matrix-free two-scale assembly equals  $J^T Q J$  ( $5.2 \times 10^{-16}$ ) and the graded overlap operator equals  $J$  ( $5.2 \times 10^{-14}$ ), reducing to the uniform odd overlap when both partitions are uniform;  $S_{\text{graded}} - S_{\text{uniform}} = R^T X^{-1} R$  as a *matrix* identity on all 43 grids ( $2.1 \times 10^{-14}$ ) with a PSD defect on every grid, so the compression error is one-sided *by* the identity (controls: unnormalised merge weights  $8.4 \times 10^{-1}$ ; dropping the  $-B$  term in  $R$   $4.0 \times 10^{-1}$ ).
- **(7)–(9) Sandwich, Perron implication, counting chain [E].** The secular sandwich  $\varepsilon/(\|u\|^2 + \varepsilon/\mu_1) \leq \lambda_{\min}(Q) \leq \varepsilon/\|u\|^2$  holds on all 48 pole-free sections at which  $A > 0$  and  $\varepsilon > 0$  are *checked* (relative width  $\leq 1.4448$ ), with Albert’s sign equivalence verified in *both* directions (6 rescaled controls with  $\varepsilon < 0$  all give  $\lambda_{\min}(Q) < 0$ );  $S^{-1} > 0$  entrywise (measured 48/48)  $\Rightarrow$  sign-constant *and* simple ground vector as an *implication* (tridiagonal control: hypothesis fails, conclusion fails with it);  $n_{\text{run}} \leq n_{\text{cross}} + 1 \leq s_2 + 2$  and  $\sum_j |\Delta p_j - \overline{\Delta p}| = \text{TV}(P)$  with 0 exceptions on a non-vacuous battery ( $s_2$  up to 31).
- **Named limits as load-bearing content.** Every statement is *per instance on small windows* — nothing is uniform in the zone index; the  $\kappa$  law is *not* claimed (T129 found it false in general), only the chain underneath it; (7) assumes nothing — positivity of the pole-free section at depth (the open M25d) is not extended and no floor value is carried; (8) has a *measured* hypothesis; (6) says nothing about the *size* of the defect (the open T130 exponent). Abel / H lder / Schur–Haynsworth 1968 / Albert 1969 / C ea 1964–Strang / Yserentant 1986 / Perron–Frobenius–Ostrowski 1937 named classical; Weil 1952 cited, never used as a criterion; zero-firewall AST-checked; NOT “almost RH”.
- **Hardening of v379 (SEAM.S3.RP.01).** T133 (CERT.FLOOR) certified that the  $10 \times 10$  Hankel matrix v379 assembles has, *as a double-precision object*, a negative direction at  $-7.75 \times 10^{-18}$  (60 digits) — inside the module’s old `min_eig > -1e-10` tolerance. The mathematical matrix is fine and the exact rescue is now in the module: the RP Gram is built explicitly as a *positive mixture*  $M = V^T \text{diag}(1/1600) V$ ,  $V_{ki} = e^{-\varepsilon_k \tau_i}$ , so  $x^T M x = \sum_k w_k (v_k \cdot x)^2$  is a *sum of squares* and  $M \succeq 0$  holds exactly; the mixture identity is zero at the rounding level ( $1.1 \times 10^{-15} \leq 64u|M|$ ) and the sum-of-squares route is nonnegative on 510/510 test directions where the assembled float form is negative on 2. The eigenvalue check is *retained and relabelled a diagnostic measurement*. Ledger row SEAM.S3.RP.01 reworded accordingly (method text: exact positive-mixture Gram identity, eigenvalue check as diagnostic); **marker stays [E] — no upgrade**. 7 checks (was 5), `run_all` green.
- **Diary Teile 132/133 (sandbox).** `bd_seam_probe.py` (contract BD.SEAM, 21/21, verdict SPECTRUM-ONLY): the model boundary-Dirac and the marks kernel share their spectrum to  $7.5 \times 10^{-13}$  but are *not* the same operator — the gap sits in the killing mass 0.1746 ( $N$ -stable), the killing measure equals the marks profile pointwise, the Markov remainder is exactly zero with a closed jump-weight form (odd lags positive, even lags zero — which explains T126’s 86.1%), and the  $D_4$  lattice symmetries leave the triad exactly invariant; ceiling: the DtN is a model DtN, the gate is *not* closed. `cert_floor_probe.py` (contract CERT.FLOOR, 23/23, verdict MIXED, 840 control validations): the v379 finding above plus a recommendation table — the hardening is implemented in this same change. Series counter: 133 probes, 3352/3352 sandbox

checks. Sandbox only: no ledger row for the probes, no marker move, not RH evidence.

- **Ledger/papers/website.** New ledger row PRIME.MARGIN.IDENT.01; \veri citations in `tfpt_1`, `tfpt_3`, `tfpt_research_contracts` and the promotion place of `tfpt_prime_front` (verified layer now *eight* modules; phase 2 extended to T126–T133); website mirrors (`/prime-front` feed entries T132/T133, status surfaces Teile 11–133 / 3352 checks, verification DAG, papers data) updated in the same change.

**2026-07-27 (IV) (prime front: T111 deep ladder maps three walls, crossing measured at  $n \approx 462$  — CROSSING-CONFIRMED; paper + diary + website sync; NO marker moves)**

- **Diary Teil 111 (sandbox).** `experiments/tfpt-discovery/deep_zone_stress_probe.py` (contract DEEP.ZONE.STRESS, 23/23 checks, 140 s, verdict CROSSING-CONFIRMED): the probe stops extrapolating and measures — the T110 grid is reproduced bit-exactly, then the ladder is extended to every prime-power zone  $n \leq 1000$  plus a thin tail to 3001 (199 zones; chain ladder  $n = 2 \dots 521$  with 117 handovers). The ratio  $m_k/\text{need109}$  falls  $179.59 \rightarrow 0.0174$ ; the crossing is *measured*: last zone  $\geq 1$  at  $n = 461$ , first below at  $n = 463$  (0.935) — T110’s  $n \approx 170$  was a fit artefact, but the wall is real, and a depth-preserving resolution study makes  $n^* \approx 462$  an *upper* bound. The failure follows the prime branch (prime powers 512/529/625/729 survive longer). New and harder: the *ladder wall* — the nested step needs a lag gap  $> 2D$ , and the twin-prime pair  $521 \rightarrow 523$  (gap  $0.003831 < 2D = 0.005617$ ) kills the ladder arithmetically (arithmetic scan to 40 000: 4052/4284 pairs endangered; co-moving  $D \sim 1/n$  forces  $h \sim n \log n/4$ ). The no-reserve finding softens ( $f_{\text{crit}} \sim n^{-0.39}$ , the reserve opens with depth); the circle breaks first at  $k = 109$ ,  $n = 463$  — it tears at the ratio, not at a step: all 117 handovers certify at retention 1.000000 (largest single loss  $5.96 \times 10^{-8}$ ), atom entry free 117/117. Three separate walls: margin  $n \approx 462$  (measured), ladder  $521 \rightarrow 523$  (twin primes), requirement  $n = 727$  ( $\omega_{\text{cert}} \geq 1$ , need109 vacuous on 46 zones). Reinterpretation: the operating variable is the window *depth*, not  $n$  (cheek doubling moves the wall  $463 \rightarrow 47$ ); a proof must beat the exponent gap  $-0.974 = -0.681 - 0.293$  and couple  $\delta_0$  to the local prime gap. Follow-up probe T112 (`adaptive_scaling_probe.py`, contract ADAPTIVE.SCALING, the  $n$ -adaptively scaled frame) is running. Series counter: 111 probes, 2781/2781 sandbox checks. Sandbox only: no ledger row, no marker move, not RH evidence.
- **Paper `tfpt_prime_front`.** Certification-arc section extended to T102–T111 (new subsection “The three walls (T111)”); the T110 three-gaps box and the  $n \approx 170$  mentions corrected to the measured  $n^* \approx 462$  (fit artefact noted); reduction-cascade caption, two-doors figure chain, three-way status split and asymptotic-mountain section updated (the three walls as the measured formulation; depth-not- $n$  reinterpretation; T112 running); abstract and probe-index appendix updated (T111 row; series counter 111/2781; T112 recorded as running and deliberately absent).
- **Website.** `/prime-front`: T111 feed entry (`lib/primeFront.ts`, new verdict enum CROSSING-CONFIRMED), status line in the two-doors convergence panel (T111 + T112 running), hero counter Teile 11–111 / 2781 sandbox checks, program-status callout sentence for T111.

**2026-07-27 (III) (prime front: T110 margin propagation closes the measured-zone circle — MARGIN-PROPAGATES; paper + diary + website sync; NO marker moves)**

- **Diary Teil 110 (sandbox).** `experiments/tfpt-discovery/margin_propagation_probe.py` (contract MARGIN.PROPAGATION, 28/28 checks, 7.6 s, verdict MARGIN-PROPAGATES with a precision statement): the induction circle closes end to end on the measured zones — the base case  $n = 2$  is certified by an explicit Cholesky at three resolutions ( $\lambda_{\min} \geq 6.93 \times 10^{-2}$ , factor 88–180 over need), the graded Loewner minorant holds all 15 handovers with retention 1.0000 (the strictly scalar variant closes 0/15), and the atom entry costs nothing — it *raises*  $\lambda_{\min}$  on 15/15

(first-order Kato term with helping sign; at  $n = 29$  the atom-free window is not even positive,  $-9.2 \times 10^{-3}$ ). Margin map:  $m_k \geq \text{need109}$  on 16/16 zones (ratio 2.09–179.6); the growth step splits exactly (embedding error  $\leq 2.6 \times 10^{-14}$ ;  $\max |\mu N| = 0.0$  on 15/15); scalar routes killed (bordered Weyl 0/15; Schur/Friedrichs cap vacuous 15/15). Three sharp gaps remain: no reserve ( $f_{\text{crit}} = 1.00$  at the first handover; retention  $1 - 2 \times 10^{-7}$ ), no scalar step law (structurally excluded by the boundary layer), no  $k$ -uniformity (trend  $m_k \sim n^{-1.93}$  against  $\text{need109} \sim n^{-0.98}$ , extrapolated crossing  $n \approx 170$ ). Follow-up probe T111 (`deep_zone_stress_probe.py`, contract `DEEP.ZONE.STRESS`, the deep ladder to  $n \sim 200$ ) is running. Series counter: 110 probes, 2758/2758 sandbox checks. Sandbox only: no ledger row, no marker move, not RH evidence.

- **Paper `tfpt_prime_front`.** Certification-arc section extended to T102–T110 (new subsection “The margin propagates: the circle closes on the measured zones” with the three gaps as a key box); reduction-cascade caption and two-doors figure chain extended to T110; kill list extended by the T110 scalar-step-law kill; three-way status split and asymptotic-mountain section sharpened (the three gaps as the precise formulation; the  $n \approx 170$  warning; T111 running); abstract and probe-index appendix updated (T110 row; T111 recorded as running and deliberately absent); symbol-table row for  $m_k/\text{need109}$ .
- **Website.** `/prime-front`: T110 feed entry (`lib/primeFront.ts`, new verdict enum `MARGIN-PROPAGATES`), status line in the two-doors convergence panel (T110 + T111 running), hero counter Teile 11–110 / 2758 sandbox checks, program-status callout sentence for T110.

**2026-07-27 (II) (prime front: T109 boundary certificates close (R) conditionally — BOUNDARY-CERTIFIED; paper certification arc T102–T109 + diary + website sync; NO marker moves)**

- **Diary Teil 109 (sandbox).** `experiments/tfpt-discovery/boundary_decay_probe.py` (contract `BOUNDARY.DECAY`, 29/29 checks, 47.2s, verdict `BOUNDARY-CERTIFIED`): both remaining scalars of (R) are certified —  $\omega$  *unconditionally* via a graded matrix cap in PSD order (one level per soft direction, Cholesky-verified,  $\omega_{\text{cert}} = 0.2561\text{--}0.7569$  against measured  $0.2366\text{--}0.7531$ , factor  $1.004\text{--}1.124$ , no hypothesis input; the T108 compression Schur distance shrinks  $0.18\text{--}0.52 \rightarrow 0.0024\text{--}0.039$ ), the boundary value via a residual / goal-oriented certificate (Prager–Synge) that *carries* the cancellation (sharpness  $1.0000\text{--}2.4937$ , 16/16 in budget). Mechanism finding: no decay — *cancellation* ( $44\text{--}4.6 \times 10^4$  on 15/16 zones); the Combes–Thomas route is refuted with the exact Green row (1/16). The full chain closes 16/16 (margin  $1.3\text{--}10.4$ ,  $M$ -stable to 3000) conditional on exactly *one* input: the strict positivity margin  $\lambda_{\min}(Q|_{\text{odd}}) \geq 1.7 \times 10^{-6}\text{--}7.5 \times 10^{-3}$ , i.e.  $10^2\text{--}10^6$  below  $\mu/2$ . Follow-up probe T110 (`margin_propagation_probe.py`, contract `MARGIN.PROPAGATION`) is running. Series counter: 109 probes, 2730/2730 sandbox checks. Sandbox only: no ledger row, no marker move, not RH evidence.
- **Paper `tfpt_prime_front`.** Certification-arc section extended to T102–T109 (new subsection “Boundary certificates: cancellation, not decay”); the reduction cascade figure now ends at the one strict-positivity margin input; two-doors figure chain extended to T109; kill list K8 extended by the T109 Combes–Thomas kill; three-way status split (certified / measured / classical) updated; abstract and probe-index appendix updated (T109 row; T110 recorded as running and deliberately absent); symbol table rows for  $x/h/x_0$  and  $\omega/\omega_{\text{cert}}$ .
- **Website.** `/prime-front`: T109 feed entry (`lib/primeFront.ts`, new verdict enum `BOUNDARY-CERTIFIED`), status line in the two-doors convergence panel, hero counter Teile 11–109 / 2730 sandbox checks, program-status callout sentence for T109.

**2026-07-27 (I) (new companion `tfpt_prime_front` — the prime / zeta line documented end to end as RESEARCH DOCUMENTATION; NO new claims, NO**

## marker moves)

- **New companion paper `tfpt_prime_front` (P).** 22 pages, 9 sections plus two appendices, 8 TikZ figures. It documents the complete prime / zeta line of the project from the first  $\mu_4 \Rightarrow \chi_4$  observation to the current residual — explicitly a *documentation of a research route*, not a results paper. Registered in `tex-artefacts/tfpt_docset.tex` as the fifth companion.
- **Framework interpretation stated once, up front.** Section 3, “The big picture”, collects the reading that motivated the line and labels it as interpretation: the  $E_8$  shell count  $N(2n) = 240\sigma_3(n)$  is the Dirichlet series  $240\zeta(s)\zeta(s-3)$ ; the  $\mu_4$  glue carries  $\chi_{-4}$ , so the  $4k+1 / 4k+3$  split is a fibre property of the compiler and the signed channel is the only pole-free one; Hecke structure is emitted by lattice adjacency (`v535`) rather than put in;  $E_8$  is unimodular, so Poisson summation gives the theta sum an exact functional equation whose fixed axis is the critical line — the same reflection that reappears as  $J_1 \circ J_{1/2} = e^{\pm u}$  (the transport operator, T83) and as the parity split of the Weil pole (T106). Explicitly typed as a picture, not a reformulation, and not used as a premise anywhere.
- **Verified layer restated, not extended.** Sections 4.1–4.7 restate the seven load-bearing modules `v535`–`v541` with their ledger status markers quoted verbatim (Hecke from geometry, Eichler trace layer, half-integral bridge, relative-trace identity, Weil structure family, amplitude linear carrier, matching-lemma / transport ledger). The `\veri` citations in this paper are the only citations it makes; no ledger row is added, moved or upgraded.
- **Sandbox layer marked as *sandbox* throughout.** The localization program (I5, the Connes–Consani dictionary, the tightness map, the Sonin crossing, the blind prime demo), the handover induction and the certification arc are carried as clearly tagged *sandbox* content with a per-section provenance marker; every quantitative statement carries its probe label (Tn) and is reproduced verbatim from the research diary. All measured quantities are declared as measured, all window results as window results, all hypothesis inputs as hypothesis inputs.
- **Kill list as a first-class section.** Section 8 records the refuted readings and dead routes of the program (essential-singularity reading, the  $C/g$  proxy law, the naive margin route, the density chain, the Landau–Widom anchor, the accumulated invariant with self-propagation, the symbol route for the odd channel, three certificate candidates for the avoidance norm), each with the probe that killed it.
- **Explicit RH fence.** The paper states in the abstract, in the scope-and-status statement, in the reduction figure and in the status section that *no progress on the Riemann Hypothesis is claimed*; the  $I5 \Leftrightarrow$  Weil positivity  $\Leftrightarrow$  RH relation is carried, as in the ledger, as an equivalence *typing* only. `ZETA.HP.CARRIER` untouched.
- **Appendices.** Probe index T11–T108 (98 parts, 99 runs, 2509 green checks; the diary’s own series counter at T108 reads 108 probes / 2701 checks including the ten parts that predate the current diary file) with contract, verdict and check count per probe, plus a symbol table for the sandbox objects used in the figures. T109 (`boundary_decay_probe.py`) was running when the document was written and is deliberately absent.

## 2026-07-25 (XIX) (`v541` — matching lemma and transport ledger package: two named limits as content; NO marker moves)

- **New module `v541` (RTF.GNS.LEDGER.01).** Promotion of the discovery proof package T78/T79/T80/T81/T82/T85 into one load-bearing module (33 checks,  $\sim 10$ s); every core check is *recomputed* from scratch, not cited. Companion to `v538/HECKE.GEOM.RTF.01`, `v539/RTF.GNS.WEIL.01` and `v540/RTF.GNS.AMP.01`. Classical Gronwall 1913 / Robin 1983 *unconditional* / Cohen 1975 / Hecke 1918–1920 / Cornacchia / Fermat–Gauss / Dirichlet  $L(1, \chi)$  / Mertens-AP / Landau 1908 / Legendre duplication / Weil 1952–Guinand–Bombieri

/ Alaoglu–Erdős named classical — NEW is the compiler-native consolidation and the machine-checked window proofs.

- **(A) Window proof [E].** The matching lemma is *proved exact-integer* on  $[4, 10^6]$ :  $7S(j) < 40A(j)$  at every atom (0 violations over 939 870 clash atoms; full enumeration, no-overflow proven, big-integer rechecks); exact margin  $X = 22242575693/270725400000 = 0.082159$  at  $j^* = 554400$ ;  $\rho_{\text{crit}} = 1.1440 > 21/20$ ; maximal-greedy identity  $\Rightarrow$  uniform in the target set and the greedy weights. Structure laws at 0 tolerance: 8-law ( $n \leq 250000$ ), seed-tower +  $\psi$   $w$ -table ( $n \leq 50000$ ), Cohen seeds  $\{-48, -80, -8, -8\}$  ( $s \leq 2000$ ), bracket  $\Theta \leq 2|\psi| \leq 3\Theta$  on  $10^6$ .
- **(B) Transport ledger [E].**  $Q_{\text{Weil}} = Q_{\text{cert}} + \Delta_{\text{arch}} + \Delta_2$  closes with  $\Delta_{\text{pole}} \equiv \Delta_{\text{conv}} \equiv 0$  *proven* (sympy kernel collapse; quadrature  $2.8 \times 10^{-16}$ ; exact scale invariance); ledger residual  $4.4 \times 10^{-16}$  on 24 battery rows; the odd-prime side of the Weil functional equals the certified plus combination  $P_{\zeta}(g_-) + P_{\zeta}(g_+)$  exactly (rel  $3.9 \times 10^{-16}$ ).
- **(C) Character-exact signed envelope [E].**  $-\psi = (\chi_{-4} + \frac{1}{4}\chi_8 + \frac{1}{4}\chi_{-8}) \cdot \Theta$  coefficient-exact on all odd  $n \leq 50000$  (both builds); signed certificate 0 violations, margin factor  $8.86\times$ ; confinement *set equality* on  $10^6$ : zero-credit clash atoms =  $\chi_{-4}$ -coherent odd composites.
- **(D) Archimedean term internal [E].** Legendre duplication bridge  $(2\pi)^{-s}\Gamma(s) = \frac{1}{2}\Gamma_{\mathbb{R}}(s)\Gamma_{\mathbb{R}}(s+1)$  (sympy +  $1.8 \times 10^{-41}$ ); kernel identity  $k_{\zeta} = K_{\text{fam}} - K_{\text{shift}}$  pointwise  $7.9 \times 10^{-31}$  on 51 points;  $\Delta_{\text{arch}}(h) = A_{\text{fam}}(h) - A_{\text{shift}}(h)$  on a 10-row battery subset at rel  $9.9 \times 10^{-16}$ .
- **(E)  $\lambda$ -equivariant channel [E].** CM carrier  $g_{\lambda} = \sum_{\mathfrak{a}} \lambda_1(\mathfrak{a})q^{N_{\mathfrak{a}}}$  exact in two independent routes ( $\mathbb{Z}[i]$  lattice vs Cornacchia reconstruction, 0 mismatches  $n \leq 10^4$ ); CM laws exact;  $\text{supp}(c_1) = \{\mathbb{Z}[i]\text{-norms}\}$  as a set equality; canonical phase mean  $\mu_1 = c_1/(c_0 d^2) \in [-1, 1]$  exact;  $\lambda$ -window certificate 0.0653 vs 0.2365 unlifted ( $3.6\times$ ), exact Fraction rechecks.
- **(F) Frontier facts [C].** Unlifted coherent chain crossing  $k^* = 14$  (exact Fractions;  $\log_{10} N^* = 23.1$ ); absolute Robin tail factor 6.16 at  $J = 10^6$  and divergent — the constant route stays open (window constants with declared classical all- $n$  extension).
- **Named limits as load-bearing content.** (i) *One* classically-shaped open lemma remains — the correlated-cancellation lemma on credit-rich non-coherent tail atoms (uniform form); provably-shaped  $\neq$  formal proof. (ii) I5 in *one-family* form ( $Q_{\text{cert}} + \Delta_2 + A_{\text{fam}} - A_{\text{shift}} \geq 0$ ) is the single remaining TFPT-specific object — by the closed ledger *equivalent* to Weil positivity  $\iff$  RH (an equivalence *typing*, no progress claim). NOT “almost RH”; ZETA.HP.CARRIER untouched; no marker moves.
- **Ledger/papers/website.** New ledger row RTF.GNS.LEDGER.01; \code{veri} citations in `tfpt_1`, `tfpt_3`, `tfpt_research_contracts`; Wolfram extension mirrors the exact algebraic identities; website mirrors (/prime-front feed, verification DAG, papers data) updated in the same change.

**2026-07-25 (XVIII) (v540 — amplitude route and positive linear carrier: open boundary  $\lambda^*$  as content; NO marker moves)**

- **New module v540 (RTF.GNS.AMP.01).** Promotion of the discovery amplitude chain T67/T68/T69/T70/T71/T72 into one load-bearing module (34 checks,  $\sim 3$  s). Companion to v538/HECKE.GEOM.RTF.01 and v539/RTF.GNS.WEIL.01: the machine-checked route out of the square plane behind the two v539 obstructions. Classical Cohen 1975 / Siegel–Weil / Jacobi–Fricke / Landen / Hecke split-Mellin / Cauchy–Littlewood / Möbius square sieve / Shimura / Shintani / Weil 1952 / Farkas named classical — NEW is the compiler-native consolidation.
- **(A) Amplitude Dirac [E].**  $D = \begin{pmatrix} 0 & V \\ V^{\top} & 0 \end{pmatrix}$  with  $V[d, m] = \sqrt{w_d} b(d) \chi_d(m)$ :  $D = D^{\top}$ ;  $D^2 = \text{diag}(VV^{\top}, K)$ ,  $V^{\top}V = K$  (rel  $2.7 \times 10^{-16}$ ); spectrum =  $\pm$  singular values; Hecke intertwining exact ( $p = 3, 5, 7$ ); Shimura  $b(dp^2) = b(d)(a_p - \chi_d(p)p)$  integer-exact on 644 pairs.

- **(B) Geometric polarisation + Cohen seed [E].**  $b = N_+ - N_-$ ,  $\Theta = N_+ + N_-$  lattice-exact;  $N_{\pm} \geq 0$  and  $b^2 = \Theta^2 - 4N_+N_-$  for all  $n \leq 50000$ ;  $\Theta$  is a *pure* Siegel–Weil Eisenstein  $\sigma_3$ -eigenform ( $p \leq 13$ , cusp component zero); Cohen seed  $\Theta(d) = -48 L(-1, \chi_d)$  exact-rational on 159 live  $d$ .
- **(C) Deletion universal = square-class double counting [E].** Cauchy–Littlewood window lemma (free Satake symbols,  $k \leq 6$ ): numerator  $1 - stuv X^2$ , hence  $1 - p^6 X^2$  pair-independent at  $st = uv = p^3$  (five channels); and fam  $[1, 0, 2, 0] + 2b [0, 2, 0, 2] + \delta_{k1} = [2, 2, 2, 2]$  generating-function exact (Möbius square sieve exact  $n \leq 2000$ ).
- **(D) Positive linear carrier + plus balance [E].**  $\ell^2(d, \mu)$  with  $\mu = 48 |L(-1, \chi_d)| |d|^{-a}$ : full weights  $\lambda_k = 1 + p^{3k} - (\chi p)^k$  (layer  $[1, 1, 1, 1]$ );  $Q = Q_{\zeta}(g_-) + Q_{\zeta}(g_+)$  (rel  $2 \times 10^{-15}$ ); the  $b$ /doubling kernel enters *with plus* (sign inverted vs the v539 square-plane minus).
- **(E) Functional equation exact [E].**  $\Theta(i/(8y)) = 8y^{5/2} \Theta^{\dagger}(iy)$  (rel  $6 \times 10^{-41}$ , Fricke closed);  $\Lambda_{\Theta}(s) = 8^{1-s} \Lambda_{\Theta^{\dagger}}(5/2 - s)$  in the strip (rel  $1.4 \times 10^{-40}$ , independent split points); poles  $5/2 / 0$  swapped; centre  $5/4$ , plus-locus offset exactly  $1/4$ .
- **(F) Cone facts + named open boundary [C].** Guaranteed cone and library hull FE-self-dual, gap functional  $\lambda^*$  FE-covariant (36/36 sampled); mirror sign law  $\text{sign } \psi(n) = (-1)^{\lfloor n/2 \rfloor + 1}$  exact ( $n \leq 50000$ ); constraint class  $C_{L3} = \{n \equiv 6 \pmod{8}\}$ ;  $\lambda^*$  exists with one exact Farkas witness ( $\lambda_{L3}^* = 0.0511$  at  $n^* = 6$ ); coverage saturates — the residual distance to the Weil cone *is*  $\lambda^*$  on  $n \equiv 6 \pmod{8}$ .
- **Fences.** Euler-region positivity only (edge  $L$ -values, not the central line); not “almost RH”;  $\lambda^*$  is the named open boundary *inside* the claim; ZETA.HP.CARRIER untouched; no marker moves.
- **Surfaces.** Ledger RTF.GNS.AMP.01; \veri in tfpt\_1/tfpt\_3/tfpt\_research\_contracts; Wolfram extension +5 (613  $\rightarrow$  618); website papers.ts + VerificationDag E8 + primeFront.ts + ScriptIndex via bash build.sh gen.

## 2026-07-25 (XVII) (v539 — Weil structure of the compiler family: two isolated obstructions; NO marker moves)

- **New module v539 (RTF.GNS.WEIL.01).** Promotion of discovery probes T55/T61/T62/T63 (Route-B chain) into one load-bearing module (25 checks,  $\sim 6$  s). Companion to v538/HECKE.GEOM.RTF.01: the compiler family carries a fully identified Weil structure up to *two explicitly isolated obstructions* (obstructions are prominent verified content, not a footnote). Classical Weil 1952 / Gelfand / direct integrals / SU(2)–Chebyshev / Sato–Tate /  $2^{\omega}$ -series / GNS named classical — NEW is the compiler-native identification and the machine-checked isolation of the two obstructions.
- **(A) GNS fibre structure [E].** Metric  $\ell^2(d, b^2/|d|)$  diagonal; fibres  $\sigma(d) = (\chi_d(3), \dots, \chi_d(13))$  partition the live family (1191 fund.  $d \leq 12000$ );  $\Lambda_{\text{fam}}$  exact constant on each fibre; twist-mix per fibre  $\leq 5.037 \times 10^{-16} < 10^{-12}$ .
- **(B) Isotype / GL(1) core [E].** Character expansions  $\Phi_1 = \chi_0 + \chi_2$ ,  $\Phi_2 = \chi_4 - \chi_2$ ,  $\Phi_3 = 2\chi_0 - \chi_4 + \chi_6$ ,  $\Phi_4 = \chi_8 - \chi_6$ ; core series  $c_{k,0} = [1, 0, 2, 0, 2, 0]$ ;  $G_0 = (1 + Y)/(1 - Y) = \zeta_p(w - 3)^2 / \zeta_p(2w - 6)$ ; global  $\zeta(u)^2 / \zeta(2u) = \sum 2^{\omega(n)} n^{-u}$  (Dirichlet exact through  $n \leq 2000$ ).
- **(C) Weil linear relation [E].** Prime<sub>F</sub> =  $2P_{\zeta}(g) - 2P_{\zeta}(g_b)$  (max rel  $9 \times 10^{-16}$ );  $Q_{\text{fam}} = Q_F + \text{Corr}$  (rel  $2 \times 10^{-16}$ ).
- **Obstructions as verified content [E].** (1) Doubling enters with a *minus* (family positivity does *not* imply  $Q_{\zeta} \geq 0$ ). (2) Plancherel Corr non-automorphic: polar/arch absorption FAIL; finite Euler factor FAIL;  $e^{-\sum p^{-u}}$  PASS.
- **Finite-class positivity [C].**  $Q_{\text{fam}} \in [4.369, 11.486]$  on the finite test class; dense-class positivity open / RH-adjacent — *not* claimed.

- **Fences.** Not “almost RH” — the claim structurally shows why family positivity does not imply RH; ZETA.HP.CARRIER untouched; no marker moves.
- **Surfaces.** Ledger RTF.GNS.WEIL.01; \veri in tfpt\_1/tfpt\_3/tfpt\_research\_contracts; Wolfram extension +5 (608 → 613); website papers.ts + VerificationDag E8 + primeFront.ts + ScriptIndex via bash build.sh gen.

## 2026-07-25 (XVI) (v538 — compiler relative-trace identity: three projections of one finite RTF; NO marker moves)

- **New module v538 (HECKE.GEOM.RTF.01).** Promotion of discovery probe T53 (relative\_trace\_compiler\_probe.py) into one load-bearing module (18 checks, ~ 14s). Companion synthesis of v535/HECKE.GEOM.01, v536/HECKE.GEOM.EICHLER.01 and v537/HECKE.GEOM.HALFINT.01: these are three *projections* of one finite relative-trace identity  $[E_8/\text{glue orbital counting}] = [\text{Newform/Waldspurger spectral sum}]$ , not three separate results. Classical Jacquet-type RTF / Eichler / Witt / Waldspurger / theta correspondence named classical — NEW is the compiler-native realisation and the machine-checked bilateral independence.
- **(R1) Trace identity, both sides independent [E].**  $\text{Tr}_V(\nu_p \circ \pi)$  geometric ( $a_p = -c(p)/8$  from the glue census, no modular input) equals spectral ( $a_p$  from  $\eta(2\tau)^4\eta(4\tau)^4$ , no geometry) for all  $(p, \pi) \in \{3, 5, 7\} \times \{\text{Id}, \pi_{\text{Eis}}, \pi_{\text{cusp}}, \pi_{E_4}, \pi_{f_8}\}$ . Anchors (Id, Eis, cusp,  $E_4$ ,  $f_8$ ):  $p=3$ : (6304, 5600, 704, 1120, 352);  $p=5$ : (105848, 98280, 7568, 19656, 3784);  $p=7$ : (727680, 688000, 39680, 137600, 19840).
- **(R2) Orbit dictionary [E].** The Eichler split  $\lambda_{\text{geom}} = \lambda_{\text{Eis}} + a_p^2$  is the RTF orbit decomposition (principal orbit  $a \cdot \text{Id} \rightarrow \lambda_{\text{Eis}} = L - \sigma_3^2$ ; elliptic  $b \cdot T_p \rightarrow a_p^2$ ; cross terms cancel). Anchors  $\lambda_{\text{Eis}} = 336/3780/19264$ ,  $a_p^2 = 16/4/576$ .
- **(R3) Period side is lattice counting [C].**  $g$  is the quaternary form  $n = (x^2 + y^2)/2 + 2z^2 + u^2 + 2w^2$  ( $x, y$  odd; sign  $(-1)^{|u|+|w|}$ );  $b_{\text{rep}} = b$  exactly through  $n \leq 200$ ; then  $[\text{signed lattice count}]^2 = R \cdot |d|^{3/2} \cdot L(f_8 \times \chi_d, 2)$  with  $R = 23.1873585645$  (rel  $\sim 10^{-12}$ ). Verdict ONE-FORMULA.
- **Fences.** Identity is *finite* ( $p \in \{3, 5, 7\}$ , finite  $d$ -windows); the *infinite* RTF (sum over all  $p$  and  $d$  + archimedean term) is the named open step; no RH; not ZETA.HP.CARRIER-relevant; no marker moves.
- **Surfaces.** Ledger HECKE.GEOM.RTF.01; \veri in tfpt\_1/tfpt\_3/tfpt\_research\_contracts; Wolfram extension +5 (603 → 608); website papers.ts + VerificationDag E8 + primeFront.ts + ScriptIndex via bash build.sh gen.

## 2026-07-24 (XV) (v537 — half-integral bridge: unique Shimura preimage + $T(p^2)$ Hecke equivariance + Kohnen scope fence + Waldspurger/Baruch–Mao constancy; NO marker moves)

- **New module v537 (HECKE.GEOM.HALFINT.01).** Consolidated promotion of discovery probes T38/T41/T44/T45 into one load-bearing module (20 checks, ~ 90s). Companion to v535/HECKE.GEOM.01 and v536/HECKE.GEOM.EICHLER.01 without duplicating weight-4 Kneser/Eichler checks. Classical Shimura/Kohnen/Waldspurger/Baruch–Mao named classical — the claim is the unique signed-scale monoid preimage, the in-suite chain, and the *measured* constant  $R$ .
- **(A) Unique Shimura preimage at scale  $-8$  [E].** In the 70-element compiler weight-5/2 theta-monoid exactly one object satisfies  $\text{Sh}_{t=2, \psi=1}(g) = -8 f_8$  (witness  $g = \vartheta_2(q^2)^2 \vartheta_3(q^2) \vartheta_4 \vartheta_4(q^2)$ , verified through  $n \leq 120$ ); three further monoid elements lift to scales  $+8, -16, +16$ .
- **(B) Hecke equivariance [E].** Exact  $T(p^2)$ -eigenform, trivial nebentypus, with  $\lambda = a_p(f_8)$  for  $p = 3, 5, 7, 11, 13$  ( $-4, -2, 24, -44, 22$ ); trial FE  $N = 512$ ,  $\varepsilon = +1$  (centre 5/4).

- **(C) Kohnen scope fence [E].**  $U_4(g) = 0$  (mass mod 4 on  $\{1, 2\}$ ); level  $32 = 4 \cdot 8$  outside Kohnen 1982 — plus-route structurally closed (negative result part of the claim).
- **(D) Waldspurger/Baruch–Mao constancy [C].** AFE twist  $L$ -values ( $\varepsilon_d = \chi_d(8)$ );  $R(d) = 23.1873585645\dots$  constant on ten fundamental  $d \equiv 1 \pmod{8}$  (spread  $\leq 10^{-12}$ , including  $d = 89$ );  $d \equiv 5 \pmod{8}$  vanishes.
- **Fences.**  $GL(2)$  centre  $s = 2$ , not the  $\xi$ -line; no RH; ZETA.HP.CARRIER untouched; no marker moves.
- **Surfaces.** Ledger HECKE.GEOM.HALFINT.01; `\veri` in `tfpt_1/tfpt_3/tfpt_research_contracts`; Wolfram extension +5 (598  $\rightarrow$  603); website `papers.ts` + VerificationDag E8 + ScriptIndex via `bash build.sh gen`.

**2026-07-24 (XIV) (v536 — Eichler trace layer at the  $E_8$  lattice: Witt  $\lambda_{\text{Eis}}$  + two-sided Eichler identity + Type-A/B local densities + signed  $a_p$ ; NO marker moves)**

- **New module v536 (HECKE.GEOM.EICHLER.01).** Consolidated promotion of discovery probes T33/T36/T37/T42 into one load-bearing module (23 checks,  $\sim 30$  s). Builds on v535/HECKE.GEOM.01 (Kneser counts,  $\nu_p$  law, oldforms) without duplicating those checks. The claim is *not* RH-related: the Eichler/Witt package of the  $\mu_4$ -glue census is lattice-native and two-sided in-suite.
- **(A) Witt layer [E].**  $\lambda_{\text{Eis}} = \sigma_3(\#\mathbb{P}^3 - \sigma_3) = L - \sigma_3^2$  identically (sympy) and for all 24 odd primes  $p \leq 100$ ; anchors  $\lambda_{\text{Eis}}(3, 5, 7) = (336, 3780, 19264)$ .
- **(B) Eichler identity [E]/[C].**  $\lambda_{\text{geom}} = \lambda_{\text{Eis}} + a_p^2$ ; type-(i)  $N_{\perp} = (p^6 - 1)/(p - 1)$  live constant on all 240 roots at  $p = 3, 5$ ; type-(ii) live at  $p = 3$  ( $352 = 364 - 12$ ) with Kneser filter = identity;  $p = 5$  T36-validated freeze ( $3784 = 3906 - 122$ ,  $R = 4$ ; Shell(25) re-enumeration budget-gated); assembler  $R(p) = \sigma_3 - 1 - c(p^2)/8 = a_p^2$ ;  $R(7) = 576$  prognosis (Shell(49) out of scope).
- **(C) Local densities [E].** Type-A = isotropic cosets representing the integer  $p$ ;  $N_A = \min(240(1 + p^3), p^7 + p^4 - p^3 - 1)$ ,  $N_B = (\#\text{iso} - 1) - N_A$ ; B empty iff  $p^4 < 241$  (only  $p = 3$ );  $p = 7$  live 82560/743040; injectivity at  $p = 11, 13$ ; fibre refinement scoped to  $\{5, 7, 11\}$ .
- **(D) Signed extraction [E].**  $a_p = -c(p)/8$  from geometric Shell( $p$ ):  $(-4, -2, +24)$  at  $p = 3, 5, 7$ ;  $b = \sigma_3 + a_p \in \{24, 124, 368\}$  (hooks the  $\nu_p$  law of v535).
- **(E)  $O(1)$  assembler [E].**  $\lambda_{\text{geom}}(p)$  and  $R(p) = a_p^2$  for all odd  $p \leq 31$  against the eta product  $f_8$  (Ramanujan bound included).
- **Fences.** Classical Eichler/Siegel/Witt named classical — the claim is in-suite two-sidedness (mod- $p$  geometry left, eta product right) and the closed compiler densities; no RH statement; Shell( $p^2$ ) for  $p \geq 7$  out of scope; weight-4  $\rightarrow GL(1)$  stays [O]; no marker moves of any pre-existing contract.
- **Surfaces.** Ledger HECKE.GEOM.EICHLER.01; `\veri` in `tfpt_1/tfpt_3/tfpt_research_contracts`; Wolfram extension mirrors (exact Witt/density/assembler/ $a_p$  identities); website `papers.ts` + VerificationDag E8 + ScriptIndex via `bash build.sh gen`.

**2026-07-24 (XIII) (v535 — Hecke from geometry: Kneser correspondence + affine neighbour-sum + 2-adic oldform recovery on the  $E_8$  census channels; NO marker moves)**

- **New module v535 (HECKE.GEOM.01).** Consolidated promotion of discovery probes T27/T31/T32 into one load-bearing module (25 checks,  $\sim 11$  s). The claim is *not* RH-related: the Hecke structure of the  $\mu_4$ -glue census is lattice-native.

- **(A) Kneser carries Hecke [E].**  $\#iso\_pts = p^7 + p^4 - p^3$  and  $\#iso\_lines = \sigma_3(p) \# \mathbb{P}^3(\mathbb{F}_p)$  identically and by full enumeration at  $p = 2, 3, 5, 7$  (135/1120/19656/137600 lines) — the Eisenstein eigenvalue  $\sigma_3(p) = 1 + p^3$  is an exact factor of a lattice-native correspondence count.
- **(B) Neighbour-sum = affine Hecke element [E].**  $\nu_p = a \text{Id} + b T_p$  exactly (overdetermined, residual 0) with  $b = \sigma_3 + a_p$ ,  $a = \#lines - b\sigma_3$ ; cusp rule  $a_p = b - \sigma_3$  recovers  $a_3 = -4$ ,  $a_5 = -2$  ( $p = 3$  live orbit-reduced;  $p = 5$  T31-validated freeze of  $S[1]$  with  $\lambda_{\text{odd}} = 3784$ );  $p = 7$  consistency  $\lambda_{\text{odd}} = \#lines - \sigma_3^2 + a_p^2 = 19840$ ; placebo fails.
- **(C) Redundancy = 2-adic oldform structure [E].**  $\dim_{\mathbb{Q}} V = 7 = 5$  ( $E_4$ -copies, levels 1, 2, 4, 8, 16) + 2 ( $f_8$ -copies, levels 8, 16); every  $\Theta_j$  an explicit  $\mathbb{Q}$ -linear combination; recovery  $\pi_{\text{cusp}} = (28 - T_3)/32$ ,  $\pi_{\text{Eis}} = (T_3 + 4)/32$ , newform via  $(1 - V_2 U_2)$ ; new quotients each dimension 1; level census purely 2-adic.
- **Fences.** Classical theorems (Kneser, Hecke, Atkin–Lehner, multiplicity one) named classical — the TFPT content is the in-suite mechanics. No RH statement. Weight-4 restriction named:  $GL(1)/\text{weight-drop transport stays [O]}$ . No marker moves.
- **Surfaces.** Cited in `tfpt_1`, `tfpt_3`, `tfpt_research_contracts`; ledger row `HECKE.GEOM.01`; Wolfram extension +5 exact mirrors (588  $\rightarrow$  593); website sync.

**2026-07-24 (I) (v534 — the straddle cone executed: THE FIRST DYNAMICAL SELECTION OF THE ALIGNMENT BIT — reflection positivity keeps exactly ONE member of the interacting mark family alive,  $\delta = \pi/2$  with positive coupling; the literal leading-order cone formalization dies; NO marker moves)**

- **New module v534 (SEAM.STRADDLE.CONE.01).** The 2026-07-23 review-round contract (recorded “not promoted” in entry (X)) is *executed*, double-verdicted. **V2 = ERFOLG** — **the selection law (the central datum):** with independent couplings  $\lambda_q$  on the four mark quartets of  $M(\delta = m\pi/8)$  and the equivariant (uniform) direction, reflection positivity holds on *every* admissible cut over the whole grid  $g \in \{1/32..8\}$  for *exactly one* member  $\times \text{sign}$ :  $\delta = \pi/2$  ( $m = 4$ ) with *positive* coupling — all four Grams PD (37, 0, 0) *including the previously untested straddled clock axes* 7/15, minimum eigenvalue *lifted* off the free RP boundary ( $1.78 \cdot 10^{-6} \rightarrow 5.5 \cdot 10^{-5}$  near  $g = 1/4$ : the  $\pi/2$  interaction pushes *away* from the boundary), Gibbs  $\beta = 2$  identical [E]. The asymmetric members die:  $m = 2/6$  with  $s = +$  already at  $g = 1/32$ , with  $s = -$  from  $g = 1/8$ . The v529 straddle law *refines*: asymmetric straddling kills RP, the *symmetric* straddling of the self-mirror member protects it.
- **V1 = KILL (the literal cone formalization).** The leading-order soft-block cone  $K_{\text{RP}} = \text{Fix}(G_\delta) \cap \bigcap_c \{\lambda : G_c^{\text{lead}} \succeq 0\}$  is *empty* for every member and both signs (first-order PT via the exact reduced resolvent, Hermitian to  $10^{-15}$ , central-difference regression  $2 \cdot 10^{-8}$ ) — the linear pencil cannot carry the selection; the protection at  $\pi/2$  is *nonperturbative*. Recoverable leading-order content: the straddled-cut drift signs are exact ( $\pi/2$ :  $+1.4 \cdot 10^{-4}$  for  $s = +$ ,  $-3.7 \cdot 10^{-2}$  for  $s = -$ ; asymmetric members reversed) — necessary, not sufficient.
- **The exact geometry behind it [E].**  $\delta = \pi/2$  is the *unique symmetric-straddle member*: stabilizer = the full clock+mirror group (rotations  $\{0, 4, 8, 12\}$ , reflection axes  $\{3, 7, 11, 15\} =$  four admissible bond cuts, straddled  $\{7, 15\}$ ), the four quartets *partition* the 16-circle, and every straddling quartet is reflection-*closed* across its cut with exact sign  $+1$ ;  $m = 2/6$  straddle asymmetrically; odd  $m$  carry no bond axis (the typed v521/v525 placement artifact).  $\text{Fix}(G_\delta)$  is 1-dimensional throughout: uniform on  $m = 2..6$ , NS-wrap-*twisted*  $(1, -1, 1, 1)/(-1, -1, -1, 1)$  on the tight members  $m = 1/7$ . Typed cross-link: the RP-selected member  $m = 4$  is exactly the carrier of the v528 twist-class order parameter  $O = 1/2$  — the dynamical selector and the kinematic definition point at the *same* member.
- **Honest fences [C].** One toy, one interaction class (FK quartets), flat-band parent,  $\deg \leq 2$  OS bases, finite coupling grid — toy-level evidence *for* the bit’s dynamical origin, *not* a

derivation; nothing about the continuum  $\mathcal{A}_{\text{hol}}$  is claimed; WOIT.OS.TWISTOR.01 stays Open, NO marker moves. Transparency: the run-1 preregistered verdict logic conflated formalization and substance (UNENTSCHIEDEN, mismatch typed); run 2 split V1/V2 with no tolerance change and no number changed — note kept verbatim in the probe. Discovery provenance: `experiments/tfpt-discovery/seam_straddle_cone_probe.py` (15/15, 2026-07-24).

- **Wolfram mirror.** Four exact mirrors (stabilizer census,  $\pi/2$  partition + straddle incidence, reflection-closure signs, equivariant Fix rays incl. the wrap-twisted  $m = 1/7$  rays) in `wolfram/tfpt_readouts_extension.wl`, verified engine run 2026-07-24,  $584 \rightarrow 588$  (the 256-dim Fock spectra, PT corrections and finite- $g$  scans stay Python-only, flagged); suite  $527 \rightarrow 528$  modules.

**2026-07-23 (X) (v530–v533 — the external review round promoted: center quotient compiler, Coxeter Stage A tensor clock, dual-degree modular checksum 744, sheet-transfer disc  $-7$  norm line; every identity machine-confirmed exactly before promotion (probe 76/76); NO marker moves)**

- **New module v530 (DIAMOND.CENTER.QUOTIENT.01).** The Sheet-Diamond center  $C = M(1, 0)$  carries the fixed primitive line  $Cv = v$  ( $v = (-1, 1, 0)$ ), and the induced operator on  $\mathbb{Z}^3/\mathbb{Z}v$  is *exactly* the atom matrix  $A = \begin{pmatrix} 8 & 2 \\ 5 & 3 \end{pmatrix} = \begin{pmatrix} \text{rank } E_8 & |\mathbb{Z}_2| \\ g_{\text{car}} & N_{\text{fam}} \end{pmatrix} [\mathbf{E}]$  with  $\text{tr } A = 11$ ,  $\det A = 14$ ,  $\text{disc} = 65 = 5 \cdot 13$ ,  $\text{perm } A = 34 = p_5(a)$ ,  $\prod A_{ij} = 240$ ,  $\det(A - I) = 4$ . The center *self-codes*: row sums  $C\mathbf{1} = (7, 11, 13) = (\det A/|\mathbb{Z}_2|, \text{tr } A, \text{disc}/g_{\text{car}})$ ; with the fixed eigenvalue 1 this is  $(1, 7, 11, 13) =$  the CP-halved  $E_8$  Coxeter exponents. The v414 resolvent ladder factorises:  $\det(C + kI) = (k+1)\det(A + kI)$  identically, quotient ladder  $(14, 26, 40, 56)$ , differences  $(12, 14, 16)$ , endpoint  $224 = 248 - 24$ . *Honest correction of the review*: eigenvalue 1 is NOT unique to  $C$  ( $K, Q$  share it) — uniqueness holds for the *conjunction* fixed line + atom quotient + CP self-code; the full selector-stratum ablation stays  $[\mathbf{O}]$ . New paragraph in `tfpt_2` (Sheet Diamond section).
- **New module v531 (COX.STAGEA.TENSOR.01).** The `tfpt_1` two-stage Coxeter audit (Stage A: a natural order-30 operator with  $\chi = \Phi_{30}$  on the even-flip atlas *or its 8-dim primitive-character space*) is *closed on the character-space alternative*:  $T_{30} = \text{Comp}(\Phi_5) \otimes \text{Comp}(\Phi_6)$  has exact order 30 and  $\chi_{T_{30}} = \Phi_{30} [\mathbf{E}]$  (CRT,  $\gcd(5, 6) = 1$ : carrier clock  $\times$  family clock = Coxeter clock as *one integer operator*); Ramanujan trace tomography  $\text{tr}(T_{30}^k) = c_{30}(k) = c_5(k)c_6(k)$  for all  $k$ , signed table  $(-1, 1, 2, 4, -2, -4, -8, 8)$ , atom readout  $(2, 4, 8) = (|\mathbb{Z}_2|, |\mu_4|, \text{rank } E_8)$ ; honest kill scope ( $\Phi_{15}$  passes the unsigned triple — the *signed* table discriminates;  $\Phi_{16}/\Phi_{20}/\Phi_{24}$  die on the triple). The flip-atlas intertwiner (COX.FLIP.INTERTWINER) and Stage B stay  $[\mathbf{O}]$ ; `tfpt_1` Stage A prose updated.
- **New module v532 (E8.DEGREE.MODULAR.01, audit-only).** The dual products of the  $E_8$  invariant degrees,  $(60, 192, 240, 252)$ , satisfy simultaneously  $\sum = 744 = 3 \cdot 248$  (= the  $j$ -function constant, recomputed from the  $q$ -series  $E_4^3/\Delta$ ),  $\prod = |W(E_8)|$ ,  $\gcd = 12$ ,  $\frac{1}{12}(\cdot) = (g_{\text{car}}, \dim S^+, \det L, 7N_{\text{fam}}) [\mathbf{E}]$ . *Mandatory fence*:  $D_{16}$  passes the same checksum ( $1488 = 3 \cdot 496 = 2 \cdot 744$  — heterotic, NOT  $E_8$ -exclusive); six non-heterotic controls fail. *Prime-totative fingerprint (audit)*:  $\{2, 3, 5\} \cup (\text{Exp}(E_8) \setminus \{1\}) =$  all primes below 30, and 30 is the *largest* integer whose totatives are all 1 or prime (scan to  $10^4$ ) —  $h_{E_8}$  as the maximal prime-totative Coxeter number; per the no-free-pattern rule nothing here is load-bearing. New keybox in `tfpt_3`, cross-note in the `tfpt_1` 496 box.
- **New module v533 (FTR.DISC7.NORM.01).** The branch-preserving transfer path  $J, K, C, F = M(1, -2..1)$  is a *norm line*:  $\det M(1, t) = 4t^2 + 14t + 14 = ((4t+7)^2 + 7)/4 = N(\alpha_t)$  with  $\alpha_t = (4t+7+\sqrt{-7})/2$  integral in  $\mathbb{Z}[(1+\sqrt{-7})/2]$ , the path one translation  $\alpha_{t+1} = \alpha_t + 2$ , norms  $(2, 4, 14, 32)$ , constant discriminant  $-7$ , vertex  $7/4 > 0$   $[\mathbf{E}]$ ; 2 splits in  $\mathbb{Q}(\sqrt{-7})$  ( $-7 \equiv 1 \pmod{8}$ , class number 1); controls: winding axis linear, anchor block disc +105.  $[\mathbf{C}]/[\mathbf{O}]$  preregistered next test: do the four external transfer solvers share *one* fractional-linear action on  $\alpha_t$  (kill:

four incompatible actions)? `GATE.FTRANSFER` unchanged. New paragraph in `tfpt_2` + frame note in `tfpt_research_contracts`.

- **Wolfram mirror.** All four modules mirrored exactly in `wolfram/tfpt_readouts_extension.wl` (20 new checks, verified engine run 2026-07-23, 564 → 584); suite 523 → 527 modules.
- **Not promoted (honest fences).** The review’s remaining proposals stay research contracts, not verifications: the straddle-cone selector on the interacting FK quartets (the highest-leverage attack on the alignment bit after v529) and the metrology-torsor formalisation of  $v_{\text{geo}}$  are recorded as next steps in `experiments/next.txt`; the primes-below-30 identity stays an audit fingerprint exactly as the no-free-pattern rule demands.

## 2026-07-23 (IX) (website entry redesign: narrative home, honesty band, proof strip, guided-tour route — editorial only, NO new modules, NO marker moves)

- **Website home.** Slim one-viewport hero (proof strip: 523 machine-checked modules · Wolfram mirror · Lean carrier · 0 external replications yet; copyable `python run_all.py` → ALL CHECKS PASSED; CTA trio film / browser check / kill board). Narrative order: Hero → IntroVideo (`#intro-video` kept) → In one breath → WhyThisMatters (plain-English column) → TrustContract → ClaimStack → HonestyBand (honest negatives: RXTE/NICER nulls, 2DCS/v529, ten bit tests) → Safeguards (four core stats) → Overview (TheoryUnpacking/ThreeDecoderMap behind progressive disclosure) → PathChooser → Downloads teaser.
- **Archive routes.** Full PapersSection → `/papers`; PredictionsSection → `/predictions`; ReconstructionChain + HorizonStory + Gravity → `/architecture`; RosettaLexicon → `/orientation`; OpenGates → `/verification`; EcosystemSection off home (footer adjacent link). Legacy hash redirects on home for old anchors.
- **Nav / design.** Primary nav: Start · Tour · Verification · Kill board · Replication · Papers · FAQ (More dropdown for secondary). SuiteBadge (static per release from `suite.ts/version.ts`). Design pass: drop purple-gradient pills/glow in hero and accent tokens; lab-notebook proof strip as above-the-fold visual.

**2026-07-23 (VIII) (prominence pass — the v519–v529 breakthroughs lifted into the FRAME texts of the papers and the website data layer; purely editorial, NO new modules, NO marker moves — (1) `tfpt_5_redteam` gains the seam round Target G: the ten-test side-blind scoreboard on the alignment bit (v512/v521/v525/v529) and the first *firing* kill test of the programme — the interacting straddle law (v529), documented as a named threat AND the first hard selection principle for  $\mathcal{A}_{\text{hol}}$ , fenced at toy level; (2) the bit-reduction statements in `tfpt_1`, `tfpt_2` and `origin_theory` extended from the v512 state to the current state (ten side-blind tests, the twist-class definition with gauge-robust order parameter, “in principle interferometrically readable”, v528 — the bit stays formal input); (3) the thermal seam lifted into the `origin_theory` horizon chapter (new keybox:  $T_{\text{seam}} = 4c_3$  measured, temperature the third leg of  $c_3$ , v526) and into the `tfpt_horizon_readouts` head keybox; (4) `tfpt_3` title/keybox now carry the thirty-step story and “the measure chain is derived” (v520/v523) prominently; (5) `tfpt_research_contracts` title/intro keybox now show the executed stages of CELEST.SEAM.01 (M1–M3,  $\delta$ -chains decided,  $w_m$  derived) and WOIT.OS.TWISTOR.01 ( $\alpha/\beta_1/\beta_2$  executed,  $\beta_3$  next under the straddle constraint); both contracts stay Open)**

- **Papers.** `tfpt_5_redteam` (Target G + method/outcome/title honesty pass); `tfpt_1`, `tfpt_2` and `origin_theory` (bit-reduction statements → ten tests + twist-class definition); `origin_theory` (thermal-seam keybox, horizon chapter); `tfpt_horizon_readouts` (head

keybox third leg); `tfpt_3` (title + keybox: thirty-step story, measure chain derived); `tfpt_research_contracts` (title + intro keybox: executed contract stages).

- **Website.** `papers.ts` (redteam Target G, horizon third leg, audit/contracts abstracts, architecture reduction state); `orientation/StatusMatrix.tsx` (v512  $\rightarrow$  current state); `glossary.ts` (new terms: twist-class choice, thermal seam, straddle law, side-blind scoreboard); new visual components `ThermalSeamLegs` (three legs of  $c_3$ : geometry v58 + anomaly/modular v208 + temperature v526) and `WoitBridgeProgress` ( $\alpha/\beta_1/\beta_2$  done,  $\beta_3$  open, straddle warning) on the home and verification pages.
- **Counters.** Predictions counter made consistent with `predictions.ts` (27 falsifiable test surfaces): `README.md` badge/hero/prose and the intro-film description (23  $\rightarrow$  27; the film narration already says twenty-seven).
- **Honesty.** No marker moves anywhere; `SEAM.EQUIV.01` and `WOIT.OS.TWISTOR.01` stay Open; the bit stays input; all fences (“derived at probe level”, “at the free level”, “toy model”) preserved verbatim.

**2026-07-23 (VII) (v529 — the interacting Fidkowski–Kitaev seam toy promoted as the ledger row `SEAM.INT.FKTOY.01`: the FIRST firing Kill-Test-2 shadow — triple verdict  $F1 = \text{KILL} / F2 = \text{KILL} / F3 = \text{ERFOLG}$ ; the celestial/twistor story in `tfpt_3` gains Step 30 and the header goes twenty-six  $\rightarrow$  thirty steps — (1)  $F1 = \text{KILL} [E]$ :  $\Theta$  exists exactly on the interacting model ( $U_r^2 = 2^8$ ,  $\Theta^2 = +1$ , clock-tower inversion,  $[\Theta, H_g] = 0$ ) so Kill-Test 1 does *not* fire; but Kill-Test 2 *fires at toy level*: RP breaks for every  $g > 0$  — inertia ladder  $(37, 0, 0) \rightarrow (33, 4, 0) \rightarrow (31, 6, 0) \rightarrow (29, 8, 0)$ ; Gibbs  $\beta = 2$  same ladder; mechanism = interference; STRADDLE LAW (24/24); [C] fence one toy / one interaction class / flat-band parent — narrows the Woit route, does *not* close it; `WOIT.OS.TWISTOR.01` stays Open with the toy shadow documented as a named Kill-Test-2 threat; (2)  $F2 = \text{KILL} [E]$ : the TENTH side-blind test (9  $\rightarrow$  10) —  $\Theta_{15}$  maps  $H(m) \rightarrow H(8-m)$  EXACTLY despite  $\delta$ -seeing spectra; (3)  $F3 = \text{ERFOLG} [E]$ : grading +  $\mathbb{Z}_8$  tower + FK anchor + constructive OS quotient at  $\pi/2$ ; Wolfram extension 552  $\rightarrow$  564 (verified engine run); suite 519  $\rightarrow$  523; *NO marker moves*)**

- **Suite.** New module `v529_seam_interacting_toy_fk.py` (24 checks,  $\sim 97$  s); registered in `run_all.py` + `script_registry.csv`; suite 523 modules green.
- **Ledger.** New row `SEAM.INT.FKTOY.01`; `WOIT.OS.TWISTOR.01` text-precised (Kill-Test-2 toy shadow + straddle law); scoreboard rows text-precised (9  $\rightarrow$  10); no status moves.
- **Papers.** `tfpt_3` Step 30; `tfpt_research_contracts` (Kill-Test-2 toy threat + bit scoreboard 10); introduction (tenth side-blind test).
- **Wolfram.** Three exact mirrors ( $\mathbb{Z}_8$  identities; straddle combinatorics; FK binomial ladder).
- **Website.** `papers.ts` (story 30 steps + scoreboard 10), generated indices via build; `README.md`, `zenodo_description.html` (523 modules, highest ID v529, Wolfram 564/564), `next.txt` notes.

**2026-07-23 (VI) (v528 — the alignment bit as an exact measurable TWIST-CLASS choice promoted as the ledger row `SEAM.BIT.TWISTCLASS.01`: the physical definition — verdict SUCCESS; the celestial/twistor story in `tfpt_3` gains Step 29 — (1) Facets #14 ( $\eta$ -flip  $\Leftrightarrow \delta = \pi/2$ ) and #15 (twist-frustration  $\Leftrightarrow \delta = \pi/2$ ) both directions exact [E] (cardinality lemma  $|A_+| = |A_-| \Rightarrow m = 4$ ;  $g$ -class census 224 axes: flip class exactly at  $\{15, 31\}$ ); (2) order parameter  $O = 1/2$  at  $m = 4$  else 0 [E], gauge-robust; (3) unification #14=#15=flag-transitivity = ONE  $\mathbb{Z}_2$  invariant [E] (v512 web 13  $\rightarrow$  15); Kramers = POSITION half (derived), twist**

class = MODULUS half (the input); (4) [C] measurement sketch (interferometric in principle); the bit remains formal input; *NO marker moves*)

- **Suite.** New module v528\_seam\_bit\_twist\_class\_definition.py (18 checks,  $\sim 46$  s).
- **Ledger.** New row SEAM.BIT.TWISTCLASS.01; SEAM.TAU.FLAG.01 text-precised (web 13  $\rightarrow$  15); no status moves.
- **Papers.** tfpt\_3 Step 29; contracts + introduction (“symmetry-lift bit” now also “the twist-class choice, in principle interferometrically readable”).
- **Wolfram.** Three exact mirrors (cardinality lemma; 224-axis census;  $O$ -values).

**2026-07-23 (V) (v527 — the silver clock eigenvalue EXPLAINED and DEMYSTIFIED, promoted as the ledger row SEAM.CLOCK.SILVER.01: double verdict STRUCTURE = SUCCESS / CONNECTION = KILL (anti-numerology style); the celestial/twistor story in tfpt\_3 gains Step 28 — (1) STRUCTURE SUCCESS [E]:  $\{1, \sqrt{2} - 1\}$  closed via diagonal Hankel pencil at  $N = 8$ ,  $\tan(\pi/8) = C(3)/C(1)$ , doubling = exact second quantisation, carrier = silver-midpoint axes, gap =  $\operatorname{arcsinh}(1) = \operatorname{gd}^{-1}(\pi/4)$ ;  $N$ -scaling = pure  $N = 8/16$  lattice datum (not continuum); positive clock only for  $N \equiv 0 \pmod{8}$ ; (2) CONNECTION KILL [E]: preregistered 11466-candidate census over 16 frozen numbers: 0 hits above null; placebos comparable; (3) cross-link v526:  $\sqrt{2} - 1 =$  thermal kernel ratio; *NO marker moves*)**

- **Suite.** New module v527\_seam\_clock\_silver\_spectrum.py (22 checks,  $\sim 10$  s).
- **Ledger.** New row SEAM.CLOCK.SILVER.01; no status moves.
- **Papers.** tfpt\_3 Step 28; contracts thermal/silver follow-on.
- **Wolfram.** Three exact mirrors ( $\tan(\pi/8)$  pencil;  $N$ -scaling witness; gap =  $\operatorname{arcsinh}(1)$ ).

**2026-07-23 (IV) (v526 — the thermal seam promoted as the ledger row SEAM.THERMAL.KMS.01: KMS of the reconstructed free OS quotient CLOSED onto the Nariai/BH temperature chain — the THIRD LEG of  $c_3$ ; verdict SUCCESS; the celestial/twistor story in tfpt\_3 gains Step 27 — (1)  $\beta$  MEASURED =  $N$  at both levels [E] ( $N = 8$ :  $\omega = \operatorname{arcsinh}(2^{-3/4})$ ,  $\kappa = q^{-8}$ ;  $N = 16$ : pairing exponent  $(N/2) + 1 = 9$ ; wrong  $\beta$  excluded); (2)  $\beta_{\text{angle}} = 2\pi$  EXACT =  $1/(4c_3)$ ,  $T_{\text{seam}} = 4c_3 =$  Bisognano–Wichmann/Hawking normalisation (v208) [E]; (3) modular Hamiltonian closed [E] ( $\nu_1 = \cos(\pi/24)$ ,  $\nu_2 = \cos(7\pi/24)$ ; Tomita KMS- $\beta = 1$  on all 256 pairs); (4)  $c_3$  third-leg class-1 selection [E]/[C] (Hawking/Nariai chain; v104 cross-link); (5) entropy  $S(8/16/32) = 0.52187/0.63510/0.75006$ ,  $\Delta S/\ln 2 \rightarrow 1/6$ , horizon fractions do NOT transfer [E]; (6) [C]-5 silver =  $C(3)/C(1)$  thermal kernel ratio; non-compact  $\Rightarrow T = 0$ ; *NO marker moves*)**

- **Suite.** New module v526\_seam\_thermal\_kms\_nariai\_bridge.py (22 checks,  $\sim 8.4$  s).
- **Ledger.** New row SEAM.THERMAL.KMS.01; no status moves.
- **Papers.** tfpt\_3 Step 27; tfpt\_horizon\_readouts (third leg of  $c_3$ ); tfpt\_research\_contracts (thermal follow-on); introduction.
- **Wolfram.** Three exact mirrors (Q-polynomial/ $\beta$  measurement;  $2\pi = 1/(4c_3)$  chain;  $\nu$  char-poly).

2026-07-23 (III) (v525 — the twist-state attack on the alignment bit DECIDED and promoted as the ledger row SEAM.BIT.TWISTBLIND.01: “mark-decorated states do not see  $\delta$ ” — an honest, decided NEGATIVE result (v508 style), the NINTH side-blind test on the v512 scoreboard ( $8 \rightarrow 9$ ); the free-plus-twist class (everything Wick-computable) is EXHAUSTED; the celestial/twistor story in tfpt\_3 gains Step 26 and the header goes twenty-three  $\rightarrow$  twenty-six steps — (1) ALL LIFTINGS WELL-DEFINED AND BLIND [E]: (a) the  $\sigma$ -gauge arc decoration ( $\mathbb{Z}_2$ ) satisfies  $\sigma \circ r = \sigma$  on the swap axes  $\Rightarrow$  the twisted Gram is a diagonal unitary congruence  $DMD$ , spectra IDENTICAL to untwisted (dev 0.0); (b) the Kadanoff–Ceva 4-twist insertion  $\omega(D \cdot)/\omega(D)$  is well-defined on the  $N = 32$  ladder ( $\omega(D)$  real nonzero; the  $N = 16$  odd- $m$  degeneration is the  $C(\text{even}) = 0$  checkerboard artifact) and its RP SPECTRA are the FIRST free-class data of the programme that depend on  $\delta$  (pairwise up to 1.05) — yet the INERTIA is  $(16, 0, 0)$  PD with the same  $\eta = +i$  for EVERY  $m = 1..7$ : the decoration sees  $\delta$ , positivity does not; (c) the  $\mu_4$  twist at  $\beta = \pi/2$  is pure plane gauge; (c') the genuine Bogoliubov defect at  $\beta = \pi/4$ :  $\Theta$ -compatibility selects exactly 2 of 16 sign patterns (NS-wrap-forced), the winner genuinely non-monomial — still PD, spectra identical; STRUCTURE THEOREM (mechanism exact): the defect planes never straddle a cut bond, so every  $\Theta$ -compatible quasi-free mark decoration is RP-spectrally INVISIBLE; (2) SUB-RESULTS [E]: the defect energy  $E_{\text{def}} = 1.2752872$  for EVERY  $m$  (spread  $4 \times 10^{-40}$ ) —  $\delta$ -blind; the 4-twist Casimir  $\ln |\omega(D)|$  measures  $\delta$  ( $-0.890.. - 1.348$ ) but exactly mirror-symmetric under  $m \leftrightarrow 8-m$  — a family gauge, never a side selector; Kramers  $\langle \Gamma \rangle_{\text{tw}} = +1$  for every  $m$ ; (3) the HARVEST [E]: two bit-presupposing  $\pi/2$  structures (the  $\eta$  flip  $+i \rightarrow -i$  on the mark-fixing axes via  $\sigma \circ r = -\sigma$ ; the twist frustration of the mark-fixing mirror,  $(4, 4, 0)$  indefinite for both  $\eta$  — the lattice shadow of the Ising twist field’s mirror-oddness) — both formulable only on the fixing axes existing iff  $\delta = \pi/2$  (the v512 facet class); (4) NEGATIVE CONTROLS [E]: the twist-free limit = v519/v521 exactly; the site cut keeps failing; the v512 counterwitness is arc-equivalent at  $N = 16$  to the blind member  $\pi/4$  — no state selector can exclude it; a single twist has  $\omega(D) = 0$  identically; [O]: the door narrows to a genuinely INTERACTING  $\mathcal{A}_{\text{hol}}$  whose OS data are NOT Pfaffian-reducible to the chiral vacuum; Wolfram extension (with v523/v524)  $540 \rightarrow 552$  (verified engine run 2026-07-23,  $552/552 + \text{base } 116/116$ ); suite  $516 \rightarrow 519$  modules; *NO marker moves — the alignment bit remains genuine discrete input*)

- **Suite.** New module v525\_seam\_bit\_twist\_blind.py (20 checks,  $\sim 80$  s; Pfaffian-exact insertion machinery cross-validated three ways, 40-digit certificates); registered in run\_all.py + script\_registry.csv; suite 519 modules green.
- **Ledger.** New row SEAM.BIT.TWISTBLIND.01 (KILL, honest negative, preregistered); SEAM.BIT.RPBLIND.01 and SEAM.TAU.FLAG.01 text-precised (scoreboard  $8 \rightarrow 9$ , “the free-plus-twist class is exhausted”); no status moves.
- **Papers.** tfpt\_3 Step 26 + the story header (twenty-six steps) + the Step-22 gap sentence; tfpt\_research\_contracts (the QGEO.REALIZE.01 reduction paragraph  $8 \rightarrow 9$  and the WP2 relocation note extended by v525); introduction (the ninth side-blind test added to the reduction-state sentence).
- **Wolfram.** Three exact mirrors (the  $N = 32$  integer frame + the  $\sigma \circ r = \pm\sigma$  congruence mechanism; the  $\pi/2$  existence solveset + the  $(3 + 4i)/5$  root-of-unity census; the  $\delta \leftrightarrow \pi - \delta$  mirror equivariance); the Pfaffian spectra, the 2-of-16 pattern census and  $E_{\text{def}}$  stay Python-only.
- **Website.** papers.ts (tfpt\_3 story mirror 26 steps + scoreboard 9), generated indices

(ScriptIndex.tsx, changelog.ts, maps) via build; README.md, zenodo\_description.html (519 modules, highest ID v525, Wolfram 552/552), next.txt notes.

**2026-07-23 (II) (v524 — the WOIT- $\beta_2$  milestone of the OS twistor bridge WOIT.OS.TWISTOR.01 executed and promoted as the ledger row WOIT.BETA2.OS.01: “the OS quotient of the free system made explicit” — verdict SUCCESS per the frozen preregistration, [C]-typed per contract precision (iii); the celestial/twistor story in tfpt\_3 gains Step 25 — (1)  $\mathcal{H}_{\text{phys}}$  EXPLICIT [E]:  $N = 16$  ( $\deg \leq 2$ ): the  $37 \times 37$  bond-cut OS Gram exactly Hermitian, parity-block-diagonal, inertia  $(37, 0, 0)$  PD (min eigenvalue  $1.7801 \times 10^{-6}$  at 40 digits), null space  $\{0\}$ ,  $\dim 37 = 29 \oplus 8$ ;  $N = 8$  (complete half algebra):  $(16, 0, 0)$  PD (min  $3.3471 \times 10^{-3}$ ),  $\dim 16 = 8 \oplus 8 = 4^2$  — compact euclidean time reconstructs a THERMAL (KMS) representation; exact certificate  $\sin^2(3\pi/8) - \sin(\pi/8)\sin(5\pi/8) = 1/2$ ; (2) the TRANSFER SEMI-GROUP (Klein–Landau) [E]:  $\tau_k$  exactly Hermitian on all domains ( $k = 1..7$ , dims  $29, 22, 16, 11, 7, 4, 2$ ), chain identities exact, vacuum fixed; the positivity pattern = the site/bond dichotomy: even  $k$  PSD via the exact square identity  $T(2j) = A_j^* A_j$  ( $((22, 0, 0)/(11, 0, 0)/(4, 0, 0))$ , odd  $k$  indefinite ( $((10, 12, 7)/(5, 6, 5)/(2, 2, 3))$ ) — the one-step transfer is NOT positive (the chirality datum, kill-3 shadow sharpened), its even powers are; (3) THE CLOCK =  $T^{N/4}$  [E]:  $N = 16$ :  $\tau_4$  PD  $(11, 0, 0)$  (min  $6.6821 \times 10^{-5}$ ), compressed spectrum  $[0.0194517, 1.64144]$ , vacuum eigenvalue exactly 1, spectral projections  $\sim 10^{-40}$ ;  $N = 8$ : spectrum EXACTLY  $\{1, \sqrt{2} - 1\} = \{1, 1/\delta_{\text{Silber}}\}$  in both parity sectors — the silver axes of the v519 torsor return as the clock eigenvalue; the reconstructed rotation group  $U(s) = e^{isH}$  unitary with the group law; (4) the v522 RESOLUTION [E]: the census  $(745, 96, 0)$  reproduced; EVERY anti-matching entry is a wrap overlap, every overlap-free entry matches — the pre-quotient non-Hermiticity is EXACTLY the domain/wrap artifact; on the quotient the clock has spectral calculus; pre-registered KMS deviation: no contraction on the compact circle ( $C(1)/C(3) = 1 + \sqrt{2} = \delta_S$  exactly,  $\det(G - \tau_4) < 0$ ); (5) GSO/ $\Theta$  [E]: the grading survives; the perpendicular torsor mirror descends anti-unitarily ( $\eta_{\perp} \in \{\pm i\}$ ),  $\Theta_{\text{phys}}^2 = +1$  on every sector (Kramers-free),  $\theta_{\text{cut}} \circ \theta_{\perp} = \alpha_{N/2}$  exactly; (6) CONTROLS [E]: site cut  $(3, 3, 1)/(2, 2, 4)$  — the kill branch fires there; family A  $(4, 4, 0)$ ; anti-chiral  $(8, 8, 0)$  no quotient; wrong direction non-Hermitian (all mismatches wrap-overlap); KILL TESTS: (1)/(2) strengthened, (3) shadow sharpened, none fires, all seven stay live on  $\mathcal{A}_{\text{hol}}$ ; TRANSPARENCY: run 1 =  $18/20$  (a sympy simplify failure on a TRUE  $\pi/16$  product formula, fixed by a Laurent-polynomial certificate; no criterion changed); *NO marker moves — WOIT.OS.TWISTOR.01 stays Open,  $\beta_3$  is next*)**

- **Suite.** New module v524\_woit\_beta2\_os\_quotient.py (20 checks,  $\sim 5$  s; exact sympy + Pfaffian-Wick + 40-digit inertia certificates); registered in run\_all.py + script\_registry.csv.
- **Ledger.** New row WOIT.BETA2.OS.01 (SUCCESS, [C]-typed per precision (iii)); WOIT.OS.TWISTOR.01 claim/canonical status extended ( $\beta_2$  executed,  $\beta_3$  next); no status moves.
- **Papers.** tfpt\_3: Step 25 plus the Woit-bridge paragraph with the  $\beta_2$ -stage status. tfpt\_research\_contracts: a full “ $\beta_2$  stage — executed” block with \veri; the  $\beta/\gamma$  roadmap entry marked executed; the scope-fence status sentence extended.
- **Wolfram.** Four exact mirrors (the  $\alpha_1^N = (-1)^F$  wrap arithmetic + the certificate identity; the Klein–Landau one-particle transfer forms with odd- $k$  zero diagonal and the even- $k$  square identity; the exact  $N = 8$  clock spectrum  $\{1, \sqrt{2} - 1\}$  + the transparency  $\pi/16$  product formula certified symbolically; the KMS silver witnesses); the  $\mathcal{H}_{\text{phys}}$  Grams, inertia spectra and the rotation group stay Python-only.

- **Website.** `papers.ts` (`tfpt_3` + `contracts mirrors`), generated indices via `build`.

**2026-07-23 (I) (v523 — the constructive  $w_m$  derivation executed and promoted as the ledger row CELEST.WP5E.WM.01: “ $1/\det_j$  is the Atiyah–Bott / zeta-determinant fixed-point factor, COMPUTED from three independent sources” — verdict ERFOLG per the frozen preregistration; the residual [O] of v516/v520 is closed at the fibre-zero-mode level and narrows to the GLOBAL BCOV integral on  $\mathbb{PT}/\mathbb{Z}_4$  beyond the fibre zero-mode factor; the celestial/twistor story in `tfpt_3` gains Step 24 — (1) ATIYAH–BOTT (route i) [E]: the equivariant mode ledger of the fibre  $\mathbb{C}^2$  equals the closed form  $1/((1 - q i^j)(1 - q i^{-j}))$  exactly in  $\mathbb{Q}(i)$  for all  $n \leq 120$ ; REGULAR at  $q = 1$  with Abel value  $(1/2, 1/4, 1/2) = 1/\det_j$ ,  $\det_j = \det(1 - g^j) = (2, 4, 2)$ ;  $(C, 2)$  Cesàro convergence in exact arithmetic; fixed-point splitting at EVERY truncation level  $d \in \{7, 12, 25\}$  for the phased (skeleton) AND the v520-forced unphased (completion) trace — the Abel-limit  $\text{contact}_j = (Q^{(0)} - Q^{(j)})/\det_j$  is the v516 contact vector COMPUTED, not declared; (2) ZETA/QUILLEN (routes ii/iii) [E]:  $\det_\zeta$  of the  $g^j$ -twisted circle Laplacian  $= 4 \sin^2(\pi j/4) = \det_j$  EXACTLY (Lerch + reflection formula symbolically, Hurwitz certificates  $\sim 10^{-41}$ ); Quillen split  $\det \Delta = \det_j^2$ ,  $|\det \bar{\partial}| = \det_j$ , the real positive holomorphic section UNIQUE via the  $SU(2)$  conjugation pairing; the  $\delta_{1f} f_1 f_3$  block at the  $(0, b)$  nodes carries the exact constant term  $1/\det_b$  (deviations  $\leq 5.2 \times 10^{-28}$ ); (3) the CONSISTENCY CHAIN [E]:  $w_m = (0, \frac{3}{2}, 2, \frac{3}{2}) = 4h_m = |\mu_4|h_m = -4\text{ch}_2(T_m)$ ;  $T_5 = 0$  at every scale;  $h_2 : h_1 = 4 : 3$ ; the  $T_3$  budget  $32 - 8c$  forces  $c = 4 = |\mu_4|$  uniquely (leftovers  $24/16/8/0/-8$ ); per-sector Okubo squares  $(18, 9, 18)\langle x, x \rangle^2$ ; total  $45\langle x, x \rangle^2$ ;  $\Phi_{T_3} : 32 \rightarrow 0$ ,  $\Phi_P : 72 \rightarrow 0$ ;  $\psi(A_{\text{fix}}) = +64$ ,  $\psi(\text{contact}) = -64$ ; (4) NEGATIVE CONTROLS [E]:  $1/\det^2$  gives  $w' = (0, 5/8, 1, 5/8)$ , leftovers  $(2, 12)$ , Quillen contradiction;  $1/|1 - i^j|$  gives  $w''_1 = 1 + \sqrt{2} \notin \mathbb{Q}$ ; the  $\mathbb{Z}_2/\text{Eguchi–Hanson}$  anchor PASSES with the same derivation ( $\det = 4$ ,  $w = 1/2 = |\mathbb{Z}_2|h^{A_1}$ ,  $9\langle x, x \rangle^2 = \frac{1}{4} \times 36$ ,  $c = 2 = |\mathbb{Z}_2|$ ); the  $SO(16)/D_8$  KILL fires;  $\text{diag}(i, i)$  breaks the zeta = AB identity itself;  $k = 3, 5$ :  $w_m = m(k - m)/2 = k h_m$  — the weights wander correctly, only  $k = 4$  carries the  $E_8$  chain; TYPED PREMISES [C] (the mandatory fence): TP-REG (Abel/Cesàro/zeta regularisation = the definition of the equivariant character), TP-Q (Quillen structure of BCOV  $F_1 = \text{v520 TP-4}$ ), TP-NUM (unphased numerator = v520 ERFOLG-A), TP-CH (insertion channel  $(0, j)$ ); [O]: the global BCOV integral beyond the fibre zero-mode factor, the resolved-ALE route not built, SEAM.EQUIV.TWISTOR.01 stays Open; *NO marker moves*)**

- **Suite.** New module `v523_celestial_wp5e_wm_derived.py` (26 checks,  $\sim 1$  s; exact  $\mathbb{Q}(i)/\mathbb{Q}(\sqrt{2})/\text{cyclotomic arithmetic}$  + 40-digit Hurwitz/block certificates); registered in `run_all.py` + `script_registry.csv`.
- **Ledger.** New row CELEST.WP5E.WM.01 (ERFOLG); the v516 row CELEST.WP5E.M2.01 and the v520 row CELEST.WP5E.MEASURE.01 text-precised (“the  $w_m$  normalisation is now DERIVED at probe level; the residual [O] narrows to the global BCOV integral beyond the fibre zero-mode factor”); no status moves.
- **Papers.** `tfpt_3` Step 24 + the “What remains, honestly” narrowing; `tfpt_research_contracts` (the WP5e measure-question block extended by the executed derivation with `\veri`, the M2 fence, the celestial summary blocks and the section status list); SEAM.EQUIV.TWISTOR.01 stays Open.
- **Wolfram.** Five exact mirrors (the mode ledger = closed form with Abel value; the zeta/Quillen route with the symbolic reflection formula and the  $\text{diag}(i, i)$  control; the derived-weight contact vectors + success table +  $\mathbb{Z}_2/\text{EH}$  anchor on the 240 glue roots; the consistency chain + wrong-

weight controls; the  $k = 3, 5$  cyclotomic wandering controls); the Cesàro/Hurwitz/stripped-block numerics stay Python-only.

- **Website.** `papers.ts` (`tfpt_3` + contracts mirrors), generated indices via `build`.

**2026-07-22 (XIII)** (HFQPO scorecard extension — the NICER/MAXI J1820+070 third-tooth null and the AstroSat/LAXPC block integrated as a new evidence-scorecard row (search target, never a claim): the preregistered single-QPO extension of the H3 ladder discriminator (frozen hypotheses/`hfqpo_ext_v1.yaml` BEFORE any photon download) ran on 6 NICER ObsIDs (13.22 GB,  $2013 \times 16$  s = 32.2 ks stack at 19.05 kc/s) — sanity gate PASS (the published 55.12 Hz anchor reproduced at 55.03 Hz, 0.94% rms,  $3.8\sigma$ ), injection calibration PASS ( $\geq 90\%$  recovery at 0.93% rms), the geometric tooth at 82.68 Hz NOT detected ( $0.69\sigma$  single-trial;  $3\sigma$  upper limit 0.75% rms BELOW the detected anchor strength  $\Rightarrow$  `null_with_sensitivity`), the integer line at  $\sim 110.6$  Hz at  $3.82\sigma$  after trials — BELOW the preregistered  $4\sigma$  threshold, documented as a sub-threshold excess consistent with the weak second harmonic of ATel #11951, NOT a hit; AstroSat/LAXPC (GRS 1915+105, scientifically the strongest channel) `infrastructure_blocked` (ISSDC login-gated); the RXTE + NICER nulls now cover five sources on two instruments — the tooth channel is recommended DORMANT; anti-double-counting: the parent RXTE row and the new extension row share `independence_group hfqpo_ladder_tooth` (one hypothesis, two instruments, never two independent nulls); scorecard regenerated (121 rows), `predictions.txt` HFQPO row extended (“Kanal ruhend”); experiment-only integration — no verification claim, no marker moves)

- **Scorecard.** New row “BH HFQPO third-tooth extension on non-RXTE archives” (NICER/MAXI J1820+070 plus AstroSat/LAXPC; `hfqpo-ladder` extension) in the generator `experiments/build_evidence_scorecard.py` (status `null`, stage `search_target`, signature `Sle`); OVERRIDES give both HFQPO rows the shared `independence_group hfqpo_ladder_tooth`; generator re-run: `evidence_scorecard.json` 121 rows, README stats block refreshed.
- **Experiment surfaces.** `experiments/hfqpo-ladder/extension-nicer-laxpc/` (preregistration, frozen manifest, download log, per-ObsID caches, stack/sanity/injection/scan reports, `results_ext.json`, README) — executed 2026-07-22; `predictions.txt` HFQPO row updated with the NICER finding and the dormant-channel recommendation; `next.txt` round notes.

**2026-07-22 (XII)** (v522 — the  $\text{WOIT-}\beta_1$  milestone of the OS twistor bridge  $\text{WOIT.OS.TWISTOR.01}$  executed and promoted as the ledger row  $\text{WOIT.BETA1.GSO.01}$ : “the clock is time-like, GSO is the gauge datum” — a TYPED result (verdict UNDECIDED per the frozen preregistration, the mechanism identified exactly, gauge-fixed RP established under the corrected typing; the celestial/twistor story in `tfpt_3` gains Step 23) — (1) the HERMITICITY WITNESS [E]: the  $\mu_4$  clock average violates Hermiticity EXACTLY (the one-step insertion  $T_1 = \langle \theta(e_a), \alpha_S(e_b) \rangle$  has entry  $-i/(8 \sin(5\pi/16))$  where the conjugate transpose demands  $+i/(8 \sin(5\pi/16))$ ); entrywise census 745 Hermitian-matching, 96 ANTI-matching, 0 violations — the pairing is complex-SYMMETRIC, strictly weaker than Hermitian: “positivity after gauge fixing” is not well-posed for the clock reading); (2) TIME-LIKE TYPING [E]: ALL 16 dihedral reflection axes of the NS seam circle invert the clock lift ( $RSR^{-1} = S^{-1}$ ), NONE commutes — the  $\mu_4$  clock IS Woit’s euclidean rotation; the gaugeable ( $\theta$ -compatible) part of the

$\mathbb{Z}_8$  Fock tower is EXACTLY its 2-torsion  $\{1, (-1)^F\}$  = the GSO/fermion-parity  $\mathbb{Z}_2$  (Hermiticity census over all powers;  $\alpha^4$  = the full wrap = the  $2\pi$  rotation of the seam = fermion parity); (3) the POSITIVE PART [E]: GSO-fixed RP holds EXACTLY —  $N = 16$  even  $\deg \leq 2$  parity-projected bond-cut Gram  $(29, 0, 0)$  PD (min eigenvalue  $1.78 \times 10^{-6}$  at 40 digits),  $N = 8$  complete half algebra: all odd rows/columns killed exactly, even block  $(8, 0, 0)$  PD ( $3.35 \times 10^{-3}$ ), no degree truncation; the site-cut defect SURVIVES gauge fixing  $((7, 9, 6)$  — the bond placement is gauge-invariant information); family A keeps failing  $((17, 12, 0)$  indefinite vs family D's  $(29, 0, 0)$ ); Kramers:  $\Theta_t^2 = \text{id}$  on  $\mathcal{A}^G$ ,  $\Theta_{\text{Fock}}$  restricts with  $\Theta^2 = +1$ ; Ramond control: the wrap +1 “true  $\mathbb{Z}_4$ ” lift fails state invariance — the NS/ $\mathbb{Z}_8$  tower is forced; (4) KILL-TEST BOOKKEEPING: kill test (2)’s free shadow does NOT fire (no legitimate Hermitian gauge-fixed form has a negative direction), kill test (1) stays discharged also gauge-invariantly; both stay formally live on  $\mathcal{A}_{\text{hol}}$ ; (5) TRANSPARENCY: the first frozen run scored 6/13 (the clock-reading checks failed exactly as the mechanism predicts) and the preregistered deg-2 invariant dimension 30 was a combinatorial error (correct census  $28 + 4 = 32$ ) — documented in full, the verdict by the frozen [C-3] logic (Hermiticity failure  $\Rightarrow$  UNDECIDED); CONTRACT PRECISION (iii) (prose, no marker): “gauge-invariant subalgebra” at the free level means the GSO/fermion-parity  $\mathbb{Z}_2$  — the only  $\theta$ -compatible part of the  $\mu_4$  tower; the clock is the euclidean rotation itself (timelike), its equivariant statement moves to  $\beta_2$  (OS quotient first, then the clock as reconstructed transfer operator); Wolfram extension (with v520/v521)  $522 \rightarrow 540$  (verified engine run); *NO marker moves* — *WOIT.OS.TWISTOR.01 stays Open*,  $\beta_1$  proper re-routed via  $\beta_2$ )

- **v522\_woit\_beta1\_gso\_gauge.py** (WOIT.BETA1.GSO.01, 16 checks,  $\sim 5$  s; exact sympy Pfaffian–Wick + exact  $\text{Cl}(16)$  Fractions; definiteness via 40-digit spectra, tolerance  $10^{-25}$ ; deterministic). G1 the gauge datum (Z8/NS tower, Ramond control, invariant dimensions  $120 > 64 > 32$ ); G2  $\Theta$  normalises the tower and restricts as an involution; G3 the  $(-1)^F$  class does not penetrate  $\mathcal{A}^G$ ; R1 the tower average is OS-symmetric, the odd sector dies; M1 the time-like census; M2 the exact Hermiticity witness; M3 the GSO 2-torsion census; R2 GSO-fixed RP ( $N = 16/N = 8$ ); C1–C3 site-cut/wrong-projection/family-A controls; V1 the frozen verdict. Registered in `run_all.py` (suite 516 modules, green), `script_registry.csv` (cluster *frontier*), ledger row WOIT.BETA1.GSO.01 (new).
- **Contract updated (markers unchanged).** WOIT.OS.TWISTOR.01: the  $\beta_1$  stage moved to *done* (typed UNDECIDED) with its own block, precision (iii) added, the  $\beta_1$  roadmap bullet annotated *executed*, the closing status line carries the  $\beta_1$  state; ledger WOIT.OS.TWISTOR.01 text extended (beta-1 executed via v522).
- **Paper integration.** `tfpt_3` (new Step 23 “the clock is time-like, GSO is the gauge datum”; the Woit-bridge paragraph extended by the  $\beta_1$  status); veri citations in `tfpt_research_contracts` and `tfpt_3`.
- **Wolfram.** Six exact mirrors (the Z8/NS tower  $S^4 = -1$  + the Ramond witness  $1/(4 \sin(5\pi/16))$ ); the invariant-dimension chain by exact ranks; the time-like reflection census; the exact Hermiticity witness with the full 745/96/0 census via exact Pfaffian–Wick; the GSO 2-torsion census; the family-A tower-normalisation failure) added to `tfpt_readouts_extension.wl`; the GSO-fixed RP inertia certificates stay Python-only, flagged in the `.wl` comment.
- **Provenance.** Discovery probe `woit_os_beta1_gauge_invariant_theta_probe.py` (16/16,  $\sim 3$  s) in `experiments/tfpt-discovery/` (2026-07-22); website mirrors regenerated; `next.txt` round notes.

2026-07-22 (XI) (v521 — the RP/ $\Theta$  attack on the alignment bit DECIDED and promoted as the ledger row SEAM.BIT.RPBLIND.01: “free OS positivity does NOT see  $\delta$ ” — an honest, decided NEGATIVE result in the v508 style (the celestial/twistor story in `tfpt_3` gains Step 22): does free reflection positivity (v519) or anti-linear  $\Theta$  existence break for  $\delta \neq \pi/2$  on the deformed mark family  $M(\delta) = \{\pm 1, \pm e^{i\delta}\}$  — which would DERIVE the v512 alignment bit from OS positivity? NO — (1) the CENSUS [E] (exact solvesets, symbolic  $\delta$ ): for EVERY  $\delta$  exactly the two mark-SWAPPING reflections  $\Theta_{\pm e^{i\delta}}$  exist (axes  $\delta/2, \delta/2 + \pi/2$ ; all 20 cut/mark incidence solvesets EMPTY); mark-FIXING reflections exist exactly at  $\delta = \pi/2$  (solveset  $\cos \delta = 0$ , lost immediately at  $\pi/2 - \varepsilon$ ); the v510/v512 counterwitness  $\{\pm 1, \pm(3+4i)/5\}$  carries the swap pair exactly; (2) RP IS  $\delta$ -BLIND [E]: bond-cut Gram inertias  $(8, 0, 0)$  PD for all tested  $\delta$  with the full sorted spectra IDENTICAL to 40 digits (deviation 0.0) — the ENTIRE free-RP spectral datum carries ZERO  $\delta$  information; the  $N = 16$  odd- $m$  failure (site axes,  $(3, 3, 1)$ ,  $\det = 0$ ) is a lattice-PLACEMENT artifact: at  $N = 32$  ALL  $m = 1.7$  have mark-compatible BOND axes with PD inertia  $(16, 0, 0)$ ; continuum: the OS kernel in axis-adapted coordinates is EXACTLY  $1/\sin((s+t)/2)$  — the axis position drops out identically (Cauchy–Stieltjes PD); the counterwitness PASSES free RP (6-point Gram  $(6, 0, 0)$ ); (3)  $\Theta$  EXISTENCE IS  $\delta$ -BLIND [E]:  $\Theta^2 = +1$ , deck normalisation and Fock implementability ( $U^2 = +2^7/2^8 > 0$ , exact  $\text{Cl}(16)$ ) for every  $\delta$ ; the ONLY  $\delta$ -sensitive v519 clause ( $\Theta\rho\Theta = \rho^{-1}$ ) is for  $\delta \neq \pi/2$  not violated but NOT FORMULABLE (a mark-compatible order-4 rotation exists iff  $\delta = \pi/2$ ) — it PRESUPPOSES the bit; same for the clock-tower clause; (4) the battery HAS TEETH at every  $\delta$  [E]: family A/antipode fails RP structurally (zero diagonal,  $(4, 4, 0)$ ), the chirality- $\eta$  pinning persists  $((0, 8, 0)$  ND) — RP separates the two real-structure FAMILIES, never the SIDES; (5) precondition controls: hexagonal (marks not concyclic — no seam circle), silver (deck not free), lifting audit (B = index relabelling,  $\delta$ -blind by construction; B’ fails the  $\pi/2$  negative control — discarded); [C] FACE #13 of the v512 equivalence web: the group generated by the admissible OS reflections is  $V_4$  for every  $\delta < \pi/2$  and closes to  $D_4$  exactly at  $\delta = \pi/2$  ( $\Theta_i \circ \Theta_1 = \text{the clock}$ ) —  $12 \rightarrow 13$  faces, a typed REFORMULATION of the bit, not a derivation; [O] the honest gap: marks decorating the STATE (defect/twist insertions, interacting  $\mathcal{A}_{\text{hol}}$ ) = exactly v519’s [O] scope; VERDICT: KILL exactly as preregistered — the free RP/ $\Theta$  battery is the EIGHTH side-blind test on the v512 scoreboard ( $7 \rightarrow 8$ ); the alignment bit is NOT derivable from OS positivity and remains genuine discrete input; *NO marker moves — the bit stays the one [C] carrier input*)

- `v521_seam_bit_rp_blind.py` (SEAM.BIT.RPBLIND.01, 25 checks, ~25 s; exact sympy solvesets + exact Gram entries + exact  $\text{Cl}(16)$  Fractions; inertia via 40-digit spectra, tolerance  $10^{-25}$ ; deterministic). S the anti-linear census (family D/A, counterwitness, face #13 closure, cut incidence, side conditions); L the lattice  $\delta$  ladder (frame,  $\pi/2$  reproduction,  $\pi/4$  PD, odd- $m$  artifact,  $N = 32$  refinement, site-axis control, spectra blindness, family-A rejection, chirality pinning); C the continuum axis elimination + counterwitness; LB the lifting audit; T Fock implementability + the clock-tower clause; W scoreboard + hex/silver controls + the preregistered KILL verdict. Registered in `run_all.py`, `script_registry.csv` (cluster core), ledger row SEAM.BIT.RPBLIND.01 (new).
- **Ledger/paper text precisions (markers unchanged).** SEAM.TAU.FLAG.01: scoreboard text extended ( $7 \rightarrow 8$  side-blind tests, face #13, web  $12 \rightarrow 13$ ); `tfpt_research_contracts` (the v512 reduction paragraph now carries the v521 kill and the honest [O] gap; the WP2 residual sentence extended), introduction and `tfpt_3` Step 2 ( $12\text{-fold} \rightarrow 13\text{-fold}$ , the eighth

side-blind test); veri citations in `tfpt_research_contracts`, introduction and `tfpt_3`.

- **Wolfram.** Six exact mirrors (the census solvesets incl. the exact-rational counterwitness; the  $V_4 \rightarrow D_4$  closure / face #13 as symbolic composition identities; the 20 incidence solvesets + the eps jump + the integer lattice frame; the continuum axis elimination + Cauchy–Stieltjes minors + the counterwitness cut ordering; the clock-tower clause + hex/silver controls; the  $\delta$ -blind side conditions) added to `tfpt_readouts_extension.wl`; the 40-digit lattice inertia spectra stay Python-only.
- **Provenance.** Discovery probe `seam_alignment_bit_rp_selector_probe.py` (25/25,  $\sim 24$  s) in `experiments/tfpt-discovery/` (2026-07-22); website mirrors regenerated; `next.txt` round notes.

**2026-07-22 (X) (v520 — the WP5e BCOV measure question DECIDED at probe level and promoted as the ledger row CELEST.WP5E.MEASURE.01: “single-valuedness is derived, the completion reading wins” — the  $\delta_{1e} + \delta_{1f}$  consolidation in the v518 style, ERFOLG-A on the preregistered decision layer (the celestial/twistor story in `tfpt_3` gains Step 21): which of the two exact, contradictory measures on  $\mathbb{PT}/\mathbb{Z}_4$  is the true BCOV measure — (A) the DECLARED completion reading (v516) or (B) the DERIVED chiral measure (v518)? — (1) the INVARIANT SUBSPACE [E]:  $\dim_{\mathbb{Q}}\{v : \rho(T)v = v, \rho(S)v = v\} = 8 = 4 \times 2$  in the dressed 16-component Weil system (exact rational nullspace); EVERY kernel vector lies in the  $\mathbb{Q}(\zeta_8)$ -span of  $\{e_H, e_{H'}\}$  — the two Lagrangian gluings, both  $\theta = E_4$ , level-1 quartic  $= Q^{(0)}$  = the UNPHASED trace; strict trivial-character solutions at physical weight  $(m, k) = (0, 0)$ : ZERO for every dressing (11 combinations); (2) SOURCE 1 — the F-LEMMA [E]: the only strict chiral families are  $\chi_4$  (dims (3, 3)) and  $\chi_{10}$  (dims (1, 1)), both with nontrivial  $T$ -character EXACTLY; the canonical column-matched member obeys  $G(\gamma\tau) = \chi_4(\gamma)G(\tau)$  pointwise (certificate  $3.9 \times 10^{-16}$ ,  $|G| \geq 59.6$ ); EXACT change of variables  $\Rightarrow$  fundamental-domain independence forces  $f = 0$  for EVERY chiral integrand: the v518 route was never a well-defined nonvanishing moduli integral (the kill is SHARPENED — its unfolded values are convention bookkeeping); (3) SOURCE 2 — the QUILLEN PAIRING [E]:  $|t\bar{t} - 1| < 8.3 \times 10^{-40}$  on all 15 pairs and both generators;  $|\chi_4|^2 = 1$  exactly vs one-sided  $\chi_4^2 = e(2/3) \neq 1$  ( $T$ -fixed:  $640 \not\equiv 0 \pmod{960}$ , 15/15 defective); the doubled 256-dim hol $\times$ antihol system closes STRICTLY at trivial character and physical weight (nullspaces  $(28, 17) \geq 11$ ), contains the unphased diagonal ( $\sim 2 \times 10^{-15}$ ) AND the physical columns  $w_b \otimes \bar{w}_b$  ( $\sigma$   $4.9/8.3 \times 10^{-16}$ ); negative control hol $\otimes$ hol: EMPTY nullspace (0, 0); (4) the CONSISTENCY CLOSURE [E]: the forced unphased numerator in the v518 scaffold reproduces  $\sum_m V \propto (1, 2, 1, 0, 0) = \langle x, x \rangle^2$  EXACTLY on every level  $n \leq 8$ ;  $J$ -weighted spreads  $\sim 5 \times 10^{-41}$ ,  $T_5/T_3 \sim 7 \times 10^{-42}$ , zero negative cells;  $\psi(\text{skeleton}) = +0.2302 = -\psi(\text{contact})$  — the  $J$ -weighted mirror image of  $\psi(A_{\text{fix}}) = +64/\psi(\text{contact}) = -64$ ; linearity  $4.6 \times 10^{-41}$ : the circle to v516 is CLOSED; (5) D3 — the  $\mathbb{Z}_2/\text{EH}$  ANCHOR [E]: only the declared reading hits ( $9\langle x, x \rangle^2 = \frac{1}{4} \times 36$ ,  $\psi = -8$ , scale tooth  $c = 1/2 = |\mathbb{Z}_2|h^{A_1}$ ); all three derived instantiations fail (spreads 1.12/0.895/0.516,  $\psi$  sign wrong); (6) D1 — COLUMN CANONICALISATION [E]: the physical AB column  $\sum_m i^{bm} e_{(m,m)}$  is contained EXACTLY in the  $\chi_4$  family ( $\sigma$   $1.8/1.7 \times 10^{-16}$ , one scalar per orbit — canonicalising  $(N_1, N_2)$ ); the canonical member STILL fails every tester; NEGATIVE CONTROLS:  $SO(16)/D_8$  ( $T_5 = 20/T_3 = -40$  exact,  $\mu_2$ -only), wrong form ( $|\text{Gauss}|^2 = 64$ ), wrong weight (dev 0.78), shuffle ( $T_5 = -14/T_3 = 36$ ), planted family detected ( $1.5 \times 10^{-16}$ ), hol $\otimes$ hol empty; [C] TYPED PREMISES (the mandatory fence): TP-1 (F-independent moduli integral), TP-2 (nonvanishing — must source  $\psi = 64$ ), TP-3 (v518 kernel**

convention), TP-4 (Quillen structure of BCOV  $F_1$ ); [O] the constructive BCOV derivation of the  $w_m$  normalisation itself; WIRKUNG (text only, no marker moves): the v516 fence and the v518 tension now read “decided at probe level for the declared reading under TP-1..TP-4 (v520)”;  
SEAM.EQUIV.TWISTOR.01 stays Open; *VERDICT ERFOLG-A per the preregistered criteria — both constructive sources fire independently, the consistency closure carries*)

- **v520\_celestial\_wp5e\_measure\_decision.py** (CELEST.WP5E.MEASURE.01, 39 checks, ~20 s; exact  $\mathbb{Q}(\zeta_8)/\text{Fraction}/\text{sympy}$  rational nullspaces + 1/960-grid phase arithmetic + mpmath 30-digit kernels + numpy double SVD, deterministic). S0 exact foundations (roots, ledgers, A-core +  $\mathbb{Z}_2$ -anchor targets, Weil relations, characters); S1 transports + derived families + the  $k = 0$  slice; S2 D1 column canonicalisation + the sharpened kill; S3 D2 the exact invariant subspace; S4 D3 the  $\mathbb{Z}_2/\text{EH}$  anchor at the integral level; S5 source-1 (the F-lemma); S6 source-2 (the Quillen pairing, doubled 256-dim system); S7 the point-4 consistency closure; S8 negative controls; S9 the ERFOLG-A verdict + typed premises. Registered in `run_all.py`, `script_registry.csv` (cluster *frontier*), ledger row CELEST.WP5E.MEASURE.01 (new).
- **Ledger/paper text precisions (markers unchanged).** CELEST.WP5E.M2.01 (the v516 fence: the completion reading is now DERIVED at probe level from two independent constructive sources under TP-1..TP-4; residual [O] = the constructive  $w_m$  derivation), CELEST.WP5E.DELTA1.01 (the v518 kill sharpened, the tension decided); `tfpt_research_contracts` (the WP5e measure-question prose updated throughout: object diagram, WP4 arc,  $\varepsilon_1/\text{M2}$  fences, the measure-question target block now carries the executed decision, the closing status lines); `tfpt_3` (Step 17/19 fences updated, new Step 21, the “what remains” block narrowed to the constructive  $w_m$  derivation); `tfpt_4_frontier` (celestial keyboard updated). veri citations in `tfpt_research_contracts` and `tfpt_3`.
- **Wolfram.** Six exact mirrors (the 16-component Weil data with both Lagrangian gluings; the invariant-subspace forcing  $\dim = 2 = \text{span}\{e_H, e_{H'}\}$  by exact NullSpace + rank; the character arithmetic of the two sources  $|\chi_4|^2 = 1$  vs  $\chi_4^2 = e(2/3)$  with the exact grid defects; the A-core +  $\mathbb{Z}_2/\text{EH}$ -anchor targets with the scale teeth; the level-1 Okubo identity of BOTH gluings (240 norm-2 vectors each, summed quartic  $36\langle x, x \rangle^2$ ); the sol6/shuffle/diag( $i, i$ ) controls) added to `tfpt_readouts_extension.wl`; the Harvey–Moore integrals, SVD memberships and  $J$ -weighted closures stay Python-only, flagged in the `.wl` comment; combined engine run with the v521/v522 mirrors: extension 522  $\rightarrow$  540 (verified, Wolfram Engine 14.3; GATE.WOLFRAM.02, Wolfram README, `tfpt_safeguards`, website counters and `zenodo_description.html` synced 116/116 + 540/540).
- **Provenance.** Discovery probes in `experiments/tfpt-discovery/` (2026-07-22): `celestial_seam_wp5e_delta1e_bcov_measure_decision_probe.py` (28/28, ~15 s) and `celestial_seam_wp5e_delta1f_single_valuedness_probe.py` (23/23, ~17 s); suite 513  $\rightarrow$  516 modules (green); website mirrors regenerated (ScriptIndex, DAG *qft/P2* nodes, `papers.ts` story counters, glossary, changelog mirror); `next.txt` round notes.

**2026-07-22 (IX)** (v519 — the WOIT- $\alpha$  milestone of the OS twistor bridge WOIT.OS.TWISTOR.01 executed and promoted as the ledger row WOIT.THETA.FREE.01: “the real structure exists — and free reflection positivity picks the same family”; the celestial/twistor story in `tfpt_3` is now TWENTY steps — (1) THETA CLASSIFICATION COMPLETE [E]: exactly TWO families of anti-linear structures on  $\mathbb{C}^2$  normalise the clock  $\rho = \text{diag}(i, 1)$  — family D ( $z \mapsto \mu\bar{z}$ ,  $|\mu| = 1$ ):  $\Theta\rho\Theta = \rho^{-1}$  EXACTLY at matrix level,  $\Theta^2 = +1$  (−1 IMPOSSIBLE:  $MM = \text{diag}(|\mu|^2, 1)$  — Kramers-free), inverts the deck, fixed set = the great circle through the deck poles; family A ( $z \mapsto \mu/\bar{z}$ ): centralises the clock projectively (NEVER inverts it),  $\Theta^2 = \pm 1$  per  $\mu$ , fixed set  $|z| = 1$  resp. empty; ROLE SEPARATION: family

A DEFINES the euclidean section (Woit's  $\rho_{\text{tw}}$  side:  $\rho_{\text{tw}}^2 = -1$ , no real points — replicated exactly on  $\mathbb{C}^4$ ), family D REFLECTS it (the OS conjugation,  $\sigma_{\text{std}} = \text{conjugation with real points } \mathbb{RP}^3$ ); (2) MARK PINNING [E]:  $z \mapsto \mu \bar{z}$  preserves the  $\mu_4$  marks iff  $\mu \in \mu_4$  — a 4-element  $\mu_4$ -TORSOR (2 mark axes, 2 silver-midpoint axes), clock conjugation  $\rho \Theta_\mu \rho^{-1} = \Theta_{-\mu} \Rightarrow 2$  clock orbits; (3) SPIN/FOCK [E]: the  $\mathbb{Z}_8$  plane has no phase leaks ( $\bar{\zeta}_8^k = \zeta_8^{-k}$ ); exact  $\text{Cl}(16)$ :  $\Theta_{\text{Fock}} = U_r \circ K$  with  $\Theta_{\text{Fock}}^2 = 2^7 > 0$  (normalised +1), inverts the clock tower ( $V \mapsto 4096 V^{-1}$ ) and normalises the deck; the KRAMERS DICHOTOMY: the deck-induced candidate  $\Theta_t = \tilde{U} \circ K$  has  $\Theta_t^2 = (-1)^F$  — *the v510 split/nonsplit dichotomy IS the  $\Theta^2 = +1$  vs  $(-1)^F$  dichotomy*; the deck does NOT furnish  $\Theta$ , the seam-circle REFLECTION does; (4) KILL TEST 1 (free level) [E]: does NOT fire —  $\Theta$  exists with  $\Theta^2 = +1$  AND  $\Theta \rho \Theta = \rho^{-1}$  and satisfies all three side conditions simultaneously (marks, deck normalisation, spin/Fock); (5) FREE RP [E]: the NS vacuum  $C(d) = (2/N)/\sin(\pi d/N)$  has  $C^2 = -1$  exactly; preconditions exact ( $RCR^T = -C$ ,  $SCS^T = C$ ,  $RSR^{-1} = S^{-1}$ ; the deck FAILS anti-invariance); BOND CUT: one-particle Gram  $(8, 0, 0)$  PD (min eigenvalue  $1.888 \times 10^{-3}$ , 40 digits), even  $\deg \leq 2$  sector  $(29, 0, 0)$  PD,  $N = 8$  complete half-sided algebra PD (even + odd each  $(8, 0, 0)$ ) — RP with NO degree truncation; twist  $\eta = +i$  FORCED ( $\eta = 1$  non-Hermitian); CONTROLS: the site cut fails exactly ( $\det = 0$ , inertia  $(3, 3, 1)$ ); continuum control: Cauchy–Stieltjes strictly positive — a lattice-PLACEMENT artifact, RP selects the site placement relative to the marks); the clock-CENTRALISING family-A structure fails RP structurally  $((4, 4, 0))$  — *RP and  $\Theta \rho \Theta = \rho^{-1}$  select the SAME family*; bonus: the anti-chiral state flips the odd sector to  $(0, 8, 0)$  ND (the free shadow of kill test 3); [C]/[O] NOT shown: the interacting algebra  $\mathcal{A}_{\text{hol}}$  (kill test 1 stays live there), gauge-fixed RP (kill test 2), OS reconstruction, chirality without mirrors (kill tests 3/6),  $\mu_4$  incidence (kill test 7) — the WOIT- $\beta/\gamma$  roadmap named ( $\beta_1$   $\Theta$  on the gauge-invariant subalgebra + gauge-fixed RP on the equivariant  $\text{SDYM}(E_8)$  sector;  $\beta_2$  the OS quotient of the free system explicit;  $\beta_3$  the  $\text{PT} \leftrightarrow \text{PT}^*$  duality vs  $\sigma_{\text{std}}$  typed;  $\gamma$  the chirality theorem + the mark incidence); TWO CONTRACT PRECISIONS (prose, no marker): (i) “ $\Theta$  induced by the seam reflection” means the seam-circle REFLECTION, not the deck — the deck furnishes the  $(-1)^F$  Kramers class (the v510 bridge); (ii) the marks sit RP-side on the BOND MIDPOINTS of the 16-Majorana circle; Wolfram extension  $516 \rightarrow 522$  (verified engine run); *NO marker moves — WOIT.OS.TWISTOR.01 stays Open: kill test 1 is discharged on the FREE level only and stays formally live on the interacting algebra*)

- `v519_woit_theta_rp_free.py` (WOIT.THETA.FREE.01, 34 checks,  $\sim 20$  s; exact sympy + exact  $\text{Cl}(16)$  Fractions; RP definiteness via 40-digit spectra of exact Hermitian matrices, tolerance  $10^{-25}$ ; deterministic). Part W: W-S1 the two-family classification (symbolic solve); W-S2 family D (relations, Kramers-free, mark pinning, seam action, torsor); W-S3 family A (equatorial/ $\rho_{\text{tw}}$ ); W-S4 the  $\mathbb{C}^4$  twistor level ( $\rho_{\text{tw}}$  vs  $\sigma_{\text{std}}$ ); W-S5 the  $\mathbb{Z}_8$  spin level; W-S6 the  $\text{Cl}(16)$  Fock level ( $\Theta_{\text{Fock}}^2 = 2^7$ ,  $V \mapsto 4096 V^{-1}$  via the implementer/centre route — a consolidation of the 105-s discovery probe to  $\sim 20$  s with identical claims and tolerances — deck normalisation, Kramers dichotomy,  $\tilde{U}V = V\tilde{U}$  control); W-S7 verdict. Part R: R1 the exact preconditions; R2 the site cut fails; R3 the bond cut passes (incl. the complete  $N = 8$  half-algebra); R4 the continuum control; R5 controls (twist forced, antipode fails, chirality flip); R6 verdict. Registered in `run_all.py` (suite 513 modules, green), `script_registry.csv` (cluster *frontier*), ledger row WOIT.THETA.FREE.01 (new).
- **Contract updated (markers unchanged).** WOIT.OS.TWISTOR.01: the  $\alpha$  stage moved to *done* with the core numbers and `veri{v519}`; the two precisions added as their own sentences

(the input definition of  $\Theta$  now names the seam-circle reflection, with the deck carrying the  $(-1)^F$  Kramers class; the marks at the bond midpoints of the 16-Majorana circle); the  $\beta/\gamma$  roadmap bullets added with success and kill criteria; the object-diagram OS row and the closing status line carry the  $\alpha$ -stage state. Ledger: WOIT.THETA.FREE.01 new; WOIT.OS.TWISTOR.01 text extended (“alpha stage executed via v519 ... kill test 1 does NOT fire at the free level and stays live on the interacting algebra; beta/gamma milestones named”). NO marker moves anywhere.

- **Paper integration.** `tfpt_3` (new Step 20 “the real structure exists — and free reflection positivity picks the same family” — the celestial/twistor story is now twenty steps; the Woit-bridge paragraph extended by the  $\alpha$  status: “the missing global object now exists at the free level; the interacting statement stays [O]”). `v519_woit_theta_rp_free.py` is cited via `veri` in `tfpt_research_contracts` and `tfpt_3`.
- **Wolfram.** Six exact mirrors (the two-family classification; mark pinning + the  $\mu_4$  torsor with the clock-orbit map and the cut-point census; the  $\mathbb{C}^4$  level  $\rho_{\text{tw}}$  vs  $\sigma_{\text{std}}$ ; the  $\mathbb{Z}_8$  tower conjugation; the  $\text{Cl}(16)$  dichotomy  $2^7$  vs  $(-1)^F$  on explicit Jordan–Wigner gammas incl.  $U_r V U_r V = 2^{19}$  and  $\tilde{U} V = V \tilde{U}$ ; the free-RP preconditions + site-cut degeneration + the Cauchy–Stieltjes continuum control) added to `tfpt_readouts_extension.wl` and verified with the active engine,  $516 \rightarrow 522$ ; the 40-digit PD inertia certificates of the bond-cut Grams stay Python-only, flagged in the `.wl` comment; GATE.WOLFRAM.02, Wolfram README and the website counters synced ( $116/116 + 522/522$ ).
- **Provenance.** Discovery probes `woit_os_theta_realstructure_probe.py` (21/21,  $\sim 105$  s) and `woit_os_theta_rp_freelevel_probe.py` (13/13,  $\sim 5$  s) in `experiments/tfpt-discovery/` (2026-07-22); website mirrors regenerated (ScriptIndex, DAG *qft* node + v519, `papers.ts` “20 steps” + the WOIT-contract mirror, glossary  $\Theta$ /Woit-bridge entries, changelog mirror); `next.txt` round notes.

**2026-07-22 (VIII) (v518 — the WP5e  $\delta_1$  chain ( $\delta_{1b} + \delta_{1c} + \delta_{1d}$ ) DECIDED and promoted as the ledger row CELEST.WP5E.DELTA1.01: “the derived chiral measure kills the  $\delta_1$  route” — an honest, decided NEGATIVE result in the v508 style, with the v516 tension stated: does the Harvey–Moore/BCOV fundamental-domain integral on  $\mathbb{PT}/\mathbb{Z}_4$ , evaluated under a measure DERIVED from blockwise  $SL(2, \mathbb{Z})$  covariance of the dressed 16-component Weil system (solved for — not declared, not scanned), deliver the twisted contact vector  $-A_{\text{fix}} = -(9, -30, -15, 0, 32)$  or the  $\psi = 64$  slice? NO — (1) the 16-COMPONENT WEIL COMPLETION ( $\delta_{1c}$ ): the discriminant module of  $D_5 \oplus A_3$  is  $\mathbb{Z}_4 \times \mathbb{Z}_4$  with  $q = (5x^2 + 3y^2)/8$ , Gauss sums  $2\zeta_8^5 \times 2\zeta_8^3 = 4 = \sqrt{16}$  (signature  $0 \bmod 8 = \text{rank } E_8$ ), the Weil relations hold EXACTLY in  $\mathbb{Q}(\zeta_8)$  ( $S^2 = C$ ,  $(ST)^3 = S^2$ ,  $S^4 = 1$ ,  $T^8 = 1$ , tensor split  $S = S_{D_5} \otimes S_{A_3}$ ), the diagonal  $H$  and anti-diagonal  $H'$  are the two invariant Lagrangians (both counts =  $E_4$ : two  $E_8$  gluings),  $P_H S P_H = \frac{1}{4} J$  with  $3/4$  of every  $S$ -image weight on the 12 shifted classes, and the 16-vector theta  $S$ -covariance CLOSES: the  $\delta_{1b}$  E1.5 residual  $2.91 = O(1)$  (naive 4-character rule) drops to  $\sim 10^{-39}$ ; (2) the OBSTRUCTION IS A FINITE  $\mu_4$  CHARACTER, IDENTIFIED ( $\delta_{1d}$ ): the dressed gauge blocks  $M = G[a, b] \eta^{-8}$  carry constant transport  $(t_p, s_p)$  (28+ digits, recognised as exact  $1/960$ -grid phases at  $10^{-25}$ ), the  $T$ -fixed pairs  $(0, b)$  carry  $t = e(-1/2) = -1$  EXACTLY (the naked level-matching defect), the transport-groupoid loops land in  $\Gamma_1(4)$  resp.  $\Gamma_0(2)^\pm$  with joint holonomy eigenspaces  $\dim 8/16$  resp.  $6/16$  and every eigen-phase of order  $\leq 4$ ; the KOBOUNDARY TEST: the  $SL(2, \mathbb{Z})$  relation defects  $c_{S^4}$ ,  $c_{(ST)^3 S^{-2}}$ ,  $c_{[S^2, T]}$  are  $(1, 1, 1)$  EXACTLY on all 15 pairs — the multiplier system is a CHARACTER of the orbit stabilisers, NOT a genuine 2-cocycle, and on  $\Gamma_1(4)$  it is  $\lambda(\gamma) = i^{2B+C/4}$ ; (3) the CANCELLATION**

EXISTS AND IS THE TWISTED FIBRE BLOCK  $f_1 f_3$ , NOT THE THREE SPHERE AXIONS:  $G[a, b] = f_1 f_3$  is an exact identity (30+ digits; the axion  $T$ -phases are v502 vacuum-energy data), the  $T$ -fix mechanism is one line of exact phase arithmetic —  $(-1) \times e(-1/6) = e(1/3) = \chi_4(T)$  at every  $T$ -fixed node — and the exact CANCELLATION TABLE reads: none  $\rightarrow$  residual order 4,  $f_1 f_2 f_3$  (the three sphere axions)  $\rightarrow$  4,  $f_2 \rightarrow 6$ ,  $f_1 f_3 \rightarrow 1$  (hol AND anti), wrong weights  $(1, 1, 2)/(2, 2, 2) \rightarrow 4$ ; the strict solutions are exactly  $\chi_4$  (dims  $(3, 3)$ ) and  $\chi_{10}$  (dims  $(1, 1)$ ) at  $\eta^0$ , certified on the dressed functions at a fresh  $\tau$  ( $10^{-15}$ , NO residual multiplier); (4) the DECISION — ALL THREE PREREGISTERED TESTERS FAIL: under both derived solutions the integral (30-digit kernels, negative-exponent cell audit, cut-6 vs cut-8 scan) fails  $T_5(V) = 0$  (fractions  $0.5/0.83$ ), fails  $W_2:W_{13} = 4:3$  ( $W_{13} = 0$ ), fails  $V = -A_{\text{fix}}N$  (spreads  $1.05/1.47$ ) and the  $\psi = -64N$  slice, and the honest  $(N_1, N_2)$  orbit rebalance rescues nothing in the positive cone ( $\chi_4$ :  $T_5 = 0$  only at  $\psi > 0$ ;  $\chi_{10}$ : only at  $N_2/N_1 = -4 < 0$ ) — a GENUINE KILL on the derived surface; (5) NEGATIVE CONTROLS:  $\mathbb{Z}_2/\text{Eguchi-Hanson}$  (gauge-only residual order 2 — the  $\mu_4$  defect is  $\mathbb{Z}_4$ -specific — and the SAME fibre dressing cancels it to 1),  $SO(16)/D_8$  (only even sectors, all axion  $b$ -characters in  $\mu_2$ : the order-4 supply structurally absent), wrong form  $(x^2+y^2)/4$  ( $|G|^2 = 64 \neq 16$ , its  $S$ -rule fails at  $O(1)$ ), wrong signature ( $S^2 = e(-1/2)C$  exactly); [C] the  $f_1 f_3 = \text{KS-weight}$  identification (deck weights  $(1, 3) =$  the twistor fibre weights forced by the v514 incidence ledger — exact match, NOT a complete BCOV derivation) + the kernel-family convention + the discrete  $\chi_k$  choice (both evaluated, both fail); the MANDATORY TENSION FENCE: the DECLARED completion reading (v516/M2) delivers the  $\psi = 64$  slice and cancels  $32 T_3$ ; the DERIVED chiral measure (this module) fails all three testers — both are exact; the sharp open question is which of the two is the true BCOV measure — neither result is hidden behind the other; [O] the BCOV-integral derivation on  $\text{PT}/\mathbb{Z}_4$  that decides between the two measures; Wolfram extension  $510 \rightarrow 516$  (verified engine run); *NO marker moves — the  $\delta_1$  route is decided negative under the derived measure, the cubic-GS question stays dissolved, SEAM.EQUIV.TWISTOR.01 stays Open, SEAM.EQUIV.01 untouched*)

- **v518\_celestial\_wp5e\_delta1\_derived\_kill.py** (CELEST.WP5E.DELTA1.01, 30 checks,  $\sim 20$  s; exact  $\mathbb{Q}(\zeta_8)/\text{Fraction}/1\text{-}960\text{-grid}$  phase arithmetic + mpmath 30-digit kernels + numpy double SVD, deterministic). S0 conventions + 16-class ledgers + targets; S1 the 16-component Weil system (exact) + the E1.5 closure; S2 gauge transport + the  $\mu_4$  multiplier system (koboundary test, character identified); S3 the cancellation ( $G = f_1 f_3$ ,  $T$ -fix mechanism, exact table); S4 the strict solutions (constructive scan,  $m \in \{0, 8, 16\}$ ); S5 the decision (function certificates, integrals, three testers,  $(N_1, N_2)$  scan); S6 negative controls; S7 verdict + the tension fence. Registered in `run_all.py` (suite 512 modules, green), `script_registry.csv` (cluster *frontier*), ledger row CELEST.WP5E.DELTA1.01 (new — “DECIDED: kill under the derived measure, tension with v516 stated”).
- **Contract updated (markers unchanged).** CELEST.SEAM.01: the  $\delta_1$  strand moves from “UNDECIDED at exploration level” to DECIDED by v518; the named remaining target is now the BCOV MEASURE QUESTION (the declared v516 completion reading vs the derived v518 chiral measure — both exact, which is the true BCOV measure is the named [O]); the cubic-GS question stays dissolved (v518 does not revive the  $d$ -channel). SEAM.EQUIV.TWISTOR.01 dependencies extended by v518 (preparation range now v492–v518; the route stays Open). CELEST.WP5E.M2.01 cross-referenced (its [O] pointer “ $\delta_{1b/1c/1d}$  territory” is executed and decided). NO marker moves anywhere.
- **Paper integration.** `tfpt_research_contracts` (the  $\delta_1$  stage moved to *done*, DECIDED

with the four exact results and the mandatory tension fence as its own paragraph; “WP5e proper” renarrowed from the  $\delta_{1d}$  derivation to the declared-vs-derived measure question; the object-diagram row, the honest status line and the per-ID status list extended by CELEST.WP5E.DELTA1.01); `tfpt_3` (new Step 19 “the derived measure decides — and disagrees with the declared one” — the celestial story is now nineteen steps; the Step-16/17 fences and “what remains” updated to the measure question); `tfpt_4_frontier` (keybox synced). `v518_celestial_wp5e_delta1_derived_kill.py` is cited via `veri` in `tfpt_research_contracts` and `tfpt_3`.

- **Wolfram.** Six exact mirrors (the 16-component Weil system with Gauss sums and tensor split; the two invariant Lagrangians  $+ P_H S P_H = \frac{1}{4}J$  + the 3/4 shifted-class weight; the koboundary test  $(1,1,1)$  on all 15 pairs from the recorded exact transport phases; the  $\Gamma_1(4)$  character formula  $\lambda(\gamma) = i^{2B+C/4}$  on the recorded loops; the T-fix mechanism  $(-1)e(-1/6) = e(1/3) = \chi_4(T)$  uniform for  $f_1 f_3$  vs inconsistent  $\{\chi_3, \chi_4\}/\{\chi_5, \chi_6\}$  for  $f_1 f_2 f_3/f_2$ ; the wrong-form/wrong-signature controls) added to `tfpt_readouts_extension.wl` and verified with the active engine,  $510 \rightarrow 516$ ; the Harvey–Moore integrals, SVD contraction solves, theta covariance certificates and block transport measurements are numerical and stay Python-only, flagged in the `.wl` comment; GATE.WOLFRAM.02, Wolfram README and the website counters synced  $(116/116 + 516/516)$ .
- **Provenance.** Discovery probes `celestial_seam_wp5e_delta1b/1c/1d_*.py` — the Harvey–Moore unfolding  $(30/30)$ , the Weil-16 completion  $(31/31)$  and the order-4 multiplier cancellation  $(36/36)$  — in `experiments/tfpt-discovery/` (2026-07-22); website mirrors regenerated (`ScriptIndex`, DAG *qft* node + v518, `papers.ts` “19 steps”, changelog mirror); `next.txt` round notes.

2026-07-22 (VII) (v517 — WP5e back-reaction milestone M3 “the  $a_0$  uplift” executed (SUCCESS on the preregistered v514 S8.3 criterion), promoted as the ledger row CELEST.WP5E.M3.01: the  $(4, \pm 4)$  GH multipole (= the  $a_0$  direction) is uplifted to the twistor-potential kernel  $\chi = \log P_4$  — the LOG of the v515 family polynomial — and the induced Kähler-potential correction is a log-type correction COUPLED TO THE CENTRE COUNT ON FOUR SCALES — (1) the KERNEL BRIDGE: the  $O(2)$  section is a NULL coordinate ( $\Delta_{3d}G(\eta_p) = 0$  identically for an ARBITRARY kernel  $G$ : the uplift is well-typed), the RESIDUE IDENTITY  $\partial_x[\text{transform}(\log \eta_p)] = 1/r_p$  holds EXACTLY (residue  $1/(2r)$ ), so the GH potential  $V = \sum_p 1/r_p$  IS the  $x$ -derivative of the contour transform of  $\chi$ ; flux  $-4\pi$  per centre, source charge  $4 = |\mu_4|$  (the  $V$ -ledger matched); (2) the ASYMPTOTIC KERNEL LOG:  $\chi = 4 \log \eta + a_2/\eta^2 + (a_0 - a_2^2/2)/\eta^4 + \dots$  with LOG COEFFICIENT  $4 = |\mu_4|$  = centre count (the CPS “ $N \log$ ” analogue at kernel level); at the clock-fixed seam fibre  $\zeta = 0$  the  $a_2$  slot closes and the first correction is EXACTLY  $a_0/\eta^4$  = the  $(4, \pm 4)$  multipole in kernel clothes; exact  $m$ -grading ( $m = \pm 4$  pieces PURE  $a_0$ ,  $m = 0$  piece  $-6t_0^4 \varphi_0$ -free: the  $U(1) \rightarrow \mathbb{Z}_4$  breaking sits exclusively in the  $a_0$  pieces); NO  $\log \times$  power terms (the v514 honest zero preserved — the  $a_0$  correction is power-law at infinity); the once-integrated GLT tower  $\sum_p \eta_p \log \eta_p = 4\eta \log \eta + p_4/(12\eta^3) + p_8/(56\eta^7) + \dots$  with  $p_n = 0$  unless  $n \equiv 0 \pmod 4$  and  $p_{4k} = 4(-a_0)^k$ : EVERY correction carries the prefactor  $4$  = centre count; (3) the EXCEPTIONAL-LOCUS LOG:  $\chi(\eta = 0) = \log a_0 = 4 \log t_0 + i(4\varphi_0 + \pi)$  ( $t_0$ -coefficient  $4 = |\mu_4|$ , clock phase  $4i$ ); the modulus response  $a_0 \partial_{a_0} \chi = a_0/(\eta^4 + a_0)$  equals 1 AT the exceptional locus and falls off as  $a_0/\eta^4$  at infinity (no constant, no log): the  $a_0$ -log lives at the exceptional locus, exactly as v514 S6.3 demanded; (4) the PERIOD RESPONSE:  $\Pi_j(\lambda)$  is LINEAR in  $t_0$  (uniform first-order shift: lockstep + phase ratios  $(1, i, i^2)$  preserved exactly),  $d \log \Pi_j / d \log a_0 = 1/4 = 1/|\mu_4|$  uniformly, and the integrated shift around the  $a_0$  circle is the monodromy  $e^{2\pi i/4} = i$  = ONE Coxeter clock step (the v493 monodromy REPRODUCED from perturbation theory; the finite monodromy is the cyclic centre permutation  $q_p \rightarrow q_{p+1}$ ); the node  $\lambda$ -support is  $t_0$ -FREE (the  $\mathbb{Z}_8$  seam bridge is  $a_0$ -rigid), all 6 pair discriminants  $\sim t_0^2 \neq 0$  ( $K_4$  connectivity: the forced uniform flux vector persists), the BM kernel is modulus-free with  $\int_{S^3} K = (2\pi i)^2$  (the quantised fluxes cannot move); (5) NEGATIVE CONTROLS: the  $(4, 0)$  multipole is  $\varphi_0$ -independent in  $V$  AND in the kernel (clock-invariant, breaks nothing); the  $\mathbb{Z}_2$ /Eguchi–Hanson analogue reads  $2 = |\mathbb{Z}_2|$  on EVERY dial ( $(2, \pm 2)$  amplitude  $\frac{3}{2} \sin^2 \theta$ , kernel log 2, exceptional log  $2 \log t_0$ , response  $1/2$ , monodromy  $-1$ ); wrong centre counts  $k = 3, 5$ : kernel log coefficient  $k$ , modulus slot  $O(6)/O(10) \neq O(8)$  — the observable MOVES with the centre count; forbidden family  $Z^4 - Z$ :  $p_3 = 3 \neq 0$  (selection rule broken),  $e_4 = 0$  (no  $a_0$ -log) — fails BOTH uplift dials; [C] the GLT dictionary ( $\eta_p \log \eta_p$  = the GH Kähler-potential kernel, Lindström–Roček / Ivanov–Roček); [O] the full nonlinear Kähler potential of the resolved  $A_3$  ALE; with v516 the v514 fence M1–M3 is FULLY WORKED OFF; Wolfram extension  $504 \rightarrow 510$  (verified engine run); NO marker moves — the KILL (decoupling from the centre count) does NOT fire, SEAM.EQUIV.TWISTOR.01 stays Open (preparation, not construction), SEAM.EQUIV.01 untouched)

- v517\_celestial\_wp5e\_m3\_a0\_uplift.py (CELEST.WP5E.M3.01, 23 checks,  $\sim 7$  s; exact sympy, no floats; deterministic). S0 replication (GH multipole ledger + the twistor family at general orbit phase, v515 slice); S1 the kernel bridge (null coordinate, residue  $V$ -ledger identity, source flux); S2 the uplift (asymptotic log 4, seam-fibre correction  $a_0/\eta^4$ , exact  $m$ -grading, GLT tower + selection rule); S3 the exceptional-locus log; S4  $\Omega_N$  consistency (period response

1/4, monodromy  $i$ ,  $a_0$ -rigid nodes and fluxes); S5 negative controls; S6 verdict. Registered in `run_all.py` (suite 511 modules, green), `script_registry.csv` (cluster *frontier*), ledger row CELEST.WP5E.M3.01 (new, SUCCESS).

- **Contract updated (markers unchanged).** CELEST.SEAM.01 now reads: M2+M3 executed via v516/v517 — the v514 fence M1–M3 is fully worked off: the twisted KS measure (declared completion reading,  $\text{ch}_2$ -weighted) cancels the  $32 T_3$  residual and delivers the  $\psi = 64$  slice without the cubic  $d$ -channel, and the  $a_0$  uplift couples to the centre count on four scales; what stays [O]: the BCOV-integral derivation of the completion reading ( $\delta_{1d}$ ) and SEAM.EQUIV.TWISTOR.01. SEAM.EQUIV.TWISTOR.01 dependencies extended by v516/v517 (preparation; the route stays Open).
- **Paper integration.** `tfpt_research_contracts` (the M3 bullet moved to *done* with the four coupled scales and the GLT fence; “what remains” narrowed to  $\delta_{1d}$  + the BCOV derivation); `tfpt_3` (new Step 18 “the  $a_0$  uplift: four coupled centre-count scales” — the celestial story is now eighteen steps; “what remains” updated). `v517_celestial_wp5e_m3_a0_uplift.py` is cited via `veri` in both.
- **Wolfram.** Six exact mirrors (multipole ledger + family at general phase; kernel bridge (null coordinate, residue  $1/(2r)$ , flux  $-4\pi$ ); asymptotic log +  $m$ -grading + GLT tower; exceptional-locus log; period response  $1/4$  + monodromy  $i$  +  $a_0$ -rigidity; negative controls) added to `tfpt_readouts_extension.wl` and verified with the active engine,  $504 \rightarrow 510$ ; GATE.WOLFRAM.02, Wolfram README and the website counters synced ( $116/116 + 510/510$ ).
- **Provenance.** Discovery probe `celestial_seam_wp5e_m2m3_a0_uplift_probe.py` (23/23) in `experiments/tfpt-discovery/` (2026-07-22); website mirrors regenerated (`ScriptIndex`, DAG *qft* node + v517, `papers.ts` “18 steps”, changelog mirror); `next.txt` round notes.

2026-07-22 (VI) (v516 — WP5e back-reaction milestone M2 “the twisted Kodaira–Spencer measure” executed (SUCCESS on the preregistered v514 S8.2 criterion, ON THE DECLARED COMPLETION MEASURE — verdict B), promoted as the ledger row CELEST.WP5E.M2.01: the one-loop box coefficient per sector on  $\mathbb{P}T/\mathbb{Z}_4$  under the completion contact term  $\text{contact}_j = (Q^{(0)} - Q^{(j)})/\det_j$  — the twisted-sector LOOP contributes the UNPHASED sector trace with the  $g^j$  zero-mode normalisation  $1/\det_j$ , of which the Atiyah–Bott skeleton kept only the phase-weighted insertion part — (1) the COMPLETION-WEIGHT IDENTITY (the centrepiece): per glue class  $w_m = \sum_j (1 - i^{jm})/\det_j = (0, \frac{3}{2}, 2, \frac{3}{2}) = 4h_m = |\mu_4|h_m = -4\text{ch}_2(T_m)$  EXACTLY ( $h_m = m(4-m)/8$  the v502 sector Casimir energies,  $\text{ch}_2(T_m)$  the McKay/Kronheimer charges, v505 S4.3 index bridge): THE THREE SPHERE AXIONS PAIR THROUGH THEIR OWN  $\text{ch}_2$  CHARGES — an exact identity, NO free scale, no fit; (2) PARAMETER-FREE LOCKS:  $T_5 = 0$  for ANY scale  $c$  (the  $\text{ch}_2$  pattern alone kills  $T_5$ ); ratio  $h_2 : h_1 = 4 : 3 =$  the  $\delta_1$ -forced leading ratio (REPRODUCED, not fitted); the  $T_3$  budget fixes  $c = 4 = |\mu_4|$  uniquely (leading system  $\det 256$ , unique weights  $(\frac{3}{2}, 2) = (4h_1, 4h_2)$ ); (3) the SUCCESS TABLE:  $\text{skeleton}_j + \text{contact}_j = Q^{(0)}/\det_j = 36\langle x, x \rangle^2/\det_j$  — a PERFECT Okubo square ( $T_5 = T_3 = 0$ ,  $\tilde{\lambda}^2 = 36$  in EVERY twisted channel:  $(18, 9, 18) \times \langle x, x \rangle^2$ ); total  $A_{\text{fix}} + \text{contact} = 45\langle x, x \rangle^2 = \frac{5}{4} \times 36\langle x, x \rangle^2 = \text{Dedekind} \times \text{Okubo} =$  the UNIQUE quartic-free weighting of v505 S3.8 (the rigidity theorem is exactly what the measure lands on); both v508 certificates KILLED ( $\Phi_{T_3}: 32 \rightarrow 0$ ,  $\Phi_P: 72 \rightarrow 0$ ); the v511 slice  $\psi = 64$  SUPPLIED EXACTLY ( $\psi(\text{contact}) = -64$ ) — NO cubic  $d$ -channel needed ( $c_d$  free = 0); remainder inside the exchange span ( $45\langle x, x \rangle^2 = \frac{45}{16}E_{22} = \frac{1}{80}E_{00}$ ,  $\Phi_P = 0$ ): the BULK axion cancels every channel with the SAME  $\lambda^2 = 36$ , the twisted propagator carrying the AB zero-mode factor  $1/\det_j$ ; the v514 measure chain stays  $\mu$ -blind; (4) NEGATIVE CONTROLS: wrong scale  $c \in \{1, 2, 3, 5\}$  leaves  $T_3 = (24, 16, 8, -8)$  — only  $c = 4 = |\mu_4|$  clears the budget; the  $h_1 \leftrightarrow h_2$  shuffle breaks  $T_5 = -14$  AND  $T_3 = 36$ ;  $SO(16)$  glue keeps  $T_5 = 20$  /  $T_3 = -40$  — the KILL branch FIRES there (Okubo failure inherited:  $E_8$  doubly special);  $\text{diag}(i, i)$  degenerates to the  $\mathbb{Z}_2$  target and breaks the conjugation pairing + the index bridge; the  $\mathbb{Z}_2$ /Eguchi–Hanson anchor PASSES with the same mechanism ( $A_2 + \text{contact} = 9\langle x, x \rangle^2 = \frac{1}{4} \times 36\langle x, x \rangle^2$ , coupling scale  $2 = |\mathbb{Z}_2| =$  centre count); (5) the MULTIPLIER NOTE (cross-reference to the parallel  $\delta_{1d}$  strand): the contact weights  $4h$  are REAL RATIONALS with no multiplier system of their own; the  $\delta_{1c}$  residual phases are the SQUARES of the  $A_3$  discriminant  $T$ -phases  $e(3a^2/8)$  — orders  $(1, 8, 2, 8)$  squaring to  $(1, 4, 1, 4)$ , a finite  $\mu_4$  system; Gauss sums  $2e(3/8)$ ,  $2e(5/8)$ , full Weil sum 4; [C] the MANDATORY FENCE: the completion reading (the loop of twisted states carries the UNPHASED sector trace with the same zero-mode normalisation) is DECLARED, supported by the  $\delta_1$  modular-completion finding, NOT derived from the BCOV integral; [O] the BCOV-integral derivation ( $\delta_{1b}/\delta_{1c}/\delta_{1d}$  territory); Wolfram extension  $498 \rightarrow 504$  (verified engine run); NO marker moves — the KILL (leftover independent quartic in ANY sector) does NOT fire, the completion reading stays [C]-declared, SEAM.EQUIV.TWISTOR.01 stays Open, SEAM.EQUIV.01 untouched)

- v516\_celestial\_wp5e\_m2\_twisted\_ks\_measure.py (CELEST.WP5E.M2.01, 23 checks,  $\sim 1$  s; exact sympy/Fraction, no floats; deterministic). S0 replication (roots, class quartics,  $A_{\text{fix}}$  two routes, bulk Okubo); S1 the index bridge + the completion weights (no free scale, parameter-free locks); S2 the contact term (two forms, genuine-contact certificate); S3 the success table (per-sector Okubo squares, Dedekind  $\times$  Okubo, certificate kills, exchange span,  $\mu$ -blind chain); S4 negative controls; S5 the multiplier note (cross-reference to  $\delta_{1d}$ ); S6 verdict. Registered

in `run_all.py` (suite 511 modules together with v517, green), `script_registry.csv` (cluster *frontier*), ledger row CELEST.WP5E.M2.01 (new, SUCCESS on the declared measure).

- **Contract updated (markers unchanged).** CELEST.SEAM.01: the M2 bullet moves to *done*; the v511 burden paragraph is closed out — the  $\psi = 64$  slice is DELIVERED by the declared KS measure (v516); the cubic  $d$ -channel is not needed — its physical-justification question dissolves unless  $\delta_{1d}$  revives it. The completion reading itself stays [C]-declared; its BCOV-integral derivation stays [O] ( $\delta_{1d}$ ).
- **Paper integration.** `tfpt_research_contracts` (the M2 bullet moved to *done* with the weight identity and the success table; the v511 fence paragraph extended); `tfpt_3` (new Step 17 “the twisted KS measure: every sector the same Okubo square” — on the way to the eighteen-step story). `v516_celestial_wp5e_m2_twisted_ks_measure.py` is cited via `veri` in both.
- **Wolfram.** Six exact mirrors (completion-weight identity + index bridge; root/quartic replication; parameter-free locks; contact term + success table + certificates +  $\mu$ -blind chain; negative controls; multiplier note) added to `tfpt_readouts_extension.wl` and verified with the active engine,  $498 \rightarrow 504$ ; GATE.WOLFRAM.02 and the Wolfram README synced.
- **Provenance.** Discovery probe `celestial_seam_wp5e_m2m3_twisted_ks_measure_probe.py` (23/23) in `experiments/tfpt-discovery/` (2026-07-22); website mirrors regenerated (ScriptIndex, DAG *qft* node + v516, changelog mirror); `next.txt` round notes.

2026-07-22 (V) (v515 — WP5e back-reaction milestone M1 “the  $A_3 \Omega_N$ ” executed (SUCCESS on the preregistered v514 S8.1 criterion), promoted as the ledger row CELEST.WP5E.M1.01: the back-reacted holomorphic 3-form  $\Omega_N$  is CLOSED FORM on the  $A_3$  twistor family, every  $S^3/\mathbb{Z}_4$  period is  $(2\pi i)^2$ -integral with the lockstep flux vector, and the lens geometry FORCES the source charge  $4 = |\mu_4|$  — (1) the RESIDUE FORM DERIVED: on  $F = XY - P(Z)$  all three ambient representatives satisfy  $dF \wedge \omega_2 = -dX \wedge dY \wedge dZ$  exactly and pull back to  $\omega_2 = dX \wedge dZ / X$  (Poincaré residue, derived not assumed); orbifold-cover pullback at  $a_0 = 0$ :  $\omega_2 = 4 dz_1 \wedge dz_2 = |\mu_4| \times (\text{flat form})$  — the residue normalisation CARRIES the source charge 4; clock  $\omega_2 \rightarrow i \omega_2$  (v492 S5 replicated); weight ledger  $4+4+2-8 = 2$ ,  $\Omega \in O(4) = -\deg K_{\mathbb{P}^T}$ ; (2) the TWISTOR FAMILY CLOSED: from the four  $O(2)$  centre sections  $q_p(\lambda) = i^p t_0 - i^{-p} t_0 \lambda^2$  exactly  $XY = Z^4 + 4t_0^2 \lambda^2 Z^2 - t_0^4 (1 - \lambda^4)^2$  ( $e_1 = e_3 = 0$  identically;  $\lambda = 0$  gives exactly the v493 family;  $a_2 \in O(4)$  opens only off-seam), CY-compatible clock lift  $\gamma : (Z, \lambda) \mapsto (iZ, -i\lambda)$ ,  $\gamma^4 = 1$ ,  $\Omega \rightarrow +\Omega$ ; (3)  $\Omega_N$  CLOSED + PERIOD TABLE:  $\Omega_N = \Omega_0 + \sum_p N_p K_p$  with CPS/Bochner–Martinelli kernels on the centre twistor lines,  $dK = 0$  off-source; boundary  $S^3/\mathbb{Z}_4$  at the orbifold point  $(2\pi i)^2 \times 1$  (minimal invariant charge  $4 = |\mu_4|$  — the lens fundamental domain FORCES  $N \in 4\mathbb{Z}$ ), deformed phase  $(2\pi i)^2 \times 4$ , linking  $S^3$  uniformly  $(2\pi i)^2 \times N$ , per-sphere flux vector  $(2\pi i)^2 \times N(1, 1, 1)$  (forced uniform: clock orbit +  $K_4$  connectivity), seam-fibre phase vector  $2\pi i t_0 (i - 1)(1, i, i^2)$ ;  $\Omega_0$  carries ZERO 3-flux (all flux sourced);  $(2\pi i)^2 = \text{fibre residue} \times \text{transversal residue}$  (density  $-\sin 2\theta$ ); (4) CLOCK COVARIANCE:  $\Pi_{j+1}(-i\lambda) = i \Pi_j(\lambda)$  for ALL  $\lambda$ ; the 12 collision nodes sit exactly on the 8 eighth roots of unity ( $8 = 2|\mu_4|$ , the  $\mathbb{Z}_8$  bridge), clock orbits  $4+4+4$ , honest conifold points (Hessian  $\det 512t_0^6$ , quadric rank 4); (5) the HONEST KILL PROBE: forbidden family  $Z^4 - Z$ :  $0/24$  lockstep (vs  $8/24$ ),  $\lambda$ -support 12 twelfth roots  $\neq \mathbb{Z}_8$ , no antipodal pairing — BUT the  $(2\pi i)^2$  quantisation holds there TOO: integrality alone is NOT the discriminator, the discriminator is the lockstep phase structure + the clock forcing; the v509 dial survives; (6) ANCHORS + CONTROLS: EH/ $\mathbb{Z}_2$  (charge  $2 = |\mathbb{Z}_2|$  forced), CPS original (every integer  $N$  allowed),  $\text{diag}(i, i)$  ( $\Omega_0 \rightarrow -\Omega_0$ , collapse at step zero), fractional charges  $N = 1, 2, 3$  excluded; [C] global kernel patching / Hitchin small resolution / CPS brane dictionary; [O] M2 (twisted KS measure), M3 ( $a_0$  uplift), the quantised BCOV coefficient; Wolfram extension  $490 \rightarrow 498$  (verified engine run); *NO marker moves — M2/M3 and BCOV stay [O], SEAM.EQUIV.TWISTOR.01 stays Open (v515 is preparation, not construction), SEAM.EQUIV.01 untouched*

- `v515_celestial_wp5e_m1_omega_n.py` (CELEST.WP5E.M1.01, 30 checks,  $\sim 1$  s; exact sympy, no floats; deterministic). S1 the residue form (three routes, cover factor 4, weight ledger, EH anchor, wrong-weight control); S2 the twistor family (closed form,  $O(2)$  reality, clock lift, section-level dichotomy); S3 periods (generic reduction, lockstep, covariance, collision ledger on  $\mathbb{Z}_8$ ); S4  $\Omega_N$  (BM anchor, lens descent, source curve, conifold nodes, M1 criterion); S5 the KILL probe (honest); S6 negative controls; S7 fences + verdict. Registered in `run_all.py` (suite 509 modules, green), `script_registry.csv` (cluster *frontier*), ledger row CELEST.WP5E.M1.01 (new, SUCCESS).
- **Contract updated (markers unchanged).** CELEST.SEAM.01 now reads: M1 executed via v515 — the back-reacted  $\Omega_N$  is closed-form on the  $A_3$  twistor family, all  $S^3/\mathbb{Z}_4$  periods are  $(2\pi i)^2$ -integral with the lockstep flux vector and clock covariance, and the lens geometry FORCES the source charge  $4 = |\mu_4|$ ; honest fence: integrality alone does not discriminate — the discriminator is the lockstep phase structure ( $0/24$  on the forbidden family); M2/M3 stay [O]. SEAM.EQUIV.TWISTOR.01 dependencies extended by v515 (preparation; the route stays Open).
- **Paper integration.** `tfpt_research_contracts` (the M1 bullet moved to *done* with the core

numbers; the object-diagram row  $\mathbb{C}^2/\mathbb{Z}_4 \rightarrow \mathbb{P}\mathbb{T}/\Gamma$  extended by v515 with the M1 kill retired to  $\delta_1/\text{M2-M3}$ ; WP4 handover, WP5 narrative, honest status line and status inventory synced); `tfpt_3` (new Step 16 “the back-reacted  $\Omega_N$ : closed form, integral periods, forced charge 4” — the celestial story is now sixteen steps; “what remains” updated: M2/M3 are the remaining preregistered milestones); `tfpt_4_frontier` stage list synced. `v515_celestial_wp5e_m1_omega_n.py` is cited via `veri` in both.

- **Wolfram.** Eight exact mirrors (residue-form identities; twistor family closed; period reduction + lockstep + covariance; collision ledger on  $\mathbb{Z}_8$ ; BM anchor + lens forcing  $N \in 4\mathbb{Z}$ ; source curve  $K_4$ ; conifold nodes; KILL probe +  $\text{EH}/\text{diag}(i, i)$  controls) added to `tfpt_readouts_extension.wl` and verified with the active engine, 490  $\rightarrow$  498; GATE.WOLFRAM.02, Wolfram README and the website counters synced (116/116 + 498/498).
- **Provenance.** Discovery probe `celestial_seam_wp5e_m1_a3_omega_n_backreaction_probe.py` (30/30) in `experiments/tfpt-discovery/` (2026-07-22); website mirrors regenerated (ScriptIndex, DAG *qft* node + v515, papers.ts “16 steps”, changelog mirror); `next.txt` round notes.

**2026-07-22 (IV) (external review integrated: route renamed, object diagram, group-extension definition, SEAM.EQUIV route split (MMST/TWISTOR ids, no status change),  $A_3$  role table, WOIT.OS.TWISTOR.01 contract — a pure structure/text integration, NO new physics claims, NO marker moves, suite unchanged at 508)**

An external expert review of the celestial/twistor branch confirmed the substance and asked for *semantic* precision — role separation, status clarity, and the exact naming of what is missing. Six changes plus one new central research contract, all editorial/structural:

- **(Ä1) Route renamed + the Woit bridge.** The `tfpt_3` celestial section and the contract heading are now “**The celestial and twistor continuum route**” (the old “from the  $\mu_4$  clock to the  $(E_8)_1$  boundary shadow” survives as the narrative-arc subtitle). A new paragraph block in `tfpt_3`, “*The Woit bridge: chiral Wick rotation and the real structure*” [O], introduces Woit’s Euclidean Twistor Unification (arXiv:2104.05099) as a *named external programme* (chiral Wick rotation problematic for purely chiral theories; Euclidean-holomorphic theory on  $\mathbb{P}\mathbb{T}$ , Lorentzian theory via conjugation structure/boundary values), lists TFPT’s existing ingredients (RP v379, OS reconstruction,  $\rho$  = clock v492,  $\sigma$  = deck/seam reflection v510), and names exactly what is missing: a global anti-linear  $\Theta : \mathcal{A}(\mathbb{P}\mathbb{T}/\Gamma) \rightarrow \mathcal{A}(\mathbb{P}\mathbb{T}/\Gamma)$  with  $\Theta^2 = 1$ ,  $\Theta\rho\Theta = \rho^{-1}$ , plus RP of the interacting BCOV+SDYM functional — “until that exists, the Woit connection is a strong analogy, not a mathematical integration”. No namedropping as confirmation.
- **(Ä2) The object diagram.** `tfpt_research_contracts` (CELEST section, up front) now opens with the chain  $(\Sigma, \mu_4, \rho, \Theta) \rightarrow \mathbb{P}^1 \setminus \mu_4 \rightarrow \mathbb{C}^2/\mathbb{Z}_4 \rightarrow \mathbb{P}\mathbb{T}/\Gamma \rightarrow \mathcal{A}_{\text{hol}} \rightarrow_{\text{OS}} \mathcal{A}_{\text{Mink}}$  as a table with one row per arrow (claim id, status, load-bearing modules v492/v493/v497–v514, kill test); the last two arrows are honestly [O] with WOIT.OS.TWISTOR.01 as the target — where mathematics ends and physics begins is visible at a glance.
- **(Ä3) The group extension, exactly.** A definition block:  $\Gamma_{\text{ALE}} = \langle \text{Deck} \rangle \cong \mathbb{Z}_4 \subset SU(2)$  (triholomorphic, preserves  $\Omega$ ; the quotiented  $\mathbb{Z}_4$ );  $\rho$  = Clock =  $\text{diag}(i, 1) \in U(2)$  (Kähler, *not* triholomorphic,  $\det = i$  rotates  $\Omega$  by  $i$  — v492 S5; normalises, is not quotiented); the global object is cyclic  $\langle \rho \rangle$  with  $\rho^2 = \text{Deck}$  (projectively  $\mathbb{Z}_4$ , spin lift  $\mathbb{Z}_8$ ; v492/v506/v507), plus the action table on  $K_{\mathbb{P}\mathbb{T}}$ ,  $\Omega$ , BCOV fields, open-string fields and boundary states, each entry [E] (arithmetic) or [O] (quantisation question); “ $\mathbb{P}\mathbb{T}/\mathbb{Z}_4$ ” is defined once for the whole contract (quotient by  $\Gamma_{\text{ALE}}$ , normalised by  $\langle \rho \rangle$ ) with a cross-reference sentence in `tfpt_3`.
- **(Ä4) SEAM.EQUIV route split (ledger restructuring, no status change in substance).** New ledger rows SEAM.EQUIV.MMST.01 (the conditional scaling-limit route — closed

modulo cited theorems, typing taken verbatim from the pre-split rows/prose; residual  $[O]$  = the cited continuum existence, v336) and SEAM.EQUIV.TWISTOR.01 (the open Costello–Li construction,  $[O]$ ; dependencies on CELEST.SEAM.01/WP modules and WOIT.OS.TWISTOR.01). SEAM.EQUIV.01 *remains* as the parent target with the OR logic as claim text: *closed if MMST.01 OR TWISTOR.01 closes; currently: MMST conditional-closed modulo cited theorems, TWISTOR open  $\Rightarrow$  parent stays  $[O]$  as an unconditional claim.* Every surface that said “SEAM.EQUIV.01 closed modulo cited theorem” (status cards in introduction, tfpt\_1/3/4, the central status card, README, origin\_theory, tfpt\_2, tfpt\_5, website StatusPyramid/OpenGates/StatusMatrix/papers.ts/glossary/VerificationDag) now attaches the statement to the MMST route id — the assertions are unchanged, only the ids are clean.

- **(Ä5) The  $A_3$  role table.** In the contract:  $A_3^{\text{family}}$  (lattice family factor,  $\mathfrak{su}(4)$  flavour),  $A_3^{\text{ALE}}$  (ADE type of  $\mathbb{C}^2/\mathbb{Z}_4$ ), the three exceptional spheres (Coxeter/PL structure),  $D_5^{\text{carrier}}$ , and — as external reference rows — Woit’s  $SU(3)_{\text{color}}$  and  $SU(2)_{\text{weak}}$ : per pair either the connecting functor with its vN witness (McKay type-level; Kronheimer–Nakajima quiver v479) or an explicit “*not identical — no identification claimed*”; in particular  $A_3^{\text{family}} \not\cong SU(3)_{\text{color}}$ , and the Woit rows are pure programme references.
- **(Ä6) Anti-number-hunting programme note.** experiments/next.txt (dated 2026-07-22 (IV)): the next real gain lies at positivity, incidence, locality and the  $\Theta$  real structure — *not* at further factorisations; recorded as working discipline per the review.
- **New central contract WOIT.OS.TWISTOR.01 (Open, research contract; new ledger row).** In tfpt\_research\_contracts after CELEST.SEAM.01: input  $X_+ = (\mathbb{PT}/\Gamma)_+$ ,  $\mathcal{A}_{\text{hol}}$  (interacting open+closed twistorial algebra),  $\rho$  (order-4 clock),  $\Theta$  (seam-reflection-induced anti-linear real structure),  $\mu_{\text{BCOV}+\text{SDYM}}$ ; target theorem  $\Theta^2 = 1$ ,  $\Theta\rho\Theta = \rho^{-1}$ , RP of the gauge-invariant algebra, OS quotient  $\Rightarrow (\mathcal{H}, \Omega, U(\mathcal{P}_+^\uparrow), \mathcal{A}_{\text{Mink}})$  with positive Hilbert metric, positive energy, local causality, *one chiral generation without mirror doubling*, the TFPT charge lattice, internal  $SU(3) \times SU(2) \times U(1)$  action, and the  $(E_8)_1$  seam net as an *actual* boundary net; *seven* preregistered kill tests ( $\Theta$ -clock incompatibility; RP failure after gauge fixing; vector-like mirror generation; Penrose transform only free/self-dual; right character but wrong OPE/net equivalence;  $SU(2)$  stays a spacetime factor; the four  $\mu_4$  marks not incidence-compatibly extendable); typing “this is the actual bridge from compiler to physics — everything in CELEST.SEAM.01 is preparation for it”; honest scope fence (self-dual  $\neq$  full 4D physics: both helicities, generic amplitudes, local matter, full Einstein dynamics, EWSB, confinement are explicit non-claims of the celestial branch). Mirrored on the website (papers.ts section, glossary entries “Woit bridge” and “Real structure  $\Theta$ ”).
- **No marker moves; no new vN modules; suite unchanged (508 modules, green).**

2026-07-22 (III) (v514 — WP5e- $\varepsilon_1$  “the  $O(-2)$  bulk-axion slot” executed (verdict B), promoted as the ledger row CELEST.WP5E.EPS1.01: the  $s = 0$  tower field of the celestial route is now a CONSTRUCTION,  $\tilde{\lambda} = 6$  is pinned three ways, and the first honest back-reaction step is taken in Gibbons–Hawking form — (1) the EQUIVARIANT  $O(-2)$  SLOT: the pushforward ledger  $\pi_* O(-2)$  on  $\mathbb{P}T' = \text{Tot}(O(1) \oplus O(1) \rightarrow \mathbb{P}^1)$  gives per degree  $d$  exactly  $(d+1)^2$  classes = the exact wave-operator nullspaces (1, 4, 9, 16, 25, 36 for  $d = 0..5$ ), the v492 action on the fibre coordinates  $(\mu_1, \mu_2) \mapsto (i\mu_1, i^{-1}\mu_2)$  with incidence-forced column weights  $(+1, +1, -1, -1)$  closes the equivariant bookkeeping BLOCK BY BLOCK (all  $d \leq 6$ , all 4 characters); character series  $P_0 = 1 + 3t^2 + 15t^4 + 21t^6 + 45t^8 + \dots$ ,  $P_1 = P_3 = 2t + 8t^3 + 18t^5 + \dots$ ,  $P_2 = 6t^2 + 10t^4 + 28t^6 + \dots$ ; the  $d = 0$  slot has multiplicity (1, 0, 0, 0) — THE BULK AXION SURVIVES THE PROJECTION; Molien invariant ring  $= (1 - t^8)/((1 - t^4)^2(1 - t^2))$  —  $Z \in O(2)$ ,  $X, Y \in O(4)$ , relation degree 8 = the  $a_0$  weight (v492/v493 bridge); E3 bijection sharpened (twisted minimal content  $(2t, 6t^2, 2t) =$  the Coxeter eigenvalues,  $\det(A - 1) = -4$ ); graviton control:  $O(+2)$  invariant only from fibre degree 4 with multiplicity  $3 = \{X, Z^2, Y\}$ ; (2)  $\tilde{\lambda} = 6$  TRIPLY PINNED: Okubo  $\text{Tr}_{\text{adj}} X^4 = 36\langle x, x \rangle^2 = (6\langle x, x \rangle)^2$  on the 240 glue roots ( $36 = h^\vee + 6$ ,  $\tilde{\lambda} = 6 = |\mathbb{Z}_2|N_{\text{fam}}$ ); the measure chain cancels  $\mu = 1/4$  EXACTLY (vertex<sup>2</sup>  $\mu^2$ , propagator  $1/\mu$ , anomaly  $\mu$ ;  $\tilde{\lambda}_{\text{eff}}^2 = 36$  for arbitrary  $\mu$ ; wrong bookings EXCLUDED: anomaly-only  $\Rightarrow 3$ , missing propagator renormalisation  $\Rightarrow 12$ ); flux (minimal CPS quantum  $N = 1$ , single channel iff  $k = 1$ ,  $(\kappa/c_3)^2 = 12$  exact); (3) the GH/ $A_3$  BACK-REACTION STEP: invariant 4-centre configurations = exactly 2 branches, the free orbit re-derives the v493 family ( $\prod_p (Z - i^p z_0) = Z^4 - z_0^4$ ,  $a_0 = -z_0^4$ , monodromy = one clock step), the charge- $k$  GH point is exactly  $\mathbb{R}^4/\mathbb{Z}_k$  ( $k = 4$ : seam boundary  $S^3/\mathbb{Z}_4$ ), periods  $\Pi_j = 4\pi t_0(i - 1)(1, i, i^2)$  LOCKSTEP (v509 counter-check), Coxeter clock from geometry ( $\Pi \cdot A = i\Pi$ ), source charge  $4 = |\mu_4|$ , EH asymptotic log coefficient EXACTLY 0 (the CPS log is an exceptional-locus statement; Burns contrast  $\det g \neq 1$ ), multipole rule  $m \equiv 0 \pmod 4$  with  $(4, \pm 4)$  carrying  $-a_0$ ; controls  $\mathfrak{so}_8$  ( $\tilde{\lambda}^2 = 12$  irrational, squares only  $\{9, 36\}$ ),  $\text{diag}(i, i)$  (Veronese),  $k = 2$  (three channels); [C] conditional on Costello’s flat- $\mathbb{P}T$  matching; [O] the quantised BCOV coefficient and the twisted channels ( $32 T_3$ ) — M1–M3 preregistered (the  $A_3 \Omega_N$  / the twisted KS measure / the  $a_0$  uplift, each with success + kill); Wolfram extension  $482 \rightarrow 490$  (verified engine run); *NO marker moves* — WP5e stays [O], SEAM.EQUIV.01 untouched, CELEST.SEAM.01 stays Open)

- v514\_celestial\_wp5e\_eps1\_axion\_slot.py (CELEST.WP5E.EPS1.01, 34 checks,  $\sim 4$  s; exact sympy/Fraction, no floats; deterministic). S1 Penrose accounting; S2 equivariant ledger + character series + Molien + E3 sharpening + graviton control; S3 the  $\tilde{\lambda} = 6$  residue (Okubo / measure / flux) with honest gaps; S4–S5 GH centre dichotomy, periods, Coxeter clock; S6 the CPS “ $N$  log” ledger (kernel charge, multipoles, EH log); S7 negative controls; S8 the M1–M3 roadmap; S9 verdict. Registered in run\_all.py (suite 508 modules, green), script\_registry.csv (cluster frontier), ledger row CELEST.WP5E.EPS1.01 (new, verdict B).
- Contract updated (markers unchanged). CELEST.SEAM.01 now reads:  $\varepsilon_1$  executed via v514 — the slot is a construction,  $\tilde{\lambda} = 6$  pinned by three exact ledgers, the GH/ $A_3$  step re-derives the v493 family and the Coxeter clock from geometry; the quantised BCOV coefficient and the twisted channels stay [O] (M1–M3 preregistered).
- Paper integration. tfpt\_research\_contracts (the  $\varepsilon_1$  stage moved to done with its own stage block; the named remaining targets now carry the M1–M3 milestones verbatim; WP4 handover and honest status line synced); tfpt\_3 (new Step 15 “the bulk axion is a construction;  $\tilde{\lambda} = 6$  is pinned three ways” — the celestial story is now fifteen steps; “what remains” updated).

v514\_celestial\_wp5e\_eps1\_axion\_slot.py is cited via veri in both.

- **Wolfram.** Eight exact mirrors (equivariant Penrose ledger; character series + bulk-axion survival; Molien hypersurface + Veronese control; Okubo perfect square + `so8` control; measure cancellation excluding 3 and 12 + flux dial; GH centre dichotomy; period lockstep + Coxeter clock; multipole rule + EH log ledger) added to `tfpt_readouts_extension.wl` and verified with the active engine, 482  $\rightarrow$  490; GATE.WOLFRAM.02, Wolfram README and the website counters synced (116/116 + 490/490).
- **Provenance.** Discovery probes `celestial_seam_wp5e_eps1_o2_bulk_axion_probe.py` (20/20) and `celestial_seam_wp5e_eps1_gh_a3_backreaction_probe.py` (14/14) in `experiments/tfpt-discovery/` (2026-07-22); website mirrors regenerated (ScriptIndex, DAG *qft* node, `papers.ts` “15 steps”, changelog mirror); `next.txt` round notes.

**2026-07-22 (II) (v513 — the  $c_d$  negative certificate, promoted as the ledger row CELEST.DTERM.NONDERIV.01 (anti-numerology certificate in the v508 style, honest rigid NEGATIVE result): can geometry derive  $c_d = 1920$ ? NO — the v511 [C] fence “ $1920 = |W(D_5)|$ ” does NOT survive the honest four-route battery with its PRE-DECLARED look-elsewhere ledger: (1) the CONVENTION TABLE gives  $c_d \in \{1920, 120, 256, 32, 69120\}$  across five normalisations (probe `tr-fund/e8-Killing`; Gell-Mann; `su(4)-Killing`; plain trace; `ch3-normalised d/6`) — only 1 of 5 hits  $|W(D_5)|$ , while the convention-STABLE content is the factorisation  $c_d = \Phi_{T_3}(A_{\text{fix}}) \times 2h^\vee(E_8) = 32 \times 60$ , both factors [E] (Killing propagator 60 from  $\sum_\alpha \langle \alpha, e \rangle^2 = 120$  over the actual roots;  $T_3$ -slot fixing via  $\det(\mathbf{P}\text{-block}) = -64 \neq 0$ ); (2) K1 DEAD: the charge-legal flux pairings (1, 3)/(2, 2) give 2048/1800 — they BRACKET 1920 and both MISS; the only flux hit  $64 \cdot 60/2$  is the FORBIDDEN (1, 2) pairing (charge 3 mod 4, v511 innerness); look-elsewhere: 924 expressions from 21 [E] numbers, 11 hits on 1920, controls  $1800 \rightarrow 8$ ,  $2048 \rightarrow 5$ ,  $2016 \rightarrow 0$ ; (3) K2: quantisation delivers only lattices ( $60\mathbb{Z}$  with 1920 the 32nd multiple, no minimality; strict `ch3` units  $2160\mathbb{Z}$  EXCLUDE 1920); the cubic quantum is lattice-dependent (6 on coroots,  $3/8$  on weights), the integer 32 has 12 factorisations; (4) K3 PROVENANCE CLASH (the death blow): cubic indices  $A_m = (0, 16, 0, -16)$  with  $\sum_m i^m A_m = 32i$  numerically  $= \Phi_{T_3}(A_{\text{fix}})$  — but the quartic class decomposition (10, -4, 30, -4) comes from classes 0/2 vs the cubic 16+16 from classes 1/3: same number, disjoint mechanism = coincidence; the Weyl side is WRONG ( $d$ -channel is  $A_3$ ,  $|W(A_3)| = 24$ ;  $D_5$  has no cubic;  $|W(E_8)|/|W(A_8)| = 1920$  a second accidental 1920); (5) K4 controls ( $SO(16)$ ,  $k = 2$  doubling 7680, false  $\mathfrak{g}_0$  with six cubics, `so(10)`) separate consistently; VERDICT: NUMEROLOGY as a derivation claim — the v511 [C] fence stays a fence, now typed look-elsewhere-loaded; the honest [E] core is  $c_d = 32 \times 60$ ; Wolfram extension 476  $\rightarrow$  482 (verified engine run); *NO marker moves — the fence stays [C], the physical generation of the 32 stays [O], CELEST.SEAM.01 stays Open*)**

- `v513_celestial_dterm_nonderivation.py` (CELEST.DTERM.NONDERIV.01, 24 checks,  $\sim 2$  s; exact `sympy/Fraction`, no floats; deterministic). N0 baseline (240 roots (52, 64, 60, 64),  $A_{\text{fix}} = (9, -30, -15, 0, 32)$ ,  $Q_{dd}$  two routes,  $c_d$  pinned by the  $T_3$  slot alone); N1 convention table + the stable factorisation  $32 \times 60$ ; K1 flux products + look-elsewhere ledger; K2 quantisation lattices; K3 cubic/quartic provenance clash + Weyl data; K4 negative controls. Registered in `run_all.py` (suite 507 modules, green), `script_registry.csv` (cluster *frontier*), ledger row CELEST.DTERM.NONDERIV.01 (new).
- **Fence sharpened (text only, marker unchanged).** CELEST.WP5E.DELTA2.01 and the CELEST.SEAM.01 contract prose now carry: “the  $1920 = |W(D_5)|$  reading is look-elsewhere-loaded (11/924 vs. 8/924 for the control target 1800) and convention-contingent — v513”; the

honest remaining target of the cubic-GS item is the physical generation of the factor 32, not the Weyl fingerprint.

- **Paper integration.** `tfpt_research_contracts` ( $\delta_2$  fence environment, named remaining targets, status list — new CELEST.DTERM.NONDERIV.01 status line); `tfpt_3` (Step-14 paragraph + the “what remains” sentence); `tfpt_4_frontier` (celestial-route clause). `v513_celestial_dterm_nonderivation.py` is cited via `veri` in the  $\delta_2$ /fence environments.
- **Wolfram.** Six exact mirrors (propagator anchor +  $T_3$ -slot pinning; five-way convention table; flux enumeration + look-elsewhere counts; K2 lattices; K3 provenance clash + Weyl data; K4 controls) added to `tfpt_readouts_extension.wl` and verified with the active engine, 476  $\rightarrow$  482; GATE.WOLFRAM.02, Wolfram README and the website counters synced (116/116 + 482/482).
- **Provenance.** Discovery probe `celestial_seam_wp5e_dterm_coefficient_derivation_probe.py` in `experiments/tfpt-discovery/` (2026-07-22, 24/24); website mirrors regenerated (ScriptIndex, DAG *qft* node, `papers.ts`, changelog mirror); `next.txt` round notes.

**2026-07-22 (I) (v512 — the flag-transitivity equivalence web promoted as SEAM.TAU.FLAG.01: the  $\tau = i$  attack executed on the free RP seam circle — BARE MARK-TRANSITIVITY IS FALSIFIED as a selector (on the test family  $M(\delta) = \{\pm 1, \pm e^{i\delta}\}$ ,  $\delta \in (0, \pi/2]$ , the symmetry group is the pair-exchanging  $V_4 = \{z, -z, b/z, -b/z\}$  for EVERY  $\delta$ , simply transitive on the four marks; complete census: deck-commuting Möbius maps are diagonal/antidiagonal; hardening: ALL undirected per-mark data are identical for every  $\delta$  — no local jet sees the bit), and THE SHARP SELECTOR is FLAG transitivity (marks AND their two sides indistinguishable)  $\iff \delta = \pi/2 \iff \tau = i \iff$  the clock — stabiliser jump  $1 \rightarrow 2$ , flag/arc orbits  $2 \rightarrow 1$ , order-4 rotation ( $b^2 = -1$  solveset  $\{\pi/2\}$ ), mark mirror  $z \mapsto 1/z \iff \cos \delta = 0$ ; welded into a 12-FOLD EQUIVALENCE WEB (every link exact): clock rotation,  $D_4$ , mark mirror, flag transitivity, three spectral degeneracies (odd-doublet split  $= (2/\pi) \ln \cot(\delta/2)$ , zero only at  $\pi/2$  — counterwitness exactly  $(2/\pi) \ln 2$ ; arc-Laplacian degeneracy  $4/\pi$ ; the v215-K3 indicator in CLOSED FORM  $(2\sqrt{2}/\pi) |\ln \tan(\delta/2)|$  — the v503 lattice curve analytically),  $\rho$ -twist existence  $((m^2 - n^2)(t - 1/t) = 0 \iff t = 1$ ; Atiyah–Bott rationality  $\delta$ -BLIND —  $\text{Tr}(\sigma e^{t\Delta}) = 1$  for all  $\delta$ ,  $j_{\text{cw}} = 148176/25$  rational non-CM: rationality selects NOTHING), cusp-4-cycle implementability (H4 rigidity:  $V_4$  with orders  $\{1, 2, 2, 2\}$  cannot induce it; centraliser scalar,  $U^2$  nonscalar  $\Rightarrow$  no projective involution — projective order exactly 4,  $\mathbb{Z}_8$ -shadow consistent v510), harmonic,  $j = 1728$ ; NEGATIVE CONTROLS: hexagonal (not concyclic, fails preconditions), silver (harmonic but topologically dead), 3 marks (impossible on the free circle: 0 vs 28 invariant subsets; bare  $n = 3$  transitivity already forces equal spacing — “4 counts”); [C] THE REST: the last input reduces from a continuous modulus to ONE discrete symmetry-lift bit ( $V_4 \rightarrow D_4$ : side-indistinguishability of the four equally-decorated marks), v216 local equality does NOT deliver it, its negation is now concretely measurable; scoreboard: the counterwitness passes all 7 established unsided tests, fails all 9 clock-equivalent ones; Wolfram extension 470  $\rightarrow$  476 (verified engine run); *NO marker moves — the bit stays [C], QGEO.REALIZE.01 and P2.TYPING.01/AX.P2.01 text-precised only, SEAM.BIT.FREEDOM.01 cross-referenced)***

- `v512_seam_tau_flag.py` (SEAM.TAU.FLAG.01, 29 checks,  $\sim 6$  s; sympy-exact + mpmath 30 digits + one numpy lattice commutator; deterministic). S0 family frame (counterwitness inside the family:  $\lambda = 5/4$ ,  $j = 148176/25$ , cross-ratio  $-1/4$ ); H1 complete symmetry census (census basis, rotation solvesets, stabiliser jump, flag/arc orbits, v216 separation, five-member

table); H2 spectral criteria in closed form; H3 heat-trace rationality ( $\delta$ -blind) + twist existence; H4 monodromy rigidity (Schur); NC negative controls; SYN the 12-fold web + counterwitness scoreboard + the typed sentence. Registered in `run_all.py`, `script_registry.csv` (cluster *core*), ledger row `SEAM.TAU.FLAG.01` (new).

- **Reduction state re-stated (text only, markers unchanged).** `AX.P2.01` and `P2.TYPING.01` synthesis notes now read: “four marks + one discrete symmetry-lift bit (flag transitivity  $\iff \tau = i) + \pi$ ”; `QGEO.REALIZE.01` precision: “the remaining input can equivalently be stated as the flag-transitivity postulate — `v512`; the search space for a future `[E]` closure is narrowed to the RP positivity of the odd sector”; `SEAM.BIT.FREEDOM.01` cross-referenced (the modulus half gains the discrete form).
- **Paper integration.** `tfpt_1` (the P2 reduction keybox re-dated 2026-07-22 + Theorem-7 skeleton keybox), `introduction` (narrative chain, consolidated reduction state, status-card addendum), `tfpt_3` (celestial Step 2 + the honest-status reduction bullet), `tfpt_research_contracts` (QGEO kill-test and marks-derived blocks, WP2 note); secondary mirrors `tfpt_2/origin_theory` aligned. `v512_seam_tau_flag.py` is cited via `veri` in all four primary documents.
- **Wolfram.** Six exact mirrors ( $V_4/D_4$  census table incl. flag/arc orbit counts; solveset equivalences + family identities; spectral faces in closed form; twist mode identity + AB  $\delta$ -blindness +  $j$ -rationality control; H4 Schur rigidity; negative controls) added to `tfpt_readouts_extension.wl` and verified with the active engine, 470  $\rightarrow$  476; the `mpmath` heat-trace numerics and the `numpy` lattice commutator stay Python-only (flagged).
- **Provenance.** Discovery probe `seam_tau_i_established_selector_probe.py` in `experiments/tfpt-discovery/` (2026-07-22, 29/29); website mirrors regenerated (`ScriptIndex`, DAG *P2/masses* nodes, `papers.ts` reduction-state mirrors, `StatusMatrix`); `next.txt` round notes.

2026-07-21 (XXIII) (v511 — WP5e- $\delta_2$  of CELEST.SEAM.01 “the full-tensor ledger” promoted as CELEST.WP5E.DELTA2.01 (intermediate verdict, exact): the named escape route of the v508 exchange no-go executed on the FULL adjoint tensor structure of  $\mathfrak{e}_8$  — the COLLAPSE IS CONFIRMED full-tensorially AND arity-crossing (the INNERNESS THEOREM:  $\mathfrak{g}_0 = \mathfrak{d}_5 \oplus \mathfrak{a}_3$  semisimple with NO  $\mathfrak{u}(1)$  (class-0 roots 40+12, root span rank 8),  $h$  in the Cartan OF  $\mathfrak{g}_0 \Rightarrow$  every invariant tensor carries total charge 0 mod 4: the bilinear Hom table is nonzero ONLY at  $j' = -j$  (dims 2/1/1/1) and ALL 15 non-neutral trilinear sector triples have Hom = 0), BUT ONE CUBIC DOOR OPENS: of the neutral trilinears ( $\{0, 0, 0\}: 3 = 1$  sym + 2 antisym,  $\{0, 1, 3\}: 2$ ,  $\{0, 2, 2\}: 2$ ,  $\{1, 1, 2\}: 1$ ,  $\{2, 3, 3\}: 1$ ; all cross-sector ones pure brackets) the UNIQUE totally symmetric survivor is the  $\mathfrak{su}(4)$   $d$ -symbol ( $\mathfrak{so}(10)$  has no cubic Casimir,  $\text{Inv}(S^3 45) = 0$  protects  $T_5$ ), whose exchange quartic  $Q_{dd} = (0, 0, -\frac{1}{240}, 0, \frac{1}{60}) = \frac{1}{60}(T_3 - P_3/4)$  (TWO independent routes, Killing propagator  $2h^\vee = 60$  from the roots) has  $\Phi_{T_3}(Q_{dd}) = \frac{1}{60} \neq 0$  — the v508 master kill (product theorem) does NOT extend to cubic vertices; the pairing stays OBSTRUCTED in the charge reading with the genuinely WEAKER certificate  $\{\Phi_{T_5}, \psi = \Phi_P - \Phi_{T_3}/4\}$ ,  $\psi(A_{\text{fix}}) = 72 - 8 = 64$  ( $M = [E_{13}, E_{22}, E_{00}, Q_{dd}]$  rank 3, augmented 4), and becomes EXACTLY solvable relaxed:  $A_{\text{fix}} = -u + 8v + 2w + 1920 Q_{dd}$  (residual 0); [C] fences  $64 = \dim \mathfrak{g}_1 = 2^6$ ,  $1920 = |W(D_5)| = 8 \cdot 240$ ; controls:  $SO(16)/D_8$  has NO symmetric cubic and no  $A_3$  block ( $E_8/\mu_4$  doubly special), false  $\mathfrak{g}_0$ /sector assignments separate; the  $\delta_1$  burden SHARPENED to the  $\psi = 64$  slice or the selection-rule relaxation — and the  $\delta_1$  exploration run itself stands UNDECIDED at exploration level (strict holomorphic  $q^0$  reading REFUTED at the  $\mathbb{Z}_2$ /Eguchi–Hanson anchor = method boundary, not a kill; contact term = the MODULAR COMPLETION of the AB data, a Harvey–Moore  $\tau$ -integral with forced leading  $(T_5, T_3)$  ratio 4:3; massive coset levels  $n \geq 2$  mandatory); Wolfram extension  $463 \rightarrow 470$  (verified engine run); *NO marker moves* — WP5e proper stays [O], SEAM.EQUIV.01 stays [O])

- `v511_celestial_wp5e_delta2_tensor_ledger.py` (CELEST.WP5E.DELTA2.01, 41 checks,  $\sim 4$  s; integer-scaled weight multisets + Kostant/Weyl alternating sums + sympy/Fraction, no floats; deterministic). D0/D1 structure: 240 roots, class split (52, 64, 60, 64);  $\mathfrak{g}_0 = \mathfrak{d}_5 \oplus \mathfrak{a}_3$  SEMISIMPLE without  $\mathfrak{u}(1)$  (40+12 class-0 roots, no mixed root, root span rank 8); sectors  $\mathfrak{g}_1 = (16_s, 4)$ ,  $\mathfrak{g}_2 = (10, 6)$ ,  $\mathfrak{g}_3 = (\mathfrak{g}_1)^*$  as multiplicity-free minuscule Weyl-orbit products, each irreducible (64/60/64); the grading is INNER ( $h = (2, 2, 2, 2, 2; 0^4)$  in the  $\mathfrak{g}_0$  Cartan). D3/D4 Hom tables: bilinear nonzero ONLY at  $j' = -j$  (dims 2/1/1/1; all 12 charged pairs zero — the inner-torus theorem at arity 2); trilinear: ALL 15 non-neutral triples Hom = 0 (arity 3), neutral dims (3, 2, 2, 1, 1) with all cross-sector trilinears ANTISYMMETRIC in their repeated legs — the unique symmetric survivor is the  $\mathfrak{su}(4)$   $d$ -symbol ( $\text{Inv}(S^3 45) = 0$ ,  $\text{Inv}(S^3 \mathfrak{a}_3) = 1$ ). D5 the new channel:  $Q_{dd} = \frac{1}{60}(T_3 - P_3/4)$  by Killing-propagator contraction AND the hand route  $|x^2 - (\text{tr } x^2/4)\mathbf{1}|^2/60$ ;  $\Phi_{T_3}(Q_{dd}) = \frac{1}{60} \neq 0$ ; brackets/tadpoles/ $d \times f$  die exactly; no  $T_5$  analogue. D6 decision: strict reading unreachable (rank 2 vs 3); generous reading STILL unsolvable (rank 3 vs 4) with the weaker certificate  $\psi(A_{\text{fix}}) = 64$ ; relaxed reading EXACTLY solvable with  $c_d = 1920 = 32 \times 60$ . D7 controls: Killing control = v508 replication ( $A_{\text{fix}}, E_{13}, E_{22}, E_{00} = 225E_{22}$ );  $SO(16)/D_8$  bilinears (1, 0, 1), trilinears (1, 0, 1, 0),  $\text{Inv}(S^3 \text{adj}_{D_8}) = 0$ ; false  $\mathfrak{g}_0 = \mathfrak{d}_5 + \mathfrak{a}_2 + \mathfrak{u}(1)$ :  $(5, 2, 2) \neq (2, 1, 1)$ ,  $\text{Inv}(S^3 \mathfrak{g}_0) = 6 \neq 1$ ; false sector content  $(16_s, \bar{4})$  kills every pairing. Discovery provenance: `experiments/tfpt-discovery/celestial_seam_wp5e_delta2_full_tensor_ledger_probe.py` (41/41), same day.
- The  $\delta_1$  fence (exploration-level, NOT promoted — verdict UNDECIDED). The twisted BCOV contact-term probe (`celestial_seam_wp5e_delta1_bcov_contact_probe.py`, 27/27, stays sandbox; v-numbered on demand) contributes its four exact sharpenings as fence

text: (i) the strict holomorphic  $q^0$  reading is REFUTED at the  $\mathbb{Z}_2$ /Eguchi–Hanson anchor (method boundary, not a kill); (ii) the forced leading  $(T_5, T_3)$  ratio is 4:3 (sector 2 : sectors 1+3); (iii) the  $SL(2, \mathbb{Z})$  orbit decomposition  $15 = 12 + 3$  identifies the contact term as the MODULAR COMPLETION of the AB data (a Harvey–Moore-type  $\tau$ -integral), not a free number; (iv) massive coset levels  $n \geq 2$  are mandatory. The remaining  $\delta_1$  target is the  $\tau$ -integral, evaluated against the  $\delta_2$ -sharpened criterion ( $\psi = 64$  slice or selection-rule relaxation).

- **Ledger + papers + Wolfram + website.** Ledger: new row CELEST.WP5E.DELTA2.01 ([E] Hom tables/innerness/ $Q_{dd}$ /certificate; [C] the number fences; [O] arity  $\geq 4$ , cubic GS term,  $\delta_1$ ); CELEST.SEAM.01 updated (delta-2 executed via v511 — the full-tensor collapse is confirmed (innerness), but the cubic  $d$ -symbol channel opens  $T_3$  with the weaker certificate  $\psi = 64$ ; delta-1 stands UNDECIDED at exploration level). Papers: `tfpt_research_contracts` WP5e bullet gains the executed  $\delta_2$  stage + the  $\delta_1$  exploration fence, WP5e-proper targets renamed ( $\tau$ -integral, cubic GS term,  $\varepsilon_1$ , back-reaction); `tfpt_3` celestial section extended to the FOURTEEN-step story (new Step 14 “the full-tensor ledger: the collapse is real, and one cubic door opens”) with the “What remains” block updated. Wolfram: seven mirrors (innerness count +  $\mathfrak{g}_0$  structure; the bilinear and trilinear Hom tables by exact Kostant alternating sums; the unique symmetric survivor; the  $SO(16)$ /false- $\mathfrak{g}_0$ / false-sector controls;  $Q_{dd}$  by two routes; the  $\psi$  certificate + the relaxed solution  $(-1, 8, 2; 1920)$  verified with the active engine the same day ( $463 \rightarrow 470$ , ALL WOLFRAM EXTENSION CHECKS PASSED). Registry + `run_all` (505 modules, suite green: ALL CHECKS PASSED). No gate closes, no marker moves — WP5e proper stays [O].

**2026-07-21 (XXII) (P2 consolidation — the axiomatics reduction state written as ONE coherent statement across the papers (editorial, NO new claims, NO marker moves): the compiler’s dimensionless inputs reduce to *four marks* (derived from P1-side topology, v216), *ONE square-modulus datum* ( $\tau = i$  — equivalently: the clock / the CM-fixed half-period / the non-split extension class on the free seam circle; v491/v499/v506/v507/v510) and  $\pi$ , plus the [C] identifications R1/R2 of v499; AX.P2.01 stays a DECLARED axiom and QGEO.REALIZE.01 stays [C]/[O] — the consolidation describes the *reduction state*, not an abolition)**

- **What is consolidated (and why now).** The P2 state of affairs was spread over five module rounds: the partition corollary (v491:  $4 = 2+1+1$  unique  $\Rightarrow g_{\text{car}} = e_2 = 5$ ), the weight typing (v499: Deligne + Birkhoff–Grothendieck + Mehta–Seshadri force  $E = \mathcal{O}(-2) \oplus \mathcal{O}(-1)^2$ , residuals R1/R2 [C]), the residual convergence (v503: QGEO-R1  $\equiv$  P2-R1), the clock rigidity and bit origin (v506/v507: one alignment bit, order fermionically forced, deck class derived) and the freedom theorems (v510: the position half is topology, the edge class excluded). The honest consolidated statement, now in one place: *the compiler’s dimensionless inputs reduce to four marks (derived from P1-side topology), one square-modulus datum ( $\tau = i$ , equivalently: the clock / the CM-fixed half-period / the non-split extension class on the free seam circle), and  $\pi$  — plus the [C] identifications R1/R2 of v499.* No new claims; conservative wording; no “axiom abolished” language anywhere.
- **Papers + ledger + website.** `tfpt_1_architecture_e8`: new keybox “The current reduction state of P2” next to the foundational-postulates keybox, with v491/v499/v506/v507/v510 citations. `tfpt_3`: mirror bullet “The current reduction state” in the honest-status section. **introduction**: the anchor paragraph gains the consolidated one-sentence state and the status card row  $a = (1, 1, 2)$  now names the reduction state. Ledger: SYNTHESIS NOTE appended to AX.P2.01 (“reduction state 2026-07-21: four marks + one square-modulus datum ( $\tau = i$ ) +  $\pi$ ”), marker *Axiom unchanged*. Website: `papers.ts` mirrors of the changed sections. No gate closes, no marker moves.

**2026-07-21 (XXI) (v510 — the freedom attack on the v506/v507 alignment bit promoted as SEAM.BIT.FREEDOM.01: the POSITION half of the bit is TOPOLOGY — the collar deck is a covering deck (the  $\mathbb{Z}_2$  seam of the Möbius double), and covering decks act FREELY; the DIHEDRAL THEOREM (on the seam circle the free involution is UNIQUE = the antipode: 17 involutions in  $D_{16}$ , exactly 1 free;  $\text{Aut}(C_{16}) = D_{16}$  complete census, brute force 8! at  $N=8$ ; quotient topology circle/antipode = CIRCLE  $\chi=0$  vs circle/reflection = INTERVAL  $\chi=1$  — the  $6\pi$  Gauss–Bonnet budget needs closed circles, so freeness IS the established P1 datum); NONSPLIT  $\iff$  FREE over the complete 17-involution census in exact  $\text{Cl}(16)$  (free  $\Rightarrow \tilde{U}^2 = 256 \gamma_1 \cdots \gamma_{16} = (-1)^F$  nonsplit with 2 roots = the  $\mathbb{Z}_8$  tower; ALL 16 reflections  $\Rightarrow U^2 = \text{scalar } \{2^7, 2^8\}$ , split, 0 roots); the REAL-STRUCTURE classification (deck-commuting families = exactly two;  $\mu_4 \rightarrow$  equatorial, deck FREE; silver  $\rightarrow$  real axis THROUGH the deck poles, NOT free — the edge class is the restriction of the BRANCHED pillowcase involution; hexagonal  $\rightarrow$  NO mark-fixing real structure, RP-incompatible); HARMONIC + FREE  $\Rightarrow \mu_4$  exactly ( $b = \pm i$ ); honest counterwitnesses (“nonsplit  $\Rightarrow$  clock” is FALSE:  $\{0, 1, 8, 9\}$  and  $\{\pm 1, \pm(3+4i)/5\}$  with  $j = 148176/25 \neq 1728$ ); the alignment bit REDUCES from “harmonic AND central” to the square-modulus datum  $\tau = i$  alone; Wolfram extension 457  $\rightarrow$  463 (verified engine run); NO marker moves — QGEO.REALIZE.01 stays  $[C]/[O]$ , AX.P2.01 stays an axiom, P2-R1 halved)**

- **v510\_seam\_bit\_freedom.py (SEAM.BIT.FREEDOM.01, 30 checks, < 5 s; sympy + exact  $\text{Cl}(16)$  Fractions, no floats; deterministic).** Part A (dihedral/fermionic): the involution census in  $D_N$  on the refined  $2N$ -point seam circle ( $N = 16$ : 17 involutions, EXACTLY ONE free = the antipodal shift by 8; all 16 reflections have exactly 2 circle fixed points — sites for even axis, edge midpoints for odd; robustness  $N = 8, 12$ ; odd control  $N = 9$ : ZERO free — an even carrier count is needed, consistent with  $16 = 2^{g_{\text{car}}-1}$ ); the census is COMPLETE because  $\text{Aut}(C_{16}) = D_{16}$  exactly (brute force over all  $8! = 40320$  permutations at  $N = 8$ , propagation at  $N = 16$ ); quotient topology: circle/antipode = CIRCLE ( $\chi = 0$ , closed), circle/reflection = INTERVAL ( $\chi = 1$ , two boundary corners) — a fixed-point deck would break the  $6\pi = 3 \times 2\pi$  closed-circle count, so “deck free on the seam circle” is exactly the established “the seam is a closed circle”, not a new premise; the complete Fock census: ALL 17 involutions implemented in exact  $\text{Cl}(16)$  — NONSPLIT  $\iff$  FREE with no exception, root dichotomy 2 vs 0; the chain: free  $\Rightarrow$  shift 8  $\Rightarrow$  crossing (all 28 invariant 4-sets)  $\Rightarrow U^2 = (-1)^F \Rightarrow$  central class  $\Rightarrow$  the edge/silver arrangement is EXCLUDED; Klein control (covering decks are free regardless of the gluing); branch control (the pillowcase involution has exactly 4 fixed points = the marks and restricts to a REFLECTION: the silver arrangement mistakes the mark-creating branch  $\mathbb{Z}_2$  for the seam deck). Part B (Möbius/RP): the cylinder-double-cover deck is free on all 120 model points and restricts to the antipode on the  $\mathbb{Z}_2$  seam fibre; one-sidedness TYPES the  $\mathbb{Z}_2$  as the free antipode (quotient  $\mathbb{RP}^2 = \text{the } |\mathbb{Z}_2| \text{ of } c_3 = 1/(|\mathbb{Z}_2|2\pi\chi)$ ); the deck-commuting real structures are EXACTLY two families ( $\mu\bar{z}$ : fix line through the poles, NOT free;  $\mu/\bar{z}$ ,  $\mu > 0$ : fix circle  $|z| = \sqrt{\mu}$ , FREE;  $\mu < 0$ : empty); the per-configuration table ( $\mu_4$  equatorial/free, silver real-axis/not free, generic-unit free without harmonicity, hexagonal RP-incompatible); harmonic + free solves EXACTLY to  $b = \pm i$  (on free circles the silver family “harmonic AND edge” is EMPTY — harmonicity becomes sufficient); honest counterwitnesses: the discrete  $\{0, 1, 8, 9\}$  and the continuous  $\{\pm 1, \pm(3+4i)/5\}$  (free, central,  $j = 148176/25$ , no clock) — the Fock dichotomy measures the CLASS, not the modulus. Premise typing: (P-i) deck = Möbius covering deck  $\Rightarrow$  free [standard topology; the one-sidedness strand `origin_theory/v456`]; (P-ii) marks on the  $\Theta$ -fix seam circle = existing REALIZE [C] content (QGEO.REALIZE.02 hypotheses, v264) — NO new input. Audit finding documented: “circle-freeness” was nowhere typed before (only mark-freeness, v216); v507 F1.1 explicitly admitted the reflection reading of the edge class — v510 closes that door (a

reflection is not a covering deck). The [C] rest, verbatim: *the alignment bit reduces from “harmonic AND central” to the square-modulus datum  $\tau = i$  alone; the position half is topology (edge class excluded), the modulus half stays the one carrier input.* Discovery provenance: `experiments/tfpt-discovery/qgeo_realize_deck_freedom_dihedral_probe.py` (14/14) and `qgeo_realize_rp_fixcircle_moebius_probe.py` (16/16), same day.

- **Ledger + papers + Wolfram + website.** Ledger: new row SEAM.BIT.FREEDOM.01 ([E] dihedral theorem + complete Fock census + real-structure classification + harmonic-free collapse + counterwitnesses; [C] the square-modulus datum); TEXT precisions (markers unchanged) in SEAM.CLOCK.RIGIDITY.01, SEAM.BIT.ORIGIN.01, QGEO.REALIZE.01 (bit  $\rightarrow$  square-modulus datum; edge class topologically excluded) and P2.TYPING.01 (R1 halved). Papers: `tfpt_3` celestial/QGEO passage extended (“the position half is topology”), `tfpt_research_contracts` QGEO kill-test + QGEO.MARKS.03 + celestial-contract passages extended, one-sentence precisions in `introduction`, `tfpt_1`, `tfpt_2`, `origin_theory` where v506/v507 are cited. Wolfram: six mirrors (dihedral census + quotient topology; Aut completeness incl. the 8! brute force; the Cl(16) dichotomy census over all 17 involutions on  $256 \times 256$  Jordan–Wigner gammas; the real-structure table; harmonic + free  $\Rightarrow \pm i$ ; the counterwitnesses) verified with the active engine the same day (457  $\rightarrow$  463, ALL WOLFRAM EXTENSION CHECKS PASSED; base file re-verified 116/116). Registry + `run_all` (504 modules, suite green: ALL CHECKS PASSED). No gate closes, no marker moves.

**2026-07-21 (XX) (v509 — WP5e- $\varepsilon_2$  of CELEST.SEAM.01 “the CPS level-from-flux dial” promoted as CELEST.WP5E.EPS2.01, verdict B: the Costello–Paquette–Sharma Burns-holography mechanism (“level = flux quantum  $\times$  Dynkin index”) executed on the lockstep spheres — the CPS SKELETON exact ( $S^3$  period  $(2\pi i)^2 N$  with pullback density  $-\sin 2\theta$ , exceptional-sphere Kähler flux  $2\pi N$ , boundary level magnitude  $2N$  by the symplectic-boson contraction,  $T(\text{vector}) = 1$ ); the PAIRING MATRIX pinned, not postulated (ch<sub>2</sub> ledger + integrality + unimodularity + effectivity  $\Rightarrow P_{mi} = \delta_{mi}$  by complete enumeration: rows 8/6/8, 48 unimodular, exactly 2 effective = identity + diagram flip), fluxes  $F_i = (64, 60, 64) = \dim \mathfrak{g}_i$  with uniformly ONE quantum per charged current; “level = total flux” KILLED by the lockstep test itself ( $F_i$  unequal = flavour multiplicity), the per-current reading (1, 1, 1) anchored by the lattice current count (240, 0, 0, 0) and embedding index 1; ord-4-vs-level-1 RESOLVED (the clock order counts fractional flux sectors:  $\mu = 16 = \text{ord}^2$ , Lagrangian  $\mu_4$  glue, KLM 16  $\rightarrow$  1) with the NEW sector-counter dial  $\#_{\text{prim}}((E_8)_k) = (1, 3, 5, 10, 15, 27)$  for  $k = 1..6$  — exactly ONE sector iff  $k = 1$ ; the LOCKSTEP THEOREM lifts the v505 “one scale  $\Rightarrow$  one level” to a THEOREM of clock invariance ( $|\Pi_j| = \sqrt{2}t$  equal, monodromy = pure phase  $i$ ,  $\det(A-1) = -4$ : no invariant flux vector; NEW falsifier  $Z^4-Z$ : 0/24 orderings lockstep); honest limit: uniform doubling keeps the lockstep (equality, not the value); controls  $k=2/SO(16)/A_2$  separate honestly; Wolfram extension 450  $\rightarrow$  457 (verified engine run); *NO marker moves — WP5e stays [O] (the CPS dictionary on  $\mathbb{P}T/\mathbb{Z}_4$  typed [C], the type-I B-model back-reaction [O]), SEAM.EQUIV.01 stays [O])***

- `v509_celestial_wp5e_eps2_level_from_flux.py` (CELEST.WP5E.EPS2.01, 28 checks,  $< 1$  s; sympy/Fraction exact, no floats; deterministic). Target: derive the boundary level  $k = 1$  from flux quantisation on the lockstep spheres, following the Costello–Paquette–Sharma Burns-holography mechanism (PRL 130, 061602; arXiv:2306.00940). *CPS skeleton replicated exactly*: the  $S^3$  period of the brane back-reaction 3-form is  $(2\pi i)^2 N$  (their eq. 3.33; large-gauge invariance quantises  $N$ ), the exceptional-sphere Kähler flux is  $2\pi N$  exactly ( $K = |u|^2 + N \log |u|^2$ , the flat part dies at  $t \rightarrow 0$ ), and the boundary level magnitude is  $2N$  via the symplectic-boson double contraction ( $N = 1, 2, 3$ ; their eq. 9.4), Dynkin normalisation  $T(\text{sog vector}) = 1$ ,  $T(\text{adj}) = 6$ . *Pairing matrix pinned*: the v505 ch<sub>2</sub> ledger ( $f(m) = -(C^{-1})_{mm}/2 = (-\frac{3}{8}, -\frac{1}{2}, -\frac{3}{8})$ )

+ integrality + unimodularity + effectivity force  $P_{mi} = c_1(T_m) \cdot [\Sigma_i] = \delta_{mi}$  by COMPLETE enumeration (row candidates 8/6/8; 48 unimodular solutions of  $PC^{-1}P^T = C^{-1}$ ; with  $P \geq 0$  exactly 2: identity +  $A_3$  diagram flip); fluxes  $F_i = (64, 60, 64) = \dim \mathfrak{g}_i$ , total  $188 = 248 - 60$ , flip-invariant; flux per charged root current uniformly ONE quantum through exactly one sphere. *The level formula, honest*: “level = total open-string flux” is KILLED by the lockstep test itself ( $k_i = (64, 60, 64)$  unequal —  $F_i$  is the FLAVOUR multiplicity, the brane matter content); what works:  $k = \text{flux per adjoint current} = F_i / \dim \mathfrak{g}_i = (1, 1, 1)$ , anchored non-circularly by the lattice current count  $(240, 0, 0, 0)$  for  $k = 1.4$  (v505 S4.8) and the embedding index  $j(D_5 \subset E_8) = j(A_3 \subset E_8) = 1$  ( $60 = 2h^\vee$ ). *Ord-4-vs-level-1 resolved*: the clock order counts FRACTIONAL flux sectors, not the level ( $\mu = \det(D_5) \det(A_3) = 16 = \text{ord}^2$ ; the  $\mu_4$  glue diagonal isotropic + Lagrangian; KLM  $16 \rightarrow 16/4^2 = 1$ , zero residual fractional flux); NEW DIAL:  $\#\text{primaries}((E_8)_k) = (1, 3, 5, 10, 15, 27)$  for  $k = 1.6$  from the affine marks — exactly ONE sector iff  $k = 1$  (independent of the CPS transplantation). *The lockstep theorem* (the lifting of v505 S4.10): clock invariance  $P(iZ) = P(Z)$  forces  $a_3 = a_2 = a_1 = 0$  sharply, the four branch points are ONE  $\mu_4$  orbit,  $|\Pi_j| = \sqrt{2}t$  EQUAL on all three spheres, the monodromy is the pure phase  $i$  and  $\det(A-1) = -4 \neq 0$  (no invariant flux vector) —  $k_1 = k_2 = k_3$  IS A THEOREM of clock invariance; NEGATIVE PROBE (new falsifier): the clock-forbidden family  $Z^4 - Z$  (smooth, disc  $-27$ , branch points  $\{0, 1, \omega, \omega^2\}$  not a  $\mu_4$  orbit): over ALL 24 orderings ZERO chains with three equal period moduli ( $1$  vs  $\sqrt{3}$ ); HONEST LIMIT: uniform flux doubling keeps the lockstep — the theorem proves EQUALITY, not the value ( $1$  from the counters and the CPS minimal quantum  $N = 1$ ). *Negative controls*:  $k = 2$  (current count 0, residual 360, prefactor  $-31/48$ ,  $\#\text{prim} 3$  — while the lockstep dial honestly does NOT fire);  $SO(16)$  (fluxes  $(0, 60, 0)$ : spheres  $1/3$  FLUX-DARK, defect  $-30 \neq -78$ , condensation stuck at  $4 = \det \text{Cartan}(D_8)$ );  $A_2$  ( $\det 3 \neq 4$ , glue diagonal  $23/24 \notin \mathbb{Z}$ ,  $\mu = 12$  no perfect square: NO Lagrangian subgroup, Coxeter  $3 \neq 4$ ). Typed honestly: the CPS dictionary itself on  $\mathbb{PT}/\mathbb{Z}_4$  stays [C] (their geometry is the  $\mathbb{C}^2$  blow-up with one  $(-1)$ -sphere and  $\mathfrak{so}(8)$  at negative level — not the  $A_3$  ALE); the type-I B-model back-reaction stays [O] (the same fence as v502/v505).

- **Ledger + papers + Wolfram + website.** Ledger: new row CELEST.WP5E.EPS2.01 ([E] CPS skeleton/pairing/lockstep/sector arithmetic + [C] CPS dictionary + [O] back-reaction); CELEST.SEAM.01 updated (“ $\varepsilon_2$  executed via v509: the one-level statement is now a THEOREM (lockstep from clock invariance, falsifier  $Z^4 - Z$ ), level = 1 carries the new sector-counter dial  $\#\text{prim}((E_8)_k) = 1 \Leftrightarrow k = 1$ ; the  $\mathbb{PT}/\mathbb{Z}_4$  back-reaction stays [O]”); the v505 “one scale  $\Rightarrow$  one level” sentence LIFTED to theorem status in ledger and prose (text move, no marker move). Papers: `tfpt_research_contracts` WP5e section extended by the  $\varepsilon_2$  block (done, verdict B; roadmap now  $\delta_1/\delta_2/\varepsilon_1$  + back-reaction); `tfpt_3` celestial section becomes a THIRTEEN-step story with Step 13 “one level is a theorem; the sector counter pins  $k = 1$ ” and the “What remains” paragraph updated. Wolfram: seven mirrors (the CPS period integrals; the  $2N$  contraction + Dynkin normalisation; the pairing enumeration  $48 \rightarrow 2$ ; the fluxes  $(64, 60, 64)$ ; the  $\#\text{prim}$  table  $(1, 3, 5, 10, 15, 27)$ ; the lockstep periods + the  $0/24$  negative probe; the  $SO(16)/A_2$  controls) verified with the active engine the same day ( $450 \rightarrow 457$ , ALL WOLFRAM EXTENSION CHECKS PASSED). **VerificationDag**: v509 on the QFT/celestial node; registry + `run_all` (503 modules, suite green). No gate closes, no marker moves — WP5e proper stays [O].

**2026-07-21 (XIX) (e8-ladder-bed v2 — the 2DCS extension executed with an honest preregistered kill\_2d verdict at the DETECTOR level: the known-answer map recovers 8/8 rungs incl.  $m_7/m_8$  via F3 difference coordinates (MC  $p = 2 \times 10^{-4}$ ), but the 2D control battery breaks (scramble 27%, jitter 18% at the destructive 30% level) — the v1 1D results (BCVO 8/8,  $\text{CoNb}_2\text{O}_6$  6/8) are untouched; scorecard row updated, status/stage unchanged; *experiments level only — no verification module, no gate closes, no marker moves*)**

- **experiments/e8-ladder-bed v2 (2DCS): amendment preregistered, run, and killed by its own controls.** Trigger: arXiv:2512.16829 (Watanabe, Trebst, Hickey) — a THEORY-only paper (four-kink/ED localized-limit spectra at  $B_y = 0\text{--}3\text{ T}$ , not the E8 point; no measurements); as of 2026-07-21 *no experimental 2DCS measurement of  $\text{CoNb}_2\text{O}_6$  exists anywhere* (web-checked). **amendment\_v2** was frozen in **hypotheses/e8\_ladder\_bed\_v1.yaml** BEFORE any 2D recovery run (detector, catalog F1/F2/F3, gates, null model, verdict criteria); **amendment\_v2\_1** discloses a null-region correction (add **n\_matched** to the false-alarm region) found on synthetic diagnostics, timing disclosed. Runs (**results/results2d.json**, deterministic): known-answer KA2D 8/8 rungs with  $m_7/m_8$  recovered ONLY via F3 difference coordinates ( $\chi^2/\text{dof} = 1.30$ , MC  $p = 2.0 \times 10^{-4}$ , 0/5000) — the mechanism that pulls  $m_7/m_8$  inside a low-pass window works; the NEG2D localized-limit map (the paper’s own non-E8 closed forms) is correctly rejected ( $p = 0.029$ ); offset and wrong-ladder controls rejected. BUT **scramble2d** fires at  $54/200 = 27\%$  and **jitter2d** shows no dose-response decay (29/21/17/18%; 18% at the destructive 30% level)  $\Rightarrow$  preregistered verdict **kill\_2d**: permuting the detection axis preserves the pump-axis marginal (which carries the full 1D ladder) and the 120-entry catalog is dense enough that rung coverage follows from the marginal alone — the v2 detector does not test the 2D *pairing*; failure kills the detector, not TFPT. The  $\text{CoNb}_2\text{O}_6$  leg stays 6/8 ( $p = 0.0038$ ) per the preregistered **m7\_m8\_upgrade\_rule**; a future v3 would need coincidence-based coverage and a sparser catalog, preregistered BEFORE real data exists.
- **Surfaces synced (scorecard row updated; counters unchanged, 120 rows).** Scorecard row “E8 mass-ladder blind recovery bed”: **data\_value** extended (“ $\text{CoNb}_2\text{O}_6$  6/8 unchanged; 2DCS extension killed at detector level (scramble 27%, jitter 18% @30%); KA 8/8 incl.  $m_7/m_8$  via F3,  $p=2\text{e-}4$ ; no experimental 2DCS data exists (arXiv:2512.16829 theory-only)”) and **status\_note** extended (amendment v2/v2.1 preregistered+disclosed; real-data leg blocked; v3 criteria) via **build\_evidence\_scorecard.py**; status/stage UNCHANGED (**consistent/not\_applicable**, **analog\_positive\_control** basket — the v1 detector validation stands). Experiment README v2 section, hypotheses YAML amendments, **experiments/next.txt** note; data-request plan on record (Armitage lab/JHU, Wang lab/Ames; 2D peak table at  $B \parallel b$ , 4.7–5.5 T, sub-Kelvin). No claim; firewall intact.

2026-07-21 (XVIII) (v508 — WP5e- $\gamma$  of CELEST.SEAM.01 “the sphere-axion pairing check” promoted as CELEST.WP5E.GAMMA.01, an honest rigid **NEGATIVE** result: the VERTEX-SPACE THEOREM — the  $W(D_5) \times W(A_3)$ -invariant quadratics on the 8-dim glue Cartan are **EXACTLY**  $\text{span}\{s_5, s_3\}$  (dim 2) and the quartics **EXACTLY**  $\text{span}\{P_1, P_2, P_3, T_5, T_3\}$  (dim 5), with the **PRODUCT THEOREM** as the master kill (any two invariant quadratics multiply into  $\text{span}\{P_1, P_2, P_3\}$  — zero  $T_5/T_3$  content, so  $\Phi_{T_3}$  annihilates **EVERY** exchange image while  $\Phi_{T_3}(A_{\text{fix}}) = 32 \neq 0$ ); the strict two-index  $\mathbb{Z}_4$  rule **COLLAPSES** (the sphere axions have no invariant bilinear Cartan vertex at all), the twist-insertion channels give a rank-2 exchange matrix with  $\text{rank}([M | A_{\text{fix}}]) = 3$  and the annihilator certificate  $(\Phi_{T_5}, \Phi_{T_3}, \Phi_P)(A_{\text{fix}}) = (0, 32, 72)$ ; side discovery  $K^{(0)} = -15 K^{(2)}$ ; naturalness **DISSOLVES** (ch<sub>2</sub>-natural and AB-weight couplings both certify (0,0) vs required (32,72) — scale-independent); controls:  $SO(16)$  has no sphere partners **AND** uncanceled  $T_5$ ,  $D_8$  no  $T_3$  structure; the O-fence is **SHARPENED**: the exchange sub-branch is dead, the 32  $T_3$  residual **MUST** be carried by twisted BCOV contact terms ( $\delta_1$  binary: coefficient exactly (9, -30, -15, 0, 32), computed not fitted; escape  $\delta_2$ : the full-tensor ledger); Wolfram extension 444  $\rightarrow$  450 (verified engine run); ***NO marker moves — CELEST.SEAM.01 stays Open, the slot bijection untouched, the preregistered level-kill still does not fire, SEAM.EQUIV.01 stays [O]***

- **v508\_celestial\_wp5e\_gamma\_pairing.py** (CELEST.WP5E.GAMMA.01, 27 checks,  $\sim 4$  s; **sympy/Fraction exact, no floats; deterministic**). Question: can the three twisted sphere axions  $\eta_m$  (Coxeter eigencharacters  $\{i, -1, -i\}$  on  $H^2(\text{ALE})$ , v505 S5.8) cancel the rigid 32  $T_3$  residual of the Atiyah–Bott ledger ( $A_{\text{fix}} = Q^{(1)}/2 + Q^{(2)}/4 + Q^{(3)}/2 = 9P_1 - 30P_2 - 15P_3 + 32T_3$ , v505 S3.7) by Green–Schwarz-type EXCHANGE with sector-compatible quadratic vertices? *No, with a certificate. Vertex-space theorem:* exact Weyl nullspace arithmetic over all bidegrees gives quadratics dim 2 (bidegrees (1,0,1)) and quartics dim 5 (bidegrees (2,0,1,0,2)) — the vertex basis is **COMPLETE**; *product theorem* (the master kill):  $(as_5 + bs_3)(cs_5 + ds_3) = acP_1 + (ad + bc)P_2 + bdP_3$  identically — every exchange image (any axion count, any rule, any couplings) is annihilated by  $\Phi_{T_3}$ , and  $\Phi_{T_3}(A_{\text{fix}}) = 32 \neq 0$ . *Selection enumeration:* the strict two-index reading collapses ( $C_{j,j'} \neq 0$  only for  $j' = -j$ , forcing  $m = 0$ : 16 raw triples, 4 nonzero, all  $m = 0$ ); twist insertion:  $\eta_1 \leftrightarrow K^{(3)}$ ,  $\eta_2 \leftrightarrow K^{(2)}$ ,  $\eta_3 \leftrightarrow K^{(1)}$ ,  $\eta_0 \leftrightarrow K^{(0)}$ ; channels  $E_{13} = (16, -96, 144, 0, 0)$ ,  $E_{22} = (16, 32, 16, 0, 0)$ ,  $E_{00} = 225 E_{22}$ ; **SIDE DISCOVERY:**  $K^{(0)} = -15 K^{(2)}$  (the even-sector quadratics are **PARALLEL** — this collapse drives the  $\Phi_P$  obstruction). *Unsolvability with certificate (the core):*  $M$  is  $5 \times 2$ , rank 2, kernel 0;  $\text{rank}([M | A_{\text{fix}}]) = 3$ ; annihilator certificate  $(\Phi_{T_5}, \Phi_{T_3}, \Phi_P)(A_{\text{fix}}) = (0, 32, 72)$  — two independent obstructions (32  $T_3$  structural, rule-independent;  $\Phi_P = 72$  selection-rule-driven);  $\ker M = 0$  forces the unique non-disturbing coupling  $c = 0$  (the sphere axions are **RIGIDLY SILENT**);  $\Phi_P(Q^{(0)}) = 0$  (the sector-0 Okubo mechanism undisturbed); relaxed control: dropping charge conservation reaches the  $P$ -block ( $\det = -64$ ) but  $T_5 = T_3 = 0$  persists. *Naturalness resolution:* ch<sub>2</sub>-natural couplings (9/64, 1/4) give (25/4, -11/2, 97/4, 0, 0) and AB-weight couplings (1/2, 1/4) give (12, -40, 76, 0, 0) — **BOTH** with certificate (0,0) vs required (32,72): the failure is scale- and coupling-INDEPENDENT (no “unnatural factor”). *Negative controls:* invariant-only = v505 S3.9 replicated + sharpened; flipped rule  $m+j-j'$ :  $\Phi_{T_3}$  kill flip-invariant,  $\Phi_P$  rule-dependent;  $SO(16)$ : odd classes **EMPTY** and AB sum (15, -18, -15, -4, 40) —  $T_5$  does not even cancel ( $E_8$  doubly special);  $D_8$ :  $16T_8 + 12s_8^2$ , no  $T_3$  structure — the question cannot even be posed. Number fences **[C]** only:  $32 = (h, h) + (h', h') = 20 + 12$ ,  $72 = 2\tilde{\lambda}_{e_8}^2$ .
- **Ledger + papers + Wolfram + website.** Ledger: new row CELEST.WP5E.GAMMA.01 (**[E]** vertex space/selection/certificate + **[C]** fences + **[O]** contact terms); CELEST.SEAM.01 stays Open with the WP5e targets **UPDATED** (“the exchange sub-branch is closed by v508; the named remaining targets are the twisted BCOV contact terms ( $\delta_1$ , binary test) and the full-tensor ledger ( $\delta_2$ )”). Papers: **tfpt\_research\_contracts** WP5e section extended by the  $\gamma$  block ( $\gamma$

done — negative result with certificate; the  $\delta_1/\delta_2/\varepsilon_1/\varepsilon_2$  roadmap with success/kill criteria); **tfpt\_3** celestial section becomes a TWELVE-step story with Step 12 “the exchange no-go and the contact-term criterion” and the “What remains” paragraph updated. Wolfram: six mirrors (the vertex-space dims 2/5 by generator-wise nullspace arithmetic; the twisted quadratics with  $K^{(0)} = -15 K^{(2)}$ ; the product theorem, symbolic; the rank-3 certificate  $(0, 32, 72)$  with channels and naturalness images; the  $SO(16)$  AB sum  $(15, -18, -15, -4, 40)$ ; the  $D_8$  quartic  $16 T_8 + 12 s_8^2$ ) verified with the active engine the same day ( $444 \rightarrow 450$ , ALL WOLFRAM EXTENSION CHECKS PASSED). VerificationDag: v508 on the QFT/celestial node; registry + run\_all (502 modules, suite green). No gate closes, no marker moves — the O-fence is only sharpened.

**2026-07-21 (XVII) (v507 — the tautology attack on the v506 alignment bit, executed and refuted, promoted as SEAM.BIT.ORIGIN.01: the ORIGIN THEOREM — marks = fixed points of the hyperelliptic involution = the 4 half-periods  $\Lambda/2\Lambda$  of the seam double, and EVERY mark-free collar deck IS a half-period translation (in all three stabiliser types  $(4, 3, 3)/(8, 5, 3)/(12, 3, 3)$  the free involutions are exactly the three  $\sigma_c$ : the CLASS of the deck is forced, its POSITION is not); the CORE SOLVE (symbolic  $\lambda$ ): a marks-preserving root of  $\sigma_c$  exists  $\iff \lambda \in \{-1, 2, \frac{1}{2}\}$  (one per  $c$ ; intrinsic: the deck pairs separate harmonically), union = the harmonic orbit  $\iff j = 1728 \iff \tau = i$ ; the EQUIVALENCE TETRAD “Uhr existiert  $\iff (\tau=i$  AND deck = CM-fixed half-period)  $\iff$  (Stab =  $D_4$  AND deck central)  $\iff$  deck pairs harmonic  $\iff$  NS Fock lift NONSPLIT”;** the FERMION DICHOTOMY (the new physical face): central  $\tilde{U}^2 = 256 \gamma_1 \cdots \gamma_{16} \Rightarrow U^2 = (-1)^F$  nonsplit vs edge  $\tilde{U}_e^2 = +2^7 \Rightarrow U^2 = +1$  SPLIT (2 vs 0 roots — the bit IS a Fidkowski–Kitaev-type extension class, distinguished but not derived); the silver set = the in-family misalignment witness; Wolfram extension  $438 \rightarrow 444$  (verified engine run); *NO marker moves — the bit stays [C], SEAM.CLOCK.RIGIDITY.01/QGEO.REALIZE.01/P2.TYPING.01 text precised only*

- **v507\_seam\_bit\_origin.py** (SEAM.BIT.ORIGIN.01, 26 checks,  $\sim 2.6$  s; **Part A** sympy-exact torus/Möbius arithmetic + **Part B** exact  $Cl(16)$  Fractions; deterministic). Conjecture under attack: marks and deck both descend from the SAME torus structure (the seam double  $T_\tau^2 \rightarrow S^2$ , v214/v260), so the common origin would force the central alignment automatically and the v506 bit would be a tautology. *Origin theorem (die Herkunft)*: the 4 marks are the fixed points of  $w \mapsto -w$  on  $\mathbb{C}/(\mathbb{Z} + \tau\mathbb{Z}) =$  the half-periods  $\Lambda/2\Lambda$  (a  $\frac{1}{3}$ -coefficient never qualifies);  $t_c$  descends iff  $2c \in \Lambda$  — exactly the three half-period translations (control  $c = \frac{1}{3}$  fails); the descended maps are the EXPLICIT rational involutions  $\sigma_i(x) = e_i + (e_i - e_j)(e_i - e_k)/(x - e_i)$  with the exact curve-lift identity  $\prod(\sigma_i - e_l) = \psi_i^2 \prod(x - e_l)$ ,  $\psi_i$  RATIONAL (the  $\wp$ -addition formula as pure algebra), closing to Klein  $V_4$ ; CONVERSELY the freely-acting involutions in ALL three stabiliser types (generic  $V_4$   $(4, 3, 3)$ , harmonic  $D_4$   $(8, 5, 3)$ , hexagonal  $A_4$   $(12, 3, 3)$ ) are EXACTLY the three  $\sigma_c \Rightarrow$  every mark-free collar deck (v216) IS a half-period translation. *Core solve (symbolic  $\lambda$ )*:  $\sigma_c$  admits a marks-preserving  $PGL_2$  root  $\iff \lambda = -1$  ( $\sigma_1$ ),  $2$  ( $\sigma_2$ ),  $\frac{1}{2}$  ( $\sigma_3$ ) — per half-period exactly ONE solution, intrinsically “the deck pairs separate harmonically” (pair cross-ratio  $-1$ ); generic modulus:  $V_4$  abelian, ALL  $\sigma_c$  central yet clockless (“central alone is empty” — the bit is the CONJUNCTION  $D_4 \wedge$  central); at  $\tau = i$  the stabiliser jumps to  $D_4$  with  $\sigma_1$  central (2 roots = the clock pair),  $\sigma_2/\sigma_3$  edge-type (0 roots); the v506 seam deck lands on the central  $\sigma_1 = -1/x$  under all 8 mark bijections; CM dictionary: mult-by- $i$  fixes exactly  $c^* = (1+\tau)/2$  (mult-by- $\rho$  3-cycles all three), the clock lift needs the CM unit  $i$  ( $\psi = 2i/(x+1)^2$ ,  $t^2+1$  irreducible over  $\mathbb{Q}$ ) while half-period decks lift rationally. *Equivalence tetrad + circularity audit*: clock exists  $\iff (\tau=i$  AND deck = the CM-fixed half-period)  $\iff$  (Stab =  $D_4$  AND deck central)  $\iff$  deck pairs harmonic  $\iff$  NS Fock lift nonsplit; the v493 converse independently re-verified — two independently computed converses of ONE equivalence, no circle. *Fermion dichotomy (Part B)*: central arrangement (shift by

8):  $\tilde{U}^2 = 256 \gamma_1 \cdots \gamma_{16} \Rightarrow U^2 = (-1)^F$ , nonsplit  $\mathbb{Z}_4$ , 2 roots in the 32-element dihedral lift (the  $\mathbb{Z}_8$  tower); edge arrangement (reflection  $j \mapsto 4-j$ , fixed sites = non-marks = the silver  $\{0, \infty\}$  between neighbour marks): the implementer EXISTS but  $\tilde{U}_e^2 = +2^7 \cdot \mathbf{1} \Rightarrow U^2 = +1$ , the extension SPLITS, 0 roots, no  $\mathbb{Z}_8$ ; one-particle  $(\lambda^2+1)^8$  vs  $(\lambda^2-1)^8$ ; census: all 28 shift-invariant 4-sets cross (central type), all 21 mark-free reflection-invariant sets do not; robust at 8 sites; R-sector edge control splits too — the bit IS a physical extension class (Fidkowski–Kitaev-type): distinguished, but not derived (WHICH class nature realises stays the input). *Silver placement*:  $\{\pm 1, \pm(3+2\sqrt{2})\} = “\tau = i$  with deck = a non-CM-fixed half-period translation” (all 8 bijections  $\rightarrow \sigma_2/\sigma_3$ , never  $\sigma_1$ ) — the in-family witness: TAUTOLOGY REFUTED. *The one [C] rest (verbatim)*: the alignment bit remains a carrier datum; “no tautology — the origin forces the CLASS of the deck (half-period translation), not its POSITION”. No overclaim.

- **Ledger + papers + Wolfram + website.** Ledger: new row SEAM.BIT.ORIGIN.01 ([E] origin theorem/core solve/tetrad/fermion dichotomy/silver placement + [C] the bit); TEXT precisions at *unchanged* markers: SEAM.CLOCK.RIGIDITY.01 (“the bit is now fully characterised as a tetrad ... and carries a physical face: the non-split NS Fock extension class — v507”), QGEO.REALIZE.01 (“the class of the deck IS derived (half-period translation, v507); only its position stays the input”, stays [C]/[O]), P2.TYPING.01 (origin cross-reference on the common R1 bit). Papers: `tfpt_3` celestial Step-1/2 environment + QGEO paragraph, contracts marks/kill-test paragraph, `introduction/tfpt_1/tfpt_2/origin_theory` one-sentence updates where v506 is cited. Wolfram: six mirrors (half-period descent  $2c \in \Lambda$  with the rational  $\sigma_i$  lift identity; the core solve  $\lambda \in \{-1, 2, \frac{1}{2}\}$ ; the stabiliser counts  $(4, 3, 3)/(8, 5, 3)/(12, 3, 3)$  via the complete 24-triple enumeration; the CM fixed point  $c^*$  + the lemniscatic  $\psi = 2i/(x+1)^2$ ; the Cl(16) dichotomy  $256 \gamma$ -monomial vs scalar  $2^7$  on explicit  $256 \times 256$  Jordan–Wigner matrices + the 2-vs-0 root census in the 32-element dihedral lift; the silver bijection census) verified with the active engine the same day ( $438 \rightarrow 444$ , ALL WOLFRAM EXTENSION CHECKS PASSED); the v493 audit direction and the 28/21 arrangement census stay Python-side (flagged). `VerificationDag`: v507 on the P2/carrier and QGEO nodes; registry + `run_all` (501 modules, suite green). No gate closes, no marker moves.

**2026-07-21 (XVI) (v506 — seam clock rigidity, Route B of the v503 classification, promoted as SEAM.CLOCK.RIGIDITY.01: TWO exact theorems reduce the order-4 clock input to ONE alignment bit — Theorem A (Möbius): the deck  $z \mapsto -z$  has EXACTLY TWO square roots  $z \mapsto \pm iz$  in  $\text{PGL}_2(\mathbb{C})$ , both *automatically* order 4, and a marks-preserving root exists  $\iff$  the deck is the CENTRAL involution of the mark- $D_4$  (counterexample  $\{\pm 1, \pm(3+2\sqrt{2})\}$ : harmonic with full  $D_4$  but deck non-central  $\Rightarrow$  0 roots — harmonicity alone insufficient); Theorem B (Fock, the core): the NS deck implementation is FORCED to order 4 —  $\tilde{U}^2 = 256 \gamma_1 \cdots \gamma_{16}$ , i.e.  $U^2 = (-1)^F$  EXACTLY, a nonsplit  $\mathbb{Z}_4$ , with the  $\mathbb{Z}_8$  tower  $V^2 \propto U$ ,  $V^4 \propto (-1)^F$  ( $8 = 2|\mu_4|$  fermionically explained) and the R-sector control split ( $U_R^2 = +1$ ); the one [C] rest: “the collar deck is the central involution of the mark- $D_4$ ” — the fermionic  $\mathbb{Z}_4$  forces the ORDER, not the spatial alignment; Wolfram extension  $434 \rightarrow 438$  (verified engine run); *NO marker moves — QGEO.REALIZE.01 stays [C]/[O] (text precised), P2.TYPING.01 R1 mapped onto the same bit)***

- `v506_seam_clock_rigidity.py` (SEAM.CLOCK.RIGIDITY.01, 34 checks,  $\sim 3.5$  s; **Part A** sympy-exact Möbius/divisor arithmetic + **Part B** exact Cl(16) Fractions; **deterministic**). v503 localised the one remaining identification input (“why does the raw seam carry the order-4 clock”, QGEO-R1  $\equiv$  P2-R1); v506 reduces it. *Theorem A (Möbius side, complete)*: Cayley–Hamilton + the complete case analysis give  $g^2 = \text{deck} \Rightarrow \text{tr } g \neq 0$  forces  $g$  diagonal with  $(a/d)^2 = -1$ , and  $\text{tr } g = 0$  forces  $\det g = 0$  (impossible) — exactly two roots  $z \mapsto \pm iz$ , both projective order 4, mutually inverse ( $\text{Aut}(\mathbb{Z}_4)$ ), conjugation-covariant; marks filter: a deck-invariant deck-free 4-set  $\{u, -u, w, -w\}$  admits a marks-preserving root  $\iff$

$w = \pm iu \iff \text{marks} = \text{one } \mu_4 \text{ orbit} \Rightarrow \text{cross-ratio class } \{2, \frac{1}{2}, -1\}, j = 1728, \tau = i, \text{ free } 4\text{-cycle}, \text{ deck} = \text{clock}^2, \text{ only the scale free; group form: } |\text{Stab}(\mu_4)| = 8 = 2|\mu_4| \text{ with order profile } \{1:1, 2:5, 4:2\} = D_4, \text{ center} = \{\text{id}, \text{deck}\}, \text{ exactly 2 order-4 elements} = \text{the clocks} \text{ — “deck central in the mark-}D_4\text{”} \iff \text{“clock exists”}; \text{ the counterexample } \{\pm 1, \pm(3+2\sqrt{2})\} \text{ is deck-free AND harmonic } (j = 1728, \text{ cross-ratio } \frac{1}{2}) \text{ with full } D_4, \text{ but the deck sits non-centrally (the central involution crosses the } \pm \text{ pairs)} \Rightarrow 0 \text{ roots; the RP filter passes BOTH roots (the } \text{Aut}(\mathbb{Z}_4) \text{ ambiguity is irreducible, v492 NC3). } B3/B4 \text{ (} n\text{-arithmetic + controls): free orbit } \Rightarrow n \mid 4; \text{ transitivity } \Rightarrow n = 4; \text{ faithfulness kills } n \geq 5 \text{ (kernel argument through } S_4); n = 3 \text{ fixes a mark; } n = 6 \text{ dies independently at the cusp degree 3 (v117); the deck is modulus-blind (v503 B5 exact); } n = 8 \text{ impossible on 4 marks; hexagonal stabiliser } A_4 \text{ (0 order-4 elements, squares only orders } \{1, 3\}); \text{ generic stabiliser } V_4 \text{ (exponent 2); stabiliser trichotomy } V_4/D_4/A_4 \text{ with order-4 counts } (0, 2, 0). \text{ Theorem B (Fock side, the core result): } S^2 \text{ has } b_1 = 0 \Rightarrow \text{unique spin structure } \Rightarrow \text{NS/bounding on every circle (established P1 datum); one-particle: } D^2 = -1 \text{ (charpoly } (\lambda^2+1)^8), \text{ quarter-shift } C^2 = D, C^8 = 1 \text{ (charpoly } (\lambda^4+1)^4, \zeta_8 \text{ spectrum); Fock (exact Cl(16)): } \tilde{U} = \prod_j (1 - \gamma_j \gamma_{j+8}) \text{ implements the deck on all 16 generators and } \tilde{U}^2 = 256 \gamma_1 \cdots \gamma_{16} \Rightarrow U^2 = (-1)^F \text{ EXACTLY; } (cU)^2 = c^2(-1)^F \text{ is never 1 (center of Cl(16) = scalars): the } \mathbb{Z}_4 \text{ lift is FORCED, } 1 \rightarrow \mathbb{Z}_2^{(-1)^F} \rightarrow \mathbb{Z}_4 \rightarrow \mathbb{Z}_2^{\text{deck}} \rightarrow 1 \text{ nonsplit; the quarter-shift lift } V \text{ (unique even block implementer) has } V^2 \propto \tilde{U}, V^4 \propto (-1)^F, \text{ projective order } 8 = 2|\mu_4| \text{ — the v492 } \mathbb{Z}_8 \text{ spin bridge realised on the 16-Majorana Fock space; R control } U_R^2 = +1 \text{ (split — the forcing rests on NS = established geometry); 16-fold way } \theta_\nu = 1 \text{ at } \nu = 16; \text{ robustness: the forcing already holds at 8 sites (16 = the carrier count } 2^{g_{\text{car}}-1}, \text{ not a tuning). The one [C] rest (verbatim): “the collar deck is the central involution of the mark-}D_4\text{” (} \iff \text{square-root existence on the mark data) — ONE alignment bit; the fermionic } \mathbb{Z}_4 \text{ forces the ORDER, not the spatial alignment. No overclaim.}$

- Ledger + papers + Wolfram + website.** Ledger: new row SEAM.CLOCK.RIGIDITY.01 ([E] both theorems + [C] the alignment bit); TEXT precisions at *unchanged* markers: QGEO.REALIZE.01 (“the rest is now ONE bit: the central position of the deck in the mark- $D_4$ ; order/uniqueness/orbit/ $\tau=i$  follow — v506”, stays [C]/[O]), P2.TYPING.01 (R1 mapped onto the same bit, fermionic halving), QGEO.EMERGE.LIGHT.01 context sharpened. Papers: `tfpt_3` celestial Step 1 (fermionic  $U^2 = (-1)^F$  sentence) + Step 2 (the one-bit reduction), contracts QGEO kill-test paragraph + marks paragraph + CELEST hypothesis note, `introduction/tfpt_1/tfpt_2/origin_theory` one-sentence updates at the v499/v503 citation sites. Wolfram: four mirrors (the complete square-root solve;  $D_4$  center + order-4 counts + the silver counterexample; the Cl(16) identity  $\tilde{U}^2 = 256 \gamma_1 \cdots \gamma_{16}$  on explicit  $256 \times 256$  Jordan–Wigner matrices + the split R control; the  $n$ -arithmetic + generic kill) verified with the active engine the same day (434  $\rightarrow$  438, ALL WOLFRAM EXTENSION CHECKS PASSED); the 24-triple stabiliser enumeration and the  $A_4$  profile stay Python-side (flagged). **VerificationDag:** v506 on the P2/carrier node; registry + `run_all` (500 modules, suite green). No gate closes, no marker moves.

2026-07-21 (XV) (v505 — WP5e- $\beta$  of CELEST.SEAM.01 “the equivariant anomaly ledger on twistor space” promoted as CELEST.WP5E.BETA.01: the Atiyah–Bott/Lefschetz fixed-point skeleton of the one-loop twistor inflow is EXACT — denominators  $(2, 4, 2)$  with Dedekind sum  $\frac{5}{4} = (|\mathbb{Z}_4|^2 - 1)/12$ , equivariant characters  $(248, 0, -8, 0)$  by two routes, invariant Okubo  $36\langle x, x \rangle^2$  with the twisted sectors NOT reducing and the AB sum leaving the RIGID residual  $32T_3$ , the index bridge  $f(m) = \text{ch}_2(T_m)$  exact with glue defect  $-78$  both routes, the level dials  $k = 1$  geometric (current count  $240/0/0/0$ ), and the honest E3 REFUTATION:  $a_0$  fills the GRAVITON slot  $O(2)$ , NOT the axion slot  $O(-2)$  — the three sphere classes fill the three twisted axion slots instead; the preregistered kill (“inflow demands a level  $\neq 1$ ”) does NOT fire on the equivariant skeleton; WP5e:  $\alpha$  (CFT side, v502) +  $\beta$  (equivariant twistor skeleton, v505) executed, the global BCOV quantisation stays [O]; Wolfram extension  $429 \rightarrow 434$  (verified engine run); SEAM.EQUIV.01 stays [O], no marker moves)

- **v505\_celestial\_wp5e\_beta\_equivariant\_ledger.py** (CELEST.WP5E.BETA.01, 47 checks,  $\sim 1.3$  s; sympy/Fraction exact, no floats; deterministic). The twistor side of WP5e pushed as far as exact arithmetic reaches: the Atiyah–Bott/Lefschetz skeleton of the one-loop box anomaly (Costello; Bittleston–Sharma–Skinner) on  $\mathbb{C}^2/\mathbb{Z}_4$  with the  $E_8$   $\mu_4$ -glue monodromy. *AB ledger*:  $\det(1 - g_j^{-1}) = (2, 4, 2)$ ;  $\sum_j 1/\det_j = \frac{5}{4} = (|\mathbb{Z}_4|^2 - 1)/12$ ;  $\text{tr}_{g_j}(\text{Ad } E_8) = (248, 0, -8, 0)$  two routes; invariant average  $60 = \text{carrier}$ ; Frobenius  $61568$ ;  $\text{tr}_{g_2} = -8 = 120 - 128$ . *Okubo per sector (honest sharpening)*: the invariant sector is exactly Okubo-quadratic ( $Q^{(0)} = 36\langle x, x \rangle^2$ ,  $36 = \tilde{\lambda}_{e_8}^2$  — v495 re-derived through the glue grading); the twisted sectors do NOT reduce ( $T_5$ :  $(-8, 32, -8)$ ,  $T_3$ :  $(48, -64, 48)$ ); the AB sum cancels the  $D_5$  quartic exactly and leaves the rigid  $32T_3$  (the only quartic-free weighting is the  $\mathbb{Z}_4$  projector; the graded GS exchange is rank-obstructed in sectors 1–3); heterotic identities ( $\text{Tr}_{120}, \text{Tr}_{128}, \text{Tr}_{E_8} = 9(\text{tr}^2)^2$ ) exact. *Index bridge*:  $f(m) = \frac{1}{4} \sum_j (i^{jm} - 1)/\det_j = (0, -\frac{3}{8}, -\frac{1}{2}, -\frac{3}{8}) = \text{ch}_2(T_m) = -(C^{-1})_{mm}/2$  EXACT (E1 = E2 arithmetic); glue  $\text{ch}_2$  defect  $-78$  both routes (classes  $(-24, -30, -24)$ ); Killing sums  $(320, 320, 240, 320)$ , total  $1200 = 2h^\vee(h, h)$ .  $k = 1$  *geometric*: lattice current count  $240$  at  $k=1$ , exactly  $0$  at  $k = 2, 3, 4$ ; embedding residual  $47k^2 + 219k - 266 = (0, 360, 814, 1362)$ ; integrality alone fixes nothing (honest, as v502); one scale  $\Rightarrow$  one level (v493 periods). *E3 refutation (prominent)*:  $a_0 \in O(8)$  fills the GRAVITON slot  $O(2)$ , NOT the BSS axion slot  $O(-2)$  ( $X = -2 \neq +2 = Y$ ; mismatch  $4 = |\mu_4|$  as a [C] coincidence) — “the theory brings its GS axion as  $a_0$ ” is REFUTED; instead the three  $H^2(\text{ALE})$  classes carry exactly the Coxeter eigencharacters  $\{i, -1, -i\} =$  the three twisted slots (bijection); slot count  $1+3 = 4 = |\mu_4|$  soft. *Negative controls*:  $\text{diag}(i, i)$  breaks the ledger four ways ( $\sum 1/\det = \frac{1}{4}$ ,  $\omega$ -weight  $-1$ , 5 generators);  $SO(16)$  glue: characters  $(120, 0, +120, 0)$ , defect  $-30 \neq -78$ , bulk Okubo fails (independent 4th Casimir);  $k = 2$ : current count  $0$ ,  $c$ -prefactor  $-31/48 \neq -\frac{1}{3}$ . [C] dictionaries ( $\text{CS}(\rho) \bmod 1 = \text{discriminant weight}$ ;  $32 = (h, h) + (h', h')$ ;  $-30 = -h^\vee$ ;  $\Omega$ -contraction). [O] the global BCOV/KS quantisation, non-fixed-point contributions, the  $O(-2)$  bulk axion, CPS level-from-flux, the pairing  $32T_3 \leftrightarrow$  sphere axions. VERDICT B; the kill test does not fire on the equivariant skeleton.
- **Ledger + papers + Wolfram + website.** Ledger: new row CELEST.WP5E.BETA.01; CELEST.SEAM.01 updated to “WP5e:  $\alpha$  (CFT side, v502) +  $\beta$  (equivariant twistor skeleton, v505) executed; the global BCOV quantisation stays [O] — with the rigid  $32T_3$  twisted residual and the sphere-axion pairing as the named remaining targets” at *unchanged* Open status. Papers: contracts WP5e bullet  $\rightarrow$  “ $\alpha + \beta$  done” with the  $a_0$  refutation as an honest finding (kill-test and honest-success paragraphs consistent); **tfpt\_3** celestial section becomes the *eleven*-step story (new Step 11 “the equivariant anomaly ledger on twistor space”, “what remains” = the global BCOV quantisation with named targets). Wolfram: five mirrors (AB denominators + characters two routes; the index bridge +  $-78$  defect; the per-sector Okubo sums over the  $240$  glue roots with the  $32T_3$  residual; the lattice current counts + embedding residuals; the

axion-slot weight arithmetic + twisted-slot bijection) verified with the active engine the same day (429  $\rightarrow$  434, ALL WOLFRAM EXTENSION CHECKS PASSED). VerificationDag qft node gains v505; registry + run\_all (499 modules at this step; 500 with v506, suite green). No gate closes, no marker moves — SEAM.EQUIV.01 stays [O].

**2026-07-21 (XIV) (v504 — WP5d- $\beta$  of CELEST.SEAM.01 “split + strong additivity from the lattice” promoted as CELEST.WP5DB.01: the two remaining KLM legs of complete rationality witnessed for the orbifold prescription — strong additivity ALGEBRAICALLY EXACT with the shared boundary Majorana ( $\text{Even}(A) \vee \text{Even}(B) = \text{Even}(A \cup B)$ ), GF2 spans full, rank 32/32; disjoint exactly index 2 = the v501  $\ln 2$  bit), the entropic touching defect BOUNDED  $< \ln 2$  with the Ising  $\frac{1}{4}$ -exponent approach ( $p = 0.2444$ ) while the preregistered  $U(1)$ /Dirac control DIVERGES (Klich–Levitov pinned, infinite index), the split property at the elliptic-nome rate  $\pi K(1-x)/K(x)$  (1.3–2.0%) with EXACT orbifold inheritance ( $C \rightarrow -C$  + the  $\Lambda^2 C$  compound — Longo heredity), and Pimsner–Popa  $E(a) - a/2 = PaP/2$  identically ( $\lambda = \frac{1}{2} = 1/[F:F_{\text{even}}]$ ,  $\lambda_{E_4} = \frac{1}{4} = 1/\mu$ ,  $e^{\Delta_\infty} = 2 = 1/\lambda_{\text{PP}}$  — two independent index routes); with v501 ALL THREE KLM ingredients carry lattice witnesses; the continuum uplift stays [O] — “finite-group orbifolds of completely rational nets are completely rational” is XU’S THEOREM, cited not claimed; Wolfram extension 424  $\rightarrow$  429 (verified engine run); SEAM.EQUIV.01 stays [O], CELEST.SEAM.01 stays Open, no marker moves)**

- **v504\_celestial\_wp5d\_klm\_completeness.py** (CELEST.WP5DB.01, 37 checks,  $\sim 12$  s; numpy/scipy Gaussian machinery ED-validated + exact GF2/integer algebra; deterministic). Complete rationality (Kawahigashi–Longo–Müger) = split + strong additivity + finite two-interval  $\mu$ -index; v501 (WP5d- $\alpha$ ) witnessed the index, v504 witnesses the other two legs on the same critical Majorana seam edge, with the  $U(1)$ /Dirac number-conserving subnet as the preregistered negative control (the current net, where strong additivity famously *fails* in the continuum — Buchholz–Schulz–Mirbach). *Strong additivity, exact algebra*: “touching = shared boundary Majorana” gives  $\text{Even}(A) \vee \text{Even}(B) = \text{Even}(A \cup B)$  EXACTLY (GF2 spans full 64/64, 256/256, 512/512, 1024/1024; literal matrix rank 32/32); disjoint intervals give exactly HALF (rank 16/32, index 2 — the missing sector odd  $\otimes$  odd is the v501  $\ln 2$  bit, localised at the split point); the neutral  $U(1)$  algebras do NOT generate the union even with the shared site (gaps 2/10/52, growing). *Entropic witness*: the fermionic touching MI carries the pure UV point-gap law  $((c/3) \ln 2$  increments, 0.115525, dev  $1.2 \times 10^{-6}$  — present for every net, no discriminator); the honest discriminator is *bounded vs divergent*: the  $\mathbb{Z}_2$ /orbifold deficit rises  $0.4045 \rightarrow 0.5899$  ( $N = 64 \rightarrow 4096$ ), stays  $< \ln 2$ , and  $(\ln 2 - \Delta_2) \sim N^{-p}$  with  $p = 0.2444$  vs  $2\Delta_\mu = \frac{1}{4}$  (Ising disorder exponent); split-point scan reflection-symmetric to  $3.9 \times 10^{-14}$ ; ED vN same direction; gap continuity to the v501  $x \rightarrow 1$  endpoint; honest note: “defect  $\rightarrow 0$ ” would be *false* at the sharp lattice split — bounded  $\Leftrightarrow$  finite index (Longo–Xu) is the correct statement; the  $U(1)$  control bursts  $\ln 2$  from  $L = 128$  and grows as  $\frac{1}{2} \ln \text{Var } Q_A$  (Klich–Levitov slope 0.10134 vs  $1/\pi^2 = 0.10132$ ). *Split/nuclearity*: coupling-block  $\sigma_k$  fall exponentially at the elliptic-nome rate  $\pi K(1-x)/K(x)$  to 1.3–2.0% (rates 2.29/3.24/4.82/5.81 at  $d = 16/64/256/512$ );  $\sigma_1 \sim x^{0.5044}$  (vs  $\frac{1}{2}$ ); trace norm summable; dimerised control exponential. *Orbifold inheritance exact*:  $P_A$  flips  $C \rightarrow -C$  ( $\sigma_k$  identical,  $8.3 \times 10^{-17}$ ) and the even-bilinear coupling Gram is the second compound  $\Lambda^2 C$  (singular values  $\{\sigma_a \sigma_b\}$ ,  $2.8 \times 10^{-14}$ ) — the same ladder at doubled rate (Longo heredity, [C]). *Pimsner–Popa, exact*:  $E(a) - a/2 = PaP/2$  identically; attainment in integers (all 16384 (even, odd) monomial pairs at  $N = 8$ : 8192 extremal with  $M^2 = 4$ ,  $\text{Tr } M = 0$ ,  $PMP = -M$ , 0 violations; 2048/2048 at  $N = 12$ ; pencil min  $\lambda = 0.5000000000$ );  $\lambda_{E_4} = \frac{1}{4} = 1/\mu$ ; index consistency  $e^{\Delta_\infty} = 2.000000000 = 1/\lambda_{\text{PP}}$  and  $e^{2\Delta_\infty} = 4 = \mu$ ;  $U(1)$ :  $\lambda = 1/(m+1) \rightarrow 0$ . *Controls + honesty*: dimerised phase trivial (drift  $3.6 \times 10^{-15}$ ); wrong prescription  $\{1, P_{AB}\}$  deficit exactly 0;  $\mathbb{Z}_2$  finite-size honesty (14.9% below

$\ln 2$  at  $N = 4096$  — witness, not proof). VERDICT: with v501 all THREE KLM ingredients of complete rationality are witnessed on the lattice; the continuum theorem for finite-group orbifolds is *Xu's* (cited, not claimed), the concrete seam quotient net and the condensed  $(E_8)_1$  net stay WP5e/Costello–Li.

- **Ledger + papers + Wolfram + website.** Ledger: new row CELEST.WP5DB.01; CELEST.SEAM.01 updated to “WP5d:  $\alpha$  ( $\mu$ -index) +  $\beta$  (split + strong additivity) executed via v501/v504 — all three KLM ingredients witnessed on the lattice; continuum uplift stays [O] (*Xu's* theorem cited)” at *unchanged* Open status. Papers: contracts WP5d bullet  $\rightarrow$  “both stages done” with the  $\beta$ -stage numbers + Xu fence (kill-test and honest-success paragraphs consistent); `tfpt_3` celestial section becomes the *ten*-step story (new Step 10 “the KLM completion on the lattice”, “what remains” = WP5e twistor inflow only); `tfpt_4` keystone note updated. Wolfram: five mirrors (GF2 span theorem; Pimsner–Popa identity + attainment on explicit gamma matrices; two-interval group  $\lambda_{E_4} = \frac{1}{4}$  + index consistency; the  $\Lambda^2/\text{Minors}$  compound with Cauchy–Binet and  $\text{Minors}[-C, 2] = \text{Minors}[C, 2]$ ; the  $U(1)$  dephasing  $\lambda = 1/(m+1)$  family) verified with the active engine the same day (424  $\rightarrow$  429, ALL WOLFRAM EXTENSION CHECKS PASSED); the lattice curves stay Python-only (flagged). VerificationDag qft node gains v504; registry + `run_all` (498 modules, suite green). No gate closes, no marker moves — SEAM.EQUIV.01 stays [O].

**2026-07-21 (XIII) (v503 — the v215 deferred emergence test (lever 7) executed *light* and promoted as QGEO.EMERGE.LIGHT.01: NO dynamical selection of the square modulus — the free field's global  $\det'$  winner is the *hexagonal* torus,  $\tau=i$  is a saddle, criticality at the square is Palais symmetry; the square is pinned by *rigidity* alone (the raw DtN is mod-4 block-diagonal *iff*  $\tau=i$ , an isolated zero); and the residual *converges*: QGEO-R1  $\equiv$  P2-R1 = one common rest, “why does the raw seam carry the order-4 clock”; Wolfram extension 420  $\rightarrow$  424 (verified engine run); NO marker moves — QGEO.REALIZE.01 stays  $[C]/[O]$ , AX.P2.01 stays a declared axiom)**

- `v503_qgeo_emergence_light.py` (QGEO.EMERGE.LIGHT.01, 17 checks; mpmath 40 digits + v280-style lattice DtN + sympy exact). The v215 lever 7 (“the decisive kill is deferred: compute the raw seam-collar DtN *without* inputting the marks and check whether four square marks *emerge*”) executed in its first honest *light* version. *No dynamical selection*:  $\log F = \log(\tau_2|\eta|^4)$  (Osgood–Phillips–Sarnak) is exactly  $S$ -invariant ( $\text{dev} < 10^{-35}$ ), so  $t=1$  is symmetry-forced critical on the rectangular family  $\tau = it$ ; there  $g''(1) = -1.298 < 0$  (strict maximum) but the  $\tau_1$ -direction has  $h''(0) = +0.298 > 0$ :  $\tau=i$  is a *saddle* of the full moduli space; the only critical points are the CM points ( $E_2^*(i) = E_2^*(\rho) = 0$  at 40 digits, generic  $0.163 \neq 0$ ); the *global*  $\det'$  winner at fixed area is the *hexagonal* torus ( $\log F(\rho) = -1.03352 > \log F(i) = -1.05469$ , diff 0.021169); criticality at the square is Palais symmetric criticality (the independent heat trace is equally critical). *Sharp rigidity*: the DtN mod-4 off-character indicator (the v215-K3 lever as a *modulus function*) vanishes to  $8 \times 10^{-15}$  exactly at  $\tau=i$  and rises strictly monotonically both ways (min  $8.4 \times 10^{-2}$  away) — an isolated zero;  $R^2 = \mathbb{Z}_2$  exact as permutation matrices (the v492  $\mathbb{Z}_8$  bridge on the lattice);  $j(it) \geq 1728$  with equality only at  $t=1$ . *The  $\mathbb{Z}_2$  chain is modulus-blind*: 4 marks +  $[\mathbb{Z}_2, \Delta_t] = 0$  on *every* torus (v216 confirmed without mark input); pillowcase  $\det' = \frac{1}{2}$  torus  $\det'$  exactly ( $\pm$ -pair structure) — only the order-4 clock distinguishes  $\tau=i$ ;  $\text{Aut}(E_\rho) = \mathbb{Z}_6$  has *no* order-4 element (the dynamical winner cannot carry the clock); given the clock the modulus freezes automatically ( $j = 1728$  identically, v493 mini). *Residual convergence (the core finding)*: the order-4 element on  $H^1$  is a Coxeter element of  $W(A_3)$  (char poly  $\lambda^3 + \lambda^2 + \lambda + 1$ , order 4 =  $|\mu_4| = N_{\text{fam}} + 1$ ) = the P2 cusp-class carrier datum of v499 R1  $\Rightarrow$  QGEO-R1  $\equiv$  P2-R1, one common residual; chain: marks derived (v216), square derived *given* the clock (v493), clock = carrier input  $[[C]]$ , rest = its dynamical realisation [O]; route classification (v347 analogue): B-rigidity in the sharpened form “RP seam data  $\Rightarrow$  order-4 clock”

is the live route. v215 lever 7 thereby moves from “deferred, not executable” to “light version executed: rigidity, not dynamical selection” — documented in v503, v215 itself unchanged.

- **Ledger + papers + Wolfram + website.** Ledger: new row QGEO.EMERGE.LIGHT.01; QGEO.REALIZE.01 text *precised* (rest = the order-4 clock, identical with the P2 typing rest R1; square + mark count derived) at *unchanged* marker [C]/[O]; cross-reference added in P2.TYPING.01 (convergence theorem). Papers: contracts kill-test paragraph (lever-7 light result + the residual-convergence sentence, both with v503), tfpt\_3 Step 2 (light-converse sentence), origin\_theory necessity paragraph, introduction/tfpt\_1/tfpt\_2 one-residual cross-references. Wolfram: four mirrors (CM-point identities  $E_2(i) = 3/\pi$ ,  $E_2(\rho) = 2\sqrt{3}/\pi \Rightarrow E_2^* = 0$  exact,  $j/\text{DedekindEta}$  special values; the no-dynamical-selection witnesses; the pillowcase halving as an exact algebraic identity; clock  $\Rightarrow$  square + Coxeter) verified with the active engine the same day (420  $\rightarrow$  424, ALL WOLFRAM EXTENSION CHECKS PASSED); the lattice DtN curves stay Python-only (flagged). VerificationDag masses node gains v503; registry + run\_all (497 modules, suite green). No gate closes, no marker moves.

**2026-07-21 (XII) (v502 — WP5e- $\alpha$  of CELEST.SEAM.01 “prefactor and level bookkeeping” promoted as CELEST.WP5E.ALPHA.01: the  $q^{-1/3}$  prefactor of  $E_4/\eta^8$  IS exact  $\mu_4$  vacuum-energy bookkeeping (the clock is inner  $\Rightarrow$  shift orbifold  $\theta = 0^8$ , common  $-c/24 = -\frac{1}{3}$ ; discriminant form = Casimir via spectral flow AND 16-Majorana free fermions), and  $k = 1$  is forced THREE independent ways (current condition; conformal embedding; central charge = the prefactor itself; plus the WP5b singular-vector dial  $31124 - 27000 = 4124$ ); honest sharpening: glue- $h$  integrality holds for ALL  $k$  and fixes nothing; the BCOV/Costello–Li twistor inflow stays [O] — WP5e is subdivided ( $\alpha$  done, WP5e proper open); Wolfram extension  $416 \rightarrow 420$  (verified engine run); SEAM.EQUIV.01 stays [O], no marker moves)**

- **v502\_celestial\_wp5e\_prefactor\_level.py (CELEST.WP5E.ALPHA.01, 33 checks, exact sympy/Fraction, no floats).** The two exactly computable consistency anchors of the WP5e twistor uplift established. *Part A — the prefactor:* Hurwitz- $\zeta$  exact ( $\zeta(-1, \theta) = -B_2(\theta)/2$ ;  $E_b(\theta) = -\frac{1}{24} + \theta(1 - \theta)/4$ ; Ising shift  $\frac{1}{16}$  per Majorana); the clock is *inner* ( $h = (2, 2, 2, 2; 0^4)$  and  $h' = (0^5; 1, 1, 1, -3)$  both read the class mod 4 on all 240 roots;  $(h, h) = 20$ ,  $(h', h') = 12$ , sum 32)  $\Rightarrow$  twist  $\theta = 0^8$  in all four sectors — a *shift* orbifold, NOT a rotation orbifold — so  $E_{\text{osc}} = -\frac{1}{3} = -c/24$  is sector-independent ( $c = 8 = \text{rank} = \text{Sugawara } 248/31$ ; one common  $\eta^8$ ); sector ground exponents  $-\frac{1}{3} + (0, 1, 1, 1)$ ; *discriminant form = Casimir energy* two ways (spectral flow  $j^2(h, h)/32 \bmod 1$ ; free fermions exact:  $D_5 = 10$  Majorana  $\rightarrow \frac{5}{8}$ ,  $A_3 = D_3 = 6 \rightarrow \frac{3}{8}$ , glue diagonal  $= 16 = 2^{g_{\text{car}}-1} = \text{the WP5d seam carrier} \rightarrow 1$ ); the rotation reading *fails* (deck  $\frac{3}{16} \neq \frac{3}{8}$  — “clock  $\neq$  deck” at the Casimir level); the four sector characters sum exactly to  $E_4/\eta^8 = q^{-1/3}(1, 248, 4124, 34752)$ . *Part B — the level:*  $k = 1$  forced three independent ways — (i) the current condition  $h(J) = k = 1$  (adjoint closure  $60+64+60+64 = 248$  at spin 1); (ii) the conformal embedding ( $47k^2 + 219k - 266 = 47(k-1)(k+266/47) = 0$ ;  $D_8: 128k(1-k) = 0$ ); (iii)  $c = 8 \Leftrightarrow 240(k-1) = 0$  (= the prefactor); plus the v498 singular-vector dial ( $\|(E_{-1}^\theta)^m 0\|^2 = m!k(k-1)\cdots(k-m+1)$ , singular vector at grade  $k+1$ ;  $31124 - 27000 = 4124$  at  $k=1$  only,  $\langle s|s \rangle = +4$  at  $k=2$ ). *Honest sharpening (mandatory):* glue- $h$  integrality  $h(J^j; k) = k(0, 1, 1, 1)$  is integer for ALL  $k = 1..8$  — the originally conjectured integrality route fixes *nothing*; the three other dials carry. *Negative controls:*  $D_8/SO(16)_1$  (same  $c = 8$ , SAME  $q^{-1/3}$  prefactor, but  $h = (0, \frac{1}{2}, 1, 1)$ : the discriminator is glue- $h$  integrality, not the prefactor;  $\Theta_{D_8} + \Theta_s = E_4$ );  $\mathbb{Z}_2$  rotation ( $E = +\frac{1}{6}$ ,  $h = \frac{1}{2}$ );  $\mu_4$  rotation (sector-dependent  $(-\frac{1}{3}, \frac{1}{24}, \frac{1}{6}, \frac{1}{24})$  — the common prefactor breaks); wrong  $k \in \{2, 3, 4\}$  fails all five dials coherently (residuals  $360/814/1362$ ;  $\langle s|s \rangle = 4/12/24$ ;  $31124 \neq 4124$ ). [[C]] Kac generation for  $k \geq 3$  ( $k = 1, 2$  machine-anchored by WP5b/WP5c). [O] the BCOV/Costello–Li twistor inflow itself (Costello’s one-loop axion  $\lambda_g/(8\pi\sqrt{3})$  on  $\mathbb{P}T$ , the Costello–Paquette–Sharma level-from-flux mechanism) — v502 anchors the *CFT side* of WP5e only; the KILL branch (“inflow demands a level  $\neq 1$ ”) is not triggered

on the CFT side, its twistor side is untested.

- **Contract + ledger + Wolfram + website.** `tfpt_research_contracts`: the WP5e bullet *subdivided* — “WP5e- $\alpha$  (done) [E]/[C]: the CFT-side prefactor + level pinning” with the core numbers and `v502_celestial_wp5e_prefactor_level.py`; “WP5e proper [O]: the twistor inflow”; summit header, honest-success paragraph and status roster updated (“WP5a–WP5d plus WP5e- $\alpha$  landed; WP5e proper the remaining open piece”). `tfpt_3` celestial section: new Step 9 (prefactor + level, one paragraph); `tfpt_4` keystone note updated. Ledger: new row `CELEST.WP5E.ALPHA.01`; `CELEST.SEAM.01` status  $\rightarrow$  “WP5e: alpha stage (CFT-side prefactor + level pinning) executed via v502; the BCOV/twistor inflow itself stays [O]”. Wolfram: four exact mirrors (Hurwitz- $\zeta$  vacuum energies; discriminant = Casimir with the deck failure; the  $k$ -fixing equations incl. the “fixes nothing” integrality table; the character sums +  $D_8$  control) verified with the active engine the same day (416  $\rightarrow$  420, ALL WOLFRAM EXTENSION CHECKS PASSED). `wolfram/README.md`, `GATE.WOLFRAM.02`, `tfpt_safeguards` Layer-6 count and the website counters (116/116 + 424/424 after (XIII)) updated; `VerificationDag` qft node gains v502; registry + `run_all` (497 modules, suite green). `SEAM.EQUIV.01` untouched; no marker moves.

**2026-07-21 (XI) (v501 — WP5d- $\alpha$  of CELEST.SEAM.01 “the two-interval index from the lattice” promoted as CELEST.WP5D.01: the KLM Haag-duality discriminator measured entropically on the seam edge — the 16-layer carrier saturates the  $\ln 2$  offset at machine precision,  $\mu_{\text{gauged}} = 4 =$  the v490 census, and the condensation arithmetic  $16 \rightarrow 4 \rightarrow 1$  closes with  $\mu = 1$ ; the preregistered KILL (“ $\mu$ -offset  $\neq 0$  after condensation”) does NOT fire; contract now “WP1–WP5d executed; WP5e the single remaining milestone”; Wolfram extension 412  $\rightarrow$  416 (verified engine run); SEAM.EQUIV.01 stays [O], no marker moves)**

- `v501_celestial_wp5d_two_interval_index.py` (CELEST.WP5D.01, 39 checks, numpy/scipy Gaussian machinery + exact Fractions). The WP5d roadmap discriminator —  $(E_8)_1 \Rightarrow \mu=1$  (Haag-dual two-interval net, one sector) vs  $SO(16)_1 \Rightarrow \mu=4$  — executed at the  $\alpha$  (lattice-index) stage on the v367 seam edge (16 critical Majorana layers,  $c_- = 8$ , v450-style ring; Peschel-exact, every Gaussian formula ED-validated to  $\leq 1.5 \times 10^{-15}$ ; the superselection identity  $[\rho, P_{AB}] = 0$  machine-exact). *The  $\mu = 1$  reference*: the fermionic (fixed-spin-structure) two-interval mutual information is *extensive* —  $c$  fit 0.5000,  $\max |F(x)| = 5.97 \times 10^{-5}$  at  $N=512$ , falling monotonically to  $1.3 \times 10^{-6}$  at  $N=1024$  (Casini–Huerta extensivity as a lattice witness). *The  $\mu$ -offset*: Rényi-2 exactly  $\Delta_2 = -\ln[(1+R)/2]$ ; the 16-layer seam carrier saturates  $\ln 2$  at machine precision ( $|\Delta_2 - \ln 2| = 1.1 \times 10^{-15}$  at  $N=512$ ;  $R \sim N^{-0.500}$ ); index reading  $[F : F_{\text{even}}] = 2.000000$ ,  $\mu_{\text{gauged}} = 4.000000 =$  the v490 parity census (*two independent lattice witnesses*); per-layer gauging costs  $16 \ln 2$  (the offset measures  $\ln[\text{index}]$ , not a blob). *Haag-duality indicator*: fermionic  $S(E) = S(E')$  to  $8.2 \times 10^{-12}$ ; the orbifold *breaks* complementarity ( $D(E) - D(E') = 0.427$  at  $a=64$ ); the complementary-pair sum is exchange-symmetric and  $\leq \ln 4 = \ln \mu(SO(16)_1)$  with max 1.2637 at the self-dual  $x = \frac{1}{2}$  — a *double-limit* statement (the first naive fixed- $x$  check was wrong and was replaced — honestly typed in the module). *Condensation arithmetic anchored at both measured ends*:  $\det \text{Cartan}(D_5) \cdot \det \text{Cartan}(A_3) = 16$ ; KLM/Longo–Rehren  $16/4^2 = 1$  and  $16/2^2 = 4 \rightarrow 4/2^2 = 1$  (both routes exact);  $\Sigma d_i^2 = (4, 4, 1)$ ;  $\theta_v = 1$  exactly at  $\nu = 2c_- = 16$ , rivals  $\neq 1$  — so  $\mu = 1$  after condensation ([C] the condensed net is interacting, the v490 fence): KILL not triggered. *Negative controls*:  $\nu=1$  single Majorana (offset non-removable,  $\theta_v = e^{i\pi/8} \neq 1$  — the discriminator has teeth); trivial phase (all  $\leq 10^{-7}$ ); wrong sector sum  $\{1, P_{AB}\}$  (offset exactly 0). [O] WP5d- $\beta$  (continuum Haag duality / local normality / split of the quotient net; runs as exploration) and WP5e (Costello–Li) stay open.
- **Contract + ledger + Wolfram + website.** `tfpt_research_contracts`: the WP5d bullet  $\rightarrow$  “ $\alpha$  stage done” with the core numbers and `v501_celestial_wp5d_two_interval_index.py`; status line and honest-success paragraph now “four of five milestones landed; WP5e (twistor

uplift) is the remaining open piece”. **tfpt\_3** celestial section: new Step 8 (one paragraph, conservative); **tfpt\_4** keystone note updated to WP1–WP5d. Ledger: new row CELEST.WP5D.01 (fine types Formal + Lattice/Numerical + [[C]] dictionary); CELEST.SEAM.01 status → “WP1–WP5d executed; WP5e [O] (single remaining milestone)”. Wolfram: four exact mirrors (Cartan determinants; KLM/Longo–Rehren quotients;  $\Sigma d^2$  + index-reading arithmetic; 16-fold-way  $\theta_v$ ) verified with the active engine the same day (412 → 416, ALL WOLFRAM EXTENSION CHECKS PASSED); the lattice curves stay Python-only (flagged). wolfram/README.md, GATE.WOLFRAM.02, tfpt\_safeguards Layer-6 count and the website counters (116/116 + 416/416) updated; VerificationDag qft node gains v501; registry + run\_all (495 modules, suite green). SEAM.EQUIV.01 untouched; no marker moves.

**2026-07-21 (X) (v500 — WP5c of CELEST.SEAM.01 “the GNS limit state” promoted as CELEST.WP5C.01: the quasi-free family  $\omega_w$  exists, is positive, stabilises exactly at the WP5a threshold, and its limit carries the null ideal in its GNS kernel — the full 9361-block exact level-2 Gram has rank 4124 exactly with kernel 27000 =  $V(2\theta)$  weight by weight;  $|s\rangle$  IS the GNS zero vector (the WP5b S4.10 tension resolved at the state level); a CCR obstruction proves the family formulation *necessary*; preregistered SUCCESS criterion met, KILL not triggered; Wolfram extension 408 → 412 (verified engine run); SEAM.EQUIV.01 stays [O], no marker moves)**

- **v500\_celestial\_wp5c\_gns\_limit\_state.py** (CELEST.WP5C.01, **35 checks, exact integer/Fraction, no floats, deterministic**). WP5a (v497) made the boundary limit precise, WP5b (v498) built the deleting operator; WP5c constructs the *state*.  $\omega_w$  is quasi-free: loop sector = the affine  $k=1$  vacuum  $n$ -point functions via the *machine-determined* compact anti-involution  $\theta(e_\alpha) = -e_{-\alpha}$  (the unique sign that is an anti-automorphism on all 61504 basis pairs; contravariance exact); radial sector = oscillator pairings  $x^{wr}$  (the exact Gibbs regulator,  $x$ -adic). *Convergence sharp*:  $\omega_w = \omega_\infty \bmod x^{N+1}$  for all  $w \geq N+1$  ( $N=8$ ), not yet at  $w = N$  — the WP5a threshold  $w = n+1$  holds *identically at the state level*; positivity of every  $\omega_w$  checked. *The kernel (core, fully machine)*: the complete exact level-2 Gram — 31124 monomials in 9361 weight blocks (profile 164:1, 37:240, 7:2160, 1:6960) — has level-1 rank 248 positive definite (Cartan block = the  $E_8$  Cartan matrix,  $\det 1$ ), level-2 rank 4124 *exactly* (the preregistered target check), kernel 27000, every block PSD; rank per Weyl orbit (0, 0, 1, 8, 44); kernel =  $V(2\theta)$  *weight by weight* (Freudenthal cross-check on all 9361 weights); highest-weight norms (3844, 49, 2, 0) at (0,  $\omega$ ,  $\theta$ ,  $2\theta$ ); direct ideal checks  $F_i|s\rangle$ ,  $F_j F_i|s\rangle$  norm 0.  $\mu_4$ -equivariance:  $\omega_w \circ \text{clock} = \omega_w$  for all  $w$ ; the clock descends to GNS; level-2 rank split by clock class (1036, 1024, 1040, 1024) =  $\Theta_{C_j}/\eta^8$  at  $q^2$  — the two-routes identity at the *state level*. *Resolution of WP5b S4.10*:  $|s\rangle$  is the zero vector of  $\text{GNS}(\omega_\infty)$  — the twisted two- $C_1$ -mode floor (6 quarters =  $q^{3/2}$ ) concerns only pre-limit objects (GNS has no state there,  $6 \bmod 4 \neq 0$ ). *Negative controls*:  $k=2$  ( $\langle s|s\rangle = +4$ , dominant blocks full rank — no ideal in the kernel);  $k=0$  (level-1 Gram  $\equiv 0$  — no 248 layer without the central extension);  $D_8$  (rank 2076 = 7380 – 5304, kernel =  $V(2\theta_{D_8})$  PSD, but  $h = (0, \frac{1}{2}, 1, 1)$ : one block of four — one-block completeness stays  $E_8/\mu_4$ -specific); wrong family  $E'_w = mw+r$  (the 248 layer erased,  $710955 \neq 248$ ); no damping ( $897266 \neq 248$ ); *CCR obstruction*: at fixed radial pairing  $c > 0$  positivity forces  $\omega(bb^*) \geq c$  — no  $w$ -uniform damping state exists, the family formulation is *necessary* and the family *exists*: the preregistered KILL is probed and not triggered. [C] quasi-free extension to the full  $*$ -algebra; Kac positivity beyond level 2. [O] local normality, net structure/Haag duality (WP5d), Costello–Li (WP5e).
- **Contract + ledger + Wolfram + website**. **tfpt\_research\_contracts**: the WP5c bullet → done with the core numbers (state-level threshold  $w = N+1$ , the full 9361-block Gram rank 4124/kernel 27000 weight by weight, the clock-class split, the CCR necessity of the family), citing v500\_celestial\_wp5c\_gns\_limit\_state.py; the WP5b handover marked answered. **tfpt\_3** celestial section: new Step 7 (one paragraph, conservative). Ledger: new row CELEST.WP5C.01 (fine type Formal/Identity [E] + [C] identifications + [O] net questions). Wolfram: four exact

mirrors (level-2 block census from the root data; rank/kernel arithmetic with  $\dim V(\omega) = 3875$  by Weyl; the glue-frame clock-class split (1036, 1024, 1040, 1024); threshold + level-dial +  $D_8$  census) verified with the active engine the same day (408  $\rightarrow$  412); the exact Gram/PSD computation stays Python-side (flagged). **VerificationDag** qft node gains v500; registry + **run\_all** (494 modules at this step, suite green). **SEAM.EQUIV.01** untouched; no marker moves.

**2026-07-21 (IX) (v499 — the P2 weight-typing postulate HARDENED: the anchor  $a = (1, 1, 2)$  typed as the Birkhoff–Grothendieck splitting exponents of the Deligne canonical extension, so integrality/positivity/sum-4 become theorems given the [E] anchors; the v491 check-9 residual sharpened, the tfpt\_2 “balanced” selection typed as stability; Wolfram extension 403  $\rightarrow$  408 (verified engine run); no gate closes, no marker moves — AX.P2.01 stays a declared axiom, QGEO.SYM stays [O])**

- **v499\_p2\_weights\_deligne\_bg.py** (P2.TYPING.01, 23 checks, sympy exact, no floats). v491 reduced P2 to “3 positive integers summing to 4”; its residual was the *typing* (why integers, why positive, why sum 4). Now typed: given the [E] anchors (4 marks, v216; rank  $3 = \# \text{marks} - 1$ , v137/v228; cusp-class local monodromy  $\lambda^3 - 1$  and exact irreducibility  $|\langle U, M_0 \rangle| = 24 = |S_4|$ ,  $\sum_g |\chi|^2 = 24$ , v117; unitarity (U)), the Deligne canonical extension  $E$  of the flavor connection satisfies:  $\deg E = -4$  as a theorem (residue exponents  $\{0, \frac{1}{3}, \frac{2}{3}\}$  the unique lift to  $[0, 1)$ , trace 1 per mark; par-deg 0;  $\infty$  apparent, monodromy product = 1 exactly);  $h^0(E) = 0$  as a theorem (Mehta–Seshadri stability; all 324 section-saturation cases; BG dictionary  $h^0=0 \Leftrightarrow a_i \geq 1$ ); integrality = Birkhoff–Grothendieck. Hence  $\{a\} = \{1, 1, 2\}$  unique and  $g_{\text{car}} = e_2 = 5$ . *Selector typed*: the Schur–Horn permutohedron has integer points exactly  $\{(2, 2, 0), (2, 1, 1)\}$  (the two options of the tfpt\_2 “Global computation” theorem) and  $h^0=0$  kills  $\{0, 2, 2\}$  — the “balanced/minimal-variance” heuristic is now stability/positivity. *The  $h^1$  discrepancy, real and resolved*:  $h^1(\mathcal{O}(-2) \oplus \mathcal{O}(-1)^2) = 1 \neq 3$ , so “generations =  $H^1$  of the rank-3 bundle” is FALSE (negative control: forcing it demands  $\deg -6$  and loses uniqueness,  $e_2 \in \{9, 11, 12\}$  never 5); correct two-object typing:  $L = \mathcal{O}(-D)$  carries the generations ( $h^1(L) = 3 = N_{\text{fam}}$ , the v137 basis =  $H^0(\Omega^1(\log D))$ ),  $E$  carries them as its fiber. *Biswas explicit* (independent witness): the weight denominator  $3 = N_{\text{fam}}$  forces the  $\mathbb{Z}_3$  cusp cover  $y^3 = (z-1)(z-i)(z+1)^2(z+i)^2$  (genus 2; NOT the pillowcase double, genus 1,  $j = 1728$  — the deck side);  $p_*\mathcal{O}_Y = \mathcal{O} \oplus \mathcal{O}(-2)^2$ ; the regular-representation model lands on  $\{0, 2, 2\} =$  the *unstable* Schur–Horn companion ( $h^0 = 1$ ): integrality and the sum rule come out of the correspondence, stability is the selector. *Negative controls*: no positivity  $\rightarrow$  4/31 solutions;  $\deg -3/ -5$  breaks;  $n$ -mark scan  $e_2 = (n-2)(n+1)/2 = 5$  ONLY at  $n = 4$  (the value, not the uniqueness, is the  $n=4$  content); MS-rational  $e_2 = \frac{16}{3}, \frac{44}{9}$ . *Honest residuals* (P2 stays an axiom): R1 module identification [C] (QGEO.SYM territory), R2 (U) [C] (v33/v40), R3 analytic monodromy (Numerical, v117 check 4); non-circular (neither v23’s splitting input nor v115’s  $\text{diag } A_0 = a/4$  input used); the v491 direction-reversal note carries over. Ledger: new row P2.TYPING.01; the v491/P2.PARTITION.01 row records “typing residual sharpened by v499” (v491 stays active, no supersede). Registry + **run\_all** (493 modules, suite green).
- **Wolfram second path + papers + website**. Five exact mirrors added to the extension file **tfpt\_readouts\_extension.wl** (partition enumeration under the full typing + naive-typing control +  $h^1$  discrepancy; Deligne residue-trace degree with the  $A_0^*$  witness char poly; Schur–Horn lattice points; Biswas  $\mathbb{Z}_3$ -cover degree arithmetic incl. deck double  $j = 1728$ ; the  $n$ -mark formula), verified with the active engine the same day (403  $\rightarrow$  408, ALL WOLFRAM EXTENSION CHECKS PASSED); wolfram/README.md + GATE.WOLFRAM.02 + **tfpt\_safeguards** Layer-6 count + website counters (116/116+408/408). Papers (one sentence each, conservative): **introduction** anchor paragraph, **tfpt\_1** (5,3) keybox, **tfpt\_2** H2 winbox and the “Global computation” theorem (the balanced selection typed as stability), **origin\_theory** (5,3) keybox — all citing v499\_p2\_weights\_deligne\_bg.py; no status card changes. Website: **VerificationDag** P2

node gains v499; generated indices via build; no marker moved anywhere.

**2026-07-21 (VIII) (Editorial: dedicated celestial-route section in Paper 3 + website mirror — the scattered celestial sentences consolidated into one coherent six-step section; *no new scripts, no status changes, no marker moves, SEAM.EQUIV.01 untouched*)**

- **tfpt\_3\_e8\_audit\_bootstrap: new section “The celestial route: from the  $\mu_4$  clock to the  $(E_8)_1$  boundary shadow”.** The celestial-contract material that had accumulated as one long paragraph inside “The Möbius origin of the irreducible  $\pi$ ” is consolidated into a dedicated section (placed after the bootstrap audit, before “Honest status”), telling the executed WP1–WP5b story in six typed steps: (1) setup — the  $E_8$   $\mu_4$ -glue as a flat  $\mathbb{Z}_4$  monodromy on the  $A_3$  ALE space,  $\text{clock}^2 = \text{deck}$  ( $\text{spin } \mathbb{Z}_8, 8 = 2|\mu_4|$ ) [E] (v492); (2) deformation —  $XY = Z^4 + a_0$  pure scale,  $j = 1728$  frozen,  $\text{clock} = \text{Picard–Lefschetz/Coxeter monodromy}$  [E] (v493); (3) consistency honestly demoted —  $(\kappa/c_3)^2 = 12$  exact but the alignment format passes 8/8: alignment survives, selectivity does not [E]/[C] (v495); (4) type mismatch + boundary limit — the  $(E_8)_1$  character is not a jet-grading conformal block (growth  $n^{2/3}$  vs  $n^{1/2}$ ) but the  $\mu_4$  period sum 248 holds exactly: a boundary-limit shadow of SEAM.EQUIV.01/MMST shape [E] (v496); (5) the null ideal quantitative — stabilisation threshold  $w = n+1$ ,  $27000 = (h^\vee)^3$  derived, quotient = sector sum = 4124,  $SO(16)_1$  contrast [E] (v497); (6) the deleting operator explicit —  $|s\rangle = (E_{-1}^\theta)^2|0\rangle$ ,  $J_1^a|s\rangle = 0$  on all 248 (tally 190/57/1), level dial  $2(1-k)$ , one-block closure  $E_8/\mu_4$ -specific [E] (v498); plus the WP5c–e roadmap and the honest status line (the route does *not* close SEAM.EQUIV.01; the GNS/limit question stays [O]). All pre-existing script citations (v492–v498) are carried into the new section; the Möbius section now points to it with one sentence. **tfpt\_research\_contracts:** a reader’s note at the head of the CELEST.SEAM.01 contract cross-references the new Paper-3 section (the contract stays the full technical reference). **tfpt\_4\_frontier:** one sentence in the metric-sector keybox records the celestial route as a third, quantitative approach to the keystone (not closing it).
- **Website mirror. papers.ts:** the Paper-3 section list gains the celestial-route section (six-step summary in the existing style) and a “Celestial route” highlight; the contracts mirror gains the same cross-reference sentence. **glossary.ts:** three new entries (celestial route, ALE space, null ideal) in the established technical style. Generated indices via build; no status component changes (no marker moved anywhere).

**2026-07-21 (VII) (CELEST.SEAM.01 WP5b executed and promoted: v498 — “the singular vector as an operator”, the deleting object of the v497 null ideal constructed explicitly and machine-verified singular; contract now WP1–WP4 + WP5a + WP5b done, WP5c–e [O] (roadmap); Wolfram extension 398  $\rightarrow$  403 (verified engine run); *no gate closes, no marker moves, SEAM.EQUIV.01 untouched*)**

- **v498\_celestial\_wp5b\_singular\_vector.py (CELEST.WP5B.01, 53 checks, exact integer/Fraction, deterministic, no floats).** Success on the preregistered WP5b criterion: the deleting object exists and is explicit. *Construction:*  $|s\rangle = (E_{-1}^\theta)^2|0\rangle$  (weight  $2\theta$ , level  $2 = 8$  quarter units =  $q^2$ , an *integer* level) in a machine-built Chevalley/Frenkel–Kac basis — bimultiplicative 2-cocycle with asymmetry verified on all 57600 ordered root pairs, the  $[e_\alpha, e_{-\alpha}]$  sign *forced* by the Jacobi identity ( $\text{SGN} = -1$ , not chosen), the invariant form  $\kappa$  derived ( $\kappa(\theta^\vee, \theta^\vee) = 2$ ), Jacobi exact on all 57600  $\mathfrak{sl}_2$ -slice triples + 6000 seeded + all 496  $\theta$ -adjacent, all  $N_{\alpha\beta} = \pm 1$ . *Singularity (machine):*  $J_1^a|s\rangle = 0$  for all 248 basis elements (0 violations), with the case classification 190 (first bracket) / 57 (second) / 1 ( $a = F^\theta$  via the central-term cancellation — the *only* case that sees the level  $k$ ); plus  $J_2^a|s\rangle = 0$ ,  $E_0^a|s\rangle = 0$ , exact weight, Shapovalov norm 0; the affine PBW engine unit-tested on all 61504 basis pairs. *Level dial:* the  $F_1^\theta$  coefficient

is  $2(1-k) \rightarrow 2/0/-2$  at  $k = 0/1/2$ : without the central extension the deletion operator does not exist.  $\mu_4$  data: glue class  $j(\theta) = 1$  (machine-built  $h$ -adapted chamber,  $\langle \theta, h \rangle = 5$ , height  $29 = h^\vee - 1$ ), clock phase  $i^{2j} = -1$ , class  $2(\theta) = 2$  sheet-even; the deleting  $\theta$ - $\mathfrak{sl}_2$  crosses the sheet-odd classes  $(1, 3)$  while its square is sheet-even; cross-table  $\theta$ -grading  $\times$  glue class consistent (rows  $(1, 56, 126, 56, 1)$ , columns  $(52, 64, 60, 64)$ ); 8 quarters  $= q^2$  via the per-period dictionary. *Level-2 match*: weight  $2\theta$  has multiplicity 1 in the level-2 Fock, so  $U(\mathfrak{g})|s\rangle$  is the  $27000 = (h^\vee)^3$ ; the direct  $\mathfrak{g}_0$ -orbit BFS matches the Freudenthal multiplicities of  $V(2\theta)$  exactly through depth 4 ([C] Weyl complete reducibility beyond); quotient  $31124 - 27000 = 4124$ ; lattice reading  $(2\theta, 2\theta)/2 = 4 > 2$ :  $|s\rangle$  is the kernel generator of  $\text{Fock} \rightarrow E_4/\eta^8$ . *Negative controls (honest separation)*: the level-1 current state  $E_{-1}^\theta|0\rangle$  (one witness,  $\rightarrow -k|0\rangle$  — the 248-current state that must remain) and 4 generic level-2 states (14–33 witnesses) are *not* singular; at  $k=2$  the *cube* is (generic Kac  $(E^\theta)^{k+1}$  mechanics, honestly typed);  $SO(16)_1$  has the same singular vector but keeps three extra level-1 primaries ( $h = \frac{1}{2}$  breaks one-block fusion;  $5304 \neq 14^3$ ) vs  $E_8$  (comarks all  $\geq 2$ , glue weights  $(0, 1, 1, 1)$  integer) — the *one-block closure* is the  $E_8/\mu_4$ -specific part. *Honest [O]*: in the twisted quarter-slot moding two sector- $C_1$  modes never sum to 8 quarters (minimum  $6 = q^{3/2}$ ) — the per-period dictionary (v496), not the per-slot identification, carries  $|s\rangle$  to  $q^2$ : exactly the WP5c GNS/limit-state question; WP5c-e stay open; SEAM.EQUIV.01 untouched.

- **Contract + surfaces synced.** `tfpt_research_contracts`: the WP5b bullet moved to *done* with the key numbers (case tally 190/57/1; level dial  $2(1-k)$ ;  $j(\theta) = 1$ , clock phase  $-1$ ; orbit BFS through depth 4; the  $D_8$  contrast) and the twisted-slot tension as the explicit WP5c handover; honest status line and K4 paragraph updated (“WP1–WP4 + WP5a + WP5b executed; WP5c-e [O]”). `tfpt_3_e8_audit_bootstrap`: the celestial-contract paragraph extended by the WP5b sentence. Ledger: CELEST.WP5B.01 (new) + CELEST.SEAM.01 (“WP1–WP4 + WP5a + WP5b executed; WP5c-e [O] (roadmap)”). Wolfram: 5 exact v498 mirrors added to `tfpt_readouts_extension.wl` and *verified* with the active engine the same day ( $398 \rightarrow 403$ , ALL WOLFRAM EXTENSION CHECKS PASSED); `wolfram/README.md` + GATE.WOLFRAM.02 counters advanced; `tfpt_safeguards` Layer-6 count + website mirrors 116/116 + 403/403. Website: `papers.ts` contract mirror (“WP1–WP4 + WP5a + WP5b done, WP5c-e roadmap”), DAG qft node + v498, generated indices via build. Suite now 492 scripts, all green; SEAM.EQUIV.01 stays [O].

**2026-07-21 (VI) (CELEST.SEAM.01 WP5a executed and promoted: v497 — “the null ideal from the limit”, the v496 boundary-limit shadow made a precise coefficientwise limit and the level-2 null ideal derived from root data; contract WP5 subdivided, now WP1–WP4 + WP5a done, WP5b-e [O] (roadmap); Wolfram extension 394  $\rightarrow$  398 (verified engine run); no gate closes, no marker moves, SEAM.EQUIV.01 untouched)**

- `v497_celestial_wp5a_null_ideal.py` (CELEST.WP5A.01, 34 checks, exact integer/Fraction, no floats). Three results make the WP4 limit reading precise. *Stabilisation theorem*: the one-parameter family  $\chi_w$  of graded Fock characters on the chiral jet generators (loop energy  $m$ , radial number  $r$ ;  $E_w = m + wr$  in quarter units  $u = q^{1/4}$ ) contains the chiral jet grading as its  $w=2$  member (generator counts 64, 120, 128, 180, 192, 240, 256, 300 per degree), and its  $u^n$  coefficient equals the quarter-moded loop Fock for *all*  $w \geq n+1$ , strictly larger for every  $w \leq n$  (verified  $n \leq 8$ ,  $w \leq 10$ ) — the v496 “boundary-limit shadow” is a precise *coefficientwise* limit  $\varepsilon = 1/w \rightarrow 0$  with explicit threshold  $w = n+1$ , not a slice (convention clarified:  $897266 =$  the quarter-moded loop Fock at  $u^4 = \text{integer level } 1$ ). *The null ideal derived, not cited*: Freudenthal multiplicities + the Weyl dimension formula + generic character peeling give  $\text{Sym}^2(248) = V(2\theta) \oplus V(\omega) \oplus V(0) = 27000 + 3875 + 1$  with residual exactly zero (independent Weyl-orbit cross-check);  $27000 = 30^3 = (h^\vee)^3$  exactly;  $C_2 = (124, 96)$ , Dynkin indices 6750/750; level-1 integrability ( $\langle 2\theta, \theta^\vee \rangle = 4$ ,  $\langle \omega, \theta^\vee \rangle = 2$ ) leaves

the vacuum as the only level-1 primary (Kac, used only through its counting consequence); level-2 quotient  $31124 - 27000 = 4124 = 1 + 248 + 3875$  exactly. *Two routes, one number*: the  $\mu_4$  theta-split sector sum at  $q^2 = (1036, 1024, 1040, 1024)$  gives the *same* 4124 — the untwisted null-ideal quotient and the boundary/twisted lattice presentation count one space ( $q^3: 34752 = \chi_3$ ). Negative controls:  $SO(16)_1$  through the same pipeline gives *four* components ( $5304 + 1820 + 135 + 1$ ),  $5304 \neq 14^3$  (the  $h^{\vee 3}$  identity is not generic), quotient  $2076 = \Theta_{D_8}/\eta^8$  at  $q^2$  (the recipe validated on a second algebra), block weights  $(0, \frac{1}{2}, 1, 1)$  break one-block fusion (vs  $E_8$ 's integer  $(0, 1, 1, 1)$ ); only  $P = 4 = |\mu_4|$  periodisation reproduces the 248 layer; swapped glue breaks the zero-mode anchor ( $64 \neq 60$ ). *Honest [O]*: the limit does *not* generate the truncation — the singular vector as an *operator* in the limit net (WP5b), the GNS limit state (WP5c) and local normality/Haag duality (WP5d) stay open; the celestial route now has the same two-step shape as MMST (limit + maximal ideal), with the ideal quantitatively fixed; SEAM.EQUIV.01 untouched.

- **Contract + surfaces synced.** `tfpt_research_contracts`: WP5 subdivided — WP5a moved to *done* with the key numbers (stabilisation threshold  $w = n+1$ ;  $27000 = (h^\vee)^3$  derived; quotient = sector sum = 4124;  $SO(16)_1$  contrast), WP5b–e added as roadmap bullets with success/kill criteria (WP5b singular-vector operator in the quarter-moded current Fock; WP5c GNS limit state, kernel  $\supseteq$  ideal; WP5d local net/Haag duality via the KLM route; WP5e BCOV/Costello–Li twistor uplift, partition function =  $E_4/\eta^8$  incl. the  $q^{-1/3}$  prefactor); K4 paragraph and the honest-success status line updated. Ledger: CELEST.WP5A.01 (new) + CELEST.SEAM.01 (“WP1–WP4 + WP5a executed; WP5b–e [O] (roadmap)”). Wolfram: 4 exact v497 mirrors added to `tfpt_readouts_extension.wl` and *verified* with the active engine the same day ( $394 \rightarrow 398$ , ALL WOLFRAM EXTENSION CHECKS PASSED); `wolfram/README.md` + GATE.WOLFRAM.02 counters advanced; `tfpt_safeguards` Layer-6 count + website mirrors 116/116 + 398/398. Website: `papers.ts` contract mirror (“WP5a done, WP5b–e roadmap”), DAG `qft` node + v497, generated indices via build. Suite now 491 scripts, all green; SEAM.EQUIV.01 stays [O].

**2026-07-21 (V) (comb-meta-limit re-run with the PG.06b FULL Vela recovery ingested — the stale single-obs PG.06b reference (noise-peak  $F_0 = 11.19275$  Hz) corrected in the channel definitions; the new X-ray channel enters *censored* (no absolute  $\varepsilon$ ), both aggregate limits numerically unchanged; *experiments level only* — *no verification module, no gate closes, no marker moves*)**

- **experiments/comb-meta-limit channel update (methodology untouched).** The Vela X-ray channel (`channels.py`) now ingests the PG.06b FULL reduction (`pg06b_full_vela.json` in `pulsar-glitch-recovery/results`: 665 obs  $\rightarrow$  491 usable  $\rightarrow$  119 segment-coherent  $\nu(t)$  points) instead of the superseded single-obs `pg06b_vela.json`; the stale code note quoting the noise-peak “detection”  $F_0 = 11.19275$  Hz is corrected on record (true Vela  $F_0 = 11.1861692$  Hz), and the channel key is fixed from the mislabel PG.07 to PG.06b. Censoring handled exactly by the existing Tier-B rule (nothing invented): the comb observable is a *linear*  $\delta\nu(\tau)$  residual ( $\mu$ Hz, not a fractional ln-flux modulation), both legs sit below the 2.8-period gate (2019: 2.17, 2021: 1.38), and the surrogate-calibrated injection never reaches  $\varepsilon_{90}$  below  $\varepsilon = 1$  ( $\varepsilon_{50} = 0.55$  in 2019 only)  $\Rightarrow$  no absolute per-channel  $\varepsilon$ -UL exists for the DL/HKSJ combination; the channel stays enumerated-only (“no constraint”), carrying the normalised context amplitude  $\sqrt{2g} = 0.40$  (leg gains 0.046/0.114, min  $p = 0.014$  — the 2019 audit-level chance feature: Bonferroni  $\times 10 = 0.14$ , shuffle  $p = 0.40$ ).
- **Full re-run (seed 0, deterministic): both limits numerically unchanged.** Boundary/horizon-scoped: `data_limited` (still no horizon channel with an absolute  $\varepsilon$ ); all-channel universal-DSI 95% UL  $\varepsilon < 0.11954$  ( $\approx 12.0\%$ ; HKSJ  $t_2 = 2.92$ ,  $\rho = 0.3$ -inflated 0.137;  $6.91\times$  the predicted  $\varepsilon = 0.0173$ ), still driven by the surface log-flux channels A1/A4/A5 (A4 GRB the anchor, per-channel UL 0.108); now 10 channels enumerated / 3 usable. Injection self-consistency unchanged (inject 5%  $\rightarrow$  recover 4.8%, UL 5.5%;

coverage 0.92/150 trials). Surfaces synced: experiment README (tier table, ingest list, PG.06b FULL bullet), results/results.json, scorecard row data\_value/source via build\_evidence\_scorecard.py (status stays data\_limited; counters unchanged, 120 rows), experiments/README.md catalog row. No claim; firewall intact.

**2026-07-21 (IV) (Wolfram engine reactivated — the deferred second-path mirrors VERIFIED: base 116/116 + extension 394/394 (378 → 394, the 16 mirrors accumulated since 2026-07-14 now counted); GATE.WOLFRAM.01/02 and all counter surfaces synced; *no gate closes, no marker moves*)**

- **Verified engine run 2026-07-21 (Wolfram Engine 14.3, license reactivated).** wolframscript -file verification/wolfram/tfpt\_readouts.wl → 116 passed, 0 failed, ALL WOLFRAM CHECKS PASSED; tfpt\_readouts\_extension.wl → 394 passed, 0 failed, ALL WOLFRAM EXTENSION CHECKS PASSED. The 16 exact mirrors added while the engine was unactivated all enter the counted total (378 → 394): v479 Kronheimer bridge (2), v491 partition corollary (3), v493 celestial WP2 clock deformation (4), v495 celestial WP3 Green–Schwarz alignment (3), v496 celestial WP4 character shadow (4).
- **Counter surfaces synced.** verification/wolfram/README.md (current-state block + the five deferral notes resolved; v484/v485/v486/v487/v488/v492 stay mirrorable-but-unmirrored, Python-exact, so they are *not* counted); ledger GATE.WOLFRAM.01 (re-verified 2026-07-21) + GATE.WOLFRAM.02 (394/394, UPDATE note); tfpt\_safeguards Layer-6 count + website papers.ts mirrors (116/116 + 394/394); v479 registry/run\_all deferral note resolved. No numeric claim changes; the second computational path is simply verified-current again.

**2026-07-21 (III) (CELEST.SEAM.01 WP3 + WP4 executed and promoted: v495 Green–Schwarz alignment (verdict B, honestly non-selective) + v496 character-vs-S-algebra type mismatch (verdict B(ii), boundary-limit shadow); contract now WP1–WP4 done, WP5 [O]; *no gate closes, no marker moves, SEAM.EQUIV.01 untouched*)**

- **v495\_celestial\_wp3\_gs\_alignment.py (CELEST.WP3.01, 25 checks, exact Fraction/sympy).** The pre-registered WP3 “alignment test only”: the Okubo coefficient  $\text{Tr}_{\text{adj}} X^4 / (\text{Tr}_{\text{adj}} X^2)^2 = 5/(2(\dim \mathfrak{g} + 2))$  is *derived* as an exact polynomial identity for all 8 algebras on Costello’s list ( $\mathfrak{sl}_2, \mathfrak{sl}_3, \mathfrak{g}_2, \mathfrak{so}_8, \mathfrak{f}_4, \mathfrak{e}_6, \mathfrak{e}_7, \mathfrak{e}_8$ ;  $\mathfrak{e}_6/\mathfrak{e}_7$  carved out of the v492  $E_8$  root set; negative control:  $\mathfrak{sl}_4$  fails,  $5/32$  vs  $3/32$ ); the closed form  $\tilde{\lambda}^2 = 10h^\vee/(\dim + 2) = h^\vee + 6$  holds exactly across the Deligne series;  $E_8$ :  $\tilde{\lambda} = 6$  (unit-trace) resp.  $\lambda_{\text{fund}} = 1/10$  (adjoint-trace) exact, so  $(\kappa/c_3)^2 = 12 = |\mu_4|N_{\text{fam}}$  resp.  $1/300$  — with  $\kappa/c_3$  itself *irrational* ( $2\sqrt{3}$ ). Byproduct: the printed  $\lambda_{\text{so}_8}^2 = 3/2$  in Costello’s appendix A is a factor-2 slip (his own weight content gives 3). **Look-elsewhere (the hard fence):** the alignment format passes 8/8 across the whole list (zero selectivity),  $\tilde{\lambda}$ -integrality  $2/8$  (shared with  $\mathfrak{sl}_3$ ), only  $g_{\text{car}}=5 \mid h^\vee$  is  $\mathfrak{e}_8$ -selective ( $1/8$ ), the isolating conjunction is post hoc. Kill test K3 does *not* fire but is scope-demoted: *alignment survives; selectivity does not* — compatibility, not  $\mathfrak{e}_8$ -selective evidence. Verdict B.
- **v496\_celestial\_wp4\_character\_shadow.py (CELEST.WP4.01, 25 checks, exact integer/Fraction/sympy).** WP4, the type mismatch head-on: the  $(E_8)_1$  character  $E_4/\eta^8 = (1, 248, 4124, 34752, 213126, \dots)$  is *not* a conformal block of the glue-equivariant  $E_8[\mathbb{C}^2]$  S-algebra in its own jet grading — localised three ways: (a) spin  $1 - d/2$  unbounded below and 248 never a tower dimension ( $d \leq 100$ ); (b) growth  $n^{2/3}$  vs  $n^{1/2}$  (quadratic cumulative count  $31s^2 + 92s + 60$ ,  $31 = k + h^\vee$ ;  $f_n/\chi_n$  strictly increasing  $n = 1..12$ ); (c) the level-2 null ideal deletes exactly  $27000 = 30^3 = (h^\vee)^3$  out of  $\text{Sym}^2(248) = 1 + 3875 + 27000$  — no jet analogue. *But* the boundary/period reading holds exactly at the current stratum (zero modes =  $60 = C_0$  currents, jet slice cycles (60, 64, 60, 64), one  $\mu_4$  period sums to 248, glue weights (0, 1, 1, 1) integer; free loop Fock  $897266 \gg 248$ , so the truncation must come from a limit) — the same MMST

scaling-limit shape as SEAM.EQUIV.01 (v336/v449). K4 fires only against the exact-block reading; the constructive limit question passes to WP5. Verdict B(ii).

- **Contract + surfaces synced.** `tfpt_research_contracts` WP3/WP4 bullets moved to *done* with the look-elsewhere demotion in the text, K3/K4 marked evaluated, status line now “WP1–WP4 executed (verdicts B, B, B, B(ii)), WP5 [O]”; one conservative sentence added to the `tfpt_3` celestial paragraph; ledger rows CELEST.WP3.01 + CELEST.WP4.01 (new) and CELEST.SEAM.01 updated; v495/v496 Wolfram mirrors added to `tfpt_readouts_extension.wl` (3 + 4 exact checks; count with the next verified engine run — engine still unactivated); website `papers.ts` contract highlight “WP1–WP4 done” + DAG `qft` node extended. Suite now 490 scripts, all green; SEAM.EQUIV.01 stays [O].

**2026-07-21 (II) (PG.06b FULL: the complete NICER Vela archive reduced — the first X-ray  $\nu(t)$  leg of the recovery-comb search, verdict `data_limited`; the old PG.06b “pipeline proven” fold corrected on record; *experiments level only — no verification module, no gate closes, no marker moves*)**

- **experiments/pulsar-glitch-recovery PG.06b FULL executed (`tfpt-pulsar vela-full`).** The full NICER Vela-pulsar (PSR B0833-45) L2 archive is reduced, with the manifest frozen *before* the bulk download (`manifest_full_reduction.json`: all 665 ObsIDs, preregistered 2019/2021/2024 glitch windows, 20-GB abort gate): 665/665 obs downloaded (2.6 GB, raw gitignored)  $\rightarrow$  491 usable  $\rightarrow$  240 per-obs detections  $\rightarrow$  **119 segment-coherent  $\nu(t)$  points** (108 fully coherent), median  $\sigma_\nu \approx 0.6 \mu\text{Hz}$ ; barycentring validated against PINT to  $3.4 \mu\text{s}$  — the *first X-ray  $\nu(t)$  probe* of the search program. Sanity gate PASS:  $\Delta\nu/\nu$  2019 =  $2.39 \times 10^{-6}$  vs  $2.5012 \times 10^{-6}$  published, 2021 =  $1.272 \times 10^{-6}$  vs  $1.26 \times 10^{-6}$ ; the 2024 glitch is not reducible (only  $\sim 150$ -s snapshots, 9.7 ks — documented).
- **Frozen-kernel comb ( $\omega = 2.5827$ ,  $\varepsilon_{\text{pred}} = 0.0173$ ): verdict `data_limited` (range- and amplitude-limited).** Reach 2019: 2.17 comb periods ( $<$  the 2.8-period gate; beats the PG.08 public Vela reach 1.83), 2021: 1.38; the 2019 raw  $p = 0.014$  has the kernel as smallest- $p$  battery member BUT Bonferroni  $\times 10 = 0.14$  and shuffle  $p = 0.395 \Rightarrow$  chance-level by the preregistered escalation rule (audit flag, NOT a candidate); 2021  $p = 0.731$ ;  $\ln \tau$  stack  $p = 0.049$ /shuffle 0.307. Surrogate-calibrated injection at the real sampling and noise:  $\varepsilon_{50} \approx 0.55$  (2019 only),  $\varepsilon_{90}$  never reached below  $\varepsilon = 1$ , 0% power at the predicted  $\varepsilon = 0.0173$  — a weaker amplitude probe than the PG.08 radio residuals, but it independently reproduces the PG.06 range-blindness on real data. Second off-kernel flag:  $\lambda = e$  in the 2021 leg (raw  $p = 0.002$ , Bonferroni 0.018; the same non-kernel periodicity class as PG.08 J0742–2822 — flagged, not claimed). The sub-day early window that would clear the gate is absent from the archive (first usable obs 3.4 d / 7 d post-glitch); next decider:  $<3$ -d post-glitch X-ray observations of a future Vela glitch (eXTP class).
- **Correction on record.** The old PG.06b single-obs “pipeline proven” detection at  $F_0 = 11.19275 \text{ Hz}$  ( $H = 18.4$ ) was a noise peak; the true Vela frequency at that epoch is  $F_0 = 11.1861692 \text{ Hz}$  ( $H = 99.7$ , consistent with the phase-connected PuMA ephemeris). The pipeline remains proven — by the corrected fold. Scorecard row updated to “recovery comb on the full NICER Vela X-ray  $\nu(t)$  (PG.06b FULL)” (status stays `data_limited`; counters unchanged, 120 rows: 49 consistent / 9 tension / 26 null / 33 data-limited / 3 parked); `experiments/README.md` catalog + §1 prose, pulsar README, `experiments/next.txt` synced. No claim; firewall intact.

**2026-07-21** (the two 2026-07-20 prediction-of-record rows promoted to the suite: **v494** —  $\alpha_s$  running +  $\mu$ -distortion branch discriminator machine-checked, ledger rows **COSMO.RUNNING.01/COSMO.MUDIST.01**, prediction surfaces synced; *no gate closes, no marker moves*)

- **v494\_cosmo\_running\_mu\_record.py (23 checks, all green)**. The  $\alpha_s$ -running and  $\mu$ -distortion scorecard rows of 2026-07-20 (IV) get their load-bearing module so the new paper/website statements are machine-backed. [E] leading-order identities (sympy exact):  $\alpha_s = dn_s/d \ln k = -2/N_\star^2 = -r/6$ ,  $\beta_s = -4/N_\star^3$  (NOT  $-8/N_\star^3$ ); LO band  $N_\star \in [50, 60]$ :  $[-8.0, -5.6] \times 10^{-4}$ . [E] exact slow roll on the Einstein-frame Starobinsky potential (third-order Lyth–Riotto + finite-difference cross-check  $< 1\%$ ):  $\alpha_s(51.4) = -7.09 \times 10^{-4}$ ,  $\beta_s(51.4) = -2.70 \times 10^{-5}$ . [C] data: Planck 2018 ( $-0.0045 \pm 0.0067$ )  $+0.57\sigma$ ; record leg P-ACT-LB ( $+0.0062 \pm 0.0052$ )  $-1.33\sigma$  (watch channel: positive central value; kill: robust  $\alpha_s < -5 \times 10^{-3}$  or  $> +3 \times 10^{-3}$  at  $\geq 5\sigma$ ; any robust POSITIVE running kills the plateau branch). [C]  $\mu$ -distortion (Chluba window,  $+21\%$  calibrated on the  $\Lambda$ CDM anchor): sharp branch  $N_\star=51.4$   $\mu = 1.6 \times 10^{-8}$ , profiled record  $N_\star=56.1$   $\mu = 1.94 \times 10^{-8}$ , band  $1.5\text{--}2.3 \times 10^{-8}$ ; branch split  $\Delta\mu = 3.0 \times 10^{-9} \Rightarrow 3\sigma$   $A_s/N_\star$  branch decision at  $\sigma(\mu) = 10^{-9}$  (Voyage-2050 class; PIXIE baseline  $9 \times 10^{-8}$  does not reach; kill: robust  $\mu < 0.9 \times 10^{-8}$  or  $> 4 \times 10^{-8}$ ). [E] comb blindness (honest S15): the frozen log-comb ( $\omega = 2.5827$ ,  $\varepsilon = 0.0173$ ) is structurally invisible in the  $\mu$  window ( $|F(\omega)| = 0.033$ ,  $\delta\mu = 7.4 \times 10^{-12} = 0.007\sigma$  even at Voyage-2050). Fences:  $N_\star$  stays an external reheating input (GATE.NSTAR.01); the frozen band  $[50, 60]$  stays the surface of record (v84/v86). Ledger rows **COSMO.RUNNING.01** and **COSMO.MUDIST.01** added.
- **Prediction surfaces synced (the missing mirror of 2026-07-20 (IV))**. `predictions.txt` gains the two rows ( $\alpha_s$  [C] with the  $-1.33\sigma$  P-ACT-LB pull and the positive-running kill;  $\mu$ -distortion [C] data-limited with the Voyage-2050 branch split); website `lib/predictions.ts` gains the matching cards `alpha-s-running` and `mu-distortion` (confrontation + experiment blocks pointing at `experiments/tfpt-discovery`); the cosmology DAG node lists **v494** and states the running/ $\mu$  readouts.
- **Paper bodies**. `tfpt_1` (Inflation interface) gains the running consistency-relation paragraph ( $\alpha_s = -2/N_\star^2 = -r/6$ , exact slow roll, pulls, positive-running kill) with `\veri{v494_cosmo_running_mu_record.py}`; `tfpt_2` (inflation-pivot keybox) gains the  $\mu$ -distortion branch-decider sentence (sharp  $1.6 \times 10^{-8}$  vs profiled  $2.0 \times 10^{-8}$ ,  $3\sigma$  at Voyage-2050) with the same `\veri`. No marker of any existing claim changes. (v494 is numerical/quadrature, Python-only by the suite convention, flagged in the docstring; GATE.WOLFRAM.01/02 untouched.)

**2026-07-20 (VIII)** (E8 mass-ladder blind-recovery bed **EXECUTED**: verdict validated at the detector level — the Cartan–PF ladder pipeline blindly recovers the published Zamolodchikov–E8 spectra of  $\text{BaCo}_2\text{V}_2\text{O}_8$  (8/8) and  $\text{CoNb}_2\text{O}_6$  (6/8 in-window) and rejects every control class; *experiments level only — confirms Zamolodchikov E8, not TFPT; no verification module, no gate closes, no marker moves*)

- **experiments/e8-ladder-bed/ executed (preregistered hypotheses/e8\_ladder\_bed\_v1.yaml)**. The blind ladder detector (assignment DP + scale fit + 5000-ladder Monte-Carlo null; the same detector class TFPT uses on search surfaces, e.g. `hfqpo-ladder`) on the only public *known-answer* mass-ladder dataset:  $\text{BaCo}_2\text{V}_2\text{O}_8$  (Zou+ PRL **127**, 077201; INS at the 4.7 T 1D QCP, Fig. 3a extraction QA-validated, 17 peaks blind) recovers **8/8 rungs**,  $\chi^2/\text{dof} = 0.35$ , MC  $p = 0.0056$  (singles-only run  $p = 0.0004$ ;  $\sigma$ -variant robust);  $\text{CoNb}_2\text{O}_6$  (Amelin+ PRB **102**, 104431; verbatim THz values) recovers **6/8 in-window rungs**,  $\chi^2/\text{dof} = 0.52$ ,  $p = 0.0038$  ( $m_7/m_8$  lie outside the 0.6-THz low-pass window; the broad 0.43-THz ( $m_1+m_2$ )-continuum onset is correctly left unassigned). Cartan–PF self-test: the sorted PF eigenvector of  $2I - A(E_8)$  equals the Zamolodchikov ladder to  $1.5 \times 10^{-15}$  ( $m_2/m_1 = \varphi$  to  $2.2 \times 10^{-16}$ ; lowest Cartan

eigenvalue  $2 - 2\cos(\pi/30)$  to  $4.5 \times 10^{-14}$ ).

- **Control battery — every class rejected.** Scramble 0.5%/1.5% false-positive rate (200 log-gap permutations, near-duplicate accounting); offset  $\pm\{0.3, 0.5\}m_1$ : 0/8 detections; jitter dose-response  $\leq 5\%$  at the destructive 30% level; wrong ladders on the real data (harmonic 1..8,  $A_3$ -PF,  $D_8$ -PF, Airy confined-spinon) all beaten by E8; the synthetic Airy bed (the true zero-field physics of these compounds) yields **0** valid rungs. The one disclosed v1.1 criteria amendment (jitter as dose-response, scramble near-duplicates,  $\chi^2$ -gated wrong-ladder comparator) is fully documented in the hypotheses YAML with all v1 numbers preserved; the recoveries themselves were never touched.
- **Integration (firewall intact).** Scorecard row “E8 mass-ladder blind recovery bed” moves parked  $\rightarrow$  consistent (stage not\_applicable, tier analog\_positive\_control, leakage detector\_control — the executed-analog basket of qc-recovery-kernel and qgeo-eit-soff); counters 120 rows: 49 consistent / 3 parked; experiments/README.md catalog row + stats (generator); website predictions.ts EXPERIMENTS\_AUDIT counters; experiments/next.txt. This validates the ladder-detector pipeline — it confirms Zamolodchikov E8, *not* the TFPT axioms: no [E], no \veri, no ledger row.

**2026-07-20 (VII) (v493 — WP2 of CELEST.SEAM.01 executed: the clock-invariant deformation  $XY=Z^4+a_0$  is a PURE SCALE with the  $\tau=i$  pillowcase frozen, the  $\mu_4$  clock IS the Picard–Lefschetz/Coxeter monodromy of the seam deformation, and the BHS deformed-algebra pattern transfers  $\mathbb{Z}_2 \rightarrow \mathbb{Z}_4$  with no sector leak; verdict B, contract updated to WP2 done, no gate closes, no marker moves, SEAM.EQUIV.01 and the v216 residual stay [O])**

- **v493\_celestial\_wp2\_clock\_deformation.py (CELEST.WP2.01).** WP2 of the celestial contract, sympy exact (47 checks), promoted from the experiments/tfpt-discovery/probe celestial\_seam\_wp2\_clock\_deformation\_probe.py. *Geometry:*  $P(iZ)=P(Z)$  forces  $a_3=a_2=a_1=0$  sharply and two-sided ( $\prod_k (Z - i^k w) = Z^4 - w^4$ );  $XY=Z^4+a_0$  is smooth iff  $a_0 \neq 0$  (disc =  $256 a_0^3$ ); the branch points  $Z^4 = -a_0$  are one free  $\mu_4$  orbit in *every* fibre with cross-ratio 2; the binary-quartic invariants of  $w^2=Z^4+a_0$  are  $I=12a_0$ ,  $J=0$  *identically*, so  $j=1728$  and the  $\tau=i$  pillowcase shape (v168/v214) is *frozen* for all  $a_0$  —  $a_0$  is a pure scale, no shape modulus survives clock-invariance. *Periods/monodromy:*  $\Pi = t(i-1)(1, i, i^2)$ , all three vanishing spheres carry the same volume  $\sqrt{2}t$  (lockstep); the clock acts on  $H_2$  as a *Coxeter element* of  $W(A_3)$  (char  $x^3+x^2+x+1$ , eigenvalues  $\{i, -1, -i\}$  = the  $A_3$  exponents,  $h(A_3)=4=|\mu_4|=N_{\text{fam}}+1$ ), fixes no cycle ( $\det(A-1)=-4$ ), and *is* the Picard–Lefschetz monodromy of the family ( $a_0 \rightarrow e^{2\pi i} a_0$  gives the 4-cycle); the surviving direction is the weight-1  $\chi_1$ -Fourier diagonal  $(1, i, i^2)$ , *not* the naive invariant diagonal; the versal weights  $(a_3, a_2, a_1, a_0) = (1, 2, 3, 0)$  are the regular  $\mu_4$  representation. *Algebra (BHS  $\mathbb{Z}_2 \rightarrow \mathbb{Z}_4$ , arXiv:2305.09451):* mode dictionary  $\{w[m, n] : m \equiv n \pmod{4}\}$ ; the fibre bracket  $-4\text{Nambu}(XY - Z^4 - a_0)$  anchored at the  $a_0=0$  orbifold ( $\{X, Y\}=16Z^3$ , Jacobi + Casimir exact, uniform clock-degree  $-1$ ); S-algebra corrections exactly *linear* in  $a_0$  at the  $\mathbb{Z}_4$  wrap, conserving the  $\mu_4$  grade — no sector leak, the v492 equivariant sector deforms consistently;  $a_0 \leftrightarrow$  BHS’s weight-0  $c^2$  slot; matrix factorisations deform along all orbit splittings, the clock fixing exactly the antipodal 2+2 sector (= the EH/ $\mathbb{Z}_2$  middle node,  $4=2 \times 2$ ). *Negative controls:*  $a_1 Z \Rightarrow j=0$  (hexagonal); pure  $a_2 Z^2$  stays singular ( $A_1$ );  $a_2 \neq 0 \Rightarrow j=1556068/81 \neq 1728$ ; explicit grading leak — doubly sharp. *Verdict B* via three named [C] identifications ( $-4\text{Nambu}$  = the  $k=4$  CCA bracket; period = root difference; seam-scale reading via  $\text{clock}^2=\text{deck}$ ); the v216 residual is *typed*, not moved (given the order-4 clock the square is automatic — a relocation of the same input, no new derivation).
- **Contract updated to WP2 done.** tfpt\_research\_contracts CELEST.SEAM.01 section: WP2 bullet “finite, next”  $\rightarrow$  *done (verdict B)* with the two sharpenings (clock = Coxeter/Picard–Lefschetz monodromy; surviving direction =  $\chi_1$ -diagonal,  $a_0$  pure scale); kill-test paragraph K1

“survived by WP1 and WP2”; status line CELEST.WP2.01 [E]/[C] executed, WP3–WP5 stay [O]. Ledger rows CELEST.WP2.01 (new) + CELEST.SEAM.01 (WP2 done) + GATE.WOLFRAM.02 (four exact v493 mirrors added to `tfpt_readouts_extension.wl`, counted with the next verified engine run — engine unactivated, same convention as v491); registry + `run_all` (487 modules, suite green); website mirrors (`papers.ts` contract section, generated script index). SEAM.EQUIV.01 stays [O]; no marker moves anywhere. Exact (sympy).

**2026-07-20 (VI) (HFQPO “third tooth”: the preregistered archival RXTE PCA scan has RUN — a well-powered null with limits; scorecard row S1.e data\_limited→null, no marker moves)**

- **H3 stage 2 executed** (`experiments/hfqpo-ladder/results/archival/`). The preregistered archival scan (`hypotheses/hfqpo_v1.yaml`, `archival_design`) ran end-to-end on public HEASARC data: 77 ObsIDs / 4.42 GB across all four 3:2-pair sources (GRO J1655–40, XTE J1550–564, GRS 1915+105, H1743–322). Sanity gate PASS (11/12 published pair lines reproduced, e.g. 450.3 Hz at 5.37% rms  $4.4\sigma$ , 280.1 Hz at 3.71%  $9.9\sigma$ ); injection calibration PASS in all four sources (90%-recovery sensitivities 5.11/2.68/0.89/1.65% rms at the tooth frequencies).
- **Blind-scan verdict: NULL with limits.** Neither the geometric tooth  $\nu_3 = 1.5\nu_u$  (661.5/414/252/363 Hz) nor the integer control line  $4\nu_0$  (588/368/224/322.7 Hz) is detected in any source ( $\sim 0\sigma$  single-trial; trials correction  $N=8$  as preregistered).  $3\sigma$  rms upper limits (tooth, primary bands): 3.06/1.61/0.53/1.26% — in every source at or below the strength of the detected upper pair line. The ladder reading gains no support but is not killed (the `harmonic_hit` branch did not fire either; the anti-kernel record 92=184/2, 34/68 stands); GR parametric resonance stays the favorite. Caveats: GRS 1915+105 scanned in a proxy state (the 113/168 Hz epochs were never published at ObsID level); H1743’s program is 80146 (YAML said 80138); GRO hard-band sensitivity 5.1%.
- **Integration.** Scorecard row S1.e (hfqpo-ladder) `data_limited→null` (120 rows: 26 null / 33 `data_limited`), next decider eXTP-class sensitivity below the 0.5–3.1% rms limits; HFQPO search-target paragraph updated in `tfpt_horizon_readouts` (+ the `papers.ts` mirror); website `predictions.ts` (hfqpo-tooth entry + EXPERIMENTS\_AUDIT counters); `predictions.txt`; `experiments/README.md`; `experiments/next.txt`. No verification/ module, no ledger row, no marker move (the search target stays [O]).

**2026-07-20 (V) (v492 + the new research contract CELEST.SEAM.01 — the celestial-holographic route: the  $E_8$   $\mu_4$ -glue is the flat  $\mathbb{Z}_4$  monodromy of the equivariant celestial chiral algebra on the  $A_3$  ALE space, with the exact correction  $\text{clock}^2=\text{deck}$  (spin bridge  $\mathbb{Z}_8$ ,  $8=2|\mu_4|$ ); verdict B, no gate closes, no marker moves, SEAM.EQUIV.01 stays [O])**

- **v492\_celestial\_z4\_orbifold.py (CELEST.WP1.01).** WP1 of the new contract, sympy exact (36 checks): the v128  $\mu_4$ -glue grading of  $E_8$  ( $240 = 52+64+60+64$ , additive on all 6720 root pairs) is *inner* —  $h = (2, 2, 2, 2, 2; 0^4)$  reads the glue class mod 4 on all 240 roots, so the glue is  $\text{Ad}(\exp(2\pi i h/4))$ , an order-4 inner torus automorphism = a flat  $\mathbb{Z}_4$  monodromy in the Kronheimer–Nakajima sense; the  $A_3$ -side detector  $h' = (0^5; 1, 1, 1, -3)$  reads the same class = the diagonal (1, 1) of the discriminant group  $\mathbb{Z}_4 \times \mathbb{Z}_4$  (the v92/v125 Lagrangian glue).  $\mathbb{C}^2/\mathbb{Z}_4$  is verified as the  $A_3$  singularity  $XY=Z^4$  (Hilbert series  $(1+t^4)/((1-t^2)(1-t^4))$ , Molien exact); the glue-equivariant SDYM( $E_8$ ) sector *closes* with graded dimensions  $60(d+1)/64(d+1)$  — possible only because  $\dim \mathfrak{g}_j = (60, 64, 60, 64)$  — with zero modes = the carrier  $D_5 \oplus A_3 + \text{Cartan} = 60$  and density  $1/4 = 1/|\mathbb{Z}_4|$ ; the four glue-sector characters sum exactly to  $\chi_{(E_8)_1} = 1+248q+4124q^2+34752q^3$  (v377), with sector weights = the discriminant form  $(5x^2+3y^2)/8$  and integer glue-diagonal  $h = (0, 1, 1, 1)$  (= locality of the  $(E_8)_1$  extension). *Critical correction*

(*verdict B*): the  $A_3$  deck acts on the celestial sphere as  $z \mapsto -z$  (order 2), *not* as the order-4 clock; the clock is the  $U(2)$  phase  $\text{diag}(i, 1)$  (normalising the deck,  $\det = i$  rotating the holomorphic symplectic form by the  $\mu_4$  generator), with the exact spin bridge (spin clock)<sup>2</sup> = deck ( $\mathbb{Z}_8$ ,  $8 = 2|\mu_4|$ , the  $c_3 = 1/(8\pi)$  winding integer). Clock-invariance selects the 1-parameter  $A_3$  deformation  $XY=Z^4+a_0$  whose four branch points are one  $\mu_4$  orbit with cross-ratio 2 (v168/v214 pillowcase). Negative controls: the false action  $\text{diag}(i, i)$  killed three ways, the false glue breaks additivity on 3112/6720 pairs, rigidity =  $\text{Aut}(\mathbb{Z}_4)$ .

- **New research contract CELEST.SEAM.01** in `tfpt_research_contracts` (own section, the fourth contract alongside `Uwall/Gmetric/CONTRACT.F.01`): hypothesis (seam = glue-equivariant  $\mathbb{Z}_4$ -orbifold sector of the CCA on the  $A_3$  ALE; twistorial bulk = the *self-dual* sector only), typing/non-circularity (the continuum  $(E_8)_1$  existence — the SEAM.EQUIV.01 target — is *not* an admissible input), work packages WP1 (done, verdict B) through WP5 (second continuum route, Costello–Li grade), pre-registered kill tests K1–K4 (K1 survived by WP1), and an honest success estimate (full success low, partial findings high). Ledger rows CELEST.WP1.01 + CELEST.SEAM.01; website mirror (`papers.ts` section, script index). SEAM.EQUIV.01 stays [O]; no marker moves. Exact (sympy).

**2026-07-20 (IV) (evidence scorecard: five new typed rows — the  $\alpha_s$  running prediction of record, the  $\mu$ -distortion  $A_s/N_\star$  branch discriminator, and three parked analog beds (bosonic-E8 thermal Hall, E8 mass-ladder recovery, Efimov DSI tuning); *experiments level only — no verification module, no gate closes, no marker moves*)**

- **$\alpha_s$  running (prediction of record, consistent).** The Starobinsky branch fixes the running of the spectral index:  $\alpha_s = -2/N_\star^2 = -7.6 \times 10^{-4}$  at  $N_\star=51.4$  (exact slow roll  $-7.09 \times 10^{-4}$ ; band  $N_\star \in [50, 60]$ :  $[-8.0, -5.6] \times 10^{-4}$ ) and  $\beta_s = -4/N_\star^3$  (leading order; NOT  $-8/N_\star^3$ ) — the identity  $\alpha_s = -r/6$  holds only at leading order ( $\sim 7\%$  off exact). Data: Planck 2018  $-0.0045 \pm 0.0067$  ( $+0.57\sigma$ ), P-ACT  $+0.0060 \pm 0.0055$  ( $-1.22\sigma$ ), record leg P-ACT-LB  $+0.0062 \pm 0.0052$  ( $-1.33\sigma$ , Calabrese et al. 2025 JCAP);  $\beta_s$  (Planck run+runrun)  $0.010 \pm 0.013$  ( $-0.77\sigma$ ). Forecast: LiteBIRD+CMB-S4  $\sigma(\alpha_s) \approx 4.9 \times 10^{-3}$  ( $0.14\sigma$ ); CMB-S4+DESI+Euclid statistical  $\sim 3.3 \times 10^{-4}$  ( $2.2\sigma$ , systematics-limited). Same `N_star_reheating` independence group as the  $n_s/r/A_s$  rows (anti double-counting: never an independent hit); kill: robust  $\alpha_s < -5 \times 10^{-3}$  or  $> +3 \times 10^{-3}$  at  $\geq 5\sigma$ ; any robust POSITIVE running kills the plateau branch. Probe: `experiments/tfpt-discovery/alpha_s_running_probe.py`.
- **$\mu$ -distortion (prediction of record, data-limited).** Chluba window integral (calibrated +20% to the full computation): sharp branch  $N_\star=51.4$  ( $A_s=1.76 \times 10^{-9}$ )  $\mu \approx 1.6 \times 10^{-8}$ , profiled record  $N_\star=56.1$  ( $A_s=2.10 \times 10^{-9}$ )  $\mu \approx 2.0 \times 10^{-8}$ , band  $1.5\text{--}2.3 \times 10^{-8}$ ; the 23% branch split ( $\Delta\mu = 3.0 \times 10^{-9}$ ) gives a  $3\sigma$   $A_s/N_\star$  branch decision at  $\sigma(\mu) = 10^{-9}$  (Voyage-2050 class; the PIXIE baseline  $|\mu| < 9 \times 10^{-8}$  does NOT reach — factor 4–7 above the band; FIRAS  $|\mu| < 9 \times 10^{-5}$ ). Honest S15 typing: the frozen log-comb ( $\omega = 2.5827$ ,  $\varepsilon = 0.0173$ ) is STRUCTURALLY BLIND in the  $\mu$  window — window attenuation  $|F(\omega)| = 0.033$ ,  $\delta\mu = 7.4 \times 10^{-12} = 0.007\sigma$  even at Voyage 2050 — the theory itself predicts where it is invisible, so NO separate comb-search row is opened. Shares the `Nstar_branch` alternative group with the two  $A_s$  rows (one branch question, several readings). Kill: robust  $\mu < 0.9 \times 10^{-8}$  or  $> 4 \times 10^{-8}$ . Probe: `experiments/tfpt-discovery/mu_distortion_probe.py`.
- **Three parked analog beds (parked\_analog; honest typing: they validate E8 edge/spectrum physics or the comb DETECTOR, never the TFPT axioms).** (i) Bosonic E8 phase / thermal Hall:  $\kappa_{xy}/T = 8 \times (\pi^2 k_B^2/3h)$  with no charge Hall (seam edge  $c_-=8$ , v450/v451/v489); no engineered material yet ( $\alpha\text{-RuCl}_3$   $c_-=1/2$  contested: ultraclean half-quantized plateau, npj QM 2025, vs phonon-dominated  $\kappa_{xy}$ , PRX 12, 021025). (ii) E8 mass-ladder validation bed ( $\text{CoNb}_2\text{O}_6/\text{BaCo}_2\text{V}_2\text{O}_8$ ): all 8 one-particle masses published (Zou et

al. 2021 PRL 127, 077201, NMR+INS; Amelin et al. 2020 PRB 102, 104431, THz  $m_1$ – $m_6$ ; Coldea et al. 2010 Science  $m_2/m_1=\varphi$ ) — blind Cartan-PF ladder recovery on the published spectra (8/8 masses + rejection of scrambled controls) as detector validation; confirms Zamolodchikov E8, not TFPT. (iii) Efimov DSI tuning:  $s_0 = \pi/(6 \ln(3/2)) = 1.2914 \Leftrightarrow \lambda = (3/2)^6 = 11.391$  at  $M/m = 8.67$  (2 resonant pairs); best candidate  $^{52}\text{Cr}_2$ – $^6\text{Li}$  ( $M/m=8.635$ ,  $s_0=1.2890$ ,  $\lambda=11.44$ ,  $+0.44\%$  vs  $(3/2)^6$ , gap deviation  $+0.18\%$ ); alternatives  $^{39}\text{K}_2$ / $^{41}\text{K}_2$ – $^6\text{Li}$  ( $\lambda = 11.59/11.20$ , needs double Feshbach control),  $^{176}\text{Yb}_2$ – $^{23}\text{Na}$  ( $+5.9\%$ ); solver validated on the Naidon–Endo anchors (homonuclear 22.69,  $\text{Cs}_2\text{Li}$  4.877); first end-to-end HARDWARE injection of the comb detector at tunable  $\lambda \approx 11.39$ ; kill condition honestly “none on TFPT axioms — detector validation only”. Probe: `experiments/tfpt-discovery/efimov_s0_massratio_probe.py`.

- Scorecard regenerated (`experiments/build_evidence_scorecard.py`): 120 rows — 48 consistent, 9 tension, 25 null, 34 data-limited, 4 parked; website mirror (`EXPERIMENTS_AUDIT` in `predictions.ts`) synchronised. Experiments level only; the verification suite, ledger and papers are untouched.

**2026-07-20 (III) (v491 —  $g_{\text{car}}=5$  as the partition COROLLARY of the four marks: 4 into exactly 3 positive parts is uniquely  $\{1, 1, 2\}$ , sharpening v53; the residual is the weight-typing postulate — AX.P2.01 stays a declared axiom; *no gate closes, no marker moves*)**

- `v491_p2_partition_corollary.py` (P2.PARTITION.01). Sympy-exact: a multiset of exactly 3 POSITIVE INTEGERS summing to 4 is UNIQUE —  $\{1, 1, 2\}$  — so  $e = (e_1, e_2, e_3) = (4, 5, 2)$  and hence  $g_{\text{car}} = e_2 = 5$ ,  $|\mathbb{Z}_2| = e_3 = 2$ ,  $N_{\text{fam}} = e_2 - e_3 = 3$  are *corollaries*, not inputs (consistency fixed point  $(2^4 - 1)/5 = 3 = \text{rank } H^1$ ). [E] negative controls, each assumption load-bearing: drop positivity  $\rightarrow$  4 solutions,  $e_2$  ambiguous  $\{0, 3, 4, 5\}$ ; sum 3  $\rightarrow e_2=3$ ; sum 5  $\rightarrow$  not unique; 2/4 parts  $\rightarrow e_2 \in \{3, 4\}/\{6\}$ ; Mehta–Seshadri RATIONAL weights  $\rightarrow e_2 = \frac{16}{3}$  resp.  $\frac{44}{9} \neq 5$  (integrality = the Birkhoff–Grothendieck splitting-type typing, not MS weights). [E] the v53 sharpening: v53 pins  $(1, 1, 2)$  with ALL THREE targets  $(4, 5, 2)$ ; the partition needs only  $e_1=4$ . [E] sum rule:  $4 \times \frac{N-1}{2} = 4$  iff  $N=3$ ; the v115 route  $|a|_1 = |\mu_4| \text{tr}(A_0) = 4$  agrees. Independence chain: #marks = 4 from v216 (P1-side, no  $g_{\text{car}}$  input); rank  $H^1 = 3$  pure topology (v137/v228);  $|a|_1 = 4$   $g_{\text{car}}$ -free (v115). [C] honest residual: P2 is NOT made axiom-free — the weight-TYPING postulate remains (integrality/positivity  $h^0=0/\text{sum}$  “given (U)”); AX.P2.01 stays a declared axiom, and the  $N_{\text{fam}}$  direction reversal vs `tfpt_constants` is typed as an over-determination check (never double-counted). Integrated in `introduction` (anchor half-forced paragraph), `tfpt_1` ((5, 3) keybox), `tfpt_2` (H2 winbox) and `origin_theory` ((5, 3) keybox); ledger row P2.PARTITION.01; three exact Wolfram mirrors added to `tfpt_readouts_extension.wl` (counted with the next verified engine run, engine currently unactivated); website mirror (DAG node P2). Exact (sympy).

**2026-07-20 (II) (v490 — the  $T^2$  spin-structure parity census:  $\mu_{\text{pre}}=4$  witnessed on the lattice, the 16-fold-way bridge  $\theta_v=1$  on measured numbers, and the honest TEE non-discriminator fence; *no gate closes, no marker moves*)**

- `v490_seam_parity_census.py` (SEAM.MU.01). The  $T^2$  ground-state fermion parity of the v367  $p+ip$  collar, resolved per spin structure ( $AA, AP, PA, PP$ ) and determined TWICE — Read–Green TRIM product AND real-space Majorana Pfaffian (Schur), relative to the trivial  $M=3$  phase, with the twisted real-space spectra matching the Bloch bands to  $9 \times 10^{-15}$ . [E] per layer  $(+, +, +, -)$  (the  $\nu=1$  Ising signature, TRIM  $\equiv$  Pfaffian in every sector); [E] the 16-layer carrier has parity<sup>16</sup> = + in ALL FOUR sectors  $\Rightarrow$  gauged torus GSD = 4 =  $\mu(\text{SO}(16)_1) = |\det \text{Cartan}(D_8)|$  — the lattice ground-state witness of “ $\mu_{\text{pre}}=4$  before condensation” (step 1 of the v378 KLM chain); [E] the 16-fold-way bridge on MEASURED numbers:  $\theta_v = e^{2\pi i \nu/16} = 1$  exactly at  $\nu = 2c_- = 16$  (v489), rivals  $\nu = 1, 2, 8$  not holomorphically condensable; [E]

honest fence: the fermionic Kitaev–Preskill TEE is identically zero for BOTH  $M=1$  and  $M=3$  ( $\gamma_f=0.000001/0.000000$ ) — it does NOT discriminate  $(E_8)_1$  vs  $SO(16)_1$ . [C] the KLM step  $\mu : 4 \rightarrow 4/2^2 = 1$  stays the cited theorem (the condensed model is interacting); the numerics pin only the premise package  $\{c_-=8, \mu_{\text{pre}}=4, \theta_v=1\}$ . [O] SEAM.EQUIV.01 stays [O] (continuum existence, MMST v336). Integrated in `tfpt_research_contracts` (new S3-stack paragraph); ledger row SEAM.MU.01; website mirror (DAG node `qft`). Python (numpy + scipy Schur + sympy exact  $\theta_v$ ).

**2026-07-20 (v489 — the chiral central charge  $c_-=8$  measured DIRECTLY from the bulk ground state via the Kim–Shehab–Kim modular commutator: the first ground-state witness for premise (i) of SEAM.EQUIV.01, independent of the Chern route (v367) and the KLM count (v378); *no gate closes, no marker moves*)**

- **v489\_seam\_modular\_commutator.py (SEAM.CCC.01).** The Kim–Shehab–Kim modular commutator  $J(A, B, C) = i \text{Tr}(\rho_{ABC}[K_{AB}, K_{BC}]) = (\pi/3) c_-$  (PRL 128 (2022); arXiv:2110.06932/2110.10400) is evaluated on the EXACT BdG ground state of the v367  $p+ip$  seam collar model, entirely from the Peschel Majorana covariance ( $W_X = -2 \text{artanh } \Gamma_X$ ,  $J = \frac{1}{4} \text{Tr}([W_{AB}, W_{BC}] \Gamma_{ABC})$ ), on a disk about the torus centre cut into three  $120^\circ$  wedges. [E] ground-state purity  $\text{eig}(i\Gamma) = \pm 1$  to  $6 \times 10^{-15}$ ; [E] per-layer  $c_- = 0.499998$  at  $L=32$  (convergence  $0.499793$  at  $L=16 \rightarrow 0.500000$  at  $L=48$ , deviation  $-1.8 \times 10^{-7}$ ); [E] controls: trivial  $M=3 \rightarrow 0.000000$ , orientation flip  $M=-1 \rightarrow -0.499986$ ; [E] layer additivity exact ( $1.4 \times 10^{-14}$ ); [E] the 16-layer carrier lands on  $c_-=8=g_{\text{car}}+N_{\text{fam}}$  (direct  $7.9967$  at  $L=16$ , additivity route  $8.0000$ ); [E] clip invariance  $5 \times 10^{-8}$ . The FIRST ground-state witness for premise (i) of SEAM.EQUIV.01 ( $c_-=8$ ) that is INDEPENDENT of both the Chern-number route (v367) and the counting route (v378) — read from entanglement data alone. [O] honest fence: does NOT close SEAM.EQUIV.01; the continuum scaling limit (MMST, v336) and the cited KLM condensation step are untouched. Integrated in `tfpt_research_contracts` (new S3-stack paragraph) and `tfpt_1` (the SEAM.EQUIV.01 package paragraph); ledger row SEAM.CCC.01; website mirrors (DAG node `qft`, papers/glossary/FAQ inventories). Numerical (numpy), Python-only by nature.

**2026-07-19 (editorial — positioning section *TFPT in the light of algorithmic physics: Zuse–Schmidhuber–Wolfram* added to introduction + website mirror; *no result, no status change*)**

- **introduction.tex — new section “TFPT in the light of algorithmic physics”.** Positions the compiler against the Zuse–Schmidhuber–Wolfram algorithmic-physics lineage: three shared instincts (minimum description length as the selection principle, deterministic resource-bounded generation, compressibility as the aesthetic) and three deliberate divergences (one *forced* program vs. an ensemble of all universes; an algebraic Lie/lattice/VOA description language vs. Turing bit-strings; and falsifiability — dated, kill-tested predictions — as the load-bearing difference). Explicitly typed as *positioning*: it introduces no formula and moves no status marker; its only machine-checked hooks are already in the suite (the grammar-internal null model and the icosahedral/McKay bedrock).
- **Website mirror.** A matching curated section was added to the introduction entry in `website/lib/papers.ts`.

**2026-07-15 (III) (v488 — the 3/5 Clebsch door ENUMERATED AND CLOSED: no minimal Pati–Salam/ $SO(10)$  Majorana channel can produce 5/3 — the v482 escape is decided dead, the v481 window candidate stands as the honest endpoint; a clean negative result, no gate closes)**

- **v488\_majorana\_clebsch\_door.py (FLAV.NUSCALE.04).** v482 named ONE escape for the integer seesaw rung  $M_R = c_3^3 \bar{M}_{PI}$ : a third-generation Majorana factor  $r \approx 1.6696$ , 0.18% from  $g_{car}/N_{fam} = 5/3$  — declined for lack of mechanism. The mechanism search is now EXECUTED: [E] NO RENORMALISABLE CHANNEL —  $\text{sym}^2(\mathbf{16}) = 136 = \mathbf{10} + \mathbf{126}$  and the  $\mathbf{126}$  is NOT in the  $E_8$  hull  $\{1, 10, 16, 45\}$  (v247), so  $M_R$  only via the  $d=5$  operator  $(\mathbf{16}_F \mathbf{16}_H)^2/\Lambda$ ; channels  $\mathbf{16} \times \mathbf{16} = \mathbf{1} + \mathbf{45} + \mathbf{210}$  with only singlet +  $\mathbf{45}$  admissible. [E] CHARGE TABLE:  $(B-L, T_{3R}, Y)(\nu^c) = (+1, -\frac{1}{2}, 0)$ ,  $\text{Tr}_{16} Y^2 = \frac{10}{3}$  (v485). [E] THE  $\{2, 3\}$ -SMOOTH THEOREM (the door closes): every nonzero  $\nu^c$  channel weight  $\{1, \frac{1}{4}, 1, -\frac{1}{2}, \frac{3}{8}\}$  is  $\{2, 3\}$ -smooth, and the multiplicative closure of  $\{2, 3\}$ -smooth rationals never contains  $\frac{5}{3}$  (no factor 5); the UNIQUE natural 5 of the embedding,  $k_Y = \frac{5}{3}$  (Ginsparg, v470), is a full-multiplet trace whose direction has  $Y(\nu^c) = 0$  EXACTLY — structurally decoupled from the Majorana operator. [E] GENERATION-BLINDNESS: Clebsches are family-space scalars;  $\text{diag}(1, 1, \frac{3}{5})$  cannot arise from group theory by construction. [O] compiler-flavor inventory typed and declined: exact  $\frac{3}{5}$ 's only in wrong slots ( $K$  lepton gen2/gen1;  $R + L$  lepton gen3/gen2; never gen3 vs a degenerate base). NET: the rung+5/3 rescue is DEAD as a mechanism candidate; the v481 one-parameter window candidate ( $M_R$  free inside the PS window, the  $\Sigma m_\nu = 0.059$  eV floor as kill test) stands. Contract provenance: `experiments/theory-contracts/clebsch01_majorana_35_door.py`. Integrated in `tfpt_2_standard_model` (new paragraph closing the seesaw arc); ledger row FLAV.NUSCALE.04 + update note on .03; Python-only (Fraction/sympy exact), mirrorable, deferred to the next verified engine run.

**2026-07-15 (II) (v487 — the lazy split FORCED by the resummed clock: the walk's one-step spectrum is the complete v124 clock ladder below the wall, deck parity = rung index, and the two comb tones are one bend apart — the v486 [O] “split selection” closes onto established structure; no gate closes, no marker moves)**

- **v487\_transfer\_clock\_rungs.py (HYP.REWRITE.03).** v486 left the split selection open (“ $(\frac{1}{2}, \frac{1}{6})$  vs uniform  $(\frac{1}{3}, \frac{1}{3})$  is selected by the physical spectrum, not yet derived”; named candidate: the vague “sheet-symmetric choice”). Closed: [E] RUNG READING — the lazy rule's one-step spectrum is EXACTLY the complete resummed-clock ladder below the wall (v124:  $\lambda_n^{1/p_2} = 1 - n/N_{fam}$ , pole at  $n=N_{fam}$ ):  $\text{eig}(M) = \{1, \frac{2}{3}, \frac{1}{3}\} = \{1 - n/3 : n = 0, 1, 2\}$ , while v327's uniform rule puts its second pair mode ON the wall ( $0 = 1 - 3/3$ ) — the v486 degeneracy re-read as “odd channel collapsed to the wall”. [E] DECK PARITY = RUNG INDEX:  $[B, \sigma] = 0$ ; the  $\sigma$ -even pair mode carries rung  $n=1$  (survival  $\frac{2}{3}$ ), the  $\sigma$ -odd mode rung  $n=2$  (survival  $\frac{1}{3}$ ) — each clock step flips the deck parity; this is the structural home of the v486 parity assignment ( $\omega_1$  even-channel,  $\omega_2$  odd-channel) that `experiments/frb-parity-comb` tests. [E] FORCING: demanding the faithful ladder (no rung missing, no physical mode on the wall — definitional for a full-spectrum generator) +  $\mathbb{Z}_2$  equivariance + non-negative rates forces UNIQUELY both the split  $(\frac{1}{2}, \frac{1}{6})$  AND the rung assignment even $\rightarrow$ 1, odd $\rightarrow$ 2 (the swapped assignment needs  $h = -\frac{1}{6} < 0$ ); the “sheet-symmetric choice” candidate is superseded by this proof. [E] BEND COROLLARY:  $\omega_1/\omega_2 = \text{rate}(2)/\text{rate}(1) = \log_{3/2} 3 = 2.70951\dots$  — the parity-comb tone ratio IS the v107/v124 bend of the pulsar-clock search. [O] what remains open is exactly what was open before v486: the arity  $\{2, 3\}$  (anchor input) and the clock's semiclassical derivation (R1); no independent choice remains at the walk level. Contract provenance: `experiments/theory-contracts/lazy01_clock_rung_forcing.py`. Integrated in `origin_theory` (new paragraph after the v486 block); ledger row HYP.REWRITE.03 + update note on .02; Python-only (sympy exact), mirrorable, deferred to the next verified engine run.

**2026-07-15 (v486 — generator hunt: the FULL transfer spectrum  $\{1, (2/3)^6, (1/3)^6\}$  from ONE forced local rule — the unique lazy  $\mathbb{Z}_2$ -pair walk (stay, hop, leak) =  $(\frac{1}{2}, \frac{1}{6}, \frac{1}{3})$ ; the previously generator-less  $(1/3)^6$  mode gets its origin; v327's uniform rule superseded as full-spectrum generator — *no gate closes, no marker moves*)**

- **v486\_transfer\_full\_rule.py (HYP.REWRITE.02).** The verified seam transfer spectrum is  $\{1, (2/3)^6, (1/3)^6\}$  (v54/v221), but the only local generator in the suite (v327) produces  $\{1, \frac{2}{3}, 0\}$  — its zero mode persists under every power, so no iterate reaches the third mode. [E] UNIQUE-NESS: for the symmetric 3-channel rule  $M(s, h) = [[1, 0, 0], [0, s, h], [0, h, s]]$  (one absorbing family channel +  $\mathbb{Z}_2$  pair) the survival spectrum is  $\{s+h, s-h\}$ ; demanding the PHYSICAL pair  $\{\frac{2}{3}, \frac{1}{3}\}$  forces UNIQUELY (stay, hop, leak) =  $(\frac{1}{2}, \frac{1}{6}, \frac{1}{3}) = (1/|\mathbb{Z}_2|, 1/(|\mathbb{Z}_2|N_{\text{fam}}), 1/N_{\text{fam}})$  — the lazy  $\mathbb{Z}_2$ -pair walk, every rate an atom expression (honest direction: spectrum  $\rightarrow$  rates). [E] SIX-HAND GENERATION: over the order-6 =  $2N_{\text{fam}} = |R^+(A_3)|$  hand,  $\text{eig}(B^6) = \{\frac{64}{729}, \frac{1}{729}\}$  exactly — both decay gaps now have one recursive generator:  $6 \ln \frac{3}{2}$  (recovery, v76) and  $6 \ln 3$  (subdominant, previously generator-less). [E] SUPERSESSION: the uniform rule's zero mode is power-persistent; its  $\lambda_2$  arity story ( $\frac{2}{3} = |\mathbb{Z}_2|/N_{\text{fam}}$ ) is unchanged. [O] the arity  $\{2, 3\}$  stays the anchor input; the lazy-vs-uniform split is SELECTED by the physical spectrum, not yet first-principles-derived (named candidate: stay =  $1/|\mathbb{Z}_2|$  as the sheet-symmetric choice). Discovery provenance: `experiments/tfpt-discovery/generator_hunt_full_transfer_rule.py` (plus a typed anchor-bilinear coverage inventory — known hits  $a^T K a = 41$ , Koide 53, and the recorded-only  $1^T Q a = 16$  — and the lepton-exponent observation  $(5, 3, 2) = K\text{-row} = \text{atom triple}$ ; no unforced claims, v354/v355 discipline). Integrated in `origin_theory` (new paragraph after the v327 rewrite block); ledger rows HYP.REWRITE.02 + update note on .01; Python-only (sympy exact), mirrorable, deferred to the next verified engine run.

**2026-07-14 (IV) (v485 — the merged analytic target's computable side COMPLETED: the diagonal self-energy is renormalised to ZERO exactly at the KMS seam scale, the mark contact calculus resums in closed form (BFK route), and the 48/41 multiplicities are ONE state set under two response weights — the residual collapses onto the SEAM.EQUIV.01 face alone; *no gate closes, no marker moves*)**

- **v485\_contact\_diagonal\_closed.py (SEAM.CONTACT.UNIT.02).** The v484 merge left one analytic step: the diagonal (self-energy) zeta-renormalisation at the marks + the multiplicity matching. Settled at the computable level: [E] DIAGONAL ZERO AT THE KMS SCALE — the renormalised coincident-point Green function of  $|D|$  on circumference  $\ell$  is  $G_{\text{reg}}(0; \ell) = \frac{1}{\pi} \ln(\ell/2\pi)$  EXACTLY (symbolic limit + 30-digit checks); it vanishes iff  $\ell = 2\pi = 1/(4c_3)$ , the v239 KMS seam circumference — the seam scale is the unique zero-self-energy scheme point, and the log is the pure SCALE channel (matching the structure:  $F_{U(1)}$  carries exactly ONE log term, the transport  $8b_1 c_3^6 \ln(1/\varphi_{\text{seam}})$ , while the  $\varphi_0^{\text{ret}}$  split is log-free). [E] CLOSED-FORM RESUMMATION (the BFK/gluing route already in the suite, v151): with the diagonal zero, the finite-rank mark determinant resums exactly —  $\det(I - uC) = (1 - 4u)(1 + 2u)^2$ ,  $u = \varepsilon c_3 \ln 2$ , all orders, the linear term ABSENT precisely because the diagonal vanishes ( $\text{Tr } C = 0$ ); the series matches v484's  $\text{Tr}(C^k) = 4^k + 2(-2)^k$  term by term. [E] WEIGHT DICHOTOMY: the SAME  $48 = 3 \times 16$  admissible states carry BOTH multiplicities — counted flat they give the  $\varphi_0^{\text{ret}}$  puncture count  $48 = \Omega_{\text{adm}}$ ;  $Y^2/\text{loop}/\text{GUT}$ -weighted they give the alpha budget  $41 = 10b_1 = 40 + 1$  ( $\text{Tr}_{16} Y^2 = \frac{10}{3}$  exact,  $\nu^c = 0$ ; Ginsparg  $(3/5)(41/6) = 41/10$ , v470) — one state set, two response weights, no new number. [C]/[O] CONSEQUENCE: with v484 EVERY finite/computable piece of the merged ALPHA.QUILLEN.EXACT.01 + HYP.PHIO.PUNCTURE.01 target is proven; the single remaining [O] is the identification of the abstract seam  $\zeta$ -det with this glued mark determinant — a face of SEAM.EQUIV.01, exactly the v382 typing, now substantiated computationally. Integrated in `tfpt_1_architecture_e8` (eighth step of the ALPHA.QUILLEN arc) and `origin_theory`; ledger rows SEAM.CONTACT.UNIT.02 + update

notes on .01 and both targets; Python-only by nature, flagged in the Wolfram README.

**2026-07-14 (III) (v484 — the two open analytic targets MERGE: the shared “ $c_3$  per boundary insertion” rule of ALPHA.QUILLEN.EXACT.01 and HYP.PHI0.PUNCTURE.01 is the KMS seam unit  $2\pi = 1/(4c_3)$  with  $\frac{1}{4} = 1/|\mu_4|$ , derived on the seam circle for the finite cycle sector — *no gate closes, no marker moves*)**

- **v484\_seam\_contact\_unit.py (SEAM.CONTACT.UNIT.01).** The v483 round left the two analytic targets sharing one rule (“weight  $c_3$  per boundary contact insertion”: the  $F_{U(1)}$  ladder  $\{0, 3, 6\}$ , the  $\varphi_0^{\text{ret}}$  puncture  $\{1, 4\}$ ). This module shows the shared rule is NOT a new unknown: [E] the unit chain  $\frac{1}{2\pi} = 4c_3 = |\mu_4|c_3 = |\mathbb{Z}_2|\chi(S^2)c_3$  — one *bare* boundary insertion divided by the orbit order  $|\mu_4|=4$  is exactly  $c_3$ ; this is the v239 KMS seam unit and the v58/v73 one-sided Gauss–Bonnet in a single identity. [E] the mechanism is DERIVED on the actual seam circle for the finite cycle sector: the bare Green function  $G(d) = -\frac{1}{\pi} \ln |2 \sin(d/2)|$  (polylog identity, 40 digits) takes INTEGER multiples of  $c_3 \ln 2$  at the  $\mu_4$  separations ( $G(\pi/2) = -4c_3 \ln 2$ ,  $G(\pi) = -8c_3 \ln 2$ ; mark matrix  $-4c_3 \ln 2 \cdot \text{circ}(0, 1, 2, 1)$ , spectrum  $\{4, -2, 0, -2\}$ ), so the  $\mu_4$ -invariant (orbit-averaged) insertion carries  $-(c_3 \ln 2)C$  exactly and the log-det contact expansion is exactly graded in  $c_3$  per insertion ( $\text{Tr } C^k = 4^k + 2(-2)^k$ ). [E] the grammar is suite-wide: the  $\Lambda$  prefactor  $\frac{3}{4\pi^2} = (8\pi)^2 d_{\text{top}} = 48c_3^2$  (v60) carries the SAME multiplicity  $\Omega_{\text{adm}}=48$  as the  $\varphi_0^{\text{ret}}$  puncture at TWO insertions; the  $g-2$  seam loop divides by the same  $2\pi = 1/(4c_3)$ . [C]/[O] THE MERGE: the two tracked [O] targets collapse into ONE named analytic step — the diagonal (self-energy) zeta-renormalisation at the marks + the state-multiplicity matching ( $48 = \Omega_{\text{adm}}$ ,  $41 = 10b_1$ ); the finite half is proven, no target coefficient is claimed (the integers are forced by the square configuration, v354/v355-clean). Integrated in `tfpt_1_architecture_e8` (seventh step of the ALPHA.QUILLEN arc) and `origin_theory` ( $\varphi_0^{\text{ret}}$  split paragraph); ledger rows SEAM.CONTACT.UNIT.01 + update notes on both targets; Python-only by nature (mpmath polylog), exact identities flagged in the Wolfram README.

**2026-07-14 (II) (v482–v483 — the deep-dive round: the  $\nu$ -scale decision computation EXECUTED (the integer  $c_3^3$ -rung excluded unstructured, the  $5/3$ -proximate structure escape recorded and declined) and the  $\varphi_0^{\text{ret}}$  puncture target HALVED (the geometric side proven exact: all twisted pillowcase heat traces are  $t$ -independent rationals, so the  $\pi^{-4}$  must sit in the per-mark coupling weight) — *no gate closes, no marker moves*)**

- **v482\_seesaw\_rung\_decision.py (FLAV.NUSCALE.03).** The decision computation NAMED by v481 is EXECUTED in bracketed (assumption-robust) form: the integer rung  $M_R = c_3^3 \bar{M}_{P1}$  needs a rescue factor  $\times 1.670$  in  $m_3$  (0.0301 vs 0.0503 eV), while  $m_t(\overline{\text{MS}}) \pm 3 \text{ GeV}$  ( $> 3\sigma$ ),  $\alpha_3^{-1} \mp 0.10$  ( $> 3\sigma$ ), a  $\pm 10\%$  band on the ADKLR  $\kappa$  run-down and a PS-leg Yukawa-beta envelope  $\times [0.5, 1.5]$  move it by at most  $\times 1.10$  singly and  $\times 1.165$  ALL-favourably combined — the *unstructured* rung is EXCLUDED conditional on the carrier premise  $y_\nu = y_t$  (the v481 1-loop miss is not an RG artifact; no exact higher-loop coefficients trusted, pure envelope argument). [C] The only escape is a third-generation Majorana STRUCTURE factor  $r \sim 1.67$ ; its post-hoc 0.18% proximity to  $g_{\text{car}}/N_{\text{fam}} = 5/3$  is recorded and DECLINED per v354/v355 (no forcing mechanism; the  $t_0 \sim 30$  discipline of v478). [O] the v481 one-parameter window candidate (NO floor  $\Sigma m_\nu = 0.059 \text{ eV}$ ) never used the integer rung and STANDS. Integrated in `tfpt_2_standard_model` ( $\nu$ -scale paragraph) and `tfpt_5_redteam`; ledger rows FLAV.NUSCALE.03 + .02 update note; Python-only by nature.
- **v483\_pillowcase\_exact\_traces.py (HYP.PHI0.GEOM.01).** The GEOMETRIC side of the  $\varphi_0^{\text{ret}}$  puncture target (HYP.PHI0.PUNCTURE.01/v408) is EXACT: every twisted/orbifold heat trace of the flat  $\tau=i$  pillowcase is a  $t$ -INDEPENDENT RATIONAL —  $\text{Tr}(\gamma e^{t\Delta}) = 1$  for

$\gamma = \sigma, \rho, \rho^3$  at every tested  $t$  (position-space Gaussian image sums  $< 10^{-9}$ ; mode-space proof: only  $n=0$  is  $\gamma$ -diagonal), matching the Atiyah–Bott fixed-point counts  $4 \times \frac{1}{4}$  ( $\det = 4 = |\mu_4|$ ) and  $2 \times \frac{1}{2}$  ( $\det = 2 = |\mathbb{Z}_2|$ ); the pillowcase contact term is EXACTLY  $\frac{1}{2} = 4 \times \frac{1}{8}$  per mark and the clock-equivariant trace exactly 1, for all  $t$ . THEREFORE the  $\pi^{-4}$  in  $d_{\text{top}} = 48c_3^4 = \frac{3}{256\pi^4}$  cannot come from flat orbifold geometry at ANY order: it must sit in the per-insertion COUPLING normalisation ( $4 = |\mu_4|$  marks of weight  $c_3$  give  $c_3^4$ , multiplicity  $\Omega_{\text{adm}} = 48$ ; the bare  $1/(2\pi)$ -propagator 4-cycle differs by the exact rational  $\frac{3}{16}$ ). The target NARROWS from “derive the term” to ONE rule — “per-mark insertion weight =  $c_3$ ” (the same residual class as the EM-Ward  $c_3$ -ladder, EM.WARD.02) — and stays [O]. Complements the experiments-side Donnelly probe (honest negative). Integrated in `origin_theory` ( $\varphi_0^{\text{ret}}$  split paragraph); ledger rows HYP.PHIO.GEOM.01 + HYP.PHIO.PUNCTURE.01 update note; Python-only by nature (quadrature verification of exact statements).

**2026-07-14 (website + figure refresh — the /verification journey views become live HTML (the stale timeline/residual-chain images replaced by interactive components wired to the in-browser reproducer), the prediction surface’s empirical-audit wall of text restructured into dated, expandable rounds, and the generated timeline/residual-chain figures extended to the current suite — *presentation only, no content or marker moves*)**

- **Website /verification: live journey views.** The static `script_timeline.png` and `residual_chain.png` images (stale at the v303–v407 figure state, with alt text and captions still describing the old eleven-phase/v237 state) are replaced by two interactive components, `SuiteTimeline.tsx` (17 phases, v1–v481) and `ResidualChain.tsx` (the reduction chain through the certification round to the bedrock): every phase/step lists representative vN scripts that run live in the browser via the existing Pyodide reproducer, exactly like the dependency graph. Stale counts fixed (“~295-script journey”, “260 scripts” → the generated suite total). The dependency graph itself was verified current (v479–v481 present; all 288 referenced scripts exist in the registry and the website mirror).
- **Prediction surface: the empirical-audit narrative restructured.** The single ~13 000-character audit paragraph in `lib/predictions.ts` is split into twelve dated, verdict-badged rounds (AuditRound: date, title, verdict, two-sentence summary, full unabridged record on expand) plus a domain-chip row and the anti-double-counting note — same content, readable surface. Long-token overflow fixed (**break-words**) in the dependency-graph detail panel and the universal-gap-lab residual cards; the over-long ALPHA.QUILLEN residual note tightened to its current state (v470/v472, residual unchanged).
- **Generated figures + captions synced.** `make_figures.py`: the script timeline gains the v408–v481 certification-round phase (title now ~475 scripts) and the residual chain gains the v458–v480 certification step (Kronheimer v479, four-interval v480, crossed-product v469) with sub-text wrapping fixed; figures regenerated (PDF + website PNG). The stale figure captions updated: `introduction` (timeline caption now spans the full arc), `tfpt_research_contracts` (residual-chain caption now states the SEAM.EQUIV.01 bedrock and certification round; QFT-skeleton caption now names the one [O] residual instead of the outdated “two open items”). No numbers, statuses or claims change anywhere.
- **KaTeX macro fix (broken formula rendering).** Formulas copied from the document set that use the shared TeX macros (`\gcar`, `\Nfam`, `\cthree`, ...) rendered as red KaTeX errors on the site (e.g. the  $\varphi_0^{\text{tree}}$  icosahedral-combinatorics formula on the Origin-Theory page). A shared macro map (`lib/katex-macros.ts`, mirroring the `changelog.tex` preamble) is now passed to every KaTeX render; an automated scan of all 19 site routes finds zero **katex-error** nodes.
- **Readability pass (text walls clamped, content unchanged).** A reusable Expandable component clamps long mirror prose with a fade + “Show more” toggle — applied to the

paper-section bodies and front-box bullet lists (`PaperSection.tsx`, the home set pages and `/papers/[slug]`), the dependency-graph node summaries and the prediction-card descriptions. Nothing is deleted; the full text expands in place.

- **Hylaeon ecosystem card corrected.** The `hylaeon.ai` card spoke of “Field AI” with structure tags that did not describe the project. It now speaks consistently of *Hylaeon* and is grounded in the project documentation (`_documentation/core-physics/big-picture.md`) and the site’s own status log: one cognitive substrate behaving like a physical material, knowledge = attractor valleys, question = boundary condition, answers committed only on physical landing, capability claims gate-checked against a machine-checked claim registry (honest negatives published) — borrowing TFPT’s structural discipline, not its physics predictions. The Agentic-Commerce card reference updated accordingly.
- **Live-site correctness pass (counts + reading order).** Sitewide audit against the deployed page found stale counts and fixed them: the document set is stated everywhere as *10 documents = 1 reading guide + 5 numbered papers + 4 companions* (the safeguards companion was missing from the papers/downloads headers, the JSON-LD collection and the `tfpt_docset.tex` box, which still said “three companions” next to “four”); the stale “23 falsifiable predictions” claims (site metadata, OG image, `llms.txt`) now state the live count of the prediction surface (25 test surfaces, sourced from `predictions.length` where possible), while the interactive compiler page is honestly labelled as its own *23-readout prediction board* (the standalone app’s board genuinely has 23 entries; the intro film keeps its spoken “twenty-three”); and the `/orientation` reading-order lists are fixed to the canonical numbering (Red Team = Doc 5 added, Appendix H = 6, Origin = 7, contracts = 8, safeguards = 9 — the old list mislabelled 6/7 and omitted three documents). Presentation/consistency only; no physics content changes.

**2026-07-14 (v479–v481 — the bird’s-eye “overlooked tools” round: R3’s graph→geometry bridge discharged to Kronheimer 1989; the four  $\mu_4$  marks realised as the exactly solvable *four-interval* multilocal modular geometry with  $\omega \circ \rho = \omega$  manifest; and the absolute  $\nu$ -scale collapsed to a one-parameter, PS-window-bracketed candidate under the carrier normalisation  $y_\nu = y_t$  — *no gate closes, no marker moves*)**

- **v479\_kronheimer\_quiver\_bridge.py (SEAM.KRONHEIMER.01).** The R3 residual of the  $\det K=1$  keystone (v344: “the rewrite attractor graph IS the du Val singularity”) is an ESTABLISHED THEOREM — Kronheimer’s ALE hyper-Kähler quotient construction + Torelli classification (J. Diff. Geom. 29 (1989) 665 and 685) — with all finite input data already compiler outputs, machine-checked [E]: the Kac marks  $\{1, 2, 2, 3, 3, 4, 4, 5, 6\}$  are the null vector of the affine- $E_8$  Cartan matrix AND the Perron vector  $A\delta=2\delta$  of the rewrite attractor (v312); the hyper-Kähler quotient dimension count lands exactly on 4 (the  $\mathbb{C}^2/\Gamma$  geometry;  $\sum_{\text{edges}} d_i d_j = \sum d_i^2 = 120$  forced by the null identity); the gauge datum is  $|2I|=120=|\mu_4|h(E_8)$ ,  $h(E_8)=30=2 \cdot 3 \cdot 5$ ; the resolution carries the unimodular  $-E_8$  form ( $\det=1$ , SNF = identity — the Poincaré link) with  $3 \cdot 8=24$  deformation parameters. [C] the theorems are cited, not re-proved (import class of NPW26/MMST/AGT); [O] the residual premise *transforms* to “the raw seam supplies ALE hyper-Kähler data asymptotic to the  $(2, 3, 5)$  cone” — the same single arrow as SEAM.EQUIV.01 (v346 L2). Integrated in `origin_theory` (R3 keybox + new Kronheimer paragraph) and `tfpt_research_contracts` (Kummer/du Val); ledger rows SEAM.KRONHEIMER.01 + SEAM.DETK.01 update note; Wolfram mirror added to the extension file (+2 exact checks, counted with the next verified engine run — the local engine is currently unactivated; the identical statements are sympy-exact in v479). LEAN: the finite input data are kernel-checked in `TfptCarrier/KronheimerMarks.lean` (marks = null/Perron vector over  $\mathbb{Z}$ ,  $\sum \delta_i^2 = 120 = 4 \cdot 30$ ,  $30 = 2 \cdot 3 \cdot 5$ , dimension count = 4; lake build clean, `#print axioms` = the three standard kernel axioms only).

- **v480\_multilocal\_four\_interval.py** (QGE0.MULTILOCAL.01). The seam circle minus its four marks IS the  $n=4$  multi-interval region of a chiral free fermion — the class whose multi-interval modular theory is exactly known (Casini–Huerta mixing term; Longo–Xu; Rehren–Tedesco multilocal fermions; KLM: holomorphic  $\Leftrightarrow$  trivial multi-interval inclusion,  $\mu=1$  — the same one bit as  $\det K=1$ ); until now the suite used only the single-ball CHM boost (v358). On the exactly solvable antiperiodic critical ring: [E] the quarter-turn clock is a signed permutation with  $\rho^4=-1$  EXACTLY (the fermionic  $\mu_4$  clock is the order-8 double-cover lift — the same  $\mathbb{Z}_2$ -lift pattern as  $2I$  over  $I$ ); [E] its primitive-8th-root character projectors decouple the 4-interval covariance at  $10^{-15}$ , each sector a single-interval quarter-ring kernel with boundary twist ( $10^{-14}$  spectra); [E]  $\rho C \rho^{-1}=C$  at  $10^{-14}$ , so  $\omega \circ \rho = \omega$  holds *manifestly* at the 4-interval state level, and a one-site displacement re-couples the sectors at  $10^{-2}$  (negative control); [E] the cross-interval modular coupling peaks on the conjugate-point diagonal (the Casini–Huerta mixing shadow). [C] continuum theory cited; [O] the raw-seam premise (QGE0.SYM.01/SEAM.EQUIV.01) unchanged. Integrated in **tfpt\_research\_contracts** (marks keybox + rigidity contract + v478 leg A), **origin\_theory** (BW paragraph), **tfpt\_4\_frontier** (v358 companion note); Python-only by nature.
- **v481\_seesaw\_carrier\_ladder.py** (FLAV.NUSCALE.02, **CANDIDATE** class, **genre v467/v468**). FLAV.NUSCALE.01's  $y_\nu=1$  probe was not the carrier normalisation: one  $SO(10)$  **16** per family + the minimal Yukawa sector ( $\mathbf{10}_H$  / PS  $(1, 2, 2)$ ) forces  $y_\nu=y_t$  at the matching scale (named premise). With explicit 1-loop RG (gauge/ $y_t/\lambda$  up; ADKLR Weinberg-operator running down, hep-ph/0108005) the  $(y_\nu, M_R)$  trade-off collapses to  $M_R$  alone: the observed  $m_3=0.0503$  eV demands  $M_R=9.3 \times 10^{13}$  GeV — INSIDE the PS window  $[4.2 \times 10^{13}, 2.4 \times 10^{15}]$  (v249) at  $\log_{c_3}(\bar{M}_{\text{Pl}}/M_R)=3.15$  (the  $y_\nu=1$  inversion gives the structureless 2.58). [X] honesty gate: the integer rung  $M_R=c_3^3 \bar{M}_{\text{Pl}}$  predicts  $m_3=0.030$  eV — 40% low at 1-loop ( $\kappa$  run-down 0.794) — the ladder pin is DECLINED per v354/v355; the decision computation is named (2-loop SM + explicit PS thresholds). What survives: a *one*-parameter, window-bracketed [O] candidate with the falsifiable band  $m_3 \in [0.002, 0.115]$  eV and the NO floor  $\Sigma m_\nu=0.059$  eV; frozen record untouched. Integrated in **tfpt\_2\_standard\_model** (new paragraph after v272), **introduction** status card (third named candidate), **tfpt\_5\_redteam**; ledger rows FLAV.NUSCALE.02 + FLAV.NUSCALE.01 update note; Python-only by nature.
- *Honest negative (experiments-only, no promotion)*: the Donnelly/Atiyah–Bott equivariant heat-kernel route for the  $\phi_0$  puncture target (HYP.PHI0.PUNCTURE.01) types the KIND of term (order-4 equivariant contact datum; the exact weights  $\sum_{k=1}^{n-1} \frac{1}{4 \sin^2(\pi k/n)} = \frac{n^2-1}{12}$  verified) but does NOT force the value  $48c_3^4$  (pure rationals vs  $\pi^{-4}$ ) — recorded in **experiments/theory-contracts/phi0\_donnelly\_equivariant\_probe.py**; the target stays [O].

**2026-07-13 (v478 — first computable steps on the two remaining legs of the entropic-action bridge: the state-side modular data flows to the CHM/Bisognano–Wichmann geometric form (Calabrese–Cardy  $c_{\text{est}} \rightarrow 1$ , CHM parabola, even bands exactly zero) — the bridge's modular side meets TFPT's Einstein-derivation input; and the measure condition reduced to one exact KMS time  $t_0 = \ln \frac{72}{17} + 9 \ln(8\pi) = 30.461$  with the  $h(E_8)=30$  near-miss explicitly declined; both legs stay [O], *no status moves*)**

- **v478\_entropic\_continuum\_legs.py** (GRAV.ENTROPIC.LEGS.01). *Leg A (AP2 continuum, exhibited)*: on the exactly solvable critical chain (infinite-volume sinc kernel, 50-digit precision) the compressed interval state has  $S(L)=(c/3) \ln L + k$  with  $c_{\text{est}}=1.00089/1.00038/1.00021$  (monotone  $\rightarrow$  the Dirac  $c=1=2 \times \frac{1}{2}$ , the v450 reading; gapped control  $\sim 6 \times 10^{-8}$ , area law); the Peschel modular Hamiltonian  $K=\ln((1-C_A)/C_A)$  organises on the CHM weight  $\beta(x)=x(L-x)/L$  (corr 0.87 $\rightarrow$ 0.99, shrinking plateau drift; constants deliberately not asserted), the even-range bands

vanish identically (particle-hole,  $\sim 10^{-36}$ ) and the odd tail is 1.8% with parabolic envelope — the state-side  $\Delta_\Sigma$  (the reading forced by v476) flows to the CHM/BW geometric form, exactly the object TFPT's Einstein derivation consumes (v323/v358): the bridge's modular side and the gravity side meet in the continuum limit; a lattice exhibit, NOT a continuum proof — AP2 stays [O]. *Leg B (the measure)*: [E] single-scale corollary —  $d\mu=\delta(t-t_0)dt$  gives  $\mu_2/\mu_1^2=e^{t_0}$  exactly, so the v477 one-moment condition forces  $t_0=\ln\frac{72}{17}+9\ln(8\pi)=30.461$  (one dimensionless KMS time, closed form); the near-miss to  $h(E_8)=30$  (difference 0.461) is explicitly DECLINED per the v354/v355 no-free-pattern rule — deriving the measure from the entropic structure stays OPEN. Keybox extended in `tfpt_research_contracts`; high-precision numerical module, Python-only by nature (Wolfram count unchanged, 378/378; flagged in the Wolfram README); ledger row `GRAV.ENTROPIC.LEGS.01`; both legs stay [O], no marker moves.

**2026-07-13 (v477 — the scale-flow representation of the entropic action: the surviving  $R^2$  route typed as *one moment condition*  $\mu_2/\mu_1^2 = \frac{72}{17}(8\pi)^9$  with the exact closure  $\frac{4608\pi^2/17}{(72/17)(8\pi)^9} = c_3^7$  — the kill-test gap is a scale-measure datum, satisfied by the same KMS moment TFPT already fixes (v36); zero new dials, *no status moves*)**

- `v477_entropic_scaleflow.py` (`GRAV.ENTROPIC.SCALEFLOW.01`). [E] The Frullani identity upgrades to a scale-flow representation:  $-\ln a = \int \frac{dt}{t}(e^{-ta}-e^{-t}) = 2 \int \frac{dx}{x}(e^{-a/x^2}-e^{-1/x^2})$  (symbolic +  $10^{-20}$  quadrature) — Bianconi's log action IS the *flat* scale-integral of relative heat-kernel actions, while TFPT's  $S_{\text{rel},\chi}$  evaluates *one* KMS scale with a weight: the bridge between the two action principles is a choice of SCALE MEASURE. [E] Moment dictionary: a weight  $w(t) dt/t$  gives  $3\beta R \mu_1 + \frac{17}{24}\beta^2 R^2 \mu_2$ ; the flat weight has  $\mu_1=\mu_2=1$  and reproduces the v475 raw coefficients identically. [E] The ONE moment condition: demanding  $m^2=c_3^7 \bar{M}_{\text{P}1}^2$  forces exactly  $\mu_2/\mu_1^2=\frac{72}{17}(8\pi)^9$  — the v475 kill-test number IS a moment ratio — and the closure identity  $\frac{4608\pi^2/17}{(72/17)(8\pi)^9}=c_3^7$  holds identically: the “13 orders” are a scale-measure datum, not a bridge mismatch. [E] v36 consistency:  $f_0=6(4\pi)^2/c_3^7 \Leftrightarrow M^2/\bar{M}_{\text{P}1}^2=c_3^7$  — one KMS moment, two parametrisations, zero new dials. [C] A consistency statement, not a derivation (the KMS weight is TFPT input; the entropic action supplies the Einstein +  $\Lambda$  level and the scale-flow form, TFPT supplies the measure). Negative side-record: an exact lattice-BW commutator attempt for AP2 (Peschel–Eisler simple forms) did NOT hold on the half-filled open chain — kept out of the suite, logged in the research notes. Keybox extended in `tfpt_research_contracts`; Wolfram mirrors both exact blocks (378/378); ledger row `GRAV.ENTROPIC.SCALEFLOW.01`; the  $R^2$  gate and the v359 typing stay [O], no marker moves.

**2026-07-13 (v476 — the AP2 compression conjecture of the entropic-action bridge made *well-posed* on an exactly solvable model: the literal operator-side reading is ill-posed on a pure bulk, the state-side reading is forced, the mismatch is exactly quadratic in the cross-cut correlations and gap-suppressed; AP2 stays [O], *no status moves*)**

- `v476_entropic_compression_toy.py` (`GRAV.ENTROPIC.COMPRESS.01`). The last open work package of the v473 round, exercised on a gapped staggered-mass chain ( $N=16$ ). The seam situation is reproduced (pure bulk: the ground-state covariance is a projection to  $10^{-15}$ ; thermal boundary:  $\text{spec}(PCP) \subset (0,1)$ , Peschel  $K=\ln((1-C_A)/C_A)$  finite,  $S>0$  — v155 in the bridge context). **The reading is decided:** on the pure bulk  $f(x)=x/(1-x)$  is singular on  $\text{spec}(C)=\{0,1\}$ , so the literal operator-side reading  $Pf(C)P$  of the v473 conjecture is *ill-posed* while the state-side  $f(PCP)$  is finite — the compression identity can only mean “build  $\Delta_\Sigma$  from the *compressed* relative metric”, which matches Bianconi's own local-metric construction (AP2 retyped, not weakened). [E] Block algebra (symbolic):  $[C^2]_{AA}-(C_{AA})^2=BB^t$ ,  $[C^3]_{AA}-(C_{AA})^3=ABB^t+BB^tA+BDB^t$  — first order in the cross block vanishes identically,

so for any analytic  $f$  the two readings agree to *second* order in the cross-cut correlations: the modular (entanglement) content lives exactly in that quadratic correction. Gap control at the bounded (covariance) level: the Bianconi-shaped bulk identity  $f(C_T)=e^{-h/T}$  holds to  $10^{-11}$  relative; the mismatch obeys the quadratic law  $d \leq x^2$  ( $d/x^2 \in [0.4, 1.0]$ ) and falls with the gap in the gapped regime, the decoupled cut agrees exactly — the two readings converge in the gap-dominated regime, where TFPT operates ( $\Delta_{\text{eff}} > 0$ , v76/v337). Keybox extended in `tfpt_research_contracts`; numerical module, Python-only by nature (Wolfram count unchanged, 376/376; the block identity flagged in the Wolfram README); ledger row GRAV.ENTROPIC.COMPRESS.01; AP2 (the continuum seam statement) stays [O], no marker moves.

**2026-07-13 (v475 — the  $R^2$  kill test of the entropic-action bridge, *executed*: exact tensorial factors on the maximally symmetric background, raw entropic scalaron  $m^2 = \frac{4608\pi^2}{17}\bar{M}_{\text{Pl}}^2$  ( $\approx 51.7\bar{M}_{\text{Pl}}$ , trans-Planckian)  $\Rightarrow$  the light-trace-mode reading is dead, KMS renormalisation the only surviving  $R^2$  route; plus the explicit Lorentzian-positivity witness; *no status moves*)**

- **v475\_entropic\_scalaron\_gate.py** (GRAV.ENTROPIC.SCALARON.01). Work package 5 of the v473 bridge. [E] On a maximally symmetric background the relative curvature operator  $\tilde{\mathcal{R}}\tilde{g}^{-1}$  has eigenvalues *exactly*  $\{R; R/4 \times 4; R/6 \times 6\}$  (Bianconi's Appendix-B flattening verified symbolically, incl. the  $6 \times 6$  two-form block), giving the exact vacuum action  $\mathcal{L}_B(R) = 3\beta R + \frac{17}{24}\beta^2 R^2 + O(R^3)$  — the 3 re-derives her Eq. (45) (= the v473 pin),  $\frac{17}{24} = \frac{1}{2}(1 + \frac{1}{4} + \frac{1}{6})$  is the exact tensorial  $R^2$  factor. [E] With the pinned  $\beta' = c_3/6$  both mass routes (Starobinsky matching and  $f'/3f''$ ) agree: the raw entropic scalaron is  $m_{\text{raw}}^2 = \frac{4608\pi^2}{17}\bar{M}_{\text{Pl}}^2$ , i.e.  $m_{\text{raw}} \approx 51.7\bar{M}_{\text{Pl}} \approx 1.3 \times 10^{20}$  GeV — TRANS-PLANCKIAN, not light; v473's interpretation 2 (the TFPT scalaron as the light  $\mathcal{G}$ -trace mode) is *dead in naive form*; a viable mechanism must supply exactly  $\frac{72}{17}(8\pi)^9 \approx 1.69 \times 10^{13}$  in mass<sup>2</sup> (the kill-test number now WITH tensorial factors). [E] Domain: first log pole at  $R=1/\beta=384\pi^2\bar{M}_{\text{Pl}}^2$  (trans-Planckian) — the obstruction is the mass, not the domain. [E] Lorentzian-positivity witness: a timelike gradient gives the  $Gg^{-1}$  eigenvalue  $1 - \alpha v^2 \leq 0$  for  $v^2 \geq 1/\alpha$  (ln undefined), spacelike/Euclidean stay positive — the v473 [C] caveat exhibited, the OS route sturdier. [C] Verdict: interpretation 1 survives (Einstein +  $\Lambda$  level), 2 is dead naive, 3 (KMS-spectral renormalisation) is the ONLY surviving  $R^2$  route. Keybox extended in `tfpt_research_contracts`; Wolfram mirrors all three blocks (376/376); ledger row GRAV.ENTROPIC.SCALARON.01; no gate closes, no marker moves.

**2026-07-13 (v474 — the operator level of the entropic-action bridge: the  $D_5$  Clifford/spinor structure exhibited on the carrier Fock space, the Hodge fold identified as the  $5 \rightarrow \bar{5}$  conjugation, and the v473  $Q$ -target *decided* — supports exactly  $\{|\mathbb{Z}_2|, \text{rank } E_8, 2^{g_{\text{car}}}\}$ , the naive pair-block reading killed; *no status moves*)**

- **v474\_entropic\_hodge\_carrier.py** (GRAV.ENTROPIC.HODGE.01). Work packages 1 and 4 algebra of the v473 bridge, machine-checked on the finite carrier Fock space  $\Lambda^\bullet \mathbb{C}^5$  ( $\dim 2^{g_{\text{car}}} = 32$ ). [E] CAR  $\Rightarrow$  Clifford: the ten  $\Gamma$ 's from  $(a_k^\dagger, a_k)$  satisfy  $\{\Gamma_a, \Gamma_b\} = 2\delta_{ab}$  exactly (all 100 pairs, integer matrices); the 45 bilinears  $[\Gamma_a, \Gamma_b]/4$  commute with  $\gamma = (-1)^N$ , span exactly  $45 = \dim \mathfrak{so}(10)$  and leak nothing from the 16-dim even subspace — the carrier half-spinor *is* the even exterior algebra *with* its spinor structure (the operator level of v2/v44/v197). [E] the Hodge–Dirac symbol:  $c(\xi) = a^\dagger(\xi) - a(\xi)$  satisfies  $c(\xi)^2 = -|\xi|^2 \text{Id}$  for symbolic  $\xi$  — Bianconi's  $D = d + \delta$  is Clifford multiplication at symbol level. [E] the fold is the  $5 \rightarrow \bar{5}$  conjugation: the traceless  $\mathfrak{u}(5)$  weights of  $\Lambda^4$  are the negatives of those of  $\Lambda^1$ , so her fiber  $1+5+10$  becomes the GUT half-spinor  $16 = 1 + \bar{5} + 10$  (the SM carrier decomposition). [E] honest obstruction:  $\Omega^{\leq 2}$  is *not* Clifford-invariant ( $\Lambda^2 \rightarrow \Lambda^3$  leak) — the bridge must go through the fold, never through degree truncation (a recorded constraint on the AP2 compression). [E] the  $Q$ -target enumeration (the v473 [C] conjecture decided at the algebraic level):  $\text{Tr } Q^2 = d k^2 c_3^4 = 32 c_3^4$  with integer  $k$  has exactly the supports

$(d, k) = (2, 4), (8, 2), (32, 1) = \{|\mathbb{Z}_2|, g_{\text{car}} + N_{\text{fam}} = \text{rank } E_8, 2^{g_{\text{car}}} = \dim \Lambda^\bullet\}$  — minimal uniform solution  $q = c_3^2 = 1/(64\pi^2)$  per Fock mode, bonus identity  $\text{Tr } Q^2/\delta_{\text{top}} = 32/48 = 2/3 = \dim \Lambda^\bullet/\Omega_{\text{adm}}$ ; the two-form block 10 (the problem-note “pair sector” conjecture) and the fiber 16 need irrational multipliers  $\sqrt{16/5}, \sqrt{2}$  — *killed* in uniform form. [C] the physical choice among the three admissible supports; [O] unchanged (AP2 compression + the  $R^2$  mechanism). Keybox extended in `tfpt_research_contracts` (Research Contract 2, the v473 keybox); Wolfram mirrors the enumeration and the weight-conjugation identity (373/373); ledger row GRAV.ENTROPIC.HODGE.01; no gate closes, no marker moves.

**2026-07-13 (v473 — the entropic-action bridge: Bianconi’s *Gravity from entropy* (Phys. Rev. D 111, 066001 (2025); arXiv:2408.14391) quantified as an *external candidate* for the missing action level of the parameter-free Einstein equation; her free constant pinned  $\beta'_B = c_3/6 = 1/(48\pi)$ , the  $R^2$  gap  $3(8\pi)^9$  pre-registered as kill test; *no status moves*, the v359 equation-of-state typing stays [O])**

- **v473\_entropic\_action\_bridge.py (GRAV.ENTROPIC.ACTION.01).** TFPT’s classical field equation  $G_{ab} + \Lambda g_{ab} = c_3^{-1} T_{ab}$  is derived thermodynamically (v358/v359) and honestly typed “equation of state, not a from-action quantisation” [O]. Bianconi’s entropic action  $S_B = -\text{Tr}_F \ln(\tilde{G}\tilde{g}^{-1})$  (the quantum relative entropy between the spacetime metric and the matter-induced metric, Araki-related via  $\Delta^{1/2} = \tilde{G}\tilde{g}^{-1}$ ) is a local covariant action whose variation gives Einstein + an emergent nonnegative  $\Lambda$  at low coupling — an explicit *external candidate* for exactly that missing variational layer. Machine-checked: [E] the carrier Hodge count (her  $\Omega^{0,1,2}$  fiber on the five-slot carrier  $\mathbb{C}^5$  counts  $1+5+10=16=2^{g_{\text{car}}-1} = \dim S_{D_5}^+$ , while on 4d spacetime it is only 11 — the 16 *requires* the carrier); [E] the one quantitative pin — matching her weak-coupling  $3\beta R$  to  $1/(16\pi G) = c_3/(2G)$  fixes her free constant *exactly*,  $\beta'_B = c_3/6 = 1/(48\pi)$ ; [E] her  $\Lambda_G = \frac{1}{2\beta} \text{Tr}(\mathcal{G} - I - \ln \mathcal{G})$  is nonnegative and quadratic at the fixed point, and with  $\varepsilon_\star = e^{-\alpha^{-1}} Q$  it reproduces the closed v60 branch  $(3/4\pi^2)e^{-2\alpha^{-1}}$  with the exact target  $\text{Tr } Q^2 = 32c_3^4 = 1/(128\pi^4) = \frac{2}{3}\delta_{\text{top}}$  ([C] until  $Q$  is constructed); [E] the  $R^2$  kill test — the raw log expansion sits *exactly*  $3c_3^{-9} = 3(8\pi)^9 \approx 1.2 \times 10^{13}$  below the TFPT Starobinsky coefficient  $1/(12c_3^7)$ , so a direct identification is *false* without a mechanism (pre-registered); [C] the compression conjecture  $P_\Sigma(\tilde{G}\tilde{g}^{-1})P_\Sigma = \Delta_\Sigma^{1/2}$  (the bridge proper) stays open, with Lorentzian positivity flagged (TFPT’s OS route is the sturdier construction). New keybox in `tfpt_research_contracts` (Research Contract 2, after the v359 keybox); residual pointers updated in `introduction/tfpt_4_frontier/status` card. Wolfram mirrors the three exact blocks (371/371); ledger row GRAV.ENTROPIC.ACTION.01; *no* gate closes, *no* marker moves, zero TFPT knobs added.

**2026-07-03 (surfacing round — three already-computed unique readouts promoted to the public prediction surfaces: the axion–photon coefficient  $g_{a\gamma\gamma} = -4c_3$ , the free-seam Higgs band  $m_H \approx 129\text{--}134\text{ GeV}$ , and the never-run BH HFQPO  $\times 1.5$  ladder-tooth search; no new vN, no status moves)**

- **Uniqueness scan follow-up (no new results — presentation only).** A structure scan for “what does TFPT predict that nobody else pins?” found three concrete, falsifiable readouts that were fully computed and ledgered but missing from the public prediction surfaces; they are now surfaced, with their existing typing unchanged. (i) **Axion–photon coefficient:** the determinant-line anomaly coefficient  $g_{a\gamma\gamma} = -4c_3 = -1/(2\pi)$ ,  $y^2 = 16c_3^2 = 1/(4\pi^2) \approx 0.0253$  (v207, the [E] leg of GRAV.ASYMP.01; the [C] caveat — coefficient, not a parameter-free  $g_{a\gamma\gamma}$  in  $\text{GeV}^{-1}$  — stated verbatim) joins the axion card in `website/lib/predictions.ts` and the frontier dark-matter mirror in `papers.ts`; the TeX source (`tfpt_4_frontier`) already carried it. (ii) **Higgs free-seam band:** the double-criticality prediction  $\lambda(\bar{M}_{\text{Pl}}) = \beta_\lambda(\bar{M}_{\text{Pl}}) = 0 \Rightarrow m_H = 133.5\text{ GeV}$  (band 129–134,  $m_t/\alpha_s$ -sensitive; measured 125.25 sits a few GeV below — the known slight

metastability, Buttazzo NNLO  $\lambda(M_{P1}) = -0.0143 \pm 0.0057, 2.5\sigma$ ) becomes a prediction card ([C]; v166, HIGGS.FREESEAM.01, kill: settled- $(m_t, \alpha_s)$  RGE pull off the surface at  $> 5\sigma$ ). (iii) **BH HFQPO ladder tooth**: the one discriminating HFQPO test — a third tooth at  $\nu_3 = (3/2)\nu_u$  (661.5/414/252/363 Hz), integer harmonics forbidden — that *no published search ever targeted* although the teeth lie in the searched RXTE band (experiments/hfqpo-ladder, preregistered): added as an [O] search-target bullet to the tfpt\_horizon\_readouts “Search targets (not claims)” gapbox (firewall language verbatim: non-canonical mapping, selection null 18.5%, J1859+226 +9.2 $\sigma$ , GR resonance favored; even a tooth hit would be [C]) and as an [O] prediction card. Website mirrors updated in the same change (predictions.ts 23  $\rightarrow$  25 cards, papers.ts frontier + horizon sections incl. the new search-targets mirror section); scorecard, ledger and all markers untouched.

**2026-07-03 (v472 — the determinant line over the  $U(1)$ -twist moduli of the collar model carries curvature = the inflow level: the v470 bridge lemma exhibited at the finite/model level; ALPHA.QUILLEN.EXACT.01 stays [O]. Plus: bird’s-eye rounds — electron-sector/Cabibbo whiteness probes, Calderón UV-universality, parameter-free flat-budget closure (sandbox); the flat-budget closure enters the evidence scorecard)**

- **v472\_quillen\_detline\_moduli.py (ALPHA.QUILLEN.DETLINE.01).** ALPHA.QUILLEN.EXACT.01 names the  $\alpha^3$  level as the curvature of the *determinant line over the  $U(1)$  seam moduli* (Quillen 1985 / Bismut–Freed CMP 106 (1986) curvature; Dai–Freed JMP 35 (1994) section); v470 computed the bulk Chern invariant only over the Bloch Brillouin zone (the translation-invariant shortcut). v472 computes the Quillen-shaped object itself: the same collar Hamiltonian ( $h(k) = \sin k_x \sigma_x + \sin k_y \sigma_y + (M - \cos k_x - \cos k_y) \sigma_z$ , v367/v470) in real space on an  $L \times L$  torus with *twisted* boundary conditions — the twist torus  $(\theta_x, \theta_y)$  is the moduli space of flat  $U(1)$  connections — and the Fukui–Hatsugai–Suzuki curvature of the *determinant line of the occupied frame* over that torus:  $\oint = 2\pi \cdot 1$  exactly (integer to  $10^{-9}$ ) at  $M=1$  for  $L=4$  AND  $L=6$  (size-independent); controls: trivial collar  $M=3 \rightarrow 0$ , orientation flip  $M=-1 \rightarrow -1$ ; the twist-moduli integer equals the Bloch-BZ integer for all three  $M$  (Niu–Thouless–Wu PRB 31 (1985), exhibited on the S3 collar model); the Fermi gap ( $= 2.0$ ) stays open over the whole twist torus, so the Dai–Freed section has no zero. **Reading [C]:** det-line holonomy over the  $U(1)$  moduli = inflow level  $k_0 = |C| = 1$  — the v470 bridge lemma “ $\delta \log \det_\zeta(\text{seam}) = \text{the inflow response}$ ” holds *verbatim at the finite/model level*, where  $\log \det$  of the occupied frame replaces  $\log \det_\zeta$ . **Not closed [O]:** the *continuum*  $\zeta$ -determinant identification on the abstract seam (= the SEAM.EQUIV.01 face); ALPHA.QUILLEN.EXACT.01 stays [O];  $\alpha^{-1}$  stays [E]. Structurally the same manoeuvre as v471 for SEAM.THEOREM.01’s replica side. 7/7 checks; suite green through v472 (466 modules); numerical (dense eigh + FHS twist grid), Python-only by nature. Papers updated: tfpt\_1 (sixth honest step in the EM closure), tfpt\_research\_contracts (residual certification), tfpt\_safeguards ( $\alpha$ -form lever), root README, zenodo\_description.html. Exploration original: experiments/tfpt-discovery/quillen\_detline\_twist\_moduli.py (same day).
- **Bird’s-eye whiteness rounds (sandbox, experiments/tfpt-discovery/, nothing else promoted).** (i) **electron\_sector\_cabibbo\_probe.py**: the  $\alpha$  metrology triangle (TFPT  $\alpha^{-1} = 137.0359992168$  vs Rb-2020  $+0.99\sigma$  / CODATA22  $+1.90\sigma$  /  $a_e$ -route  $+3.39\sigma$  / Cs-2018  $+6.33\sigma$  — TFPT takes the Rb side of the  $5.5\sigma$  metrology split; new honest pressure point recorded); the v204 seam-vertex universality catch (a flavor-universal reading of  $a_\mu^{\text{seam}} = 2.879 \times 10^{-9}$  is excluded by the electron at  $\sim 2 \times 10^4\sigma$ ; linear scaling at  $\sim 10^2\sigma$ ; quadratic survives — the [C] bridge must scale at least quadratically in  $m_\ell$ ); the Cabibbo-angle anomaly as a third dissolution watchdog ( $\lambda_C = 0.2243762$  exact sits  $+0.08\sigma$  on the PDG-2026 rescaled kaon average, between  $K_{\ell 3}$  and  $K_{\mu 2}$ ; exact unitarity puts  $V_{ud} = 0.97450$  — superallowed is the outlier at  $+2.6\sigma$ , all neutron/pion routes agree). (ii) **kernel\_calderon\_uv\_universality.py**: the raw-collar

Calderón/DtN datum is UV-universal (mode-by-mode first-order convergence to the exact Bessel symbol  $mI'_\nu(mR)/I_\nu(mR)$ , Richardson residual  $3 \times 10^{-4}$ ;  $N_\theta$ -independent;  $|k|$  principal symbol;  $\mu_4$  block structure at machine precision; both spec- $T$  masses; 3-fold mesh-defect negative control) — the operator equality demanded by QGEO.KERNEL.01 has a well-defined discretization-independent left-hand side (gate stays [O]). (iii) `flat_budget_closure.py` + scorecard row: see next bullet.

- **Flat-budget closure enters the evidence scorecard (experiments-only, pattern candidate).** The zero-dial closure  $\{\Omega_b = \varphi_0^{\text{ret}}(1 - \frac{1}{4\pi})$  [frozen],  $\Omega_c = \frac{2}{7}(1 - \frac{1}{4\pi})$  [the 2026-07-02 numerology-flagged Nebenbefund],  $\rho_\Lambda/\bar{M}_{\text{Pl}}^4 = (3/4\pi^2)e^{-2\alpha^{-1}}$  [frozen],  $\Sigma m_\nu = 0.0588 \text{ eV}$  [v468 route]] + spatial flatness determines the whole flat  $\Lambda$ CDM budget:  $H_0 = 67.15 \text{ km/s/Mpc}$  ( $-0.39\sigma$  Planck18,  $-5.67\sigma$  SH0ES22 — the parameter-free Planck side of the Hubble tension),  $\Omega_m = 0.3133$  ( $-0.27\sigma$ ),  $t_0 = 13.86 \text{ Gyr}$ ; honest stresses recorded:  $\omega_b -2.02\sigma$  (BBN-only  $-0.73\sigma$ ),  $\omega_c -1.19\sigma$ , DESI-DR2  $H_0 -2.16\sigma$  (one correlated low- $h$  direction). Scorecard row `cosmo / flat budget closure` (consistent, composite: shares the `phi0_seed` group (and the  $\rho_\Lambda$  leg the  $\alpha$  engine),  $\Omega_c$  leg in `alternative_group=axion_branch` vs the axion-spine DM reading — never double-counted); nothing promoted to `verification/`.

**2026-07-02 (v471 — the Seam–Horizon replica chain exercised numerically on the discretized collar: the kernel-identification premise of SEAM.THEOREM.01 discharged at the finite/model level; residual retypes to the continuum MMST leg + the anchor; gate stays [O]. Plus: the GW ringdown-echo channel hardened — 10 loudest events incl. GW250114, redshift-corrected templates, stacked search and a 12-variant signature battery, all null)**

- **v471\_seam\_horizon\_replica.py (SEAM.EHMODEL.04).** The residual isolated by v150/v151/v152 — “the RP seam kernel *is* this BFK Calderón datum” — is exercised numerically on a discretized seam collar with the seam’s *own* kernel and *real* replica sheets: the matched conical deficit has slope  $2C(\gamma)$  on deficit cones (0.01–0.05%) and replica sheets  $n=2$  (0.8%),  $n=3$  (1.5%); the EH coefficient is cutoff-independent (0.5% drift) while the  $\zeta$ -scale  $\mu_{\text{lat}}$  drifts — the v152 anchor exhibited on data, with *no* canonical  $\ln(m/\mu)=3/4$  emerging (recorded honestly); the BFK/Schur split is exact ( $\sim 10^{-16}$ ) with the Calderón factor conically clean *measured on the kernel* (pure cut-edge law, tip term 0, power-checked) and the halves carrying  $2C_{\text{cone}}$  via Kac doubling; the v302 transfer masses  $6\ln(3/2)$ ,  $6\ln 3$  sit on the calibrated EH line ( $\leq 0.05\%$ ); and the Perron/attractor mode ( $m=0$ ) is IR-divergent — **the recovery gap  $\Delta=6\ln(3/2)$  is what makes the induced-gravity coefficient finite** (the same gap that makes the attractor unique makes  $1/G$  finite). SEAM.THEOREM.01 stays [O]; its residual retypes to the *continuum* scaling limit (the MMST class — the same single residual as SEAM.EQUIV.01, v336) plus the one declared anchor. 16/16 checks; suite green through v471 (465 modules); numerical (sparse lattice determinants), Python-only by nature. Papers updated: `origin_theory` (new collar paragraph in the Seam–Horizon gapbox + residual retyping), `tfpt_research_contracts` (R3 premise), `tfpt_4_frontier` (3/4 audit note), root README, `verification/README`, `zenodo_description.html`. Exploration original: `experiments/theory-contracts/seam_horizon_replica.py` (17/17, same day).
- **GW ringdown-echo channel hardened (experiments/gw-ringdown-echo, scorecard row `data_limited`→`consistent`).** (i) Strain coverage extended to the *10 loudest* ringdown events incl. GW250114 (network SNR 78.6, O4b; new `fetch_strain_4096.py` crops the 4096 s GWOSC archive segments). (ii) **Redshift correction:** the GWTC catalogue reports source-frame masses, the observed ringdown is at  $f_0/(1+z)$  — all templates moved to detector-frame  $M(1+z)$  (before the fix GW190521,  $z=0.56$ , was filtered  $\sim 1.6\times$  too high); all stages re-run. (iii) Stage 1b *stacked* search over 23 detector streams: stacked  $p=0.262$ , kernel-consistent streams 0/23 — STACKED NULL. (iv) Stage 1c *signature battery* (Bonferroni  $\times 12$ ): ratio semantics (amplitude  $(2/3)^6$  vs *energy reading*  $(2/3)^3$  vs step  $2/3 \times \mu_4$  per-bounce phases  $\{0, \pi/2, \pi, 3\pi/2\}$

(the boundary-birefringence analogue — path birefringence from parity violation  $c_- = 8$  cancels in the inter-echo ratio; only a per-reflection phase does not)  $\times$  lags 0.5–350 ms (ECO through Planckian window), with joint (2, 2, 0)+(2, 2, 1) subtraction and a template-agnosticism control: NO\_VARIANT\_ECHO on all 10 events — the null is robust against alternative signature readings. Search-target firewall unchanged: upper bound, no detection, no tension claim.

## 2026-07-02 (v295 robustness patch — external-review follow-up: the numeric $a_2$ -kernel tolerance must dominate its own finite-difference floor)

- **v295\_flataway\_a2\_exact.py** (FLATAWAY.A2.01). Alessandro’s Git-baseline audit (commit 8cb8ad0) hit a platform-marginal failure in check 3: the exact divided-difference  $a_2$  kernel is compared against an  $\varepsilon = 10^{-3}$  second finite difference whose *own* truncation error is  $O(\varepsilon^2) \sim 10^{-6}$ , yet the tolerance was  $10^{-6}$  — on his machine  $k=3$  landed at  $1.1 \times 10^{-6}$  (ours:  $2.9 \times 10^{-7}$ ). Fix: tolerance widened to  $5 \times 10^{-6}$  so it *dominates* the control’s error floor, with the reasoning recorded in the check text; the load-bearing anchor remains the *closed form* at machine precision (v296), the numeric witness stays as the independent cross-check. A numerical-robustness patch, no claim or status change; ledger row updated; suite green (ALL CHECKS PASSED through v470).

## 2026-07-02 (v469/v470 — the closure-route round on the two named targets: the SEAM.EQUIV.01 extension leg re-founded on 1995–2001 peer-reviewed crossed-product theorems + the realisation axiom reduced to invariant level; the $\alpha^3$ level = the *computed* bulk Chern invariant + the seam $F$ -normalisation = the embedding index $k_Y = 5/3$ ; both targets stay [O])

- **v469\_seam\_crossedproduct\_route.py** (SEAM.EQUIV.CROSSEDPRODUCT.01). The 128-spinor extension leg of SEAM.EQUIV.01 no longer rests on 2024/2025 preprints: [E] the  $SO(16)_1$  spinor has  $h_s = N/16 = 16/16 = 1 \in \mathbb{Z}$ , so its statistics phase is +1 — *exactly* the Longo–Rehren locality criterion — and the  $\mathbb{Z}_2$  simple-current crossed product  $SO(16)_1 \rtimes \{1, s\}$  is a *local* index-2 extension by the peer-reviewed subfactor package (Longo–Rehren, Rev. Math. Phys. 7 (1995); the level-1  $so(N)$  net with its DHR sectors from even-CAR: Böckenhauer, Rev. Math. Phys. 8 (1996); simple-current extensions of nets: Böckenhauer–Evans, Comm. Math. Phys. 197 (1998)); KLM (Comm. Math. Phys. 219 (2001)) gives  $\mu(B) = 4/2^2 = 1 \Rightarrow$  holomorphic  $\Rightarrow (E_8)_1$  by the v463 pin. The AGT/AMT lattice-VOA route (v459) is *demoted to an independent second witness*. [E] the index-4  $\mu_4$  glue runs on the same integer:  $h(J^k) = \{1, 1, 1\}$  for  $k=1, 2, 3$  (the v125 isotropy  $q(k(1, 1)) = k^2$  at net level),  $\mu = 16/4^2 = 1$ , same endpoint. [E] the realisation input is reduced from model fiat to *invariants* (R1’): quasi-free [C] (v155/v160) + gap [E] (v302) + class D +  $c_- = 8$  [E] (v456, from P1), with the invariants *computed* — per-copy FHS Chern  $|C| = 1$  vs 0 (control),  $\nu = 16 \times |C| = 16$ ,  $c_- = \nu/2 = 8$  — and  $\nu \equiv 0 \pmod{16}$  is exactly Kitaev’s class-D phase whose edge admits a purely bosonic description = the  $(E_8)_1$  level-1 state (Ann. Phys. 321 (2006), sixteen-fold way); the phase-invariance of  $c_-$  is cited (Kapustin–Spodyneiko, PRB 101 (2020); Kim et al., PRL 129 (2022)). [O] *not closed*: the cited theorems stay cited; SEAM.EQUIV.01 stays [O].
- **Lean hardening**. TfptCarrier/SeamResidualAxiom.lean gains the *parallel* peer-reviewed derivation `seamResidualClosed'`: the invariant-level axiom `collar_invariants` (R1’) + *one* combined 1995–2001 package `crossedproduct_route_theorem`; the new joints are kernel facts with *no* axioms — `lr_locality_integer` ( $h_s = 16/16 \in \mathbb{Z}$ ), `glue_locality_integers` ( $\mu_4 h(J^k) \in \mathbb{Z}$ ), `klm_holomorphic` ( $4/2^2 = 1$ ,  $16/4^2 = 1$ ), `sixteenfold_pin` ( $\nu = 16 \equiv 0 \pmod{16}$ ,  $c_- = 8$ ), bundled as `crossedproduct_arithmetic` ([propext] only); `#print axioms seamResidualClosed'` = exactly four; the original `seamResidualClosed` (AGT/AMT witness) is kept unchanged as the independent second route; lake build OK (3285 jobs).
- **v470\_alpha\_inflow\_level.py** (ALPHA.QUILLEN.INFLOW.01). Both residuals left

on ALPHA.QUILLEN.EXACT.01 after v433–v435 are upgraded. [E] target re-verified ( $\alpha^{-1}=137.0359992168$ ; Quillen split  $<10^{-35}$ ). [C] *the level is computed, not minimal*: the  $a^3$  coefficient  $k_0=1$  equals the FHS bulk Chern invariant  $|C|=1$  of the same p+ip collar that realises S3 (v367/v461) — “why exactly 1” is answered by computation plus the cited inflow identification (TKNN PRL 49 (1982) / Avron–Seiler–Simon PRL 51 (1983) quantisation; Callan–Harvey, Nucl. Phys. B250 (1985) inflow; APS 1975 / Alvarez-Gaumé–Della Pietra–Moore 1985 / Witten, Rev. Mod. Phys. 88 (2016)  $\eta$ =CS part of  $\delta \log \det$ ; Quillen 1985 / Bismut–Freed 1986 determinant-line integrality), replacing v435’s minimality assumption. [E]  $k_Y = \text{tr}(Y^2)/\text{tr}(T_3^2) = (5/6)/(1/2) = 5/3$  (Ginsparg, Phys. Lett. B197 (1987)):  $(3/5) \times \frac{41}{6} = \frac{41}{10} = b_1$  — the “GUT 3/5” is the inverse embedding index;  $\frac{41}{6} = \frac{20}{3} + \frac{1}{6}$  (carrier decomposition). [C] the seam  $F$ -normalisation (the one [C] of v434) is *retyped*: once the seam net is  $(E_8)_1$  level 1,  $\langle JJ \rangle = k/(z-w)^2$  is fixed by  $k_Y$  — level-1 current-algebra rigidity, zero independent content, a face of SEAM.EQUIV.01 (per the v382 typing). [E] *structural unification*: one invertible phase, two quantised responses —  $c_- = 8$  (gravitational, feeds SEAM.EQUIV.01/S3) and  $C=1$  ( $U(1)$ , feeds EM1) — so closing R1’ feeds *both* named targets. [O] *not closed*: the bridge lemma “ $\delta \log \det_\zeta(\text{seam}) = \text{the inflow response}$ ” stays the named cited step; ALPHA.QUILLEN.EXACT.01 stays [O];  $\alpha^{-1}$  stays [E].

- **Integration.** Both registered in `run_all.py` + `script_registry.csv` (cluster frontier); ledger rows SEAM.EQUIV.CROSSEDPRODUCT.01 + ALPHA.QUILLEN.INFLOW.01; FORM.SEAM.RESIDUAL.01 updated (the `seamResidualClosed'` route); the exact joints Wolfram-mirrored (365  $\rightarrow$  368: the LR/glue locality integers, the KLM  $\mu$ -arithmetic, the sixteen-fold pin, the  $k_Y=5/3/b_1$  identities); cited in `tfpt_1` ( $\alpha$  fixed point + EM-Ward residual prose), `tfpt_research_contracts` (seam residual + Lean section), `introduction` (closure ledger), `tfpt_5_redteam` (Target A residual wording) and the README/website status surfaces; exploration provenance `experiments/tfpt-discovery/closure_route_*.py` (2026-07-02 note in `experiments/next.txt`). Suite ALL CHECKS PASSED through v470. Curated version bumped to **v5.4** (Zenodo deposit updated).

**2026-07-02 (v467/v468 — two named post-hoc [O] flavor candidates, record unchanged: the three  $\sim 2\sigma$  mixing tensions as *one* third-generation  $-\varphi_0$  pattern, and the splitting ratio  $\Delta m_{21}^2/\Delta m_{31}^2 = |J_{\text{PMNS}}|$ )**

- **v467\_thirdgen\_mixing\_pattern.py (FLAV.THIRDGEN.PATTERN.01).** The suite’s three  $\sim 2\sigma$  mixing tensions ( $\sin^2 \theta_{13} + 2.0\sigma$ ,  $\sin^2 \theta_{23} + 1.76\sigma$ ,  $|V_{cb}| + 1.79\sigma$ ; v307) are exactly the three *third-generation* channels and share *one* relative deviation  $-5.36\% \pm 1.67\%$  — non-zero at  $3.2\sigma$ ,  $-0.03\sigma$  from  $-\varphi_0$ . [E] vocabulary identities:  $\lambda_C^2 = \varphi_0(1-\varphi_0)$  (the frozen Cabibbo);  $\sin^2 \theta_{23} = (1-\varphi_0)/2 \Leftrightarrow \cos 2\theta_{23} = \varphi_0$ ; the suppression size  $\varphi_0 = 5.3\%$  sits inside the seam-gapped band  $(2/3)^6 = 8.8\%$  (v393). [N] the ladder-base forms  $\lambda_C^2 e^{-5/6} = 0.021880 / (1-\varphi_0)/2 = 0.473414 / \lambda_C^2 / (1+\lambda_C) = 0.041119$  drop the joint four-observable  $\chi^2$  from 10.4 to 0.11 with *zero* new parameters;  $|V_{ub}|$  (no bare  $\varphi_0$ ) and the 1–2 channels stay untouched and correct (dressing  $\lambda_C$  would fail at  $-24\sigma$  — scope checked, not chosen); Daya Bay alone already prefers the pattern form ( $\Delta\chi^2 = 4.3$ ). [C] **honest**: post-hoc (5/16 equally-simple dressings fix the two PMNS tensions;  $(1-\varphi_0)/(1-c_3)/\text{resolvent}$  degenerate  $<1.4\%$ ). [O] no seam-transport derivation (the Bernoulli-variance reading  $\lambda_Y^2 = \text{Var}[\text{Bernoulli}(\varphi_0)]$  is an exact identity but only an interpretation) — the frozen record (v84/REG.FREEZE.01) is *unchanged*. [X] pre-registered, symmetric:  $\theta_{13}$  stable at 0.0231 kills the pattern, at  $\sim 0.0219$  (sub-1%) kills the record’s bare- $\varphi_0$  reading, exclusive  $|V_{cb}| \sim 0.0395$  kills both — JUNO-era reactors + Belle II decide. Python-only.
- **v468\_dm2\_ratio\_jarlskog.py (FLAV.DM2RATIO.01).** The splitting ratio  $\Delta m_{21}^2/\Delta m_{31}^2$  was previously *unpredicted*; the frozen PMNS matrix (v9/v268/v270,  $\delta = 4\pi/3$ ) has one canonical derived CP invariant, and [N]  $|J_{\text{PMNS}}| = 0.029653$  meets the NuFIT 6.0 measured ratio  $0.029805 \pm 0.000792$  at  $-0.19\sigma$  (equivalently  $m_2/m_3 = \sqrt{|J|} = 0.17220$  vs  $0.17264(229)$ ) — zero

dials. [N] the informal `tfpt_2` table heuristic  $m_2/m_3 \sim \pi\varphi_0 = 0.16704$  (a “ $\sim$ ” entry, never frozen) is now  $-2.44\sigma$  and is *superseded as the comparator*. [C] spectrum consistency:  $\Sigma m_\nu = m_3(1 + \sqrt{|J|}) = 0.0588$  eV reproduces the documented 0.0586 eV (v272); no absolute number is added. [O] no mechanism (the  $\mu\tau$  texture gives  $\theta_{13}=0$  at leading order). [X] JUNO shrinks the window  $\sim 4\times$ ; adopting the v467 pattern would shift  $|J| \rightarrow 0.028849$  ( $-1.21\sigma$  today) — the two candidates *cross-discriminate* at JUNO precision. Python-only.

- **Integration.** Both registered in `run_all.py` and `script_registry.csv` (cluster neutrinos); ledger rows `FLAV.THIRDGEN.PATTERN.01` and `FLAV.DM2RATIO.01` (both [O] candidates); cited in `tfpt_2` (§ Neutrinos: pressure-point block + new candidate paragraphs, the  $m_2/m_3$  table row retyped to the  $|J|$  comparator, the octant prose kept honest — record does not select, the candidate would), `introduction` (closure ledger), `tfpt_4_frontier` (live kill-test board) and `tfpt_5_redteam` (seed-hyperplane caption); v328’s “single outlier” wording scoped to the seed observables; exploration provenance `experiments/tfpt-discovery/pattern_hunt_pmns_dressing.py` (2026-07-02 note in `experiments/next.txt`); Python-only (flagged in the Wolfram README).

**2026-07-01 (v466 — a sixth seed channel from a new measurement sector: the charged-lepton mass ratio  $m_e/m_\mu = \frac{12}{7}\varphi_0^2$  back-solves the same axiom seed to  $-0.11\%$ , extending the empirical over-determination to *five* sectors)**

- **v466\_seed\_leptonmass\_channel.py** (`SEED.LEPTONMASS.01`). The seed over-determination of v306/v465 used five observables spanning three measurement sectors (neutrino mixing, CMB, quark mixing). This module adds the *charged-lepton mass* sector: the ratio  $m_e/m_\mu = \frac{12}{7}\varphi_0^2$  is a sixth independent readout of the *one* axiom seed  $\varphi_0 = 1/(6\pi) + 48c_3^4$ . [E] plumbing: the axiom  $\varphi_0$  reproduces the frozen  $m_e/m_\mu = \frac{12}{7}\varphi_0^2 = 0.00484673$  to  $10^{-9}$ . [N] **new sector (headline):** back-solving  $\varphi_0$  from the measured *pole* ratio  $m_e/m_\mu = 0.004836$  (PDG 2024) gives  $\varphi_0 = 0.053113$ , only  $-0.11\%$  from the axiom — a fourth measurement sector beyond {neutrino mixing, CMB, quark mixing}. [C] **transport caveat:** the formula is the *source* ratio and the datum the *pole* ratio; the source→pole shift is the seam-gapped correction class (bound  $(2/3)^6 = 8.78\%$ , v393) but largely cancels for  $m_e/m_\mu$ , so the 0.11% residual sits far inside the band — the channel is added as a *consistent new sector*, not a sharper pin. [N] **extends:** the six-channel cluster (v306’s five + this) has  $\chi^2/\text{dof} < 1$  and the error-weighted mean stays at the axiom to  $< 0.1\%$ ; with the corroborating heavy-quark  $m_t/m_b$  ( $-0.23\%$ ) it is a seven-channel, five-sector agreement. [N] **power:** a data↔formula shuffle explodes the cluster  $\chi^2$  by  $> 100\times$ , and the scheme-ambiguous  $m_c/m_s = (34/47)/\varphi_0$  (FLAG 11–14) is deliberately not counted. [C] **honest:** theoretically still the *one* seed (v305 multiplicity 1) — an *empirical* strengthening, not theoretical independence;  $\varphi_0$  is fixed by the axioms, so it upgrades *no* prediction and closes *no* gate. Python-only (floats).
- **Integration.** Registered in `run_all.py` and `script_registry.csv` (cluster registry); ledger row `SEED.LEPTONMASS.01`; cited in `tfpt_5_redteam` (the seed-cross-validation firewall winbox, alongside v305/v306/v465); Python-only (flagged in the Wolfram README).

**2026-07-01 (v465 — the cross-sector one-parameter seed test: a CMB birefringence angle and a quark Cabibbo angle point to the same seed within  $0.01\sigma$ , the  $\theta_{13}$ -independent externally-runnable slice of v306)**

- **v465\_seed\_crosssector\_joint.py** (`SEED.CROSSECTOR.01`). The sharpest falsifiable signature of a *shared-origin* theory is a correlation no non-shared-origin model predicts. In the Standard Model the cosmic-birefringence rotation  $\beta$  (a CMB *EB/TB* polarisation observable) and the Cabibbo angle  $\lambda_C = |V_{us}|$  (a quark-flavour observable) are unrelated; TFPT reads both as functions of the *one* axiom seed  $\varphi_0 = 1/(6\pi) + 48c_3^4$ . [E] plumbing: the axiom  $\varphi_0$  reproduces the frozen  $\lambda_C$ ,  $\sin^2 \theta_{12}$ ,  $\beta$  to  $10^{-9}$ . [N] **cross-sector (headline):** back-solving  $\varphi_0$

*independently* from  $\beta$  (CMB; ACT DR6 + Planck PR4 combined) and from  $\lambda_C$  (quark; PDG 2024) gives the same seed within  $0.01\sigma_{\text{comb}}$ . [N] **one-parameter fit**: the error-weighted mean of the three back-solved  $\varphi_0$  reproduces the axiom to 0.06%; the joint  $\chi^2$  at the axiom (zero free parameters, 3 dof) is 0.007,  $\chi^2/\text{dof} < 1$ , combined pull  $0.08\sigma$ . [N]  $\theta_{13}$  **exclusion declared**: adding  $\sin^2 \theta_{13}$  raises  $\chi^2$  by  $\sim 4$  (the documented  $\sim 2\sigma$  pressure point, v328/v338), so this slice is the honest  $\theta_{13}$ -independent statement, not a hidden dilution. [N] **power**: a data $\leftrightarrow$ formula shuffle explodes the cluster  $\chi^2$  by  $> 5 \times 10^4$ . A consistency / out-of-sample statistic (a subset of v306);  $\varphi_0$  is fixed by the axioms, so it upgrades *no* prediction and closes *no* gate. Python-only (floats).

- **Integration.** Registered in `run_all.py` and `script_registry.csv` (cluster registry); ledger row `SEED.CROSSECTOR.01`; cited in `tfpt_5_redteam` (the seed-cross-validation firewall winbox, alongside v305/v306) and in the `origin_theory` birefringence dependency chain; Python-only (flagged in the Wolfram README).

**2026-06-30 (v463/v464 + Lean + Wolfram — the  $c=8$  holomorphic-uniqueness pin and the one-particle realisation rigidity: the SEAM.EQUIV.01 residual reduced to *entirely certification* on both halves)**

- **The bird’s-eye reframing.** The recent G-block modules all sharpened the *target* of the scaling limit; this round attacks the two halves that actually remain. The *continuum existence* is already citable (v458 MMST + v459 AGT/AMT); what was implicit was (B) *why* a  $c=8$  holomorphic limit must be  $(E_8)_1$ , and (A) the *realisation* input  $R_1$  that the abstract seam IS the concrete quasi-free collar. v463 closes B; v464 reduces A to its one-particle data.
- **v463\_seam\_c8\_holomorphic\_uniqueness.py (SEAM.EQUIV.UNIQ.01).** The  $c=8$  holomorphic-uniqueness pin. [E]  $c=8$  is NOT unique at level 1:  $c = \dim \mathfrak{g}/(1+h^\vee)=8$  has THREE simple solutions  $A_8$  ( $\dim 80$ ),  $D_8=\mathfrak{so}(16)$  ( $\dim 120$ ),  $E_8$  ( $\dim 248$ ), so the  $\det K=4$   $SO(16)$  rival is a genuine same- $c$  competitor. [E] HOLOMORPHY FORCES  $\dim V_1=248$ : the only weight-0 form with leading  $q^{-1/3}$  and integer coefficients is  $j^{1/3}=E_4/\eta^8$ ,  $q^1$  coefficient 248 (the 1-dim candidate space, vacuum  $a_0=1$ ). [E]  $\dim 248+$  level-1  $c=8 \Rightarrow E_8$  uniquely ( $248/(1+30)=8$ ); tower  $\{1, 248, 4124, 34752, 213126\}=E_4/\eta^8=j^{1/3}$ . [C] Dong–Mason/Schellekens: the holomorphic  $c=8$  VOA is unique  $= V_{E_8}$  (even unimodular rank-8 lattice unique  $= E_8$ , Mordell–Witt) — so the identification half is classification-forced.
- **v464\_seam\_oneparticle\_rigidity.py (SEAM.EQUIV.ONEPARTICLE.01).** The one-particle rigidity attacking  $R_1$ . The seam is quasi-free (v113/v160/v161), so by Araki self-dual CAR the net is the second quantisation of its one-particle symbol  $P$ . [E] RIGIDITY: the gapped ground state is the UNIQUE quasi-free state with  $P=$  projection onto  $K<0$  ( $P^2=P$  to  $< 10^{-12}$ ),  $K$  fixed by gap + chirality + reflection (v426). [E] ONE-PARTICLE LIMIT EXISTS: the kernel is Cauchy on a fixed window ( $\max |P_{2N}-P_N| \sim 1/N \rightarrow 0$ ). [E] the limit is  $c=1$  Dirac: entanglement  $S(L)=(c/3) \ln L$  gives  $c \rightarrow 1$ , 16 copies  $\Rightarrow c_-=8$  (ties v450). [C] Shale–Stinespring makes the Bogoliubov map implementable, so  $R_1$  is upgraded from a bald axiom to “the unique quasi-free realisation, modulo the named CAR functor”. Python-only (numpy).
- **Lean (SeamResidualAxiom.lean).** Adds the v463 kernel facts `c8_three_candidates` ( $A_8/D_8/E_8$  each  $\dim=8(1+h^\vee)$ ) and `holomorphy_selects_e8` (forced  $\dim V_1=248=E_8 \neq 80, 120$ ) to `residual_arithmetic` (still [propext]-only); v464 is numerical. `#print axioms seamResidualClosed` still lists exactly FOUR axioms.
- **Wolfram + integration.** v463 mirrored (the three candidates, the forced 248, the  $E_4/\eta^8$  tower); the extension count is now 365/365. Integrated across the contracts paper (attacks (xxx)/(xxxi)), README, zenodo, ledger and website. Net effect: across (v)–(xxxi) the SEAM.EQUIV.01 residual is now *entirely certification* — a named, hypothesis-audited package

(MMST, AGT/AMT, Dong–Mason, Araki, Shale–Stinespring) with no open internal mechanism — yet SEAM.EQUIV.01 stays [O] because it still rests on cited continuum-existence theorems we do not re-prove.

**2026-06-30 (v461/v462 + Lean axiom merge — the strict-locality obstruction made explicit, the 128-spinor extension exhibited at character level, and the SEAM.EQUIV.01 residual reduced to ONE realisation axiom + ONE combined cited theorem)**

- **v461\_seam\_strict\_locality.py (SEAM.S3.LOCALITY.01).** Sharpens v460’s verdict from a *cited* statement to an *exhibited* topological obstruction on the same v367  $p+ip$  collar. [E] CHERN INTEGER: the FHS Chern is  $C=+1$  (chiral  $M=1$ ), 0 (trivial  $M=3$ ),  $-1$  (opposite mass). [E] WILSON-LOOP / WANNIER-CENTRE WINDING = CHERN: the hybrid Wannier centre  $\theta(k_y)$  winds by  $|C|=1$  (chiral) and 0 (trivial) over the BZ; exponentially-localised Wannier exist IFF the winding is 0 (Brouder–Panati et al., PRL 98 046402; Thouless), so the chiral collar has NONE. [E] OBSTRUCTION INTEGERS TIED: winding =  $|C|=1 \neq 0$  and  $c_- = 8 = g_{\text{car}} + N_{\text{fam}} = \text{rank } E_8 \neq 0$  (the same 8 as  $c_3$ ’s one-sided count, v456). [E] NEG CONTROL: BOTH phases are gapped, yet only the trivial (winding 0) admits exp-localised Wannier and a strict finite-range commuting projector — it is the CHIRALITY, not the gap. [C][O] Kapustin–Fidkowski (arXiv:1810.07756): a strictly finite-range commuting projector would force winding 0,  $C=0$ ; since  $C=1 \neq 0$  NONE exists, so v424 sub-step (i)’s realisation is PROVABLY the quasi-local NPW26 LTO net (strict locality topologically forbidden, not a missing premise). SEAM.EQUIV.01 stays [O]. Python-only (numpy); the winding/Chern integers Wolfram- and Lean-mirrored.
- **v462\_seam\_spinor\_continuum.py (SEAM.EQUIV.SPINOR.01).** The constructive companion of the v458 (MMST audit) / v459 (lattice-VOA) residual: the 128-spinor extension  $SO(16)_1 \rightarrow (E_8)_1$  exhibited three ways. [E]  $(E_8)_1$  TOWER FROM  $SO(16)_1$  SECTORS: the exact Jacobi/ $E_8$  identity  $\theta_2^8 + \theta_3^8 + \theta_4^8 = 2E_4$  gives  $\chi_{(E_8)_1} = E_4/\eta^8 = \chi_o^{SO16} + \chi_s^{SO16}$  — the holomorphic  $(E_8)_1$  character IS the  $SO(16)_1$  vacuum + spinor sector sum. [E] THE 128 SPINOR WEIGHT-1 STATES FILL  $120 \rightarrow 248$ :  $\chi_o$  gives 120,  $\chi_s$  gives 128, summing to the exact tower  $\{1, 248, 4124, 34752\}$  ( $248=120+128$ ). [E] FINITE 16-MAJORANA FOCK:  $C(16, 2)=120$  NS bilinear currents and  $2^{16/2}=256$  Ramond ground states = 128 spinor + 128 cospinor. [E] FINITE- $L \rightarrow$  CONTINUUM CONVERGENCE: the critical free-fermion ring Casimir gives  $c_- \rightarrow 8$  and the lowest scaled NS gap  $x_1 \rightarrow 1$  with  $|x_1 - 1| \sim 1/L^2$  ( $L=64..1024$ ). [C][O] AGT/AMT (v459) supply the rigorous extension; SEAM.EQUIV.01 existence (v336) stays [O]. Mixed exact ( $\theta$  identity + tower + 120/128/256, Wolfram-mirrored) + numerical (finite- $L$ , Python-only).
- **Lean (TfptCarrier/SeamResidualAxiom.lean, FORM.SEAM.RESIDUAL.01).** The two former cited axioms `mmst_existence`  $\circ$  `agt_lattice_extension` are MERGED into ONE combined external axiom `seam_realisation_theorem` (dropping the intermediate `ChiralCFT_c8`), so `#print axioms seamResidualClosed` now lists exactly FOUR axioms (down from six). Two more kernel facts are added axiom-free: `strict_locality_forbidden` (v461: winding =  $|C|=1 \neq 0$ ,  $c_- \neq 0$ ) and `fock_sectors` (v462:  $C(16, 2)=120$ ,  $2^8=256$ , spinor 128,  $120+128=248$ ); `residual_arithmetic` (now including both) depends only on `[propext]`. `lake build` OK. So the residual is reduced to ONE realisation axiom + ONE combined cited theorem.
- **Wolfram. tfpt\_readouts\_extension.wl** adds 3 exact mirrors (total 363/363): the v461 obstruction integer-logic and the v462  $\theta$ -identity +  $120+128=248$  sector / Fock counts.

**2026-06-30 (v460 — the explicit flat-band commuting-projector (LTO) realisation of the S3 input: advancing the SEAM.EQUIV.01 sub-step (i) from abstract existence to an explicit gap-protected quasi-local projector net carrying the  $\mathbb{Z}_2$  reflection and the  $\mu_4$  clock)**

- **v460\_seam\_s3\_lto.py (SEAM.S3.LT0.01).** NPW26 (arXiv:2605.10693) prove the intrinsic Bisognano–Wichmann lemma for *commuting-projector* (local-topological-order) models with a  $\mathbb{Z}_2$  reflection; v424 reduced the keystone BW residual to that theorem modulo two open sub-steps, of which (ii) was settled by v426 and the continuum lift of BW by the uniform-in- $N$  tower v455. This module attacks the *commuting-projector* side of sub-step (i) on the same explicit v367  $p+ip$  collar. [E] FLAT-BAND PARENT: the flattened Bloch  $Q=h/|h|$  has  $Q^2=I$  (flat  $\pm 1$  bands) and  $P=(I-Q)/2$  is an exact projector EQUAL to v367’s occupied ground-state projector — a frustration-free commuting-projector parent with the same gapped ground state (machine precision over a  $k$ -grid). [E] GAP-PROTECTED QUASI-LOCALITY: the gapped projector kernel  $|P(n,0)|$  decays rapidly ( $\xi \approx 1.14$  at  $M=1$ ,  $\exp R^2=0.995$ ,  $|P(20,0)| < 10^{-6}$ ) and  $\xi$  grows as the gap closes (2.7 at gap 0.02); at the gapless point it decays as a clean power law  $|R|^{-2.1}$  ( $R^2=0.999$ ) — locality needs the gap. [E]  $\mathbb{Z}_2$  REFLECTION +  $\mu_4$  INHERITED: the flattened modular Hamiltonian  $K_{\text{flat}}=\text{sign}(K)$  inherits  $\Theta K_{\text{flat}} \Theta = -K_{\text{flat}}$  ( $u_\Theta=J$ ) and  $[\rho, K_{\text{flat}}]=0$  EXACTLY (sign is odd;  $\rho$  commutes with any  $f(K)$ ), with side-even and off- $\mu_4$  neg controls firing. [C][O] the chiral  $c_-=8 \neq 0$  (FHS Chern  $|C|=1$ ) obstructs STRICT finite-range commuting-projector locality (Kapustin–Fidkowski, arXiv:1810.07756;  $C \neq 0$  forbids exponentially-localised Wannier), so the realisation is the quasi-local LTO net NPW26 use — the form sub-step (i) asks for. Advances v424 sub-step (i); SEAM.EQUIV.01 stays [O] (the continuum theorem MMST v336 is the backbone). Python-only (numpy); registered in `run_all.py`, the registry, the ledger and cited in `tfpt_research_contracts` (route (xxvii) of the convergent attacks).

**2026-06-30 (v454–v459 + Lean + Wolfram — the post-F “G-block”: the lattice Sugawara current algebra, a uniform-in- $N$  Tomita–Takesaki tower, chirality forced by  $P_1$ , an  $(E_8)_1$  character kill test, the exact MMST citation audit and the complementary lattice-VOA route)**

- **v454\_seam\_edge\_virasoro.py (SEAM.EQUIV.EDGE.VIRASORO.01).** Tests the cited MMST “Koo–Saleur lattice generators  $\rightarrow$  continuum Virasoro with the correct  $c$ ” fact on our own free-fermion edge (G2). [E] the Affleck–Cardy Casimir energy fits  $c_{\text{Dirac}}=1.000$  ( $\frac{1}{2}$  per Majorana,  $L=64..1024$ ); [E] the lattice  $U(1)$  current closes its affine level exactly ( $\langle 0|J_n J_{-n}|0\rangle=n$ ,  $n=1..8$ ; a gapped control vanishes); [E] the level-1 Sugawara  $c=\dim G/(1+h^\vee)=8$  for *both*  $SO(16)_1$  (120/15) and  $(E_8)_1$  (248/31). [C]  $16 \cdot \frac{1}{2}=8=g_{\text{car}}+N_{\text{fam}}$  is the level-1 current-algebra  $c_-$ ; SEAM.EQUIV.01 stays [O]. Sugawara Wolfram-mirrored.
- **v455\_seam\_bw\_uniform.py (SEAM.EQUIV.BW.UNIFORM.01).** A uniform-in- $N$  Tomita–Takesaki tower for the intrinsic Bisognano–Wichmann condition  $u_\Theta=J$  — the inductive-limit companion to v426, advancing v424 sub-step (i) (G1). [E]  $\|\Theta K \Theta + K\|=0$  exactly at every  $N$  ( $N=8..128$ ); [E] an  $N$ -independent gap  $6 \log \frac{3}{2}$ ; [E]  $[\rho, K]=0$  exactly at every  $N$ ; [E] an  $N$ -independent neg-control margin  $2\|K\|$ . [C][O] the inductive limit inherits  $u_\Theta=J$ ; SEAM.EQUIV.01 stays [O]. Python-only.
- **v456\_seam\_chirality\_from\_c3.py (SEAM.S3.FROM-P1.01).** The edge chirality ( $c_- \neq 0$ , the S3 phase) is *forced* by the one-sidedness that defines  $c_3=1/(8\pi)$  (P1) (G6). [E]  $c_3$ ’s “8” is the one-sided count  $8=|\mathbb{Z}_2| \cdot (fK/\pi)=2 \cdot 4$ ; [E] a reflection sends the Chern integer  $C \rightarrow -C$  ( $+1 \rightarrow -1$  on the  $p+ip$  model); [E] a two-sided boundary forces  $C=-C=0$ . [C] the one-sided seam (no reflection) forces  $C \neq 0$ , magnitude  $c_-=8=\text{rank } E_8$  — S3 is a consequence of  $P_1$ ; SEAM.EQUIV.01 existence stays [O]. The arithmetic is Wolfram-mirrored.
- **v457\_seam\_character\_killtest.py (SEAM.EQUIV.KILLTEST.01).** An explicit charac-

ter/sector kill test of  $(E_8)_1$  against  $SO(16)_1$  (G7). [E] the  $(E_8)_1$  character  $\chi=E_4/\eta^8$  with tower degeneracies  $\{1, 248, 4124, 34752, \dots\}$ ; [E] the weight-1 count  $248=120+128$  (not the free 120); [E] the sector count  $|\det \text{Cartan } E_8|=1$  (not  $|\det \text{Cartan } D_8|=4$ ). [X] a measured 120, four sectors, or a wrong coefficient would *falsify*  $(E_8)_1$ ; the data give  $248/1/\{1, 248, 4124\}$  (passed). SEAM.EQUIV.01 stays [O]. Wolfram-mirrored.

- **v458\_seam\_mmst\_citation\_audit.py** (SEAM.EQUIV.MMST.AUDIT.01). The exact hypothesis-by-hypothesis citation audit of MMST (arXiv:2107.13834) against the collar (G3, the “exakter Zitat-Abgleich”). [C] H1 CAR class, H2 massless limit, [E] H3 per-copy  $c=\frac{1}{2}$  (Thm 4.11), [C] H4 decoupled additivity to  $c=8$ , H5 the WZW range  $8 \leq 8 \leq 16$  with the 120 bilinear currents all match; [O] the single residual H6 is the 128-spinor extension  $SO(16)_1 \rightarrow (E_8)_1$  ( $\det K 4 \rightarrow 1$ ), outside MMST’s bilinear method. SEAM.EQUIV.01 stays [O], its open part narrowed to this one holomorphic extension.
- **v459\_seam\_lattice\_voa\_route.py** (SEAM.EQUIV.LATTICEVOA.01). The second, independent route to  $(E_8)_1$  supplying exactly the residual of v458 (G5). [E]  $E_8$  even unimodular ( $\det=1$ ); [E] the weight-1 space  $248=8+240$ ; [E] the 240 roots split 112 integer + 128 half-integer (the spinor,  $248=120+128$ ). [C] the lattice net  $A_{E_8}$  (Adamo–Giorgetti–Tanimoto, arXiv:2506.01008; OS arXiv:2407.18222) constructs the holomorphic extension; [O] the two routes leave one shared realisation input (collar =  $A_{E_8}$ , tied to  $P_1$  by v456). SEAM.EQUIV.01 stays [O]. Wolfram-mirrored.
- **Lean** (FORM.SEAM.RESIDUAL.01, `TfptCarrier/SeamResidualAxiom.lean`). Kernel-checks the whole G-block arithmetic (the one-sided  $8=|\mathbb{Z}_2| \int K/\pi$ , the reflection  $C \rightarrow -C$ ,  $c_-=8$ , the MMST range  $8 \leq 8 \leq 16$ , decoupled additivity,  $248=8+240=120+128$ ,  $240=112+128$ ,  $\det K 1$  vs 4) by `decide`, reducing the SEAM.EQUIV.01 residual to ONE named TFPT-internal realisation axiom (`CollarRealizedAsLatticePhase`) plus two cited external theorems (MMST v458, AGT v459); lake build OK.
- **Wolfram + integration**. Six exact mirrors added to `tfpt_readouts_extension.wl` (Sugawara  $c=8$  for 120/15 and 248/31; the one-sided  $8=2 \cdot 4$  and reflection  $C=-C \Rightarrow 0$ ; the  $(E_8)_1$  tower  $\{1, 248, 4124, 34752\}$  with  $248=120+128$  and  $|\det \text{Cartan}| 1$  vs 4; the root split  $112+128$  and  $248=8+240$ ) — extension total now 360/360. Registry, ledger, master index, website mirror and manifests updated in the same change.

**2026-06-30 (v449–v453 + Lean + Wolfram — the edge CFT named end-to-end: uniform-in- $N$  convergence, three independent reads of  $c_-=8$ , the Ising operator tower, the  $(E_8)_1$  modular data, and  $\mu_4$ -symmetry derived from the four marks)**

- **v449\_seam\_mmst\_uniform.py** (SEAM.EQUIV.MMST.UNIFORM.01). Strengthens the cited MMST hypothesis from convergence at a single cross-ratio (v444/v448) to the *uniformity* the theorem actually asserts, on the same v367/v368 collar. [E] cross-ratio A  $Q(1, 2, 4, 7)/12 \rightarrow 2+\sqrt{3}$ , the deviation shrinking monotonically; [E] an *independent* cross-ratio B  $Q(1, 3, 5, 9)/12 \rightarrow$  its own value 2, so convergence is not a tuned point; [E] a uniform  $1/N$  rate  $|Q(N)-Q_\infty| \cdot N \leq 4$  for *both* observables at every  $N_x \in \{36, 48, 60\}$  (one  $N$ -independent constant); [E] the bulk control is off-scale. [C] the MMST uniform-convergence hypothesis is empirically satisfied; SEAM.EQUIV.01 stays [O].
- **v450\_seam\_edge\_entanglement.py** (SEAM.EQUIV.EDGE.ENTANGLEMENT.01). A *third* independent reading of the edge central charge (after the correlator v444 and the topology v447). [E] the Calabrese–Cardy entanglement law gives  $c_{\text{Dirac}}=1.000$  for the critical free-fermion chain; [E] so  $c$  per Majorana =  $1/2$ ; [E] a gapped control saturates ( $c=0$ ). [C] bulk–edge:  $c_-=N_{\text{Maj}} \cdot \frac{1}{2}=16 \cdot \frac{1}{2}=8=g_{\text{car}}+N_{\text{fam}}$ . SEAM.EQUIV.01 stays [O].
- **v451\_seam\_edge\_cardy\_tower.py** (SEAM.EQUIV.EDGE.CARDY.01). Names the edge CFT by its *operator content*. [E] the Cardy finite-size spectrum gives the spin dimension  $h_\sigma=1/16$  (the

periodic–antiperiodic ground-energy split) and [E] the energy dimension  $h_e=1/2$  (the lowest NS pair excitation), both [E]  $L$ -independent (conformal  $1/L$  scaling). [C]  $\{c, h_\sigma, h_e\}=\{\frac{1}{2}, \frac{1}{16}, \frac{1}{2}\}$  is the chiral Ising (free-Majorana) minimal model  $M(3, 4)$ ; 16 copies give  $c_-=8$ . SEAM.EQUIV.01 stays [O].

- **v452\_seam\_e8\_modular.py** (SEAM.EQUIV.E8.MODULAR.01). The  $(E_8)_1$  torus modular signature, extending v156's  $q$ -expansion to the transformation data. [E] the character  $\chi=E_4/\eta^8$  is  $S$ -invariant  $\chi(-1/\tau)=\chi(\tau)$  (one primary,  $S=[1]$ ); [E]  $T$ -phase  $\chi(\tau+1)/\chi(\tau)=e^{-2\pi i/3}=e^{-2\pi i c/24}$  ( $c=8$ ); [E] leading power  $\chi q^{1/3} \rightarrow 1$  so  $q^{-c/24}=q^{-1/3} \Rightarrow c=8$ ; [E] holomorphic  $c=8 \equiv 0 \pmod{8}$  with  $T$ -phase order  $24/\gcd(8, 24)=3$ . [C] the  $(E_8)_1$  identity is now pinned on the torus *and* the edge. SEAM.EQUIV.01 stays [O].
- **v453\_seam\_mu4\_from\_marks.py** (SEAM.RIGIDITY.MU4FROMMARKS.01). Folds the last structural premise of the rigidity chain (v446's residual QGEO.SYM.01) into the existing realisation premise. [E] the four Gauss–Bonnet marks *are*  $\mu_4$ = roots of  $z^4-1$ ; [E] the order-4 clock  $z \rightarrow iz$  is a 4-cycle permuting them (the v445 clock); [E] it preserves the cross-ratio  $\lambda=2$  (in the  $D_4$  stabiliser, v168); [E] the form basis  $\omega_k=z^{k-1}dz/(z^4-1) \rightarrow i^k\omega_k$  is  $\mu_4$ -graded (weights  $(1, 2, 3)=A_3$  exponents). [C] so QGEO.SYM.01 follows from marks= $\mu_4$  + the existing QGEO.REALIZE.01 — no new premise; the chain (v453→v446→v445) closes modulo realisation. SEAM.EQUIV.01 stays [O].
- **Lean: TfptCarrier/SeamEdgeChern.lean** (FORM.SEAM.EDGE.CHERN.01). Kernel-hardens the integer backbone the four edge reads converge on: the FHS Chern integers  $(+1/0/-1)$ , the bulk–edge channel count  $|C|=1$ ,  $N_{\text{Maj}}=2^{g_{\text{car}}-1}=16$ , the assembly  $2c_-=N_{\text{Maj}} \cdot |C|$  (so  $c_-=8$ ),  $c_-=8=g_{\text{car}}+N_{\text{fam}}=\text{rank } E_8$ , holomorphic  $c \equiv 0 \pmod{8}$  and the order-3 modular  $T$ -phase — all by kernel `decide` (`edge_kernel_facts`, axioms `propext` only). `lake build` OK, `#print axioms` clean.
- **Wolfram mirror**. `tfpt_readouts_extension.wl` gains nine exact checks (v452 modular data  $\times 4$ , v450 entanglement assembly, v451 Ising Kac weights, v453 marks= $\mu_4 \times 3$ ): 345 → 354 checks, all passing (GATE.WOLFRAM.02).  $E_4$  is evaluated via its  $q$ -series since `EisensteinE` stays symbolic for complex  $\tau$ .
- **Papers**. The adversarial audit (`tfpt_5_redteam`, Target A) and the research-contracts note record the edge CFT as named end-to-end — central charge  $c_-=8$  from three independent observables (correlator v444, topology v447, entanglement v450), the Ising operator tower (v451), the  $(E_8)_1$  torus modular data (v452), and the  $\mu_4$  premise derived from the marks (v453) — while the only residual that stays [O] remains the cited continuum-existence theorem (v336).

**2026-06-30 (v446/v447/v448 + Lean + Wolfram — deriving clock-invariance from  $\mu_4$ -symmetry, an independent topological reading of the edge  $c_-=8$ , and pinning the one external MMST fact at four-point order)**

- **v446\_seam\_clock\_invariance.py** (SEAM.RIGIDITY.CLOCKINV.01). Closes v445's single residual one notch further: the operator condition  $[\rho, \Lambda_\Sigma]=0$  (clock-invariance) is shown to be a *consequence* of a structural symmetry, not an independent assumption. [E] the OS canonical transfer  $\Lambda=C(1)C(0)^{-1}=e^{-h}$  (fixed by RP+reflection, v54) of a  $\mu_4$ -invariant RP covariance satisfies  $[\rho, \Lambda]=0$  (swept  $N=4..32$ ); [E] a  $\mu_4$ -breaking covariance gives  $[\rho, \Lambda] \neq 0$  (iff-sharp neg control); [E]  $\Lambda$  is a positive contraction (spectrum in  $(0, 1]$ ); [E] the free modular Hamiltonian  $K=\log((1-C)C^{-1})$  (v258) of the invariant state has  $[\rho, K]=0$  (Tomita–Takesaki), broken control  $\neq 0$ . [C] so with v445 (commute  $\Leftrightarrow$  block-diagonal), RP-definability *plus* the four marks *force* the block-diagonal  $\Lambda_\Sigma$ ; the residual reduces from an operator commutation to “the seam RP data is  $\mu_4$ -symmetric” (QGEO.SYM.01). SEAM.EQUIV.01 stays [O].
- **v447\_seam\_edge\_chern.py** (SEAM.EQUIV.EDGE.CHERN.01). An *independent* (topological)

reading of the edge central charge, corroborating **v444**'s correlator route from a different observable. [E] the  $M=1$  collar has integer Chern number  $C=+1$  (Fukui–Hatsugai–Suzuki link-variable method, no fit); [E]  $C=0$  in the trivial phase and  $C=-1$  at  $M=-1$  (opposite chirality); [E] the open strip carries exactly  $|C|=1$  chiral edge branch per boundary; [E] the Li–Haldane entanglement spectrum has in-gap modes at  $1/2$  only in the topological phase. [C] bulk–edge:  $c_- = N_{\text{Maj}} \cdot \frac{1}{2} \cdot |C| = 16 \cdot \frac{1}{2} = 8 = g_{\text{car}} + N_{\text{fam}}$  — the same  $c_- = 8$  as **v444**. SEAM.EQUIV.01 stays [O].

- **v448\_seam\_edge\_fourpoint.py** (SEAM.EQUIV.EDGE.FOURPOINT.01). Pins the one external MMST fact (continuum existence of the chiral scaling limit) empirically at the next order. [E] the ground-state correlation matrix is an exact projector  $G^2=G$  ( $\sim 10^{-13}$ ), so the state is quasi-free (the MMST CAR hypothesis holds); [E] the edge four-point cross-ratio  $Q=[G_{01}G_{23}]/[G_{02}G_{13}]$  at commensurate positions is Cauchy-convergent under refinement; [E]  $Q$  converges to the free-fermion conformal cross-ratio  $\sin(5\pi/12)/\sin(\pi/12)=2+\sqrt{3} \approx 3.732$ ; [E] the bulk control is off-scale (exponential). [C] so the one external fact is supported at *both* two-point (**v444**) and four-point order. SEAM.EQUIV.01 stays [O].
- **Lean: TftCarrier/SeamRigidityForcing.lean** (FORM.SEAM.RIGIDITY.FORCING.01). Kernel-hardens **v445** (and **v442**'s uniformity): the entrywise forcing iff  $\text{clock4 } a = \text{clock4 } b \Leftrightarrow a=b$  (i.e.  $i^a=i^b \Leftrightarrow a \equiv b \pmod{4}$ , residue-only hence uniform in  $N$ ), the four distinct eigenphases, the exact commutant dimension  $\sum n_s^2=64<256$  and the order discriminator ( $128>64$ ) are all proved by kernel **decide** (**forcing\_kernel\_facts**, no domain axioms); only the operator-level lift to the full  $L^2$  is the named premise. lake build OK, #print axioms clean.
- **Wolfram mirror. tfpt\_readouts\_extension.wl** gains three exact **v445** checks (entrywise forcing for all  $a, b \in 0..31$ ; commutant dimension  $64<256$ ; order-2 commutant  $128>64$  swept  $N=4..64$ ):  $342 \rightarrow 345$  checks, all passing (GATE.WOLFRAM.02).
- **Papers.** The adversarial audit (**tfpt\_5\_redteam**, Target A) records the five-direction narrowing of the SEAM.EQUIV.01 keystone (edge scaling **v444**, forcing converse **v445** + Lean, derived clock-invariance **v446**, independent topological  $c_- = 8$  **v447**, four-point pinning **v448**) explicitly as “not promoted”: the only residual that stays [O] is still the cited continuum-existence theorem (**v336**). The research-contracts note lists the same as “eleven convergent attacks”.

**2026-06-30 (v445 — the rigidity FORCING converse: commuting with the order-4 clock is *exactly*  $\mu_4$ -block-diagonality, upgrading the rigidity step from “permit” to “force”)**

- **v445\_seam\_rigidity\_forcing.py** (SEAM.RIGIDITY.FORCING.01). The rigidity route (**v398/v442**) showed a  $\mu_4$ -block-diagonal  $K$  *commutes* with the order-4 clock  $\rho = \text{diag}(i^n)$ . This module proves the *converse* and the exact structure, isolating the single remaining physical premise. [E] FORCING (entrywise iff):  $[\rho, K]=0$  iff  $K_{ij}=0$  whenever  $i \not\equiv j \pmod{4}$ , verified entrywise and swept  $N=4..128$  — commuting *forces*  $\mu_4$ -block-diagonality, it does not merely permit it. [E] EXACT COMMUTANT DIMENSION:  $\dim\{K : [\rho, K]=0\} = \sum_s n_s^2$  (the  $\text{ad}_\rho$  nullspace), a proper subspace of the full  $N^2$  algebra ( $N=16$ :  $64<256$ ) — a genuine quantitative restriction. [E] OFF-BLOCK  $\Leftrightarrow$  NON-COMMUTING: every off-block matrix unit has  $\|[\rho, E_{ij}]\| \geq \sqrt{2}>0$  (zero slack). [E] THE ORDER IS THE DISCRIMINATOR: the order-4 clock gives 4 sectors; the order-2 clock leaves a strictly larger commutant ( $N=16$ :  $64$  vs  $128$ ) — the four Gauss–Bonnet marks (order  $4=|\mu_4|=h(A_3)$ ) force strictly more structure than two, and only the index-4 extension is  $(E_8)_1$  ( $\det=1$ ; **v281/v154/v351**). [E] NONDEGENERACY: exactly 4 distinct eigenvalues  $\{1, i, -1, -i\}$  for all  $N \geq 4$ . [C] SYNTHESIS: commuting with the order-4  $\mu_4$  clock is *equivalent* to  $\mu_4$ -block-diagonality, so RP-definability (**v54/v398**) + clock-invariance *force* the block-diagonal  $\Lambda_\Sigma$  — **v442**'s “permit” upgraded to “force”.

[O] the single residual is now precisely that the physical transfer lies in that commutant (clock-invariance = QGEO.SYM.01, v323/v335); SEAM.RIGIDITY.01/SEAM.EQUIV.01 stays [O]. Python-only (numpy; det facts Wolfram-mirrored via v89/v281/v422). Registered in `run_all.py`, `script_registry.csv`, the ledger, and cited in `tfpt_research_contracts` (route (xii) of the convergent attacks, and the sharpened residual in the Seam State Rigidity contract).

## 2026-06-30 (Lean kernel-hardening of two of the seven SEAM.EQUIV.01 routes — the Borchers/Wiesbrock standard-pair relations and the applicability-ledger count, machine-verified with clean axioms; rigor without new mathematics)

- **Context.** Two of the seven convergent attacks on the keystone residual are turned from numerical witnesses into kernel-checked facts, in the audit-contract style of `SeamScalingLimit.lean` / `BWKeystone.lean`: the load-bearing algebra/bookkeeping is proved by the Lean kernel, while the cited theorems and the one open continuum input remain named axioms. `lake build` OK; `#print axioms` clean. Tracked as two new FORM.\* ledger rows; `\veri`-context in `tfpt_research_contracts`; Lean manifest refreshed.
- **FORM.SEAM.BW.HSMI.01 (TfptCarrier/SeamStandardPair.lean) — standard-pair relations, kernel-checked (hardens v438).** The four Borchers/Wiesbrock relations on the centred 5-mode ladder over  $\mathbb{Z}$  are proved by kernel `decide` (axioms `propext` / `Classical.choice` / `Quot.sound` only, no domain axiom):  $[K, P]=P$ ,  $J^2=1$ ,  $JKJ=-K$ ,  $JPJ=P_+$  (and  $\text{tr } K=0$ ). The cited Borchers/Wiesbrock theorem and the one continuum input (the continuum standard pair) are named axioms; the composition pins face (ii) of SEAM.EQUIV.01 to exactly those two — intrinsic Bisognano–Wichmann, no rotation-covariance presupposed.
- **FORM.SEAM.APPLICABILITY.01 (TfptCarrier/SeamApplicabilityLedger.lean) — the audit count, kernel-checked (hardens v441).** The headline bookkeeping is itself machine-verified: of the 12 cited-theorem hypotheses (MMST/NPW26/Adamo, mirrored in order) exactly 11 are internal and 1 external (`total_count=12`, `internal_count=11`, `external_count=1`), plus `mmst_range` ( $8 \leq 8 \leq 16$ ) and `npw_outside_bucket` ( $\det K=1 \neq 4$ ) — ALL with NO axioms (pure kernel `decide`). The single external fact (continuum scaling-limit existence) is the named axiom; the keystone reduction pins its external dependence to exactly that one fact.
- **Net.** Both modules are imported by the root `TfptCarrier.lean`, axiom-audited in `AxiomCheck.lean`, and signature-locked in `AuditContract.lean`; full `lake build` succeeds. No new mathematics — the same facts v438/v441 verify numerically are now kernel-proved; SEAM.EQUIV.01 stays [O].

## 2026-06-30 (the chiral-edge scaling motor v444 — the edge half of the one SEAM.EQUIV.01 continuum residual: the lattice edge already carries the conformal data of a chiral $c=1/2$ Majorana CFT and converges under refinement, grounding the EDGE the way v439 grounded the BULK; the keystone stays [O])

- **Context.** v439 grounded the *bulk* half of the cited MMST existence (uniform gap, exponential clustering, finite light cone) and explicitly narrowed the one open analytic step to the chiral massless *edge* scaling limit. v444 attacks exactly that residual on the SAME explicit v367/v368  $p+ip$  collar. vN-registered, runs in `run_all.py` (suite ends ALL CHECKS PASSED), `\veri`-cited in the body of `tfpt_research_contracts` (the  $G_{\text{net}}$  contract). Python-only (numpy; no new exact identity to Wolfram-mirror).
- **v444\_seam\_edge\_scaling.py (SEAM.EQUIV.CONTINUUM.EDGE.01) — chiral-edge motor.** The lattice edge carries the kinematic and conformal data of a chiral  $c=1/2$  Majorana CFT: [E] a linear, chiral in-gap dispersion (velocity  $v \approx 0.98$ ,  $R^2=0.9999$ , with *opposite* per-edge

velocities — one chiral Majorana per edge); [E] an *algebraic* equal-time edge two-point  $\sim r^{-\eta}$  with  $\eta \approx 1$ , versus the *exponential* bulk row and the exponential trivial-phase ( $M=3$ , no edge) control; [E] a Cauchy-convergent scaling-collapse under refinement ( $\eta=1.001 \pm 0.007$ , successive profile spread  $0.028 \rightarrow 0.013$  across  $N_x=36..60$ ) — the numerical hallmark of an existing scaling limit. [C]  $\eta=1 \Leftrightarrow$  one chiral Majorana ( $c=1/2$ ) per edge, so the 16-copy carrier has a chiral  $c_-=8$  ( $E_8$ )<sub>1</sub> edge CFT ( $c_-=g_{\text{car}}+N_{\text{fam}}$ ). [O] the rigorous uniform-convergence theorem (cited MMST, v336) is not supplied.

- **Net.** With v439 (bulk) and v444 (edge), *both halves* of the one continuum residual are numerically grounded on our explicit model and pinned to a single cited existence theorem; SEAM.EQUIV.01 stays [O]. Ledger row, `script_registry.csv` index and the generated surfaces (`verification.tex`, `ScriptIndex.tsx`) updated in the same change; manifest refreshed last.

**2026-06-30** (six convergent attacks on the one SEAM.EQUIV.01 continuum residual — intrinsic Bisognano–Wichmann, a Lieb–Robinson continuum motor, the  $\beta$ -homotopy, an applicability audit, uniform rigidity, and falsifiable kill tests; none closes the keystone, together they reduce it to a single analytic fact and make the claim falsifiable)

- **Context.** The keystone SEAM.EQUIV.01 is “closed modulo a cited theorem”: its one open input is the continuum existence of the seam’s chiral massless scaling limit. Six new modules attack that residual from independent directions; each is honest about *not* closing it. All six are vN-registered, run in `run_all.py` (suite ends ALL CHECKS PASSED), and are \veri-cited in the body of `tfpt_research_contracts` (the  $G_{\text{net}}$  contract). Python-only (no new exact identity to Wolfram-mirror).
- **v438\_seam\_hsmi\_borchers.py** (SEAM.EQUIV.BW.HSMI.01) — **intrinsic BW.** The seam supplies a Longo *standard pair* / +half-sided modular inclusion (positive lightlike translation  $P \geq 0$ , boost  $K$  with  $[K, P]=P$ , reflection  $J$  with  $JKJ=-K$ ,  $JPJ=P_-$ ), so by Borchers’ theorem the modular flow is geometric *intrinsically* — *without* presupposing rotation-covariance of the vacuum (complementary to Adamo–Giorgetti–Tanimoto). [E] ladder algebra exact,  $sl(2, \mathbb{R})$  positivity (min eig  $> 0$ ); [C] synthesis; [O] the continuum standard pair.
- **v439\_seam\_lieb\_robinson.py** (SEAM.EQUIV.CONTINUUM.LR.01) — **continuum motor.** The explicit gapped  $p+ip$  collar (v367) has a uniform-in- $N$  gap, exponential clustering with  $N$ -independent  $\xi$ , and a finite Lieb–Robinson light cone, so by Nachtergaele–Sims its *bulk* thermodynamic limit exists — grounding the bulk half of the cited MMST existence in our model and narrowing the open step to the chiral *edge* scaling limit. [E]/[C], [O] the edge CFT.
- **v440\_seam\_lto\_rp\_beta.py** (SEAM.EQUIV.LTORP.BETA.01) —  **$\beta$ -homotopy.** The (LTO-RP)/BW data ( $\Theta K \Theta = -K$ ,  $[\rho, K]=0$ ) are  $\beta$ -independent, and along the whole path  $C_\beta=1/(1+e^{\beta K})$  from the NPW26-proven *trace* state ( $\beta=0$ ) to the seam KMS state ( $\beta=1$ ) every covariance is reflection-related,  $\mu_4$ -invariant and valid — a norm-continuous, obstruction-free homotopy, upgrading the single point of v426 to a path argument for sub-step (ii) of v424.
- **v441\_seam\_applicability\_ledger.py** (SEAM.EQUIV.APPLICABILITY.01) — **audit.** A hypothesis-by-hypothesis ledger of the three cited theorems (MMST / NPW26 / Adamo–Giorgetti–Tanimoto) finds 11 of 12 hypotheses are TFPT-internal computed facts and exactly one is external — the continuum existence of the chiral scaling limit; covariance is an Adamo consequence (defuses the v215 circularity), not an input. The keystone’s external dependence is pinned to that one analytic fact.
- **v442\_seam\_rigidity\_uniform.py** (SEAM.RIGIDITY.UNIFORM.01) — **uniform rigidity.**

The band-limited  $\mu_4$ -character rigidity of v398 is uniform in  $N$ : exact 4-sector commutation to  $N=256$  (not merely the  $N \leq 64$  sweep), a uniform gap, and an  $N$ -independent clock-breaking lower bound  $\|[\rho, V]\|=2$  — so it is not a small- $N$  artifact (the full- $L^2$  forcing stays [O]).

- **v443\_seam\_e8\_discriminator.py (SEAM.E8.DISCIMINATOR.01) — kill tests.** Mode D of the closure-mode classification (v347) as concrete *falsifiable* discriminators against the nearest  $c=8$  rival  $SO(16)_1$ : torus ground-state degeneracy 1 vs 4, level-1 currents 248 vs 120, lattice theta 240 vs 112 norm-2 roots;  $c=8$  alone cannot decide (v277), only  $\det K=1$  (the genus-1 statement, v90) selects  $E_8$ . A seam with  $GSD>1$  would falsify  $(E_8)_1$ . [X]/[C].
- **Status discipline.** SEAM.EQUIV.01 stays [O] throughout; the six modules reduce, audit and falsify the residual but do not supply the one cited continuum-existence theorem. Ledger rows, the `script_registry.csv` index, and the generated surfaces (`verification.tex`, `ScriptIndex.tsx`) are updated in the same change; manifest refreshed last.

**2026-06-30 (shadow-export coherence — the tfpt5-shadow mirror is now reproducible *as shipped*: the audit is shadow-aware, the content maps are non-destructive on the website-free subset, and an export gate blocks any incoherent mirror)**

- **Context (reviewer finding, Alessandro).** On the exported `tfpt5_shadow.zip` the independent surfaces reproduced (`build.sh notes` → 11 ok; Wolfram base/extension; Red Team; Lean transcript), but the *package* layer did not: `make_manifest.py -check` reported a stale `verification/website_map.csv` and `build.sh audit` crashed because it unconditionally required `website/lib/release.ts`, which the shadow subset deliberately excludes.
- **make\_docs\_map.py is now shadow-aware.** `website_map.csv` mirrors the `website/` sources; on a tree without `website/` the generator now *skips* it instead of regenerating it down to a single `README.md` row. This removes the silent corruption that made `build.sh notes/gen` leave `website_map.csv` stale against the manifest on the export (it now matches the established skips in `make_script_index.py` and `make_changelog_web.py`).
- **audit\_sync.py is now shadow-aware.** When `website/` is absent it skips the website mirror (section D), the website version stamps (E) and the website Wolfram-count card (F), and the website generated-surface freshness in (A), printing exactly which sections were skipped. `build.sh audit` therefore runs to a clean AUDIT OK on the shadow subset instead of raising `FileNotFoundError`.
- **Export coherence gate.** `scripts/export-shadow.sh` now re-runs `make_manifest.py -check` and the shadow-aware `audit_sync.py` *on the filtered tree* before the mirror is published. The manifest check catches the earlier v235–v298 symptom at the source: a module referenced by `run_all.py/script_registry.csv/manifest.sha256` but not committed (and thus dropped by `git ls-files`) now fails the export loudly instead of shipping a package that says ALL CHECKS PASSED while `run_all.py` stops at the first missing script.
- **Manifest refreshed.** `manifest.sha256` was regenerated as the last release step so the shipped digests match the committed `changelog.tex` and `website_map.csv`; the full Python suite reproduces end-to-end on the regenerated export. No theory content, status marker or vN module changed — this is an export/infrastructure fix only.

**2026-06-29 (discoverability — an SEO/GEO pass on the public website: a generated `/llms.txt`, explicit AI-crawler `robots.txt`, query-targeted metadata, and the first real backlinks from the README and the Zenodo deposit)**

- **New `/llms.txt` context file** for AI / answer engines (ChatGPT, Perplexity, Claude, Gemini), generated at build time from the same data layer as the site (`papers.ts`, `predictions.ts`,

`version.ts`) so it never drifts. It carries the one-line summary, the “claims / does not claim” split, the document set with PDF links, the headline predictions, and an explicit *disambiguation* from the unrelated Brouwer–Lefschetz “topological fixed point theory” of mathematics.

- **robots.txt** now lists the major search and AI crawlers (Googlebot, Google-Extended, Bingbot, GPTBot, ChatGPT-User, OAI-SearchBot, PerplexityBot, ClaudeBot, anthropic-ai, ...) explicitly as allowed; the site-wide meta description and keywords are reworded toward natural search queries (“derivation of the fine-structure constant”, “Standard Model from first principles”, “where does 137 come from”) and carry the same physics/mathematics disambiguation.
- **Backlinks (the missing piece).** The root `README.md` and the Zenodo deposit description (`zenodo_description.html`) previously contained *no* outbound links; both now link to `fixpoint-theory.com` and the source repository, and the `README` gains a citation block with the archived deposit DOI 10.5281/zenodo.20846087.
- Website / documentation infrastructure only — no theory, suite, ledger or status change.
- **New module v437 (E8.DEGREE.JOINT.01) — a consolidation of v6/v66/v355/v431 that derives more from the  $E_8$  Casimir degrees *without* crossing the v354/v355 anti-numerology line.** It adds *no* new per-degree coincidence — only one joint lock and a reconstruction.
  - **[E] all of  $E_8$  is fixed by its degree multiset.** From  $\deg(E_8) = \{2, 8, 12, 14, 18, 20, 24, 30\}$  alone:  $\text{rank} = \# \deg = 8$ ,  $h = \max \deg = 30$ ,  $\# \text{positive roots} = \sum \exp = 120$ ,  $\# \text{roots} = 240$ ,  $\dim E_8 = 248$ ,  $|W(E_8)| = \prod \deg = 696,729,600$ ,  $\sum \deg = 128 = \dim S^+$ .
  - **[E] the joint quadratic (the new lock).** The two TFPT structural integers ( $g_{\text{car}}, N_{\text{fam}}$ ) are the *unique* root pair of  $x^2 - (\text{rank } E_8)x + (h/2) = x^2 - 8x + 15 = (x-3)(x-5)$ : sum of roots =  $\text{rank } E_8 = 8$  (the rank-fill, v6), product =  $h/2 = 15 = g_{\text{car}}N_{\text{fam}}$  ( $2g_{\text{car}}N_{\text{fam}} = h$ , v431);  $\{3, 5\}$  is the unique positive-integer factor pair of 15 summing to 8. So  $g_{\text{car}} = 5$  and  $N_{\text{fam}} = 3$  are forced *together* by two  $E_8$  degree-invariants, not chosen one at a time. The three primary-readout degrees  $\{2, 8, 30\}$  are exactly  $\{\min, \text{rank}, \max\}$ .
  - **[E] discipline upheld:** no new per-degree mining beyond the joint quadratic; the five non-primary degrees keep their v431 family homes; the v354/v355 line is not crossed.
- **Surfaces updated.** Registered in `run_all.py`, `script_registry.csv` and `status_ledger.csv` (E8.DEGREE.JOINT.01); cited in the `tfpt_5` degree-audit winbox beside v431; mirrored in `wolfram/tfpt_readouts_extension.wl` (extension 339  $\rightarrow$  342). Exact integer/lattice; no new physical number, no status change.

**2026-06-29 (v436 — hardening the unconditional “is it numerology?” floor to an assumption-minimal counting floor; plus a literature note grounding the  $\alpha^3$  Chern level)**

- **New module v436 (OVERDET.FLOOR.02) — it *hardens* OVERDET.FLOOR.01 (v432) so the numerology verdict no longer rests on any subjective chance assignment.** v432’s nominal  $\sim 10^{-10}$  multiplies the counting census by hand-chosen plausibilities (input-“8” independence  $0.1^3$ , seed 0.012, kaon 0.15) a skeptic can contest; v436 shows the conclusion survives without them.
  - **[E] assumption-minimal floor.** The  $\alpha$  cubic-class census (v100 layer D) is *pure counting*: of  $N = 94500$  complexity-matched  $F_{U(1)}$  variants only *one* lands in the  $4 \times 10^{-8}$  CODATA window — the TFPT instance, reproducing  $\alpha^{-1} = 137.0359992$  — so  $p_\alpha \leq 1/94500 = 1.06 \times 10^{-5} \sim 4.40\sigma$  with *no* subjective probability, prior or multi-observable grammar.

- **[E] monotone concession ladder.**  $-\log_{10} p$  from generous to adversarial: full scorecard+ $\alpha \sim 30.7 \rightarrow$  scorecard-only  $\sim 25.8 \rightarrow \alpha$ -census-only  $\sim 4.98$ ; each rung uses strictly less evidence and the “not chance ( $> 4\sigma$ )” verdict holds at the *most adversarial* rung, so it never rests on one contestable piece. Stripping *all* of v432’s subjective factors ( $\sim 1.8 \times 10^{-6}$ ) leaves the floor at  $4.40\sigma$  — they only sharpen it.
- **[C] honest  $5\sigma$  gap.** The conventional  $5\sigma$  line ( $p = 5.7 \times 10^{-7}$ ) is crossed unconditionally by the census times exactly *one* further independent factor  $\leq 0.054$  (the input-“8” four-fold over-determination, or one foreign witness) — stated, not hidden. **[C]** Firewall unchanged: bounds only “is it numerology?”, not the physics bridge (v187).
- **Literature note (research log, not a status change).** A `next.txt` entry grounds the  $\alpha^3$  Chern-level reading of v435 in named theorems — Dai–Freed ( $\eta$ -invariants and determinant lines, the exponentiated  $\eta$ -invariant as a section of the inverse Quillen determinant line), Bismut–Freed (determinant-line curvature = the  $\alpha^2$  Calderón side) and Redlich/Stora–Zumino (the cubic  $\text{Tr } F^3$  anomaly descends to a Chern–Simons level, quantised to  $\mathbb{Z}$ ). It records that the EM1 proof obligation reduces to constructing the seam Dirac family — i.e. it is a *facet* of SEAM.EQUIV.01 — and remains **[O]** external math (no fabricated closure).
- **Surfaces updated.** Registered in `run_all.py`, `script_registry.csv` and `status_ledger.csv` (OVERDET.FLOOR.02); cited in the `tfpt_safeguards` “unconditional floor” keybox beside v432; mirrored on the website. Python-only; no new physical number, no status change.

**2026-06-29 (v435 — a fourth honest step on the  $\alpha^{-1}$  Quillen target: a  $\pi$ -power test isolates the cubic  $\alpha^3$  as the unique metric-independent (topological) rung, and types its coefficient as a conditional integer level)**

- **New module v435 (ALPHA.QUILLEN.PROGRESS.04) — it attacks the *single* remaining **[O]** left after v434 (residual (2), the cubic  $\alpha^3$  Chern/Maxwell moment) and *sharpens* it; it does *not* close it.** The target ALPHA.QUILLEN.EXACT.01 (v382) stays **[O]**;  $\alpha^{-1} = 137.0359992$  stays **[E]**.
  - **[E]  $\pi$ -power / metric-independence test.** The three EM-closure coefficients carry  $\pi$ -powers exactly  $\{0, 3, 6\}$ , equal to their  $c_3$ -orders: Maxwell  $k_0 = 1 \rightarrow \pi^0$ , Calderón  $-2c_3^3 = -1/(256\pi^3) \rightarrow \pi^3$ , transport  $-8b_1c_3^6 = -41/(327680\pi^6) \rightarrow \pi^6$ . So the Maxwell  $a^3$  is the *unique*  $\pi^0/c_3^0$  (metric-independent) term. A heat-kernel boundary coefficient always carries the  $(4\pi)^{-d/2}$  measure; a metric-independent term carries none — the signature of a *topological* term. The test thus partitions the closure into *one* topological level ( $a^3$ ) + *two* heat-kernel boundary terms, confirming v342/v434’s “ $a^3$  is a level, not a Seeley–DeWitt curvature integral” from a *test* rather than a claim.
  - **[C] conditional level quantisation.** If the  $a^3$  term is the seam  $U(1)$  Chern–Simons/ $\eta$  boundary level (the EM1 reading, v48), large-gauge invariance quantises the level to  $\mathbb{Z}$ , so  $k_0 = 1$  is the *unit* (minimal) level — an integer, not a tunable real. This converts “the coefficient happens to be 1” into “the coefficient is an integer level and 1 is the unit”.
  - **[O] not closed:** the EM1 lemma itself — the from-first-principles Chern–Simons boundary proof that  $\partial_\tau \log Z_{\text{Maxwell}} = a^3$  — stays external math (type A, v384); the seam  $F$ -normalisation stays the one **[C]** (residuals 1 & 3, v434);  $\alpha^{-1}$  stays **[E]**.
- **Surfaces updated.** Registered in `run_all.py`, `script_registry.csv` and `status_ledger.csv` (ALPHA.QUILLEN.PROGRESS.04); cited in the `tfpt_1` EM-closure subsection beside v433/v434; mirrored on the website residual-gap lab. Python-only; no new physical number.

**2026-06-28 (v434 — a third honest step on the  $\alpha^{-1}$  Quillen target:  $b_1$  is the  $a_4$  heat-kernel coefficient, and the three residuals collapse to one [C] + one [O])**

- **New module v434 (ALPHA.QUILLEN.PROGRESS.03)** — it settles the status of the three residuals named after v433 and shows they are *not* three independent open problems. The target ALPHA.QUILLEN.EXACT.01 (v382) stays [O];  $\alpha^{-1} = 137.0359992$  stays [E].
  - [E]  $b_1$  *is* the  $U(1)$   $a_4$  heat-kernel coefficient (the  $\beta = a_4$  theorem). The one-loop  $\beta$  function of a gauge coupling *is* the  $a_4$  (Seeley–DeWitt  $a_2$  in  $d=4$ ) heat-kernel coefficient of the charged-field operator (Vassilevich). Computed from the carrier hypercharge content with the standard spin weights ( $\frac{2}{3}$  per Weyl fermion,  $\frac{1}{3}$  per complex scalar),  $\sum \frac{2}{3}Y_f^2 + \frac{1}{3}Y_s^2 = 41/6$  (SM) and  $(3/5) \cdot \text{that} = 41/10 = b_1$  (GUT). So the  $b_1$  in the counterterm  $8b_1c_3^6$  is the  $a_4$  coefficient of the carrier  $U(1)_Y$  content (the same number as the  $\beta$ , v159/v246), the *same kind* of heat-kernel object as the  $\alpha^2$  Calderón  $a_4$  (v342) — not a fitted multiplicity.
  - [E] **the geometry factors off:**  $8b_1c_3^6 = b_1 \cdot (\text{rank } E_8) \cdot c_3^6$ , with  $\text{rank } E_8 = 8$  the  $\chi=2$  seam boundary count (v216) and  $c_3^6$  the six boundary insertions (the  $\pi$ -power test, v342).
  - [C] **residual (1)  $\equiv$  residual (3).** With  $b_1$  and the  $c_3$ -powers heat-kernel-forced, the only freedom left in the exact seam  $a_4$  value is how the gauge curvature couples to the boundary measure — the seam  $F$ -normalisation. So “the actual seam  $a_4 = 8b_1c_3^6$ ” and “the seam  $F$ -normalisation” are *one* [C] obligation, not two.
  - [E] **residual (2) located (not closed):** the determinant-line variation factors  $a^3 - 2c_3^3a^2 = a^2(a - 2c_3^3)$ ;  $a^2$  is the Gilkey gauge-curvature (Quillen/Chern) density (v342), so the leading  $a^3 = a \cdot a^2$  is the *first moment* of the Chern density — a topological level at boundary order  $c_3^0$ , not a Seeley–DeWitt curvature integral. Its from-first-principles (Chern–Simons-type) origin stays [O].
  - [E] **net reduction:** the EM-Ward residual is *one* [C] (residuals 1 & 3) + *one* [O] (residual 2), not three independent problems — a genuine narrowing of the open surface. [O] ALPHA.QUILLEN.EXACT.01 is *not* closed (the  $\alpha^3$  Chern level and the from-first-principles proof stay external, type A, v384).
- **Surfaces updated.** Registered in `run_all.py`, `script_registry.csv` and `status_ledger.csv` (ALPHA.QUILLEN.PROGRESS.03); cited in the `tfpt_1` EM-closure subsection beside v433; mirrored on the website residual-gap lab. Python-only; no new physical number.

**2026-06-27 (v433 — a second honest step on the  $\alpha^{-1}$  Quillen target: a solvable 4D model reaches the  $a_4$  order, connecting v391 and v342)**

- **New module v433 (ALPHA.QUILLEN.PROGRESS.02)** — a second honest step on the external target ALPHA.QUILLEN.EXACT.01 (v382); it *grounds and connects*, it does *not* close it. The target stays [O] (external math, type A, v384); the value  $\alpha^{-1} = 137.0359992$  stays [E]. The earlier honest attempt v391 gave a solvable model on a 1D circle whose  $\zeta$ -determinant variation reached only the low Seeley–DeWitt orders  $a_0, a_1$ ; separately, v342 grounded the quadratic Calderón term in the Gilkey gauge-curvature coefficient  $a_4$  ( $30/360 = \frac{1}{12}$ ,  $\Omega^2/12$ ). This module supplies the missing link between the two.
  - [E] **a solvable 4D model reaches the  $a_4$  order:** the flat operator  $\Delta = -\partial^2 + m^2$  on a 4-torus of volume  $V$  has heat trace  $\text{Tr } e^{-t\Delta} = V/(4\pi)^2 t^{-2} e^{-tm^2}$ , with Gilkey coefficients  $a_0 = V/(4\pi)^2$ ,  $a_2 = -Vm^2/(4\pi)^2$  and  $a_4 = Vm^4/(2(4\pi)^2)$ ; the  $t^0$  (conformal-anomaly) order  $a_4$  is *non-zero* — exactly the order v391’s 1D circle toy could not reach.
  - [E] **it is the order v342 uses:** so v391 (“a  $\zeta$ -determinant variation *is* closed heat-kernel data”) and v342 (“the seam term *is* the Gilkey  $a_4$ ”) are unified — the variation is heat-kernel data *at the  $a_4$  order*, not only the low orders v391 reached.

- **[E] the measure is  $c_3$ -built:** the universal 4D heat-kernel measure  $(4\pi)^{-2} = 1/(16\pi^2) = (2c_3)^2 = 4c_3^2$  in the seam’s one-sided 2D-boundary normalisation  $c_3 = \frac{1}{2}(4\pi)^{-1}$ ; and the v382 split is re-verified at  $\alpha^{-1} = 137.0359992$ .
- **[O] not closed:** computing the *actual* seam  $U(1)$   $\zeta$ -determinant  $a_4$  coefficient ( $= 8b_1c_3^6$ ) on the  $\mu_4$  seam moduli, and the origin of the cubic  $\alpha^3$  Maxwell/Chern moment, stay open (external math, type A, v384); the exact seam  $F$ -normalisation fixing the coefficient *value* stays **[C]** (v342). A second honest step that grounds and connects, it does not close.
- **Surfaces updated.** Registered in `run_all.py`, `script_registry.csv` and `status_ledger.csv` (`ALPHA.QUILLEN.PROGRESS.02`); cited in the `tfpt_1` EM-closure subsection beside v382/v391; mirrored on the website (the residual-gap lab). Python-only (like v391); no new physical number.

## 2026-06-27 (v431, v432 — the “unmapped” $E_8$ degrees are forced spine+flavor structure; the unconditional over-determination floor)

- **New module v431 (E8.DEGREE.LADDER.01) — the reverse-audit “unmapped”  $E_8$  Casimir degrees are NOT diffuse overhead but a forced two-family decomposition.** The structural complement of the reverse audit (v354/v355) and its sheet/deck complement (v430):  $\deg(E_8) = 6 \cdot \text{spine}\{2, 3, 4, 5\} \sqcup (\{2\} \cup \det\text{-ladder}\{8, 14, 20\})$ . **[E]** the two-family split is *forced* by  $h(E_8) = 30 = 2 \cdot 3 \cdot 5$  — the exponents are the  $\varphi(30) = 8$  totatives of 30, hence coprime to 6, hence  $\equiv \pm 1 \pmod 6$ , so the degrees occupy only the residue classes  $\{0, 2\} \pmod 6$ . **[E]** the  $6k$  family  $\{12, 18, 24, 30\}/6 = \{2, 3, 4, 5\}$  is the v91 spine; the  $6k+2$  family  $\{8, 14, 20\} = (\det R, \det C, \det L)$  is the winding line  $\det M(s, 0) = 6s + 8$  of the flavor diamond (v135). **[E]** “ $18 = 6N_{\text{fam}}$ ” is not a holdout (spine family), and the clean split is special to  $E_8$  among the simply-laced exceptionals ( $E_6/E_7/D_5$  fail). **[C] honest v355 line:** the arithmetic decomposition is exact, but the *functorial* claim “ $6k+2$  Casimir stratum = flavor sheet operators” needs a representation-theoretic map and stays **[C]** (12, 24 admit two canonical readings each) — a structure theorem, not a forced functorial flavor map. It reclassifies the v354 overhead to zero orphan degrees and closes no gate.
- **New module v432 (OVERDET.FLOOR.01) — the unconditional improbability floor and the  $L1 \leftrightarrow L2$  bridge.** The complement of the conditional null model (v100) and the multiply-vs-compress framework (v427/v428): the defensible over-determination number that survives even if a skeptic *rejects* the declared formula grammar. **[E]** compression removed (v428), the genuinely multiplicative pieces are the input “8” forced four independent ways plus the foreign witness  $\alpha^{-1} \sim 137$ . **[C]** multiplying only across disjoint pieces (input over-determination  $\times$  the  $\alpha$  census  $1/94500 \times$  the  $\varphi_0^{\text{ret}}$ -seed cluster counted once  $\times$  one downstream witness) gives a floor  $\leq 10^{-6}$  across a grid of generous assignments, nominal  $\sim 10^{-10}$ . **[E]** that floor sits  $\sim 20$  orders *above* the v100 conditional  $10^{-30.7}$ ; the gap is exactly the evidential value of proving the forms forced, so the  $\alpha$  form-derivation (`ALPHA.QUILLEN.EXACT.01`, v382) and this floor are the *same* lever. **[C] honest scope:** bounds only the “is it numerology?” question, not the physics bridge (firewall v187); a conservative floor, not a posterior.
- *Surfaces synced in the same change:* v431/v432 added to `run_all.py`, `script_registry.csv` (master index + website `ScriptIndex`), `status_ledger.csv`, `tfpt_5_redteam` (reverse-audit winbox + Casimir-degree figure) and `tfpt_safeguards` (the over-determination layer); v431 mirrored in `wolfram/tfpt_readouts_extension.wl` (v432 is a conservative bound, Python-only).

## 2026-06-27 (v430 — the seam’s “other side” is forced-disjoint from $E_8$ ’s unmapped Casimir degrees)

- **New module v430 (E8.OTHERSIDE.AUDIT.01) — the sheet/deck complement of the reverse audit (v354/v355).** The reverse audit found five  $E_8$  Casimir degrees  $\{12, 14, 18, 20, 24\}$  with no primary readout (“hull overhead”); the double cover has an *other side* (the one-sided  $\mathbb{Z}_2$  collar / conjugate half-spinor  $S^-$ ), so the sharp adversarial follow-up is “*is the leftover where the second sheet lives?*” The answer is a disciplined **negative**. [E] the deck is the degree-2 invariant: the one-sided cover contributes  $|\mathbb{Z}_2| = 2$  (the  $\frac{1}{2}$  in  $c_3 = 1/(|\mathbb{Z}_2| 4\pi) = 1/(8\pi)$ , v58/v216), and  $2 = \min \deg(E_8)$  is the quadratic/metric — one of the *matched* primary readouts  $\{2, 8, 30\}$ , never one of the unmapped degrees. [E] the two sheets are the 128-spinor:  $\dim S^+ = \dim S^- = 16$ , the spinor block  $128 = \text{rank } E_8 \cdot \dim S^+ = 2^{\text{rank}-1} = \sum \deg$ , the spinor half of  $248 = 120 + 128$ ;  $S^-$  enters 248 as the conjugate  $(\overline{16}, \overline{4})$  ( $128 = 2 \cdot 64$ ), not a spare singlet. [E] forced-disjoint: the sheet/deck set  $\{2, 16, 32, 128\}$  has empty intersection with  $\{12, 14, 18, 20, 24\}$ , its only degree contact the matched 2. [O] each unmapped degree, “explained” from sheet atoms, needs an unforced coefficient and admits  $\geq 2$  readings (the v355 discriminator), so there is no forced sheet→unmapped-degree identity.
- **A structural negative, not a status change.** It consolidates the  $S^-$ -dark-matter downgrade (v227) and the no-spare-singlet WIMP no-go of the frontier  $E_8$  scan (tfpt\_4\_frontier): the other side is matched conjugate matter, the five unmapped degrees are the hull overhead, and the two do not meet. The only live “other side” question — whether the 128 phase/glue channel hosts a dark sector (v227) — concerns *matched* structure, not the unmapped five. Closes no gate, upgrades nothing.
- **Surfaces synced.** run\_all.py, script\_registry.csv, status\_ledger.csv; the reverse-audit Outcome winbox + degree-audit table caption and the Target D winbox of tfpt\_5\_redteam, the magnitude/phase keybox of tfpt\_3\_e8\_audit\_bootstrap, the WIMP-no-go passage of tfpt\_4\_frontier; the E8 node of the website verification DAG; the Wolfram extension/README/ledger counts and the website verification count; manifests refreshed.

## 2026-06-26 (v429 — the ‘unmapped’ golden $E_8$ structure is the geometry of the one external input $\theta_i$ )

- **New module v429 (DM.AXION.PENTAGON.01) — a geometric re-reading answering “why does so much idle  $E_8$  structure exist?” for the one case where the idle structure has a job.** The reverse audit (v354) had argued the golden ratio  $\varphi$  is *unmapped* (it appears only internally — the affine- $E_8$  attractor angle v312, the icosian lattice v348 — and in no physical readout), and used exactly that to conclude  $\varphi$  cannot be numerology. [E] this sharpens that statement: because  $N_{\text{fam}} = g_{\text{car}} - 2$ , the axion misalignment *spine* angle (v211, DM.AXION.SPINE.01) is  $\theta_i = \pi N_{\text{fam}} / g_{\text{car}} = (g_{\text{car}} - 2)\pi / g_{\text{car}}$ , which is *exactly* the interior angle of the regular  $g_{\text{car}}$ -gon — for  $g_{\text{car}} = 5$  the pentagon,  $\theta_i = 3\pi/5 = 108^\circ$ . [E] its cosine is golden,  $\cos(3\pi/5) = (1 - \sqrt{5})/4 = -1/(2\varphi)$  (partner  $2\cos(2\pi/5) = 1/\varphi = \varphi - 1$ ,  $\varphi = 2\cos(\pi/5)$ ). [E] the golden character is *unique* to  $g_{\text{car}} = 5$ : among the small regular  $n$ -gons only  $n = 5$  has an irrational interior-angle cosine ( $n = 3, 4, 6$  give  $1/2, 0, -1/2$ ), so  $\theta_i$  is golden *because* the carrier is 5 (P2).
- **An honest [C] bridge, not a status change.** It connects two standing facts —  $\varphi$  is the unmapped  $g_{\text{car}} = 5$  signature (v354/v313) and  $\theta_i = 3\pi/5$  is the one external cosmological input (v211) — and *refines* v354 from “ $\varphi$  in no physical readout” (it is the unmapped  $g_{\text{car}} = 5$  signature, v313) to “ $\varphi$  touches *only* the [C] spine misalignment angle” (conditional, branch-level, never a frozen prediction). It is a geometric motive for the spine over the hilltop  $\theta_i \approx 170^\circ$  (whose cosine is not a clean pentagon value), consistent with — not replacing — the

finite- $T$  solver that decides the branch (v185/v211). It does *not* derive  $\theta_i$ , does *not* upgrade DM.AXION.SPINE.01 (stays [C]) and closes no gate.

- **Surfaces synced.** `run_all.py`, `script_registry.csv`, `status_ledger.csv`, the dark-matter section of `tfpt_4_frontier`; the Wolfram extension/README/ledger counts (GATE.WOLFRAM.02) and the website verification count; manifests refreshed.

## 2026-06-25 (External-works references appendix — web-verified)

- **New shared appendix `tex-artefacts/tfpt_references.tex`** (“External works built upon”), `\input` into the six math-heavy documents (`tfpt_1`, `tfpt_2`, `tfpt_4`, `tfpt_horizon_readouts`, `origin_theory`, `tfpt_research_contracts`). The active papers named the foundational results they build on by author only, with no reference list; the appendix supplies web-verified locators (journal, volume, year / arXiv) grouped by topic: modular theory & AQFT (Tomita–Takesaki LNM 128 1970; Bisognano–Wichmann J. Math. Phys. 16/17; Osterwalder–Schrader CMP 31/42; Doplicher–Haag–Roberts CMP 23/35; Haag–Ruelle; Longo CMP 126/130; Kawahigashi–Longo–Müger CMP 219), conformal nets/VOAs/lattices (Frenkel–Kac Invent. Math. 62; Segal CMP 80; MMST; Adamo–Moriwaki–Tanimoto; Adamo–Giorgetti–Tanimoto 2506.01008/2508.07109; Naaijken–Penneys–Wallick), lattices/quadratic forms (Conway–Sloane; Siegel Ann. Math. 36 1935), singularities (Brieskorn Invent. Math. 2 1966; Milnor AMS 61 1968) and spectral geometry/gravity/thermal time (Chamseddine–Connes CMP 186; Connes–Rovelli CQG 11; Wald PRD 48; Casini–Huerta–Myers JHEP 05 2011). Added to the `make_manifest.py` TEX list. No claim or status change — a provenance/citation completeness pass.

## 2026-06-25 (External AQFT citations + the four-fold $\mu_4$ identity sharpened)

- **Two recent Adamo–Giorgetti–Tanimoto theorems cited where the keystone builds on them.** The open SEAM.EQUIV.01 continuum/realisation legs (v336) now cite, alongside MMST and Adamo–Moriwaki–Tanimoto (arXiv:2407.18222, already in use): **(a)** the construction and classification of two-dimensional conformal nets from even lattices (arXiv:2506.01008, 2025) — “the lattice is a conformal net” is a theorem of the same operator-algebraic lineage, strengthening the lattice-realisation leg; and **(b)** automatic positive-energy/diffeomorphism covariance of conformal-net representations (arXiv:2508.07109, 2025), which defuses the “Bisognano–Wichmann presupposes the covariance it should produce” circularity (v215) on the intrinsic-BW residual (v424). Added to v336 docstring/registry, the SEAM.EQUIV.CONTINUUM.01 ledger `external_data`, and `tfpt_research_contracts`.
- **The four-fold  $\mu_4$  identity made explicit (v422).** v422 already proved that the seam deck/divisor  $\mu_4$ , the carrier Galois group  $\text{Gal}(\mathbb{Q}(\zeta_5))$ , the discriminant  $\text{disc}(D_5)=\text{disc}(A_3)=\mathbb{Z}/4$  and the  $(E_8)_1$  simple-current glue are ONE cyclic  $\mathbb{Z}/4$  (with the cyclic-vs-Klein negative control). It is now named the *four-fold  $\mu_4$  identity* and typed, per v428, as a *compression* (one object read four ways), NOT four independent witnesses — in v422, the registry, the SEAM.GALOIS.NET.01 ledger row and `origin_theory`. (No new module: a separate v429 would have duplicated v422 and violated the very compression discipline.)

## 2026-06-25 (v428 — honest self-correction of the over-determination map)

- **v428 (OVERDET.WITNESS.RECLASS.01) — applying v427’s own multiply-vs-compress test to v427 itself.** A pattern search and the right objection exposed that calling the seven arithmetic witnesses “disjoint grammars that *multiply*” overstates independence. [E] by the Brieskorn classification (v236) the  $(2, 3, 5)$  singularity is the ONE generator behind the order-30 clock (Milnor  $\mu=(2-1)(3-1)(5-1)=8=\text{rank } E_8$ ,  $30=2\cdot 3\cdot 5=h(E_8)$ ,  $\varphi(30)=8$ ), so

each witness is a facet of that same object (primes 2, 3, 5 of 30, the clock 30, or  $E_8$ ): all seven collapse to one generator  $\Rightarrow$  *compression*, not seven multiplications. [E] what genuinely multiplies is the input forced four independent ways — the “8” in  $c_3=1/(8\pi)$  from rank  $E_8$ ,  $h(D_5)=2(5-1)$ ,  $\varphi(30)$  and the Milnor number — plus the foreign witness  $\alpha^{-1}\approx 137$ . [C] refines and partly walks back v427, consistent with v236/v305/v313.

- **Surfaces corrected (same change).** v427’s prose, the ledger row (OVERDET.WITNESS.MAP.01, now noting the refinement), tfpt\_safeguards Layer 3 (the seven compress one object; a second keybox states what multiplies), the witness-map figure (redrawn), the website (papers.ts, Safeguards.tsx, release.ts, the intro-video description) and the Zenodo description. The over-determination claim is now the narrower, harder-to-attack “one richly over-determined input + a foreign readout”.

## 2026-06-25 (Safeguards visuals + paper expansion + the website safeguards band)

- **Five safeguards figures** (make\_figures.py, in manifest.sha256): the layered defense, the  $\alpha^{-1}$  uniqueness scan (one  $F_{U(1)}$  variant of 94,500 in the CODATA window), the reverse audit (3/8 vs 5/8), the witness-independence map (seven disjoint grammars), and the look-elsewhere null model (13/13 vs  $\leq 5/13$ ).
- **tfpt\_safeguards expanded.** Layer 5 now states the null model explicitly ( $\prod p_i=10^{-25.8}$ ,  $2\times 200k$  pseudo-theories  $\leq 5/13$ ,  $\varphi_0^{\text{ret}} \pm 10\% \rightarrow 6/13$ , shuffle 1.2) and the  $\alpha$  uniqueness ( $\leq 10^{-30.7}$ ,  $\sim 102$  bits, v100), with the data watchdog (v307) and kill board (v321); the five figures are embedded.
- **Website: a prominent animated Safeguards band** (components/Safeguards.tsx) in the home flow (after the trust beat) — headline statistics, the seven layers, the figures, links to the paper and the suite — to address the coincidence/numerology suspicion proactively.
- **Koide flow-time investigation (honest negative).** Post-v425: the flow-time is *not* native and does *not* reduce to one scale — v99 already rejects a scale reading as look-elsewhere; the only theory-side content is the conditional integer reading  $t=N_{\text{fam}}=3 \Rightarrow m_\tau=1776.94$  MeV, a Belle II kill test. No new module.

## 2026-06-25 (Three closure-strategy modules v425–v427 + the Safeguards paper)

- **v425 (DYN.TRANSFER.UNIVERSAL.01) — the single-flow reduction of  $F_{\text{transfer}}$ .** The four frontier transfers do not each need a separate external solver: their *dynamics* is the ONE native flow (the seam modular/recovery semigroup, v238/v240, gap  $6 \ln(3/2)$ ) restricted to each sector. [E] the v99 Koide generator linearised at  $q=2$  has rate  $-\Delta$ , time-1 contraction  $(2/3)^6$ ; Boltzmann washout is that contraction (CP phase  $4\pi/3$  native, v320); QCD rate  $b_3=-7$ . [C] only the anchors ( $v_{\text{geo}}$ ,  $C_p$ ,  $M_R$ ) stay external;  $F_{\text{relic}}$ ’s cosmological IC  $\theta_i$  is the one wall. A reduction (extends v383 static $\rightarrow$ dynamic), not internalisation; the firewall holds.
- **v426 (SEAM.EQUIV.LTORP.KMS.01) — the invertible  $\beta=1$ -KMS variant of (LTO-RP).** The corner NPW26 leave open (they prove it for topologically-ordered models; the seam is invertible + KMS, not a trace). [E] a gapped reflection-symmetric  $\mu_4$ -even collar at  $\beta=1$  KMS satisfies  $\Theta K \Theta^{-1} = -K$  ( $u_\Theta = J$ ) and  $[\rho, K]=0$ , so (LTO-RP) holds by direct Tomita–Takesaki;  $|\det \text{Cartan}(E_8)|=1$  (no anyons), state not a trace; neg controls have teeth. Advances v424 sub-step (ii); SEAM.EQUIV.01 stays [O] (continuum sub-step (i) remains).
- **v427 (OVERDET.WITNESS.MAP.01) — the honest over-determination accounting.** A reviewer correction made machine-explicit: witnesses from *disjoint* grammars multiply, projections of the *same* generator compress. [E] seven disjoint-grammar witnesses each reproduce a carrier-skeleton integer from their own structure — Gauss  $N(3+2i)=13$ , Eisenstein  $N(3+2\omega)=7$ , cyclotomy  $N(3+2\zeta_5)=55$ , Galois  $|(\mathbb{Z}/5)^\times|=4$ ,  $|\det \text{Cartan}(E_8)|=1$ , Pascal

$\binom{4}{0} + \binom{4}{1} + \binom{4}{2} = 11$  ( $2^4 = 16$ ), Coxeter  $h(E_8) = 30 = 2 \cdot 3 \cdot 5$  with  $\varphi(30) = 8$  — and the anchor  $a = (1, 1, 2)$  is the compression generator. Extends v305.

- **New paper `tfpt_safeguards` (companion S).** A dedicated methodology paper collecting, in one language, every mechanism by which TFPT defends a claim against coincidence and numerology: the status calculus + ledger + audit, no-free-pattern + reverse audit, the over-determination map (v427), the firewall + No-Unit theorem, frozen predictions + null model, the independent Wolfram and Lean paths, and the red team — with a final kill-test summary. Registered in `build.sh`, `make_docs_map.py`, `make_manifest.py` and the document-set box; the document set is now **eleven** deliverables (five numbered papers + four companions + introduction + changelog).
- **Surfaces synced.** `run_all.py` (421 modules), the registry, the ledger, `tfpt_research_contracts` (the  $F_{\text{transfer}}$  and keystone sections), `tfpt_safeguards`; manifests refreshed.

## 2026-06-25 (The (LTO-RP) reduction of the keystone `SEAM.EQUIV.01`: v424)

- **New module v424 (`SEAM.EQUIV.LTORP.01`).** An honest MMST-style reduction of the one open keystone — `SEAM.EQUIV.01`'s intrinsic Bisognano–Wichmann lemma (the v308/v309 residual) — to the recent commuting-projector theorem of Naaijken–Penneys–Wallick ([arXiv:2605.10693](#), [NPW26]), with the central mapping *corrected*. [E] the  $\mu_4$  clock is a modular *symmetry* ( $[\rho, K] = 0$ , v309/v199), not the flow —  $\sigma^\psi = \rho$  is type-wrong (a one-parameter group versus a finite order-4 element). [E] NPW26's axiom (LTO-RP) means  $u_\Theta = J$  (reflection = modular conjugation) = the intrinsic BW condition  $\theta K \theta = -K$ , *distinct* from  $\omega \circ \rho = \omega$ : a positive character-block-diagonal  $K$  has  $[\rho, K] = 0$  yet never  $\theta K \theta = -K$ , so clock-invariance does *not* imply (LTO-RP) (refuting that identification). [O] make-or-break: NPW26 prove (LTO-RP) for *topologically ordered* models (Toric Code  $|\det K| = 4$  via a trace; Levin–Wen anyonic), while the seam is invertible ( $|\det \text{Cartan}(E_8)| = 1$ , no anyons) and a  $\beta = 1$  KMS state (not a trace) — in neither proved bucket. [C] verdict: the BW residual reduces to NPW26 Theorem A modulo two open sub-steps (realise the seam as a  $\mathbb{Z}_2$ -reflection commuting-projector LTO; extend (LTO-RP) to the invertible KMS case) — a reduction, *not* a closure; `SEAM.EQUIV.01` stays [O]. Cited in `tfpt_research_contracts`; Python-only (numpy; the det discriminators mirrored via v89/v281).
- **Validation provenance.** The three pivotal references of the closure-strategy note were independently verified real and correctly described: NPW26 ([arXiv:2605.10693](#)), the RP–Yang–Mills complete-monotonicity reconstruction (Int. J. Geom. Meth. Mod. Phys. 2026), and Marolf ([arXiv:2203.07421](#)).
- **Surfaces synced.** `run_all.py`, the script registry, the status ledger and the `tfpt_research_contracts` keystone section; manifests refreshed.

## 2026-06-25 (No-Ambient-Dependency: no frozen readout is computed from `QG.AMB.01`: v423)

- **New module v423 (`QGAMB.NODEP.01`).** Sharpens the v369/v384 prose redundancy (“`QG.AMB.01` is a [C] redundancy, not dynamics”) into a machine-checked DAG+script fact, told honestly. [E] gap-decoupling margin  $\Delta_{\text{eff}} = 6 \ln \frac{3}{2} - 31/(4\pi^2) \approx 1.648 > 0$  ( $31 = 2^{g_{\text{car}}} - 1$ ), susceptibility  $\chi = 729/665$  finite (negative control gap  $\rightarrow 0 \Rightarrow \chi \rightarrow \infty$ ); [E] `QG.AMB.01` is a certification *sink* (every claim that lists it is a gravity-gate/QG row, never a frozen prediction) and *not a value source* (no producing script, while  $\alpha^{-1}$ , the mixing angles and cosmology are each *computed* by a compiler script, v3/v9/v268/v7); [E] the one transitive link is architectural — of the frozen readouts only  $\theta_{13}$  references it, via the 4d-QFT contract spine (`PS.DIRAC`→`CONTRACT.QFT4D`→`QG.G2`→`GATE.METRIC`), a coherence edge, not a numerical input. [C] verdict: `QG.AMB.01` is gap-decoupled certification — v369 as a DAG/script fact;

honest scope — this does *not* discharge SEAM.EQUIV.01, on which the readouts still depend. Cited in `tfpt_4`; Python-only (gap algebra mirrored via `v337`).

- **Paper sharpening (seam selector, up front).** The `tfpt_4` quantum-gravity keybox now states explicitly that the open SEAM.EQUIV.01 premise is the *holomorphic*  $|\det K|=1$  point that selects  $(E_8)_1$  — the central charge  $c=8$  alone does *not*, being shared by the non-holomorphic  $SO(16)_1$  counterexample (`v277/v281`).
- **Surfaces synced.** `run_all.py`, `script_registry.csv`, `status_ledger.csv`, `tfpt_4`; manifests refreshed.

## 2026-06-25 (The Galois $\leftrightarrow$ Net bridge: `v422`)

- **New module `v422` (SEAM.GALOIS.NET.01).** The seam  $\mu_4 = \text{Gal}(\mathbb{Q}(\zeta_5))$  is the SAME cyclic  $\mathbb{Z}/4$  as the  $(E_8)_1$  simple-current glue — not a mere order-4 coincidence — bridging the Galois gearbox of `v419` to  $G_{\text{net}}/\text{SEAM.EQUIV.01}$ . [E]  $\text{disc}(A_3) = \text{disc}(D_5) = \mathbb{Z}/4$  (one Smith invariant factor  $4 = \text{cyclic, det } 4$ );  $D_n$  disc is  $\mathbb{Z}/4$  for  $n$  odd,  $\mathbb{Z}_2 \times \mathbb{Z}_2$  for  $n$  even, so the carrier  $D_5$  (rank  $5 = g_{\text{car}}$ , odd) is cyclic while  $D_8$  (rank  $8$ , even) is Klein; [E]  $(\mathbb{Z}/5)^\times = \langle 2 \rangle$  is cyclic order 4 (only the carrier prime 5 gives order 4); [E] in  $\mathbb{Z}_4 \times \mathbb{Z}_4$  ( $16 = \dim S^+$ ) the glue  $(1, 1)$  is order 4 with Lagrangian quotient  $16/4^2 = 1 = (E_8)_1$  (`v89/v154`); [E] negative control: the Klein / order-2  $\langle (2, 2) \rangle$  gives  $16/2^2 = 4 = |\text{disc}(D_8)| = SO(16)$  (det 4), NOT  $E_8$  (det 1) — cyclic  $\mathbb{Z}/4 \rightarrow E_8$ , Klein  $\rightarrow D_8$ . [C] the bridge: one cyclic  $\mathbb{Z}/4$  threads the seam deck  $\mu_4$  (`v419`),  $\text{Gal}(\mathbb{Q}(\zeta_5))$  and the  $(E_8)_1$  glue — the Galois gearbox  $(\mathbb{Z}/30)^\times$  IS the holomorphic net's glue, cyclicity forced by 5, Klein excluded. Cited in `origin_theory`; Wolfram-mirrored (`324 \rightarrow 327`).
- **Surfaces synced.** `run_all.py`, `script_registry.csv`, `status_ledger.csv`, `origin_theory`; the Wolfram extension/README/ledger counts (`327/327`) and the website verification count; manifests refreshed.

## 2026-06-25 (F\_pole external: the Koide source $\rightarrow$ pole transfer is one-loop EM: `v420`)

- **New module `v420` (FR.POLE.QED.01).** The honest external-physics follow-up to `v93` and the clock No-Go: the Koide source $\rightarrow$ pole transfer is a one-loop *electromagnetic* effect, not an algebraic step. [E] Koide  $Q$  is rescaling-invariant ( $Q(cm) = Q(m)$ ), so flavor-universal running leaves it fixed and the source $\rightarrow$ pole shift is purely flavor-split; [C] the tree deficit  $\frac{2}{3} - Q_{\text{src}} = 0.00220$  is one-loop-EM-sized ( $= \varphi_0/24$  to 0.6%,  $= \alpha/\pi$  to  $\sim 5\%$ ), the PDG pole sits at  $2/3$  (the IR attractor); [C] a 1-loop QED pole $\leftrightarrow \overline{\text{MS}}$  per-lepton-log estimate shifts  $Q$  by EM-loop order toward  $2/3$  (the `v82` direction); [O] the exact flow time  $t \approx 2.84$  (`v93`) is scheme/scale-dependent external QED, only sized. [C] verdict:  $F_{\text{pole}}$  is a one-loop EM effect — the lens forces the rate  $(2/3)^6$  (`v327`), external EM supplies the flow; **F\_pole** stays [C], firewalled (`v187`), so  $F_{\text{transfer}}$  is confirmed genuinely external — the open theme sharpened (QED), not closed. Cited in `tfpt_4`; numerical (mpmath), Python-only (no Wolfram check, by the suite convention like `v93/v371`).
- **Surfaces synced.** `run_all.py`, `script_registry.csv`, `status_ledger.csv`, the Koide section of `tfpt_4`; manifests refreshed.

## 2026-06-25 (The seam $\mu_4$ is the carrier's Galois group: `v419` + figure)

- **New module `v419` (SEAM.GALOIS.01).** A positive resolution of the cyclic/Galois asymmetry (`v409`, RES.COXETER.SYMMETRY.01): why the seam  $\mu_4$  sits on the Galois side, not on the cyclic  $\zeta_{30}$  dial. [E]  $h(E_8) = 30$  is squarefree, so  $\mathbb{Z}/30$  has element orders  $\{1, 2, 3, 5, 6, 10, 15, 30\}$  — *no* order 4, so the seam  $\mu_4$  cannot be a rotation hand and is forced onto the Galois side; [E] by CRT  $(\mathbb{Z}/30)^\times = (\mathbb{Z}/2)^\times \times (\mathbb{Z}/3)^\times \times (\mathbb{Z}/5)^\times = \mu_4 \times \mathbb{Z}_2$  (order  $\varphi(30) = 8 = \text{rank } E_8$ ),

and the  $\mu_4$  ( $\mathbb{Z}/4$ ) factor IS ( $\mathbb{Z}/5$ ) $^\times$  (the carrier prime 5), the  $\mathbb{Z}_2$  IS ( $\mathbb{Z}/3$ ) $^\times$  (family 3 = CP conjugation, v316); [E] so the seam  $\mu_4 = \text{Gal}(\mathbb{Q}(\zeta_5))$ , realised on the carrier clock  $C_5$  (v418) by the Frobenius  $\zeta_5 \mapsto \zeta_5^2$  — the explicit order-4 operator  $G$  with  $GC_5G^{-1} = C_5^2$ ,  $G^4 = I$ ,  $G^2C_5G^{-2} = C_5^{-1}$ . [C] this is the positive content of v409: the  $2^2=4$  is the order of ( $\mathbb{Z}/5$ ) $^\times$ , and “no contraction rate” is because the seam is an *automorphism*, not a hand; so  $\mathbb{Q}(\zeta_{30})$  carries both  $E_8$  invariants (cyclic  $h=30$ , Galois rank=8). Cited in `origin_theory`; Wolfram-mirrored (321  $\rightarrow$  324).

- **New figure `coxeter_galois.pdf`.** The order-30 cyclic dial with the four flavor hands (sheet/family/carrier/CP) beside the Galois gears ( $\mu_4 = \text{Aut}(\zeta_5)$  the seam,  $\mathbb{Z}_2 = \text{Aut}(\zeta_3)$  the CP conjugation); added to `make_figures.py`, the manifest figure list and `origin_theory`.
- **Surfaces synced.** `run_all.py`, `script_registry.csv`, `status_ledger.csv`, the Wolfram extension/README/ledger counts (324/324) and the website verification count; manifests refreshed.

## 2026-06-25 (The cyclotomic norm triple + the carrier-5 clock: v418)

- **New module v418 (ARITH.NORMTRIPLE.01).** The carrier split (3,2) read in the three atom-rings gives (7,13,55)=(scalaron,  $\Delta_Q$ , quark numerator), and the carrier-5 clock missing from v417 is found as the  $4 \times 4$   $\Phi_5$ -companion  $C_5$ . [E]  $C_5^5=I$ ,  $\text{tr } C_5=-1$ ,  $\det C_5=1$  (the four primitive 5th roots), with the golden ratio as its real-part data ( $2 \cos 72^\circ=1/\varphi$ ,  $2 \cos 144^\circ=-\varphi$ ; output-side, v313/v349; dim 4, not 3 — this refines v417’s gap from “no operator” to “no  $3 \times 3$  operator”); [E]  $N_{\mathbb{Z}[\zeta_5]}(3+2\zeta_5)=\det(3I+2C_5)=55=5 \cdot 11=g_{\text{car}} \parallel \text{Pl}(K) \parallel_1$ , the  $c_u/c_d$  numerator (v410/v411); [E] the triple  $\det(3I+2\text{Comp}(\Phi_n)) = (7,13,55)$  for  $n=3,4,5$  over  $\mathbb{Z}[\omega], \mathbb{Z}[i], \mathbb{Z}[\zeta_5]$  (atoms 3,2,5, v416); [E] NEG control: the anchor (5,4)  $\rightarrow$  (21,41,461) reaches named values only in the  $\omega$ - and  $i$ -rings ( $21=N_\omega(5,4)$ ,  $41=10b_1$ , v222) but 461 (prime) in the carrier ring, so the carrier ring separates (3,2)  $\rightarrow$  55 from (5,4)  $\rightarrow$  461. [C] honest: (3,2) is the *forced* carrier split (v14), not the unique clean one — a scan finds  $\{(2,1) \rightarrow (3,5,11), (3,2) \rightarrow (7,13,55)\}$ , a small ladder. The sharper discriminator is *cross-sector*: only (3,2) spans three physics sectors (gravity 7, flavor 13, masses 55), the unique cross-sector seed, while (2,1) is all compiler-core. Cited in `origin_theory`; Wolfram-mirrored (318  $\rightarrow$  321).
- **Surfaces synced.** `run_all.py`, `script_registry.csv`, `status_ledger.csv`, a triple paragraph in `origin_theory`, the Wolfram extension/README/ledger counts (321/321) and the website verification count; manifests refreshed.

## 2026-06-25 (The Eisenstein/CP operator + the order-30 clock: v417)

- **New module v417 (DIAMOND.EISEN.01).** The hexagonal dual of v415: the  $\tau=\omega$  flavor side put on operators and unified with the seam into one order-30 Coxeter clock. [E] the family rotation  $P = (1\ 2\ 3)$  is the order-3 Eisenstein deck ( $P^2+P+I = \text{ONES}$ , so  $\omega^2+\omega+1=0$  on the plane  $\perp (1,1,1)$ ), the order-3 dual of  $J^2=-I$ ; [E] it realises the hex CM norm as  $(3I+2P)(3I+2P^2) = 7I+6\text{ONES}$ ,  $\text{Spec}\{7,7,25\}$  ( $7=N_{\mathbb{Z}[\omega]}(3+2\omega)=\text{scalaron}$ ,  $25=g_{\text{car}}^2$ ) — the dual of v415’s  $\{13,13,9\}$ ; [E] the CP clock  $W=-P^2$  ( $\rho$  of v233) has  $W^2=P$ ,  $W^3=-I$ ,  $\text{Spec}\{-1, \zeta_6, \zeta_6^{-1}\}$ , giving  $\delta_{\text{CKM,lead}}=\arg \zeta_6=\pi/3$  and  $\delta_{\text{PMNS}}=\arg \zeta_6^4=4\pi/3$  (v316/v320); [E] all flavor clocks are powers of  $\zeta_{30}$  ( $\zeta_{30}^{15}=-1$ ,  $\zeta_{30}^{10}=\omega$ ,  $\zeta_{30}^6=\zeta_5$ ,  $\zeta_{30}^5=\zeta_6$ ) while the seam  $\mu_4=i$  is the Galois side ( $\mathbb{Z}/30$ ) $^\times$  of order  $\varphi(30)=8=\text{rank } E_8$  — so  $\mathbb{Q}(\zeta_{30})$  carries both  $E_8$  invariants ( $h=30$  and  $\text{rank}=8$ ). [C] honest gap: the carrier  $\mu_5$  has no  $3 \times 3$  operator (the golden  $\varphi$  is an output-side trace, v313/v349); negatives: no  $U, V$ -diamond operator carries  $\zeta_6$ ,  $[D, P] \neq 0$  (no  $\mu_{12}$ ),  $N(3+2\zeta_6)=19 \neq 7$ . Cited in `tfpt_2`; Wolfram-mirrored (315  $\rightarrow$  318).
- **Surfaces synced.** `run_all.py`, `script_registry.csv`, `status_ledger.csv`, the Sheet-Diamond keybox of `tfpt_2`, the Wolfram extension/README/ledger counts (318/318) and the website verification count; manifests refreshed.

## 2026-06-25 (The Gaussian operator + the atom trichotomy: v415, v416)

- **New module v415 (DIAMOND.GAUSS.01).** The square CM-norm dictionary of v222/v230 is *operator-realised*, and the  $\mu_4$  deck becomes an intrinsic diamond object. [E] the integer commutator  $J := \frac{1}{3}[U, V] = \begin{pmatrix} -1 & 1 & 0 \\ -2 & 1 & 0 \\ -3 & 1 & 0 \end{pmatrix}$  has  $\text{Spec}(J) = \{i, -i, 0\}$ ,  $J^2 = -I$  on the rank-2 image and  $J^3 + J = 0$  — a  $\mu_4$  quarter-turn born from the binary  $V$  and ternary  $U$  compilers (v410/v95); [E]  $3I + 2J$  and  $5I + 4J$  have spectra  $\{3+2i, 3-2i, 3\}$ ,  $\{5+4i, 5-4i, 5\}$ , so the Gaussian integers  $3+2i$  (norm  $13=\Delta_Q$ ) and  $5+4i$  (norm  $41=10b_1$ ) are *literal eigenvalues*; [E]  $(aI+bJ)(aI-bJ)$  reads  $(a^2+b^2)$  on the  $\mu_4$ -plane and the real part<sup>2</sup> on the kernel ( $\{13, 13, 9\}$ ,  $\{41, 41, 25\}$ ); [E] the intrinsic order-4 deck  $D = -I + J - J^2$  ( $D^4 = I$ ,  $\chi_D = (x+1)(x^2+1)$ ) has  $\chi_2$ -line  $\ker[U, V] = a - 1$ . [C] the geometric-deck identification sharpens (does not close) the v140  $\mathbb{Z}_3$  residue. Cited in `tfpt_2`; Wolfram-mirrored.
- **New module v416 (ARITH.TRICHOTOMY.01).** The atoms  $\{2, 3, 5\}$  are ramified/inert/split in the two exceptional CM rings. [E] in  $\mathbb{Z}[i]$  (square/seam)  $2=|\mathbb{Z}_2|$  ramifies,  $3=N_{\text{fam}}$  is inert,  $5=g_{\text{car}}$  splits as  $(2+i)(2-i)$ ; [E] in  $\mathbb{Z}[\omega]$  (hex/flavor)  $3=N_{\text{fam}}$  ramifies,  $2, 5$  are inert; [E] the ramified prime of each ring is its own atom; [E] the three quadratic facets  $\mathbb{Q}(i), \mathbb{Q}(\sqrt{-3}), \mathbb{Q}(\sqrt{5})$  of  $K = \mathbb{Q}(i, \sqrt{-3}, \sqrt{5})$  (v403) ramify over exactly one atom each, product  $2 \cdot 3 \cdot 5 = 30 = h(E_8)$ , two imaginary (CM/dynamic) + one real (RM/static). [C] the seam/flavor/golden reading. Cited in `origin_theory`; Wolfram-mirrored.
- **Wolfram extension 308  $\rightarrow$  315 + a pre-existing fix.** Added the seven exact checks for v415/v416 and corrected a one-character sign bug in the v412 `CharacteristicPolynomial` check (Wolfram returns  $-\chi$  for odd rank), so `tfpt_readouts_extension.wl` now runs genuinely clean at 315/315 (GATE.WOLFRAM.02); README and ledger counts updated.
- **Surfaces synced.** `run_all.py`, `script_registry.csv` and `status_ledger.csv` updated; a new keybox paragraph in the Sheet-Diamond section of `tfpt_2` and a trichotomy sentence in `origin_theory`; manifests refreshed.

## 2026-06-24 (The sheet generator $V$ as a binary internal compiler: v410–v414)

- **New module v410 (SHEET.GEN.BINARY.01).** The rank-2 sheet axis  $V = Q \text{diag}(0, 1, 1)$  of the centered cross (v95/v218) is a binary internal compiler: its powers print the carrier spine. [E] the closed form  $V^n = \begin{pmatrix} 0 & 2^{n-1} & 0 \\ 0 & 2^n & 0 \\ 0 & 2^{n+1}-2 & 1 \end{pmatrix}$  (proved by the inductive step  $F(n)V=F(n+1)$ ), so  $V^n \mathbf{1} = (2^{n-1}, 2^n, 2^{n+1}-1) = (1, 2, 3), (2, 4, 7), (4, 8, 15), (8, 16, 31)$ ; [E] the four bilinear families  $\mathbf{1}^T V^n \mathbf{1} = 7 \cdot 2^{n-1} - 1$ ,  $\mathbf{1}^T V^n a = 7 \cdot 2^{n-1}$ ,  $a^T V^n a = 11 \cdot 2^{n-1}$ ,  $a^T V^n \mathbf{1} = 11 \cdot 2^{n-1} - 2$  give  $(6, 7, 9, 11)$  at  $n=1$  and  $\mathbf{1}^T V^2 \mathbf{1} = 13 = \Delta_Q$ ,  $\mathbf{1}^T V^3 \mathbf{1} = 27 = \mathbf{1}^T R a$ ,  $\mathbf{1}^T V^4 \mathbf{1} = 55$ ,  $\mathbf{1}^T V^4 a = 56 = \dim \mathbf{56}_{E_7}$ ; [C] the binary spine is FORCED by  $\text{Spec}(V) = \{0, 1, 2\}$ , so the carrier/Lie matches (rank  $E_8=8$ ,  $\dim S^+=16$ ,  $248/8=31$ ) and the theta cross-link ( $\sigma_3(2)=9=a^T V \mathbf{1}$ ,  $\sigma_3(3)=28=\mathbf{1}^T V^3 a = \det(I+R)$ ,  $\sigma_3(5)=126$ ) are audit-typed (the v95  $G_2$ -center typing).
- **New module v411 (FLAV.UD.VPOWER.01).** The quark ratio is a pure  $V$ -power readout,  $c_u/c_d = (\mathbf{1}^T V^4 \mathbf{1}) / ((a^T V \mathbf{1})(\mathbf{1}^T V^2 \mathbf{1})) = 55/(9 \cdot 13) = 55/117$ , identical to the established  $g_{\text{car}} \|\text{Pl}(K)\|_1 / (N_{\text{fam}}^2 \Delta_Q) = 5 \cdot 11/(9 \cdot 13)$  (v94). [E] the identity; [C] an exact *re-encoding* (no new physical input) — the value and the  $u/d$  identification stay conditional, coupled to the  $(U_{\text{wall}})$  readout rigidity.
- **New module v412 (DIAMOND.SHEET.SOURCE.J.01).** Names the operator on the  $\mathbb{Z}_2$  wall ( $t=-2$ ,  $\det=2$ , v135),  $J=M(1, -2)=C-2V$ , completing the sheet axis  $J, K, C, F=M(1, -2..1)$ . [E]  $\chi_J = \lambda^3 - 6\lambda^2 + 3\lambda - 2 \Rightarrow (\text{tr}, e_2, \det)=(6, 3, 2)$ ;  $a^T J a = 30 = h(E_8)$ ,  $\det(I+J)=12 = \dim \mathfrak{g}_{\text{SM}}$  and  $\det(2I+J)=40=|R(D_5)|$ ; [C] the Lie readings audit-typed like the  $G_2/F_4$  shadows.
- **New module v413 (DIAMOND.SHEET.CALCULUS.01).** On  $M_t=C+tV$  the elementary-symmetric coefficients encode the atoms as difference orders:  $e_1=3t+12$ ,  $e_2=2t^2+15t+25$ ,

$e_3=4t^2+14t+14$  give  $\Delta e_1=3=N_{\text{fam}}$ ,  $\Delta^2 e_2=4=|\mu_4|$ ,  $\Delta^2 e_3=8=\text{rank } E_8$  (the  $e_3$  curvature is the v218/v224 sheet quadratic; the full triple is new); [E] the anchor energy  $a^T M_t a = 52 + 11t$  ( $30 \rightarrow 41 \rightarrow 52 \rightarrow 63$ , step  $11 = a^T V a$ ); [C] atom/Lie readings audit-typed.

- **New module v414 (DIAMOND.CENTER.RESOLVENT.01).** The center  $C=M(1,0)$  is a resolvent portal: [E]  $\det C=14=\dim G_2$ ,  $\det(I+C)=52=\dim F_4$ ,  $\det(2I+C)=120=|R^+(E_8)|$ , the exceptional shell ladder  $G_2 \rightarrow F_4 \rightarrow E_8^+$  as  $\det(C+kI)$ ,  $k=0,1,2$  (with  $\text{tr } C=12=\chi_{E_8}(i)$ ,  $\sum C=31=2^{g_{\text{car}}}-1$ , v95); [C] the portal reading.
- **Surfaces synced.** Integrated into the Sheet-Diamond section of `tfpt_2` (new keybox, five \veri citations) and the  $E_8$  theta-raster of `tfpt_3`; `run_all.py`, `script_registry.csv` and the ledger updated; Wolfram extension mirror  $303 \rightarrow 308$ .

## 2026-06-24 (Two new tracked targets + axion-branch consistency: v408, v409)

- **New module v408 (HYP.PHI0.PUNCTURE.01).** Names the  $\varphi_0$  puncture term  $48c_3^4$  as the order- $|\mu_4|=4$  local boundary heat-kernel (Seeley–DeWitt) contact term of the  $\mu_4$  puncture divisor, summed over  $\Omega_{\text{adm}}=N_{\text{fam}} \dim S^+=48$  admissible states — the analytic complement of the icosahedral tree term (v396). [E] the exact split  $\varphi_0=1/(6\pi)+48c_3^4$ , the puncture count, the closed form  $3/(256\pi^4)$  and the hypergraph negative (reused from v396/ARCH.QUAD.01); [O] the from-first-principles heat-kernel proof; [C] a FACE of the seam boundary analysis (the  $c_3$  Gauss–Bonnet datum), not a new gate. Cited in `origin_theory`; no new number.
- **New module v409 (RES.COXETER.SYMMETRY.01 closed as a structural lemma).** prime-2 (the seam sheet involution) is *necessary* but *not autonomous*:  $\sigma=\text{diag}(1,-1,-1)$  has spectrum  $\{\pm 1\}$  and projector  $(I+\sigma)/2$  spectrum  $\{0,1\}$ , none in  $(0,1)$ , so prime-2 carries no contraction rate, whereas the genuine rates  $2/3$  and  $(1/3)^6$  lie in  $(0,1)$  and carry prime-2 only as a factor. [E] no prime-2-only attractor exists; [C] dynamics begins only after coupling to family-3 or carrier-5 ( $|\mu_4|=2^2$  the Galois bridge). Resolves the v394 open question’s falsifiable corollary; cited in `tfpt_research_contracts`; the ledger row RES.COXETER.SYMMETRY.01 retyped [O]  $\rightarrow$  [E]/[C].
- **Axion branch consistency (tfpt\_4\_frontier, website predictions.ts, v185).** The dark-matter section’s opening + keybox are realigned with the finite- $T$  result already stated later in the same section: the robust dominant-DM branch is the *spine* angle  $\theta_i=\pi N_{\text{fam}}/g_{\text{car}}=3\pi/5=108^\circ$  (untuned, in band), while the seam hilltop  $\theta_i \approx 170^\circ$  over-produces ( $\sim 5.4\times$ ) and is disfavoured; the misalignment angle is retyped from a closed [E] input to a [C] branch decision. The website prediction card now leads with the spine angle.
- **Lean closure of the prime-2 corollary (FORM.COXETER.PRIME2.01).** The v409 result is now also machine-proved in Lean 4 / Mathlib (new module `TfptCarrier/CoxeterPrime2.lean`): an involution eigenvalue ( $r^2=1$ ) is  $\pm 1$ , a projector eigenvalue ( $r^2=r$ ) is 0 or 1, neither lies in the open interval  $(0,1)$ , while  $2/3$  does — so *no prime-2-only attractor exists*. Built clean (`lake build`); the new theorems are axiom-pure (`#print axioms = propext`, `Classical.choice`, `Quot.sound` only) and pass the `AuditCheck/AuditContract` signature locks. This lifts RES.COXETER.SYMMETRY.01 to the machine-proof tier.
- **Verified, no change (Plücker 11).** The quark constant  $11=\|Pl(K)\|_1=\det(1|a|\sigma)$  is already typed honestly as an [E] identity with “why 11” the lone [O] residue (v407, `tfpt_1`); the repo carried no overstatement, so no edit was required.

## 2026-06-24 (Cross-surface consistency sweep: papers, website, gates, figures realigned)

- **Ledger hygiene.** Resolved a duplicate active `claim_id`: GATE.UWALL.06-09 was carried by two distinct chains. It is an ID collision (not a supersede): the monodromy/residue chain

v114–v117 keeps .06–.09 (as asserted by `v123_inventory_update.py` and `run_all.py`); the older exterior-leg chain v42–v45 is renumbered to GATE.UWALL.15–.18 (internal `depends_on` carried along). No claim content changed.

- **Website.** `papers.ts` document count  $8 \rightarrow 9$  (the Red-Team paper was missing from the tally); Red-Team subtitle/highlight *Targets A–E*  $\rightarrow$  *A–F* with a new Target F section + contribution bullet (the abstract already carried F); `StatusMatrix.tsx` *Doc 7*  $\rightarrow$  *Doc 8* for the research contracts and the residual interfaces re-typed  $v_{\text{geo}}$  [O],  $G_{\text{net}}$  [C],  $F_{\text{transfer}}$  [C]; *objections Paper 1*  $\rightarrow$  *Doc 1*.
- **Papers (marker/wording alignment, no canonical status change).** `tfpt_1` Strong CP [E]  $\rightarrow$  [E]/[C] (given RP) and  $\theta_{12}$  “genuinely open”  $\rightarrow$  “conditionally derived”; `tfpt_2` Starobinsky  $A_s$  [E]  $\rightarrow$  [E]/[C] ( $M$  exact,  $A_s$  a band over  $N_\star \in [50, 60]$ ) and the Part-II abstract “opens the one remaining gap”  $\rightarrow$  “closes the last structural gap”; `tfpt_research_contracts` the QFT4D “nonperturbative measure stays open” line and the “only  $U_{\text{point}}$  stays open” headline reconciled with the [C] redundancy /  $v_{\text{geo}}$  reduction stated alongside; `origin_theory` class C re-typed to the ambient-measure [C] redundancy; `introduction` ( $U_{\text{wall}}$ )  $\rightarrow$   $v_{\text{geo}}$ . Forbidden old-paper attributions (“Papers 1–7”, “Papers 2/6”, “Paper-7”, “Paper-1”) replaced by current document / neutral wording (the `tfpt_3` cascade bridge stays exempt).
- **Suite README.** The residual-gates table (Alessandro-5.0 vintage) brought to the current model ( $v_{\text{geo}}$  [O];  $G_{\text{net}}$ /SEAM.EQUIV.01,  $F_{\text{transfer}}$ ,  $N_\star$ , QG.AMB.01 [C]; Wolfram 116/116 + 303/303).
- **Figures.** Regenerated `status_heatmap` (ledger-driven) and extended the narrative figures `script_timeline/residual_chain/qft_skeleton` through the v303–v407 arc (typed solvers, parameter-free local Einstein equation v359, QG.AMB.01 [C] redundancy, seam closed modulo cited theorems); the `qft_skeleton` “Stelle ghost” open box is retyped (resolved truncation artefact). Added `qft_skeleton/qft_unification` to the manifest figure list.

## 2026-06-24 (The $\tau=\omega$ dual keystone + the flavor residuals it homes: v405–v407)

- [E]/[C]/[O] `v405_seam_equiv_omega.py` (SEAM.EQUIV.02): the *dual keystone* at  $\tau=\omega$  – the family/flavor sector as the order-3 Eisenstein/ $A_2$  face of  $E_8$ , dual to the seam  $= (E_8)_1$  at  $\tau=i$  (v404). A named contract (genre of v286/v398), NOT a closure. [E]  $\chi_{E_8}(\omega) = 0$  (the family CM point degenerates  $E_8$ ) vs  $\chi_{E_8}(i) = 12$ ; [E]  $A_2$  is the  $\tau=\omega$  Eisenstein lattice (Cartan  $\det = 3 = N_{\text{fam}}$ ,  $h = 3 = N_{\text{fam}}$ ); [E]  $R$  lives in  $E_6 \times A_2$  ( $248 = 78 + 8 + 2 \cdot 27 \cdot 3$ ,  $\det R = 8 = \dim A_2 = \text{rank } E_8$ ); [E]  $\delta_{\text{PMNS}} = 4\pi/3$  is the  $\tau=\omega$  CM phase; [E] the cusp weights  $\{0, 1/3, 2/3\}$  are the order-3 deck  $= N_{\text{fam}} = \det Q$ . [C] the family sector is the order-3 Eisenstein/ $A_2$  deck; the two keystones sit at the only two elliptic points of  $\text{PSL}(2, \mathbb{Z})$ . [O] residual = the canonical  $\mathbb{Z}_3$  deck map (v140) + the  $v_{\text{geo}}$ -class scale. Cited in `origin_theory`; Python-only.
- [E]/[C]/[O] `v406_pmns_phase_origin.py` (FLAV.PMNS.CLOSE.01): the PMNS dynamics closed *modulo* the  $\tau=\omega$  keystone – the CP-phase ORIGIN is the family CM point and the absolute scale is one  $v_{\text{geo}}$ -class unit, so PMNS has *zero free flavor dials beyond*  $v_{\text{geo}}$ . [E] the three angles (v9/v268)  $\rightarrow$  unitary  $U_{\text{PMNS}}$  (v270); [C]  $\delta_{\text{PMNS}} = 4\pi/3$  is the  $\tau=\omega$  hexagonal CM phase (v405), upgrading v270’s [O] “assembled input” to a keystone consequence; [C]  $J_{\text{PMNS}} = -0.0297$  derived; [E] the absolute scale = one seesaw ratio =  $v_{\text{geo}}$ -class (v272/v153). [O] residual = the seesaw operator realisation ( $M_R$ , v184) + the  $\Sigma m_\nu$  floor kill test. Cited in `tfpt_2_standard_model`; Python-only.
- [E]/[C]/[O] `v407_dn_pairings_omega.py` (FLAV.SELECTOR.CLOSE.01): the  $R4'$  selector residue folds into the  $\tau=\omega$  keystone – the three  $(d, n)$  pairings ARE the  $\tau=\omega$  family-slice atoms. [E] the frame  $(1, a, \sigma)$  has  $\det = 11 = \|Pl(K)\|_1$  and  $n = (5, -9, 6)$  is the unique covector with  $n \cdot 1 = 2$ ,  $n \cdot a = 8$ ,  $n \cdot \sigma = 121 = 11^2$  ( $\Rightarrow (d, n) \Rightarrow R$ , v136/v139); [E]

$n \cdot a = 8 = \text{rank } E_8 = \dim A_2 = \det R$  (the  $A_2$  block reading  $R$  in  $E_6 \times A_2$ , v405); [E] the pairings  $\{2, 8, 121\} = \{|Z_2|, \dim A_2 = \det R, \|Pl(K)\|_1^2\}$ . [C] deriving  $(d, n)$  folds into SEAM.EQUIV.02 ( $d$  anchor-derived, v139); [O] the lone residue “why  $11 = \|Pl(K)\|_1$ ”. Cited in `tfpt_1_architecture_e8`; Python-only.

## 2026-06-24 (The arithmetic hull + the $E_8$ -character CM duality: v403, v404)

- [E]/[C] `v403_facet_compositum.py` (ARITH.HULL.01): the three single-prime number-field facets of the order-30 clock (v390/v394, used so far only one at a time) *compose* to one field  $K = \mathbb{Q}(i, \sqrt{-3}, \sqrt{5})$ . [E] the squarefree parts  $-1, -3, 5$  are independent in  $\mathbb{Q}^\times/(\mathbb{Q}^\times)^2$ , so  $[K : \mathbb{Q}] = 2^3 = 8 = \text{rank } E_8 = g_{\text{car}} + N_{\text{fam}}$  with Galois group  $(\mathbb{Z}/2)^3$  (order 8); [E] exactly 7 quadratic subfields  $\{-1, -3, 5, 3, -5, -15, 15\}$ , splitting 4 imaginary  $= |\mu_4|$  (CM/dynamic) + 3 real  $= N_{\text{fam}}$  (RM/static); [E] ramified *only* over  $\{2, 3, 5\} = \text{the atoms} = \text{the primes of } h(E_8) = 30$ ; [E]  $2^{\omega(h)} = \text{rank}$  is  $E_8$ -special (the  $E_7$  control  $h = 18$  gives  $2^2 = 4 \neq 7$ ). [C]  $K$  is the abelian  $(\mathbb{Z}/2)^3$  *arithmetic* hull dual to the  $E_8$  *lattice* hull; the 7 quadratic subfields are a candidate for the 7  $E_8$  audit slices. A synthesis composing the established facets, no new number; cited in `origin_theory`; Python-only by the synthesis-module convention (cf. v390/v394).
- [E]/[C] `v404_e8_char_cm_duality.py` (E8.CMDUAL.01): the  $(E_8)_1$  vacuum character  $\chi_{E_8} = E_4/\eta^8 = j^{1/3}$  read at *both* CM points, extending v282 ( $\chi_{E_8}(i) = 12$ ) to the second elliptic point. [E]  $\chi_{E_8}^3 = j$  (q-series); [E] SEAM  $\tau=i$  ( $\mathbb{Q}(i)$ , order 4): cross-ratio  $2 \Rightarrow j = 1728 \Rightarrow \chi_{E_8}(i) = 1728^{1/3} = 12$  ( $E_8$  present, v214/v282); [E] FAMILY  $\tau=\omega$  ( $\mathbb{Q}(\sqrt{-3})$ , order 6): the equianharmonic  $\lambda$  ( $\lambda^2 - \lambda + 1 = 0$ )  $\Rightarrow j = 0 \Rightarrow \chi_{E_8}(\omega) = 0$  ( $E_4$  vanishes, v220/v267); [E]  $\tau=i, \tau=\omega$  are the *only* elliptic points of  $\text{PSL}(2, \mathbb{Z})$ , so the values are forced. [C] the seam/flavor rigidity duality: the seam is rigid where  $E_8$  is present ( $\chi = 12$ , the keystone SEAM.EQUIV.01 closes) while the flavor/CP sector is the soft residual where  $E_8$  degenerates ( $\chi = 0$ :  $U_{\text{wall}}, v_{\text{geo}}$ , the CP texture), a candidate dual keystone at  $\tau=\omega$ . Cited in `origin_theory`; core values  $j = 1728/j = 0$  already mirrored (v214/v220/v267/v282), Python-only.

## 2026-06-24 (The five TOE/QFT closure contracts v398–v402 + a ledger rescope)

- [E]/[O] `v398_seam_state_rigidity.py` (SEAM.RIGIDITY.01, Paper A): the *Seam State Rigidity Theorem* as ONE named target, consolidating the scattered state-invariance reductions (v177/v199/v308/v309/v335) and *hardening* the finite-block evidence to a multi-mode sweep. [E] RP-definability hinge (OS canonical transfer  $T=e^{-H}$  from RP + reflection alone, v54, quasi-free v155, no  $\mu_4$  normal form); [E] band-limited character-block-diagonal core  $[\rho, |k|]=0$  swept  $N=4.64$  (beyond the 3-dim  $H^1$ ); [E] holomorphy  $\Rightarrow (E_8)_1$  ( $c=8$ ,  $|\det \text{Cartan}|=1$  vs 4, index  $4=|\mu_4|$ ). [O] the one residual:  $[\rho, \Lambda_\Sigma]=0$  on the full  $L^2$  (intrinsic Bisognano–Wichmann). NOT a closure of SEAM.EQUIV.01 (continuum = cited MMST v336).
- [E]/[C]/[O] `v399_gravity_complete.py` (GRAVITY.COMPLETE.01, Paper B): the explicit verdict *Gravity-complete* = *Boundary-complete* + *Ambient-redundancy*, combining the parameter-free local equation (v358/v359) with the ambient redundancy (v369). [E] the five gravitational readout classes (Newton  $8\pi$ ,  $\Lambda$ , scalaron  $c_3^7$ , horizon  $1/|\mu_4|$ , recovery  $(2/3)^6$ ) are all boundary/seam/gap, so  $\mathcal{O}_{\text{phys}} \subset \mathcal{A}_\Sigma$ ; [O] residual = SEAM.EQUIV.01 + intrinsic BW (no new gate).
- [E]/[O] `v400_grav_nonlocal_action.py` (GRAV.NONLOCAL.01, Paper C): the nonlocal spectral-gravity action with the local  $R+R^2$  as its IR projection (consolidating v304/v370/v380/v386, adding the IR-matching check). [E] entire  $a=e^u \Rightarrow$  one pole (ghost-free), every truncation carries a Stelle zero (monotone, v380), and the IR order reproduces  $R+R^2$  with the atom-fixed scale  $M_{\text{scal}}^2/\bar{M}_{\text{Pl}}^2=c_3^7$  (v36/v253). [O] the entire- $a$  assumption + perturbative-only.

- **[E]/[O] v401\_metrology\_closure.py** (METROLOGY.CLOSURE.01, Paper D): the final input set  $\{a, \pi, v_{\text{geo}}\}$  certified — 0 dimensionless dials + 1 unit (consolidating v153/v274/v364/v384). **[E]** dimensional bookkeeping  $\text{mass}/1G/\Lambda/U_{\text{point}}/(m/\mu) = \text{number} \times v_{\text{geo}}^{1,2,4,1,0}$ ; **[E]** the  $G$ -vs- $\Lambda$  over-determination 0.11% (v274); **[O]**  $v_{\text{geo}}$  theorem-forbidden ( $\text{TOE}_{\text{strict}} = \text{TOE}_{\text{dimensionless}} + [v_{\text{geo}}]$ ).
- **[E]/[C] v402\_ftransfer\_suite.py** (FTRANSFER.SUITE.01, Paper E): the four frontier sectors as ONE functor  $F_{\text{transfer}}$ , the composition  $F_{\text{obs}} \circ F_{\text{thr}} \circ F_{\text{RG}}$  (consolidating the **FR.\*.SOLVE** solvers), each with an exact atom source, a committed kill test, and the firewall (physical prediction stays **[C]/[O]**, only the algebraic source **[E]**; never re-pressed into the compiler, v187).
- **Ledger rescope (repo hygiene)**: GATE.METRIC.01 no longer reads “full covariant metric-sector field equation open” — the *local* covariant equation is closed parameter-free (GRAV.NONLINEAR.01/v359); the gate is rescoped so the open item is *only* the global nonperturbative ambient measure (QG.AMB.01, a **[C]** redundancy v369), removing the stale contradiction with the status card.

## 2026-06-24 ( $\varphi_0$ leading term from icosahedral combinatorics: v396)

- **[E]/[C] v396\_phi0\_icosahedral.py** (HYP.PHI0.01): the tree-level seed  $\varphi_0^{\text{tree}} = 1/(6\pi)$  is exactly  $F/(4h\pi) = F/(|\text{Aut}| \pi) = (F/(g_{\text{car}} N_{\text{fam}})) c_3$  on the icosahedral hypergraph ( $F = 20$  triangular hyperedges,  $h = 30$ ,  $|\text{Aut}| = 120$ );  $F/h = |Z_2|/N_{\text{fam}} = 2/3$  (the same survival ratio as v327). **[E]** Gauss–Bonnet consistent with  $c_3 = 1/(|Z_2|4\pi)$  (v216/v342). **[E]** honest negative: the puncture  $48c_3^4$  is still analytic, not a hypergraph fraction. **[C]**  $\varphi_0^{\text{tree}}$  is a combinatorial reinterpretation, not a new independent injection; the puncture stays **[O]**.

## 2026-06-24 (Coupled hypergraph rewrite: v395 unifies carrier $\times$ family in one local rule)

- **[E] v395\_hypergraph\_coupled.py** (HYP.COUPLED.01): one coupled local rewrite on a  $9 \times 3$  labelled grid (affine- $E_8$  nodes  $\times$  family channels) unifies v299 (carrier growth), v327 (branching rule  $M$ ) and v324 (fiber product). Micro-step = network lazy diffusion on each cusp column +  $M$  on each node’s family vector ( $T_{\text{micro}} = T_{\text{net}} \otimes M$ ); joint attractor marks  $\otimes e_0$ ; one clock hand (6 family steps) gives recovery  $(2/3)^6$ ; v324’s  $T_{\text{net}} \otimes T_{\text{cusp}}$  emerges as the cusp-readout basis. **[O]** the 3-arm seed ( $P_2$ ) and analytic  $\varphi_0$  remain irreducible (v312).

## 2026-06-24 (A falsifiable clock probe of the frontier: v397 how external is the “external physics”?)

- **v397 (FR.CLOCK.PROBE.01)** — do the “external” frontier scheme-numbers land on the  $\{2, 3, 5\}$  clock? A falsifiable, anti-numerology probe (strict No-Free-Pattern, v100: “on the clock” means an *exact* match, not a wide-band fit). **[E]** the  $\eta_B$  decuple  $A_\Lambda = 10 = |Z_2|g_{\text{car}}$  is ON the clock (v212); **[E]** the Koide rate  $(2/3)^6 = (|Z_2|/N_{\text{fam}})^6$  is ON the clock (v303); **[E]** the axion spine angle  $\theta_i = \pi N_{\text{fam}}/g_{\text{car}} = 3\pi/5$  is ON the clock (v211/v373); **[E]** the proton  $C_p \sim 2.83 \pm 0.15$  is *not* a discriminating clock value — within its  $\pm 5.3\%$  band several simple values fit  $(2^{3/2}, 17/6, 14/5)$ , so by No-Free-Pattern it is declared EXTERNAL (the firewall v187/v374 is correct there). **[C]** verdict: 3 of the 4 frontier scheme-numbers land *exactly* on the clock, so the irreducible “external physics” residue shrinks to essentially the proton’s non-perturbative QCD number  $C_p$ . Cited in **tfpt\_4\_frontier**; a falsifiable probe whose answer could have been “none on the clock”; Python-only.

**2026-06-24 (Clockwork coherence + an open question: v394 primes = atoms = facets, and RES.COXETER.SYMMETRY.01)**

- **v394 (COXETER.ATOMS.01) — the three Coxeter primes ARE the three anchor atoms ARE the three number-field facets.** A bird’s-eye [E] re-reading tying v390 (the prime facets) to the anchor  $a = (1, 1, 2)$  and v319 (the  $5 \times 6$  clock). [E]  $a = (1, 1, 2)$  has  $e_3=2=|Z_2|$ ,  $p_0=3=N_{\text{fam}}$ ,  $e_2=5=g_{\text{car}}$ , product  $2 \cdot 3 \cdot 5 = 30 = h(E_8)$  — the clock’s primes *are* the anchor’s atoms. [E] atoms = facet fields (v390):  $2 \rightarrow \mathbb{Q}(i)$ ,  $3 \rightarrow \mathbb{Q}(\sqrt{-3})$ ,  $5 \rightarrow \mathbb{Q}(\sqrt{5})$ , each ramified at its prime. [E] the QG susceptibility is pure  $\{2, 3\}$ :  $\chi = 1/(1-(2/3)^6) = 729/665 = 3^6/(3^6-2^6)$ ,  $(2/3)^6 = 2^6/3^6$ , exponent  $6 = 2N_{\text{fam}}$ . [E] both dynamic rates carry prime-2:  $2/3 = |Z_2|/N_{\text{fam}}$  and  $(\varphi+2)/4 = (\varphi+2)/|\mu_4|$  with  $|\mu_4|=4=2^2$ . [E]  $30 = 5 \times 6 = g_{\text{car}}(2N_{\text{fam}})$  (v319). [C] the prime-2 seam is the universal dynamical engine; prime-3 (family) and prime-5 (carrier golden) the two attractors.
- **RES.COXETER.SYMMETRY.01 — a new registered open research question.** Is the prime-2-as-common-factor reading *forced* (the seam involution provably in every sector’s gap, no independent prime-2 attractor), or is there a residual asymmetry? Resolving it would either tighten v383’s “every sector is the same gapped operator” to “the same *seam-driven* operator” (with the falsifiable corollary that no prime-2-only attractor exists), or reveal a missing dynamical slot. Tracked in `tfpt_research_contracts`; a question, not a claim. Cited in `origin_theory`; Python-only.

**2026-06-24 (Correction error-bars: v393 the typed budget as actual first-correction magnitudes)**

- **v393 (CORRECTIONS.NUMERIC.01) — the typed correction budget (v388) as actual numbers.** The short-term, testable harvest of v388: instead of merely typing each prediction’s correction, this gives the magnitude, so a parameter-free central value can be quoted as “value  $\pm$  computed first correction”. [E] FIXED-POINT band (the  $\varphi_0$ -derived  $\theta_{12}, \theta_{13}, \lambda_C, \beta_{\text{rad}}$ ) = the explicit  $\varphi_0$  puncture term  $36c_3^4/c_3 = 9/(128\pi^3) \approx 0.227\%$  (exact; =  $\theta_{12}$ ’s  $\varepsilon$  first-correction,  $\varepsilon = (3/4)\varphi_0 = c_3 + 36c_3^4$ ); [E] SEAM-GAPPED band (Koide  $F_{\text{pole}}$ , recovery) =  $(2/3)^6 \approx 8.78\%$ , the QG-decoupling bound capped by  $\chi - 1 = 64/665 \approx 9.62\%$  (v337); [E] EXACT-IDENTITY band = 0 structurally ( $\det R = 8$ ,  $N_\Phi = 1$ ,  $\theta_{\text{eff}} = 0$ ); [E] EXTERNAL-RATE band = external physics, quoted ( $m_p/m_e \pm 7.5\%$  v374;  $n_s/r$  the  $N_\star \in [50, 60]$  band;  $\eta_B 1.07 \times \text{v212}$ ), fenced by v187. [C] the assembled budget gives each prediction a computed theoretical-uncertainty band. Cited in `origin_theory`, mirrored on the website prediction surface; a harvest of v388, no new number; Python-only.

**2026-06-24 (Attacking the last open lemma: v392 the scaling-limit central charge converges)**

- **v392 (SEAM.S3.SCALINGLIMIT.01) — an honest attack on SEAM.EQUIV.01’s one open lemma (continuum existence, v336).** It does *not* close v336 (the abstract existence is the cited MMST theorem); it strengthens the case with the existence-relevant observable v376 did not test: that the finite-size data *converge*. [E] the Calabrese–Cardy entanglement slope  $c_1(L)$  at  $L = 66, 130, 258, 386, 514$  is a monotone Cauchy-like sequence approaching 1 ( $|c_1(L) - 1|$  strictly decreasing,  $4.6 \times 10^{-4} \rightarrow 7 \times 10^{-6}$ ); [E] a  $1/L$  Richardson fit extrapolates to  $c_1(\infty) \approx 1.000$ , so the 16-Majorana collar  $c = 8c_1 \approx 8.00 = g_{\text{car}} + N_{\text{fam}}$  is reached *in* the limit; [E] the successive differences are themselves strictly decreasing (a numerical Cauchy sequence). [O] convergent data are evidence, not a proof of existence (the cited MMST theorem, v336), and do not pin  $(E_8)_1$  over the same- $c$   $SO(16)_1$  (the holomorphy discriminator, v377/v378). Strengthens the residual, does *not* close it. Cited in `tfpt_research_contracts`; Python-only.

## 2026-06-24 (Lean 4: FORM.SPECTRALGAP.01 extended to the finite-dimensional Perron–Frobenius case)

- **FORM.SPECTRALGAP.01 — the multi-dimensional case, reduced to the scalar core.** The Lean module `TfptCarrier/SpectralGapAttractor.lean` now also proves the *finite-dimensional* gapped-operator statement (lake build OK, no sorry): **[E]** `component_tendsto` (a non-dominant eigencomponent  $|r| < 1$  decays,  $r^n c \rightarrow 0$ ); **[E]** `deviation_tendsto` (with the dominant eigenvalue normalised to 1, the deviation vector — all non-dominant components — tends to 0, so the iterate converges to the dominant eigendirection, the unique attractor, for any start); **[E]** `deviation_unique` (two starts give deviation vectors both  $\rightarrow 0$ , so the selected eigendirection is independent of the start — parameter-freeness in finite dimensions); **[E]** `flavor_deviation_tendsto` (a concrete instance on the cusp-transfer spectrum  $\{1, (2/3)^6, (1/3)^6\}$ , v56/v82/v383). Honest scope: this is the finite-dimensional *diagonalisable* case (the genuine multi-dimensional content, reduced to the scalar core); the general non-diagonalisable / infinite-dimensional Perron–Frobenius theorem stays the cited statement. Referenced in `origin_theory`.

## 2026-06-24 (Website: an animated “Universal Gap Lab” visualising v383–v391)

- **Website — the UniversalGapLab animation on /verification.** A new client component (Framer Motion, no new dependency) with three linked, module-grounded views: (A) every sector as the gap contraction  $\text{iter}_n = x_\star + r^n(x_0 - x_\star)$  racing to its attractor at  $r = (2/3)^6$  (seam-gapped, v303/v387/v388) or the golden  $(\varphi + 2)/4$  (v312), with a gapless  $r = 1$  track that never forgets its start (the Lean theorem FORM.SPECTRALGAP.01) — the rate *is* the first-correction size (v388); (B) the entire-form-factor graviton ratio  $e^{-u}$  falling faster than any power, ghost-free and UV-soft (v386/v389); (C) the residual matrix as certification, not construction — every open item external, zero open internal mechanisms (v384). Mirror-only: every number is grounded in a module; no theory change.

## 2026-06-24 (Honest attempt at an external target: v391 reduces it, not a closure)

- **v391 (ALPHA.QUILLEN.PROGRESS.01) — an honest attempt at the external target ALPHA.QUILLEN.EXACT.01 (v382, door 5).** It REDUCES the open step to a sharper named sub-target via a solvable model but does *not* close it: ALPHA.QUILLEN.EXACT.01 stays **[O]** (external math, type A, v384). **[E]** a solvable model  $\det_\zeta(-\partial_\tau^2 + m^2)$  on a circle of circumference  $\beta$  equals  $(2 \sinh(\beta m/2))^2$ , whose  $m$ -variation  $\beta \coth(\beta m/2) = 2/m + (\beta^2/6)m - (\beta^4/360)m^3 + \dots$  is a closed heat-kernel/Laurent series (the  $m^3$  coefficient  $-1/360$  verified). **[E]** so a  $\zeta$ -determinant’s coupling-variation is structured heat-kernel data, and v382’s counterterm  $8b_1 c_3^6$  (a Gauss–Bonnet heat-kernel power) is the right *kind* of object; the v382 split is re-verified at  $\alpha^{-1} = 137.0359992$ . **[E]** the reduction: the open step now reduces to the named sub-target “the seam  $U(1)$  operator’s Seeley–DeWitt coefficient  $= 8b_1 c_3^6$ ”. **[O]** *not closed*: the actual seam-operator  $\zeta$ -determinant proof stays open, and SEAM.EQUIV.01 continuum existence (MMST, v336) stays the other external **[O]**; the value  $\alpha^{-1}$  stays **[E]**. An attempt that reduces, does not close. Cited in `tfpt_1_architecture_e8`; Python-only.

## 2026-06-24 (Lean 4: FORM.SPECTRALGAP.01 formalises the scalar core of the universal spectral-gap principle)

- **FORM.SPECTRALGAP.01 — a Lean 4 proof of the gap  $\Rightarrow$  unique-attractor  $\Rightarrow$  parameter-free core (door 4).** New module `TfptCarrier/SpectralGapAttractor.lean` (lake build OK, no sorry, only the three standard kernel axioms), formalising the meta-theorem of v303/v383/v387 at the one-dimensional reduction — the affine gap contraction  $x \mapsto x_\star + r(x - x_\star)$  with multiplier  $r = \lambda_2/\lambda_1$ . **[E]** `iter_sub`: the closed form  $\text{iter}_n - x_\star = r^n(x_0 - x_\star)$  (general  $n$ ); **[E]** `iter_tendsto`:  $0 \leq r < 1$  (the gap)  $\Rightarrow \text{iter}_n \rightarrow x_\star$  (the unique attractor);

[E] **starts\_agree**: two starts differ by  $r^n(a - b) \rightarrow 0$ , so the attractor is independent of the initial state (parameter-freeness); [E] **no\_gap\_no\_forget**:  $r = 1$  (no gap) leaves  $\text{iter}_n - x_* = x_0 - x_*$  constant — the initial condition is then a free parameter, so the gap is exactly what removes it; [E] the seam rate  $(2/3)^6$  and the golden  $(\varphi + 2)/4 = (5 + \sqrt{5})/8$  both lie in  $[0, 1)$ . Honest scope: the scalar reduction (the contraction core v303/v383 iterate numerically); the full multi-dimensional Perron–Frobenius statement stays the cited theorem. Referenced in **origin\_theory**; ledger row **FORM.SPECTRALGAP.01**.

## 2026-06-24 (Completing the Coxeter clock: v390 the prime-2 (Gaussian / $\mathbb{Z}/4$ -seam) facet of $30 = 2 \cdot 3 \cdot 5$ )

- **v390 (COXETER.PRIME2.01) — the prime-2 facet of the order-30 Coxeter clock.** v383/v387 named two number-field facets of the clock  $30 = 2 \cdot 3 \cdot 5 = h(E_8)$  — the golden  $\varphi \in \mathbb{Q}(\sqrt{5})$  (prime-5, carrier) and  $(2/3)^6 \in \mathbb{Q}$  (prime-3, family). This names the *prime-2* facet and identifies it with structure already in the theory: the Gaussian field  $\mathbb{Q}(i)$ , i.e. the CM point  $\tau = i$  and the order-4  $\mu_4 = \mathbb{Z}/4$  *square* seam deck (v282/v288/v333). [E]  $h(E_8) = 30$  factors into exactly three primes; one quadratic facet field per prime, each ramified at its prime:  $\text{disc } \mathbb{Q}(i) = -4$  (2 ramifies),  $\text{disc } \mathbb{Q}(\sqrt{-3}) = -3$  (3 ramifies),  $\text{disc } \mathbb{Q}(\sqrt{5}) = 5$  (5 ramifies); the ramified primes are  $\{2, 3, 5\}$  with product  $30 = h(E_8)$ . [E] the prime-2 (Gaussian) facet is the  $\mathbb{Z}/4$  seam deck: min. poly  $x^2 + 1$ ,  $j(i) = 1728 = 12^3$ ,  $\tau = i$  fixed by the modular  $S$  of order  $4 = 2^2$ ,  $|\mu_4| = 4 = 2^2$  — the static, even, seam-orientation facet. [C] so the three primes of the clock now each carry a facet (square/hexagon/pentagon = seam/CP/carrier); the prime set  $\{2, 3, 5\}$  is forced by  $h(E_8) = 30$ , not chosen. Cited in **origin\_theory**; a synthesis, no new number; Python-only.

## 2026-06-24 (Graviton loop finiteness: v389 the entire form factor is UV-finite at the loop level by power counting)

- **v389 (GRAV.LOOP.FINITE.01) — the entire-form-factor graviton is UV-finite at the loop level.** The honest next step past v386 (which established the tree amplitude), extending v304/v370/v380/v386 from the propagator to the superficial degree of divergence of loop graphs. This is the standard infinite-derivative / nonlocal-gravity finiteness statement (Tomboulis; Biswas–Mazumdar–Siegel; Modesto), *not* a full proof of perturbative quantum gravity. [E] super-polynomial falloff:  $p^k e^{-p^2/M^2} \rightarrow 0$  for every degree  $k$ . [E] one dressed line turns a quadratic divergence into the finite  $(M^2/2)e^{-p_0^2/M^2}$ , while the GR  $\int p^3/p^2 dp$  diverges quadratically. [E] the superficial degree  $\rightarrow -\infty$  at every loop order  $L$  (the form factor multiplies the integrand by  $e^{-(\#\text{internal})p^2/M^2}$ ; dressed cutoff-ratio  $\rightarrow 1$  vs GR  $\rightarrow 4$ ). [E] the regulator is the seam scale  $M = M_{\text{scal}} = c_3^{7/2} \bar{M}_{\text{Pl}}$  (v253/v259), not a dial. [C][O] loop *unitarity* of the nonlocal form factor (Lorentzian continuation / macro-causality, Pius–Sen / Briscese–Modesto) stays the honest open point — tree unitarity is v386; not claimed solved. Cited in **tfpt\_4\_frontier**; Python-only.

## 2026-06-24 (Typed correction budget: v388 the gap correction is keyed to the mechanism class, not a uniform band)

- **v388 (CORRECTIONS.BUDGET.01) — the gap-driven correction (v387) is a typed budget.** A machine-checked answer to “does the  $(\lambda_2/\lambda_1)^n$  correction apply to *all* predictions?” — the answer is *no*, and for one whole class a band would be *wrong*. Because the size is the distance from a gapped operator’s leading eigenvector, it is defined only where a subleading  $\lambda_2$  exists, so the prediction surface splits into four classes. [E] *seam-gapped* (Koide  $F_{\text{pole}}$  v303, recovery v221, the QG bound capped by  $\chi = 729/665$  v337): rate  $(2/3)^6$ ; the discrete compiler: golden  $(\varphi + 2)/4$  (v312). [E] *exact identity* ( $\det R = 8$ ,  $N_\Phi = 1$ ,  $\theta_{\text{eff}} = 0$ ,  $\sin^2 \theta_{13} = \varphi_0 e^{-5/6}$ ): no  $\lambda_2$ , so the correction is 0 structurally and a band would contradict the [E] status. [E]

*external-rate* ( $\eta_B$  washout, axion freeze,  $m_p/m_e$  RG): the gapped *shape* with an external rate — only  $F_{\text{pole}}$  has the seam rate (v303), firewalled (v187). [E] *fixed-point*: the texture is already explicit (e.g.  $\sin^2 \theta_{12}$ ,  $\varepsilon = (3/4)\varphi_0 = c_3 + 36c_3^4$ ). [E] the Koide first correction (door 2):  $F_{\text{pole}}$  is the one seam-gapped prediction, rate  $(2/3)^6 \approx 0.0878$  (v371), the source quotient 0.33% below  $2/3$  — the rate is the suppression *scale*, not the residual. [C] so a uniform band on all predictions is wrong; the size is keyed to the mechanism class. A typing of v387 (as v384 is of v383); cited in `origin_theory`; Python-only.

## 2026-06-24 (Harvesting the QFT/gravity machinery: v385 the UV-branch kill-test surface, v386 the graviton-exchange amplitude, v387 the gap-driven correction calculus)

- **v385 (UVBRANCH.KILLTEST.01) — the carrier-Pati–Salam UV branch on the all-order footing.** A consolidation that ties the all-order EG/BRST closure (v381) to the optional UV branch and computes the one genuinely new thing: the *proton-decay safety hierarchy*. [E] the carrier content is the SM with no new states, so the SM-non-unification (v246) is now all-order-robust; [X] the plain-SM 4D-GUT stays killed ( $\sin^2 \theta_W = 3/8$  vs 0.231); [C] the optional carrier-PS branch sits at  $M_{PS} = \kappa M_{\text{scal}} \approx 3 \times 10^{13}$  GeV ( $\kappa \in [1.0, 1.2]$ , v253); [E] minimal PS gauge bosons are  $SU(4)$  leptoquarks that mediate rare LFV ( $K_L \rightarrow \mu e$ ), *not*  $p \rightarrow e^+ \pi^0$ , so the branch clears the binding bound by  $M_{PS}/M_{\text{bound}} \sim 10^7$  – proton-safe, with *no* fake  $\tau_p$  window (v265). Cited in `tfpt_5_redteam`; Python-only.
- **v386 (GRAV.AMPLITUDE.01) — the graviton-exchange amplitude is finite and UV-soft.** Extends v304/v370/v380 from the propagator to the actual tree amplitude. [E] the dressed spin-2 propagator  $e^{-p^2/M^2}/p^2$  has its only pole at  $p^2 = 0$  (residue  $1 > 0$ , ghost-free), is exponentially softened in the UV (the ratio  $e^{-u}$  falls faster than any power) and reduces to GR + a calculable  $-p^2/M^2$  correction at low energy ( $M = M_{\text{scal}}$ , v253); [C] a single positive-residue pole gives tree unitarity, so perturbative graviton scattering  $A \sim (T \cdot P_2 \cdot T') e^{-p^2/M^2}/p^2$  is an explicit finite computation (perturbative-only; QG.AMB.01 stays a [C] redundancy, v369). Cited in `tfpt_4_frontier`; Python-only.
- **v387 (CORRECTIONS.GAP.01) — the gap-driven correction calculus.** The practical harvest of v383: the same spectral gap that makes a sector parameter-free also sets the *size* of its first correction ( $\text{correction}_n \sim (\lambda_2/\lambda_1)^n$ ). [E] flavor, horizon recovery and the QG bound all share the  $(2/3)^6$  rate (the family-3 facet; QG susceptibility  $\chi = 729/665$ ), while the discrete compiler transient decays at the golden  $(\varphi + 2)/4$  (the carrier-5 facet) — the two number-field facets of v314/v383. [C] the gap does double duty (why no free parameter *and* how big the first correction), so sub-leading magnitudes are organised, not free. Cited in `origin_theory`; Python-only.

## 2026-06-23 (Bird’s-eye synthesis: two new structural patterns — v383 the universal spectral-gap principle and v384 the residual is certification, not construction)

- **v383 (DYNAMICS.UNIVERSAL.01) — the universal spectral-gap / Perron–Frobenius principle.** A fresh bird’s-eye pass found that *every* TFPT sector is the *same* structural object: a gapped operator with a *unique* leading attractor (the physics) and a spectral gap (the reason there is no free parameter). v303 named this only for  $F_{\text{transfer}}$ ; this module extends it to the sectors made parameter-free *after* it. [E] three gaps are computed — the flavor transfer  $\{1, (2/3)^6, (1/3)^6\}$  with gap  $6 \ln(3/2)$  (v56/v82); the affine- $E_8$  adjacency with Perron eigenvalue 2 (Kac marks) and subleading  $2 \cos(\pi/5) = \varphi$ , gap  $2 - \varphi = 1/\varphi^2$  (v312/v313); the decoupling margin  $\Delta_{\text{eff}} \approx 1.648$  (v76/v337). [E] the two number-field facets (v314): golden  $\varphi \in \mathbb{Q}(\sqrt{5})$  (carrier-5, static) and  $(2/3)^6 \in \mathbb{Q}$  (family-3, dynamic) are the prime-5 and prime-3 facets of the order-30 Coxeter clock  $30 = 2 \cdot 3 \cdot 5 = h(E_8)$ . [C] the other sectors are the same structure — gravity ( $\delta S=0$  equilibrium, v358/v359), the QG measure (Gaussian

saddle, v365), the boundary QFT (holomorphic  $c=8$  RG fixed point, v157/v158), the all-order S-matrix (adiabatic limit, v381), recovery (Petz fixed point, v221),  $\tau=i$  (v333). [C] the meta-theorem: gapped  $\Rightarrow$  unique attractor  $\Rightarrow$  parameter-free, theory-wide. Cited in `origin_theory`; a synthesis (no new number), Python-only.

- **v384 (RESIDUAL.CERTIFICATION.01) — the residual is certification, not construction.** After v369/v381/v382 the *kind* of every open item has shifted: there is *no* open TFPT physics mechanism left. A ledger-CI audit classifies every residual into exactly three external kinds: [E] (A) external math proof (SEAM.EQUIV.01 continuum existence = the cited MMST/Adamo theorem v336; ALPHA.QUILLEN.EXACT.01 v382; QG.AMB.01 the constructive measure, a [C] redundancy v369), [E] (B) theorem-forbidden ( $v_{\text{geo}}$ , the No-Unit theorem v153/v364), [E] (C) external physics ( $F_{\text{transfer}}$ , v371–v374, firewalled v187). The two load-bearing facts are verified (No-Unit dimensionlessness; the gap-decoupling margin  $1.648 > 0$ ), and the count of open *internal* TFPT mechanisms is **zero**. [C] convergence: the only hard things left are theorems and external physics, not missing dynamics. Cited in `tfpt_research_contracts`; an audit (no new number), Python-only.

**2026-06-23 (External-review closure: the two genuine residual targets named — v381 the all-order Epstein–Glaser/BRST contract for  $S_{\text{pert}}$ , and v382 the Quillen determinant-line variation as a tracked target)**

- **v381 (QFT4D.EG.ALLORDER.01) — the all-order 4D perturbative-QFT closing statement.** A mathematical-physics review of the QFT/TOE layers identified exactly one genuinely new, buildable item: the *all-order* Epstein–Glaser/BRST contract for the perturbative 4D S-matrix  $S_{\text{pert}}$  (the previous documents had the  $S_{\text{pert}}$  skeleton and one-loop, v269/v271/v273/v278, but no named all-order claim). This module types the standard causal-perturbation-theory closure for the TFPT finite interaction class — it does *not* build a path integral. [E] power-counting: every gauge+matter vertex (Yang–Mills + Dirac–Yukawa + Higgs + ghost) is a mass-dimension-4 marginal operator, so the EG singular order  $\omega \leq 4$  and the counterterm space is finite at every order; [E] BRST nilpotency  $s^2=0$  verified exactly via the Jacobi identity for  $su(2)$  and  $su(3)$  (the carrier weak+colour content); [E] the seam gap  $\Delta = 6 \log \frac{3}{2} > 0$  (v302) gives the IR-safe adiabatic limit. [C] the two heavy legs are imported rigorous theorems — all-order  $T_n$  existence by causal factorisation (Epstein–Glaser 1973) and all-order BRST invariance (Piguet–Sorella / Kugo–Ojima) — applied to the TFPT class. Matter+gauge *only*: the  $R^2/\text{Weyl}^2$  gravity sub-sector is the entire-form-factor resummation (v304/v370/v380), and  $S_{\text{pert}}$  is not the ambient measure QG.AMB.01 (a [C] redundancy, v369). NET [C], prerequisites [E]; not [E] and not a path integral. Cited in `tfpt_5_redteam` and `tfpt_2`; Python-only.
- **v382 (ALPHA.QUILLEN.EXACT.01) — the Quillen determinant-line variation, named as a target.** Per the same review, the “why *this*  $F_{U(1)}$  functional” obligation (previously tracked only as EM.WARD.01’s residual, v341/v342) is elevated to its own citable target:  $\delta_\tau(\log \det_\zeta \Delta_{U(1)} + 8b_1 c_3^6 \log \varphi_{\text{seam}}) = 0$  on the determinant line over the  $\mu_4$  seam moduli. [E] the Quillen split holds at the  $\alpha^{-1}=137.0359992$  root with every coefficient a named atom (index  $8b_1$ , heat-kernel  $c_3^{3,6}$ , discriminant  $q(D_5)+q(A_3)=2$ ); [E] the value stays closed (v3); [O] the from-first-principles exactness proof is the target. [C] it is a *face* of SEAM.EQUIV.01 (v336) — the seam  $U(1)$  determinant line lives on the same holomorphic net — so it adds *no* second independent gate; it makes the existing residual trackable. No new number (reuses the Wolfram-mirrored v341 identities). Cited in `tfpt_1_architecture_e8`; Python-only.

**2026-06-23 (Track 2 of the forward plan v2 — the AMBIENT REDUNDANCY statement (v369): the holographic route to Gravity-complete that sidesteps building the general Euclidean QG measure)**

- **v369 (QGAMB.REDUNDANCY.01) — Track 2 (forward plan v2).** Instead of constructing the non-perturbative ambient measure (the conformal-factor swamp, v332/v334), assemble the established discriminators that prove the ambient sector carries *no* physical degrees of freedom relative to the holomorphic boundary algebra, so the QG measure (gate C7/QG.AMB.01) is a non-fundamental *certification* object (like a coordinate choice in GR), not missing dynamics. [E] holomorphy  $\Rightarrow$  DHR( $(E_8)_1$ ) = Vec: #primaries =  $|\det \text{Cartan}| = 1$  for  $E_8$  vs 4 for  $SO(16)$   $\Rightarrow$  no nontrivial superselection charge a bulk could carry invisibly; [E] no torus ground-state degeneracy ( $|\det K| = 1$ ); [E] finite Petz recovery rate  $(2/3)^6$  (v221); [E] the gap-decoupling margin  $\Delta_{\text{eff}} = 6 \log \frac{3}{2} - \frac{31}{4\pi^2} \approx 1.648 > 0$  (v76/v337). [C] with the Bisognano–Wichmann reconstruction map (v258/v309/v323) the redundancy statement  $\mathcal{O}_{\text{phys}}^{\text{bulk}}/\text{Diff} \cong \mathcal{A}_{\Sigma}^{\mu_4}$  is well-posed. [O] residual: the two consumed premises — SEAM.EQUIV.01 (the S3 lattice input, v366) and the BW intrinsicality of the raw collar (v329); it does *not* construct QG.AMB.01 but provides the *alternative* (holographic redundancy) route: Gravity-complete = Boundary-complete + Ambient-redundancy. Cited in `tfpt_research_contracts`; Python-only.

**2026-06-23 (Status propagation: the v365–v380 + FORM.SEAM.MMST.01 status carried into the remaining papers, the README and the Zenodo description)**

- **Deep-sync of the post-S3 status.** The canonical surfaces (`tfpt_status`, `introduction`) were already current; this pass propagated the same honest wording into `origin_theory`, `tfpt_1–tfpt_5`, `tfpt_research_contracts`, the root `README.md`, the Zenodo description and `website/lib/papers.ts`: SEAM.EQUIV.01 is *closed modulo a cited theorem* (lattice v367/v368 + the  $S_3$  stack v376–v379 + Lean FORM.SEAM.MMST.01; residual [O] = the cited MMST/OS existence only); QG.AMB.01 is a [C] *redundancy* (v369/v379), a certification object not missing dynamics; perturbative graviton unitarity is established (the Stelle ghost is a Seeley–DeWitt truncation artefact, v304/v370/v380); and the frontier transfers are typed runnable solvers with kill tests (v371–v374) + the observatory CI (v375). No new claim — a wording propagation so every surface matches the ledger; AUDIT OK.

**2026-06-23 (v380 — the KMS Entire Hessian: the Stelle ghost is EXACTLY the Seeley–DeWitt truncation, and resummation pushes it to infinity)**

- **v380 (GRAV.KMS.HESSIAN.01).** Upgrades v304/v370 from the *assumption* “the resummed graviton form factor is entire” to a derived statement. [E] the seam KMS cutoff  $f(u)=e^{-u}$  (v259) gives the dressed spin-2 propagator  $e^{-p^2/M^2}/p^2 = 1/(p^2 a)$  with  $a(u)=e^u$  entire and nowhere zero (only the  $p^2=0$  pole). [E] each finite Seeley–DeWitt truncation is a Taylor partial sum of  $a$  and carries a Stelle-ghost zero ( $T_1=1+u \rightarrow u=-1$ ;  $T_2 \rightarrow -1 \pm i$ ); the entire  $a$  has none. [E] the modulus of the nearest truncation-zero grows monotonically with the order  $(1, 1.41, \dots, 3.07 \text{ for } n=1..8) \rightarrow \infty$  — the ghost decouples, “entire” = “do not truncate”. [C]  $\Rightarrow$  perturbative graviton unitarity; [O] the exact off-shell curved-space Hessian identification is the cited heat-kernel resummation, perturbative-only. Cited in `tfpt_5_redteam` (Target F); Python.

**2026-06-23 (Lean: the S3 continuum leg formalised as “closed modulo a cited theorem” — FORM.SEAM.MMST.01)**

- **FORM.SEAM.MMST.01 — Lean 4 formalisation.** The S3 continuum leg is now machine-formalised in a new module (`SeamScalingLimit.lean`), lake build OK and `#print axioms` clean, in the audit-contract style of `SeamEquivChain.lean`. The MMST applicability hypotheses for the seam collar are *provable* by kernel `decide` (no axioms):  $D = 16$ ,  $\text{rank} = c = 8$ ,

the range  $8 \leq c \leq 16$ ,  $c_- = 8 \neq 0$ , and  $\det K = 1$  vs  $SO(16)$ 's 4. The MMST scaling-limit theorem (CMP 2022) and the Adamo–Moriwaki–Tanimoto OS reconstruction (2024) are *named cited axioms*; the conclusion  $(E_8)_1$  follows by a well-typed composition whose residual (`#print axioms`) is exactly those two published theorems plus the carrier gap — no hidden assumption, no `sorry`. Mirrors the S3 closure stack `v376–v379`; so `SEAM.EQUIV.01` is now “closed modulo a cited theorem”, the honest formalisation ceiling. Cited in `tfpt_research_contracts`; `lean_manifest` refreshed.

**2026-06-23 (S3 closure stack — the target of the seam scaling limit pinned at EVERY computable level: central charge  $c=8$  from the lattice (`v376`), the  $(E_8)_1$  character (`v377`), the genus-1 torus count = 1 (`v378`), and reflection positivity (`v379`); only the abstract continuum existence (cited MMST) stays external)**

- **`v376 (SEAM.S3.CENTRALCHARGE.01)`**. The Calabrese–Cardy finite-size entanglement-entropy scaling of the critical free-fermion content gives  $c = 1.0000$  per complex mode (Peschel correlation-matrix EE; the  $L=4m+2$  sizes avoid the half-filling zero-mode), so the 16-Majorana (= 8 complex) collar has  $c = 8 = g_{\text{car}} + N_{\text{fam}}$ , chiral ( $c_- = 8$ , `v367/v368`). Numerical  $c=8$  from the lattice. Python (numpy).
- **`v377 (SEAM.S3.E8CHARACTER.01)`**. The exact affine character  $\chi = E_4/\eta^8 = q^{-1/3}(1 + 248q + 4124q^2 + \dots)$  gives  $c = 8$  and 248 level-1 currents (one primary); the  $\mu_4/\text{GSO}$  promotion  $248 = 240 + 8 = 120 + 128$  ( $240 = 16 \cdot 5 \cdot 3$ ) pins  $(E_8)_1$  over the same- $c$  rival  $SO(16)_1$  (120 currents, 4 primaries). Python (sympy q-series).
- **`v378 (SEAM.S3.MODULAR.01)`**. The genus-1 discriminator: KLM condensation  $\mu(\text{carrier})=16 \rightarrow \mu(E_8)=16/4^2=1$ ,  $|\det \text{Cartan}(E_8)|=1$  vs  $D_8=4$ , and the single character's modular  $T$ -eigenvalue  $e^{-2\pi i/3}$  give torus ground-state degeneracy 1 (holomorphic =  $(E_8)_1$ ) — the “hard part” `v344` isolated. Python (sympy).
- **`v379 (SEAM.S3.RP.01)`**. Reflection positivity (the OS axiom) of the explicit gapped collar: the positive spectral measure makes the Hankel/reflection Gram matrix PSD (a ghost mode breaks it). With  $c=8 + (E_8)_1$  in hand, the OS inputs are complete and `C7/QG.AMB.01` is then discharged by the redundancy theorem `v369`. Python (numpy).
- **Net:** S3 is the master key — on the explicit lattice model the target net  $(E_8)_1$  is pinned exactly (central charge, character, genus-1 count, RP); the only remaining residual is the abstract *continuum existence* of the scaling limit (the cited MMST theorem `v336`), which lives outside the computational suite. Cited in `tfpt_research_contracts`.

**2026-06-23 (Track 4 — the prediction OBSERVATORY: a status-typed CI over the frozen prediction registry that makes the falsifiability surface machine-checkable, with a live data scorecard (`v375`))**

- **`v375 (OBSERVATORY.REGISTRY.01)`**. A CI over `freeze_file.csv`: [**E**] every prediction carries a kill criterion (falsifiability complete, no orphan claims); [**E**] the headline values re-derive from  $\{\varphi_0^{\text{ret}}\}$  ( $\sin^2 \theta_{12} = 1/3 - \varphi_0^{\text{ret}}/2$ ,  $\sin^2 \theta_{13} = \varphi_0^{\text{ret}} e^{-5/6}$ ,  $\beta_{\text{rad}} = \varphi_0^{\text{ret}}/(4\pi)$ ); [**C**] a live scorecard records  $\theta_{12}$  vs the JUNO first run ( $0.28\sigma$ , compatible),  $\theta_{13}$  vs NuFIT 6.0 NO ( $2.0\sigma$ , the flagged most-tensioned core prediction),  $\beta_{\text{rad}}$  vs ACT DR6 ( $0.37\sigma$ , a hint),  $r < 0.036$  (compatible). Tooling that makes the closed pieces usable and the open ones honestly fenced; no new physics. Cited in `tfpt_5_redteam`; Python (stdlib).

**2026-06-23 (Track 3 — the  $F_{\text{transfer}}$  functor promoted from contracts to TYPED runnable solvers with kill tests: Koide source→pole (v371),  $\eta_B$  Boltzmann (v372), axion finite- $T$  relic (v373),  $m_p/m_e$  uncertainty budget (v374))**

- **v371 (FR.POLE.SOLVE.01).** A clean *negative*: standard QED/EW running cannot be the Koide source→pole transfer (it pushes  $Q$  above  $2/3$ ; the required  $\epsilon$  is negative while QED's is positive), so the transfer is the non-QED  $53/54$  operator /  $(2/3)^6$  Moebius generator (v183/v82). [C], plain running excluded as the kill. Python (numpy+scipy).
- **v372 (FR.BOLTZMANN.SOLVE.01).** The integrated Buchmueller–Di Bari–Pluemacher Boltzmann ODE at the TFPT-frozen  $M_1=M_{\text{scal}}(\varphi_0^{\text{ret}})^2/A_\Lambda$  (v212) gives  $\eta_B$  within a factor  $\sim 1.1$  of  $6.1 \times 10^{-10}$  with no  $M_R$  dial; replaces the v326 toy washout. [C]/[X].
- **v373 (FR.RELIC.SOLVE.01).** The finite- $T$  axion misalignment ODE decides the branch: the frozen spine angle  $3\pi/5$  gives  $\Omega_a h^2 \sim 0.12$  (in  $[0.08, 0.16]$ , lands on  $\Omega_{\text{DM}}$  untuned) while the hilltop  $170^\circ$  over-produces ( $\sim 5.4 \times$ ); replaces the v326 toy. [C]/[X] (lattice  $\chi(T)$  input).
- **v374 (FR.QCD.BUDGET.01).** The  $m_p/m_e$  uncertainty budget: with  $b_3 = -7$  [E] and the two declared externals ( $\alpha_s(M_Z)$ ,  $C_p$ ,  $\sim 5.3\%$  each), the central  $\sim 1824$  (v262) carries a propagated  $\pm 7.5\%$  band  $[\sim 1690, \sim 1960]$  CONTAINING 1836.15; lattice tightening is the kill. The falsifiability companion to v262; [C].

**2026-06-23 (Track 1 — the S3 input made CONCRETE: an explicit gapped chiral-Majorana ( $p+ip$ ) lattice model with numerical Chern  $|C|=1$  realises the  $c_-=8$  edge (v367), and a strip shows the chiral edge exists by anomaly inflow (v368))**

- **v367 (SEAM.S3.LATTICE.01) — Track 1.** The abstract S3 residual of v356/v366 (“the collar realised as a genuine lattice chiral free-fermion invertible phase”) is turned into a RUNNABLE model. [E] the  $p+ip$  Bloch model is gapped at  $M=1$  with a numerically computed Fukui–Hatsugai–Suzuki Chern number  $|C|=1$  ( $C=0$  trivially at  $M=3$ ); [E]  $N_{\text{Maj}} = \dim S^+ = 16$  copies give  $c_- = 16/2 = 8 = g_{\text{car}} + N_{\text{fam}}$ , with the edge K-matrix  $SO(16)_1$  ( $\det 4$ )  $\rightarrow (E_8)_1$  ( $\det 1$ ) by the order-4  $\mu_4$  clock (v351/v369). [C] the  $\mu_4$  condensation (v154/v351); [O] the continuum scaling-limit existence (MMST, v336) stays open. Cited in `tfpt_research_contracts`; Python (numpy + sympy).
- **v368 (SEAM.S3.INFLOW.01) — Track 1.** The bulk-edge / anomaly-inflow leg (S4) made concrete on the v367 model: [E] a finite strip has an edge-localised in-gap chiral branch ( $\min_{k_x} |E| \sim 0$ ) in the topological phase and a clean gap in the trivial phase (neg control); [E] bulk-edge correspondence gives one chiral Majorana per edge ( $c_-=8$ ). [C] so by anomaly inflow the chiral edge EXISTS as a consequence of the invertible bulk, discharging the v336/v351 “does the edge exist?” residual on the lattice. [O] the continuum scaling limit stays open. Cited in `tfpt_research_contracts`; Python (numpy).

**2026-06-23 (Track 2 — perturbative graviton unitarity sector-by-sector: the Barnes–Rivers spin decomposition localises the Stelle ghost in the spin-2 sector, cured by the entire KMS form factor; the spin-0 mode by the GHP contour (v370))**

- **v370 (GRAV.SPIN2.UNITARITY.01) — Track 2.** Extends v304 (the scalar  $1/(p^2(p^2+M^2))$  pole algebra) to the actual tensor spin structure. [E] the Barnes–Rivers projectors are complete ( $\text{tr } P_2 = (d-2)(d+1)/2$ ,  $\text{tr } P_1 = d-1$ ,  $\text{tr } P_{0s} = \text{tr } P_{0w} = 1$  sum to  $d(d+1)/2$ ;  $d=4$  split  $5+3+1+1=10$ ), the EH propagator is  $P_2/p^2 - \frac{1}{d-2}P_{0s}/p^2$ , and the Stelle ghost is a *spin-2* state. [E] the entire KMS form factor  $e^{p^2/M^2}$  (v304/v259) makes the spin-2 sector ghost-free; [C] the wrong-sign spin-0 conformal mode (v332) is cured separately by the GHP contour

(v334). [C] so the full *perturbative* graviton is ghost-free sector-by-sector (conditional on v304's entire-analyticity assumption). [O] residual: perturbative-only + the open analyticity assumption; not the non-perturbative measure. [O] DECLINE:  $\text{tr } P_2=5$  is dimension counting, not  $g_{\text{car}}$  (v354/v355). Cited in `tfpt_5_redteam` (Target F); Python-only.

**2026-06-23 (Directions 3 & 7 — the two hard functional-analysis residuals ADVANCED: the QG measure as one-loop fluctuations around the parameter-free saddle (v365), and the seam collar verified IN MMST's free-fermion class so the scaling limit is  $(E_8)_1$  (v366))**

- **v365 (QGAMB.SADDLE.01) — Direction 3.** With the parameter-free saddle (v358/v359) the QG *measure* (gate C7/QG.AMB.01) organises as a one-loop Gaussian fluctuation determinant. [E] the fluctuation stiffness is  $1/c_3 = 8\pi$  (no free dial, so the one-loop problem is now well-posed); [E] the fluctuation operator is gapped  $M \geq \Delta_{\text{eff}} = 6 \log \frac{3}{2} - \frac{31}{4\pi^2} \approx 1.648 > 0$  (v76/v337)  $\Rightarrow$  mode-by-mode convergence, finite model  $\log\text{-det } \text{Tr}' \log M = 6 \log \frac{9}{2}$ ; [C] the conformal mode converges on the GHP contour (v334); [E] the admissible projective limit exists ( $\chi = 729/665$ , v330) with 0 new dials (v364). [O] residual: the full non-perturbative projective limit on the ambient sector. Sharpens C7 from “diffuse full QG” to a one-loop Gaussian problem around the parameter-free saddle; does not close it. Cited in `tfpt_research_contracts`; Python-only.
- **v366 (SEAM.MMST.INCLASS.01) — Direction 7.** The seam collar is verified IN MMST's free-lattice-fermion scaling-limit class hypothesis-by-hypothesis: [E]  $D = \dim S^+ = 16$  Majoranas,  $c = g_{\text{car}} + N_{\text{fam}} = 8$ ,  $\text{rank } E_8 = 8$  so  $\text{rank} \leq c \leq D$  is  $8 \leq 8 \leq 16$  (in range); [E] gapped ( $\Delta = 6 \log \frac{3}{2} > 0$ , derived v302); [C] quasi-free CAR class (v155/v160); [E] chiral ( $c_- = D/2 = 8 \neq 0$ ). [C] the MMST scaling-limit theorem then applies, and [E] the order-4  $\mu_4$  clock pins the target to  $(E_8)_1$  ( $\det K = 1$  vs  $SO(16) \det K = 4$ , v351). [O] residual: the single S3 input (the collar as a genuine lattice chiral invertible phase, the seam one-sidedness, v297/v356). Reduces the continuum side of SEAM.EQUIV.01 to one named input; ADVANCES, does not close. Cited in `tfpt_research_contracts`; Python-only.

**2026-06-23 (Consistency pass: the parameter-free *full covariant* Einstein equation (v358/v359) propagated across ALL stale paper and website surfaces; new v359 keybox; homepage gravity animation)**

- **Deep-sync of the parameter-free gravity status.** A cross-surface audit found several documents and website components still framing gravity as “exact only in the  $R + R^2$  regime” or the metric-sector field equation as “open”. These were reconciled with the canonical post-v358/v359 status: the *local* covariant field equation  $G_{ab} + \Lambda g_{ab} = c_3^{-1} T_{ab}$  is PARAMETER-FREE (both coefficients fixed); what remains open is the Jacobson equation-of-state status, the global ambient measure (QG.AMB.01/C7), and the single unit  $v_{\text{geo}}$ . Touched: `introduction`, `tfpt_2_standard_model`, `tfpt_3_e8_audit_bootstrap`, `tfpt_4_frontier` (summary table), `tfpt_research_contracts`, `origin_theory`.
- **New v359 keybox** added to `tfpt_research_contracts` (full covariant equation: fixed-volume  $\Rightarrow$  Einstein tensor, Lovelock  $\Rightarrow$  conservation an output); the v358 residual corrected from “linear $\rightarrow$ nonlinear” (stale — v359 closed it) to the equation-of-state + C7 +  $v_{\text{geo}}$  residual, including the generated script index (`script_registry.csv`).
- **Website.** The GravityEmergence animation (three origins of  $c_3$  converging to  $1/(8\pi)$ ; the parameter-free Einstein equation) was updated to the full covariant result and surfaced as a dedicated homepage section (`/#gravity`); gravity-status wording aligned across `ReconstructionChain`, `VerificationDag`, `StatusMatrix`, `WhatTfptClaims`, `ThreeDecoderMap`, `StatusPyramid`, `papers.ts`, `glossary.ts` and the review page.

**2026-06-23 (Direction 6,  $v_{\text{geo}}$  sharpened: after the parameter-free gravity, the gravity sector adds NO new dimensionful input either, so  $v_{\text{geo}}$  is the SINGLE dimensionful input of the complete theory — final tally  $\{v_{\text{geo}}, \pi\}$ , ZERO dimensionless dials: v364)**

- **v364 (VGE0.SHARPEN.01) — Direction 6.** The No-Unit Theorem (v153) already made  $v_{\text{geo}}$  a forced metrology primitive. [E] The SHARPENING after the parameter-free gravity (v358/v359/v361): the Einstein coefficient  $1/c_3 = 8\pi$  and the  $\Lambda$  prefactor  $(8\pi)^2 \cdot 48c_3^4 = 3/(4\pi^2)$  are dimensionless, so  $1/G \sim v_{\text{geo}}^2$  and  $\Lambda \sim (\text{dimensionless}) v_{\text{geo}}^4$  reduce to  $v_{\text{geo}}$  times atoms — the gravity sector adds no new scale. [E] FINAL TALLY: every dimensionful quantity = (a dimensionless ratio)  $\times$  (a power of  $v_{\text{geo}}$ ); TFPT's free content is  $\{v_{\text{geo}}, \pi\}$  with ZERO dimensionless dials, a  $\sim 26 \rightarrow 1$  reduction vs the SM. [O]  $v_{\text{geo}}$  stays irreducibly primitive (No-Unit;  $1/G$  UV-sensitive, Sakharov v68). Direction 6 is clarity — the last dimensionful anchor named and shown to be the only one. Cited in `tfpt_research_contracts`; Python-only.

**2026-06-23 (Direction 5, the matter–gravity backreaction: the carrier's gravitational anomaly is forced  $c_- = 8$ , and the matter vacuum-energy backreaction is finite ( $\rightarrow$  the forced  $\Lambda$  from  $\alpha$ , not  $M_{\text{Pl}}^4$ ) — the loop closes; no new SM-quantum-number read-out (declined): v361)**

- **v361 (GRAV.BACKREACT.01) — Direction 5.** With the parameter-free equation (v359) and the explicit J3 flux (v358), the backreaction of TFPT's carrier matter is computed. [E] the carrier's gravitational anomaly is forced:  $c_- = 16/2 = 8 = g_{\text{car}} + N_{\text{fam}}$  (the chiral central charge of the 16-Majorana content). [E] the backreaction is FINITE  $\rightarrow \Lambda$  from  $\alpha$ : the matter vacuum energy through  $G_{ab} = c_3^{-1} T_{ab}$  is gap-suppressed (Decoupling Thm v337) to  $\rho_{\Lambda}/\bar{M}_{\text{Pl}}^4 = (3/(4\pi^2))e^{-2\alpha^{-1}}$  (v60), the 123 orders =  $119.03 + 3.92$  — *not* the  $M_{\text{Pl}}^4$  catastrophe; the loop (v359+v60 + the carrier vacuum) closes. [E] J3 explicit:  $\delta\langle K_B \rangle = (8\pi^2 R^4/15)\delta\langle T_{00} \rangle$ , carrier  $T_{ab}$  = the SM's (v159). [C] the SM trace anomaly is reproduced (content = SM). [O] DECLINE: no new atom-forced read-out of individual SM quantum numbers beyond  $c_- = 8$  and  $\Lambda$  — per v354/v355 declined. The backreaction is finite and consistent, adding no new dial. Cited in `tfpt_4`; Python-only.

**2026-06-23 (Direction 2, the disciplined result: the gap-induced GR correction is the known  $R^2$  scalaron — already falsifiable via  $n_s/r$ ; a *new* forced near-term deviation does NOT survive the discriminator and is declined: v360)**

- **v360 (GRAV.GAPCORR.01) — Direction 2, run honestly.** Does the seam transfer gap  $\{1, (2/3)^6, (1/3)^6\}$  yield a NEW, calculable, FORCED deviation from GR (a near-term kill test) beyond the known  $R^2$  scalaron? With the v354/v355 discriminator the answer is *no*, and this is recorded rather than manufacturing a coefficient. [E] the leading higher-curvature term IS the  $R^2$  scalaron (spectral-action  $a_4 = R^2/72$ , v36;  $M_{\text{scal}}^2/\bar{M}_{\text{Pl}}^2 = c_3^7$ , exponent  $7 = g_{\text{car}} + N_{\text{fam}} - 1$ ) — the calculable beyond-Einstein correction, already in TFPT. [C] it is the entanglement-equilibrium NEXT order (the Wald first law  $\rightarrow f(R) = R + R^2$ , v28/v359). [E] it is ALREADY falsifiable via  $n_s = 1 - 2/N_*$ ,  $r = 12/N_*^2$  (live CMB-S4 kill test). [E] the gap  $\Delta = 6 \ln \frac{3}{2}$  is a dimensionless rate, so a distinct gap-induced curvature<sup>2</sup> term sits at  $\xi v_{\text{geo}} \sim$  the seam/Planck scale (not near-term observable). [O] DECLINE: a new dimensionless coefficient is not atom-forced (the scalaron exponent  $7 = g + N - 1$  and the gap exponent  $6 = 2N_{\text{fam}}$  are different;  $c_3^7$  and  $(2/3)^6$  unrelated) — per v354/v355 declined. Direction 2 yields no new near-term forced prediction; the answer is the existing scalaron. Cited in `tfpt_4`; Python-only.

**2026-06-23 (the FULL covariant Einstein equation, parameter-free: the fixed-volume entanglement equilibrium gives  $G_{ab} + \Lambda g_{ab} = c_3^{-1} T_{ab}$  with BOTH coefficients TFPT-fixed ( $8\pi = 1/c_3$  from v358,  $\Lambda$  from  $\alpha$  via v60); B6 closed at the local level: v359)**

- **v359 (GRAV.NONLINEAR.01) — Direction 1, the full covariant equation.** Extends v358 from the linearised (fixed-radius, Ricci-level) result to the FULL covariant equation. [C][E] Jacobson 2015: demanding stationarity of the small-ball entanglement entropy at fixed *volume* brings in the EINSTEIN TENSOR  $G_{ab}$  (not just  $R_{ab}$ ) — trace  $g^{ab}G_{ab} = (1 - d/2)R = -R$  in  $d=4$ , the structure that carries  $\Lambda$ . [E] Lovelock:  $G_{ab}$  is the unique symmetric divergence-free second-order metric tensor, so  $\nabla^a(G_{ab} + \Lambda g_{ab}) = 0$  forces  $\nabla^a T_{ab} = 0$  — conservation is an output. [E] coefficient 1:  $8\pi G \rightarrow 8\pi = 1/c_3$  fixed (v358). [E] coefficient 2: Jacobson leaves  $\Lambda$  free, TFPT fixes it —  $\rho_\Lambda/\bar{M}_{\text{Pl}}^4 = (3/(4\pi^2))e^{-2\alpha^{-1}}$ , prefactor  $(8\pi)^2 \cdot 48c_3^4 = 3/(4\pi^2)$  (v60); so BOTH coefficients are TFPT-fixed, not free integration constants. [C] all timelike directions  $\rightarrow$  the covariant tensor equation; the matter side is the assembled CHM ball modular flux (v358/v323). NET: the original B6 “full covariant Einstein-side field equation” is now PARAMETER-FREE at the local level. [O] residual: the Jacobson equation-of-state status (not a from-action quantisation) + the global ambient measure (C7/QG.AMB.01, separate, gap-decoupled);  $v_{\text{geo}}$  the one unit. Cited in `tfpt_research_contracts`; mirrored in Wolfram (303/303). (`next-plan.md` Direction 4 folded in: the  $\Lambda$  here matches the  $\alpha$ -readout.)

**2026-06-23 (the classical gravity field equation, parameter-free: the entanglement first law  $\delta S = \delta\langle K \rangle$  with TFPT’s atoms gives the LINEARISED Einstein equation with  $G$  from  $c_3$ ; the thermodynamic and geometric origins of  $c_3$  coincide via  $|\mu_4| = |\mathbb{Z}_2|\chi = 4$ ; the two formerly-open pieces (matter flux J3, entropy-density coefficient) closed/atom-fixed: v358)**

- **v358 (GRAV.ENTROPY.EQUILIBRIUM.01) — the bridge to the dynamic world, instantiated on the gravity side.** Carrying out the Jacobson-2015 / Faulkner-et-al. “first law of entanglement  $\Rightarrow$  Einstein equation” with TFPT’s atoms. [E] PARAMETER-FREE COEFFICIENT:  $\delta S = \delta\langle K \rangle$  gives  $G_{ab} = c_3^{-1} T_{ab}$  with  $c_3^{-1} = 8\pi$  fixed; all four  $c_3$ -roles consistent (Unruh  $1/(2\pi)=4c_3$ , EH  $1/(16\pi)=c_3/2$ , Einstein  $8\pi=1/c_3$ , Bekenstein  $1/4=1/|\mu_4|$ ) — no free dimensionless Newton dial. [E] THERMO = GEO COINCIDENCE (new): the first law’s coefficient  $2\pi/\eta$  equals the geometric  $|\mathbb{Z}_2|2\pi\chi$  iff  $|\mu_4| = |\mathbb{Z}_2|\chi = 4$  (v216) — the thermodynamic and geometric origins of  $c_3$  are ONE. [C] J3 SOLVED: the CHM ball modular Hamiltonian (boost via BW, v323) gives the matter flux  $\delta\langle K_B \rangle = \frac{8\pi^2 R^4}{15} \delta\langle T_{00} \rangle$ . [E] ENTROPY DENSITY ATOM-FIXED:  $1/4=1/|\mu_4|$  (v57) + central charge  $c = g_{\text{car}} + N_{\text{fam}} = 8$  (form v59). [O] residual: linear  $\rightarrow$  full non-linear (the original B6) + the  $v_{\text{geo}}$  unit. Closes the LINEARISED Einstein equation parameter-free; cited in `tfpt_research_contracts`; mirrored in Wolfram (302/302).

**2026-06-23 (the two frontier walls attacked directly: Direction A assembles the continuum chain of SEAM.EQUIV.01 theorem-by-theorem and ties its residual to the seam one-sidedness; Direction B was found ALREADY closed by v40 (harmonic metric = finite linear algebra, not a PDE): v356)**

- **v356 (SEAM.EQUIV.CONTINUUM.03) — Direction A, the continuum chain assembled.** After the reverse-audit/bandwidth work showed no hidden *discrete* lever remains, the only fundamental progress is to assemble the *continuum* chain. Link-by-link: **S1** the gapped ( $\Delta=6\ln(3/2)>0$ , v302) quasi-free 16-Majorana collar (v155/v160)  $\rightarrow$  **S2** invertible (gapped free fermions have no intrinsic topological order, Kitaev, v301)  $\rightarrow$  **S3** the *nontrivial* invertible phase ( $c_- = 8 \neq 0$ )  $\rightarrow$  **S4** a *non-gappable* anomalous chiral edge (anomaly inflow / bulk-edge), so edge *existence* is a CONSEQUENCE of S2+S3, not a separate assumption — this directly

reduces the v336/v351 “does the collar have a chiral edge?” residual  $\rightarrow$  **S5** its chiral-CFT scaling limit (MMST, v336; range rank  $\leq c \leq D$ ,  $8 \leq 8 \leq 16$ )  $\rightarrow$  **S6** the order-4  $\mu_4$  clock + the AQFT stack pin  $(E_8)_1$  (v351/v297). [E] the only non-theorem link is S3, and [C] S3 is the seam being *one-sided* (chiral,  $c_- \neq 0$ ) — the *same* one-sidedness that makes  $c_3=1/(8\pi)$  a one-sided Gauss–Bonnet ( $|\mathbb{Z}_2| \nmid K$ , v58). [O] the one remaining input is “the abstract collar is realized as a genuine lattice chiral free-fermion invertible phase” (the v297 Flat-Away). NET: SEAM.EQUIV.01 reduced to ONE realization input tied to the constant that *defines* the theory; it ADVANCES, it does not close.

- **Direction B — found ALREADY closed by v40.** The U-wall “transcendental Hitchin solve” (v31) is the worst case for a *generic* Higgs bundle; but at the *polystable* point TFPT selects, Mehta–Seshadri makes the bundle *unitary*, so the harmonic metric is the constant invariant Hermitian form (*finite linear algebra*, Higgs  $\Phi=0$ ) and  $\det R=8$  is read off — not a PDE. Only the absolute amplitude  $U_{\text{point}}$  (an anchor) remains. A redundant new B module was built, found to duplicate/contradict v40, and *deleted* (no padding). Honest outcome: B needs no new work.

**2026-06-22 (bandwidth search of the unmapped  $E_8$  region, with a strict forced-identity discriminator: the genuine overlooked structure is COLLECTIVE ( $\sum \deg = 128 = \dim S^+$ ,  $\prod \deg = |W(E_8)|$ , exponents = totatives of 30); the per-degree atom matches are unforced and DECLINED; no new physical hit – reconciling v66 and v354: v355)**

- **v355 (E8.UNMAPPED.BANDWIDTH.01) — the next step after v354, run honestly.** The obvious follow-up (“search those five degrees for hits we overlooked”) is itself the numerology temptation v354 warns about: a rich object like  $E_8$  returns a numerical match for almost anything. So the scan uses a STRICT discriminator – a candidate counts only if it is a FORCED identity (a theorem about  $E_8$ ), never a coincidence. [E] FORCED, SET-LEVEL (the genuinely overlooked structure, collective not per-number):  $\sum \deg = 128 = \dim S^+$ , the spinor half of  $248 = 120 + 128$  (theorem:  $\sum \deg = \# \text{pos roots} + \text{rank} = 120 + 8$ ) – the invariant budget of  $E_8$  equals the fermion content the carrier reads;  $\sum \exp = 120 = \# \text{pos roots}$ ;  $\prod \deg = |W(E_8)| = 696,729,600$ ; exponents = the  $\varphi(30)=8$  totatives of 30 (v223), so the unmapped degrees’ exponents are part of the forced set;  $\max \deg = 30 = h(E_8)$ . [C] SUGGESTIVE, not a theorem:  $12=h(E_6)$ ,  $18=h(E_7)$ ,  $30=h(E_8)$  are all degrees of  $E_8$  (the  $2T/2O/2I$  McKay ladder, v219), but “ $\deg(E_8)$  contains its subalgebras’ Coxeter numbers” is not a general theorem – flagged, not claimed. [O] DECLINED (the numerology temptation): the per-degree atom matches  $14=\dim G_2$ ,  $20=\det L$ ,  $24=|W(A_3)|$ ,  $12=|R(A_3)|$ ,  $18=p_4(a)$  (v66 already flagged 12, 18 as fittable) each admit  $\geq 2$  readings, and the lone extra prime in  $|W(E_8)| = 2^{14}3^55^27$  “=” the scalaron exponent is one of many readings of 7 – all declined. [E] RECONCILES v66 (the 8 degrees coincide with the load-bearing integers) and v354 (only 2, 8, 30 are primary readouts): the search finds NO new forced physical hit, and the disciplined decline IS the anti-numerology result. (This also sharpens v354: the five are “no PRIMARY readout / hull overhead”, not literally “unmapped” – they coincide with structural atoms, but unforced.) Cited in `tfpt_5_redteam` (with a full audited degree table); mirrored in Wolfram (301/301).

**2026-06-22 (the REVERSE numerology audit, and what the golden ratio means: of  $E_8$ ’s 8 Casimir invariants only 3 feed a primary readout and 5 are hull overhead; the golden ratio is UNMAPPED structure, so it cannot be numerology: v354)**

- **v354 (E8.REVERSE.AUDIT.01) — the reverse complement of v305.** TFPT’s forward discipline (v305) is one-directional: every physical READOUT must map onto an  $E_8$  structure (anti-fitting). The sharp adversarial question runs the OTHER way — how much  $E_8$  structure

maps onto NOTHING, and is the recurring golden ratio numerology? [E] THE REVERSE AUDIT: of  $E_8$ 's eight Casimir invariant degrees  $\{2, 8, 12, 14, 18, 20, 24, 30\}$  (exponents  $+1$ ), exactly THREE feed a readout — 2 (metric), 8 (rank  $\rightarrow c_3=1/(8\pi)$ ), 30 (Coxeter  $\rightarrow g_{\text{car}}=5$  and the order-30 cycle) — and FIVE carry no primary readout  $\{12, 14, 18, 20, 24\}$  (they coincide with structural atoms per v66, but unforced – reconciled in v355); so 3/8 are primary readouts, 5/8 hull overhead. [E] ECONOMY, NOT PROMISCUITY: a small fixed set (rank 8,  $h=30$ ,  $\det K=1$ ,  $D_5 \oplus A_3$ , the 16 half-spinor) generates the ENTIRE readout set, and the higher Casimirs are never reached into to fit — a numerological use would be PROMISCUOUS (mine the rich structure), TFPT is ECONOMICAL (few invariants, many readouts) and leaves most of  $E_8$  untouched, the anti-numerology signature being precisely the large unused remainder. [E] THE GOLDEN RATIO IS UNMAPPED  $\Rightarrow$  NOT NUMEROLOGY:  $\varphi = 2 \cos(\pi/5)$  appears only in internal structure (the affine- $E_8$  attractor eigenvalue v312, the icosian lattice v348) and in NO physical readout; numerology means a number FITTED to an observation,  $\varphi$  is fitted to nothing — it is the algebraic signature of the 5-fold ( $h=30=2 \cdot 3 \cdot 5$ ), the carrier announcing it is icosahedral. [O] the 5/8 unmapped is the HULL OVERHEAD ( $E_8$  is the minimal even-unimodular hull, not the world group); where a future higher-invariant readout could live, or genuine over-capacity — stated plainly, not hidden. Cited in `tfpt_5_redteam`; mirrored in Wolfram (300/300).

**2026-06-22 (the bird's-eye rethink: TFPT is a unique self-consistent LOOP with ZERO free dials, not a linear axiom system;  $\pi$  is a fixed geometric constant, not a dial; the one residual is the continuum closure: v353)**

- **v353 (TFPT.SELFLOOP.01) — refining v352.** Stepping all the way back: TFPT is not a LINEAR theory (axioms  $\rightarrow$  theorems) but a CLOSED SELF-CONSISTENT LOOP whose outputs fix its own inputs. [E] it is a loop:  $g_{\text{car}}=5 = \max \text{ prime of the output } h(E_8)=30$ , and the 8 in  $c_3 = \text{rank } E_8$  – the outputs fix the inputs (v6), with a unique fixed point (v56); “which is the axiom?” is the wrong question for a loop. [E] ZERO free adjustable dimensionless dials: every load-bearing number is forced/over-determined. [E]  $\pi$  is a FIXED geometric constant (the boundary 2-sphere,  $\oint K=4\pi$ ), not a dial. [C] this refines v352's “framework +  $\pi$ ”:  $\pi$  is not a free input and the framework is a SELECTION PRINCIPLE (the unique self-consistent loop), not a free meta-axiom – the honest characterisation is “a unique self-consistent loop with zero free dials”. [O] the one residual is the CONTINUUM closure of the loop (the chiral-edge existence, v336/v351); structural-realism (loop complete) vs constructivism (continuum open) is a philosophical, not numerical, fork – no free numbers either way. A capstone synthesis.

**2026-06-22 (two consequences: the  $c=8$  which-net ambiguity resolved by the order-4 clock, and BOTH axioms shown bootstrap-forced so the only irreducibles are the framework +  $\pi$ : v351, v352)**

- **v351 (SEAM.EQUIV.CONTINUUM.02) — the which-net ambiguity falls.** The long-standing “ $c=8$  ambiguity” ( $E_8 \det 1$  vs  $SO(16)=D_8 \det 4$ , both  $c=8$ , flagged open in v277/v344) is RESOLVED, non-circularly, by the seam's ORDER-4  $\mu_4$  clock: the index-4 simple-current extension of  $D_5 \oplus A_3$  is  $E_8$  (v154), the index-2 would be  $SO(16)$ , so the order-4 clock (the four Gauss–Bonnet marks, v216) selects  $E_8$  (full  $\mu_4$  condensation  $16 \rightarrow 4 \rightarrow 1$ ) not  $SO(16)$  (partial  $\mathbb{Z}_2$   $16 \rightarrow 4$ ). The bootstrap agrees ( $h(E_8)=30$  max prime  $5=g_{\text{car}}$  vs  $h(D_8)=14$  max prime 7). [E] so  $\det K=1$  is a consequence, not an open input. [O] the residual of SEAM.EQUIV.01 is then PURELY the chiral-edge / scaling-limit existence (v336), not which-net.
- **v352 (TFPT.IRREDUCIBLE.01) — the inputs are not axioms.** The deepest answer to “the inputs are back-determined”: BOTH P1 and P2 are bootstrap-forced fixed points. [E]  $g_{\text{car}}=5$  is forced three ways (v6); the 8 in  $c_3$  four ways (rank  $E_8 = h(D_5) = \phi(30) = \det R$ , v54); the  $E_8$  hull is forced (unique even unimodular rank-8, v83/v154). [E] so the ONLY irreducibles

are the FRAMEWORK (a discrete reflection-positive boundary with a finite carrier) and the transcendental  $\pi$ . [O] the framework is the META-AXIOM — not math-reducible, the foundational stance (like “spacetime is a manifold”), minimal, tested by its consequences not proved. TFPT’s true input is ONE framework +  $\pi$ . Wolfram mirror +2 (297  $\rightarrow$  299).

**2026-06-22 (correction: the inputs are bootstrap-forced fixed points, not free axioms –  $g_{\text{car}}=5$  IS the icosahedral 5 of  $h(E_8)=30$ , and the golden ratio is emergent from the  $\mu_4$ -glue, not external: v350)**

- **v350 (SEAM.EQUIV.BOOTSTRAP.01) — correcting v349.** An honest correction prompted by the right objection that the inputs are not axioms but are back-determined by the theory. [E] the inputs ARE bootstrap-forced (v6):  $g_{\text{car}}=5$  is over-determined three ways – rank-fill ( $g_{\text{car}}+N_{\text{fam}}=8$ ), Coxeter-match ( $g_{\text{car}}= \max \text{ prime of } h(E_8)=30$ ), Pascal ( $2^4=C(5,0)+C(5,1)+C(5,2)$ ). [E] so  $g_{\text{car}}=5$  IS the icosahedral 5-fold (the 5 in  $h(E_8)=30=2\cdot3\cdot5$ ), not the  $D_5$  rank; the atoms (2, 3, 5) ARE the primes of the icosahedral Coxeter number 30. [E] the golden ratio is EMERGENT from the glue:  $D_5$  ( $h=8$ )/ $A_3$  ( $h=4$ ) are non-golden alone, but  $D_5\oplus A_3 \rightarrow E_8$  ( $\mu_4$  index-4, v154) gives  $h(E_8)=30$  (golden) – the seam’s own  $\mu_4$  produces the 5-fold, so v349 checked the un-glued pieces, the wrong object, and its pentagon-vs-count dichotomy was false. [O] the residual relocates: the golden is bootstrap-forced, the only open thing is the physical continuum realisation (v336), NOT the golden ratio. Honest caveat: self-consistency within the framework, not from-nothing. Wolfram mirror +1 (296  $\rightarrow$  297).

**2026-06-22 (the honest test – does the raw seam carry  $\varphi$ ? – answer NO: the bare carrier gives  $\sqrt{2}$ , golden is the icosahedral input; the keystone reduces to “is  $g_{\text{car}}=5$  a pentagon or a count?”: v349)**

- **v349 (SEAM.EQUIV.GOLDEN.01).** The honest, computed test of the v348 question, and a clarifying NEGATIVE: the raw seam does *not* carry the golden ratio. [E] the golden ratio  $2\cos(\pi/5)$  is the signature of a 5-fold rotation ( $5 \mid h$ ); the carrier  $D_5$  has  $h=8$  (eigenvalue  $\sqrt{2}$ ) and  $A_3$  has  $h=4$  – no 5-fold; the dynamical data (cusp  $\{0, 1/3, 2/3\}$ , recovery  $(2/3)^6$ , seed  $\varphi_0$ ) are rational/transcendental. [E]  $5 \mid h$  holds only for  $A_4=SU(5)$  ( $h=5$ ),  $H_3$  ( $h=10$ ),  $E_8$  ( $h=30$ ) – the output/icosahedral side, so  $\varphi$  is the icosahedral INPUT. [E]  $\varphi$  alone would not suffice ( $2I \neq C_5 \times \mu_4$ ,  $20 \neq 120$ ). [O] the resolution: the keystone reduces to one question – is  $g_{\text{car}}=5$  a PENTAGON (a 5-fold rotation  $\Rightarrow$  golden  $\Rightarrow$  icosahedral  $2I = E_8$ ) or just a COUNT (the rank of  $D_5 \Rightarrow \sqrt{2}$ )? The “absurdly simple solution” is real but is a *reading* of axiom P2 (the golden ratio is the pentagon content of  $g_{\text{car}}=5$ , not derivable from  $g_{\text{car}}=5$ -as-a-count); there is no hidden  $\varphi$ . L2 is foundational: P2-as-pentagon closes it (mode C), else a rigidity theorem must produce the 5-fold (mode B); not closed. Wolfram mirror +1 (295  $\rightarrow$  296). A NEGATIVE/clarification; prevents a false closure.

**2026-06-22 (Route B carried out: rigidity + the icosian-ring identity reduce the whole keystone to ONE question – does the raw seam carry the golden ratio? – the simplest form of L2: v348)**

- **v348 (SEAM.EQUIV.RIGID.01).** Executes the rigidity route (mode B of v347) and lands on a strikingly simple statement. [E] rigidity closes the crystallographic part: the order-4 clock has the unique normal form  $z \mapsto iz$  (Nielsen/Kerckhoff, v180), Troyanov gives the unique flat pillowcase metric (v284). [E] the downstream is a LATTICE identity, not just a graph: by Conway–Sloane the ring of icosians (120 unit icosians =  $2I$ , golden-ratio quaternion norm) IS the rank-8  $E_8$  lattice. [E] the single bridge is the golden ratio: the crystallographic restriction allows only orders  $\{1, 2, 3, 4, 6\}$  (the 5-fold absent), so  $\varphi = 2\cos(\pi/5)$  is exactly what extends the crystallographic  $\mu_4$  to the non-crystallographic icosahedral  $2I$  (the quasicrystal). [O] so

the entire keystone reduces to ONE question: *does the raw seam carry the golden ratio?* – if yes,  $\mu_4 + \varphi \Rightarrow 2I \Rightarrow E_8$  and rigidity fixes the rest. [O] honest subtlety: TFPT’s  $\varphi$  is currently  $E_8$ -side (v312/v313); the raw-seam data so far are not manifestly golden, so showing the RAW seam carries  $\varphi$  non-circularly is the residual (= L2, sharpest form). Wolfram mirror +1 (294  $\rightarrow$  295). An analysis/reduction; closes nothing.

**2026-06-22 (can the one open arrow be *fully* solved? the closure-mode classification: the flat $\rightarrow$ spherical base locus, four modes, no free theorem – only rigidity or axiom  $P_3$ : v347)**

- **v347 (SEAM.EQUIV.CLOSURE.01).** A direct, honest answer to “is there any other way to fully solve the one open arrow L2?”. [E] the precise locus: the seam gives the *flat* pillowcase base  $S^2(2, 2, 2, 2)$  ( $\chi_{\text{orb}} = 2 - 4(1 - \frac{1}{2}) = 0$ , v216) while the  $2I/E_8$  structure is the Seifert base over the *spherical*  $S^2(2, 3, 5)$  ( $\chi_{\text{orb}} = 2 - (\frac{1}{2} + \frac{2}{3} + \frac{4}{5}) = \frac{1}{30}$ ), so L2 bridges two geometric types – why it needs the carrier  $(2, 3, 5)$  input. [E] as a “physics-realises-geometry” statement L2 has exactly four closure modes: A construct (the analytic wall), B rigidity (a unique-realisation theorem – the only theorem-route w/o new input, pursued v284/v300/v331), C axiom  $P_3$  (then TFPT is complete relative to 3 axioms), D empirical. [E] the “no new state” discipline (v246) blocks deriving L2 from extra physics, so no route makes it a free theorem. [E] verdict: full closure is either a rigidity theorem (B) or axiom status (C), validated either way by (D); only B closes it as a theorem. [O] NOT closed – classifies the modes, locates the difficulty. Wolfram mirror +1 (293  $\rightarrow$  294). An analysis/roadmap; closes nothing.

**2026-06-22 (the geometric bridge made explicit: the keystone is ONE arrow – raw-seam  $\Rightarrow \det K=1$  is a six-link chain, five theorems + one open orbifold realisation, the group forced by the  $(2, 3, 5)$  atoms: v346)**

- **v346 (SEAM.EQUIV.GEOM.01).** The honest capstone of the SEAM.EQUIV.01 investigation: it lays out the full path raw-seam  $\Rightarrow \det K=1$  as six links and shows exactly one is open. [E]/[C] L1 raw seam  $\rightarrow \mu_4$  + four marks (Gauss–Bonnet, v216); [O] L2  $\mu_4$  + the  $(2, 3, 5)$  atoms  $\rightarrow$  the  $2I$  orbifold  $\mathbb{C}^2/\Gamma$  (THE ONE OPEN ARROW = the geometric content of SEAM.EQUIV.01); [E] L3  $2I \rightarrow$  affine  $E_8$  McKay (v219); [E] L4  $2I \rightarrow E_8$  du Val (v232); [E] L5  $\mathbb{C}^2/2I \rightarrow$  Poincaré link  $H_1=0$  (v345); [E] L6  $H_1=0 \rightarrow \det K=1$ . [E] five of six links are theorems; [E] the group is FORCED by the seam’s own atoms (the exceptional  $SU(2)$  axis signatures  $(2, 3, 3)/(2, 3, 4)/(2, 3, 5) \rightarrow 2T/2O/2I$ ;  $(2, 3, 5)$  unique to  $2I$ , and  $(|\mathbb{Z}_2|, N_{\text{fam}}, g_{\text{car}})=(2, 3, 5)$  ARE those axes); [E] everything downstream of L2 is then forced. [O] the irreducible open content is L2 alone (“the raw seam normal bundle is an ADE orbifold point  $\mathbb{C}^2/\Gamma$ ”) = SEAM.EQUIV.01; not closed, but the bridge is now demonstrably ONE arrow with the group pinned and all downstream proven.

**2026-06-22 (executing route R3: the  $(2, 3, 5)$ -rewrite attractor link computed to be the Poincaré homology sphere ( $\det K=1$ ), the combinatorial core complete, only the geometric bridge open: v345)**

- **v345 (SEAM.DETK.02).** Carries out the combinatorial core of the hypergraph-homotopy route (R3, the most promising fresh angle from v344) as far as it computably goes. By plumbing topology (Brieskorn/Hirzebruch) the link of a tree plumbing with the ADE Cartan intersection form has  $H_1 = \text{coker}(\text{Cartan})$ . [E] the Smith normal form of the  $E_8$  Cartan matrix is the identity, so  $\text{coker}(E_8) = 0$  – the  $E_8$ -graph plumbing boundary is an INTEGRAL HOMOLOGY SPHERE (the genus-1 meaning of  $\det E_8=1$ ); among ADE only  $E_8$  does this ( $A_n \rightarrow \mathbb{Z}/(n+1)$ ,  $D_n \rightarrow \mathbb{Z}/4$ ,  $E_6 \rightarrow \mathbb{Z}/3$ ,  $E_7 \rightarrow \mathbb{Z}/2$ ). [E]  $\pi_1 = 2I = SL(2, 5)$  is PERFECT (commutator subgroup = the whole order-120 group, computed directly), so  $H_1 = 0$  and the link is THE Poincaré sphere,  $|2I| = 120 = |\mu_4| h(E_8)$ . So R3’s combinatorial half is

[E]-complete: the  $E_8$  attractor link is forced to be the Poincaré sphere ( $\det K=1$ ), uniquely among ADE. [O] what stays open is only the geometric realisation bridge (the raw rewrite seam attractor *is* this  $E_8$  du Val plumbing) = the SEAM.EQUIV.01 content; not closed, but the keystone is now reduced to that single geometric identification. Wolfram mirror +1 (292  $\rightarrow$  293).

**2026-06-22 (the one keystone bit  $\det K=1$  mapped: six equivalent faces, the genus-1 obstruction, three access routes, the hypergraph-homotopy route the most promising fresh angle: v344)**

- **v344 (SEAM.DETK.01).** A bird’s-eye, full-spectrum map of SEAM.EQUIV.01’s only open residual, the holomorphy bit  $\det K=1$ . [E] it is ONE statement with six equivalent faces (holomorphy / homology-sphere link  $H_1=0$  / one 1-dim irrep /  $\det \text{Cartan} = 1$  / full  $\mu_4$  condensation /  $\tau=i$  CM), unified by  $|\det \text{Cartan}| = |H_1(\text{link})| = \#(1\text{-dim irreps})$  across ADE ( $A_n \rightarrow n+1$ ,  $D_n \rightarrow 4$ ,  $E_6 \rightarrow 3$ ,  $E_7 \rightarrow 2$ ,  $E_8 \rightarrow 1$ ) – only  $E_8$  gives 1. [E] the GENUS-1 OBSTRUCTION explains why it is hard: every easy argument (plane vacuum, gap, free-fermion) is genus-0, necessary but not sufficient (v37);  $\det K$  is the torus degeneracy. [E] THREE genus-1 access routes: R1 continuum-modular (v336), R2 anyon-condensation (v281), R3 hypergraph-homotopy. [C] the HYPERGRAPH route is the most promising fresh angle: the  $(2, 3, 5)$  rewrite attractor is the affine- $E_8$  graph (v312) and the  $E_8$  du Val link is the Poincaré sphere (v232), so  $\det K=1$  becomes a finite combinatorial statement (no continuum limit). [O] none closes it; the gap is the combinatorial-to-geometric bridge = the SEAM.EQUIV.01 content. A synthesis/roadmap; closes nothing.

**2026-06-22 (the four black-hole-birth completion routes investigated: none closes all or is 100% provable alone; all four converge on SEAM.EQUIV.01 + the decoupled QG contour: v343)**

- **v343 (FOUR.ROUTES.01).** A direct, honest answer to “which black-hole-birth route closes ALL problems and is 100% provable”: *none individually* — but the four converge on the same two residuals as v335. [E] (A) the finite causal diamond reframes QG.AMB to a finite thermal trace (de Sitter  $S_{\text{dS}} = 32\pi^4 e^{2\alpha^{-1}}$  finite,  $\chi = 729/665$ ) but inherits the GHP contour (v334, [O]); does not touch the seam. [C] (B) Carlip–Cardy gives a second route to existence+ $c$  ( $c=8$  consistent, but the Carlip ambiguity and no holomorphy  $\det K=1$ ) — inherits SEAM.EQUIV.01. [E] (C) the modular/thermal flow (v239) is well-defined but its geometric content is QGEO.SYM.01, a corollary of SEAM.EQUIV.01 (v335) — a reduction. [E]/[O] (D) the unique attractor (v56) gives parameter-freeness but the cyclic dynamics is [C] interpretation. [E] **Verdict:** independent closures = 0; irreducible residuals after all four = 2 (unchanged from v335); the horizon premise makes both more natural but eliminates neither — focus = SEAM.EQUIV.01. An analysis module; closes nothing, fabricates nothing.

**2026-06-22 (the heat-kernel origin of the  $\alpha$  terms: the Calderón quadratic is the Gilkey gauge-curvature coefficient; the cubic Maxwell moment stays the EM.WARD.01 residual: v342)**

- **v342 (EM.WARD.02) — the heat-kernel sharpening of EM.WARD.01.** Derives the ORIGIN and STRUCTURE of the  $F_{U(1)}$  determinant-line terms from textbook Seeley–DeWitt / Gilkey coefficients, advancing the EM-Ward origin from [C] (conjectured) to [E] for the term structure; no new gate. [E]  $c_3 = 1/(8\pi)$  *is* the boundary Gauss–Bonnet coefficient ( $\oint_{S^2} K = 2\pi\chi = 4\pi$ ,  $\chi=2$ , v216). [E] the  $\alpha^2$  (Calderón) term *is* the Gilkey gauge-curvature coefficient ( $30\Omega^2/360 = \frac{1}{12}\Omega^2$ ,  $\Omega=i\alpha F \Rightarrow -(\alpha^2/12)F^2$ ), so it is forced, not an ansatz. [E] the  $c_3$ -orders  $\{0, 3, 6\}$  are an arithmetic ladder and  $2c_3^3 = 1/(256\pi^3)$  carries  $\pi^3$  (three boundary insertions). [C] the exact coefficient needs the seam  $F$ -normalisation. [O] the cubic  $\alpha^3$

Maxwell moment (a topological/Chern level) + the full “this is the seam  $U(1)$   $\zeta$ -determinant” identification stay the EM.WARD.01 residual, tied to SEAM.EQUIV.01.  $\alpha^{-1}$  stays [E] (v3). Wolfram mirror +1 (290  $\rightarrow$  291).

**2026-06-22 ( $\alpha$  as the stationarity of a  $U(1)$  determinant line: every ingredient an index / heat-kernel coefficient / discriminant form, the open EM-Ward obligation named: v341)**

- **v341 (ALPHA.QUILLEN.01).** Answers the external review’s sharpest valid point — “derive the form of  $\alpha$ , not more decimals”. The EM closure  $F_{U(1)} = \alpha^3 - 2c_3^3\alpha^2 - \frac{4}{5}c_3^6 M \log(1/\varphi_{\text{seam}}) = 0$  is rewritten as a *stationarity* of the boundary  $U(1)$  determinant line:  $[\alpha^3 - 2c_3^3\alpha^2]_{\delta \log \det \Delta_{U(1)}} = [8b_1c_3^6]_{\text{counterterm}} \cdot [\log(1/\varphi_{\text{seam}})]_{\text{seam response}}$ . [E] every ingredient is a NAMED object, not a knob:  $M=41$  is a  $U(1)$  index ( $\frac{4}{5}M = 8b_1$ ,  $b_1=41/10$  the SM  $U(1)_Y$  beta, EM Ward v48);  $c_3=1/(8\pi)$  the Gauss–Bonnet boundary coefficient ( $8=\text{rank } E_8$ );  $-5/4 = -q(D_5)$  a discriminant form and  $\varphi_{\text{base}}=1/(6\pi)=\frac{4}{3}c_3$  carries the other,  $c_3/\varphi_{\text{base}}=3/4=q(A_3)$ ;  $\delta_{\text{top}}=\Omega_{\text{adm}}c_3^4$ . The Quillen split is verified at the  $\alpha^{-1}=137.0359992$  root. **No new gate:**  $\alpha^{-1}$  stays closed [E]; v341 adds no open item — it *sharpens* the pre-existing [O]/physical-origin obligation EM.WARD.01 (“why *this* function”, already conjectural). [O] the one still-open part *is* that residual: the from-first-principles AQFT proof that  $F_{U(1)}$  is the EXACT Quillen functional (the variational principle derived, not assembled), unchanged in scope. Wolfram mirror +1 (289  $\rightarrow$  290).

**2026-06-22 ( $A_s$  reconciled with the archived derivation: predicted & dimensionless (so it does not define a mass), the tension is the reheating channel: v340)**

- **v340 (AS.RECONCILE.01).** Answers “why don’t we have  $A_s$  / the old versions derived it” and “can a ratio define mass itself”. [E]  $A_s$  *is* predicted and *dimensionless*:  $A_s = N_\star^2(M_{\text{scal}}/\bar{M}_{\text{Pl}})^2/(24\pi^2) = N_\star^2c_3^7/(24\pi^2)$  with  $M_{\text{scal}}/\bar{M}_{\text{Pl}} = c_3^{7/2} = 1.26 \times 10^{-5}$  a pure ratio, so  $A_s$  pins the e-fold count  $N_\star$ , NOT  $\bar{M}_{\text{Pl}}$  — it cannot define an absolute mass (No-Unit, v153, stands). [C] the  $-11.4\sigma$  at the slow Higgs channel ( $N_\star=51.4$ ) is the slow-vs-fast reheating dichotomy; the  $A_s$ -preferred  $N_\star=56.2 \sim$  the instantaneous bound. [C] the archived TFPT v4.5/v4.6 reported a clean  $A_s = 2.124 \times 10^{-9}$  using the algebraic  $N_\star = 3/\varphi_0^{\text{ret}} - 1 = 55.42$ ; at that  $N_\star$  the current normalisation gives  $2.047 \times 10^{-9}$  ( $-1.9\sigma$ ) — the old clean  $A_s$  is near-instantaneous reheating, not the slow channel. [E] nothing lost: the current theory replaced the posited algebraic  $N_\star$  with one derived from reheating physics and exposed honestly that  $A_s$  requires fast (pre)heating. Record band  $N_\star \in [50, 60]$  (v84) untouched.

**2026-06-22 (the first-principles boundary, mapped: can  $M_{\text{Planck}}$  be computed? can  $F_{\text{transfer}}$  be first-principles? both NO, with the exact reason: v339)**

- **v339 (FIRSTPRINCIPLES.BOUNDARY.01).** An honest boundary audit answering two head-on questions and fabricating nothing. [E] *The Planck mass cannot be computed* —  $v_{\text{geo}}/M_{\text{Planck}}$  is forbidden by theorem (No-Unit, v153): a mass is mass-dimension 1 while every TFPT datum is dimension 0, so no dimensionless map outputs an absolute scale. It is not a gap but over-determined (gravity = dark energy to 0.11%, v274); every dimensionless *ratio* is derived, only the one unit is not. [E]  $F_{\text{transfer}}$  *cannot be made first-principles* either: for  $m_p/m_e$  the structure is in place (v262, carrier  $b_3=-7$  + closed electron) but two inputs stay external. [X]  $\alpha_s(M_Z)$  is external *because of a tension* — the SM (= TFPT content, no new states) does not unify (1-loop spread  $\sim 8.8$  at  $2 \times 10^{16}$  GeV, v246). The lattice  $C_p \sim 2.83$  is parameter-free but lattice-only (v35);  $\eta_B$ /axion ride on cosmological initial conditions; Koide  $Q=2/3$  is structural [E]. [E] The residuals classify into four honest kinds: forbidden-by-theorem, external-and-tensioned, external-but-parameter-free, cosmological. A classification, not a solution.

**2026-06-22 (next-steps round: the one open lemma literature-anchored, the Decoupling Theorem, the holomorphy discriminator hardened in Lean, and the  $\theta_{13}$  budget: v336, v337, v338, FORM.CARTAN.DET.01)**

- **v336 (SEAM.EQUIV.CONTINUUM.01) — the one open lemma, literature-anchored.** SEAM.EQUIV.01’s only open input (the continuum OS reconstruction of the gapped quasi-free collar) is split into a CITABLE part and a narrow open part. [C] continuum leg: Osborne–Stottmeister (*CFT from Lattice Fermions*, doi:10.1007/s00220-022-04521-8, CMP **398** (2023) 219) give, building on the MMST (Morinelli–Morsella–Stottmeister–Tanimoto) operator-algebraic (wavelet) renormalization framework (CMP 387 (2021) 299; PRL 127 (2021) 230601), the chiral-CFT scaling limit of free lattice fermions (Koo–Saleur  $\rightarrow$  Virasoro, correct  $c$ ); WZW range  $\text{rank} \leq c \leq D$ , and  $(E_8)_1$  ( $c=8$ , rank 8,  $D=16$  Majoranas,  $SO(16)_1 \rightarrow (E_8)_1$ ) is inside it. [C] OS leg: Adamo–Moriwaki–Tanimoto (arXiv:2407.18222, 2024) prove the conformal OS axioms for unitary lattice VOAs. [E] target pinned by  $c=g_{\text{car}}+N_{\text{fam}}=8$ ,  $\det \text{Cartan}(E_8)=1$ . [O] narrowed residual = “the raw collar’s massless scaling limit is the holomorphic  $(E_8)_1$  net” ( $\det K=1$ ), not a from-scratch construction.
- **v337 (DECOUPLING.THEOREM.01) — the Decoupling Theorem.** [E] every dimensionless readout factors through the gapped admissible transfer spectrum  $\{1, (2/3)^6, (1/3)^6\}$ ; finite susceptibility  $\chi = 1/(1 - (2/3)^6) = 729/665$  gap-suppresses the ambient contribution (margin  $6 \ln \frac{3}{2} - 31/(4\pi^2) \approx 1.648 > 0$ , with  $2 \dim(E_8)c_3^2 = 31/(4\pi^2)$  exactly), so no readout depends on the ambient measure  $\Rightarrow$  QG.AMB.01 decoupled; negative control (gap closed  $\Rightarrow \chi$  diverges) makes it non-vacuous. This is the theorem that makes the QG.AMB.01 reclassification (v335) honest; consolidates v76/v311/v330/v332.
- **v338 (THETA13.BUDGET.01) — the  $\theta_{13}$  budget, fenced.** The  $+2\sigma$  pressure point (v328) quantified against *both* NuFIT 6.0 NO variants:  $+2.0\sigma$  vs the no-SK central (0.02195) but  $+1.7\sigma$  vs the with-SK best fit ( $0.02215 \pm 0.00057$ ), a  $\sim 0.3\sigma$  systematic spread; [E] 0.023108 lies inside the with-SK  $3\sigma$  band (upper 0.02388), an upper-edge pressure point not a falsification; [X] JUNO’s sub-1% precision turns it into a  $\sim 6\sigma$  kill-or-confirm.
- **Lean (FORM.CARTAN.DET.01).** CartanDeterminants.lean extended with the explicit  $D_8 = SO(16)$  Cartan matrix and a fully-proven (kernel-only) holomorphy discriminator:  $D_8$  is even and symmetric just like  $E_8$  (same  $c=8$ ) but  $4 = \det \text{Cartan}(D_8)$  is genuinely not a unit while  $\det \text{Cartan}(E_8)$  is — so unimodularity (one anyon) is the load-bearing selector. lake build green.

**2026-06-22 (keystone unification + status reframe: ONE open theorem, ONE decoupled foreign problem: v335, FORM.QGEO.BW.01 qgeoSymIsCorollary)**

- **v335 (SEAM.EQUIV.UNIFY.01) — the keystone unification.** A bird’s-eye consolidation of the v308/v323/v325/v329/v333 arc that collapses the “two open structural items” narrative. [E] QGEO.SYM.01 is a **COROLLARY** of SEAM.EQUIV.01 (no extra premise): a chiral conformal net has, by axiom, a Möbius-covariant vacuum (the unique invariant positive-energy vector); rotations  $U(1)$  are the compact subgroup of the Möbius group, so the vacuum is rotation-invariant, and the  $\mu_4$  clock = rotation by  $\pi/2$  ( $(e^{i\pi/2})^4=1$ ) fixes it  $\Rightarrow \omega \circ \rho = \omega$ . So SEAM.EQUIV.01  $\Rightarrow$  QGEO.SYM.01 with NO extra premise (the v323 “rotation-invariant vacuum” is a net axiom) — the two bedrock items collapse to ONE. [E] **the one keystone theorem:** SEAM.EQUIV.01 = a construction theorem ( $c=g_{\text{car}}+N_{\text{fam}}=8$ ,  $\det \text{Cartan}(E_8)=1$ , gap  $6 \ln \frac{3}{2} > 0$  derived) with EXACTLY one open continuum lemma (the rigorous continuum OS reconstruction + invertibility of the raw collar). [E] **QG.AMB.01 decoupled, not a gate:** margin  $\Delta - 31/(4\pi^2) \approx 1.648 > 0$  (v76/v311/v330/v332)  $\Rightarrow$  no TFPT prediction depends on the ambient measure; it is the *general* Euclidean-QG conformal-factor problem (Gibbons–Hawking–Perry 1978), inherited not TFPT-specific — reclassified from “TFPT structural frontier” to “decoupled general QG problem”. [E] **collapsed status:** open structural items

$2 \rightarrow 1$ ; live residual =  $v_{\text{geo}} + G_{\text{net}} (= \text{SEAM.EQUIV.01}) + F_{\text{transfer}}$ . [O] the one residual = SEAM.EQUIV.01's continuum OS reconstruction.

- **Lean (FORM.QGEO.BW.01, qgeoSymIsCorollary).** BWKeystone.lean extended with the cited conformal-net vacuum axiom `conformal_net_vacuum_rotation_invariant`; with it, `StateInvariance (=QGEO.SYM.01)` follows from `SeamIsE8` alone, so QGEO.SYM.01 is a COROLLARY of SEAM.EQUIV.01 with no independent open premise (`#print axioms` confirms only the chain + cited BW/net axioms). lake build green (3362 jobs).
- **Status deep-sync.** The ledger (QG.AMB.01 reclassified to “decoupled general QG; non-TFPT”; QGEO.SYM.01 marked a corollary; new SEAM.EQUIV.UNIFY.01 row), the canonical status card (`tfpt_status.tex`), introduction (status core + gate table), `origin_theory`, `tfpt_4_frontier`, `tfpt_5_redteam`, `tfpt_research_contracts`, `README`, `next.txt` and the website (`OpenGates`, `FAQ`, `papers.ts`) all moved to “one solvable theorem (SEAM.EQUIV.01) + one decoupled foreign problem (QG.AMB.01)”. No falsifiable prediction changed; this is a status-framing sharpening backed by v335 + the Lean corollary.

**2026-06-22 (closing round: the CM-selection mechanism (the last keystone residual), the conformal-mode resolution, and the Conway–Sloane even-unimodular side in Lean: v333, v334, FORM.CARTAN.DET.01)**

- **v333\_cm\_selection.py (QGEO.CMSELECT.01, [E]/[O]).** The CM-selection mechanism for the last keystone residual:  $\tau=i$  (the square) is the *unique* order-4 modulus —  $S = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$  has  $S^4=I$  and fixes  $i$ , and an order-4 deck selects  $j=1728$  (vs the order-3 hexagon  $j=0$ ) — AND the square is the Fekete attractor of symmetric mark-equilibration (four points minimizing the log-energy converge to equal  $\pi/2$  spacing). So given the  $\mu_4$  deck,  $\tau=i$  is forced; the residual sharpens from “why the square” to “the deck acts geometrically” (= QGEO.SYM.01). NOT a closure.
- **v334\_conformal\_resolution.py (QGAMB.CONFORMAL.01, [E]/[C]/[O]).** The conformal-mode resolution for the QG metric sector: the Gibbons–Hawking–Perry contour rotation  $\phi \rightarrow i\phi$  flips the wrong-sign kinetic term, making the divergent conformal Gaussian  $\int e^{+|c|\phi^2}$  into the convergent  $\int e^{-|c|\phi^2} = \sqrt{\pi/|c|}$ ; the entire IDG form factor  $e^{p^2/M^2}$  dresses the propagator pole-free (the polynomial  $R+R^2$  ghost as control). Together they give a *conditional* mode-by-mode construction of the metric-sector measure; the residual is only the contour’s nonperturbative validity (gap-decoupled, blocks no test). NOT a closure.
- **CartanDeterminants.lean extended (FORM.CARTAN.DET.01, [E]).** The Conway–Sloane lattice side: the  $E_8$  Cartan matrix is proved to be an EVEN (diagonal = 2), symmetric, unimodular ( $|\det|=1$ ) integer Gram matrix — the defining data of an even unimodular rank-8 lattice — all by kernel `decide` / the integer-inverse argument. The remaining content (Witt’s uniqueness:  $E_8$  is the only even unimodular lattice in  $\dim < 16$ ) is the cited classification residual.

**2026-06-22 (endgame round: the necessity of  $H$  reduced to the order-4 CM selection, the QG metric sector characterized as the conformal-mode problem,  $|\det E_8|=1$  and the  $\mu_4$  commutation grounded in Lean: v331, v332, FORM.MU4COMM.01)**

- **v331\_necessity\_of\_H.py (QGEO.NECESS.01, [E]/[O]).** The necessity companion to v329: a concrete Steklov (DtN) computation shows the seam  $\Lambda = |k| + M_f$  commutes with the  $\mu_4$  clock *iff* the four marks sit at the square ( $\tau=i$ ) — moving a mark off the square breaks  $\mathbb{Z}_4$  and  $[\rho, \Lambda] \neq 0$ . The square has cross-ratio 2 ( $j=1728$ ), the unique 4-config with an order-4 automorphism (v214/v267), so  $H \Leftrightarrow \text{square} \Leftrightarrow \mu_4$  and “necessity of  $H$ ” sharpens to: the raw

dynamics places the four Gauss–Bonnet marks (v216) at the order-4 CM point. Residual = that dynamical selection.

- **v332\_qgamb\_metric\_sector.py** (QGAMB.METRIC.01, [E]/[C]/[O]). The open metric sector of QG.AMB.01 is characterized precisely as the Gibbons–Hawking–Perry conformal-factor problem: the conformal mode has a negative kinetic coefficient  $-(D-1)(D-2)/4$  ( $= -\frac{3}{2}$  at  $D=4$ ), so the bare Euclidean measure diverges over it. The obstruction is in the conformal/gauge direction, gap-insulated from the admissible sector (v330/v76); the only route is the conditional IDG taming (v304), with the polynomial  $R+R^2$  ghost as the control. Open, but pinned to one named obstruction.
- **CartanDeterminants.lean extended** (FORM.CARTAN.DET.01, [E]). The rank-8 holomorphy discriminator  $|\det \text{Cartan}(E_8)|=1$  ( $E_8$  has one anyon, vs  $D_8$ ’s four) is now proved principally and kernel-only: `cartanE8 * cartanE8inv = 1` by `decide` on the  $8 \times 8$  PRODUCT gives `IsUnit (det E8)` via  $\det E_8 \cdot \det E_8^{-1} = \det 1 = 1$  — no 8! determinant, no `native_decide` (Mathlib itself leaves `E8_det = 1` a `proof_wanted`).
- **Mu4Commutation.lean** (FORM.MU4COMM.01, Lean 4, [E]). The v201/v323 mark-locality criterion as exact linear algebra over the Gaussian integers  $\mathbb{Z}[i]$ :  $i^4=1$ ,  $i^2=-1$ , and (on concrete  $6 \times 6$  matrices) an offset-4 coupling commutes with the  $\mu_4$  clock while an offset-2 coupling does not — all by kernel `decide`. Built `lake build` on Lean v4.29.1 + Mathlib.

**2026-06-22 (frontier round: the OS step discharged at the many-body level, the ambient measure built on the gap-decoupled sector, and the carrier Cartan determinants grounded in Lean: v329, v330, FORM.CARTAN.DET.01)**

- **v329\_os\_gap\_reduction.py** (QGEO.OSGAP.01, [E]/[O]). The one analytic input the v308 keystone chain still cited — “the OS-reconstructed bulk gap = the transfer gap” — is discharged at the many-body level: the OS/Euclidean Hamiltonian is the second quantization  $dT(-\log T)$ , whose spectrum is the set of sums of single-mode energies, so the bulk gap = the smallest positive single-mode energy  $= 6 \ln \frac{3}{2}$ , *independent of system size*. Together with  $H \Rightarrow \omega \circ \rho = \omega$  and  $H \Rightarrow \text{SRE} \Rightarrow (E_8)_1$ , this exhibits the rotation-invariant flat-pillowcase premise  $H$  as the *single sufficient premise* of both bedrock IDs; residual = the necessity of  $H$  on the raw collar.
- **v330\_qgamb\_admissible\_measure.py** (QGAMB.MEASURE.01, [E]/[O]). The constructive step v275’s roadmap calls for: on the gap-decoupled *admissible* sector the ambient measure is built as a bona fide OS object — reflection-positive (symmetric PSD transfer), exponentially clustering ( $C(n)/C(n-1) \rightarrow (2/3)^6$ ), with finite susceptibility  $\chi = 729/665$  so  $\text{Var}(S_\Lambda)/\Lambda \rightarrow \chi$  converges  $\Rightarrow$  uniform tightness (Prokhorov)  $\Rightarrow$  the projective limit exists; OS reconstruction gives  $H_{\text{OS}} \geq 0$  with gap  $6 \ln \frac{3}{2}$ . The full  $\text{QG}_{\text{amb}}$  (the metric sector) stays open, gap-decoupled — a genuine *partial* construction, not a closure.
- **CartanDeterminants.lean** (FORM.CARTAN.DET.01, Lean 4, [E]). The carrier piece of the holomorphy/anyon discriminator is grounded in real `Matrix.det` computations (kernel `decide`, no extra axiom):  $|\det \text{Cartan}(A_3)|=4=|\mu_4|$  ( $3 \times 3$ , `det_fin_three`) and  $|\det \text{Cartan}(D_5)|=4$  ( $5 \times 5$ , `decide`), giving the carrier  $|\det \text{Gram}|=4 \cdot 4=16$  (the v281 anyon count). The rank-8  $E_8/D_8$  case stays the Nat discriminator — Mathlib’s own `Data.Matrix.Cartan` leaves `E8_det = 1` as a `proof_wanted`. Built `lake build` on Lean v4.29.1 + Mathlib.

**2026-06-22 (next-steps round: the keystone collapsed to one shared premise (Lean + one citable theorem), the  $F_{\text{transfer}}$  solver suite, the cusp weight derived from a minimal rewrite, and the  $\theta_{13}$  pressure point: FORM.QGEO.BW.01, v325, v326, v327, v328)**

- **BWKeystone.lean** (FORM.QGEO.BW.01, Lean 4, [E]/[O]). The v323 Bisognano–Wichmann linkage formalised: the nodes `RotationInvariantVacuum`  $\rightarrow$  `GeometricModularFlow`  $\rightarrow$  `Mu4ModularSymmetry`  $\rightarrow$  `StateInvariance` (=QGEO.SYM.01) composed over the v308 chain. The theorem `bedrockReducesToRotationInvariance` shows the WHOLE bedrock QGEO.SYM.01 reduces to the SINGLE shared open premise `RotationInvariantVacuum` — the two open bedrock items collapse to one — with `#print axioms` confirming the residual = the cited chain steps + recovery gap + the named BW steps + that one premise. Plus a decide-proven discriminator  $(1 \mid 4) \wedge \neg(4 \mid 1)$  (BW geometric flow strictly stronger than v309 clock invariance). Built lake build on Lean v4.29.1 + Mathlib.
- **v325\_pillowcase\_keystone.py** (QGEO.KEYSTONE.01, [E]/[O]). The keystone as ONE citable theorem: IF the raw seam state is the flat  $\tau=i$  pillowcase (rotation-invariant) state THEN the whole bedrock closes (4 square marks,  $\mu_4$  deck,  $\omega \circ \rho = \omega$ , holomorphic  $c=8 \Rightarrow (E_8)_1$ , with the recovery gap as input); all pieces machine-checked (incl. the pillowcase  $\chi_{\text{orb}}=0$ ), the single open lemma shared by both bedrock IDs and Lean-pinned. An assembly certificate, not a closure.
- **v326\_ftransfer\_suite.py** (FR.SUITE.01, [E]/[C]). The four  $F_{\text{transfer}}$  readouts productised into one runnable harness, each a gapped relaxation to a unique attractor:  $F_{\text{pole}}$  (Koide, seam rate  $(2/3)^6$ ),  $F_{\text{Boltzmann}}$  ( $\eta_B$ , H-theorem),  $F_{\text{relic}}$  (axion adiabatic invariant),  $F_{\text{QCD}}$  ( $m_p/m_e$ , RG to the UV fixed point), plus the  $F_{\text{RareKaon}}$  bridge. Only  $F_{\text{pole}}$  carries the seam rate; the four external rates are honestly fenced (v187/v303).
- **v327\_hypergraph\_rewrite.py** (HYP.REWRITE.01, [E]/[O]). The cusp weight  $2/3$  is DERIVED from a minimal rewrite: a 3-channel/1-absorbing family rule has spectrum  $\{1, 2/3, 0\}$ , so  $2/3 = (N_{\text{fam}} - 1)/N_{\text{fam}} = |\mathbb{Z}_2|/N_{\text{fam}}$  emerges from the rule arity and  $(2/3)^6 = 64/729$  is the recovery gap; with a PROOF that  $2/3$  is not a root of the affine- $E_8$  charpoly ( $p(2/3) = -3520/19683 \neq 0$ , sharpening v312). v324’s injected datum is reduced to the anchor arity  $\{2, 3\}$ .
- **v328\_theta13\_pressure.py** (FLAV.TH13.PRESSURE.01, [C]/[X]) + **v307 hardening**. The honest weak spot named and quantified:  $\sin^2 \theta_{13} = \varphi_0^{\text{ret}} e^{-5/6} = 0.023108$  is  $+2.0\sigma$  above NuFIT 6.0 NO, a SEPARATE carrier-trace channel (v268), not relieved by any freeze-respecting correction (RG < 1%, exponent 5/6 exact), pre-registered as the kill. The v307 watchdog gains a data provenance/date check and the rare-kaon  $F_{\text{transfer}}$  bridge row.
- **Wolfram mirror** 285  $\rightarrow$  288. v325 ( $\chi_{\text{orb}}=0$ ) and v327 (rewrite spectrum  $\{0, 2/3, 1\}$ ,  $(2/3)^6 = 64/729$ , and the non-graph-spectral proof) added to `tfpt_readouts_extension.wl`; GATE.WOLFRAM.02 and the Wolfram README updated to 288/288.

**2026-06-22 (consistency + de-accretion editorial pass: current-state alignment across papers, website and ledger; no new modules)**

- **Keystone naming reconciled to SEAM.EQUIV.01**. Following v323/QGEO.BW.01 (the deck premise is downstream of the seam being a chiral net), the single named open keystone is now SEAM.EQUIV.01 *everywhere*, with QGEO.SYM.01 stated as its *downstream* conformal-deck face. The ledger row QGEO.SYM.01 was updated (it no longer calls itself “the single structural residual of the whole theory”; it now depends on SEAM.EQUIV.01/QGEO.BW.01), and the superseded “GATE.QGEO open gate” / “two research gates ( $U_{\text{wall}}$ ) and  $G_{\text{metric}}$ ” framing was replaced by the current  $\text{Rest} = v_{\text{geo}} \oplus G_{\text{net}} \oplus F_{\text{transfer}}$  with the two structural

items SEAM.EQUIV.01 + QG.AMB.01 (intro status card, `tfpt_4_frontier`, `origin_theory`, `tfpt_5_redteam`, `tfpt_research_contracts`, and the website).

- **De-accretion: the sprint-by-sprint reduction history was moved to this changelog, where it belongs.** Several reader-facing sections had grown by accretion (each round appended a clause). They were rewritten to state the *current* result once, with the round-by-round narrative left here: the introduction closure ledger (“honest bottom line”) and hard-reduction subsection; the `origin_theory` “residual inventory” (the five “Update/.../Terminal” passes collapsed to one current inventory); the `tfpt_5_redteam` Target-A Outcome; and on the website the FAQ “What is still open?”, the `OpenGates` closing-condition block, the `papers.ts` research-contracts/red-team entries, the verification-DAG `qft` node, the glossary `G_net` entry, the Rosetta lexicon, and the verification/review pages. No scientific content or `\veri` citation was dropped; only the embedded version-stamp logs were removed.
- **Stale facts corrected.** Version label 5.0→v5.2 (intro, single-sourced via `\TFPTversion`); Lean build 1886→3357 jobs (`FORM.SEAMEQUIV.01`); README Python suite v213/ 212→v324/ 322 modules; Wolfram extension 269/269 / 249/249→285/285 (README, website replication page, `zenodo_description.html`); residual “Paper-3” old-paper attributions in `tfpt_2` neutralised.

(Original 2026-06-22 entry follows.)

**2026-06-22 (external-review pass: a registry bug fix, the NA62 2026 rare-kaon update, CP-phase prose hygiene, and three firewall/visualisation enhancements: v202, v203, v305, v307, v321)**

- **Bug fix —  $\delta_{\text{CKM}}$  display value.** The predictions table (`tex-artefacts/predictions.tex`) showed  $\delta = \frac{\pi}{3} + 3\lambda^2 = 66.96^\circ$ , but that formula evaluates to  $68.65^\circ$  (the canonical frozen `DELTA_CKM_RAD` = 1.198232 rad, used by v88/v202/FLAV.CP.01 everywhere else). Corrected to  $68.65^\circ$ . An external review correctly flagged the inconsistency; it was an isolated display typo (the website was already clean).
- **v202\_rare\_kaon.py — NA62 2016–2024 update ([E]/[C]).** The charged-mode comparison is moved from the historical NA62 2016–2022 observation  $(13.0_{-3.0}^{+3.3}) \times 10^{-11}$  ( $\sim 1.2\sigma$  above) to the full 2016–2024 combination  $\text{BR}(K^+) = (9.6_{-1.8}^{+1.9}) \times 10^{-11}$  (La Thuile 2026), on which the prediction  $9.45 \times 10^{-11}$  lands at  $+0.08\sigma$ . Framed honestly: a strong *consistency* hit for the closed CKM point, but an  $F_{\text{transfer}}$  bridge — *not* a unique TFPT-vs-SM discriminator (the SM  $(8.6 \pm 0.4) \times 10^{-11}$  is also inside the NA62 error). Propagated to the ledger (`FR.RAREKAON.01`), `freeze_file` (new `rare_kaon` kill row), the watchdog (v307, a new  $F_{\text{transfer}}$  bridge row), the forward board (v321, rank 9), and `tfpt_2_standard_model` + the website.
- **CP-phase prose hygiene.** The Galois CP lock is tightened from “a relation to the *measured* quark phase” to “a relation to the *leading* ( $\pi/3$ ) component of the quark phase”: the lock is to the structural  $\pi/3$ , not the full measured  $\gamma = \delta_{\text{CKM}}^{\text{lead}} + 3\lambda_C^2 \approx 68.7^\circ$ , so  $\delta_{\text{PMNS}} = 240^\circ$  (not  $248.7^\circ$ ). Fixed in `tfpt_2_standard_model`, `origin_theory`, `papers.ts` and `predictions.ts`.
- **v305\_witness\_independence.py — the No-Golden-Dynamics rule ([E]).** The number-field split is made an explicit anti-numerology firewall rule:  $\phi = 2 \cos(\pi/5) \in \mathbb{Q}(\sqrt{5})$  is the *static* carrier ( $\mathbb{Z}/5$ ) field, every *dynamic* rate is rational ( $\mathbb{Q}$ , the  $\mathbb{Z}/6$  hand), so a dynamic rate explained by  $\phi$  is a cross-field false friend (v314/v315).
- **v203\_eht\_achromatic.py — the  $\mu_4$  Fourier fourth null ([X]).** The EHT residual pre-registration gains a fourth null: the azimuthal residual must carry power *only* at  $m \equiv 0 \pmod{4}$  (the  $\mu_4$  fingerprint of the four marks, v201); a  $\mu_4$ -orbit profile has  $\sim 0$  off-mode power while a  $\mathbb{Z}_2$  control leaks at  $m = 2$ .

- **Seed-hyperplane figure (Fig. ??).** A new figure visualises v306: every scorecard observable inverts *independently* to the one latent seed  $\varphi_0$ , the error-weighted mean matching the axiom to 0.04%, with  $\sin^2 \theta_{13}$  the  $\sim 2\sigma$  pressure point. Added to `make_figures.py`, `tfpt_5_redteam` and the manifest.

**2026-06-22 (Wolfram mirror of the arithmetic arc + three sharpening steps: the CP-lock made quantitative, the Bisognano–Wichmann keystone, the minimal hypergraph fibre: v322, v323, v324)**

- **Wolfram mirror of the cyclotomic/Galois arithmetic arc (the independent second path).** The exact algebraic results of the arc are now mirrored in `verification/wolfram/tfpt_readouts_extension.wl` (ten new `checkExact` blocks,  $275 \rightarrow 285$ , *all passing*): v313 (the affine- $E_8$  characteristic polynomial with the golden factor  $x^2-x-1$ ; the  $(2, 3, 5)$  atoms as  $2 \cos(\pi/k)$ ; the 31/30 selection), v314 (the rational kernel vs  $\mathbb{Q}(\sqrt{5})$  dynamic-rate split), v315 ( $30=5 \cdot 6$ ,  $\varphi(30)=8$ , the Galois group  $\mu_4 \times \mathbb{Z}_2$ , the  $\sqrt{5}$  Gauss sum), v316 ( $\rho=\zeta_6=\zeta_{30}^5$ ,  $\rho^4=-\rho$ ,  $\text{Gal}=\mathbb{Z}_2=\text{CP conjugation}$ ), v317 ( $N_{\text{fam}}=3$  as the  $\mu_3$  orbit, the 1+2 Galois split), v318 ( $\varphi_0^{\text{ret}}=(4/3)c_3+48c_3^4$  a pure function of  $\pi$ ) and v320 (the CP lock  $\delta_{\text{PMNS}}=240^\circ$ ). Ledger `GATE.WOLFRAM.02` and the Wolfram README updated to 285/285.
- **v322\_cp\_lock\_sharpened.py (GALOIS.CP.SHARPEN.01, [E]/[C]/[X]).** The v320 CP-lock sharpened into a quantitative, pre-registered kill test. [E] the leading phase sits on the six hexagonal nodes  $\arg \rho^k = k \cdot 60^\circ$ , and *which* one is selected is the deck order:  $\delta_{\text{PMNS}}^{\text{lead}} = |\mu_4| \delta_{\text{CKM}}^{\text{lead}} = 4(\pi/3)$  (node 4); [C] the sub-leading correction is bounded by the quark analogue  $3\lambda^2 \approx 8.7^\circ$ , so the prediction is the band  $240^\circ \pm \sim 9^\circ$ ; [C] against NuFIT 6.0 (NO,  $\delta_{\text{CP}}=212_{-41}^{+26^\circ}$ ) the leading value is  $+1.08\sigma$ , the band edge  $+0.74\sigma$  – compatible today; [X] the nearest wrong node is  $60^\circ$  away, far beyond the budget, so DUNE/Hyper-K ( $\sim 5\text{--}15^\circ$ ) discriminate cleanly. Cited in `origin_theory`; ledger `GALOIS.CP.SHARPEN.01`.
- **v323\_bw\_geometric\_modular.py (QGEO.BW.01, [E]/[C]/[O]).** The keystone pushed one honest step via Bisognano–Wichmann. [E] the  $\mu_4$  clock is a geometric rotation  $\rho=\text{diag}(i^n)=\exp(i\frac{\pi}{2}L)$ ,  $\mu_4 \subset U(1)_{\text{rot}}$ ; [E] for a rotation-covariant covariance  $C=f(L)$  the modular Hamiltonian  $K=\log((1-C)/C)=g(L)$ , so  $[K, L]=0$  (the modular flow is geometric) and the clock is a modular symmetry *for free* (the discriminator: a period-4 curvature preserves the clock but its flow is not geometric,  $[K, L] \neq 0$ ); [C] given the seam is the  $(E_8)_1$  chiral net (v308), BW/Hislop–Longo make the vacuum modular flow geometric, so `QGEO.SYM.01` ( $\omega \circ \rho = \omega$ ) is *downstream* of `SEAM.EQUIV.01` + a rotation-invariant vacuum, not an independent axiom; [O] residual: the raw collar vacuum is rotation-invariant (cleaner than v309, shared with v308). Cited in `origin_theory`; ledger `QGEO.BW.01`.
- **v324\_hypergraph\_fiber.py (HYP.FIBER.01, [E]/[C]).** The constructive v312 follow-up: the *minimal* substrate that carries both the  $E_8$  skeleton and the recovery gap is a *product*. [E] the carrier network  $T_{\text{net}}=(A+2I)/4$  (attractor = Kac marks = skeleton) fibred by the 3-node family cusp  $T_{\text{cusp}}=\text{diag}((1-w)^{2N_{\text{fam}}})$ ,  $w \in \{0, \frac{1}{3}, \frac{2}{3}\}$  (spectrum  $\{1, (2/3)^6, (1/3)^6\}$ ); the fibred  $T_{\text{net}} \otimes T_{\text{cusp}}$  has top eigenvalue 1 (eigenvector marks  $\otimes (w=0)$ ) and  $(2/3)^6$  as a genuine eigenvalue (marks  $\otimes (w=\frac{1}{3})$ ) – the  $N_{\text{fam}}=3$  cusp is the minimal fibre and the substrate = carrier  $\times$  family = the  $30=g_{\text{car}}(2N_{\text{fam}})=5 \times 6$  of v315; [C] honestly (v312) the cusp weight is still *injected*, not derived from the graph – a consistent realisation, not a derivation of  $2/3$  from a pure rewrite. Cited in `origin_theory`; ledger `HYP.FIBER.01`.

**2026-06-22 (the forward kill-test board: the decisive upcoming measurements ranked, each locked to a freeze row — incl. the new Galois-CP relation: v321)**

- **v321\_killtest\_board.py (KILL.BOARD.01, [E]).** The forward companion to v307 (the current-data board): the decisive *upcoming* measurements, ranked and each locked to a `freeze_file.csv` kill row – #1 JUNO  $\sin^2 \theta_{12}=0.306747$  (fastest), #2 CMB-S4/LiteBIRD  $r$

(kill  $r > 0.01$ ), #3 DESI/Euclid  $w$  (kill  $w \neq -1$ ), #4 DUNE/Hyper-K  $\delta_{\text{PMNS}} = 240^\circ$  (the new v320 Galois-CP relation), #5 DUNE/LEGEND/nEXO ordering +  $m_{\beta\beta}$ , #6 PSI n2EDM, #7 JUNO  $\sin^2 \theta_{13}$  ( $\sim 2\sigma$  now), #8 LiteBIRD  $\beta$ . A forward decision ledger (publicly-stated experimental reach), complementing v307; no new physics. Cited in `origin_theory`; ledger row KILL.BOARD.01.

## 2026-06-22 (Lean: the SEAM.EQUIV.01 composition chain formalised — the key-stone residual machine-pinned to the named cited steps: FORM.SEAMEQUIV.01)

- **SeamEquivChain.lean** (FORM.SEAMEQUIV.01, Lean 4, [E]/[O]). A Lean 4 mirror of the v308 closing chain: the four nodes  $\text{GapPositive} \rightarrow \text{InvertibleSRE} \rightarrow \text{HolomorphicC8} \rightarrow \text{SeamlsE8}$  assembled as a well-typed composition theorem `seamEquivChain`, with the three cited literature steps (Kitaev free-fermion invertibility; Muger / Kawahigashi–Longo–Muger holomorphy; Conway–Sloane  $E_8$  uniqueness) as named `axioms` and the carrier recovery gap (the OS step) as the single open input. [E] `#print axioms seamEquivChain` confirms the residual is exactly those named steps + `recovery_gap` (no hidden assumption); [E] a `decide`-proven discriminator  $\text{primariesE8} = 1 \neq \text{primariesD8} = 4$  (the  $|\det \text{Cartan}|$  that separates the holomorphic  $E_8$  from the same- $c$  rival  $D_8 = SO(16)$ ). Built end-to-end (lake build, 3357 jobs) on Lean v4.29.1 + Mathlib. [O] The value is the machine-checked composition + axiom audit (pinning the keystone residual), *not* a from-scratch proof — it does NOT close SEAM.EQUIV.01. Ledger row FORM.SEAMEQUIV.01; mirrors v308.

## 2026-06-22 (a NEW falsifiable prediction from the Galois structure: the two CP phases are locked, $\delta_{\text{PMNS}} = \delta_{\text{CKM}}^{\text{lead}} + \pi$ : v320)

- **v320\_galois\_cp\_relation.py** (GALOIS.CP.PREDICT.01, [E]/[C]/[X]). The v316 follow-up turned into a testable cross-prediction. [E] both leading CP phases are powers of the *one* hexagonal unit  $\rho = \zeta_6$  of the family Galois factor:  $\delta_{\text{CKM}}^{\text{lead}} = \arg \rho = \pi/3$  ( $60^\circ$ ),  $\delta_{\text{PMNS}} = \arg \rho^4 = 4\pi/3$  ( $240^\circ$ ), and  $\rho^4 = -\rho$  locks them as  $\delta_{\text{PMNS}} = \delta_{\text{CKM}}^{\text{lead}} + \pi$  (the  $\mathbb{Z}_2$  sheet flip); [C] this upgrades the previously assigned  $\delta_{\text{PMNS}} = 240^\circ$  to a Galois-forced relation to the measured quark phase (the quark and lepton leading CP phases are not independent); [X] kill test: a  $\delta_{\text{PMNS}}$  robustly incompatible with  $240^\circ$  ( $> 3\sigma$  at DUNE/Hyper-K/JUNO) falsifies the Galois-CP organisation;  $240^\circ$  is currently within the broad NuFIT 6.0 range. A genuine new falsifiable consequence of the v313–v319 arithmetic arc (also a `freeze_file.csv` `cp_phase_relation` kill row). Cited in `origin_theory`; ledger row GALOIS.CP.PREDICT.01.

## 2026-06-22 (the translation clock: the discrete $\leftrightarrow$ dynamic bridge *is* the order-30 Coxeter element, a clock with two coprime hands $5 \times 6$ , read law-inclusive (0..5) or live-only (1..5): v319)

- **v319\_translation\_clock.py** (TRANSLATE.CLOCK.01, [E]/[C]). A synthesis of the v313–v318 arc with v124/v55, answering the question “is the static $\leftrightarrow$ dynamic translation a clock running 0, 1, 2, 3, 4, 5 or 1, 2, 3, 4, 5?”. [E] the only object holding both the static carrier ( $\mathbb{Q}(\sqrt{5})$ ) and the dynamic family (rational) data is the order-30 Coxeter element (v314);  $30 = h(E_8) = g_{\text{car}}(2N_{\text{fam}}) = 5 \cdot 6$  and  $\gcd(5, 6) = 1 \Rightarrow \mathbb{Z}/30 = \mathbb{Z}/5 \times \mathbb{Z}/6$  (two coprime hands, v315). [E] the *dynamic* hand is  $\mathbb{Z}/6 = 2N_{\text{fam}}$  (ticks  $0, \dots, 5$ ): the recovery exponent is exactly  $6 = 2N_{\text{fam}}$ , rate  $(2/3)^6 = (|\mathbb{Z}_2|/N_{\text{fam}})^{2N_{\text{fam}}}$ , order-6 generator  $c^5$  ( $\zeta_6 = \zeta_{30}^5$ , v316). [E] position 0 is the *conserved law* — the resummed clock rate( $n$ ) =  $-p_2 \ln(1 - n/N_{\text{fam}})$  (v124) has rate(0) = 0 (the attractor, eigenvalue 1, v56), the live decaying modes from  $n=1$ , and the  $E_8$  live phases (totatives of 30) start at 1 ( $\{1, 7, \dots, 29\}$ ), the 0 phase not a totative — so “0, 1, 2, 3, 4, 5” is law-inclusive, “1, 2, 3, 4, 5” live-only. [E] the *static* hand is  $\mathbb{Z}/5 = g_{\text{car}}$  (ticks  $1, \dots, 5$ ): the golden field  $\mathbb{Q}(\sqrt{5})$  ( $\varphi = 2 \cos(\pi/5)$ ), field-obstructed (no rational dynamic image, v314) — static-only;  $\zeta_5 = \zeta_{30}^6$  carries no phase. [E] one full turn

$= \text{lcm}(5, 6) = 30 = h(E_8)$ , rank  $E_8 = 8 = \varphi(30)$  live phases pairing into  $|\mu_4| = 4$  invariant planes (v55). [C] verdict: the discrete→dynamic translation *is* the order-30 clock = (static 5-ring)  $\times$  (dynamic 6-ring), the dynamic hand carrying the rate (exponent  $6 = 2N_{\text{fam}}$ ), the static hand the golden carrier (no rate), and “0 vs 1” = law-inclusive vs live-only; the arithmetic is [E], the “it is one clock” reading [C]. Python-only (sympy), like the rest of the arc. New figure [figures/translation\\_clock.pdf](#) (concentric 5- and 6-hands). Cited in [origin\\_theory](#); ledger row TRANSLATE.CLOCK.01.

**2026-06-21 (capstone: the SM structural sector is  $\mathbb{Q}(\zeta_{30})$ + Galois  $\mu_4 \times \mathbb{Z}_2$ , and the complete input is  $\{a, \pi, v_{\text{geo}}\}$  — zero dimensionless free parameters: v318)**

- **v318\_arithmetic\_capstone.py (ARITH.CAPSTONE.01, [E]/[O]).** The synthesis of the v313–v317 arc. [E] the SM *structural* sector (the 3 generations, the 2 CP phases, their orbit/hierarchy) is the cyclotomic field  $\mathbb{Q}(\zeta_{30})$  with Galois group  $\mu_4 \times \mathbb{Z}_2$  (degree  $\varphi(30) = 8 = \text{rank } E_8$ ,  $30 = |\mathbb{Z}_2|N_{\text{fam}}g_{\text{car}} = h(E_8)$ ), forced by the anchor atoms  $\{2, 3, 5\} = e(a)$ ; [E] the magnitude seed  $\varphi_0 = (|\mu_4|/N_{\text{fam}})c_3 + \Omega_{\text{adm}}c_3^4 = \frac{4}{3}c_3 + 48c_3^4$  ( $c_3 = 1/(8\pi)$ ) is a pure function of  $\pi$  (no free parameters, anchor-derived coefficients), so the magnitudes ( $\sin^2 \theta_{12} = \frac{1}{3} - \varphi_0/2$ ) are  $\{a, \pi\}$ -determined, not free; [E] free inventory: *zero* dimensionless free parameters,  $\pi$  the unique transcendental primitive (ARCH.CORE.01),  $v_{\text{geo}}$  the unique dimensionful input (No-Unit Theorem) — complete input  $\{a, \pi, v_{\text{geo}}\}$ . [O] honest residual: this rigidity holds *given* the axioms and does not realise the structure on the raw seam (QGEO.SYM.01/SEAM.EQUIV.01 stay [O]). Cited in [origin\\_theory](#); ledger row ARITH.CAPSTONE.01.

**2026-06-21 (are the 3 generations a  $\mu_3$ /Galois orbit? Yes ( $\mu_3$ ), Galois-refined 1+2 with the fixed generation = the recovery attractor: v317)**

- **v317\_galois\_family.py (GALOIS.FAMILY.01, [E]/[C]).** The v316 follow-up to the generation structure. [E] the three cusp weights  $\{0, \frac{1}{3}, \frac{2}{3}\}$  (with  $\text{Spec}(Q_+) = \{1, 2, 3\}$ , the  $A_3$  exponents) are the cube roots of unity  $\{1, \zeta_3, \zeta_3^2\}$  under  $w \mapsto e^{2\pi i w}$ , and the cyclic family symmetry  $w \mapsto w + \frac{1}{3}$  is a *transitive*  $\mu_3$  orbit (the family IS the cube-root orbit); [E] the transfer eigenvalues  $(1-w)^6 = \{1, (2/3)^6, (1/3)^6\}$  attach to the weights (attractor = eigenvalue 1 at  $w=0$ ); [E] the family Galois  $\mathbb{Z}_2$  ( $\zeta_3 \mapsto \zeta_3^2 = \bar{\zeta}_3$ , = the v316 CP conjugation) refines the orbit to 1+2 — fixing  $w=0$  (rational) and swapping  $w=\frac{1}{3} \leftrightarrow \frac{2}{3}$ ; [E] so the Galois-FIXED generation is the recovery ATTRACTOR (the law) and the Galois-conjugate pair the decaying hierarchy. [C] verdict: the family STRUCTURE is the same arithmetic as the CP phase (a  $\mu_3$  orbit, Galois-refined 1+2), the mass MAGNITUDES remaining the analytic seed. Cited in [origin\\_theory](#); ledger row GALOIS.FAMILY.01.

**2026-06-21 (does the Galois  $\mu_4 \times \mathbb{Z}_2$  organize the readouts? Yes — the CP phases are the family-factor cyclotomic data, Galois  $\mathbb{Z}_2 = \text{CP conjugation}$ : v316)**

- **v316\_galois\_readout.py (GALOIS.READOUT.01, [E]/[C]).** The v315 follow-up to phenomenology: does the Galois coupling  $\mu_4 \times \mathbb{Z}_2$  reach the physics? [E] it does, through the *family* factor — the hexagonal CP unit  $\rho = \zeta_6 = \zeta_{30}^5$  is exactly the order-6 =  $2N_{\text{fam}}$  factor  $c^5$  of v315 (the carrier being  $\zeta_5 = \zeta_{30}^6$ ); [C] the leading CP phases are its cyclotomic arguments  $\delta_{\text{CKM}}^{\text{lead}} = \pi/3 = \arg \zeta_6$  and  $\delta_{\text{PMNS}} = 4\pi/3 = \arg \zeta_6^4$  (power 4 =  $|\mu_4|$ ), with  $\zeta_6^4 = -\zeta_6$  the  $\mathbb{Z}_2$  sheet flip and  $\mu_6 = \mu_3 \times \mu_2$  (inherits v231/v233); [E] the family Galois  $\text{Gal}(\mathbb{Q}(\zeta_6)/\mathbb{Q}) = \mathbb{Z}_2$  ( $\zeta_6 \mapsto \zeta_6^5 = \bar{\zeta}_6$ ) *is* CP conjugation  $\delta \mapsto -\delta$ ; [E] the carrier factor  $\zeta_5$  (golden  $\sqrt{5}$ ) carries no phase ( $\mu_4$  static), and the magnitudes ( $\sin^2 \theta_{12} = \frac{1}{3} - \varphi_0/2$ , transcendental  $\varphi_0$ ) are not cyclotomic. [C] verdict: the Galois bridge reaches phenomenology in the phase sector predicted by the v314 field split. Cited in [origin\\_theory](#); ledger row GALOIS.READOUT.01.

**2026-06-21 (couple or factorize? the order-30 Coxeter element couples the carrier and family sectors as a cyclotomic compositum, Galois =  $\mu_4 \times \mathbb{Z}_2$ : v315)**

- **v315\_coxeter\_coupling.py** (COX.COUPLE.01, [E]/[C]). Answers the v314 question — does  $30 = g_{\text{car}}(2N_{\text{fam}}) = 5 \cdot 6$  couple the carrier (5-fold, golden,  $\mathbb{Q}(\sqrt{5})$ ) and family ( $6 = 2 \cdot 3$ , rational) sectors, or just factorize? [E] the cyclic group *factorizes*:  $\gcd(5, 6) = 1 \Rightarrow \mathbb{Z}/30 = \mathbb{Z}/5 \times \mathbb{Z}/6$  (direct product, group-decoupled, in line with the v314 field obstruction); [E] the  $E_8$  exponents = totatives(30) split as totatives(5)  $\times$  totatives(6),  $\varphi(30) = 4 \cdot 2 = 8 = \text{rank } E_8$ ; [E] but the *field* couples:  $\mathbb{Q}(\zeta_{30})$  is the compositum of the carrier  $\mathbb{Q}(\zeta_5) \ni \sqrt{5}$  and the family  $\mathbb{Q}(\zeta_3)$ , degree  $8 = \text{rank } E_8$ ; [E] the Galois group is exactly  $(\mathbb{Z}/5)^\times \times (\mathbb{Z}/3)^\times = \mathbb{Z}/4 \times \mathbb{Z}/2 = \mu_4 \times \mathbb{Z}_2$  (order 8, max element order 4), so the  $\mu_4$  seam clock *is* the Galois group of the carrier's golden field and  $\mathbb{Z}_2$  that of the family field (refining v223); [E]  $\sqrt{5} = \phi + 1/\phi$  is the quadratic Gauss sum in  $\mathbb{Q}(\zeta_5)$ . [C] verdict:  $30 = 5 \cdot 6$  does not dynamically entangle the sectors — it couples them as the cyclotomic compositum, the static $\leftrightarrow$ dynamic bridge being *Galois*, not a dynamical map. Cited in *origin\_theory*; ledger row COX.COUPLE.01.

**2026-06-21 (do the discrete-kernel and dynamic rates translate? An exact number-field split  $\mathbb{Q}$  vs  $\mathbb{Q}(\sqrt{5})$ : v314)**

- **v314\_rate\_translation.py** (RATE.TRANSLATE.01, [E]/[C]). Answers, exactly, whether the discrete kernel (the affine- $E_8$  network, carrying the golden ratio) and the dynamic part (the seam transfer, carrying the recovery rate  $(2/3)^6$ ) translate. [E] FIELD SPLIT: the adjacency spectrum is the rational part  $\{0, \pm 1, \pm 2\} \subset \mathbb{Q}$  plus the golden part  $\{\pm \phi, \pm 1/\phi\} \subset \mathbb{Q}(\sqrt{5})$ , while the transfer spectrum  $\{1, (2/3)^6, (1/3)^6\}$  is *entirely* rational; [E] the family/sheet rates DO translate (the rational angles  $2 \cos(\pi/2)=0 = |\mathbb{Z}_2|$  and  $2 \cos(\pi/3)=1 = N_{\text{fam}}$  build  $(2/3)^6 = (|\mathbb{Z}_2|/N_{\text{fam}})^{2N_{\text{fam}}} \in \mathbb{Q}$ , same field, same atoms), but [E] the carrier 5-fold (golden,  $\mathbb{Q}(\sqrt{5})$ ) has NO rational dynamic image – a field obstruction ( $2/3$  is neither an adjacency nor a lazy-update eigenvalue; heat-kernel  $t = \Delta/(2 - \phi) \approx 6.37$  not clean), so the carrier is *static-only*; [E] the only object holding both is the order-30 Coxeter element  $30 = g_{\text{car}}(2N_{\text{fam}}) = 5 \cdot 6$ , and the  $\phi$ -quasicrystal  $E_8 = H_4 + \phi H_4$  ( $240 = 2 \cdot 120$ ) carries the carrier but not the (rational, 3-fold) cusp-weight family dynamics. Cited in *origin\_theory*; ledger row RATE.TRANSLATE.01.

**2026-06-21 (the golden ratio made precise: the  $(2, 3, 5)$  atoms are the affine- $E_8$  network spectral angles,  $\phi = 2 \cos(\pi/5)$  the  $g_{\text{car}}=5$  signature: v313)**

- **v313\_golden\_atoms\_spectral.py** (GOLD.ATOMS.01, [E]/[C]). A follow-up to v312's observation, made exact. [E] the affine- $E_8$  adjacency characteristic polynomial factors as  $x(x^2-4)(x^2-1)(x^2-x-1)(x^2+x-1)$ , so the golden minimal polynomial  $x^2-x-1$  divides it and  $\phi = 2 \cos(\pi/5) = (1+\sqrt{5})/2$  is an *exact* eigenvalue; [E] the non-trivial eigenvalues are  $2 \cos(\pi/k)$  for exactly  $k \in \{2, 3, 5\} = \{|\mathbb{Z}_2|, N_{\text{fam}}, g_{\text{car}}\}$  ( $2 \cos(\pi/2)=0$ ,  $2 \cos(\pi/3)=1$ ,  $2 \cos(\pi/5)=\phi$ ,  $2 \cos(2\pi/5)=1/\phi$ ), so the golden ratio is *specifically* the  $g_{\text{car}}=5$  (5-fold) signature and  $g_{\text{car}}=5$  gains a new spectral witness distinct from Pascal/Riemann–Roch (raising the v305 multiplicity); [E] the  $(2, 3, 5)$  spherical excess  $1/2+1/3+1/5 = 31/30 > 1$  (the condition selecting the finite  $2I \Rightarrow E_8$ ) ties the v63/v76 gap-margin numerator  $31 = 2^{g_{\text{car}}} - 1 = 248/8 = 1+h(E_8)$  to the Coxeter  $30 = h(E_8)$  — a unification of already-anchored quantities, NOT new evidence (honest anti-numerology, v305); [E] honest null:  $\phi$  is a structural lattice/network quantity, not a phenomenological readout. [C] direction:  $\phi$  is the  $E_8$ -quasicrystal (Elser–Sloane cut-and-project) ratio, a candidate aperiodic substrate for the v312 question. Cited in *origin\_theory*; ledger row GOLD.ATOMS.01.

**2026-06-21 (TOE-roadmap solver round: the SEAM.EQUIV.01 keystone certificate, the modular bedrock, the carrier→SM closure, the gap-decoupled measure, and the sharpened hypergraph substrate: v308–v312)**

- **v308\_seam\_equiv\_chain.py** (SEAM.EQUIV.CHAIN.01, [E]/[O]). TOE attack point 1: the keystone “the raw seam-Calderón net is the holomorphic  $c=8$   $(E_8)_1$  net” assembled into ONE well-typed logical chain  $\text{gap}>0 \rightarrow \text{invertible/SRE (Kitaev)} \rightarrow \text{holomorphic } c=8 \text{ (Müger/KLM)} \rightarrow (E_8)_1 \text{ (Conway–Sloane)}$ , with the exact discriminators  $|\det \text{Cartan}(E_8)|=1$  vs  $|\det \text{Cartan}(D_8=SO(16))|=4$ , the carrier tower  $4 \cdot 4=16 \rightarrow 4 \rightarrow 1$  and  $c=g_{\text{car}}+N_{\text{fam}}=8$  all machine-checked; the input gap is the derived recovery gap  $6 \ln(3/2)>0$ . The only undischarged premise is the cited OS step — an assembly/reduction certificate (a Lean target), not an end-to-end closure. Cited in `tfpt_research_contracts`; ledger row SEAM.EQUIV.CHAIN.01.
- **v309\_modular\_bedrock.py** (QGEO.MODHAM.01, [E]/[O]). TOE attack point 2: the bedrock  $\omega \circ \rho = \omega$  via an explicit modular Hamiltonian, non-circular — the four marks come from Gauss–Bonnet (v216) and the order-4 clock from cross-ratio 2 (v214), so  $\mu_4$  is derived; the quasi-free modular Hamiltonian  $K = \log((1-C)/C)$  round-trips to the seam operator and  $[\rho, K]=0 \Leftrightarrow \omega \circ \rho = \omega$  (a  $\mathbb{Z}_2$  profile breaks it). Reduces the bedrock to the Bisognano–Wichmann intrinsicity of the raw collar; not a closure of QGEO.SYM.01. Cited in `tfpt_research_contracts`; ledger row QGEO.MODHAM.01.
- **v310\_carrier\_sm\_anomaly.py** (CAR.SM.ANOM.01, [E]). TOE attack point 3: the carrier half-spinor  $16 = \Lambda^{\text{even}}(\mathbb{C}^5) = 1+10+5$  is exactly one anomaly-free Standard-Model generation ( $SU(5) \subset SO(10): 10+\bar{5}+1$ ), with  $\sum Y = \sum Y^3 = [SU(2)]^2 U(1) = [SU(3)]^2 U(1) = 0$  (exact) and the one-loop  $(b_1, b_2, b_3) = (\frac{41}{10}, -\frac{19}{6}, -7)$ ; a fourth family shifts  $b_1$ , so  $N_{\text{fam}}=3$  pins the RG. Closes the spectrum/anomaly/ $\beta$  sub-question; the interacting Epstein–Glaser  $S$ -matrix and the Yukawa sector stay [C]/[O]. Cited in `tfpt_research_contracts`; ledger row CAR.SM.ANOM.01.
- **v311\_gap\_decoupled\_measure.py** (QGAMB.CLUSTER.01, [E]/[C]/[O]). TOE attack point 4: the physical QG measure as a gap-decoupled, exponentially-clustering object — the transfer spectrum  $\{1, (2/3)^6, (1/3)^6\}$  gives geometric clustering at rate  $(2/3)^6$  (correlation length  $1/(6 \ln(3/2))$ , finite susceptibility  $729/665$ ), and the v76 margin  $6 \ln(3/2) - 31/(4\pi^2) \approx 1.648 > 0$  decouples the physical sector from the un-built ambient. QG.AMB.01 stays [O] (gap-decoupled, inert). Cited in `tfpt_research_contracts`; ledger row QGAMB.CLUSTER.01.
- **v312\_hypergraph\_substrate.py** (HYP.INJECT.01, [E]/[O]). TOE attack point 5: the hypergraph substrate sharpened — the affine- $E_8$  Kac marks are the Perron eigenvector ( $A m=2m$ ) and the subleading eigenvalue is exactly  $2 \cos(\pi/5) =$  the golden ratio, but the recovery rate  $(2/3)^6$  is *not* any eigenvalue and  $\varphi_0$  is not a rational edge fraction; so a full-structure rewrite must inject exactly two non-graph-spectral data — the cusp weight  $2/3=|\mathbb{Z}_2|/N_{\text{fam}}$  and the analytic seed  $\varphi_0$ . The open question is precisely delimited, not closed. Cited in `introduction`; ledger row HYP.INJECT.01.

**2026-06-21 (adversarial-review follow-ups: a structural anti-numerology firewall, an out-of-sample seed cross-validation, an operational data watchdog, and the Target-A holomorphy kill test: v305–v307)**

- **v305\_witness\_independence.py** (WITNESS.ECON.01, [E]). The *structural* anti-numerology firewall complementing the v100 phenomenological null test, against the sharpest self-deception worry (“the same integer counted many times as if independent”). Every headline integer ( $|\mu_4|=4$ ,  $g_{\text{car}}=5$ ,  $|\mathbb{Z}_2|=2$ ,  $N_{\text{fam}}=3$ ,  $\dim S^+=16$ ,  $\text{rank } E_8=8$ , 240, 248,  $|2I|=120$ ,  $h(E_8)=30$ ,  $|W(D_5)|=1920$ ,  $\Omega_{\text{adm}}=48$ , the gap exponent 6) is a documented image of the single anchor  $a=(1, 1, 2)$  with  $\pi$  the only transcendental primitive (13 atoms from 2 free primitives,  $6.5 \times$

compression made explicit), so the recurring  $2/3=|\mathbb{Z}_2|/N_{\text{fam}}$  and  $(2/3)^6$  are *one* ratio; the  $E_8$  spine is overdetermined ( $\geq 3$  structurally independent witnesses) while the pure-seed readouts are single-witness (scored jointly by v100, never multiplied); the negative control shows  $\alpha^{-1}$ 's “137” is foreign to the anchor family (an independent witness via the  $F_{U(1)}$  cubic, v3). Cited in the `tfpt_5_redteam` body; ledger row WITNESS.ECON.01. Python-only.

- **v306\_seed\_crossval.py (SEED.CROSSVAL.01, [E]).** A leave-one-out cross-validation turning the “many observables, one seed  $\varphi_0$ ” worry into a falsifiable out-of-sample test: back-solving  $\varphi_0$  from each of the five seed-grammar observables ( $\sin^2 \theta_{12}$ ,  $\sin^2 \theta_{13}$ ,  $\beta$ ,  $\Omega_b$ ,  $\lambda_C$ ; NuFIT 6.0 / ACT DR6 / Planck / PDG) and error-weighted-averaging reproduces the axiom seed  $\varphi_0 = 1/(6\pi) + 48c_3^4 = 0.0531720$  to 0.04%; leave-one-out predicts every held-out observable within  $3\sigma$  out of sample (most-tensioned =  $\sin^2 \theta_{13}$ ,  $\sim 2\sigma$ ); a data $\leftrightarrow$ formula shuffle explodes the  $\chi^2$  by  $> 100\times$ . Cited in `tfpt_5_redteam`; ledger row SEED.CROSSVAL.01. Python-only.
- **v307\_data\_watchdog.py (DATA.WATCHDOG.01, [E]).** The operational decision pipeline over the frozen registry (v84): it classifies all twenty registry entries PASS/WATCH/TENSION/KILL by live  $n_\sigma$  against the repo-documented current bests (CODATA 2022, NuFIT 6.0, ACT DR6, Planck 2018, PDG 2024, BK18). No decidable prediction is in KILL; the watch items are  $\alpha^{-1}$  ( $1.9\sigma$ ),  $s_{23}$  ( $1.8\sigma$ ) and the most-tensioned core prediction  $\sin^2 \theta_{13}$  ( $2.0\sigma$ , the honest current pressure point);  $\sin^2 \theta_{12} = 0.306747$  is compatible ( $0.02\sigma$ , the JUNO falsifier), the  $r \in [0.0033, 0.0048]$  band is below the BK18 limit and the kill, and  $w=-1$  is flagged as the DESI dark-energy watchdog. Cited in `origin_theory`; ledger row DATA.WATCHDOG.01. Python-only.
- **Target-A holomorphy kill test (freeze\_file.csv).** A new `seam_holomorphy` freeze row records the sharpened *physical* kill criterion for Target A: a raw quasi-free seam bulk showing topological ground-state degeneracy on the torus ( $|\det K| > 1$ , the same- $c$  rival  $SO(16)_1$ ) would break holomorphy and kill “seam net =  $(E_8)_1$ ” — since holomorphic  $c=8 \Leftrightarrow E_8$  is machine-proven (v83/v143) and  $\det K=1 \Leftrightarrow$  invertible/SRE (v237/v301/v302).

## 2026-06-19 (an infinite-derivative narrowing of the Stelle-ghost attack, the double-pushout figure, and the “constancy of $c$ ” framing of SEAM.EQUIV.01: v304)

- **v304\_idg\_nonlocal\_ghost.py (QGAMB.IDG.01, [C]/[O]).** A conditional, parameter-free narrowing of the red-team `rt_F/v278` Stelle-ghost attack on the perturbative gravity sector. [E] the local  $R+R^2/\text{Weyl}^2$  four-derivative propagator  $1/(p^2(p^2+M^2))$  carries a negative-residue massive spin-2 mode (the Stelle ghost) only as a *truncation artefact*; [E] for an *entire* graviton form factor  $a(p^2) = e^{p^2/M^2}$  (nowhere zero) the dressed propagator  $1/(p^2 a)$  has no extra pole (Tomboulis 1997; Biswas–Mazumdar–Siegel 2006), whereas any *polynomial* truncation has a real zero = the ghost; [C] the spectral-action cutoff is the seam KMS weight  $f(u) = e^{-u}$  (v259), so the un-truncated trace points to a nonlocal form factor with *no new parameter*. [O] It does *not* close QG.AMB.01: the trace regulator is not yet shown to resum to an entire  $a(\square)$ , and a ghost-free *perturbative* graviton is not the nonperturbative ambient measure. Cited in the red-team Target F body; ledger row QGAMB.IDG.01.
- **Double-pushout figure (tfpt\_1\_architecture\_e8).** A new shared TikZ macro (`TFPTdoublepushout` in `tex-artefacts/tfpt_figures.tex`) draws the commuting square showing that the order- $|\mu_4|=4$  glue is *one* operation in two categories: the lattice pushout  $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$  (v1) and the conformal-net / anyon-condensation extension  $(D_5)_1 \otimes (A_3)_1 \rtimes \mu_4 \cong (E_8)_1$  (v92/v281/v234/v277/v260). Expository only; no new claim.
- **Framing (introduction, tfpt\_research\_contracts).** SEAM.EQUIV.01 is now stated explicitly as the *one* irreducible structural postulate of the theory — the role the constancy of  $c$  plays in relativity (cf. v267). Prose sharpening; no status change.

2026-06-19 (the discrete→dynamic lens on **F\_transfer**: one shape, one seam-rate, v303)

- **v303\_fttransfer\_dynamics.py** asks whether the main-branch mechanism — a gapped, positivity-preserving relaxation to a *unique* attractor (the local-averaging update → the  $E_8$  marks, gap  $6\ln(3/2)$ ) — is also the shape of the four **F\_transfer** instances, or an unrelated bolt-on. It is the *same* shape: [E] **F\_pole** (Koide) *is* the seam dynamics (Möbius multiplier  $F'=(2/3)^6=\lambda_2$ , the seam-clock eigenvalue ⇒ a unique attractor at the seam rate, v82/v54); [C] **F\_Boltzmann** ( $\eta_B$ ) is a gapped positive contraction (washout  $\kappa_f \in (0,1)$ , H-theorem); [C] **F\_relic** (axion) freezes at an adiabatic fixed point; [E] **F\_QCD** ( $m_p/m_e$ ) is the 1-loop RG flow with the carrier  $b_3=-7$  (asymptotic freedom, Gaussian UV attractor). [O] Verdict: one shape underlies all four; **F\_pole** has the seam rate  $(2/3)^6$  *exactly*, the other three share only the shape with external rates (thermal/cosmological/RG). So **F\_transfer** is the readout end of the one discrete→dynamic principle, *not* a bolt-on; the theory’s simplicity is preserved because the external rates are honestly fenced (v187), not reduced to the seam. [E] suite v303 5/5, AUDIT OK.

2026-06-19 (the last lever: the seam mass gap *is* the derived Recovery gap  $6\ln(3/2)$ , v302)

- **v302\_seam\_gap.py** closes the chain on the single statement v301 had left: “the quasi-free seam bulk is gapped”. That gap is *not* a free input — it is the established *Recovery gap* of the seam transfer operator, derived from the carrier. [E] the frozen transfer spectrum is  $\text{spec}(T) = \{1, (2/3)^6, (1/3)^6\}$  (v160/v162), a simple Perron eigenvalue 1 separated from the rest; [E] the gap is  $\Delta = -\ln((2/3)^6) = 6\ln(3/2) = 2N_{\text{fam}}\ln(3/2) \approx 2.4328 > 0$  (multiplicative gap  $1 - (2/3)^6 = 665/729$ ;  $(2/3)^6$  the  $\mu_4$ -deck transfer eigenvalue forced by the carrier); [E] with the decoupling margin  $\Delta - 31/(4\pi^2) \approx 1.648 > 0$  (v76) it is robust, and Perron–Frobenius primitivity makes the leading mode simple; [C] by Osterwalder–Schrader / quasi-free clustering a transfer-operator gap  $\Delta > 0$  *is* a mass gap of the reconstructed Hamiltonian. [O] **Bottom line:** with v300/v301, **SEAM.EQUIV.01**’s entire residual is now a composition of *standard cited theorems* (OS/quasi-free clustering + Kitaev free-fermion invertibility + the v297 AQFT stack) over *established* TFPT facts (16 quasi-free Majoranas; transfer gap  $6\ln(3/2)$ ) — *no undischarged TFPT-internal assumption remains*. It stays [O] (not machine-proved end-to-end), but the open front is reduced to citable machinery over a spectral gap that is provably positive. [E] suite v302 5/5, AUDIT OK.

2026-06-19 (Route A’s last hypothesis discharged: the gapped quasi-free bulk is invertible by the free-fermion classification, v301)

- **v301\_route\_a\_invertible.py** attacks the single remaining *unconditional* gap of **SEAM.EQUIV.01**. Since v300 collapsed Route B into Route A’s rationality, that gap is Route A’s open hypothesis (v297 LIT-A): “the raw quasi-free seam bulk is invertible (SRE)”. A *gapped* free-fermion (Gaussian) 2+1D bulk cannot carry intrinsic topological order — free fermions give the integer (IQHE/Chern) phases, all *invertible*; intrinsic topological order needs interactions (Kitaev’s periodic table, class D). [E] the carrier is 16 Majoranas,  $c = g_{\text{car}} + N_{\text{fam}} = 8 = 16/2$  (v148); [E]  $SO(16)_1 \supset (E_8)_1$  ( $\dim \mathfrak{so}(16) = 120$  currents + 128 spinor = 248, v154); [C] so a gapped quasi-free bulk is invertible automatically; [E] the invariant  $|\det K| = \#\text{anyons}$  is 1 for  $E_8$  (invertible) vs 4 for the gauged  $D_8=SO(16)$  (anyons), which free fermions cannot produce. [O] LIT-A’s ‘invertible bulk’ now *follows* from ‘gapped 16-Majorana quasi-free bulk’ (v148/v160+the mass gap), so the residual changes from a topological-order statement to a spectral one. [O] **Bottom line:** with v300, the whole open content of **SEAM.EQUIV.01** reduces to the *single spectral statement* “the quasi-free seam bulk is gapped” — the topological-order obstruction (hidden anyons) is removed; the gap is not

manufactured. [E] suite v301 6/6, AUDIT OK.

## 2026-06-19 (the Flat-Away attack: a hard discrete obstruction + the pin derived from $(E_8)_1$ , v300)

- **v300\_flataway\_rigid.py** attacks the single external fact the three Flat-Away routes (v291/v293/v295) left open, on two fronts. *Harden:* the flat seam was only a *soft* minimum (“the spectral mismatch grows with  $\varepsilon$ ”); here the obstruction becomes a *discrete* on/off invariant. [E] the flat fingerprint —  $\text{spec}(|D_\theta|)$  is the integer ladder with every level  $n \geq 1$  of multiplicity exactly 2; [E] the resonant mode  $4k$  splits level  $2k$  by gap  $= g_k$  (sympy  $2 \times 2$   $[[2k, g/2], [g/2, 2k]] \Rightarrow 2k \pm g/2$ , + an  $O(g^2)$  common shift), validated against the full Toeplitz operator; [E] any  $g_k \neq 0$  lifts the resonant degeneracy ( $2 \rightarrow 1+1$ ) and changes the spectral *multiset*, so preserving the flat ladder *with* its mult-2 degeneracies forces every  $g_k = 0 \Rightarrow f = 0$  — strictly stronger than the “deviation grows” scan; [E] with the v295 convexity, flat is isolated analytically *and* discretely. *Derive the pin:* [C] the  $(E_8)_1$  character  $E_4/\eta^8 = q^{-1/3}(1 + 248q + 4124q^2 + \dots)$  has integer  $q$ -coefficients (248 at level 1;  $E_8$  even unimodular  $\Rightarrow$  integer weights), so by 2d Steklov conformal rigidity the integer ladder  $\Leftrightarrow$  flat seam, i.e. the external pin *is* the  $(E_8)_1$  rationality (Route A). [O] verdict: Flat-Away carries *zero* new open content beyond Route A — the two SEAM.EQUIV.01 routes converge on one rationality fact. [E] suite v300 6/6, AUDIT OK.

## 2026-06-19 (the hypergraph/Wolfram bridge hardened: the $(2, 3, r)$ seed reaches $E_8$ as the last finite point, v299)

- **v299\_hypergraph\_0d\_autonomous.py** hardens the Wolfram/hypergraph reading into a precisely typed bridge (builds on v219/v298): [E] the icosahedron is a 3-uniform hypergraph ( $|\text{Aut}(\text{graph})| = 120 = |I_h| = |A_5 \times \mathbb{Z}_2|$ ; the McKay group is the  $SU(2)$  binary lift  $2I$ , a *different* order-120 group whose McKay graph is affine  $E_8$ ; 30 edges  $= h(E_8)$ ); [E] the  $1 \rightarrow 4$  subdivision rewrite is equivariant (all 120 symmetries at every scale); [E] the  $2I$  fusion tensor is a weighted 3-uniform hypergraph whose 2-dim slice is the affine  $E_8$  McKay graph; [E] Smith ( $\rho \leq 2 \Leftrightarrow$  affine ADE) + Collatz–Wielandt ( $\rho \leq 2 \Leftrightarrow$  a local witness  $Am \leq 2m$ ); [E]\* the autonomous rule (diffusion-generated witness + *local* gate  $Am \leq 2m$ , Perron only as audit) runs  $E_6 \rightarrow E_7 \rightarrow E_8 \rightarrow \hat{E}_8$ ; [E] the negative controls  $\rho(A_n) = 2 \cos \frac{\pi}{n+1}$  and  $\rho(D_n) = 2 \cos \frac{\pi}{2n-2}$  are  $< 2$  for *all*  $n$  (closed form), so neither family ever reaches  $E_8$ . Honest type: [O] the  $(2, 3, \cdot)$  seed is the single irreducible input = P2 — *the rule reaches  $E_8$  as the last finite point on this seed; it is not a derivation of P2.* [E] build green, AUDIT OK.

## 2026-06-18 ( $E_8$ as a network attractor — the Wolfram-program reading, v298)

- **v298\_e8\_network\_attractor.py** is an honest audit of the “TFPT as a hypergraph/rewrite system” idea, on the v219 McKay bedrock. [E] the affine- $E_8$  Kac marks  $(1, 2, 3, 4, 5, 6, 4, 2, 3)$  are the *unique dynamical attractor* of a minimal local-averaging update  $m \leftarrow (A+2I)/4m$  on the  $(2, 3, 5)$  McKay graph (converges from any positive start;  $Am = 2m$ , top eigenvalue 2); the spherical tower terminates at  $E_8$  and the cascade is nested-subgraph coarse-graining. [C] the Wolfram reading “ $E_8$  is the universal attractor of a minimal  $(2, 3, 5)$  network” holds at the Coxeter-skeleton level. [O] a minimal hypergraph *rewrite* reproducing the *full* structure stays open; honest negatives:  $\varphi_0^{\text{ret}}$  is the analytic seed  $\frac{4}{3}c_3 = 1/(6\pi)$ , not an edge fraction, and recovery is a 1-D Möbius fixed point, not a rewrite. [E] build green, AUDIT OK.

## 2026-06-18 ( $a_2$ closed form + Route A literature stack, v296–v297)

- **v296\_flataway\_a2\_closed.py** (FLATAWAY.A2.CLOSED.01) evaluates the exact  $a_2$  coefficient in *closed form*: the operator tails telescope geometrically ( $C_k + D_k e^{-4kt} = 4kt(1 - e^{-4kt})$ ) to  $W_k^{\text{tail}} = t(1 - e^{-4kt})/(8k(1 - e^{-t}))$ , leaving a finite  $(4k-1)$ -term middle; e.g.  $W_1(t) =$

$t(2te^t + e^{3t} + 3e^{2t} + e^t - 1)e^{-3t}/8$ . Manifestly positive, small- $t$  law  $W_k = t/2 + O(t^3)$  (the  $t^2$  term cancels). Reproduces v295 to machine precision.

- **v297\_route\_a\_literature\_stack.py** (SEAM.EQUIV.A.LIT.01) writes Route A’s one import as a citable stack: LIT-A (Kitaev; Freed–Hopkins)  $\rightarrow$  LIT-B (Müger; Kawahigashi–Longo–Müger)  $\rightarrow$  LIT-C (Conway–Sloane; Dong–Xu) compose into “invertible bulk  $\Rightarrow (E_8)_1$ ”, all legs established; the single open hypothesis (seam-bulk invertibility) coincides with Route B’s Flat-Away, and the stack is non-circular with Route B (the v286 firewall holds). [E] build green, AUDIT OK.

## 2026-06-18 (Flat-Away heat route made exact + Lean-formalised, v295 / FORM.FLATAWAY.01)

- **v295\_flataway\_a2\_exact.py** (FLATAWAY.A2.01) upgrades the v292 positive-definiteness from a numerical scan to a *proof*: the first variation of  $\text{Tr } e^{-t\Lambda}$  vanishes (zero-mean off-mark curvature, Gauss-Bonnet), and since  $\Lambda \mapsto \text{Tr } e^{-t\Lambda}$  is convex (Klein/Peierls),  $\varepsilon=0$  is the *global* minimum — so  $\Delta \text{Tr} \geq 0$  for all deformations,  $= 0$  iff  $f = 0$ . The exact second-order coefficient is the divided-difference kernel of  $e^{-tx}$  (Duhamel), validated to rel. err  $< 10^{-6}$ .
- **Lean FORM.FLATAWAY.01** (SeamDeckClosure.lean Part 5): the formal  $\Delta=0 \Leftrightarrow$  flat step — a positive-weighted sum of squares is positive-definite (`heatDeviation_eq_zero_iff`); built and axiom-checked (only `propext`, `Classical.choice`, `Quot.sound`). So the heat route’s positive-definiteness is now both analytically proved and machine-formalised; only the single external fact (the seam fixes  $a_2$ ) remains. [E] build green, AUDIT OK.

## 2026-06-18 (Flat-Away attacked on all three routes, v292–v294)

- **v292\_flataway\_heat\_reduction.py** (FLATAWAY.HEAT.01): the heat route, made precise. The heat-trace deviation  $\Delta \text{Tr}(e^{-t\Lambda}) = c(t)\varepsilon^2 + O(\varepsilon^4)$  is a *positive-definite* quadratic form in the off-mark curvature ( $c_k(t) > 0$  for every  $\mathbb{Z}_4$  mode, positive on all combinations), leading invariant  $\|f\|_{L^2}$  — so Flat-Away reduces to the single fact “the seam fixes the  $a_2$  heat coefficient”.
- **v293\_flataway\_spectral\_hessian.py** (FLATAWAY.SPEC.01): the spectral route, upgraded from one direction to the *full* smooth- $\mathbb{Z}_4$  space. Mode  $4k$  splits the degenerate Steklov level  $|2k|$  at first order; the Hessian of the spectral mismatch over modes  $\{4, 8, 12\}$  is positive-definite (eigenvalues  $\{0.497, 0.500, 0.503\}$ ), so the flat seam is a strict isolated minimum — the “only one direction” objection is answered.
- **v294\_flataway\_rp\_energy.py** (FLATAWAY.ENERGY.01): the variational route. With the rigid mark divisor (4 cone angles  $\pi$ , Gauss-Bonnet  $4\pi$ ), Troyanov uniqueness gives a unique flat cone metric, and the curvature energy  $\int f^2$  is strictly convex (Hessian  $2\pi I$ ) with a unique minimiser — Flat-Away reduces to “RP+gap realises the energy-minimiser”. The three routes converge on one selection principle for the flat metric; closing any one closes Route B. [E] build green, AUDIT OK.

## 2026-06-18 (Route-B red-team: the smooth $\mathbb{Z}_4$ adversary + Flat-Away as a mini-theorem, v290–v291)

- **v290\_z4\_smooth\_curvature\_adversary.py** (SEAM.ADVERSARY.01) is the honest red-team that  $\mathbb{Z}_4$  block-diagonality is *not* mark-locality: a smooth  $\mathbb{Z}_4$  off-mark curvature  $f = \varepsilon \cos(4\theta)$  [E] passes the v288 commutator ( $\|[\rho, \Lambda]\| = 0$ , same as mark-local) yet [E] shifts the DtN spectrum ( $\max |\Delta\lambda| = 0.155$ ) and the heat trace ( $\Delta = 0.019$ ). So the commutator is necessary, not sufficient; the modulus is excluded by the spectral data, not  $[\rho, \Lambda]=0$  — which is exactly why Flat-Away cannot be read off the commutator.

- **v291\_flataway\_contract.py** (FLATAWAY.RP.01) states Flat-Away as its own named mini-theorem and types three independent proof routes: [E] spectral-rigidity and [E] heat-kernel (the flat seam is the *unique* spectral/heat match of the smooth  $\mathbb{Z}_4$  family — deviation zero only at  $\varepsilon=0$ ) and [C] RP-energy/Troyanov (constant-curvature minimiser). Closing any one closes Route B. [E] build green, AUDIT OK.

## 2026-06-18 (Route B sharpened: raw mark-locality decomposed + reviewer firewalls, v289 / v287–v288)

- **v289\_marklocal\_raw.py** (MARKLOCAL.RAW.01) decomposes the one remaining Route-B residual (‘why is the raw seam sub-principal term mark-local?’) into a 5-lemma chain and isolates the *single* open analytic lemma: [C] L1 conic parametrix (b-calculus/Melrose, imported); [O] L2 **Flat-Away** — RP+gap+4 marks  $\Rightarrow$  curvature vanishes away from the marks; [E] L3 orbit-source (v195/v216), L4 Fourier-support on  $4\mathbb{Z}$  (v264/v288), L5 full-operator  $[\rho, \Lambda]=0$  (v288). 4/5 discharged — Route B reduces to exactly L2.
- **v288 reviewer firewall**: a SCOPE FIREWALL check makes the exact-label split explicit (Fourier lemma Formal/Exact; numerical test Numerical Exact; transfer to the raw conic DtN Conditional, v264; mark-locality from the RP seam Open Selection, v289) — so it can never be read as ‘full- $L^2$  QGEO proved’.
- **v287 literature matrix**: L3 (‘invertible bulk  $\Rightarrow$  single-sector’) is typed as a composition of citable results (LIT-A Kitaev/Freed–Hopkins, LIT-B Muger/Kawahigashi–Longo–Muger, LIT-C Kawahigashi–Longo) conditional on one open hypothesis — seam-bulk invertibility, which coincides with Route B. [E] build green, AUDIT OK.

## 2026-06-18 (closing sprint: the Seam Equivalence Theorem named + both routes attacked, v286–v288)

- **v286\_seam\_equivalence\_contract.py** (SEAM.EQUIV.01) names the one open theorem — *the raw RP seam state is the holomorphic  $(E_8)_1$  net at  $\tau=i$*  — and concentrates the whole open structure on it: four objects, six arrows (2 closed, 1 conditional v282, 1 open Gral), an **import firewall** proving the two routes (v276/v280/v284 vs v277/v281/v285) never import each other (no ‘ $E_8$  into the geometry and back’ circularity), and an exact-label taxonomy. Writes `seam_equivalence.json`.
- **v287\_free\_rp\_bulk\_to\_holomorphic\_boundary.py** (SEAM.EQUIV.A01): Route A (AQFT) dependency checker — the 5-lemma chain RP+gap+chirality+Gaussian+invertible  $\Rightarrow \mu=1 \Rightarrow$  holomorphic  $\Rightarrow (E_8)_1$  is logically complete *modulo one* standard theorem: *invertible (SRE) Gaussian bulk  $\Rightarrow$  single-sector boundary* (the AQFT form of  $\det K=1$ ). 4/5 discharged.
- **v288\_full\_l2\_subprincipal\_z4.py** (SEAM.EQUIV.B01): Route B (DtN) — *proves* the full- $L^2$  lift of the sub-principal  $\mathbb{Z}_4$  block-diagonality (a mark-sourced curvature has Fourier support only on  $m \equiv 0 \pmod{4}$ , so  $\Lambda = |n| + M_f$  commutes with  $\rho = \text{diag}(i^n)$  on the full operator,  $\|[\rho, \Lambda]\| \sim 2 \times 10^{-15}$ ; generic curvature  $\rightarrow 4.44$ ), lifting v201/v284 to  $L^2$ . The residual shrinks to one sharper question: why is the raw seam sub-principal term mark-local? [E] build green, AUDIT OK.

## 2026-06-18 (the two proof routes decomposed: Route (i) and Route (ii) into lemma chains, v284/v285)

- **v284\_route\_i\_rp\_uniqueness.py** (QGEO.ROUTEI.01) turns Route (i) (RP-state uniqueness) into a 6-lemma chain: [E] 5/6 lemmas are already discharged — L1 RP  $\rightarrow$  DtN (v194), L2 marks  $\rightarrow$  pillowcase (v195/v216/v214), L3 flat Steklov  $= \sqrt{-\Delta_{\text{flat}}}$  (verified here), L4 covariance from DtN (v258), L5  $\mu_4$ -invariance (v276/Lean/v280); [C] Troyanov ( $\chi_{\text{orb}}=0 \Rightarrow$  unique flat

$\tau=i$ ) reduces the residual to one lemma. [O] the one open lemma: *the raw seam is the constant-curvature geometric state*.

- **v285\_route\_ii\_seam\_condensation.py** (QGAMB.ROUTEII.01) turns Route (ii) (seam-net condensation) into a 4-lemma chain: [E]  $3/4$  discharged (no abelian sector  $\Leftrightarrow \det K=1 \Leftrightarrow$  holomorphic  $\Leftrightarrow E_8$ ; the tower  $16 \rightarrow 4 \rightarrow 1$ ), and **its open lemma coincides with Route (i)’s** — both are “the raw seam is the  $(E_8)_1$ -at- $\tau=i$  object” (v282), so proving *either* route closes *both*. [O] the unified open lemma: a canonical equivalence between the raw seam KMS/DtN state and the holomorphic  $(E_8)_1$  net at  $\tau=i$ . Build green, AUDIT OK.

## 2026-06-18 (self-investigation round: pillowcase DtN, modular data, the E8-at- $\tau=i$ unification, live scorecard, v280–v283)

- **v280\_pillowcase\_steklov.py** (QGEO.STEKLOV.01) runs the QGEO chain on the actual flat  $\tau=i$  pillowcase orbifold: the order-4 deck permutes the 4 cone points (2 fixed + 1 swap) and the geometric chain  $[\rho, \Delta]=0 \Rightarrow [\rho, H]=0 \Rightarrow [\rho, C]=0 \Rightarrow \omega \circ \rho = \omega$  is numerically exact — route-(i) evidence for QGEO.OBLIG.01; the premise stays [O].
- **v281\_holomorphy\_modular\_data.py** (QGAMB.MODULAR.01): #anyons =  $|\det \text{Gram}|$  ( $E_8 \rightarrow 1$ ,  $SO(16) \rightarrow 4$ ), the anyon-condensation tower  $16 \rightarrow 4 \rightarrow 1$ , Gauss–Milgram  $\rightarrow c=8$ ; holomorphy ( $\det K=1$ ) is the single Tier-B discriminator.
- **v282\_e8\_tau\_i\_unification.py** (QGAMB.UNIFY.01): the candidate *simplification* —  $\chi_{E_8}=j^{1/3}$  gives  $\chi_{E_8}(i)=1728^{1/3}=12$ , so  $\tau=i$  is both the order-4 CM point (QGEO) and the  $(E_8)_1$  modulus (QG.AMB): the two obligations are two faces of one object (the seam is  $(E_8)_1$  at  $\tau=i$ ), **obligation count**  $2 \rightarrow 1$ .
- **v283\_live\_killtest\_scorecard.py** (PRED.LIVE.01): the recent round’s computable predictions vs current data — all consistent  $< 2\sigma$ , with two sharp near-term kill tests ( $\Sigma m_\nu$  floor 0.0586 eV; proton decay  $1.55 \times 10^{35}$  yr).
- Wolfram 275/275 (v281/v282 exact-mirrored); build green, AUDIT OK.

## 2026-06-18 (the Gräl: QGEO.SYM.01 as one precise lemma + the all-orders step Lean-formalised, v279/FORM.QGEO.03)

- **v279\_qgeo\_obligation\_lemma.py** (QGEO.OBLIG.01) writes the bedrock as **ONE precise constructive-QFT lemma**. Given the premise *the raw seam carries the flat  $\tau=i$  pillowcase metric* (Trojanov-unique,  $\Lambda_\Sigma = \sqrt{-\Delta_{\text{flat}}}$ ), then  $\omega_\Sigma \circ \rho = \omega_\Sigma \rightarrow$  mark-locality  $\rightarrow E_8$ . [E] the reduction DAG from ‘free RP seam’ has exactly *one* open leaf — ‘the raw seam state is the flat state’; [E] the geometric side is fixed (flat DtN diagonal,  $\rho$  commutes to all orders), so the obligation is a state-identification, with [C] two proof routes (RP-state uniqueness, or Trojanov + OS).
- **FORM.QGEO.03: the all-orders step is now machine-proved in Lean**. `SeamDeckClosure.lean` gains `diagonal_commutatesClock` and `flat_all_orders_clock`: any spectral function  $g(\Delta_{\text{flat}})$  of the Fourier-diagonal flat Laplacian commutes with the carrier clock  $\rho = \text{diag}(i^n)$  — so the v276 all-orders closure is a Lean theorem (only `propext`, `Classical.choice`, `Quot.sound`; no `sorry`; full lake build green, 3356 jobs). The premise (the seam *is* flat) stays the [O] QGEO.SYM.01 input. Build green, AUDIT OK.

## 2026-06-18 (4D QFT: the $S_{\text{pert}} \rightarrow S_{\text{phys}}$ LSZ bridge + one-loop unitarity, v278)

- **v278\_lszy\_bridge\_unitarity.py** (QFT4D.SPERT.04) connects the perturbative S-matrix to the physical asymptotic one and verifies one-loop unitarity. [E] LSZ bridge: the EG  $T$ -products of  $S_{\text{pert}}$  (v269) are the OS Wick-rotated correlators

(v240), so amputate+on-shell-residue gives  $S_{\text{pert}} \xrightarrow{\text{LSZ}} S_{\text{phys}}$ ; [C] the gap  $\Delta=6\log(3/2)>0$  gives Haag–Ruelle asymptotic states. [E] *one-loop optical theorem*: the bubble discontinuity  $\text{Im } B(s)/\pi = x_+ - x_- = \sqrt{1-4m^2/s}$  *exactly* = the two-body phase space, so  $2\text{Im } M = \sum \int d\Pi |M|^2$  (cutting rule) holds — the matter+gauge sector is unitary, with the cut turning on at the physical two-particle threshold. [O] the gravity sector’s Stelle ghost (v269/rt\_F) is non-unitary  $\rightarrow$  QG.AMB.01. Exact core mirrored in Wolfram (273/273). Build green, AUDIT OK.

**2026-06-18 (QG.AMB.01 Tier-B made concrete: the seam-Calderón  $\rightarrow (E_8)_1$  match, v277)**

- **v277\_seam\_calderon\_e8\_match.py (QGAMB.TIERB.01) reduces Tier-B to ONE bit.** [E] the full  $(E_8)_1$  matching target is computed and matches:  $c=8$ ,  $\det \text{Cartan}(E_8)=1$ , the holomorphic character  $E_4/\eta^8=j^{1/3}=q^{-1/3}(1+248q+\dots)$  with a single primary and exactly  $248=\dim E_8$  weight-1 currents (240 roots). [E] the discriminator is holomorphy — the  $c=8$  counterexample  $(D_8)=SO(16)_1$  has  $\det \text{Cartan}=4$  (non-holomorphic), so the one condition is  $|\det K|=1$ . [C] the seam side matches via v276 (flat  $\tau=i$ ) + v260 (Kummer/K3). [O] residual = the single holomorphy bit; reduced, not closed. Exact core mirrored in Wolfram (272/272). Build green, AUDIT OK.

**2026-06-18 (QGEO.SYM.01 narrowed: the flat pillowcase closes the commutator to all orders, v276)**

- **v276\_qgeo\_flat\_closes\_commutator.py (QGEO.SYM.03) collapses the bedrock to ONE geometric postulate.** The state-level residual the whole bedrock rests on is the operator equation  $[\rho, H]=0$ . [E] if  $H = f(\Delta_{\text{flat}})$  and the order-4 deck  $\rho (z \mapsto iz)$  is a flat- $\tau=i$  isometry, then  $[\rho, \Delta]=0 \Rightarrow [\rho, H]=0$  *exactly*, to all orders (verified on the discretised square torus:  $\rho^4=I$ ,  $[\rho, \Delta]=0$ ,  $[\rho, \sqrt{-\Delta}]=0$ ; generic non-isometry fails) — upgrading the leading-order v198/v201 to the full  $H$ . [C]  $\tau=i$  is forced by order-4 ( $j=1728$ ; hexagonal  $j=0$  is order-6, excluded). [O] QGEO.SYM.01 therefore collapses to the single statement “the raw seam IS the flat  $\tau=i$  pillowcase quasi-free state” — one human constructive-QFT proof, or an honest axiom. Narrows, does not close. Build green, AUDIT OK.

**2026-06-18 (Red Team Target F — adversarial audit of the v269–v275 round, rt\_F)**

- **redteam/rt\_F\_qft4d.py (REDTEAM.F.01) stress-tests the just-published round** (the perturbative  $S_{\text{pert}}$  layer, the PMNS Jarlskog, the  $\nu$ -scale, the over-determination, the QG.AMB.01 roadmap). SURVIVES (NARROWED): two attacks land and are folded in. (1) the gravity  $R^2/\text{Weyl}^2$  sub-sector is power-counting renormalizable but *non-unitary* (the Stelle ghost: the four-derivative propagator  $1/(p^2(p^2 + M^2))$  has a negative-residue massive spin-2 mode), so  $S_{\text{pert}}$  is unitary for the matter+gauge sector only — which is exactly why the non-perturbative QG.AMB.01 is the real frontier (caveat added to v269). (2) the 0.11% anchor over-determination is conditional on the  $\Lambda$ -branch readout (LAMBDA.BRANCH.01, [C]), the gravity route being unconditional (caveat added to v274). Added as Target F in `tfpt_5_redteam` + the red-team table; `run_redteam.py` green.

**2026-06-18 (Zenodo release 5.2 (rev 191) — the perturbative 4D-QFT layer + scale closure, v269–v275)**

- **Published to Zenodo.** Version 5.2 (rev 191) of the ten-PDF document set was published as a new version of the TFPT deposit — record 20743347, DOI 10.5281/zenodo.20743347. This release carries the perturbative 4D-QFT layer (the Epstein–Glaser  $S_{\text{pert}}$  skeleton v269, the

concrete one-loop quartic v271 and gauge  $\beta$ -coefficients v273), the complete complex PMNS matrix and leptonic CP strength v270, the honest absolute  $\nu$ -mass-scale typing v272, the over-determination of the single mass anchor v274, and the QG.AMB.01 obligation roadmap v275. Suite 273 modules (highest ID v275; v186/v226 skipped), Wolfram 271/271, run\_all.py green, AUDIT OK, npm build clean. The Zenodo description (zenodo\_description.html) and the public website were synced in the same release.

## 2026-06-18 (scale + frontier round: $\nu$ -mass scale, EG gauge running, over-determination, QG roadmap, v272–v275)

- **v272\_nu\_mass\_scale.py (FLAV.NUSCALE.01) types the absolute neutrino-mass scale honestly** (no fabricated  $\Sigma m_\nu$  — would contradict the v184 firewall). [E] the absolute scale reduces to *one* seesaw ratio  $m_3 = (y_\nu v_{EW})^2/M_R$  (one UV input, like  $v_{geo}$ , v153); [C] the observed  $m_3=0.050$  eV with the TFPT PS→GUT window  $M_R \in [4.2 \times 10^{13}, 2.4 \times 10^{15}]$  GeV (v249) needs  $y_\nu \in [0.26, 2.0]$  —  $O(1)$ , no tuning; [O]  $M_R$  is not a clean  $\bar{M}_{Pl}$  power, so the absolute value stays open; [X] the NO floor  $\Sigma m_\nu=0.0586$  eV is the cosmological kill test. Closes the PMNS bookkeeping (the last [O] is one seesaw ratio).
- **v273\_eg\_gauge\_running.py (QFT4D.SPERT.03) derives the SM one-loop  $b_i$  via the EG gauge self-energy.** [E] the gauge 2-point is the same scaling-degree-4 extension as the quartic (v271), giving  $(b_1, b_2, b_3) = (41/10, -19/6, -7)$  from the carrier/SM content (the same 41 as the carrier algebra, v159); [O] two-loop + all-order stay open. Exact  $b_i$  mirrored in Wolfram.
- **v274\_scale\_overdetermination.py (SCALE.OVERDET.01) shows the single mass anchor is over-determined.** [E]/[C] two independent routes to  $\bar{M}_{Pl}$  agree — gravity  $(8\pi G)^{-1/2} = 2.4353 \times 10^{18}$  GeV and cosmology (the compiler prediction  $\rho_\Lambda/\bar{M}_{Pl}^4 = (3/4\pi^2)e^{-2\alpha^{-1}}$  with  $\rho_\Lambda^{1/4} = 2.24$  meV) =  $2.438 \times 10^{18}$  GeV, to 0.11%; the seam being a conformal  $c=8$  CFT makes “no derivable absolute scale” a theorem, not a gap. Sharpens v153/v78.
- **v275\_qgamb\_roadmap.py (QGAMB.ROADMAP.01) gives the open gate QG.AMB.01 a module.** [E] Tier A gap decoupling ( $\Delta_{eff} = 1.648 > 0$ , v36/v76); [E]/[C] Tier B reduction to the holomorphic  $(E_8)_1$  boundary net (v77); [C] the perturbative layer in hand (v269/v271/v273); [O] the nonperturbative ambient measure stays the one structural frontier.
- Suite 271/271 Wolfram, run\_all green, AUDIT OK, build green.

## 2026-06-18 (4D QFT: a concrete Epstein–Glaser one-loop quartic renormalization, v271)

- **v271\_eg\_oneloop\_quartic.py (QFT4D.SPERT.02) takes the v269  $S_{pert}$  skeleton from “exists” to an actual EG number** (no path integral). [E] the first renormalization is the order-2 product  $T_2$ ; the massless propagator has scaling degree  $sd=2$  in  $d=4$ , the one-loop bubble has  $sd=4=d$ , so  $\omega=sd-d=0 \Rightarrow$  exactly one logarithmic local counterterm ( $C(\omega+d, d)=1$ , finite, no infinite tower); the loop factor  $\Omega_3/(2(2\pi)^4) = 1/(16\pi^2)$  exactly — the *same* factor that normalises the spectral-action  $a_4$  geometric quartic. [C] the  $\phi^4$  one-loop  $\beta_\lambda = 3\lambda^2/(16\pi^2)$  (symmetry  $3 = s, t, u$  channels), EG initial condition  $\lambda_\Phi(UV) = 1/(16\pi^2)$ . [O] order-2/one-loop only; the all-order EG construction and the nonperturbative measure QG.AMB.01 stay open. Exact core mirrored in Wolfram (270/270). [E] build green, AUDIT OK.

## 2026-06-18 (PMNS: the complete complex matrix + the leptonic Jarlskog CP strength, v270)

- **v270\_pmns\_jarlskog\_assembly.py (FLAV.PMNS.03) assembles the four TFPT channels** ( $\sin^2 \theta_{12} = \frac{1}{3} - \frac{\varphi_0^{\text{ret}}}{2}$ ,  $\sin^2 \theta_{23} = \frac{1}{2}$ ,  $\sin^2 \theta_{13} = \varphi_0^{\text{ret}} e^{-5/6}$ ,  $\delta=4\pi/3$ ) **into one exactly-unitary**

**complex PMNS matrix and derives the leptonic CP strength.** [E]  $U = R_{23}U_{13}(\delta)R_{12}$  is unitary; [C] the Jarlskog  $J_{\text{PMNS}} = \Im(U_{e1}U_{\mu 2}U_{e2}^*U_{\mu 1}^*) = -0.02965$  ( $J_{\text{max}} = 0.03424$ ) is a *derived* number, the leptonic analogue of the CKM  $J$  (v88); [C] data:  $J_{\text{max}}$  vs NuFIT 5.2 0.0332 ( $\sim 3\%$ ),  $J$  vs best fit  $-0.026$ , the negative CP-violating sign reproduced. [O]  $\delta$  is the  $\mu_6$ /triality phase (assembled, not from  $M_\nu/D_F$ ) and the absolute  $\nu$ -mass scale stays open — closes the CP-*strength* readout, not the phase origin nor the scale. [E] build green, AUDIT OK.

## 2026-06-18 (4D QFT: the perturbative S-matrix as an Epstein–Glaser / pAQFT skeleton, v269)

- **v269\_spert\_paqft\_skeleton.py** (QFT4D.SPERT.01) **builds the honest middle layer of the three-layer S-matrix** (the first concrete 4D-QFT step beyond the spectral-action contract), *not* a nonperturbative closure. [E] three distinct layers:  $S_{\text{top}}$  (2d DHR braiding  $|M|=1$ , statistics, v243),  $S_{\text{pert}}$  (4d Epstein–Glaser/BV S-matrix of the spectral action),  $S_{\text{phys}}$  (LSZ on the OS Wightman functions, v240). [E] the spectral-action  $a_4$  interaction is power-counting renormalizable (all operators  $\dim \leq 4$ ; 7 marginal + 3 relevant, a finite counterterm basis), so [C] by the Epstein–Glaser theorem the perturbative S-matrix exists order by order (causal factorisation + finite local extension; no path integral, no UV divergences). [C] the gap  $\Delta = 6 \log(3/2)$  gives an IR-safe adiabatic limit ( $S_{\text{pert}} \rightarrow S_{\text{phys}}$  via LSZ). [O] perturbative only — the ambient measure QG.AMB.01 stays open, the canonical 4d reading remains boundary-only (v265). [E] build green, AUDIT OK.

## 2026-06-18 (PMNS: the reactor-angle exponent closed as the carrier hypercharge trace, v268)

- **v268\_theta13\_carrier\_trace.py** (FLAV.TH13.01) **closes the  $\theta_{13}$  exponent origin.** The value  $\sin^2 \theta_{13} = \varphi_0^{\text{ret}} e^{-5/6} = 0.0231$  was already checked (v16); now the exponent is exact: [E]  $\gamma = 5/6 = \text{tr}_E Y^2$ , the carrier hypercharge-squared trace ( $Y = \text{diag}(-\frac{1}{3}, -\frac{1}{3}, -\frac{1}{3}, \frac{1}{2}, \frac{1}{2})$ , roots of  $6Y^2 - Y - 1 = 0$ ; complement  $\frac{1}{6}$  the neutrino ratio). [E] own channel: the  $\mu\tau$ -breaking  $\varepsilon = \frac{3}{4}\varphi_0^{\text{ret}}$  induces only  $\sim 1.6 \times 10^{-3} \ll 0.023$ , so  $\theta_{13}$  is a separate carrier-trace channel — correcting the tempting (and wrong) “ $\theta_{13}$  from  $M_\nu$ ” route flagged by the new script atlas. [O] the CP magnitude and the absolute  $\nu$ -mass scale remain the open PMNS items. Exact core mirrored in Wolfram. [E] build green, AUDIT OK.

## 2026-06-18 (Infrastructure: a generated causal “script atlas” so the suite is navigable at a glance)

- **verification/make\_script\_atlas.py** generates **verification/script\_atlas.md** — a single grouped, dependency-aware view fusing the existing single sources (**status\_ledger.csv** for claim→status→dependencies→supersedes, **script\_registry.csv/script\_clusters.csv** for the theme grouping, **docs\_map.csv** for where each script is cited). It lists every registered script by cluster with its claim(s), four-class marker, one-liner, resolved dependencies and citing papers, plus a *supersede map* (which old claim each new one replaces) and a dependency/most-depended-on overview — the antidote to duplicating work or forgetting superseded results. No new data, a re-projection only; wired into **bash build.sh gen**. [E] build green, AUDIT OK.

## 2026-06-18 (The bedrock in its minimal-axiom form: a rigidity theorem for QGEO.SYM.01, v267)

- **v267\_qgeo\_rigidity\_minimal\_axiom.py** (QGEO.SYM.02) **sharpens the one bedrock postulate to its irreducible kernel** (it does *not* close it). [E] the *single* bare statement “the seam admits the carrier order-4 clock as a conformal symmetry permuting its four marks”

( $= \omega \circ \rho = \omega$ ) forces *everything*: an order-4 Möbius orbit  $\{a, ia, -a, -ia\}$  has cross-ratio 2 independent of  $a$ ; cross-ratio  $2 \Leftrightarrow j=1728$  (only  $\lambda \in \{-1, \frac{1}{2}, 2\}$ )  $\Leftrightarrow$  the order-4 CM automorphism on  $y^2=x^3-x$ ; and the unique flat pillowcase metric, mark-locality,  $[\rho, \Lambda]=0$  and  $\omega \circ \rho = \omega$  all follow (v214/v201/v264/v198). [E] neg controls: generic config  $j \neq 1728$  ( $\mathbb{Z}_2$  only), hexagonal  $j=0$  ( $\mathbb{Z}_6$ , order 6) — only the square (order-4) realises the  $\mu_4$  clock. [C] second, independent reason:  $\{1, i, -1, -i\}$  over  $\mathbb{Q}$  with  $j=1728$  (CM by  $\mathbb{Z}[i]$ ), Belyi/dessins Galois-rigidity. [O] the bare order-4 symmetry stays the one foundational postulate (like  $c=\text{const}$ ) — the *reduction* is closed (axiom  $\Rightarrow$  everything), the axiom is not. [E] build green, AUDIT OK.

**2026-06-18 (Branch B decided: the explicit PS threshold model + proton-decay window — minimal content killed,  $+(15, 1, 1)$  survives, Hyper-K decisive, v266)**

- **v266\_ps\_threshold\_proton.py** (PS.PROTON.02/PS.THRESHOLD.02) decides branch B of the v265 fork. The explicit two-step Pati–Salam thresholds give  $d=6$   $\tau_p(p \rightarrow e^+ \pi^0)$  (writes `ps_threshold_proton.json`): [X] the minimal 16-Higgs content ( $M_{GUT} \sim 2.4 \times 10^{15}$ ) gives  $\tau_p \sim 4.4 \times 10^{33}$  yr < Super-K and is *killed* even at the high hadronic band; [C] the  $E_8$ -allowed  $(15, 1, 1)$  [the single 45] raises  $M_{GUT} \sim 5.9 \times 10^{15} \Rightarrow \tau_p \sim 1.6 \times 10^{35}$  yr, proton-safe *now*; [X] but that sits at the Hyper-K reach ( $\sim 1.4 \times 10^{35}$  yr) — a *dated* kill test decisive within the decade. So branch B is promoted from “well-mannered candidate” to *content-selected* (*must be*  $+(15, 1, 1)$ ), *Hyper-K-decisive UV branch*; [O] structurally marginal (only one 45 in the hull, v247) with an  $O(3)$   $\tau_p$  band. Branch A (boundary-only) is unchanged and canonical. [E] build green, AUDIT OK.

**2026-06-18 (The 4D-QFT fork frozen as a branch policy: boundary-only canonical, Pati–Salam a falsifiable UV branch, SM-only GUT killed, v265)**

- **v265\_qft4d\_fork\_freeze.py** (QFT4D.FORK.01) turns the 4D-QFT fork from an open ambiguity into a frozen decision tree (`qft4d_fork_freeze.json`), no new physics. **A (canonical)**: boundary-only is the default 4D reading — a 4D-GUT is *not claimed by default* ( $E_8$  is the audit hull). [X] SM-only 4D-GUT unification is killed (1-loop: at the  $\alpha_1=\alpha_2$  scale  $\alpha_3^{-1}$  off by  $\sim 5.6$ , spread  $\sim 8.8$  at  $2 \times 10^{16}$  GeV, re-deriving v246) — the canonical expectation, not a failure. **B (conditional)**: carrier-native Pati–Salam is a falsifiable UV branch only with the  $E_8$ -allowed content  $\{1, 10, 16, 45\}$  (no **126**, v247),  $\sin^2 \theta_W = 3/8$  (v248), the  $O(1)$   $\kappa = M_{PS}/M_s$  band (v249/v253), a threshold model and a proton-decay kill test ([X], no fake  $\tau_p$  window; carrier gauging stays the contract). [E] a prose guard (QFT4D.NO\_OVERCLAIM.01) forbids bare 4D over-claims and requires the scoped policy wording. New ledger rows QFT4D.BOUNDARY.DEFAULT.01, QFT4D.SMONLYGUT.01, PS.UVBRANCH.01, PS.THRESHOLD.01, PS.PROTON.KILL.01. A fork-policy box added to `tfpt_research_contracts`; [E] build green, AUDIT OK.

**2026-06-18 (Three open-residual advances: the  $F_{QCD}$  solver,  $M_\nu$  from  $D_F$ , and the raw-seam  $\Rightarrow$  mark-locality reduction, v262–v264)**

- **v262\_fqcd\_mp\_me.py** (FR.MPME.02) implements the one contract-only  $F_{\text{transfer}}$  pipeline. A real two-loop  $\alpha_s$  solver runs  $\alpha_s(M_Z)=0.1179$  with  $n_f$  thresholds (carrier  $b_3 = -7$ , [E]) to  $\alpha_s(2 \text{ GeV}) \sim 0.30$  and  $\Lambda_{\overline{\text{MS}}}^{(3)} \sim 0.33 \text{ GeV}$ ; with the lattice bridge  $C_p$  (external  $O(1)$ ) and the closed  $m_e$  it reproduces  $m_p/m_e \sim 1824$  vs the measured 1836.15 within the  $\sim 7\%$  external budget. A [C] transfer reproduction (firewall: no compiler integer for 1836), advancing FR.MPME.01 ([O]) to FR.MPME.02 ([C]).
- **v263\_mnu\_from\_df.py** (FLAV.PMNS.02) makes  $M_\nu$  a seesaw operator of  $D_F$ . [E]  $M_\nu = -m_D M_R^{-1} m_D^\top$  is symmetric, seesaw-suppressed and (for  $\mu\tau$ -symmetric inputs)  $\mu\tau$ -symmetric;

[C] it reproduces  $\theta_{23}=45^\circ$  and  $\sin^2 \theta_{12}=\frac{1}{3}-\frac{\varphi_0^{\text{ret}}}{2}$  (v9) under the seam misalignment; [O]  $\theta_{13}=0$  in the leading texture, the CP *magnitude* ( $240^\circ$ ) is the  $\mu_6$ /triality phase (v231/v233), and  $M_R$  is the free seesaw scale. An operator advance, not a closure of full PMNS dynamics.

- **v264\_raw\_seam\_marklocal.py (QGEO.ENERGY.03) sharpens the bedrock.** Composing v216+v201+v198 on the canonical pillowcase: the flat-pillowcase curvature lives only at the 4 marks, so the DtN sub-principal symbol is  $\mathbb{Z}_4$ -invariant (Fourier on  $4\mathbb{Z}$ , FFT-verified), block-diagonal, and  $[\rho, \Lambda]=0 \Rightarrow \omega \circ \rho=\omega$  on this metric. So the whole bedrock collapses to the single metric statement “the raw RP seam carries the uniformising flat pillowcase metric” (=QGEO.SYM.01). A reduction/sharpening, [O] — it does *not* prove the seam must carry that metric (negative control: curvature off the  $\mu_4$  orbit breaks the chain). Paper citations added in `tfpt_4_frontier`, `tfpt_2_standard_model` and `tfpt_research_contracts`. [E] AUDIT OK, build green.

**2026-06-18 (Paper consistency pass: the closure ledger and the status/QFT sections updated for the Modular Spectral Closure, with an explicit “what is *not* closed” line)**

- **Five paper sections that predated the v238–v261 round are brought up to date** (additive, no status marker moved). The introduction closure-ledger table and the “five questions” item now carry a *boundary QFT (Modular Spectral Closure)* row — one relative object  $\text{TFPT}_{\text{QFT}} = (\mathcal{A}_\Sigma, \omega_\Sigma, \Delta_\Sigma, \rho, A_F, H_F, D_F, J, \gamma, S_{\text{rel}})$  closed modulo QGEO.SYM.01 (v258–v261) — plus an explicit list of what the closure does *not* reach: the full measured pole masses, the absolute QCD scales ( $m_p/m_e$ ), the complete PMNS dynamics, and the real 4D interacting QFT / ambient measure. The `tfpt_2` “ambient closure” section now states that the cutoff of its  $S_{\text{rel}}$  is the seam KMS weight ( $f_2/f_0=1$ , v259) and  $D_{\text{rel}}$  the covariance induction (v258); the `tfpt_2` “(4) Constructive QFT closure” section adds the relative-object pointer; `tfpt_4_frontier`’s full-quantum-gravity section records that the cutoff moment  $f_0$  is pinned by the seam state (v255/v259); and `origin_theory`’s honest-status keybox notes that the whole boundary QFT collapses onto the same QGEO.SYM.01 postulate (no sixth structural class). An editorial consistency pass so every status surface agrees with the ledger. [E] build green (10 ok, 0 failed), AUDIT OK.

**2026-06-18 (Consistency pass: the Modular Spectral Closure surfaced structurally across the website and the intro status card)**

- **The v258–v261 closure is now reflected on every “what is open” surface, not just the research-contracts paper.** The homepage open-gates section and compiler-pipeline diagram, the hostile-referee FAQ, the glossary  $G_{\text{net}}$  entry, the reviewer page and the verification residual-chain caption now state that the *entire* emergent-QFT layer (Dirac, cutoff, gauging, glue, orientability) collapses onto the *same* premise QGEO.SYM.01 via the Modular Spectral Closure — the boundary QFT is one relative object that adds *no new open item*, ambient QG separate. The introduction status card (“the structural residual is one condition”) carries the same sentence. No status marker moved; this is an editorial consistency pass so the public surfaces agree with the ledger and the paper. [E] build green, AUDIT OK.

**2026-06-17 (The Modular Spectral Closure: Dirac as covariance induction, the modular cutoff, the Kummer/K3 unification, and the assembly certificate, v258–v261)**

- **v258\_dirac\_covariance\_induction.py (PS.DIRAC.03) closes the “posited Yukawa vs derived operator” gap of v250/v252.** The finite Dirac operator  $D_F$  is the modular/covariance *induction* of the boundary seam quasi-free (KMS,  $\beta=1$ ) state: a Gaussian state is fixed by its covariance  $C$  (a positive contraction) with one-particle modular Hamiltonian

$H = \log((1-C)C^{-1})$  (Araki; Casini–Huerta), inverse to the KMS occupation  $C = (1+e^H)^{-1}$ . [E] the inversion identity holds exactly; [E]  $C_F$  is a positive contraction; [E] the induced operator is chirality-odd; [E] it is unitarily equivalent to the v252  $D_F$  (same mass spectrum); [E] the Majorana/seesaw  $m_D^2/M_R$  is an off-diagonal covariance entry. [C]  $C_F = P_F C_\Sigma P_F$ , so  $D_F$  is the Kasparov shadow  $[D_\Sigma] \hat{\otimes}_{(E_8)_1} [K_{\text{car}}]$  of the boundary geometry and the v18 Yukawas are the readout of  $C_\Sigma$ .

- **v259\_modular\_cutoff\_kappa.py (PS.SPECACT.02) fixes the last open spectral-action input.** The cutoff  $f$  is the seam’s own modular/KMS weight: by Tomita–Takesaki/Bisognano–Wichmann the seam is KMS at  $\beta=1$  (v239) and  $2\pi=1/(4c_3)$  (v58) fixes  $\beta$ , so the heat-kernel cutoff  $f(u) = e^{-u}$  is *derived*. [E] its moments give  $f_2/f_0 = f_4/f_2 = 1$  exactly; [E] a generic Gaussian cutoff gives  $\sqrt{\pi}/2 \neq 1$  (the choice is non-vacuous); [E] hence  $\kappa = \sqrt{(f_2/f_0) c_{PS}/c_{\text{grav}}} = \sqrt{c_{PS}/c_{\text{grav}}}$  loses its scheme factor. [C] the open [O] cutoff freedom of v253/v255 becomes a finite trace ratio over the 96-dim  $H_F$  (the RG  $\kappa \approx 1.3$  pins  $c_{PS}/c_{\text{grav}} \approx 1.7$ ).
- **v260\_k3\_kummer\_unification.py (ARCH.K3.01) unifies seam, carrier and lattice on one surface.** The pillowcase is the elliptic-involution quotient  $T_{\tau=i}^2/(z \mapsto -z)$  (v214); squaring it gives the Kummer/K3 surface, whose three facets are three TFPT objects. [E] the  $E_8$  Cartan is even/det 1/positive-definite; [E]  $H^2(K3, \mathbb{Z}) = U^3 \oplus E_8(-1)^2$  (rank 22, det  $-1$ , even, signature  $(3, 19)$ ); [E] the seam is  $T^2/(z \mapsto -z)$  with 4 marks (orbifold Euler char 0); [E] the carrier-16 are the  $|A[2]|=2^4=16$  Kummer nodes =  $\dim S^+$  (v197),  $16=4 \times 4$ ; [E] honest note — the rank-16 Kummer lattice is *not* the  $E_8(-1)^2$  summand. [C] so the v1 “glue  $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ ” is the lattice shadow of a single object;  $E_8$  appears both locally (du Val  $\mathbb{C}^2/2I$ , v232/v236) and globally ( $H^2$  summand). A unification, not a closure.
- **v261\_modular\_spectral\_closure.py (QFT.MSC.01) is the assembly capstone.** The whole QFT round (v238–v260) is certified as *one* relative object  $\text{TFPT}_{\text{QFT}} = (\mathcal{A}_\Sigma, \omega_\Sigma, \Delta_\Sigma, \rho, A_F, H_F, D_F, J, \gamma, S_{\text{rel}})$  reduced to *one* open premise. [E] the cross-consistency invariants agree:  $4 = [B:A] = |\mu_4| = 2\chi = |(\mathbb{Z}/2)^2|$  (index = marks = fixed points = glue order), one carrier-16 (spinor = lattice = triple), one gap  $6 \log(3/2)$  (OS = recovery = Lindblad); [E] the three closures re-derived; [E] the ten-component manifest present. [O] the QFT layer reduces to QGEO.SYM.01 (seam realisation), the ambient QG measure QG.AMB.01 kept separate. [C] the *Modular Spectral Universality* master statement: the unique RP holomorphic index-4 KMS seam state induces the 96-dim KO-6 triple whose inner fluctuations give the gauged Pati–Salam spectral action — the honest sense in which the boundary QFT is complete.

**2026-06-17 (Infrastructure: the changelog is now a public website page, generated from changelog.tex and audit-gated)**

- **The canonical changelog now also drives a public /changelog website page.** A new generator `verification/make_changelog_web.py` parses this file (dated \subsection\* entries, itemize items, and the inline markup \textbf/\emph/\texttt/\vref, the status markers [E]/[C]/[O]/[X], inline math and accents) into the typed data file `website/lib/changelog.ts`, rendered server-side (math via KaTeX) by `website/app/changelog/page.tsx + website/components/Changelog.tsx`. The generator is wired into `bash build.sh gen`; its freshness is enforced by `audit_sync.py` (section A.generated), so the page can never drift from this file – editing the changelog and re-running `gen` is the only supported path (the `.ts` mirror is never hand-edited). `/changelog` is linked from the navbar, footer and sitemap. The `tfpt-workflow`, `website-sync` and `sync-maps` cursor rules were updated to record the new single-source generation step. [E] `build (npm run build green, /changelog prerendered static)` and `bash build.sh audit (AUDIT OK)`.

**2026-06-17 (Orientability + Poincaré duality: the honest, literature-confirmed status — strict fail, weakened hold, v257)**

- **v257\_quasiorient\_modpd.py (PS.NCG.ORIENT.02) corrects the slightly too-generous v256 typing.** The published result (Cacic–Stephan arXiv:0902.2068; CCM07; SM-vacuum hep-th/0601192) is that the Chamseddine–Connes–Marcolli Standard-Model finite triple (KO-dimension 6, with three right-handed neutrinos — exactly the TFPT seesaw case) *genuinely fails* strict orientability and strict Poincaré duality, and instead satisfies the physically correct weakened versions. Verified on the explicit v252 triple: [E] strict orientability fails ( $\gamma \notin \text{span}\{\lambda(a)\rho(b)\}$ ); [E] *quasi-orientability* holds (the left/right  $A$ -irrep boxes are disjoint by chirality:  $L = H$  weak-doublet,  $R = \mathbb{C}$  singlet, so  $L^{\text{LR}}(\mathcal{H}^{\text{even}}, \mathcal{H}^{\text{odd}}) = \{0\}$ ); [E] strict Poincaré duality fails (KO-6 skew intersection form, corank 1, the equal- $L/R$ -neutrino case); [C] modified Poincaré duality holds (CCM), Stephan’s alternative triple (St06) restoring strict PD with identical physics. A known property of the NCG-SM model class, not a TFPT defect — no faked closure.

**2026-06-17 (Remaining NCG obligations discharged or honestly typed: inner fluctuations, spectral action, orientability/Poincaré, v254–v256)**

- **v254\_inner\_fluctuations.py (PS.NCG.FLUCT.01) closes v250 #5 + the no-junk obligation.** The inner fluctuations  $\Omega_D^1(A_F)$  of the finite Dirac operator are computed explicitly: [E] they are purely chirality-odd (scalars), and for the SM algebra sit *exactly* in the one Higgs doublet  $(1, 2)_{1/2}$  — no junk, no triplet, no  $(10, 1, 3)/\mathbf{126}$ . [E] the  $\sigma$  ( $\nu_R$  Majorana) direction is absent for the SM algebra but present for Pati–Salam (the CCvS beyond-first-order scalar), and [E]  $\sigma$  is an SM singlet  $(1, 1, 0)$  with  $B-L = 2$ . [C]  $\sigma = \text{scalaron}$ .
- **v255\_spectral\_action\_expansion.py (PS.SPECACT.01) closes the spectral-action expansion (v252 #11).** The Seeley–DeWitt expansion  $S = 2f_4\Lambda^4 a_0 + 2f_2\Lambda^2 a_2 + f_0 a_4$ : [E] structure ( $a_0$  cosmological,  $a_2$  Einstein–Hilbert+Higgs-mass,  $a_4$  Yang–Mills+Weyl<sup>2</sup> +  $R^2$ +Higgs-quartic,  $N = 96$ ); [E]  $b/a^2 = 1/3$  (top-dominated); [E]  $\sin^2 \theta_W = 3/8$ ; [E] the  $R^2$  spin-0 mode = the Starobinsky scalaron; [C] Higgs quartic  $\rightarrow \sim 125$  GeV with  $\sigma$ ; [C]  $\kappa = \sqrt{(f_2/f_0)c_{PS}/c_{\text{grav}}}$  explicit; [O] exact decimal needs the cutoff moments.
- **v256\_orientability\_poincare.py (PS.NCG.ORIENT.01) — honest status, no faked proof.** KO-dimension 6 forces the intersection form  $\text{Tr}(\gamma\pi(p)J\pi(q)J^{-1})$  to be *antisymmetric*: [E] skew form (verified), [E] rank 2, corank 1, null  $\sim (1, 1, -1)$  on the SM K-theory  $\mathbb{Z}^3$ ; [E] the orientability local condition  $[\gamma_F, \pi(a)] = 0$ . [C] the canonical hypercharge-faithful SM/PS triple satisfies orientability and Poincaré duality (van Suijlekom; CCM); [O] a self-contained machine proof needs the complete canonical bimodule.

**2026-06-17 (Full 96-dim finite spectral triple built + NCG axioms verified, and  $\kappa$  pinned to  $O(1)$ , v252–v253)**

- **v252\_full\_finite\_triple.py (PS.DIRAC.02) advances the Dirac contract v250 from a 7-obligation contract to a concrete object discharging 5 to [E].** The full 96-dim finite spectral triple  $(A_F, H_F, D_F, J, \gamma)$  —  $H_F$  the particle $\oplus$ antiparticle doubling of three  $SO(10)$  16-plets — is built explicitly and the NCG axioms are checked by direct computation: [E] reality ( $J^2=+1$ ,  $JD=DJ$ ), grading ( $\gamma^2=1$ ,  $[\gamma, a]=0$ ,  $J\gamma=-\gamma J$ ), KO-dimension 6, order-zero, the first-order condition (HOLDS for the SM algebra including the Majorana; VIOLATED for the Pati–Salam algebra exactly by the Majorana, localized to  $\{\nu_R, e_R\}$  — the CCvS beyond-first-order mechanism on the full triple), and the spectrum of  $D_F$  reproducing the fermion masses and the type-I seesaw  $m_{\text{light}}=m_D^2/M_R$ . [C] Poincaré duality; [O] orientability, no-junk, spectral-action expansion.

- **v253\_kappa\_scalaron\_ps.py** (PS.KAPPA.01) **discharges residual 6(b) of v249.**  $\kappa$  in  $M_{PS} = \kappa M_s$  ( $M_s = c_3^{7/2} \bar{M} \approx 3.06 \times 10^{13}$  GeV, exact) is pinned by an RG scan (PDG errors  $\times$  the two  $E_8$ -allowed contents) to a tight  $O(1)$  interval ( $\approx 1.3$ ). [C] heat-kernel origin:  $\kappa = \sqrt{(f_2/f_0) c_{PS}/c_{\text{grav}}}$  is a scale-free ratio of heat-kernel traces over the 96-dim  $H_F$  ( $\Lambda$  cancels). [X] the **126**-content drives  $\kappa$  out of the  $O(1)$  window (content-selective). [O] the exact decimal needs  $f_2/f_0$  fixed.

**2026-06-17 (Spectral triple, first step: the first-order condition is violated exactly by the Majorana/ $\sigma$  — the CCvS Pati–Salam mechanism, v251)**

- **v251\_first\_order\_sigma.py** (PS.DIRAC.01) **advances the Dirac contract v250 from a contract to an exhibited mechanism.** On a faithful lepton-block model ( $H = H_F \oplus H_F^c$ ,  $H_F = \mathbb{C}^3$ ), the real even structure is checked ( $J^2=+1$ ,  $\gamma^2=1$ ,  $D\gamma=-\gamma D$ , KO-dim-6 signs), order-zero holds, the first-order condition HOLDS for the Yukawa block, and is VIOLATED *exactly* by the Majorana ( $\nu_R \nu_R^c = \sigma$ ) block — the Chamseddine–Connes–van Suijlekom “beyond first order” mechanism by which the inner fluctuations promote the SM to Pati–Salam and generate  $\sigma$ . Honest correction to the review:  $\sigma$  does *not* rescue the first-order condition; its controlled violation *is* the mechanism. [C]  $\sigma = \text{scalaron}$  (v249); [O] the full 96-dim triple + spectral action.

**2026-06-17 (Carrier-native Pati–Salam program:  $E_8$  forbids the 126, PS algebra, unification kill-test, Dirac contract, v247–v250)**

- **Four scripts turning the gauge-unification finding into typed claims with negative controls.** **v247\_e8\_branching\_no126.py** (PS.E8BRANCH.01): the  $E_8$  hull branches under  $SO(10) \times SU(4)$  as  $(45, 1) + (1, 15) + (10, 6) + (16, 4) + (\bar{16}, \bar{4})$ , so the  $SO(10)$  content is exactly  $\{1, 10, 16, 45\}$  and the **126** is *absent* — the renormalisable  $126_H$  is algebraically forbidden and  $10+16+45$  is  $E_8$ -supplied (only one  $45 \Rightarrow$  proton-marginal). **v248\_ps\_algebra.py** (PS.ALGEBRA.01):  $A_F = \mathbb{H}_L \oplus \mathbb{H}_R \oplus M_4(\mathbb{C}) \rightarrow SU(2)_L \times SU(2)_R \times SU(4)_c$ , exact hypercharges  $Y = T_{3R} + (B-L)/2$ , matching  $\alpha_1^{-1} = \frac{3}{5}\alpha_{2R}^{-1} + \frac{2}{5}\alpha_4^{-1}$ ,  $\sin^2 \theta_W = 3/8$  (colour  $SU(4)_c \subset D_5$ , distinct from the family  $A_3$ ). **v249\_ps\_unification.py** (PS.RGTEST.01): the two-step  $PS \rightarrow SO(10)$  unification kill-test with negative controls (SM-only fails; 126 breaks the scalaron match; proton decay selects content),  $M_{PS}/M_s = 1.0\text{--}1.7$ . **v250\_dirac\_triple\_contract.py** (CONTRACT.QFT4D.DIRAC.01): the spectral-triple contract ( $H_F = 3 \times 16$ , the seven NCG axioms,  $\sigma = \text{scalaron}$ ) that would turn “carrier is gauged” from a fork into a theorem. The root-level **qft.txt** carries the full analysis.

**2026-06-17 (Honest data cross-check + emergent-QFT figures: the unification tension, v246)**

- **v246\_unification\_data\_crosscheck.py** (QFT4D.RGTEST.01) — **the first hard data confrontation of the spectral-action prediction.** TFPT’s gauge-charged content *is* the Standard Model (v159) and the theory adds no new states; the  $E_8$  cascade is an arithmetic spine (v5), not a threshold tower. So the run *is* the SM run: at one loop the couplings spread  $\sim 10^{13}\text{--}10^{17}$  GeV (residual  $\Delta\alpha^{-1} \approx 9$  at  $2 \times 10^{16}$  GeV), and at two loops (RK4) the minimal spread is  $\Delta\alpha^{-1} \approx 3.2$  at  $1.7 \times 10^{14}$  GeV — the three never meet. Typed [X]/[O]: with “no new state” there is *no* admissible threshold source, so the spectral-action 4d-GUT route is in genuine tension (same status as NCG-SM), likely falsified unless extended — never a confirmation, and *not* rescued by a cascade. Two new figures (**figures/qft\_skeleton.pdf**, **figures/qft\_unification.pdf**) added via **make\_figures.py** (PDF + website PNG) and embedded in **tfpt\_research\_contracts.tex**; a root-level **qft.txt** summarises the emergent-QFT skeleton, what each script proves, the data status and the open questions.

**2026-06-17 (Spectral-action matter content: the  $SO(10)$  16 = one anomaly-free SM generation, v245)**

- **v245\_spectral\_matter\_content.py (QFT4D.MATTER.01) discharges the computable core of the 4d contract v244.** The carrier half-spinor (the  $SO(10)$  16) is verified, in exact rational arithmetic, to be one Standard-Model generation:  $\mathbf{16} = SU(5)(\mathbf{10} + \bar{\mathbf{5}} + \mathbf{1})$ , the SM multiplet dimensions sum to  $16 = \dim S^+$  (15 Weyl fermions +  $\nu_R$ ), the electric charges  $Q = T_3 + Y$  are exactly the SM values, all four gauge anomalies cancel ( $\sum Y = \sum Y^3 = [SU(2)]^2 U(1) = [SU(3)]^2 U(1) = 0$ ), the spectral-action normalisation gives  $\sin^2 \theta_W = 3/8$  ( $g_3 = g_2 = \sqrt{5/3} g_1$ ), and the Higgs is the  $(1, 2)_{1/2}$  inner fluctuation. So the matter-content + tree-relation half of `CONTRACT.QFT4D.01` is now [E]; only the seam→Dirac realisation and the run-down couplings stay [O]. Wired into `run_all.py`, the ledger, the registry, cited in `tfpt_research_contracts.tex`, mirrored to the website.

**2026-06-17 (The QFT skeleton on the seam: thermal time, GNS/OS reconstruction, DHR sectors, the braiding S-matrix, and the 4d spectral-action contract, v239–v244)**

- **Six emergent-QFT skeleton scripts (v238).** `v239_kms_thermal_time.py` (DYN.KMS.01): the modular flow is KMS at  $\beta = 1$ , and  $2\pi = 1/(4c_3)$  fixes the modular temperature as  $T_H = c_3/M$  — thermal time = horizon time. `v240_gns_os_reconstruction.py` (DYN.GNS.01): GNS reconstructs the Hilbert space from  $(\mathcal{M}, \omega)$  and the OS Hamiltonian  $H_{OS} = -\log T = -L$  is positive with gap  $\Delta$  (the v64 mass gap). `v241_dhr_sectors.py` (DHR.SECTORS.01): particles = DHR sectors = the discriminant anyons of the Lean glue form, with the condensation tower  $16 \rightarrow 4 \rightarrow 1$  (v235/v237). `v242_spin_statistics_modular.py` (DHR.MODULAR.01): spin-statistics, the modular  $(S, T)$  data, Verlinde fusion, and Gauss–Milgram  $c = 8$ . `v243_haag_ruelle_braiding.py` (DHR.SMATRIX.01): the 2-particle S-matrix = the braiding monodromy (factorised, crossing), with the v238 mass gap supplying Haag–Ruelle asymptotic states. `v244_spectral_action_contract.py` (CONTRACT.QFT4D.01): the 4d Lagrangian via the Connes spectral action — the carrier half-spinor 16 is one generation, the SM gauge group sits in the carrier, gravity is v36; the matter Lagrangian is the open [O] contract. Net: the residual to a full 4d QFT is *one premise* (QGEO.SYM.01) plus *one contract* (CONTRACT.QFT4D.01). All wired into `run_all.py`, the ledger and registry, cited in `tfpt_research_contracts.tex`, mirrored to the website.

**2026-06-17 (The static→dynamic flip: modular flow + a recovery Lindblad generator, v238)**

- **New v238\_modular\_lindblad\_dynamics.py (DYN.SEMIGROUP.01, cluster registry).** A companion to the modular round (v196/v198) and the recovery code (v221/v64) that reads the *same* seam objects *dynamically* rather than statically. [E] the modular flow  $\sigma_t = e^{it\Lambda_\Sigma}$  is a unitary one-parameter group (Tomita–Takesaki) whose conservation of the  $\mu_4$  clock,  $\rho \sigma_t \rho^{-1} = \sigma_t \Leftrightarrow [\rho, \Lambda_\Sigma] = 0$ , IS the v198 state-invariance — so the static commutator is the conservation law of modular time. [E] the recovery channel’s generator  $L = \log T$  is a doubly-stochastic/CPTP-classical Markov generator ( $e^L = T$ ,  $e^{L\tau} \rightarrow$  the democratic projector = the arrow of time). [E] keystone: the dissipative gap  $= -\log((2/3)^6) = 6 \log(3/2)$  = the OS mass gap  $\Delta$  (v64) — the static recovery bound (v221) and the dynamical mass gap (v64) are one number, one exponentiated. [E] the GKSL split  $G = i\Lambda_\Sigma + L$  (imaginary + real  $\leq 0$  spectrum). [O] the residual is the *geometric* modular flow (Connes–Rovelli “thermal time = physical time”) on the full seam algebra = QGEO.SYM.01 (v198) — a target, not a closure. Wired into `run_all.py`, `status_ledger.csv`, `script_registry.csv` and cited in `tfpt_research_contracts.tex`; mirrored to `website/public/verification`.

2026-06-17 (Experiments: black-hole-cosmology signatures, the  $\eta_B$  Boltzmann solve, the Petz/baby-universe realisation, and a real-data GW-echo pipeline)

- **Three new standalone black-hole-cosmology experiments (experiments/, all data\_limited, not load-bearing).** From the problem\_b black-hole-cosmology survey: (i) **ccbh-dark-energy** — the de Sitter seam interior ( $w_{\text{in}}=-1$ ) forces the Croker–Weiner cosmological-coupling index  $k = -3w_{\text{in}} = 3$ , hence a population  $w = -1$ ; vs. Farrah+2023  $k = 3.11 \pm 0.79$  ( $-0.14\sigma$ ), a *contested* downstream bridge and an *alternative reading* of the same  $w = -1$  the DESI watchdog tests (**alternative\_group w\_de\_eos**, never double-counted). (ii) **gravastar-compactness** — the Nariai  $Q_{\text{geom}}=3/8$  equals the Jampolski–Rezzolla (2026) maximum horizonless compactness  $\mathcal{C}=3/8$  (shared de Sitter endpoint  $1/2$ ); since  $1/3 < 3/8 < 4/9 < 1/2$  this is a light-trapping ECO, yielding an echo template (round-trip delay  $\sim 0.7$  ms at  $62 M_\odot$ , amplitude  $\leq (2/3)^6$ ) — a [C] structural echo, no  $\mathcal{C} \leftrightarrow Q_{\text{geom}}$  map proven. (iii) **cosmic-handedness** — a parity watchdog (Shamir JADES  $\sim 3.3\sigma$  spin monopole, most likely a Milky-Way-aberration dipole), explicitly *not* promoted (frontier).
- **$\eta_B$  — full BDP Boltzmann ODE solve confirms the route at the frozen scale (experiments/ftransfer/leptogenesis\_boltzmann/fboltzmann\_solve.py; tfpt\_4\_frontier, [C]).** Integrating the Buchmüller–Di Bari–Plümacher Boltzmann network (integrated efficiency  $\kappa_f = 0.092$ , validating the analytic fit) at the *frozen* heavy scale  $M_1 = M_{\text{scal}}(\varphi_0^{\text{ret}})^2/A_\Lambda \approx 8.65 \times 10^9$  GeV with  $\delta_{\text{CP}} = 4\pi/3$  gives  $\eta_B = 6.5 \times 10^{-10}$  — a factor 1.07 from the observed  $6.1 \times 10^{-10}$ , with *no* free  $M_R$  dial. This moves the leptogenesis route from “solve pending” to a consistent [C] readout (the full *flavored* density-matrix solve is the next refinement); tfpt\_4\_frontier Route C synced.
- **The Petz recovery map and the rank-one baby universe, realised numerically (experiments/recovery-channel; tfpt\_horizon\_readouts, [C]; companion to v221).** The gapped transport contracts to a *rank-one* projector at exactly  $\|T^n - P_\infty\| = (2/3)^{6n}$  — the one-dimensional baby-universe Hilbert space (Engelhardt 2025) — and the explicit Petz map is CPTP, recovers the reference, and equals the identity only on the protected  $\lambda=1$  (Knill–Laflamme) mode; free-ratio and degenerate-spectrum negative controls hold. So the Petz identification v221 deferred is now verified at the channel level (**internal\_consistency**, no new datum); the full holographic reading stays gated on the Seam–Horizon theorem.
- **GW ringdown echo — Stage-1 matched filter validated, then run on real GWOSC strain (experiments/gw-ringdown-echo; tfpt\_horizon\_readouts search-targets).** The Stage-1 machinery (Kerr-ringdown subtraction  $\rightarrow$  matched filter on the residual, ratio  $(2/3)^6$  frozen, lag free, + free-ratio control) classifies 3/3 synthetic injections; run on *real* GWOSC strain (GW150914, GW190521; PSD-whitening + off-source background): **no kernel-ratio echo coincident in  $\geq 2$  detectors**, consistent with the  $(2/3)^6$  *upper* bound (a single loud H1 excess has  $\hat{q} \approx 2$ , far from  $(2/3)^6$ , so the free-ratio control flags it as residual ringdown, not an echo). First pass: dominant-QNM subtraction, incoherent combination — full multi-mode + coherent stacking is the next step; no detection claim.
- **Scorecard + website synced.** experiments/evidence\_scorecard.json now 51 rows (26 consistent, 7 null, 13 data-limited, 4 tension, 1 parked); the website EXPERIMENTS\_AUDIT mirror and the  $\eta_B$  prediction entry are updated to match. These are standalone experiments/ surfaces (not in the verification suite or ledger); the tfpt\_4\_frontier/tfpt\_horizon\_readouts and papers.ts/predictions.ts edits are the in-same-change sync.

2026-06-17 (Red-team layer re-synced to the paper; shadow export mirrors figures/)

- **Red Team Target A re-synced to tfpt\_5\_redteam (rt\_A\_e8net.py, redteam\_table.txt, redteam/README.md, infrastructure).** The Python red-team surface still emitted the

historical *three-residual* “A reduced, not closed” form, lagging the reader-facing note. It now encodes the sharper narrowing already in the paper: the residual collapses to the *single* statement “the seam–Calderón boundary net is holomorphic with  $c = 8$ ” ( $\Leftrightarrow$  the index-4 simple-current extension), with the index-4  $(D5)_1 \times (A3)_1 \rightarrow (E8)_1$  glue realised at Lie level (v143) and its odd sectors twisted/Ramond ( $E_8 = 120_{\text{NS}} + 128_{\text{R}}$ , v148), while the  $q(A_3)$  normalisation collapses into the one declared anchor (v152). Two exact adversarial checks added ( $120+128 = 248$ ; glue index  $= |\mu_4| = N_{\text{fam}}+1 = 4$ ); status stays  $[C]/[O]$  (reduced, not closed).

- **Shadow export now mirrors figures/ but not website/ (scripts/export-shadow.sh, verification/make\_script\_index.py, infrastructure).** export-shadow.sh ships figures/\*.pdf (so make\_manifest.py --check passes literally on the exported tree) while still excluding website/ and the compiled paper PDFs; the SHADOW\_MIRROR.md table is corrected accordingly. make\_script\_index.py now detects a missing website/ and skips its ScriptIndex.tsx mirror, so build.sh notes runs on the subset without crashing.

## 2026-06-17 (CP phases as $\mu_6$ powers — a reduction of red-team Target D)

- **Both CP phases are  $\mu_6$  powers of one hexagonal CM unit, split by the sheet (v231\_cp\_mu6\_phases, tfpt\_5\_redteam,  $[E]/[C]$ ).** Sharper than v220/v225 (which only *located* the CP residual). With  $\rho = e^{i\pi/3}$  the primitive 6th root (the  $j=0$  Eisenstein CM unit):  $\delta_{\text{CKM}} \text{ leading} = \arg(\rho^1) = \pi/3$  (quark) and  $\delta_{\text{PMNS}} = \arg(\rho^4) = 4\pi/3$  (lepton, the phase lattice), with  $\rho^4 = -\rho$ , so  $\delta_{\text{PMNS}} = \delta_{\text{CKM,lead}} + \pi = (\text{CM unit}) \times (\mathbb{Z}_2 \text{ sheet}, \rho^3 = -1)$ . The two CP phases are *one* hexagonal unit on the two sheets, not two independent inputs; the dual-frame Jarlskog orientations are sheet-flipped ( $\pm 21 \sin(\pi/3)$ ). A structural reduction of Target D (removes one of its two phase inputs); the quark seam correction  $3\lambda_C^2$  stays the v88 misalignment, the deck stays  $\mathbb{Z}/4$ , and the physical CP derivation stays  $[C]$  (CP.MU6.01).
- **The seam as the  $E_8$  Kleinian singularity (v232\_e8\_kleinian\_seam, origin\_theory,  $[E]/[C]$ ).** The du Val side of the McKay correspondence (v219) gives the open seam-realisation premise QGEO.REALIZE.01 a canonical algebraic-geometric model: dropping the trivial-rep node of the affine- $E_8$  McKay graph of  $2I$  leaves the finite  $E_8$  Dynkin = the dual intersection graph of the eight exceptional  $\mathbb{P}^1$ 's of the minimal resolution of  $\mathbb{C}^2/2I$ ; the eight curves = rank  $E_8 = g_{\text{car}} + N_{\text{fam}}$  (a fourth reading of the 8), their negated intersection form is the  $E_8$  Cartan (det 1, even,  $(-2)$ -curves), and the link is the Poincaré homology 3-sphere  $S^3/2I$  ( $2I$  perfect  $\Rightarrow H_1=0$ ,  $\pi_1=2I$  order 120). The raw seam ( $\mathbb{P}^1$  with marks) is canonically a du Val exceptional curve — a model for the bedrock, not a P2 proof (it presupposes  $E_8$ ; TOP0.E8.01, bedrock stays  $[O]$ ).
- **CP is the universal family/triality phase, only sheet-split (v233\_cp\_triality\_phase, tfpt\_5\_redteam,  $[E]/[C]$ ).** One step past v231 in the  $\mu_6 = \mu_3(\text{family/triality}) \times \mu_2(\text{sheet})$  factorisation:  $\rho = \omega^2(-1)$  with  $\omega = e^{2\pi i/3}$  the  $\mathbb{Z}_3$  triality centre (the  $2/3$  cusp weight). Since  $\rho^4 = \rho^1 \rho^3$  with  $\rho^3 \in \mu_2$ , the quark ( $\rho^1$ ) and lepton ( $\rho^4$ ) CP phases share the *same* family/triality class and differ only by the sheet — so CP *is* the universal triality phase, sheet-split, not a free power choice (CP.TRIALITY.01).
- **The Seam-Holomorphy selection certificate: the structural residual is one gate (v234\_seam\_holomorphy\_selection, tfpt\_research\_contracts,  $[E]/[O]$ ).** The whole structural residual is *one* condition — “the seam carries no nontrivial abelian sector” — with three provably-equivalent faces, all forcing  $E_8$ : (i) holomorphy ( $\mu$ -index 1; Target A), (ii) the seam link is a homology 3-sphere ( $\Gamma$  perfect  $\Leftrightarrow 2I$ ; v232), (iii) exactly one 1-dim irrep (v219). All equal  $\#(\text{mark-1}) = |\Gamma^{\text{ab}}| = |H_1|$ , which is 1 only for  $E_8$ ; and holomorphic  $c=8=g_{\text{car}}+N_{\text{fam}}$  gives the unique even unimodular rank-8 lattice  $E_8$ . So P2,  $G_{\text{net}}$ , Target A and the Kleinian seam are one gate. The stated closing theorem ( $[O]$ ):  $\text{RP}+\text{gap}+\text{chirality} \Rightarrow \text{quasi-free bulk}$

(v160)  $\Rightarrow$  holomorphic boundary  $\Rightarrow E_8$ ; the single residual analytic step is “free bulk  $\Rightarrow$  holomorphic boundary” (GATE.HOLO.01).

- **The closing step in Chern–Simons language: holomorphic  $\Leftrightarrow \det K=1$**  (v235\_seam\_chern\_simons, tfpt\_research\_contracts, [E]/[O]). A genuine reduction (not a closure) of GATE.HOLO.01 into standard anyon-condensation physics: a free gapped bosonic 2+1d bulk is an even integer  $K$ -matrix with  $\#\text{anyons} = |\det K|$ , edge  $c = \text{signature}(K)$ , and the edge net holomorphic  $\Leftrightarrow |\det K|=1$ . The TFPT extension tower (v92) is the condensation tower read by determinant: carrier  $D_5 \oplus A_3$  (det 16, 16 anyons)  $\rightarrow SO(16)_1 = D_8$  (det 4)  $\rightarrow (E_8)_1$  (det 1, holomorphic) — all rank 8,  $c=8=g_{\text{car}}+N_{\text{fam}}$  — so holomorphic  $= E_8$  = the Kitaev  $E_8$  quantum-Hall state. Holomorphy  $\Leftrightarrow$  condensing the order- $|\mu_4|$  Lagrangian glue (det 16  $\rightarrow$  1, vs. a partial order-2 isotropic  $\rightarrow 4=D_8$ ). The residual narrows to one integer condition “the free RP seam condenses the Lagrangian glue (det=1)” = the sheet/QGEO selection, now with holomorphy  $\Leftrightarrow \det 1$  a theorem (GATE.HOLO.02).
- **The (2, 3, 5) Brieskorn singularity is the one generator of the skeleton** (v236\_brieskorn\_capstone, origin\_theory, [E]). Capstone of the icosahedral round: the singularity  $x^2+y^3+z^5$ , whose exponents are the atoms ( $|\mathbb{Z}_2|, N_{\text{fam}}, g_{\text{car}})=(2, 3, 5)$ , has Milnor number  $\mu=(2-1)(3-1)(5-1)=8=\text{rank } E_8$  (a *fifth* origin of the “8”), Milnor monodromy = the order-30  $E_8$  Coxeter element (eigenvalues the primitive 30th roots = the  $E_8$  exponents, charpoly  $\Phi_{30}$ ), Milnor lattice  $E_8$ , link the Poincaré sphere; the two clocks are facets of this one monodromy — the  $\mu_3$  triality (CP, v233) is  $h^{10}$  in  $\langle h \rangle = \mathbb{Z}/30$ , the  $\mu_4$  clock (v223) is the order-4 Galois automorphism of  $\mathbb{Q}(\zeta_{30})$  (TOPO.BRIESKORN.01).
- **The closing step as a physical condition: no topological degeneracy  $\Leftrightarrow \det K=1 \Leftrightarrow$  the seam is SRE** (v237\_seam\_sre\_closure, tfpt\_research\_contracts, [E]/[O]). Sharpens GATE.HOLO.02’s residual from definitional to falsifiable: the  $K$ -matrix ground-state degeneracy on a genus- $g$  surface is  $|\det K|^g$  (torus: carrier 16,  $D_8$  4,  $E_8$  1), so “no topological degeneracy on any closed surface”  $\Leftrightarrow \det K=1$ ; among rank-8 even positive-definite  $K$  ( $c=8=g_{\text{car}}+N_{\text{fam}}$ ) only  $E_8$  is short-range-entangled (the Kitaev  $E_8$  phase). A unique vacuum on the plane (RP+gap+cluster) is necessary but not sufficient (the plane cannot see  $\det K$ ); so the one open step is the physical statement “the free RP seam is SRE (no topological order)” (GATE.HOLO.03).

## 2026-06-16 (the icosahedral bedrock, CM-norm duality, and a structural-finds round)

- **McKay bedrock: why the atoms are  $\{2, 3, 5\}$**  (v219\_icosahedral\_mckay, origin\_theory, [E]).  $E_8$  is the exceptional *top* of the McKay tower of finite  $SU(2)$  subgroups ( $2T \rightarrow \hat{E}_6$ ,  $2O \rightarrow \hat{E}_7$ ,  $2I \rightarrow \hat{E}_8$ ). Built *from the group*: the 120 unit-quaternion icosians close to the binary icosahedral group  $2I$  (element orders  $\{1, 2, 3, 4, 5, 6, 10\}$  — it contains the 2, 3, 5 rotation-axis orders), 9 conjugacy classes; the 9 irreducible character degrees (Dixon, from the class algebra) are  $\{1, 2, 2, 3, 3, 4, 4, 5, 6\}$ , sum  $30=h(E_8)$ , sum of squares  $120=|R^+(E_8)|$ ; the McKay adjacency from the natural 2-dim rep *is* the affine  $E_8$  graph with Kac marks = the degrees. A backward certificate of the closed  $E_8$ , not a P2 proof (MCKAY.E8.01).
- **CM-norm duality: 41 (square) and 7 (hex)** (v222\_cm\_norm\_duality, origin\_theory, [E]). The square seam (Gaussian  $\mathbb{Z}[i]$ ,  $j=1728$ ) gives  $N(g_{\text{car}}+i|\mu_4|)=5^2+4^2=41=10b_1$  (the EM index *is* the Gaussian norm of the carrier-glue vector) and  $N(3+2i)=13=\Delta_Q$ ; the hexagonal partner (Eisenstein  $\mathbb{Z}[\omega]$ ,  $j=0$ ) gives  $N(3+2\omega)=7=\text{scalaron}$ . The (3, 2) split generates (5, 6, 7, 13) by sum/product/Eisenstein/Gauss norm; rings rigid (CMNORM.DUAL.01).
- **The  $\mu_4$  clock is the order-4 character of the  $E_8$  Coxeter cycle** (v223\_coxeter\_totative\_clock, origin\_theory, [E]).  $(\mathbb{Z}/30)^\times$  are the  $E_8$  exponents; the order-4 generator 7 ( $\langle 7 \rangle = \{1, 7, 13, 19\}$ ) is a literal  $\mu_4$  clock permuting the four conjugate planes; only  $E_8$  has  $\varphi(h)=\text{rank}$  and  $h=2 \cdot 3 \cdot 5$  (COX.CLOCK.01).

- **248=120+128 as a magnitude/phase channel typing (v227\_degree\_exponent\_channel\_split, tfpt\_5\_redteam, [E]).** Exponents sum  $120=|R^+(E_8)|$  (magnitude channel), degrees sum  $128=\text{rank} \cdot \dim S^+$  (phase/glue channel); the red-team Target D magnitude bijection lives in 120, the un-covered CP phases in 128; the dark-sector reading of 128 stays Frontier (replaces the over-strong  $S^-$  reading; E8.CHAN.01).
- **P2 as the mode space of a degree-4 seam divisor (v228\_rr\_index\_gate, origin\_theory, [E]/[C]/[O]).** A Riemann–Roch index gate:  $\deg D=4=|\mu_4|$  on  $\mathbb{P}^1$  forces  $h^0=5=g_{\text{car}}$ ,  $\text{rank } H_1=3=N_{\text{fam}}$ ,  $h^0+H_1=8=\text{rank } E_8$ ,  $\Lambda^{\text{even}}(\mathbb{C}^5)=16=\dim S^+$  — bedrock language for P2, conditional on the seam producing the degree-4 divisor (QGEO.RR.01).
- **The seam as a finite recoverability code (v221\_seam\_qecc, origin\_theory, [E]/[C]).** Code dimension  $\dim S^+=16$ ; the gapped transport is a CPTP doubly-stochastic contraction with spectrum  $\{1, (2/3)^6, (1/3)^6\}$ , recovery bound  $\|T^n \delta\| \leq (2/3)^{6n} \|\delta\|$  (Knill–Laflamme type); free ratio breaks it. Holographic/Petz reading [C], gated on the Seam–Horizon theorem (QEC.SEAM.01).
- **CP lives in the hexagonal phase fiber (v220\_cp\_hexagonal\_modulus, tfpt\_5\_redteam, [E]/[C]).** The seam deck is the square modulus  $j=1728$  ( $\mathbb{Z}/4$ ); its partner is the hexagonal  $j=0$  ( $\mathbb{Z}/6$ ,  $\arg \rho=\pi/3$ ). The frozen  $\delta=\pi/3+3\lambda_C^2=68.654^\circ$  has leading term  $\pi/3=\arg \rho$ ; the deck stays  $\mathbb{Z}/4$ , CP is the  $\mathbb{Z}/6$  fiber (CP.MOD.01).
- **The dual normal frame  $(d, n)$  with CP as orientation (v225\_dual\_normal\_frame, tfpt\_5\_redteam, [E]/[C]).**  $d=a^\top R^{-1}=(-\frac{1}{2}, -\frac{1}{2}, 1)=-\frac{1}{2}$  Nariai;  $n=(5, -9, 6)$  reads  $\det$  on the first-generation column;  $\det(\mathbf{1}, d, n)=21=3 \cdot 7$ ;  $\text{Im } \det(\mathbf{1}, d, \rho n)=21 \sin(\pi/3)$  — CP as orientation (DUAL.FRAME.01).
- **$F_{\text{transfer}}$  as one path on the Sheet Diamond (v224\_diamond\_ftransfer\_path, tfpt\_2\_standard\_model, [E]/[C]).**  $M(s, t)=R+Q \text{diag}(s, t, t)$ :  $K, C, F$  on the sheet axis (curved, 2nd diff 8), the winding axis flat (slope 6), Plücker steps  $(1, 8, 10), (1, 8, 16)$ ; the Koide source→pole is the 1-D projection (FTR.PATH.01).
- **The charged leptons as an étale Frobenius algebra (v229\_lepton\_frobenius\_algebra, tfpt\_2\_standard\_model, [E]/[C]).**  $(c_e, c_\mu, c_\tau)=(\frac{16}{7}, \frac{4}{3}, \frac{7}{6})$ , ring closure  $c_e c_\tau=\frac{8}{3}=|\mathbb{Z}_2|c_\mu$ , product  $\frac{32}{9}$ ;  $A=\mathbb{Q}[t]/(m)$  is étale (trace-form Gram nondegenerate), a commutative Frobenius algebra over the  $\mu_6$  resolvent (LEP.FROB.01).
- **The center budget  $(7, 11, 13)$  as three local norms (v230\_center\_budget\_norms, tfpt\_2\_standard\_model, [E]).**  $C=R+Q \text{diag}(1, 0, 0)$  row sums  $(7, 11, 13)$ :  $7=N_{\mathbb{Z}[\omega]}(3+2\omega)$  (hex),  $13=N_{\mathbb{Z}[i]}(3+2i)$  (square),  $11=\binom{4}{0}+\binom{4}{1}+\binom{4}{2}$  (QBL Fock count); the two CM rings and the boundary count meet in one operator center (CENTER.NORM.01).

## 2026-06-15 (the diamond axis geometry — two axes + the transfer corner)

- **The Sheet Diamond is a discrete geometry with two axes (v218\_diamond\_axis\_geometry, tfpt\_2\_standard\_model, [E]).** Sharpens v94/v95 with three exact [E] lemmas (no new numbers). **(1) Axis curvature:** along the centered cross the determinant is linear on the winding axis  $\det(C+xU) = 14+6x$  (slope  $6=|R^+(A_3)|$ ) and quadratic on the sheet axis  $\det(C+yV) = 14+14y+4y^2$  (2nd difference  $8=\text{rank } E_8$ ); the anchor block has  $\det B(C+xU) = 2 \det(C+xU)$  and sheet curvature  $6=|R^+(A_3)|$  — two curvatures  $(8, 6) = (\text{rank } E_8, |R^+(A_3)|)$ . **(2) Plücker transfer ladder:**  $\text{Pl}(K) \rightarrow \text{Pl}(C) \rightarrow \text{Pl}(F)$  lifts in steps  $(1, 8, 10) = (N_\Phi, \text{rank } E_8, A_\Lambda)$  then  $(1, 8, 16) = (N_\Phi, \text{rank } E_8, \dim S^+)$  — the decuple then the full generation (identity [E], source→pole reading [C]). **(3) Spectral ramification:** for  $Q, K, C, F$  the cubic discriminant factors as  $q(r)^2 \text{Disc}(q)$  with squares  $(1, 3, 4, 6)$  and kernels  $(13, 48, 65, 105)$ ,  $F$  carrying  $105 = 3 \cdot 5 \cdot 7$ . The anchor-defect spectrum and pair-sum/staircase blocks are honest *audit* fingerprints ( $28=2 \dim G_2$ ,  $52=\dim F_4$  stay audit-only, not spine, as in v94/v95);  $F$  as the transfer-completion corner is a *heuristic*,

not a hard filter. Ledger DIAMOND.AXIS.01, DIAMOND.PLUCKER.01, DIAMOND.SPECTRAL.01; Wolfram-mirrored (249/249).

## 2026-06-15 (QGEO: the four marks emerge from Gauss–Bonnet — K1 executed)

- **The four marks emerge from Gauss–Bonnet — count derived, not assumed** (`v216_marks_gauss_bonnet`, `tfpt_research_contracts`, [E]; ledger QGEO.MARKS.03). The K1 lever of the v215 kill-test (“exactly four marks”) is *executed*: if the marks are  $\mathbb{Z}_2$  branch points (deficit  $\pi$ ) on the flat seam sphere ( $\chi = 2$ , the same  $\chi$  as the 8 in  $c_3$ ), Gauss–Bonnet forces  $n\pi = 2\pi\chi = 4\pi$ , so  $n = 2\chi = 4 = |\mu_4| = N_{\text{fam}} + 1$ . The closed Euclidean sphere 2-orbifolds are exactly  $(2, 3, 6)$ ,  $(2, 4, 4)$ ,  $(3, 3, 3)$ ,  $(2, 2, 2, 2)$ , and three independent criteria — all cone orders = 2 (the  $|\mathbb{Z}_2|$  sheet branch), rank  $H^1 = \# \text{marks} - 1 = 3 = N_{\text{fam}}$  (the three generations pick the 4-mark square over the 3-mark hexagonal), and a free order-4 deck — converge on the pillowcase  $(2, 2, 2, 2)$ . The numerical sibling `v217_marks_emergence_scan` ([C], ledger QGEO.EMERGE.01) confirms it on the actual DtN/state (number  $n$  and positions free, the 4-square config is the unique joint minimiser of clock-invariance + 3-family cohomology + gap  $(2/3)^6$ ). The mark *count* and *equality* move from input to derived; only the square *modulus* (cross-ratio 2, the order-4 clock) stays the one carrier input, and the from-zero raw-seam emergence QGEO.REALIZE.01 stays [O]. v216 exact (Wolfram extension  $\rightarrow 248/248$ ); v217 numerical (Python-only). Suite  $\rightarrow$  217 modules.

## 2026-06-15 (QGEO: the bedrock recast as a falsifiable kill-test)

- **QGEO.SYM.01 as a kill-test, not a proof-promise** (`v215_seam_deck_killtest`, `tfpt_research_contracts`, [E]/[O]; ledger QGEO.KILL.01). A foundational symmetry cannot be derived from nothing (Bisognano–Wichmann would presuppose the covariance it should produce), so the bedrock  $\omega \circ \rho = \omega$  is recast as four independently-falsifiable predictions about the raw seam DtN. The non-circular reduction (`v198/v199/v201`) gives  $\omega \circ \rho = \omega \Leftrightarrow [\rho, C] = 0$  with  $|k|$  commuting exactly, so the residual is the bounded sub-principal  $M_f$  (equivalently the raw Gaussian bulk form  $A$  is  $\mu_4$ -deck-invariant). Levers: **K1** exactly four marks ( $b_1 = \# \text{marks} - 1 = 3 = N_{\text{fam}} \Rightarrow \# \text{marks} = 4 = |\mu_4|$ ); **K2** the square config (cross-ratio 2  $\Rightarrow j = 1728$ ,  $\text{Aut} = \mathbb{Z}/4$ ; order 4 selects  $\mathbb{Z}/4$  not the  $j = 0$   $\mathbb{Z}/6$ ; generic  $\Rightarrow \mathbb{Z}/2$ ); **K3** mod-4 sub-principal support ( $[\rho, M_f] = 0 \Leftrightarrow f$   $\mathbb{Z}_4$ -invariant); **K4** transfer gap  $(2/3)^6 = 64/729$ . Freeze-bound via `freeze_file.csv` (`seam_deck_symmetry`), carrier-side [E] (regression guards), with the *decisive* kill deferred to QGEO.REALIZE.01 [O] (the raw collar DtN must *produce* four square marks without inputting them — an emergence test). A kill-test, not a closure. Python-only (falsification layer; the exact sub-parts  $j = 1728$  and  $(2/3)^6$  are Wolfram-mirrored via `v214/v54/v56`). Suite  $\rightarrow$  215 modules.

## 2026-06-15 (QGEO: the pillowcase reduction — two residuals become one canonical metric)

- **Pillowcase reduction of QGEO.SYM.01** (`v214_seam_pillowcase`, `tfpt_research_contracts`, [E]/[O]; ledger QGEO.PILLOW.01). The two separately-tracked open QGEO residuals — QGEO.ISO.01 (v180: the carrier clock is an order-4 *isometry* of the seam metric) and QGEO.SUBPRIN.01 (v201: the DtN sub-principal symbol is *mark-local*, the seam flat away from the  $\mu_4$  marks) — are shown to be *one* statement: the seam carries the uniformising flat orbifold (*pillowcase*) metric. **(1)** The four marks make the Euclidean orbifold  $S^2(2, 2, 2, 2)$  ( $\chi_{\text{orb}} = 2 - 4(1 - \frac{1}{2}) = 0$ ), so by Troyanov uniformisation the metric is flat (unique up to scale) with curvature only at the cone points (Gauss–Bonnet  $4\pi = 2\pi\chi$ ) — this *is* the v201 “flat away from the marks”. **(2)** New link: cross-ratio( $\mu_4$ ) = 2 (v168)  $\Rightarrow j = 1728$  ( $j(\lambda) = 256(\lambda^2 - \lambda + 1)^3/(\lambda^2(\lambda - 1)^2)$ ; all six harmonic cross-ratios give 1728)  $\Rightarrow$  the *square* modulus  $\tau = i$ , whose CM by  $\mathbb{Z}[i]$  *derives* the order-4 clock  $z \mapsto iz$  (explicit  $(x, y) \mapsto (-x, iy)$ )

on  $y^2 = x^3 - x$ ); generic cross-ratio gives  $j \notin \{0, 1728\}$  and only  $\mathbb{Z}/2$ . **(3)** Unification: “seam = uniformising flat pillowcase metric” implies *both* v201 and v180, so two residuals collapse to one canonical-metric premise. **[O]** It does *not* close QGEO.SYM.01: the choice “RP seam metric = the constant-curvature representative” stays open, milder and more canonical than the bare isometry premise. Wolfram-mirrored (the exact orbifold/ $j$ -invariant/CM arithmetic); Klein- $J$  values mpmath-numerical. Suite  $\rightarrow$  214 modules.

## 2026-06-15 (the $F_{\text{transfer}}$ functor contract CONTRACT.F.01)

- $F_{\text{transfer}}$  formalised as a typed functor with four axioms (v213\_ftransfer\_functor, tfpt\_research\_contracts, [C]). The four frontier transfers are consolidated into ONE functor  $F_{\text{transfer}} = F_{\text{observable}} \circ F_{\text{threshold}} \circ F_{\text{RG}}$ , each instance a discrete compiler kernel **[E]** composed with an external continuous solver **[C]**: **(1)**  $\mu_4$ -deck equivariance (the Koide multiplier  $\lambda_2 = (2/3)^6 = 64/729$  is the deck transfer eigenvalue); **(2)** Plücker preservation ( $53 = a^T(R+Q)\mathbf{1}$ ,  $\text{Pl}(F) = (1, 22, 30)$ ,  $\|\text{Pl}(K)\|_1 = 11$ ); **(3)** positivity/stochasticity ( $\text{spec } T$  positive,  $\kappa_f \in (0, 1)$ ); **(4)** external modules explicit ( $b_3 = -7$  a carrier output). The four instances are  $F_{\text{pole}}$  (v183),  $F_{\text{Boltzmann}}$  (v212),  $F_{\text{relic}}$  (v211),  $F_{\text{QCD}}$  (v164). CONTRACT.F.01 is the third research contract alongside ( $U_{\text{wall}}$ ) and ( $G_{\text{metric}}$ ), the structural complement of the v187 typing guard — an architectural consolidation, *not* a closure of any frontier number. Ledger CONTRACT.F.01.

## 2026-06-15 (frontier: two sharper [C] scenario branches — axion angle + leptogenesis)

- **Axion DM spine-angle branch**  $\theta_i = \pi N_{\text{fam}}/g_{\text{car}} = 3\pi/5$  (v211\_axion\_spine\_angle, tfpt\_4\_frontier, [C]). A more *robust* misalignment angle than the determinant-line hilltop  $\theta_i \approx 170^\circ$  (v185, which over-produces  $\Omega_a h^2 \approx 0.66$ ): the central spine quotient gives  $\theta_i = 3\pi/5 = 108^\circ$  exactly (no fit), the finite- $T$  solver reaches  $\Omega_{\text{DM}}$  at  $\approx 106^\circ$ , and  $108^\circ$  sits in the *mild*-anharmonic regime ( $62^\circ$  below the hill-top) — not exponentially sensitive. An alternative ansatz to  $\theta_i = \pi(1 - \varphi_\Sigma)$  (mutually exclusive, the full solver decides, DM.AXION.SPINE.01); **[X]**  $\Omega_a h^2$  outside  $\sim [0.08, 0.16]$  demotes it. A sharper scenario, not a derivation.
- **Leptogenesis scalaron-decuplet branch** (v212\_leptogenesis\_decouple, tfpt\_4\_frontier, [C]). Both Boltzmann inputs share the decuplet  $A_\Lambda = 10 = |E(K_5)|$ :  $M_1 = M_{\text{scal}}(\varphi_0^{\text{ret}})^2/A_\Lambda \approx 8.65 \times 10^9$  GeV and  $\tilde{m}_1 = m_3/A_\Lambda \approx 5$  meV.  $M_1$  uses *only*  $\{M_{\text{scal}}, \varphi_0^{\text{ret}}, A_\Lambda\}$  — no hidden seesaw scale, cleaner than v184’s  $M_R \varphi_0^4$  — and lands in the thermal window where  $\eta_B \approx 1.2 \times 10^{-9}$  brackets the observed  $6.1 \times 10^{-10}$ . The coupling mechanism is posited not derived, so  $\eta_B$  stays **[C]**; **[X]** a precise Boltzmann/RGE solve outside the band falls the route, not the theory (FR.ETAB.04; Route A independent).

## 2026-06-15 (Lean: QGEO.COHO.M.01 character grading + MODULE parity machine-proved)

- **The cohomology character grading is now Lean-formalised** (CohomologyGrading.lean, ledger FORM.QGEO.03, [O]). A new module machine-proves the COHO.M node of the QGEO proof tree (v177; AUDIT: PASS, no sorry/admit, axioms propext, Classical.choice, Quot.sound only): the three  $H^1(\mathbb{P} \setminus \mu_4)$  eigenforms  $\omega_k = z^{k-1}/(z^4 - 1)$  ( $k=1, 2, 3$ ) have pullback character  $\rho^* \omega_k = i^k \omega_k$  under the clock  $\rho : z \mapsto iz$  (each an exact rational-function identity, the denominator clock-invariant since  $i^4=1$ ), so the grading is  $(i, -1, -i)$  with weights  $(1, 2, 3) =$  the  $A_3$  exponents  $= \text{Spec}(Q_+)$ , rank  $3 = N_{\text{fam}}$ . The MODULE *parity* is also proved: the reflection  $\sigma : z \mapsto 1/z$  satisfies  $\sigma^* \omega_1 = \omega_3$ ,  $\sigma^* \omega_2 = \omega_2$ ,  $\sigma^* \omega_3 = \omega_1$  (swap  $\omega_1 \leftrightarrow \omega_3$ ,  $\omega_2$  fixed). Signature-locked in AuditContract.lean. The geometric COHO.M/MODULE-parity nodes are now **[E]**; the MARKS/KERNEL obligations and the MODULE *uniqueness* stay **[O]**.

## 2026-06-15 (Lean: QGEO.UNIFORM.01 geometric normal form machine-proved)

- **The Möbius uniformisation normal form is now Lean-formalised** (Möbius Uniformisation.lean, ledger FORM.QGEO.02, [O]). A new module machine-proves the constructive UNIFORM node of the QGEO proof tree (v177; AUDIT: PASS, no sorry/admit, axioms propext, Classical.choice, Quot.sound only): the clock  $\rho : z \mapsto iz$  has  $\rho^4 = \text{id}$  and  $\rho^2 \neq \text{id}$  (order exactly 4); the reflection  $\sigma : z \mapsto 1/z$  is an involution with  $\sigma\rho\sigma = \rho^{-1}$  (so  $\langle \rho, \sigma \rangle$  is the dihedral  $D_4$  of order 8); a non-fixed orbit  $\{a, ia, -a, -ia\}$  scales to  $\mu_4 = \{1, i, -1, -i\}$ ;  $\sigma$  permutes  $\mu_4$  (fixes 1, -1, swaps  $i, -i$ ); and the multiplier classification “ $z \mapsto \zeta z$  has order exactly 4  $\Leftrightarrow \zeta = \pm i$ ”. Signature-locked in AuditContract.lean. This formalises the geometric half of the seam realisation *given* the four marks and the clock; the raw-seam marking obligation QGEO.MARKS.01 stays [O], *not* closed.

## 2026-06-15 (Lean: the QGEO.SYM.01 conditional theorem machine-proved)

- **The mark-local  $\Rightarrow \omega \circ \rho = \omega$  implication is now Lean-formalised** (SeamDeckClosure.lean, ledger FORM.QGEO.01, [O]). A new module in the carrier-rigidity Lean project machine-proves the algebraic core of v201/v210 (AUDIT: PASS, no sorry/admit, no domain axioms — #print axioms = propext, Classical.choice, Quot.sound only): (i) the 4th-root character orthogonality  $\sum_{j<4} \zeta^j = (\text{if } \zeta=1 \text{ then } 4 \text{ else } 0)$  for  $\zeta^4=1$  (so a  $\mu_4$ -mark sum is supported only on modes  $\equiv 0 \pmod{4}$ ); (ii) the clock generator  $-i$  has order 4; (iii) markLocal\_blockDiagonal: a mark-local Toeplitz symbol connects only equal clock-characters,  $[\rho, M_f] = 0$ ; (iv) the premise is a typed structure SeamDeckPremise (the physical seam DtN *is* mark-local) — consumed, *not* proved, exactly as CalderonProjector encodes the Paper-1 analytic input; (v) SeamDeckPremise.clock\_invariant: *given* the premise, the clock commutes with the DtN, whence  $\omega \circ \rho = \omega$ . So the *implication* is now [E] (formally proved); the *premise* (the physical seam being mark-local) stays [O] — the one fundamental seam-identification postulate, *not* closed.

## 2026-06-15 (QGEO sharpening: mark-local DtN certified on realistic profiles)

- **Mark-local DtN  $\Rightarrow \omega \circ \rho = \omega$ , certified on realistic Steklov profiles and at the state level** (v210\_mark\_local\_dtn, tfpt\_research\_contracts, [O]). A numerical sharpening of QGEO.SUBPRIN.01/v201: a real von-Mises curvature bump summed over the four  $\mu_4$  marks,  $f(\theta) = \sum_{j=0}^3 g(\theta - j\pi/2)$ , builds the Steklov DtN  $\Lambda = |D_\theta| + M_f$  whose off-(mod 4) Fourier support vanishes to  $< 10^{-16}$ ,  $\|[\rho, \Lambda]\| < 10^{-15}$ , and — the genuinely new step — the quasi-free covariance  $C = \frac{1}{2}(1 + \text{sgn } H_1)$  is clock-invariant ( $\rho C \rho^\dagger = C$  to  $< 10^{-13}$ ), i.e. the actual  $\omega \circ \rho = \omega$ . Negative controls (single off-centre bump,  $\mathbb{Z}_3$  source, four generic marks) break both by  $O(1)$ ; convergence stable for  $N \in \{16, 32, 64\}$ . This certifies the *implication*; the premise that the physical seam *is* mark-local stays [O] (it does *not* close QGEO.SYM.01; net existence and full-cone RP remain the [E] of v175). Ledger QGEO.MARKLOCAL.01. Python-only (the exact mark-sum identity is Wolfram-mirrored via v201).

## 2026-06-15 (archive integration #5: six dormant results revived from the old papers)

- **Muon  $g-2$  as a seam vertex** (v204\_muon\_seam\_g2, tfpt\_4\_frontier, [C]). The carrier’s second-order topological defect  $\delta_2 = \frac{5}{4}\delta_{\text{top}}^2$  projected through the seam-loop phase gives  $a_\mu^{\text{seam}} = \delta_2/(2\pi) = 45/(524288\pi^9) \approx 2.879 \times 10^{-9}$  ( $0.81\sigma$  vs the dispersive  $\Delta a_\mu$ ;  $\sim 1.5\sigma$  vs lattice/CMD-3). The value is exact [E]; the vertex identification is the [C] bridge. Ledger FR.MUONG2.01; Wolfram-mirrored.
- **An independent gravitational  $3/4$**  (v205\_xi\_threequarter, tfpt\_4\_frontier, [E]/[C]).  $\xi = c_3/\varphi_{\text{tree}} = 3/4$  exactly (Einstein limit of  $\kappa^2 = \xi\varphi_0^{\text{ret}}/c_3^2$ ) — the same

$3/4 = q(A_3) = \ln(m/\mu)$  as the seam-determinant replica (v152), an over-determination. Ledger GRAV.XI.01; Wolfram-mirrored.

- **Quantitative  $H_0$  from the  $\Lambda$  branch** (v206\_h0\_lambda\_branch, tfpt\_4\_frontier, [C]).  $\delta_\Sigma = 48 e^{-2\pi\alpha^{-1/3}} \approx 1.085 \times 10^{-123}$ , the  $(8\pi)^2$  Planck-convention split, and  $H_0 = 66.5\text{--}67.1$  km/s/Mpc (below Planck, far below SH0ES — the Hubble tension is *not* relieved). Ledger GRAV.H0.01.
- **Asymptotic Safety as a second QG route** (v207\_asymptotic\_safety, tfpt\_4\_frontier, [O]). A complementary continuum-FRG argument (a non-Gaussian UV fixed point from the same axioms;  $y^2 = 16c_3^2 = 1/(4\pi^2)$ ) — a plausibility cross-check, *not* load-bearing, does *not* close  $G_{\text{metric}}$ . Ledger GRAV.ASYMP.01.
- **Black-hole thermodynamics from the seam** (v208\_bh\_thermodynamics, tfpt\_horizon\_readouts, [E]/[C]). Modular  $T_H = \kappa/(2\pi)$  (the  $2\pi = 1/(4c_3)$  seam unit, ties to v198/v199) and the scalaron-corrected Wald entropy  $S_W = (A/4G)(1 + R_h/3M_s^2)$  (exact from v28). Ledger HOR.BHTHERMO.01; Wolfram-mirrored.
- **The black hole as a seam fixed point** (v209\_bh\_regular\_core, tfpt\_horizon\_readouts, [C]). No singularity ( $\varphi \rightarrow \varphi_\star$  at the gapped attractor  $(2/3)^6$ ); the maximal BH = the anchor (Nariai roots  $(1, 1, -2)$ ); information via Page recovery; a Planck-scale holographic floor. The superseded RN/torsion metric is *not* resurrected. Ledger HOR.BHCORE.01.

#### 2026-06-15 (archive integration #4: Möbius $\cong$ Lorentz foundation citation)

- **The Möbius seam is the boundary Lorentz structure** (tfpt\_3, [O]). Added a paragraph to the “Möbius origin of  $\pi$ ” section:  $\text{Möb}(\hat{\mathbb{C}}) \cong \text{PSL}(2, \mathbb{C}) \cong \text{SO}^+(3, 1)$  (Penrose & Rindler, *Spinors and Space-time*, Vol. 1, §1.2–1.4), the celestial sphere of null directions via stereographic projection. This gives the self-consistency loop a spacetime meaning — the carrier clock  $z \mapsto iz$  is an elliptic  $\text{PSL}(2, \mathbb{C})$  rotation (v180) and the RP→OS causal cone is the same Lorentz structure — so “Möbius origin of  $\pi$ ” and “one Lorentzian cone” are two faces of one boundary  $\text{PSL}(2, \mathbb{C})$ . A foundation citation, not a new claim ( $\pi$  stays primitive).

#### 2026-06-15 (archive integration #3: axion coupling $g_{a\gamma\gamma} = -4c_3$ + retired small-angle reading)

- **Haloscope coupling tied to  $c_3$**  (tfpt\_4\_frontier, [C]). Made explicit: in the determinant-line normalization the axion–photon anomaly coefficient is  $g_{a\gamma\gamma} = -4c_3 = -1/(2\pi)$  ( $y^2 = 16c_3^2 = 1/(4\pi^2) \approx 0.0253$ ), so the haloscope  $F\hat{F}$  coupling is set by the *same*  $c_3$  that fixes  $\alpha$  and  $\beta_{\text{rad}} = \varphi_0/4\pi$  — a [C] structural relation (the coefficient, not a parameter-free physical  $g_{a\gamma\gamma}$ ). The earlier small-angle  $\theta_i = \varphi_0$  reading (old determinant-line note, a much higher axion mass) is *superseded* by the closed near-hilltop  $\theta_i = \pi(1 - \varphi_{\text{seam}}) = 170.4^\circ$ , recorded as a retired reading and guarded by v188 (a new sentinel forbids that stale axion mass from re-entering the active docs).

#### 2026-06-15 (archive integration #2: the EHT achromatic polarization intercept — v203)

- **HOR.EHT.01** (v203, [E]/[C]/[X]). The surviving core of the old UFE/black-hole notes (*not* the RN-type metric — superseded by the Nariai/seam=horizon readouts v101–v104/v190; only the polarization signature survives): a local achromatic polarization-rotation intercept around a horizon-scale black hole,  $\beta_{\text{BH}}(r) = 16c_3^4 Q_e^{\text{eff}} Q_m^{\text{eff}} / r^2$ . The coupling  $16c_3^4 = 1/(256\pi^4) = \delta_{\text{top}}/3$  is *exact* ([E], Wolfram-mirrored) — the *same* top-form coefficient  $\delta_{\text{top}} = 48c_3^4$  that fixes the  $\alpha$ -kernel precision-zone correction, one row from the cosmic-birefringence seed  $\beta_{\text{rad}} = \varphi_0/4\pi$ . TFPT fixes the  $1/r^2$  shape, achromaticity and sign-flip ([C]); the amplitude  $Q_e Q_m$  is an MHD/GR weight, never a pixel prediction. **Dated kill test** [X]: the residual intercept

must pass three nulls (frequency, spatial  $1/r^2$ , sign-flip) after honest GRMHD subtraction. Integrated into `tfpt_horizon_readouts` (new EHT section), the ledger, registry, Wolfram (243/243), and the website (`predictions.ts`). Suite 202 modules.

## 2026-06-15 (archive integration #1: the rare kaon sector $K \rightarrow \pi\nu\bar{\nu}$ — v202)

- **FR.RAREKAON.01 (v202, [C])**. The closed TFPT CKM point ( $s_{12}=\lambda_C$ ,  $s_{23}=\varphi_0/(1+\lambda_C)=|V_{cb}|$ ,  $s_{13}=\lambda_C^3/3$ ,  $\delta=\pi/3+3\lambda_C^2=1.19823$  rad; v18/v88, no flavor fit) feeds the cleanest FCNC probe. With standard external short-distance input (Brod–Gorbahn–Stamou  $X_t, P_c, \kappa_{\pm}$ ) it gives  $\text{BR}(K^+ \rightarrow \pi^+\nu\bar{\nu}) = 9.45 \times 10^{-11}$  and  $\text{BR}(K_L \rightarrow \pi^0\nu\bar{\nu}) = 3.33 \times 10^{-11}$  — a [C] downstream flavor readout, *not* a compiler power (the SD functions are external). Recomputed from the *current* CKM point (an earlier archive scaffold quoted  $\text{BR}(K_L) = 3.47 \times 10^{-11}$  from an  $\Im(\lambda_t)/\lambda^5$  slip, corrected here). Grossman–Nir ratio  $0.352 \ll 4.3$ ; NA62  $(13.0^{+3.3}_{-3.0}) \times 10^{-11}$  is  $\sim 1.2\sigma$  above. **Dated kill test [X]**: a stable NA62  $\text{BR}(K^+)$  outside  $[7, 12] \times 10^{-11}$ , or a KOTO-II  $\text{BR}(K_L)$  off the GN point, breaks the flavor bridge (core untouched). Integrated into `tfpt_2_standard_model` (§rare kaon), the ledger, v187 guard (new FR.RAREKAON. family), registry, and the website (`predictions.ts`). Python-only. Suite 201 modules.

## 2026-06-15 (bedrock: the $\omega \circ \rho = \omega$ residual reduced once more — v201)

- **QGE0.SUBPRIN.01 (v201, [E]/[O])**. The v199 bedrock residual — “the bounded sub-principal symbol of the raw seam DtN has no off-character (mod 4) matrix elements” — is reduced one genuine step further: block-diagonality is *not* an independent postulate. Writing the DtN as  $\Lambda = |k| + M_f$  with  $M_f$  the bounded sub-principal piece = multiplication by a curvature  $f(\theta)$  (standard Steklov structure),  $M_f$  is  $\mu_4$ -character-block-diagonal iff  $f$  is  $\mathbb{Z}_4$ -invariant ([E]; principal  $|k|$  already commutes, v198); and a  $\mu_4$ -mark sum  $f = \sum_{j=0}^3 g(\theta - 2\pi j/4)$  has  $f_m = 4g_m[m \equiv 0 \bmod 4]$  for any profile  $g$ , hence is automatically  $\mathbb{Z}_4$ -invariant ([E]). So the  $\mu_4$ -mark orbit (forced by v195) *forces* the sub-principal symbol block-diagonal; 3 marks ( $\mathbb{Z}_3$ ) and 4 generic marks break it (negative controls). The residual collapses to “the DtN sub-principal symbol is mark-local (seam flat away from the  $\mu_4$  marks = conformal deck, v177)”, structurally definitional — the narrowest form yet of QGE0.SYM.01, [O]. Registered in `run_all.py`, the ledger, `tfpt_research_contracts`, `origin_theory`; Wolfram-mirrored (the mark-sum Fourier identity is exact; extension now 242/242). Suite 200 modules.

## 2026-06-15 (F\_transfer workshop: the axion relic solver — an honest over-production tension)

- **Axion relic solver (experiments/ftransfer/axion\_relic/relic\_solver.py, [C])**. The first real  $F_{\text{transfer}}$  pipeline computation (work package C / review §6.2): a standard misalignment relic with a numerically-computed anharmonic factor  $F(\theta_i) \approx 4$  at the hilltop. Honest finding: at the *predicted*  $\theta_i \approx 170^\circ$  the estimate *over*-produces dark matter,  $\Omega_a h^2 \sim 0.6\text{--}2.3$  (robustly  $> 0.12$ ), because the large  $\theta_i^2 \approx 8.8$  times  $F$  overshoots. So the determinant-line axion as the *dominant* DM is in *mild tension* (it would prefer  $\theta_i \sim 120\text{--}130^\circ$  or extra dilution) — this *refines* v185’s optimistic “can be all-DM” toward an over-production tension. The near-hilltop regime is  $O(1)$ -uncertain, so a full finite- $T$  solve decides; over-production kills only the axion-as-dominant-DM *branch*, not the compiler. Reflected in `tfpt_4_frontier` (dark-matter section) and the v185 docstring; experiments-only, not a suite/ledger claim. No refitted exponents;  $\theta_i = \pi(1 - \varphi_{\text{seam}})$  only.
- **Full finite- $T$  solve confirms it (axion\_relic/full\_finiteT\_solve.py)**. A converged nonlinear misalignment solve  $(\theta'' + (3 + d \ln H/dN)\theta' + (m_a(T)/H)^2 \sin \theta = 0$ , lattice  $\chi(T) \propto T^{-8.16}$ , realistic  $g_*(T)$ ; normalised so  $\theta_i=1 \rightarrow \Omega_a h^2 \approx 0.03$ , the standard value) gives  $\Omega_a h^2 \approx 0.66$  at the predicted  $\theta_i \approx 170^\circ$  —  $\sim 5.5\times$  above  $\Omega_{\text{DM}} = 0.12$ , reached only at  $\theta_i \approx 106^\circ$ . The over-production is thus *confirmed* (not the optimistic “all-DM”); `tfpt_4_frontier`, v185

docstring and the website updated accordingly. Stays [C]; kills only the axion-as-dominant-DM branch.

## 2026-06-14 (work packages A+B: attacking $\omega \circ \rho = \omega$ + the variational scan — v199, v200)

- **v199\_seam\_state\_invariance.py** [E]/[O] (claim QGEO.STATE.01, work package A). Attacking  $\omega \circ \rho = \omega$  directly on the *raw* seam. (1) [E] the setup is non-circular:  $\rho$  is the Coxeter element of  $W(A_3) = S_4$  ( $\rho^4 = 1$ , v117), an automorphism of the raw CAR algebra (16 Majoranas) — both carrier-side, no seam geometry. (2) [E] for a quasi-free RP state  $C_\Sigma = \frac{1}{2}(1 + \text{sgn } H_1)$ , so  $\omega \circ \rho = \omega \Leftrightarrow [\rho, H_1] = 0$ . (3) [E]  $[\rho, H_1] = 0 \Leftrightarrow H_1$  is block-diagonal in the four  $\mu_4$ -character classes  $\{n=r \bmod 4\}$  (verified). (4) [O] the residual is sharpened to “the bounded sub-principal symbol has no off-character matrix elements” — a bounded, finite-character condition, not a diffuse geometry; not a closure. Cited in *tfpt\_research\_contracts*; Wolfram-mirrored.
- **v200\_seam\_variational\_scan.py** [C]/[O] (claim QGEO.VARI.02, work package B). The discretised variational test: with the clock angle free,  $E_{\text{fail}}(\theta)$  has its unique minimum at  $\theta = \pi/2$  (the  $\mu_4$  clock  $z \mapsto iz$ ) — order-4 *and* the three distinct characters  $i, -1, -i$  = the (1, 2, 3) grading; decoys  $(0, \pi, \pi/3, 2\pi/3)$  all lose. A numerical sign that the variational principle selects  $\mu_4$ ; the full operator-valued minimisation stays the v199 residual [O]. Cited in *tfpt\_research\_contracts*.
- *Also (this pass, experiments/ only, not the ledger): the  $F_{\text{transfer}}$  four-pipeline scaffold (experiments/ftransfer/) with one CONTRACT per interface, and confirmation that the FRB preregistration (frb\_tfpt\_v1.yaml) and its status semantics already exist. Suite now 199 modules, highest ID v200; Wolfram extension 241/241.*

## 2026-06-14 (cracking the bedrock to a state-invariance: principal symbol + Tomita–Takesaki — v198)

- **v198\_modular\_commutator\_reduction.py** [O] (claim QGEO.MODULAR.01). The decisive crack on the last residual. (1) [E] the DtN principal symbol  $|k| = \sqrt{-\partial_\theta^2} = \text{diag}(|n|)$  and the clock  $\rho: z \mapsto iz = \text{diag}(i^n)$  are both diagonal in the Fourier basis, so  $[\rho, |k|] = 0$  *exactly* on all of  $L^2$  (verified on 33 modes) — the leading-order commutation is free on the whole boundary, never the open part. (2) [E] by Tomita–Takesaki the modular operator is intrinsic to  $(M, \Omega)$ , so a state-preserving symmetry ( $\rho\Omega = \Omega$ ) commutes with  $\log \Delta \sim \Lambda_\Sigma$ : hence  $[\rho, \Lambda_\Sigma] = 0$  *follows from*  $\omega \circ \rho = \omega$ , using only the state + reflection (RP) and *no* conformal covariance — which **removes the v194 Bisognano–Wichmann circularity worry**. (3) [E] the finite  $H^1$  block is already  $\mu_4$ -invariant (v177). So the entire residual collapses to the single *state-invariance* “the raw quasi-free seam state is  $\mu_4$ -invariant” ( $\omega \circ \rho = \omega \Leftarrow$  the Gaussian bulk measure is deck-invariant) — the maximally-operational, non-circular form of QGEO.SYM.01, [O]. *Not* a full closure (a foundational symmetry cannot be derived from nothing, like  $c = \text{const}$ ), but the sharpest falsifiable form, with the circularity gone. Cited in *tfpt\_research\_contracts*; OpenGates updated. Suite now 197 modules, highest ID v198; Wolfram extension 240/240.

## 2026-06-14 (three review levers: Marks-via-Lefschetz, E\_fail functional, RR→D5 — v195–v197)

- **v195\_marks\_lefschetz\_character.py** [E]/[O] (claim QGEO.MARKS.02). The four seam marks are *forced* (not posited):  $\rho: z \mapsto iz$  fixes only  $\{0, \infty\}$ ,  $H^1 = \chi_1 \oplus \chi_2 \oplus \chi_3$  (the  $A_3$  exponents,  $\text{Tr}(\rho|H^1) = i + i^2 + i^3 = -1$ ) has no trivial component  $\Rightarrow$  no puncture at a fixed point, and  $\text{rank } H^1 = n - 1 = 3 \Rightarrow n = 4 \Rightarrow$  a single free  $\mu_4$  orbit (cross-ratio 2). So

MARKS reduces from the geometric posit “ $D_4$  marks” to the milder premise “ $H^1$  carries the  $A_3$ -exponent rep” — less circular, still [O]. Cited in `tfpt_research_contracts`.

- **v196\_seam\_energy\_functional.py** [E]/[O] (**claim QGEO.VARI.01**). The failure functional  $E_{\text{fail}} = \|[\rho, \Lambda_\Sigma]\|_{HS}^2 + \|\rho^4 - 1\|^2 + \|\Theta\rho\Theta - \rho^{-1}\|^2$ , zero locus = the QGEO.SYM.01 conditions; on the finite  $H^1$  block it vanishes at the  $\mu_4$  config [E] and is  $> 0$  for any  $\mu_4$ -breaking perturbation, so  $\mu_4$  is a true minimiser. The full-operator minimisation is the v194 Bisognano–Wichmann residual — a concrete discretisable *target* on the open bedrock, [O]. Cited in `tfpt_research_contracts`.
- **v197\_rr\_carrier\_clifford\_d5.py** [E]/[C] (**claim ARCH.RRCAR.02**). Hardens the Riemann–Roch carrier from  $g_{\text{car}}$  to  $D_5$  itself:  $\Lambda^{\text{even}}(C_{\text{car}}) = \binom{5}{0} + \binom{5}{2} + \binom{5}{4} = 16 = 2^{g_{\text{car}}-1} = \dim S^+$ , the  $\mathfrak{so}(10) = D_5$  half-spinor — so  $D_5$  is the Clifford spinor of the 5-dim mode space, closing  $\mu_4 \rightarrow g_{\text{car}} \rightarrow D_5$ . [E] arithmetic, [C] identification (as v189). Cited in `origin_theory`.
- All three are validated external-review levers (the axion-hilltop “ $\pi - 3\phi_0$ ” proposal was *rejected* as a  $\pi \approx 3$  coincidence, not built). Suite now 196 modules, highest ID v197; Wolfram extension 239/239.

## 2026-06-14 (subclaim 2 of ENERGY.02 tackled: raw-seam-DtN RP-definability — v194)

- **v194\_raw\_seam\_dtn\_rp.py** [O] (**claim QGEO.DTN.01**). The next hard lever — “is the raw RP-seam DtN definable from RP data alone?” (sub-claim 2 of QGEO.ENERGY.02) — was tackled, and the honest result is a *reduction*, not a closure. (1) [E] the commutator  $[\rho, \Lambda_\Sigma] = 0$  closes on the finite cohomology  $H^1$  ( $\rho^*w_k = i^k w_k$ , distinct  $\mu_4$  characters; v177). (2) [E] the *hinge* is met: the DtN/transfer operator is RP-canonical (Osterwalder–Schrader, v54) on a quasi-free seam state (v155), so  $\Lambda_\Sigma$  is RP-definable without naming  $\mathbb{P}^1 \setminus \mu_4$ . (3) [O] the residual relocates: the full- $L^2$  commutation  $\Leftrightarrow$  Bisognano–Wichmann geometric modular flow of the quasi-free seam state (which must be intrinsic, else it re-circulates). So QGEO.SYM.01 reduces from a definitional postulate to “intrinsic BW geometric modular covariance” — one notch sharper, still [O]; the last structural fog point sharpens, it does *not* close to [E]. Cited in `tfpt_research_contracts`; OpenGates updated. Suite now 193 modules, highest ID v194.

## 2026-06-14 (QGEO.ENERGY.02 energy-commutator + QFT-wording hardening — v193)

- **v193\_qgeo\_energy\_commutator.py** [O] (**claim QGEO.ENERGY.02**). The non-circular sharpening of QGEO.ENERGY.01: the energy commutator  $[\rho, \Lambda_\Sigma] = 0$  on the rank-8 Calderón polarisation, with three sub-claims — (1) [E]  $\rho$  is carrier-defined (Coxeter of  $W(A_3)=S_4$ , v117), not imported from the seam; (2) [O] the non-circularity hinge:  $\Lambda_\Sigma$  must be the *raw* RP-seam DtN, not the  $\mu_4$  normal form; (3) [E] the commutator forces the (1, 2, 3) grading on  $H^1$  (QGEO.COHO.01). Exact [E] rider (Wolfram-mirrored): the induced EH coefficient  $k = (\ln m)/(12\pi)$  (v150) reproduces  $c_3/2$  iff  $\ln m = q(A_3) = \frac{3}{4}$  (*family*), not  $q(D_5) = \frac{5}{4}$  (carrier, off by 5/3) — gravity is family-geometry-induced. A *proof target*; if sub-claim (2) holds, QGEO.SYM.01 becomes an energy-rigidity theorem. Cited in `tfpt_research_contracts`. Suite now 192 modules, highest ID v193; Wolfram extension 236/236.
- **QFT-wording hardening (review point 7)**. `tfpt_2_standard_model` §(4) was retitled to “Constructive QFT closure of the admissible physical sector”, and its standalone opening over-claim replaced by an explicitly scoped sentence (the unprojected ambient metric measure is not constructed;  $G_{\text{metric}}$  frontier). The two superseded standalone phrasings were added to the v188 prose guard so a frozen release cannot re-introduce them.

**2026-06-14 (geometry round: Riemann-Roch carrier, Nariai bound, branch line, energy clock — v189–v192)**

- **v189\_riemann\_roch\_carrier.py** [E]/[C] (claim ARCH.RRCAR.01). The pair  $(g_{\text{car}}, N_{\text{fam}}) = (5, 3)$  are the two canonical invariants of *one* object, the four-point seam divisor  $D = \mu_4$  on  $\mathbb{P}^1$ : the cycle side  $\text{rank } H_1(\mathbb{P}^1 \setminus \mu_4) = \deg -1 = 3 = N_{\text{fam}}$  (already QGEO.COHO.01) and the function side  $h^0(\mathbb{P}^1, \mathcal{O}(\mu_4)) = \deg +1 = 5 = g_{\text{car}}$ , matched by  $\text{rank}(D_5) = 5$ . The carrier-as-mode-space reading is [C] (geometric, does not displace the over-determined  $g_{\text{car}} = 5$ ). Cited in `origin_theory`; Wolfram-mirrored.
- **v190\_nariai\_entropy\_bound.py** [E] (claim HOR.NARIAI.02). A variation-free one-line proof of the SdS entropy floor:  $3(x^2+1) - 2\Phi_3(x) = (x-1)^2 \geq 0$ , so  $S_{\text{tot}} \geq \frac{2}{3}S_{dS}$  with equality iff  $x = 1$  (the Nariai merge). Cited in `tfpt_horizon_readouts`; Wolfram-mirrored.
- **v191\_universal\_branch\_line.py** [C] (claim HOR.BRANCHLINE.01). The affine map  $q = \frac{7}{2} + \frac{N_{\text{fam}}^2}{2}m$  carries the SdS branch points  $\pm 1/N_{\text{fam}}$  onto the flavor labels  $\{2, 5\}$ . *Honestly typed* [C] — an exact *relabeling*, not a theorem (a decoy pair  $\{3, 4\}$  admits an equally exact map; the shared content is the known  $2/3$  ramification). Cited in `tfpt_horizon_readouts`; Wolfram-mirrored.
- **v192\_qgeo\_energy\_clock.py** [O] (claim QGEO.ENERGY.01). An operator-checkable restatement of the bedrock:  $\rho^* \Lambda_\Sigma \rho = \Lambda_\Sigma$  (the clock preserves the DtN energy form). *Honestly typed*: it *relocates* QGEO.SYM.01 to a more falsifiable surface but does *not* eliminate it (plausibly equivalent unless  $\rho$  is independently defined and the invariance proven). Cited in `tfpt_research_contracts`. Suite now 191 modules, highest ID v192 (v186 skipped).
- *All four are validated external-review proposals (problem\_b); each is built with the honest typing determined on review — the branch line is recorded as an alignment, not a theorem, and the energy clock as a relocation, not a closure.*

**2026-06-14 (Zenodo release 5.1 (rev 120) — F\_transfer firewall, v184–v188)**

- **Published to Zenodo.** Version 5.1 (rev 120) of the ten-PDF document set was published as a new version of the TFPT deposit — record 20687564, DOI 10.5281/zenodo.20687564. This version carries the F\_transfer-firewall round (v184–v188): the honest  $\eta_B$  anchored-Boltzmann test (sharper scenario, not a closure), the axion hilltop-relic typing, the ledger v187 and prose v188 guards, and the Route B/Koide/dark-matter wording fixes plus the P1/P2-vs-QGEO.SYM.01 two-layer clarification. The uploader’s DEFAULT\_RECORD\_ID was advanced to the new record for the next release. Suite 187 modules (highest ID v188; v186 skipped). `run_all.py` green, audit clean, website built.

**2026-06-14 (F\_transfer round: eta\_B honest test v184, axion relic v185, discipline guard v187)**

- **v184\_etaB\_anchored\_boltzmann.py** [C] (claim FR.ETAB.03). An honest test of the external-review  $\eta_B$  proposal ( $M_1 = M_R \varphi_0^4$ ,  $\tilde{m}_1 = m_3/A_\Lambda$ ). Firewall verdict: *sharper scenario, not a closure*. The washout anchors plausibly ( $\tilde{m}_1 = m_3/10 \approx 5$  meV,  $A_\Lambda = 10$  atom, in the strong-washout window) [C]; but  $M_1$  does *not* —  $M_R$  is the seesaw scale  $\sim v^2/m_3$  and the nearest clean TFPT powers of  $\bar{M}_{\text{P1}}$  both miss it ( $c_3/\varphi_0$ -powers off  $\sim 75\%/ \sim 40\%$ ), so  $M_1 = M_R \varphi_0^4$  merely relocates the free input.  $\eta_B$  stays [C], the route viable but with one (not zero) free input. Cited in `tfpt_4_frontier`.
- **v185\_axion\_relic\_solver.py** [C] (claim FR.DM.03). Honest misalignment estimate: the determinant-line axion  $f_a = M_{\text{scal}}/128 \approx 2.39 \times 10^{11}$  GeV ( $m_a \approx 23.8 \mu\text{eV}$ ) CAN be all dark matter, but the harmonic misalignment under-produces ( $\sim 21\%$  at  $\theta \sim O(1)$ ), so the  $\theta_i \approx 170^\circ$  hilltop anharmonic enhancement is *required* — where the relic is exponentially sensitive.

Hence a [C] scenario, not a sharp prediction; the relic contract is a kill test for the BRANCH, not the theory. Cited in `tfpt_4_frontier`.

- **v187\_ftransfer\_laws.py** [E] (**claim FR.TRANSFER.01**). A CI-style firewall guard: the four continuous-transfer frontier observables (Koide,  $\eta_B$ , axion,  $m_p/m_e$ ) are read from the ledger and required to carry a [C]/[O] marker — never a primitive [E] compiler prediction (exact sub-parts like the Koide 53/54 factor or  $b_3 = -7$  may be [E], the physical prediction never is). Records the four interfaces  $F_{\text{pole}}/F_{\text{Boltzmann}}/F_{\text{relic}}/F_{\text{QCD}}$ ; prevents any future edit from quietly promoting a frontier number to a closed output. Cited in `tfpt_4_frontier`.
- **v188\_frontier\_wording\_guard.py** [E] (**claim AUDIT.WORDING.01**). A prose sentinel (no new physics): v187 protects the ledger *typing*, this guard protects the *prose*. It scans the 10 active documents and forbids five stale phrasings whose semantics earlier rounds superseded (the old [E]-typed Route B leptogenesis claim and its heavy-scale wording  $\rightarrow$  v184 [C]; the old “open relic scale” dark-matter line  $\rightarrow$  v185; the old “near, not exact” Koide line  $\rightarrow$  v183; and the module-count/ID conflation), so a frozen Zenodo release can never re-introduce wording that was true yesterday and is half-true today. The same pass updated the `tfpt_4_frontier` Route B (to [C]), the Koide and dark-matter summary lines, and added the P1/P2-vs-QGEO.SYM.01 two-layer clarification to the introduction. Suite now has **187 modules**, highest verification ID v188 (v186 was intentionally skipped as redundant —  $m_p/m_e$  is already typed [O] as a QCD matching contract, `QCD.LAMBDA.01`).
- *Review framing points (P1/P2 as projections of the seam-deck postulate,  $v_{\text{geo}}$  as unit gauge,  $G_{\text{net}}$  a corollary of QGEO.SYM.01, grammar tuning = data) were validated as already implemented in the prior rounds;  $m_p/m_e$  is already typed [O] as a QCD matching contract (QCD.LAMBDA.01), so no redundant module was added.*

## 2026-06-14 (operator origin of the Koide 53/54 factor: the missing Sheet-Diamond corner, v183)

- **v183\_koide\_f\_corner\_transfer.py** [E]/[C] (**claim FR.KOIDE.07**). An external-review proposal, validated exactly: the Koide source $\rightarrow$ pole factor  $\frac{53}{54}$  is not a bare arithmetic guess but an *operator* identity on the carrier,  $u_{\text{pole}}/u_{\text{source}} = a^T(R+Q)\mathbf{1}/(2\mathbf{1}^T Ra) = 53/(2\cdot 27) = \frac{53}{54}$ . Here  $\mathbf{1}^T Ra = 27$  is the  $E_6 \times A_2$  cubic family block ( $248 = \dim E_6 + \det R + 2(\mathbf{1}^T Ra)N_{\text{fam}} = 78 + 8 + 2\cdot 27\cdot 3$ ) and  $a^T(R+Q)\mathbf{1} = 53 = 52 + 1$ , where  $F = R+Q$  is the *missing* Sheet-Diamond corner with  $\det B_F = 52 = \dim F_4$  (v80). Hard negative control:  $a^T M\mathbf{1}$  for  $M \in \{R, Q, K, L\}$  is  $\{32, 21, 35, 56\}$  — only the absent corner  $F$  gives 53. This upgrades FR.KOIDE.03 from “near miss + conjecture” to “operator-sourced source $\rightarrow$ pole transfer”: the *factor* is [E] (exact, mirrored in Wolfram, 232/232); the *mechanism* (the leptonic source $\rightarrow$ pole transfer reads the  $F$  corner) stays [C], an  $F_{\text{transfer}}$  special case. The prediction itself stays [C]. Cited in `tfpt_4_frontier` (Koide section). Suite now 183 modules.
- **Review point 2 noted as already-present:** the claim that flavor and Nariai share one double-cover ClockFunctor (flavor rate  $(2/3)^{2N_{\text{fam}}}$  vs gravity curvature  $(2/3)^{N_{\text{fam}}}$ , exponent ratio  $|\mathbb{Z}_2|$ , “one clock, two geometries”) is exactly HOR.TRISECT.01/v103 — a rediscovery of an existing result, no new module.

## 2026-06-14 (review recommendations 1/3/4 + the four concerns A–D mapped to the named residuals)

- **v182\_reviewer\_residual\_map.py** [E]/[C] (**claim REVIEW.MAP.01**). A careful external review raised four concerns (A mapping  $2D \rightarrow 4D$ , B UV-gravity abstractness, C grammar-tuning/meta-look-elsewhere, D Koide Möbius flow). This module shows three of them *reduce*, by the same discipline as v175–v181, onto the *two already-named* residuals  $\{\text{QGEO.SYM.01}, F_{\text{transfer}}\}$ : A (Plücker-11 = QBL boundary count v174 + holographic

reduction), B ( $\equiv$  QGEO.SYM.01), D (Möbius forced by  $\text{Aut}(\mathbb{P}) = \text{PSL}(2, \mathbb{C})$  on the deck; the missing generator  $= F_{\text{transfer}}$ ). Only C is genuinely *experiment-only* (frozen + minimal + over-determined grammar; residual falls to JUNO/CMB-S4). A unification of the concern *structure*, not a closure. Cited in `tfpt_5_redteam`. Suite now 182 modules.

- **Review recommendation #3 — a “Rosetta” translation lexicon** (website `RosettaLexicon`): TFPT’s compiler-speak (seam, deck, readout, microcode, audit hull, gap, anchor,  $F_{\text{transfer}}$ ,  $v_{\text{geo}}$ , QGEO.SYM.01) mapped onto standard QFT / mathematical-physics language (Wilsonian boundary CFT, orbifold deck, index/intersection number, RG irrelevance rate, renormalization condition, fundamental postulate) — a reading aid for mainstream physicists, not a new claim.
- **Review recommendation #1 — kill-tests up front** (website `TrustContract`): the frozen pre-data predictions ( $\sin^2\theta_{12} = 0.3067$ , JUNO;  $r \approx 0.004$ , CMB-S4) and the null model ( $P \leq 10^{-30.7}$ ) are now a prominent strip directly under the hero.
- **Review recommendation #4 — the bedrock as a postulate.** QGEO.SYM.01 is now framed confidently as TFPT’s *one fundamental postulate* (the role “ $c = \text{const}$ ” plays in relativity), not a gap — in the `tfpt_research_contracts` box and the website “what is open” card; the achievement is that the whole theory is compressed onto exactly one sharply-located, falsifiable premise.

**2026-06-14 (external-review response: CSV fix,  $G_{\text{net}}$  3-level, QGEO.SYM.01 page, claim detox)**

- **Ledger CSV bug fixed + guarded.** Three rows (AX.P1.01, AX.P2.01, QGEO.ISO.01) had *unquoted commas* in a status/external field (e.g. “a=(1,1,2)”, “PSL(2,C)”); so a strict CSV parser read 11–12 columns instead of 10 — a real machine-readability defect in the single source of truth. All offending fields are now quoted (every row parses to exactly 10 columns), and `audit_sync.py` gained a `check_csv_wellformed` guard (ledger, registry, clusters, freeze must all parse to constant width) so it cannot regress.
- **$G_{\text{net}}$  typed in three explicit levels** (website “What is open” card): (1) *algebra* [E]  $(D_5)_1 \otimes (A_3)_1 \rtimes \langle (1,1) \rangle \cong (E_8)_1$ ; (2) *AQFT machinery* [E] (net existence + full-cone RP, CAR functor, v175); (3) *seam instantiation* [O] (QGEO.SYM.01). The target object’s mathematics is closed; only the physical coupling of the raw seam is open.
- **QGEO.SYM.01 now has its own one-page box** in `tfpt_research_contracts`: what is identified, why it is weaker than CONF/ISO, what follows automatically (the conditional theorem), and *what would falsify it* (genus  $> 0$  double; non-deck transport;  $N_{\text{fam}} \neq 3$ ); with the explicit conditional framing “*given* QGEO.SYM.01, the closure follows” rather than “the theory derives the seam geometry”.
- **Claim-language detox.** “no free fundamental number” softened to “no free *dimensionless compiler dial* beyond the anchor and  $\pi$  (dimensionful scale + transfer physics stay typed)” (`origin_theory`, `introduction`, website); the site title softened from “the Standard Model Derived” to “the Standard-Model *Skeleton* Derived”.
- **External-reviewer map** added to the website verification page: four boxes (exact kernel [E], conditional physics [C], open premises [O], kill tests [X]) so a reviewer sees the attack surface without reading 181 scripts. Suite unchanged at 181.

**2026-06-14 (Zenodo release 5.1 (rev 114) — v175–v181 bedrock + figures + P1/P2)**

- **Published to Zenodo.** Version 5.1 (rev 114) of the ten-PDF document set was published as a new version of the TFPT deposit — record 20686284, DOI 10.5281/zenodo.20686284.

This version carries the AQFT-closure-to-bedrock chain (v175–v181, the structural residual reduced to the single definitional premise QGEO.SYM.01), the two overview figures (residual\_chain, script\_timeline), and the P1/P2 anchor-reduction provenance on the axiom surfaces. The README and the Zenodo description were refreshed (181 modules), and the uploader’s DEFAULT\_RECORD\_ID was advanced to the new record for the next release.

## 2026-06-14 (overview figures + P1/P2 reduction reflected on the axiom surfaces)

- **Two overview figures** (make\_figures.py; PDF + website PNG). (i) `residual_chain` — the structural-residual reduction chain v175–v181 as a descending staircase from “build a quantum-gravity measure” to the single definitional bedrock QGEO.SYM.01; embedded in `tfpt_research_contracts`. (ii) `script_timeline` — the eight-phase journey of the ~181 scripts (foundations → SM readouts → seam= horizon → horizon/flavor geometry → R1–R5/premise-(A) → PyR@TE cross-checks → AQFT bridges → AQFT closure), what each did mathematically and physically; embedded in `introduction` ahead of the master script index, and both shown on the website verification page.
- **P1/P2 reduction now reflected on the axiom surfaces.** The reduction of the two axioms to the single parabolic anchor  $a = (1, 1, 2)$  plus  $\pi$  (with  $g_{\text{car}} = 5$  an over-determined bootstrap fixed point — forced three ways, v6, Pascal-unique and Lean-formalised, FORM.LADDER.01) was already established (ANCHOR.GEN.01/v23, ARCH.CORE.01/v53) and stated in the papers/glossary, but the *axiom surfaces* still read as bare “declared inputs”. Fixed: the ledger rows AX.P1.01/AX.P2.01 now carry the anchor/bootstrap reduction and point to it, and the website dependency-graph nodes P1/P2 now show the anchor input ( $c_3 = 1/(2e_1(a)\pi)$ ;  $g_{\text{car}} = e_2(a)$ , bootstrap-forced) rather than “declared axiom”. No marker change (still declared inputs), only honest provenance.
- Manifests refreshed (two new figures added to the FIG list). Suite unchanged at 181.

## 2026-06-14 (the final reduction to bedrock v181 + full non-changelog status sweep)

- **v181\_clock\_is\_conformal\_symmetry.py [E]/[O] (claim QGEO.SYM.01).** The isometry premise QGEO.ISO.01 (v180) is weakened one more genuine step — to a conformal-*symmetry* premise — via *equivariant uniformisation* (Nielsen realisation; for cyclic groups Kerékjártó + uniformisation): a *conformal* automorphism (strictly weaker than an isometry — every isometry is conformal, conformal allows any positive scale) already suffices to put the order-4 clock in the form  $z \mapsto iz$ . Since the clock is the Coxeter element of  $W(A_3) = S_4$  (v117), the  $\mu_4$  deck, and a deck transformation preserves the pulled-back conformal structure by definition, the residual collapses to the *definitional* identification QGEO.SYM.01: “the seam’s conformal structure *is* the  $\mu_4$ -deck structure of the carrier clock”. This is bedrock — a physical/definitional statement tying P2 to the seam’s conformal geometry, *not* a finite computation and not reducible further without relabeling. The chain `REALIZE(v175)→...→SYM.01(v181)` ends here; we stop honestly. Python-only. Suite now 181 modules.
- **Full non-changelog status sweep.** The current bedrock framing was propagated to every live status surface (not just the changelog): the `introduction` “latest reductions” prose, `tfpt_5_redteam` Target A, the `tfpt_research_contracts` central-theorem keybox, `origin_theory`; and on the website `OpenGates`, `VerificationDag`, `papers.ts` (Target A  $\times 2$ ,  $G_{\text{net}}$ ), the `glossary`  $G_{\text{net}}$  entry and the `faq` live-residual answer — all now end at QGEO.SYM.01.

## 2026-06-14 (the conformal premise reduces to a milder isometry premise v180)

- **v180\_clock\_is\_mobius.py [E]/[O] (claim QGEO.ISO.01).** Cracks QGEO.CONF.01 one level further: the conformal-realisation premise is *replaced*, via three classical theorems, by a strictly *milder, non-circular* premise. (1) *Uniformisation* (Poincaré–Koebe): a genus-0 Riemann surface is conformally  $\mathbb{P}$ , so the genus-0 seam double (P1, v179) with its RP-metric conformal structure *is*  $\mathbb{P}$  — the target is *produced*, not assumed; (2) *Kerékjártó* (1919; Constantin–Kolev 2003): a finite-order orientation-preserving homeomorphism of  $S^2$  is conjugate to a rotation, so the order-4 clock ( $h(A_3)=4$ ) is topologically  $z \mapsto iz$ ; (3) *isometry*  $\Rightarrow$  *conformal*  $\Rightarrow$  *Möbius*: an orientation-preserving isometry of a Riemannian surface is holomorphic,  $\text{Aut}(\mathbb{P}) = \text{PSL}(2, \mathbb{C})$ ; (4) an order-4 Möbius map is  $z \mapsto iz$  (verified: projective order 4, fixed  $\{0, \infty\}$ , multiplier  $i$ , free orbit  $\mu_4$ ). Hence QGEO.CONF.01 is *implied* by the milder premise QGEO.ISO.01: “the carrier clock is an order-4 orientation-preserving *isometry* of the RP seam metric” — which does not name  $\mathbb{P}/\mu_4$  (uniformisation produces them) and is the natural statement that the seam transport is metric-compatible (a holonomy preserving the RP inner product). That milder premise is the single, now milder, structural residual of the whole theory; its surface realisation from the raw seam stays honestly [O]. Cited in `tfpt_research_contracts`; Python-only. Suite now 180 modules.

## 2026-06-14 (the two obligations collapse to ONE geometric premise v179)

- **v179\_conformal\_realisation.py [E]/[O] (claim QGEO.CONF.01).** Investigating the two residual premises left by v178 shows they are *not* independent — they collapse onto a single geometric premise (honest unification; the premise itself stays [O]). MARKS’s residual is not three independent assumptions: (1) genus-0 *is* P1 — the one-sided Gauss–Bonnet  $c_3 = 1/(|\mathbb{Z}_2| 2\pi \chi(S^2)) = 1/(8\pi)$  uses  $\chi(S^2) = 2$  and  $\chi = 2 - 2g \Rightarrow g = 0$ , so “genus-0” is the same  $\chi = 2$  that gives the “8” in  $c_3$  (v58/v73); (2) the order-4 clock *is* the carrier — the Coxeter element of  $W(A_3) = S_4$  (v117), order  $h(A_3) = 4 = |\mu_4|$ ; (3) marks = one orbit is *forced* (the parabolic marks are not the two elliptic fixed points, a free order-4 orbit has exactly  $4 = e_1(a)$  points). KERNEL then *follows* from the same geometry (DtN symbol  $|k|$ , Lee–Uhlmann v156;  $c = 8$  rigidity v158; the cusped-surface residual  $= H^1 = \mathbb{C}^3 = N_{\text{fam}}$ ). So the *entire* structural residual of the theory is the *single* geometric premise QGEO.CONF.01: “the raw RP seam normal double is conformally  $(\mathbb{P}, \mu_4)$  with the carrier clock as the order-4 Möbius automorphism”. The two doors of v177/v178 are one geometric realisation, left honestly [O]. Cited in `tfpt_research_contracts`; Python-only. Suite now 179 modules.

## 2026-06-14 (an honest attempt at the two obligations v178 — deeper reduction, not closure)

- **v178\_marks\_kernel\_attempt.py [E]/[O] (claim QGEO.REDUCE.01).** An honest attempt at QGEO.MARKS.01 and QGEO.KERNEL.01. These are genuine constructive-geometry/AQFT statements, *not* finite computations — neither is closed. What the module does is push each as far as computation allows: it closes the finite [E] core of each and reduces each to *one sharper* premise. *Marks core* [E]: given a genus-0 double with an order-4 clock  $\rho$  (fixed points  $\{0, \infty\}$ ) and reflection  $\tau$ , the marks are forced to be a generic  $\rho$ -orbit  $\{1, i, -1, -i\} = \mu_4$  (*not* the fixed points), distinct iff  $b_1 = 4 - 1 = 3 = N_{\text{fam}}$  (any collision drops  $b_1$ ), with  $\tau\rho\tau = \rho^{-1}$  giving a faithful  $D_4$  of order 8 — so MARKS reduces to the single topological/clock premise. *Kernel core* [E]: on the 3-dim multiplicity-free  $H^1$  the  $\mu_4$  characters  $(1, 2, 3)$  are distinct, so by Schur a  $\mu_4$ -equivariant operator is forced *diagonal* and completely determined by its eigenvalue per character; two such with the forced transfer spectrum  $\{1, (2/3)^6, (1/3)^6\}$  (v162) and the unique residue-normalised basis (v177) are *equal as operators*, not merely isospectral — the spectrum-vs-operator worry *vanishes* on the finite block. So KERNEL reduces to the single full-operator-reduction premise (principal symbol  $|k|$ , Lee–Uhlmann/v156; no drift v158).

Net: neither obligation closed; both sharpened to milder premises with their finite cores [E]. Cited in `tfpt_research_contracts`; Python-only (Möbius/Schur symbolic + numeric). Suite now 178 modules.

## 2026-06-14 (QGEO proof tree v177 — the residual sharpened into two named obligations)

- `v177_seam_marking_kernel.py` [E]/[O] (claims QGEO.TREE.01, QGEO.MARKS.01, QGEO.KERNEL.01). A second external residual-class audit pointed out that the v176 statement takes the marks/ $D_4$  as a *hypothesis* — a near-circular trap. This module factors the central theorem honestly as `REALIZE = MARKS · UNIFORM · COHOM · MODULE · KERNEL · NET` and closes *four* nodes *constructively* (all validated symbolically before adoption): `UNIFORM [E]` — an order-4 Möbius map conjugates to  $z \mapsto iz$ , a non-fixed orbit scales to  $\mu_4$ , and  $\sigma\rho\sigma = \rho^{-1}$  for  $\sigma : z \mapsto 1/z$  gives a faithful  $D_4$  (order 8), so a genus-0 four-marked faithful- $D_4$  boundary *is*  $(\mathbb{P}, \mu_4)$  (strengthening v168’s cross-ratio to the full uniformisation); `COHOM [E]` —  $\rho^*\omega_k = i^k\omega_k$ , grading  $(1, 2, 3) = A_3$  exponents; `MODULE [E]` — a *unique*  $\mu_4$ -equivariant iso (multiplicity-free + residue normalisation; reflection  $\omega_1 \leftrightarrow \omega_3$ ,  $\omega_2$  fixed); `NET [E]` (v154/v175). The structural residual is thereby sharpened from one diffuse premise into *exactly two* named open obligations, neither a finite computation: QGEO.MARKS.01 (the raw RP seam collar canonically produces the genus-0 four-marked  $D_4$  boundary) and QGEO.KERNEL.01 (the raw seam Calderón operator *is* the  $\mu_4$ -equivariant free gapped contraction, as an operator). Cited in `tfpt_research_contracts` (the central-theorem keybox); Python-only (Möbius/cohomology symbolic; numbers mirrored via v168/v137/v162/v154/v175). Suite now 177 modules.

## 2026-06-13 (Seam Collar Realisation Theorem v176 — the one central proof obligation)

- `v176_seam_collar_realisation.py` [E]/[O] (claim QGEO.REALIZE.02). Following an external residual-class audit, the single remaining structural item is formulated as one central theorem and certified as a *reduction* (not a fabricated proof). *Theorem*: given an oriented one-sided RP seam collar with a sheet-odd Calderón involution, faithful  $D_4$  marks, four parabolic marks with  $\mu_4$  decomposition and admissible  $c=8$  data, its conformal boundary is Möbius-equivalent to  $\mathbb{P} \setminus \mu_4$  with  $H^1$  grading  $(1, 2, 3)$ , the Calderón contraction is the rank-8  $K_\Sigma$  polarisation and the index-4 extension is  $(E_8)_1$ . The proof decomposes into six steps, *five already* [E]: geometry/uniformisation (v168), Calderón DtN symbol  $|k|$  (v156),  $H^1$  character = generation grading  $(1, 2, 3) = A_3$  exponents (v137/v168),  $\mu_4$ -equivariant contraction with gap  $(2/3)^6$  (v162), AQFT closure  $(D_5)_1 \otimes (A_3)_1 \rtimes \mu_4 = (E_8)_1 + \text{full-cone RP all } m$  (v154/v175). The whole structural residual thus reduces to the *one* open premise QGEO.REALIZE.01: that the physical seam collar’s boundary *is* this  $\mathbb{P} \setminus \mu_4$  — a constructive-geometry statement, left honestly [O]. Cited in `tfpt_research_contracts` (the central theorem keybox); assembly/reduction certificate, Python-only (its exact identities are mirrored via v168/v156/v162/v154/v175). Suite now 176 modules. *Audit note*: the metrology residual the same review flags is already discharged — v153 (No-Unit Theorem) is the metrology-line typing ( $v_{\text{geo}} \mapsto \lambda^{-1}v_{\text{geo}}$ ,  $1/G \mapsto v_{\text{geo}}^2$ , dimensionless data invariant), and the Koide  $u \rightarrow \frac{53}{54}u$  source→pole transfer is already typed [C] (FR.KOIDE.03); neither is re-derived.

## 2026-06-13 (Zenodo release 5.1 (rev 105) — v174 + v175 + doc refresh)

- **Published to Zenodo.** Version 5.1 (rev 105) of the ten-PDF document set (introduction, origin\_theory, tfpt\_1–5, horizon, research\_contracts, changelog) was published as a new version of the TFPT deposit — record 20681451, DOI 10.5281/zenodo.20681451. This version carries the AQFT closure round (v170–v175): the seam Fock-space readings and

the heavy-artillery discharge of net existence + full-cone reflection positivity to [E]. The README and the Zenodo description were refreshed to the current state (175 modules, Wolfram 116/116 + 231/231), and the uploader's DEFAULT\_RECORD\_ID was advanced to the new record for the next release.

## 2026-06-13 (the heavy artillery v175: net existence + full-cone RP discharged to [E])

- **v175\_net\_existence\_full\_cone.py** [E]/[O] (claim QFT.NETRP.01). The two structural residuals that the premise-(A) chain had relocated to *already-open* items —  $A_2$  net existence and full-cone reflection positivity — are now *discharged to a theorem*, leaving the seam realisation as the single irreducible open premise (nothing fabricated). (1) *Full-cone RP for all  $m$* : the CAR second-quantisation functor  $\Gamma(t) = \bigoplus_m \Lambda^m(t)$  lifts the one-particle seam contraction  $t$  to the complete Fock space  $\Lambda^\bullet(\mathbb{R}^{16})$  ( $\dim 2^{16} = 65536$ ); by the exterior-power spectrum theorem every eigenvalue is a subset product of  $\text{spec}(t)$ , so all  $2^{16}$  products lie in  $[0, 1]$  (a contraction = full-cone RP), with eigenvalue-1 multiplicity  $2^8 = 256$  (wedges of the rank-8  $K_\Sigma$  polarisation) and sub-leading = the one-particle gap  $(2/3)^6$  — so full-cone RP reduces *for every  $m$*  (not only  $m \leq 6$ , v161) to the one-particle contraction (Shale–Stinespring), discharging the v160 scope premise. (2)  $A_2$  *assembled + verified*: the  $(E_8)_1$  character  $E_4/\eta^8 = 1 + 248q + 4124q^2 + 34752q^3 + 213126q^4 + 1057504q^5$  (level-1 = 248, extends the order-4 v156 check by one order), the embedding  $248 = 120 + 128$ , and the  $E_8$  Cartan matrix even with  $\det 1$  (the unique rank-8 even unimodular lattice, v83); the net *exists* by cited theorems (free-fermion net, lattice VOA, index-4 Q-system, v125). (3) Both then need the *same* input — the seam realisation QGEO.REALIZE.01 (finite half proven, v168) — which is a constructive-geometry/AQFT statement, *not* closeable by a finite computation, and is left honestly [O]. **Net**: net existence and full-cone RP become [E]; the one irreducible structural premise is the seam realisation. Cited in **tfpt\_research\_contracts** (premise (A) closure) and **tfpt\_5\_redteam** (Target A); the exact parts (Cartan unimodular, character order 5, functoriality integers) mirrored in the Wolfram extension (231/231). Suite now 175 modules.

## 2026-06-13 (seam Fock-space readings v174: CAR cone, Witten index, g-theorem)

- **v174\_seam\_fock\_readings.py** [E] (claim QFT.FOCK.01). Three more exact audit bridges from the same source note. (1) *CAR cone compression*: the even CAR algebra of the 16 seam Majoranas has  $\dim \Lambda^{\text{even}}(\mathbb{R}^{16}) = 2^{15} = 32768$  directions, but a quasi-free Bogoliubov transport reduces the whole cone to a 16-dim one-particle problem ( $2^{15}/16 = 2^{11}$ ) — the explicit count behind v161. (2) *Witten index = the Plücker norm 11*: the four  $\mu_4$ -corner fermions span a  $2^4 = 16 = \dim S^+$  Fock space; QBL ( $\deg \leq 2$ ) keeps  $\sum_{k \leq 2} \binom{4}{k} = 11 = \dim S^+ - g_{\text{car}}$ , so  $c_u/c_d = 55/117$  reads a topological state-space dimension. (3) *Affleck–Ludwig g-theorem*:  $c_{UV} = 5 + 3 = 8 = c_{IR}((E_8)_1)$ , so  $\Delta c = 0$  forces an exact isometry — the boundary-entropy form of the v157/v158 rigid fixed point. Cited in **tfpt\_2** (Exterior Leg Lemma); mirrored in the Wolfram extension (230/230). Suite now 174 modules. (The categorical reframings in the source note remain framing, not asserted claims.)

## 2026-06-13 (CFT/QFT bridges: slice compression v170, OS moment v171, trace-anomaly seed v172, Pfaffian CP v173)

- **v170\_diamond\_slice\_compression.py** [E] (claim E8.SLICE.01). E8 Slice Compression Lemma: the seven  $E_8$  audit slices are *one* projection of two alphabets — anchor power sums  $P = (3, 4, 6, 10)$  ( $p_n = 2 + 2^n$ ) and Sheet-Diamond determinants  $(3, 4, 8, 14, 20, 32)$  summing to  $81 = N_{\text{fam}}^4$ . All seven 248-slices, the  $K_4$  edge graph, and the row-budget cross  $(7, 11, 13) + s(3, 3, 3) + t(1, 2, 3)$  follow;  $78 = \dim E_6 = p_2(p_0 + p_3)$ . Cited in **tfpt\_3** (slice atlas). Exact; reading is audit (anti-numerology).

- **v171\_os\_moment\_cluster.py** [E]/[C] (**claim QFT.OSMOMENT.01**). Atomic OS Moment Lemma:  $\text{spec}(T) = \{1, 64/729, 1/729\}$  makes every correlator a positive moment sequence, so the OS Hankel matrices factorise as Vandermonde Grams (a clean RP certificate); cluster  $729/665$ ,  $\xi = 0.4111$ . Sugawara Gap Safety:  $31 = 1 + h^\vee(E_8)$ ,  $c((E_8)_1) = 248/31 = 8$ ,  $\Delta_{\text{eff}} = 6 \log \frac{3}{2} - 31/(4\pi^2) > 0$ . Cited in **tfpt\_research\_contracts** (G5).
- **v172\_trace\_anomaly\_seed.py** [E] (**claim SEED.ANOMALY.01**). The seed prefactor  $\frac{4}{3}$  is the halved  $(E_8)_1$  Weyl anomaly: over the seam  $S^2$  ( $\chi = 2$ ) the  $c = 8$  trace anomaly is  $c\chi/6 = \frac{8}{3}$ , halved by the one-sided  $|\mathbb{Z}_2|$  to  $\frac{4}{3} = 16/12$ . Plus QCD  $b_3 = 7 = \text{scalaron exponent } 48 - 41$  (confinement  $\leftrightarrow$  inflation), and the Volkov–Wipf  $-98/45 = -2 \cdot 7^2 / \dim(D_5)$  factorisation. Cited in **tfpt\_2** (seed).
- **v173\_pfaffian\_cp\_car.py** [E] (**claim FLAV.STRONGCP.02**). Strong CP  $\theta_{\text{eff}} = 0$  as Pfaffian reality in the self-dual 16-Majorana CAR model:  $\{D, \Sigma\} = 0$  pairs the spectrum,  $\gamma_5$ -Hermiticity makes the Pfaffian real, and RP picks the positive branch. Cited in **tfpt\_2** (Strong CP section).
- All four validated computationally before adoption; the exact identities (v170/v171/v172) are mirrored in the Wolfram extension (229/229). Suite now 173 modules. (The categorical reframings in the source note — compiler-as-single-functor,  $E_8 = K_0$ , GIT/monadic/Langlands — are recorded as framing, not asserted as machine-checked claims.)

## 2026-06-13 (Zenodo release 5.1 (rev 97))

- **Published to Zenodo.** Version 5.1 (rev 97) of the ten-PDF document set (introduction, origin\_theory, tfpt\_1–5, horizon, research\_contracts, changelog) was published as a new version of the TFPT deposit — record 20678568, DOI 10.5281/zenodo.20678568. The uploader’s DEFAULT\_RECORD\_ID was advanced to the new record for the next release. Also fixed **build.sh zenodo** on macOS bash 3.2 (empty-array expansion under **set -u**).

## 2026-06-13 (QGEO rigidity theorem v168 + $\eta_B$ leptogenesis interface v169)

- **v168\_qgeo\_rigidity.py** [E]/[C] (**claims QGEO.RIGID.01, QGEO.REALIZE.01**). Hardens the *finite* half of the one remaining Tier-1 hinge **GATE.QGEO** (the seam = flavour = horizon identification). Exact, machine-checked:  $\mu_4 = \{1, i, -1, -i\}$  has cross-ratio 2 and a faithful Möbius stabiliser of order  $8 = |D_4|$ ;  $H^1(\mathbb{P}^1 \setminus \mu_4)$  has rank  $4 - 1 = N_{\text{fam}} = 3$ ; and the forms  $\omega_k = z^{k-1} dz / (z^4 - 1)$  ( $k = 1, 2, 3$ ) are  $\mu_4$ -eigenforms with characters  $\chi_1, \chi_2, \chi_3$  of weights  $(1, 2, 3) = A_3$  exponents =  $\text{Spec}(Q_+)$ . So a genus-0 boundary with faithful  $D_4$ , four parabolic marks and this  $\mu_4$ -decomposition is Möbius-equivalent to  $\mathbb{P}^1 \setminus \mu_4$  — the QGEO Rigidity Theorem (QGEO.RIGID.01, [E]). The seam-collar realisation stays the one premise (QGEO.REALIZE.01, [C]). Mirrored in the Wolfram extension (226/226). Cited in **tfpt\_research\_contracts** (U0). *Not* a new structural class — it hardens the existing hinge (consistent with v167’s completeness claim).
- **v169\_etaB\_boltzmann\_interface.py** [C] (**claim FR.ETAB.02**). Operationalises the Tier-3  $F_{\text{transfer}}$  leptogenesis path. Type-I thermal leptogenesis  $\eta_B \sim 0.96 \times 10^{-2} \varepsilon_1 \kappa_f$  (sphaleron 28/79), fed by TFPT’s normal-ordered neutrino spectrum and the predicted  $\delta_{CP} = 240^\circ$ . Over natural  $M_1 \in [3 \times 10^9, 3 \times 10^{10}]$  GeV and washout,  $\eta_B$  brackets the observed  $6.1 \times 10^{-10}$  (a canonical point gives  $6.0 \times 10^{-10}$ , untuned) — the route is *viable*. But  $M_1$ /washout are scenario inputs, so  $\eta_B$  stays [C]; if a precise solve excluded the window the leptogenesis *route* falls, not the theory (the downstream  $\Omega_b$  readout is independent). Cited in **tfpt\_4\_frontier** (Route B). Python-only.

## 2026-06-13 (Higgs quartic from the free seam: SM near-criticality explained, v166)

- **v166\_higgs\_free\_seam.py** [E]/[C] (**claim HIGGS.FREESEAM.01**). A new prediction tying the Higgs mass to premise (A). The seam UV is the free chiral  $c=8$  fixed point (v158, Gaussian, no relevant/marginal interactions), so the one marginal SM scalar coupling vanishes there:  $\lambda(M_{\text{seam}}) = 0$  and  $\beta_\lambda(M_{\text{seam}}) = 0$  — the Shaposhnikov–Wetterich double-criticality, here *derived* from the free seam, not assumed. Integrating the PyR@TE-confirmed *two-loop* SM RGEs from  $M_Z$  up with the measured  $(m_H, m_t)$  gives  $\lambda(\bar{M}_{\text{Pl}}) \approx 0.002$  and  $\beta_\lambda(\bar{M}_{\text{Pl}}) \approx 0$ : the famous SM near-criticality is a *consequence* of the free seam (2-loop sharpens  $\lambda(\bar{M}_{\text{Pl}})$  from  $\sim 0.022$  at one loop to  $\sim 0.002$ ). The double condition predicts  $m_H \sim 129\text{--}134$  GeV; the same condition at the scalaron scale gives  $\sim 107$  GeV (too low), so the boundary condition lives at the reduced Planck scale (seam = horizon = Planck). Fills the Higgs-sector gap (was only  $N_\Phi = 1$ ); cited in **tfpt\_4\_frontier**.
- PyR@TE-sourced: the reproducer’s SM run supplies the two-loop  $\beta_\lambda$ , reduced to top-only (**verification/pyrate/sm\_higgs\_betas.txt**). Suite now 166 modules. Python-only (numerical 2-loop ODE).

## 2026-06-13 (PyR@TE $F_{\text{transfer}}$ follow-ups: flavor scale-tagging v163, QCD scale v164)

- **v163\_rg\_stability\_flavor.py** [E]/[C] (**claim FLAV.RGSTAB.01**). Scale-tagging of the flavor sector. The lepton-mixing seeds ( $\sin^2 \theta_{12} = \frac{1}{3} - \frac{\varphi_0}{2} = 0.3067$ ,  $\sin^2 \theta_{13} = \varphi_0 e^{-5/6} = 0.0231$ ) are *RG-stable*: the PMNS running is  $y_\tau^2$ -suppressed (the only mixing-rotating Yukawa term is  $\frac{3}{2} Y_e Y_e^\dagger Y_e$ , PyR@TE SM RGEs), so  $\kappa_\tau = y_\tau^2 / (16\pi^2) \ln(M_{\text{scal}}/M_Z) \sim 1.8 \times 10^{-5}$  and  $|\Delta \sin^2 \theta_{ij}| \lesssim 10^{-4}$  — two to three orders below precision. Seam value = measured value; the  $y_t^2$ -driven quark mass ratios ( $m_t/m_b \sim 41$ ) by contrast carry a scale tag. Cited in **tfpt\_2** (solar/reactor angle box).
- **v164\_qcd\_scale.py** [E]/[O] (**claim QCD.LAMBDA.01**). The QCD confinement scale is parameter-free from the carrier  $b_3 = -7$  (v159): running  $\alpha_s(M_Z) = 0.1179$  down (two-loop,  $n_f$  thresholds) reproduces  $\alpha_s(m_b) \simeq 0.22$ ,  $\alpha_s(m_c) \simeq 0.38$  and confinement at  $\sim 0.5$  GeV by dimensional transmutation — so the QCD-scale *numerator* of  $m_p/m_e$  is independently sourced. The  $O(1)$  proton/ $\Lambda$  factor is lattice, so  $m_p/m_e = 1836$  stays the honest “[O] — not forced” frontier ratio. Cited in **tfpt\_4\_frontier**.
- Both are PyR@TE-backed (the reproducer now also extracts the SM Yukawa  $\beta$ ’s, **verification/pyrate/sm\_yukawa\_betas.txt**). Suite now 164 modules. Python-only (no Wolfram mirror: magnitude bound / numerical running).

## 2026-06-13 (PyR@TE external cross-check of the $F_{\text{transfer}}$ gauge inputs — v159)

- **New module v159\_pyrate\_gauge\_crosscheck.py** [E]/[C] (**claim EM.BUDGET.02**). The one-loop gauge coefficients used by the transfer functor are confirmed by an *independent* third-party RGE generator. Feeding the carrier / Standard-Model field content into the textbook one-loop formula (GUT normalisation) gives  $(b_1, b_2, b_3) = (\frac{41}{10}, -\frac{19}{6}, -7)$ , and **PyR@TE 3** (**arXiv:2007.12700**, **github.com/LSartore/pyrate**) writes  $\beta_{g_1} = (\frac{41}{10})g_1^3$ ,  $\beta_{g_2} = -\frac{19}{6}g_2^3$ ,  $\beta_{g_3} = -7g_3^3$  verbatim (plus the full standard 2-loop matrix). So  $b_1 = 41/10$  is pinned *three* independent ways — carrier algebra  $10b_1 = g_{\text{car}} 2^{g_{\text{car}}-2} + 1 = 41$ , the  $U(1)$  hypercharge index  $\sum \frac{3}{5} Y^2$ , and PyR@TE — and the split  $10b_1 = 40_{(\text{ferm.})} + 1_{(\text{Higgs})} = 41$  reproduces the EM-budget Gate 1 of **tfpt\_2**. Cited there with **\veri**; the gauge sector of  $F_{\text{transfer}}$  is thus not a free knob.
- **PyR@TE reproducer (not vendored)**. **verification/pyrate/reproduce.sh** clones PyR@TE on demand into **experiments/pyrate** (git-ignored), neutralises its interactive

pdf<sub>latex</sub> end-command, runs the SM at one and two loops, and re-extracts the gauge  $\beta$  functions; the committed evidence is `verification/pyrate/sm_gauge_betas.txt`. PyR@TE is the right external tool for the transfer/RGE gates (gauge running, and — with a seesaw model — the neutrino-Yukawa running behind  $\eta_B$ ); it is *not* a tool for premise (A) (a 2d boundary-CFT question). Mirrored in the Wolfram extension; suite now 159 modules.

**2026-06-13 (terminal residual inventory: (A) no longer a structural class; one hinge + F\_transfer + irreducible core, v167)**

- **New module v167\_inventory\_post\_closure.py** [E]/[C]. The terminal residual inventory (line v105→v123→v167), re-pinned after the (A)-chain. With (A) discharged (GATE.METRIC.16–19) the v123 *five* structural classes collapse: the seam clock (R1), the index-4 net (R2) and the  $Q$ -realisation (R5) all merge into *one* Tier-1 hinge — the seam=flavour=horizon identification =  $A_2$  net existence (assembly of established net constructions) + GATE.QGEO (seam boundary =  $\mathbb{P} \setminus \mu_4$ , cohomology  $b_1=N_{\text{fam}}=3$ ); the EH step (R3) folds into  $v_{\text{geo}}$  (v152) and the ambient measure into  $G_{\text{net}}$  (v76). **Terminal structure:** Tier 0 = the irreducible core  $\{\pi, v_{\text{geo}}\}$  (a theorem, No-Unit v153); Tier 1 = one hinge; Tier 2 = folded; Tier 3 =  $F_{\text{transfer}}$ , one functor with four typed conditional interfaces (Koide,  $\eta_B$ , axion,  $m_p/m_e$ ),  $\eta_B$  carrying the most open computational room. Completeness: a sixth independent structural class — or a new irreducible — fails the script (ledger SYNTH.INVENTORY.03). A unification + re-typing, NOT an unconditional proof. Python-only (reads the ledger).
- **Bookkeeping.** Suite at 167 modules, all green; Wolfram extension unchanged at 224/224; `origin_theory` (residual-inventory keybox) and `next.txt` moved together.

**2026-06-13 (premise A maximal honest closure: A2 by assembly, R1 = GATE.QGEO, zero independent open gates, v165)**

- **New module v165\_premise\_a\_closure.py** [C]/[O]. Pins the two residual gates of the v160–v162 (A)-chain to their floor.  $A_2$  (**net existence**) **closes by assembly** of established mathematics [C]: the 16 free Majoranas give a free-fermion conformal net (Buchholz–Mack–Todorov), the holomorphic  $c=8$  lattice net is  $(E_8)_1$  (Frenkel–Kac–Segal / Dong–Xu / Kawahigashi–Longo), the  $\mu_4$  extension is a Longo Q-system of index 4 (v125),  $\mu(B)=16/16=1$ ,  $248=120+128$ , character  $E_4/\eta^8=j^{1/3}$  (v156); the one TFPT-specific input (seam Calderón kernel = free-fermion two-point function) is done via  $\text{DtN} = |k|$  (v156/v157).  $R_1$  (**seam clock**) **reduces to GATE.QGEO** [C]: a  $\mu_4$ -equivariant flat gapped generator is *unique* ( $= A_0^*$ , v115/v116), so the gap  $(2/3)^6$  is forced and the residual is the one geometric identification “seam boundary =  $\mathbb{P} \setminus \mu_4$ ” = GATE.QGEO (cohomology established, v137,  $b_1=3=N_{\text{fam}}$ ). **Final accounting:** by the No-Unit Theorem (v153) the irreducibles stay  $\{\pi, v_{\text{geo}}\}$ ; premise (A) = ( $A_2$  assembly) + ( $R_1=\text{GATE.QGEO}$ ) + the  $\{\pi, v_{\text{geo}}\}$  core, so it carries *zero independent open gates* and ceases to be its own gate — a unification and re-typing, *not* an unconditional proof (ledger GATE.METRIC.19). Python-only (numpy/sympy + stdlib).
- **Bookkeeping.** Suite at 165 modules, all green; Wolfram extension unchanged at 224/224; `tfpt_research_contracts` (premise-A keybox) and `next.txt` moved together.

**2026-06-13 (the final identification: premise A factors into the already-open A2 + R1, no new open content, v162)**

- **New module v162\_seam\_transport\_identification.py** [E]/[O]. Pushes the v161 residual as far as it rigorously goes and shows it carries *no new open content*. The one-particle symbol splits into a gap value ( $\beta$ ) and a fixed space ( $\alpha$ ). ( $\beta$ ) **is forced:** a  $\mu_4$ -equivariant generator is diagonal in the cusp basis (the deck average is the diagonal projection,

$\sum_k U^k X U^{-k} = |\mu_4| \text{diag } X$  for  $U = \text{diag}(1, i, -i)$ ; the commutant of  $U$  is exactly the diagonal algebra), so its spectrum is the cusp weights  $\text{spec}(A_0^*) = \{0, \frac{1}{3}, \frac{2}{3}\}$  (v115) and the transfer eigenvalue  $(1-\alpha)^{p_2}$  with  $p_2=6=|R_+(A_3)|$  gives  $\{1, (2/3)^6, (1/3)^6\}$ , gap  $6 \log \frac{3}{2}$  — all carrier data; the lift to the 16 Majoranas is the  $A_3$  subnet grading (v137).  $(\alpha)$  is the rank-8 Calderón polarisation of  $\omega_k=|k|$  (v113/v156). **Closure:** with v160+v161, (A) closes given only  $\mu_4$ -equivariance + admissibility, whose two non-derived inputs are the already-open A2 (net existence, Kawahigashi–Longo) and R1 (seam clock, HOR.CLOCK). So premise (A) factors exactly into  $A_2 + R_1$  — it adds *no* independent open item and is closed *conditionally* on them (a unification, not an unconditional proof; ledger GATE.METRIC.18). Python-only (numpy/sympy + stdlib).

- **Bookkeeping.** Suite at 162 modules, all green; Wolfram extension unchanged at 224/224; `tfpt_research_contracts` (premise-A keybox) and `next.txt` moved together.

**2026-06-13 (the cone reduces to one particle: premise A’s scope premise discharged to a 16-dim statement, v161)**

- **New module v161\_one\_particle\_reduction.py [E]/[O].** *Discharges* the scope premise of v160 (“the transport is gapped on the *full* Schwinger cone”) via Bogoliubov second quantisation. A quasi-free transport is  $T_\Sigma = \Gamma(t)$ , the second quantisation of a one-particle contraction  $t$  on the 16-dim Majorana space;  $\Gamma(t)$  acts on the degree- $m$  correlator sector as the exterior power  $\Lambda^m(t)$ . Machine-checked: (1)  $\text{spec}(\Gamma(t)|_{\deg m}) =$  products of  $m$  one-particle eigenvalues (compound-matrix theorem, all  $m=0..6$ ); (2) **the cone gap equals the one-particle gap** — the sub-leading eigenvalue over the whole cone is  $(2/3)^6$  exactly, so premise 5 on the cumulant space *is* premise 5 on the 16-dim space; (3) the eigenvalue-1 sector is  $\Lambda^*$  of the fixed subspace (disconnected Wick powers), with uniqueness closed by v113 (one polarisation  $\Leftrightarrow$  one vacuum); (4) quasi-freeness is *forced* — the  $120 = \dim \mathfrak{so}(16)$  chiral currents are the Bogoliubov generators and v158 makes every non-Gaussian deformation irrelevant. **Consequence:** v160’s three full-cone premises collapse to ONE finite (16-dim) one-particle statement (seam symbol = gapped Calderón Bogoliubov map, fixed = rank-8  $K_\Sigma$ , gap  $(2/3)^6$ ); all numbers supplied, only the physical identification (Kawahigashi–Longo class = the A2 net residual) stays open — the infinite-dimensional cone is eliminated (ledger GATE.METRIC.17). Python-only (numpy + stdlib).
- **Bookkeeping.** Suite at 161 modules, all green; Wolfram extension unchanged at 224/224; `tfpt_research_contracts` (premise-A keybox) and `next.txt` moved together.

**2026-06-13 (premise A as a conditional fixed-point theorem; the load relocates to one scope premise, v160)**

- **New module v160\_seam\_gaussianity\_from\_pf.py [E]/[O].** Formalises the proposed closure of premise (A) (“the seam bulk is a reflection-positive free field”) as a fixed-point theorem and types it honestly. The *exact* parts: (1) a quasi-free state has *all* higher connected cumulants  $\kappa_{2n}=0$  — verified by the *signed* set-partition/Pfaffian moment  $\leftrightarrow$  cumulant inversion on a concrete pure Majorana covariance (the v113 “one kernel = net” property at the cumulant level); (2) a kernel-preserving transport fixes the whole quasi-free state by Wick functoriality; (3) the Perron–Frobenius dichotomy (a *fixed*  $\kappa_{2n}$  is a second fixed ray  $\Rightarrow$  breaks uniqueness; a *non-fixed* one decays by the gap) forces Gaussianity; (4) the qualitative claim needs only  $\text{gap} > 0$ , the value  $(2/3)^6$  only sets the rate. **The honest residual:** v56’s gapped PF uniqueness lives on the *three-dimensional* scalar transfer spectrum  $\{1, (2/3)^6, (1/3)^6\}$ , while the 16-Majorana seam already carries  $\binom{16}{4}=1820$  and  $\binom{16}{6}=8008$  higher-cumulant directions — so (A) reduces to the single internal, exact-testable statement “the admissible TFPT seam transport is primitive + spectrally gapped on the *full* Schwinger cone, not only the

readout spectrum of v56”, which the suite does *not* yet establish (ledger GATE.METRIC.16). Python-only (numpy + stdlib); no new exact-identity content for the Wolfram mirror.

- **Bookkeeping.** Suite at 160 modules, all green; Wolfram extension unchanged at 224/224 (v160 is numerical, Python-only); `tftpt_research_contracts` (the premise-A keybox) and `next.txt` moved together.

### 2026-06-13 (changelog date-order audit + reorder pass)

- **Changelog sort enforced.** `audit_sync.py` now checks that every dated `\subsection × {YYYY-MM-DD...}` block in this file is non-increasing (newest first; range titles such as 2026-06-07 / 2026-06-08 sort by the later date). A manual prepend had left five 2026-06-12 entries above seven 2026-06-13 blocks — reordered; future mis-sorts fail `build.sh audit` as `[B.changelog]`.

### 2026-06-13 (premise A: the destination fixed point is provably stable, v158)

- **The free chiral  $c=8$  fixed point is provably stable (v158, exact dimension count).** The finite core of (A)’s “reaches-the-fixed-point” residual: the chiral Majorana has  $h=\frac{1}{2}$ , the bilinear current  $h=1$ , the quartic  $h=2$ . The relevant window  $0 < h < 1$  for bosonic deformations is *empty*, the  $h=1$  currents are chiral (v157, not  $(1,1)$ -marginal), and the lowest interaction (quartic,  $h=2$ ) is irrelevant — so the fixed point is isolated and stable against all interactions. Premise (A) reduces to a basin-of-attraction statement with a *proven* stable target (ledger GATE.METRIC.15). *Tooling note:* PyR@TE (4d gauge-Yukawa RGEs) is the wrong tool for this 2d boundary-CFT residual, but the right tool for the  $F_{\text{transfer}}$  interfaces ( $\eta_B$  leptogenesis RGEs, gauge-coupling running, the  $b_1=41/10$  cross-check).
- **Bookkeeping.** Suite at 158 modules, all green; Wolfram extension 224/224; contracts / `origin_theory` / `next.txt` / website moved together.

### 2026-06-13 (premise A reframed: freeness is a rigid boundary fixed point, v157)

- **The free-bulk premise becomes a fixed-point property (v157, simplicity-first).** Instead of postulating Gaussianity, three facts click: (i) the DtN/Calderón symbol is *universally*  $|k|$  (Lee-Uhlmann — order-1  $\Psi$ DO, principal symbol  $|\xi|_g$ ; interactions live in the irrelevant lower-order symbol), so the free dispersion is bulk-detail-independent; (ii) a holomorphic  $c=8$  theory is *rigid* — every field has  $\bar{h}=0$ , so no marginal  $(1,1)$  field exists and the fixed point is isolated, with no knob for an interaction; (iii) the same  $|\mathbb{Z}_2|$  that halves  $8\pi$  in  $c_3 = \frac{1}{8\pi}$  is the Ramond/GSO projection giving  $\mu=1$  ( $248=120_{\text{NS}}+128_{\text{R}}$ ). With  $(E_8)_1$  unique (v83), premise (A) shrinks from “prove the seam is Gaussian” to “the gapped ( $\Delta=6\log\frac{3}{2}$ ) boundary flow reaches the holomorphic fixed point” — freeness then forced by rigidity, not fine-tuned (ledger GATE.METRIC.14).
- **Bookkeeping.** Suite at 157 modules, all green; Wolfram extension 223/223; contracts / `origin_theory` / `next.txt` / website moved together.

### 2026-06-13 (Route 3: the seam net constructed and $(E_8)_1$ verified, v156)

- **The constructive route carried to its exact endpoint (v156).** “ $B \cong (E_8)_1$ ” is no longer an assertion: the chiral seam net is built explicitly and the identification *checked*. (a) The Dirichlet-to-Neumann (Calderón) operator of the 2d Laplacian is the free chiral dispersion  $\omega_k = |k|$  (harmonic extension  $e^{ikx-|k|y}$ ); (b) 16 free Majoranas give  $c = 8 = 5+3$ ; (c) the weight-1 currents are  $248 = 120_{\text{NS}} + 128_{\text{R}}$  ( $SO(16)$  bilinears + Ramond spinor, Frenkel-Kac); (d) the holomorphic character  $E_4/\eta^8 = q^{-1/3}(1 + 248q + 4124q^2 + 34752q^3 + 213126q^4) = j^{1/3}$  (verified by  $q$ -series) — the 248 at level 1 is  $\dim E_8$ , so the net *is*  $(E_8)_1$ . *Residual:* exact given

(i) the free-bulk premise ( $\text{DtN} = |k|$ , v155) and (ii) the operator-algebraic existence of the net from the seam kernel (a Kawahigashi–Longo-type theorem, not a finite computation) — where TFPT meets constructive QFT, now sharply located (ledger GATE.METRIC.13).

- **Bookkeeping.** Suite at 156 modules, all green; Wolfram extension 222/222 (the  $\text{DtN} = |k|$  identity, the  $(E_8)_1$  character  $E_4/\eta^8 = j^{1/3}$  and the  $248=120+128$  split mirrored); contracts / redteam (Target A) / `origin_theory` / `next.txt` / website moved together.

## 2026-06-13 (the seam-side identification reduced to one shared premise, v155)

- **$G_{\text{net}}$ 's last premise reduces to a single shared physical premise (v155, Python-only).** The seam-side identification of the Simple-Current Extension Theorem (v154) splits (v110) into (a) “the Calderón involution is sheet-odd” (the double-cover deck, structural) and (b) “the seam 2-point kernel is quasi-free”. Part (b) is *not* independent: the boundary marginal of a reflection-positive *Gaussian* bulk is the compression *PTP* of the covariance, and the compression of a contraction is a contraction — so the seam state is automatically quasi-free (Wick inherited). Hence (b) *follows* from “the seam bulk is a reflection-positive free field”, the *same* premise already load-bearing in the area law (v59) and the EH mechanism (v150/v151). **The entire seam-side residual is therefore one physical premise**, shared by  $G_{\text{net}}$ , the area law and R3 — not reducible by a finite computation, but no longer three separate items (ledger GATE.METRIC.12).
- **Bookkeeping.** Suite at 155 modules, all green; Wolfram unchanged at 221/221 (v155 is numerical/numpy, Python-only); contracts / `origin_theory` / `next.txt` / website moved together.

## 2026-06-13 (external review formalised: No-Unit Theorem + Simple-Current Extension Theorem, v153–v154)

- **$v_{\text{geo}}$  re-typed: the No-Unit Theorem (v153).** Every load-bearing compiler datum is mass-dimension 0 and invariant under a unit rescaling  $L \mapsto \lambda L$ , whereas a mass /  $1/G$  is not — so a dimensionless boundary compiler *provably cannot* select an absolute scale.  $v_{\text{geo}}$  is therefore an *irreducible metrology primitive*, not an open gap, and the three apparent residuals collapse onto it ( $U_{\text{point}} \sim v_{\text{geo}}$ ,  $1/G \sim v_{\text{geo}}^2$ ,  $m/\mu = e^{3/4}$  a pure ratio); the irreducibles stay {one unit,  $\pi$ } (ledger ANCHOR.VGEO.02).
- **$G_{\text{net}}$  stated as the central closing theorem (v154).** The Simple-Current Extension Theorem:  $A=(D_5)_1 \otimes (A_3)_1$  extended by the isotropic glue  $L=\langle(1,1)\rangle$  has  $[B:A]=|L|=4$ ,  $c=5+3=8$ ,  $\mu(B)=\mu(A)/|L|^2=16/16=1 \Rightarrow$  holomorphic  $\Rightarrow B \cong (E_8)_1$  ( $SO(16)_1$  excluded). Every algebraic ingredient is machine-checked (v89/v92/v125/v143/v148); the only residue is the seam-side identification. Pulled to the front of `tfpt_research_contracts`, `tfpt_5_redteam` (Target A final form) and `origin_theory` (ledger GATE.METRIC.11).
- **Status surfaces re-typed (review):** the canonical status card now reads  $v_{\text{geo}}$  as a metrology *primitive*,  $G_{\text{net}}$  as the simple-current extension  $\cong (E_8)_1$  ( $[\mathbf{E}]/[\mathbf{C}]$ ), and  $F_{\text{transfer}}$  as one functor with four typed interfaces — “not structural gaps”. The final one-line framing (dimensionless boundary compiler; irreducibles  $\{\pi, v_{\text{geo}}\}$ ; one seam-net theorem; frontier =  $F_{\text{transfer}}$ ) is in `origin_theory`, the introduction, the README and `next.txt`.
- **Bookkeeping.** Suite at 154 modules, all green; Wolfram extension 221/221; intro/README/origin\_theory/contracts/redteam/status-card/website moved together.

## 2026-06-13 (currency sweep: remaining old-series “Paper N” cross-references removed)

- **No old-version framing left in the active docs.** A consistency sweep found a residual layer of old-series “Paper N” attributions and removed them: four literal “Paper 6” references (`tfpt_4_frontier` leptogenesis/axion, `tfpt_2` seam determinant), the old “Paper 4” (OS/admissibility) and “Paper 5” (APS/Hodge) references in `tfpt_2`, the old “Paper 3” transport-kernel attributions in `tfpt_2`, and the internal “Paper 1/2/4” cross-references in `tfpt_3`, `tfpt_4` and `tfpt_horizon_readouts` — all rewritten as content-based phrasing (the flavor sector, the architecture document, the  $E_8$  decoder, the anchor characteristic polynomial, etc.). The active documents now attribute results to the current documents / axioms / scripts only, as the workflow rule requires; the machine-checked sync (scripts  $\leftrightarrow$  papers  $\leftrightarrow$  ledger  $\leftrightarrow$  website) was already current (152 scripts, AUDIT OK).

## 2026-06-13 (fifth round: the R3 normalisation is the dimensionful anchor, v152)

- **R3: the  $q(A_3)$  normalisation merges into the one anchor (v152).** The replica EH coefficient is a *log-ratio*  $k = \ln(m/\mu)/(12\pi)$  ( $\ln m$  alone is scale-ambiguous, only  $m/\mu$  is physical); pinning  $k = c_3/2$  fixes the dimensionless ratio  $m/\mu = e^{3/4}$ , and in  $d=2$  that ratio *is* the induced Newton constant  $1/G$  (v68). So  $v_{\text{geo}}$  (v78),  $1/G$  (v68) and  $m/\mu$  are *one* dimensionful anchor in three readings — the irreducibles stay {one scale,  $\pi$ }, and SEAM.THEOREM.01’s residual reduces to the kernel-identification premise alone. *Audit (not promoted)*: the log-ratio is numerically  $\frac{3}{4} = q(A_3)$ , the target value the anchor takes in seam units — not a derivation (ledger SEAM.EHMODEL.03).
- **Bookkeeping.** Suite at 152 modules, all green; Wolfram extension 219/219; intro/README/origin\_theory/next.txt/website moved together.

## 2026-06-12 (Zenodo version label: semantic + git rev)

- **Release version string.** `build.sh` now writes `tex-artefacts/release-version.txt` as `{TFPTversion}` (rev {git count}) (matches `\TFPTdatestamp` without the date). `scripts/zenodo_upload.py` and `build.sh zenodo/zenodo-publish` use this label on Zenodo; cursor rules extended for when `zenodo_description.html` must move with deposit-facing changes.

## 2026-06-12 (Zenodo API upload workflow)

- **Automated Zenodo versioning.** Added `scripts/zenodo_upload.py` and `.github/workflows/zenodo-release.yml`: creates a new draft version of record 20579275 (DOI 10.5281/zenodo.20579275), uploads the ten active paper PDFs, refreshes `zenodo_description.html`, bumps the version from `tex-artefacts/version.tex`, and optionally publishes via GitHub Actions (ZENODO\_TOKEN secret).

## 2026-06-12 (Overleaf shadow mirror: tfpt5-shadow)

- **One-way GitHub mirror for Overleaf sync.** Added `scripts/export-shadow.sh` and `.github/workflows/shadow-sync.yml`: on every push to main, a filtered copy is pushed to `sthamann/tfpt5-shadow` (papers, `tex-artefacts/`, `figures/`, `verification/`, `experiments/`; excludes `_archive/`, `website/`, all `*.pdf`, `.cursor/`).  $\sim 270$  files /  $\sim 3$  MB — within Overleaf GitHub-sync limits.

## 2026-06-12 (origin\_theory style alignment + box reduction)

- **Fewer boxes, embedded in text.** `origin_theory` from 22 to 16 boxes: the narrative/interpretation callouts (“Why the interpretation is internally consistent”, “Dependency chain of the birefringence test”, “The precise reformulation”, “The three theses honestly graded”, “The whole story”, “One-line summary”) become run-in bold prose; the boxed reformulation becomes a `keyeq`. The exact result keyboxes (v53–v56, v101, v105, v113, ...) and the master-principle / Seam–Horizon target boxes are kept.
- **Colour-marked module references + marker hygiene.** `\vref` applied to the inline vN references; stale literal `[A]/[I]/[L]` markers in running text replaced by the 4-class `[O]/[E]`.

## 2026-06-12 (tfpt\_5 / horizon / research-contracts pass + build hardening)

- **Forked document-set box removed (horizon).** `tfpt_horizon_readouts` carried a hand-written, *stale* copy of the document-set card (“five short documents”, missing `tfpt_5`) and the canonical `\input{tex-artefacts/tfpt_docset}`; the fork is deleted so the single-source card (all five papers + three companions) appears once.
- **Fewer boxes, embedded in text.** Narrative/result callouts converted to run-in bold prose (with the `\veri{}` citation moved out of the bold title into the body): `tfpt_horizon_readouts` from 22 to 7 boxes (the ten-box “clock” wall becomes prose); `tfpt_research_contracts` from 19 to 10 (the nine progress/result winboxes become prose, the named-theorem keyboxes kept); `tfpt_5_redteam` from 10 to 9 (the outcome summary; the per-target verdict boxes are kept as the structured red-team output).
- **Colour-marked module references + marker hygiene.** `\vref` applied to the inline vN references in all three documents; the horizon “Scope/typing” marker line and a stale literal `[A]` in the research contracts updated to the 4-class markers.
- **Build hardening.** `build.sh` now removes a document’s `.aux/.toc/.out` on failure, so a half-written cross-reference file from a crashed compile cannot poison the next build; artefacts are still kept on success.

## 2026-06-12 (fourth round: the Calderón transfer answered — the BFK split, v151)

- **R3: the Calderón/DtN kernel is conically clean (v151).** Reflection doubling derives that the Kac corner constant is boundary-condition independent ( $C_N = C_D$ ), so by the Burghilea–Friedlander–Kappeler gluing ledger the conical deficit of the two-sided Calderón jump determinant *vanishes identically* ( $C_{\text{cone}}(\gamma) - 2C_D(\gamma/2) = 0$ ): the seam-reduced effective action inherits the v150 EH form *verbatim*, with all of it localised in the *local* half-space determinants — the “Calderón version” demanded by v150 needs no separate derivation. SEAM.THEOREM.01 narrows to the  $q(A_3)$  normalisation plus the standing kernel premise; the gate stays `[O]` (ledger SEAM.EHMODEL.02).
- **Bookkeeping.** Suite at 151 modules, all green; Wolfram extension 218/218; intro/README/origin\_theory/next.txt moved together.

## 2026-06-12 (third round: both normals anchor data; the R3 mechanism exists, v149–v150)

- **R4': both dual normals are anchor data (v149).** In the cusp frame the torsion normal pairs to  $(6, 3, 5) = (p_2, p_0, e_2)(a)$  — one anchor invariant per  $Q_+$  line (the cusp matrix is unimodular, so this determines  $n$  uniquely); with  $d = \frac{3}{2}a - 2 \cdot \mathbf{1}$  the dual normal pair is anchor data through and through. Equivalences to the  $(2, 8, 121)$  frame data and the  $w_0$ -lift exact. R4' residue = *one* discrete assignment (ledger GATE.UWALL.14).

- **R3: the mechanism exists at the gapped-model level (v150) — the first movement on SEAM.THEOREM.01.** The conical heat-trace deficit is exact and  $t$ -independent (Cheeger image sum  $(N^2-1)/(12N)$ ); for a *gapped* operator the  $\zeta$ -regularised replica variation is *finite and cutoff-independent*,  $\Delta \log \det' = 2C(\gamma) \ln m$ , and its linearisation is the Einstein–Hilbert form  $(\ln m/12\pi) \int \sqrt{g} R$ . The v73 normalisation becomes one exact target equation:  $k = c_3/2 \Leftrightarrow \ln m = \frac{3}{4} = q(A_3)$  (audit, not asserted). Remaining: the Calderón (boundary) version and that normalisation — the gate narrows but stays [O] (ledger SEAM.EHMODEL.01).
- **Bookkeeping.** Suite at 150 modules, all green; Wolfram extension 217/217; status surfaces (intro reductions + card, `origin_theory`, contracts, `next.txt`, website) moved together.

## 2026-06-12 (R1–R5 second round: every class at its floor, v145–v148)

- **R4': the pairing values are atom identities (v145).** Reading the v139 lift structurally ( $n = w_0(\sigma) + |\mu_4|e_3$ ,  $w_0$  = the  $A_3$  exponent duality), all three pairings become atom arithmetic — including the *new* closure  $|\mu_4| + g_{\text{car}} = N_{\text{fam}}^2$  ( $4+5=9$ ,  $\sigma \perp w_0(a)$ ) and  $\|\sigma\|^2 = 110 = |\mathbb{Z}_2|g_{\text{car}}\|\text{Pl}(K)\|_1$ . R4' = derive *one* map (ledger GATE.UWALL.13).
- **R5: the Möbius  $D_4$  realisation (v146).** The *full* dihedral symmetry  $\langle z \mapsto iz, z \mapsto 1/z \rangle$  of the seam curve acts on  $H^1$  exactly as the integer model:  $\iota^*$  is the exponent duality with parity +1 on the self-conjugate line (=  $T_A$  in the cusp basis, glue-swap parity), and  $\Sigma$  is the  $\delta\iota$  class — the premise reduces to the module identification itself (ledger FLAV.QGEO.07).
- **R1: the clock is a Gaussian zero-mode integral, and the bend is  $\det'$ -clean (v147).** Variance ratio  $(1 - \alpha)$  (forced by norm = area) gives the Born-squared ratio  $(1 - \alpha)^{p_2}$ , the  $\ln$  series *is* the v127 ring sum; the two quantum weights share *one* Nariai geometry, so the bend  $\log_{3/2} 3$  carries no determinant correction — the v144 obstruction is localised to the  $\alpha = 0$  reference (ledger HOR.CLOCK.13).
- **R2: the NS/R sector census (v148).** The untwisted (Neveu–Schwarz) module carries only the *even* discriminant classes (integer weights), so the odd  $\mu_4$  simple currents have zero NS support — they are Ramond modules; the R zero-mode module (256) splits 128+128 into the odd sectors of the *two* Lagrangian glues: choosing the glue *is* choosing the R-projection ( $248 = 120_{\text{NS}} + 128_{\text{R}}$ ); the one remaining net statement is irreducibly twisted-sector (ledger GATE.METRIC.10). *Correction note (same day):* the first version of v148 graded the Ramond labels by the sum  $q_D + q_A$  (not the glue label) and called that module untwisted; the census was rewritten as above — conclusion unchanged, support corrected, the R-sector sheet split is new.
- **Bookkeeping.** Suite at 148 modules, all green; Wolfram extension 215/215 (README + GATE.WOLFRAM.02 + website counter in lock-step); status surfaces (intro card, `tfpt_1`, `origin_theory` third update, `next.txt`) moved together.

## 2026-06-12 (external review integrated: $\Lambda$ typing, marker migration, website parity)

- **$\Lambda$  normalisation made unambiguous.** The external review flagged the adjacent reduced/unreduced displays in `origin_theory` as a type trap (the formulas were individually correct —  $\bar{M}_{\text{P1}}$  vs  $M_{\text{P1}}$  — but easy to misread). Both normalisations are now shown explicitly side by side with their order counts ( $3/(4\pi^2) \Rightarrow 120.147$  reduced;  $3/(256\pi^4) \Rightarrow 122.948 = 119.028 + 3.920$  unreduced) plus a typing note.
- **Stale references cleaned.** “How each of the seven papers simplifies”  $\rightarrow$  “How the legacy layers simplify (bridge map)” with the lead-in “(Paper 6)” reference replaced by the scalaron-reheating chain (v86); “(Paper~6/7)” in the `tfpt_3` bridge table reworded; the `tfpt_1` “single remaining geometric input (H2)” passage and the `tfpt_2` “single remaining input (U)” passage

carry dated status updates pointing at the current decomposition (v118–v122, v139, v141, v142, v75).

- **Marker migration completed in the shared figures.** The claim stack, anchor cockpit,  $E_8$  glue,  $2/3$  motif and  $K_5$  figures (`tex-artefacts/tfpt_figures.tex`) now show the four public classes  $[E]/[C]/[O]$  instead of the legacy  $[A]/[I]/[L]/[B]/[P]/[R]$  badges.
- **Red-team front table pulled to the final state.** The Target-A residual column now states the one remaining statement (boundary-net holomorphy +  $c=8 \Leftrightarrow$  index-4 inclusion; v83/v87/v89, Lie-level realisation v143); the historical three-residual form is kept below, dated.
- **Website parity.** The Red Team document is now a first-class entry (`papers.ts` number 5, slug `redteam`) and the changelog PDF a download row; Appendix H / Origin Theory / Research Contracts are labelled *companions* (never “Paper 5–7”); the marker system on every website surface migrated to  $[E]/[C]/[O]/[X]$  (fine types named in prose; the script index keeps quoting the scripts’ internal fine-grained typing); the verification page counters are live (144 checks generated from the registry via `lib/suite.ts`, Wolfram 116/116 + 211/211 — now guarded by the sync audit) and the reproduce block matches the current pipeline. The `papers.ts` section bodies of the affected papers carry the v141–v144 outcomes (sheet question: deck derived;  $G_{net}$ : Lie-level realisation; resummed clock: in-family det-ratio cancellation; selector triangle: two line pairings).
- **Not adopted** (recorded for transparency): the review’s narrative reframings (anchor-algebra trilogy, “one seam clock” framing, one-page structure) and the front-matter/status-card layout changes are editorial decisions left to the authors; the claimed  $\Lambda$  error was in fact a correct-but-treacherous notation switch (see above).

## 2026-06-12 (R1–R5 attack round: four residual classes sharpened, v141–v144)

- **R5 closed at the discrete level (v141, deck selection theorem).** The v140  $\mathbb{Z}_3$  choice is *derived*: the established integer deck  $G = T_A \Sigma$  (v97/v98) pairs the  $Q_+=1$  cusp line with the self-conjugate character 2, so only the sheet-twisted assignment  $(2, 3, 1) = \text{chars}(-U)$  survives; the cusp exponential  $i^{Q+}$  is excluded (same spectrum, wrong grading pairing). Explicit conjugacy constructed; under  $k \rightarrow -k$  only the established sheet  $\mathbb{Z}_2$  flips. Consequence: Euler grading =  $Q_+ \circ \rho$ . GATE.QGEO keeps only its realisation premise — no discrete freedom left (ledger FLAV.QGEO.06).
- **R4’ narrowed: three pairings  $\rightarrow$  two (v142, frame integrality).** Integer covectors in the  $(1, a, \sigma)$  dual frame form an index-11 sublattice ( $x - 3y + z \equiv 0 \pmod{11}$ ; the index is  $\|\text{Pl}(K)\|_1$ ); with  $(x, y) = (|\mathbb{Z}_2|, \text{rank } E_8)$  the  $\sigma$ -pairing is forced  $\equiv 0 \pmod{11}$ , and primitivity forces the line parameter even. Cramer reductions for both line pairings recorded as restructuring (ledger GATE.UWALL.12).
- **R2 realised at the finite Lie level (v143, graded Frobenius).** The  $\mathbb{Z}_4$ -graded hull carries exact coset duality ( $-C_1 = C_3$ ; Killing pairing nondegenerate per  $g_k \times g_{-k}$ ), the glue average is exactly the carrier projector (index 4), and each glue sector is one Weyl orbit (simple currents, fusion =  $\mathbb{C}[\mathbb{Z}_4]$ ) — the v125 Q-system realised by the hull; residue = one conformal-net statement (ledger GATE.METRIC.09).
- **R1’s det-ratio step derived in-family (v144).**  $e_2$ -rigidity of the traceless SdS cubic gives  $r_b r_c = 1 - \Delta^2/3$  exactly; with the v132 anomaly the non-zero-mode determinant ratio is  $(1 - \Delta^2/3)^{4/3}$  — no first-order term in the horizon split (the v131 cancellation derived, not assumed); deficit coefficient  $\frac{32}{27} = |\mu_4|(\frac{2}{3})^3$  (audit). Finite-weight absorption stays  $[C]$  with the  $6 \rightarrow \frac{14}{3}$  obstruction stated sharply (ledger HOR.CLOCK.12).
- **R3 attempted, honestly unchanged.** No defensible computation landed for the seam-determinant  $\rightarrow$  EH step (SEAM.THEOREM.01); the gate keeps its  $[O]$  typing untouched.

- **Wolfram mirror extended** to v141–v144: extension file now 211/211 (GATE.WOLFRAM.02 and the wolfram README updated in lock-step); Python suite at 144 modules, all green.

## 2026-06-12 (sync infrastructure: single-source script index, content maps, one audit)

- **Single-source script index.** The master script→check table (`tex-artefacts/verification.tex`) and the website `ScriptIndex.tsx` are now *generated* from one registry (`script_registry.csv` + `script_clusters.csv` in `verification/`) by `make_script_index.py`; both targets carry a generated-file header and are never edited by hand.
- **Content maps without splitting the papers.** `verification/make_docs_map.py` generates `docs_map.csv` (every paper section with line range, the vN scripts cited there, a content hash and a last-changed date) and `website_map.csv` (which website file mentions which scripts/documents) — the machine-readable enumeration of all sync surfaces.
- **One sync audit.** `verification/audit_sync.py` bundles and extends the previous loose shell checks, now in *both* directions: suite ↔ `run_all.py` ↔ registry ↔ ledger; every script cited in a paper *body* (the master index alone no longer counts); no stale `\veri/\vref` targets; every new module mentioned in this changelog; website mirror byte-identical (PDFs, scripts, `release.ts` hashes, version stamps); wolfram counts; overfull boxes. Frozen exceptions live in `audit_baseline.json` (remove-only).
- **Build pipeline.** `build.sh` gains `gen` / `website` / `audit` / `release` steps; `notes` now regenerates the single-source surfaces first; `website` copies all active PDFs (now incl. `tfpt_5_redteam.pdf` and `changelog.pdf`, with new `release.ts` entries) and the vN scripts, and stamps version, date and git revision into `website/lib/version.ts` (shown in the site header) and `release.ts`.
- **Citation gaps closed.** The audit immediately found three scripts cited only in the master index: v17 and v85 are now cited in their `tfpt_2` sections (hexagonal resolvent; master cover), and two short `\veri{v19}` citations were expanded to the full script name. v91 (spine tetrahedron) had no paper section at all — it is now integrated: a keybox in `tfpt_3` (“Three views of one spine”: anchor closure, edge/face products, volume 120,  $K_6$  negative control, the tautology/sub-grammar caveats) and a closing note in the `tfpt_1` Spine Quotient Lemma; the ARCH.SPINE.01 ledger location is corrected to `tfpt_1;tfpt_3` and the baseline exception list is empty again.
- **Rules.** `.cursor/rules` updated (`tfpt-workflow`, `website-sync`) and a new `sync-maps` rule added describing the agentic procedure: *regenerate* the maps first, enumerate the affected surfaces from them, audit as the exit gate. The mirror map (`website_map.csv`) also covers the root `README.md` and `next.txt` (bare vN ids resolved to full script names), and both files are named explicitly as status surfaces that move with every finding.

## 2026-06-12 (tfpt\_3 / tfpt\_4 style alignment + box reduction)

- **Shared header/footer on tfpt\_3.** `tfpt_3_e8_audit_bootstrap` dropped its custom `\footmarkers` footer (the post-collapse four-[E] artefact) and inherits the shared `tfpt_style` header/footer with the version stamp; the box-orphan `needspace` guards are kept.
- **Fewer boxes, embedded in text.** Narrative/commentary callouts converted to run-in bold paragraphs (boxed one-line formulas to `keyeq`): `tfpt_3` from 39 to 27 boxes (e.g. “Purpose and discipline”, the five “(i)–(v)” audit-lemma boxes, “Net”, “The connection in one sentence”, “The unification”, “The question and the honest answer”, “Net: two axioms”); `tfpt_4` from 18 to 7 (the seven-box Koide stack and the two dark-matter conjecture boxes become prose). The per-item “Honest status of ...” keyboxes are kept — `tfpt_4` is the status authority for the frontier.

- **Colour-marked module references.** `\vref` applied to the inline `vN` references in `tfpt_3` and `tfpt_4` (TikZ node names excluded).
- **Marker hygiene.** A stale literal `[P]` in `tfpt_4` (Koide flow-time) replaced by the 4-class `[C]`; status tables checked against the ledger.

## 2026-06-12 (tfpt\_1 / tfpt\_2 style alignment + box reduction)

- **Shared header/footer everywhere.** `tfpt_1_architecture_e8` dropped its custom `\footmarkers` footer (which after the marker collapse showed four redundant `[E]`) and now inherits the shared `tfpt_style` header/footer with the version stamp, identical to every other document.
- **Fewer boxes, embedded in text.** The narrative/commentary callouts of both documents are converted to run-in bold paragraphs (and the boxed one-line formulas to the colour-marked `keyeq`): `tfpt_1` from 21 to 14 loud boxes (+3 `keyeq`); `tfpt_2` from 52 to 43 (e.g. “In one sentence”, “The one-paragraph version”, the four “Sharpening” notes, “What the simplification means net”, “The disciplined main statement”, “What improves...”, “Scope and honesty”). Only genuine results / theorems / gates keep a box.
- **Colour-marked module references.** `\vref` now also colours the inline `vN` references throughout `tfpt_1` and `tfpt_2` (TikZ node names accidentally matching the pattern were excluded).
- **Status currency.** `tfpt_2`’s inflation-pivot table gains the `v86` scalaron-reheating point ( $N_\star=51.4 \Rightarrow n_s=0.9611, r=0.0045, A_s$ -disfavoured) and states the band-of-record explicitly, matching `COSMO.NSTAR.01` and the introduction; the box is re-typed `[C]`. The other status tables were checked against the ledger and are current under the 4-class markers.

## 2026-06-12 (marker simplification to 4 classes + status reconciliation)

- **Four-class display markers.** The reader-facing marker system is collapsed from eleven symbols to four: `[E]` exact / machine-proven (identity, Lie/lattice, formalised, numerical FP), `[C]` conditional (physical / bridge / readout under named hypotheses), `[O]` open / axiom, and `[X]` kill test. `tex-artefacts/tfpt_style.tex` defines `\mE/\mC/\mO/\mX` and keeps the legacy `\tI ... \tX` names as aliases that render their class. All marker call-sites across the nine documents and the shared fragments were rewritten, collapsing same-class combinations (e.g. `[I]/[L] \rightarrow [E]`, `[N]/[P] \rightarrow [E]/[C]`). The fine-grained per-claim type (Axiom / Formal / Lattice / Numerical / Identity / Physical) is unchanged in `status_ledger.csv` — the single source of truth — so no fidelity is lost. The marker key, badge bar, README marker table and the workflow rule were updated to match.
- **Status tables reconciled against the ledger.** The introduction’s inflation/ $N_\star$  rows are made consistent with `v86/COSMO.NSTAR.01`:  $r$  and  $n_s$  are shown as the frozen *bands* over  $N_\star \in [50, 60]$  with the scalaron-reheating conditional point  $N_\star=51.4$  ( $n_s=0.9611, r=0.0045$ ), replacing the stale point values  $N_\star=55$  (predictions table) and  $N_\star \simeq 57$  (residual card); the closure-ledger inflation row is re-typed `[E] \rightarrow [C]` (conditional on  $M_{\text{Star}}=M_{\text{scal}} + \text{reheating}$ ), matching the other two tables and the ledger `[P]` typing.

## 2026-06-12 (version system, predictions/K5, readability pass B3)

- **Authors, auto-date and auto-version on every paper.** New single source `tex-artefacts/version.tex` defines `\TFPTauthors` (Stefan Hamann, Alessandro Rizzo), a curated `\TFPTversion`, and a title/footer date stamp (`\today + version + revision`). `build.sh` writes the auto build revision (git commit count) into the gitignored `tex-artefacts/version-auto.tex`, so every document’s title page and page footer carry

author, date and v5.2 (rev N) automatically. All nine active documents and `changelog.tex` now set `\author{\TFPTauthors}`.

- **Builder keeps build artefacts.** `build.sh` no longer deletes the `.aux/.log/.out/.toc` after a successful build (the `.log` is needed for the overfull-box audit and the `.aux/.toc` for correct cross-references on the next incremental build).
- **Predictions table outsourced.** The running/upcoming-experiment predictions table is moved from `introduction.tex` into the shared fragment `tex-artefacts/predictions.tex` (`\input` in the predictions section).
- **K5 figure fixed.** The `\TFPTactionkfive` mnemonic ( $K_5$  on the five carrier slots) is redrawn so the complete graph and the Pascal-row reading list sit side by side instead of overlapping.
- **Colour-marked script references.** New `\vref` macro colours inline module references (e.g. v108–v113, v49/v71) in the introduction, predictions and verification fragments, matching the blue of the `\veri` citation, so machine-checked references are visible at a glance.
- **Lighter, non-breaking, embedded boxes.** The `tcolorbox` families are restyled to a light tint + thin frame + light title band with dark coloured title (no heavy white-on-saturated bars); a neutral `navbox` carries the document map and a colour-marked `keyeq` sets off load-bearing display formulas. The introduction’s callouts use the non-breaking variants (`keyboxnb/winboxnb/gapboxnb`) so no box splits across a page, and the shared status card is non-breaking.
- **Readability / sectioning.** The dense run-in prose of the introduction is broken into `\subsection*` headings (In plain words; one anchor and  $E_8$  as a compiler hull; the companion documents; the compact status formula; the hard reduction; the Pascal action grammar; the closure ledger) with added whitespace.
- **Orphan results linked to scripts.** The Glue theorem, the “Reduction in one line” and the fermion-spectrum callouts now carry inline `\veri{}` citations (v1, v6/v14/v23, v18/v20/v46).

## 2026-06-12 (tex-artefacts refactor + introduction visual pass B2)

- **Shared imports moved to tex-artefacts/.** The shared, imported-only TeX fragments (`tfpt_style.tex`, `tfpt_figures.tex`, `tfpt_docset.tex`, `tfpt_status.tex`) now live in a dedicated `tex-artefacts/` folder; all nine active documents `\input` them via `tex-artefacts/<name>` (and `tfpt_style` loads `tex-artefacts/tfpt_figures`). `changelog.tex` stays at the repo root because it is also a standalone `build.sh` notes deliverable. The `make_manifest.py` TEX list and the workflow rule were updated to the new paths.
- **Master script index outsourced (tex-artefacts/verification.tex).** The long `vN→check` table is moved out of `introduction.tex` into the shared fragment `tex-artefacts/verification.tex` (`\input` in the verification appendix); the paper-consistency audit now also reads that file, so every registered script stays cited exactly once.
- **Introduction box reduction (B2).** The introduction’s eleven `keyboxes` are reduced to the single headline “In one sentence” card; the explanatory `keyboxes` (*In plain words*, *Sharper still*, *compact status formula*, *hard reduction*, *Cosmology Pascal*,  $E_6 \times A_2$ , *closure ledger*, *Plausibility balance*) are converted to inline run-in paragraphs, the *Glue theorem* is re-typed as a `winbox` (a result, not a claim), and the explanatory *plausible/follows next* `winboxes` become prose. The loud boxes now carry only genuinely load-bearing callouts (one `keybox`, five `winboxes`, one `gapbox`) plus the two shared orientation cards.
- **Document-count consistency.** The introduction’s document map gained the missing `tfpt_5_redteam` row; “seven/eight companion documents” and the ledger “six documents”

wording are corrected to the true count (nine active documents); `README.md` (“10 ok”, 140 scripts) and the workflow rule (“9 docs”) are reconciled.

## 2026-06-12 (visual pass: box reduction B1)

- **Box reduction (B1).** The 26 non-load-bearing aside boxes (audit / recorded / interpretation: `auditbox/audbox/intbox/resbox/roadbox`) across `tfpt_1`, `tfpt_2`, `tfpt_3` and `origin_theory` are converted to inline bold run-in paragraphs — the reviewer’s prescription that an audit fingerprint must not carry the same visual volume as a theorem. The loud box families are now only `keybox` (claim), `winbox` (result) and `gapbox` (open/caveat).

## 2026-06-12 (visual pass: shared figures, badges, marker typology)

- **Shared figure library `tfpt_figures.tex`** (\input by `tfpt_style`): reusable TikZ macros for the architecture *claim stack* (`\TFPTclaimstack[zone]`, B3), the *anchor cockpit*  $a = (1, 1, 2)$  (B4), the  $E_8$  *glue* diagram (B5), the  $2/3$  *motif map* (B6) and the  $K_5$  *action lattice* (B7). The claim stack and a marker *badge bar* are now shown near the front of *every* document, each highlighting that document’s layer; the topical figures are placed in their natural homes (cockpit +  $K_5$  in the introduction, cockpit + glue in `tfpt_1`, motif in the horizon note).
- **Marker typology (A2).** `tfpt_style` adds the sharper sub-types `[E]` (exact algebra), `[E]` (exact number), `[C]` (readout / standard physics in TFPT units) to the existing `[C]` (bridge) and `[X]` (kill test); the marker key and the new `\TFPTbadges` bar show the full typology. (Inline body markers retain `[E]` where they are genuine exact identities.)

## 2026-06-12 (review pass on list.txt)

- **Claim hygiene (review section A).** The introduction’s one-sentence claim is softened from “derives all constants” to “constructs a discrete compiler; physical readouts run through named, status-typed bridges”; the ledger zombie in `FLAV.QRATIO.01` is removed (the statement no longer says “stays [P]” — ratios are `[E]` closed on the derived stratum, only the absolute scale stays `[O]`, with an explicit forbidden-wording note); the seam misalignment is written explicitly as  $\varepsilon = \frac{3}{4}\varphi_0^{\text{ret}} = c_3 + 36c_3^4$  (leading  $c_3$ , fourth-order puncture correction exact);  $\theta_{12}$  has a single prediction of record  $0.306747$  with the seam/non-linear values typed as scheme diagnostics; JUNO’s first 59.1-day result ( $0.3092 \pm 0.0087$ , compatible, precision pending) is recorded and NuFIT 6.0 is marked the pre-JUNO baseline; the ACT DR6 birefringence hint carries a systematics caveat (not a validation target); CODATA-2022 is dated; the null-model figure is moved out of the main text and typed as a grammar-internal bound. “Complete solutions of the five open questions” becomes “Closure status of the five former open questions”.
- **Interface language unified (review sections A5/C2/C3/C6/C7/C8).** The live residual is stated everywhere as  $v_{\text{geo}} \oplus G_{\text{net}} \oplus F_{\text{transfer}}$  (historical  $U_{\text{wall}}/G_{\text{metric}}/F_{\text{frontier}}$  kept only for ledger continuity); `tfpt_research_contracts` is retitled and its recommended order rewritten (selector-triangle pairings  $\rightarrow v_{\text{geo}} \rightarrow G_{\text{net}}$ );  $F_{\text{transfer}}$  is named as one missing functor with  $\text{Koide}/\eta_B/\text{axion}/m_p/m_e$  as its four instances; full QG closure is framed as a certification layer, not a prerequisite. Document numbering aligned with file names (the Red Team paper is now in the canonical set; document  $N \equiv \text{tfpt}_N$ ). Website mirrored throughout (`papers.ts`, `predictions.ts`, `faq.ts`, `glossary.ts`, `OpenGates`, `Hero`, orientation components).

## 2026-06-12 (later)

- **Shared style and central status artifact.** Two new single-source fragments: `tfpt_style.tex` (the canonical palette, status markers, marker key, `\veri` reference, the four `tcolorbox` families with back-compat aliases, and one running header/footer) is

now \input by all eight documents — the per-paper duplicated preambles are gone; and `tfpt_status.tex` (“TFPT status at a glance”) gives one canonical overall-status card imported near the front of every document, replacing the divergent per-paper status statements. Both registered in `build.sh` and the manifest `TEX` list.

2026-06-12

- **Margin theorem (v122).** The established frozen selectors ( $D_4$  annihilator  $n = (5, -9, 6)$ ,  $\det R = 8$ ,  $\det K = 4$ ) pin  $R$  uniquely among hexagon matrices ( $64 \rightarrow 12 \rightarrow 1$ ); the atom margins become theorems — H2 introduces no new residual class.
- **Inventory update (v123).** Residual table re-pinned post v110–v122: exactly five structural classes (R1 clock, R2 index-4 net, R3 EH step, R4' selector establishment — replacing the discharged H2 dictionary — and R5  $Q$  realisation); R2 carries three loads (metric + carrier + QBL: one theorem, three doors).
- **Resummed clock + glue Q-system (v124/v125).** R1's bend in closed form:  $\text{rate}(n) = -p_2 \ln(1 - n/N_{\text{fam}})$  (pole at  $N_{\text{fam}}$  forces the three-level spectrum); the index-4  $\mu_4$  extension exists explicitly as a Longo  $Q$ -system on  $\mathbb{C}[\mathbb{Z}_4]$  (all Frobenius axioms exact) — R2 sharpens from “find” to “identify”.
- **Clock–wall bridge (v126) and ring resummation (v127).** The resummed-clock weights are the Mehta–Seshadri parabolic weights ( $\text{spec } A_0^*$ );  $\det(\mathbf{1} - A_0^*) = \frac{2}{9}$  = the entropy curvature; the clock is a log-determinant (RPA rings, one tower per hexagon site, coupling = parabolic weight).
- **Graded hull (v128).**  $E_8$  is a  $\mathbb{Z}_4$ -graded Lie algebra over the carrier (grading = glue  $Q$ -system; exact on all 6720 root-pair sums); ring multiplicity 6 = sheet-doubled Ginsparg–Perry zero modes. *Notation fix in the same round:* the clock factor 6 is  $p_2(a)$ , not  $2p_2$  — labels corrected across modules, ledger, papers, website (numbers unchanged).
- **Entropy power law  $\rightarrow$  zeta budget (v129–v133).** The clock is  $\Gamma_n \propto (S_n/S_{dS})^{p_2}$  (Gibbons–Hawking form);  $p_2 = 2h$  with  $h = N_{\text{fam}}$  from mode counting + the Born rule (v130); the zero-mode norm is the area,  $\|Y_{1m}\|^2 = A/(4\pi)$  exactly (v131); the det-ratio anomaly is the Koide constant,  $\zeta(0)|_{\det'} = -\frac{2}{3}$  (v132); both  $\zeta(0)$  budget routes computed exactly — the reduced seam route carries pure anchor ratios ( $-\frac{4}{3}$  = minus the seed gain), the naive 4d route ( $-\frac{109}{45}$ ) carries no atom (v133).
- **Dual anchor + determinant surface (v134/v135).**  $d := a^\top R^{-1} = a^\top L^{-1} = (-\frac{1}{2}, -\frac{1}{2}, 1)$  with  $(1, 1, -2) = -2d$  — the inverse flavor response is the Nariai root, invariance exact via Sherman–Morrison (third algebraic flavor–horizon bridge leg);  $\det M(s, t) = 3s(t+1)(t+2) + t^2 + 5t + 8$  with iso-volume walls at the  $\mathbb{Z}_2/\mu_4$  atoms, winding line  $(8, 14, 20)$ ,  $\det F = 32$ , row-budget axes with  $\text{rowsum}(V) = \text{Spec}(Q_+)$ ; micro-identities  $41 = 16 + 25$ ,  $52 = 16 + 36$ .
- **Dual-normal selector,  $Q_+$  cohomology, Volkov–Wipf firewall (v136–v138).**  $(d, n)$  pins  $R$  *columnwise* (each column the unique lattice point of the address box; kernel  $(6, 8, 7)$ ); the  $A_0^*$  zero mode carries the spectral normal  $\sigma = (2, -9, 5)$  with a  $W(A_3)$  orbit control excluding the naive lift;  $H^1(\mathbb{P}^1 \setminus \mu_4) = \chi_1 \oplus \chi_2 \oplus \chi_3$  with degrees  $(1, 2, 3) = \text{Spec}(Q_+)$  = the  $A_3$  exponents and cusp weights =  $\text{spec } A_0^*$ ; the external full-4d graviton/ghost value  $-\frac{98}{45}$  is pinned (arXiv:2506.02142 = Volkov–Wipf hep-th/0003081) and its *edge* part is exactly two copies of the reduced seam budget — R1 typed closed as an edge-sector object.
- **Selector triangle + canonical map (v139/v140).**  $d = \frac{3}{2}a - 2 \cdot \mathbf{1}$  is pure anchor data (first selector *derived*); the frame  $(\mathbf{1}, a, \sigma)$  has volume 11 and  $n$  is the unique covector with atom pairings  $(2, 8, 121)$  — the R4' residue is three pairings; the  $H^1 \rightarrow$  generation equivariant map exists and is Schur-rigid (only sign-twisted  $\mu_4$  actions match), so the `GATE.QGEO` residue collapses to one  $\mathbb{Z}_3$  choice.

- **Editorial and consistency passes.** Content-currency sweep (superseded “single remaining input” claims retired across contracts, paper 3, intro, website); introduction: self-explanatory title (spells out Topological Fixed-Point Theory), the old-version comparison removed throughout, redundant diagrams dropped, the unreadable status heatmap removed, the script-index master table converted to a page-breaking `xltabular` (it had silently truncated  $\sim 110$  rows), status card extended with the v134–v140 reductions, predictions table completed (CP phase, cosmic birefringence); the canonical document-set box extracted into the shared `tfpt_docset.tex`; paper 1 cleaned of all 35 old-series references, every section now opens with its machine-check script references; smaller box fonts across all eight documents; this `changelog.tex` introduced as a dual-mode (standalone + imported) document.

## 2026-06-11

- **External-review integration (v83) and release hygiene.** Manifest `-check` mode and the release rule; holomorphy pins  $(E_8)_1$  (the unique even unimodular rank-8 lattice theory).
- **Sheet diamond and centered form (v94/v95).** All four flavor operators are points of one two-parameter surface  $M(s, t)$ ; centered cross  $R, L = C \mp U$ ,  $K, F = C \mp V$  with the cofactor seam normal  $n = (5, -9, 6)$ ; documentation pass (“The Sheet Diamond and the Winding Line” in paper 3).
- **Horizon/anchor round (v101–v103).** The maximal (Nariai) black hole is the anchor: roots  $(1, 1, -2)$ , entropy bound  $\frac{2}{3} = |\mathbb{Z}_2|/N_{\text{fam}}$ ; one repeller/attractor orientation in both sectors; trisection normal form (the canonical cosine coordinate).
- **Clock/ladder round (v104–v108) + Lean hardening.** The classical Nariai clock speaks anchor ( $\chi_{\text{clock}} = (\lambda - 1)(\lambda + 2)$ ); machine-pinned residual inventory; first review validation; quantum-clock target made quantitative; Pascal Ladder Theorem ( $g = 2K + 1 \Leftrightarrow$  the closure).
- **QBL theorem chain (v109–v113).** Sheet-Pairing Lemma; Calderón-sheet selection (a scalar seam datum exists iff the involution is sheet-odd); Quadratic-Transport Theorem (degree exactly 2); Self-Counting Channel (the Pascal closure as an identity); quasi-free kernel (one 2-point kernel determines the whole net,  $\text{rank} = c$ ) — the QBL input merges with the R2/holomorphy premise; communicated across all status surfaces.
- $U_{\text{wall}}$  **made exact (v114–v118).** Torsion normal form and  $\delta = \frac{1}{2}$  theorem; exact anchor residue  $A_0^*$  ( $\text{diag} = a/|\mu_4|$ , off-weights  $(8, 0, 5)/144$ ); resonance theorem ( $M_\infty = \mathbf{1} \Leftrightarrow a_{13} = 0$ , one gauge orbit); the holonomy is the exact 24-element  $W(A_3) = S_4$ ; the hypercharge hexagon is the sign-twisted family spectrum with the lepton coefficients as resolvent determinants.
- **Second review validation + address pinning (v119–v121).** Anchor ratio triad  $(\frac{2}{3}, \frac{4}{3}, \frac{5}{3})$ ; the 121 audit lemma; the lepton words are the compiler atoms  $(8, 5, 3)$  with divmod-hexagon addresses;  $R$  is the unique hexagon matrix with atom margins and  $\det 8$  — the address table carries zero residual information.

## 2026-06-10

- **B/C/D round closed.** Koide attractor toy model (v93), glue uniqueness (v92), the Wolfram extension as the independent second verification path, Lean hardening; papers and website synced with the round.
- **Content rounds v96–v100.** Sheet conjugation bridge, discriminant dictionary, Koide flow time, and the look-elsewhere-corrected numerology null test (v100: joint null probability  $\leq 10^{-30.7}$ , conditional on the declared grammar).

## 2026-06-09

- **Blind registry frozen (v84).** Machine-enforced prediction registry (`predictions_frozen.json`): exactly one  $\theta_{12}$  number of record before JUNO; conditional entries carry their hypotheses by name. Research-contracts document added to the set.

## 2026-06-07 / 2026-06-08

- **Compiler-closure consolidation.** The TFPT development is consolidated into the current document set with the verification suite (modules v1–v82:  $E_8$  glue, carrier Pascal, EM fixed point, flavor matrix, cascade, bootstrap, gravity/cosmology, transport, masses, registry and gate audits) as the single proof layer; manifests (`manifest.sha256`, `lean_manifest.sha256`) as content identity; website mirrors the suite (interactive DAG, in-browser script reproducer, script index).

## 2026-04-27

- **Initial commit.** TFPT papers and website baseline: electromagnetic closure and flavor transport (UFE bridge, birefringence seed, dyonic intercept), PDF-download tracking, accessibility and metadata passes.